SGTX PLATFORM BLUEPRINT

Implementation Edition — *v6.3*

Full, expanded and fully detailed retype — tenant-centric, zero-cost, sovereign trade operating system

Confidential — Internal Implementation Reference

SGTX PLATFORM BLUEPRINT – IMPLEMENTATION EDITION v6.3 – TABLE OF CONTENTS

Part 0: Executive Summary & Core Governance Philosophy

0.1 Mission Statement

0.2 Core Governance Philosophy

0.3 Key Principles (G1–G4)

0.4 Example Enforcement Flow

0.5 SGTX FEES Supremacy Rule & Container Release Control

0.6 USTN Format – Deterministic & Company-Anchored

0.7 Competitive Landscape & Moat (Zero-Cost Stack)

0.8 Zero-Cost Technology Commitment

0.9 Implementation Checklist

0.10 Part 0 Status

Part 1: Constitutional & Governance Layer

1.1 Constitutional Layer (ZeroCost WasmEdge + OPA)

1.2 AI Authority Levels (Three-Tier Zero-Cost Strategy)

1.3 Human Authority Hierarchy

1.4 Jurisdiction Supremacy Rule

1.5 Governor Service – Single Point of Truth

1.5.1 Cryptographic Enforcement Layer

1.5.2 Replay Protection

1.6 ExecutionGate (WasmEdge)

1.7 Legal Disclaimer

Part 1 Implementation Checklist

Part 2: Identity, Tenants & Internal Authority

2.1 Global Trade Identity (GTID)

2.2 GTID Resolution Service

2.3 Tenant Types & Onboarding Flow

2.4 Permission System – Action-Based OPA Enforcement

2.5 Icon Shuffle & Visibility Rules

2.6 Saved Contacts / Network Feature

2.7 Tenant Onboarding Wizard & Sandbox

2.7.1 Onboarding Wizard Flow

2.7.2 Sandbox Environment

2.7.3 “Skip for Now” & Reminder Banner

2.7.4 Draft Auto-Save & Expiry

2.7.5 Backend Integration & API Endpoints

2.8 Dual-Mode Toggle for Trading Companies

2.9 Internal Tenant Organization Graph

2.10 Tenant Lifecycle Engine

Part 2 Implementation Checklist

Part 3: End-to-End Workflow with Governance Overlay

3.0 Governance Overlay

Phase 1: Trade Initiation (Buyer Portal) – Structured Container-Level Request Form

Step 1.1 Direct GTID Entry

Step 1.2 Structured Container & Commodity Entry

Step 1.3 AI Container Advisor

Step 1.4 Marketplace Relationship Detection

Step 1.5 Multi-Shipment Request (Buyer Option)

Step 1.6 Governor PreScreen & Submission

Step 1.7 Draft Auto-Save

Phase 2: Seller Quote, Packing & Logistics Orchestration

Step 2.1 Loading Origin

Step 2.2 EXW Price Lock

Step 2.3 Logistics Cost Entry (Manual or RFQ)

Step 2.4 Alternative Delivery Ports

Step 2.5 Packing & Containerisation

Step 2.6 Seller Responds to Multi-Shipment Request

Step 2.7 SGTX Fee Calculation & Display (Paid by Seller, Added to Quote)

Step 2.8 Submit Quote Package

Phase 3: Contracting, Negotiation & Fee Collection

Step 3.1 Buyer: Accept, Negotiate, or Amend

Step 3.2 Seller Response to Amendment/Negotiation

Step 3.3 Mutual Confirmation

Step 3.4 Contract Assembly with SGTX Witness Clause

Step 3.5 Upload Own Contract (with SGTX FEES Addendum)

Step 3.6 Multi-Shipment Per-Shipment Fee Collection

Step 3.7 Logistics Providers Sign Addendum

Step 3.8 Digital Signatures & Contract Lock

Step 3.9 Container Release Confirmation

Phase 4: Universal Trade Finance

Step 4.1 Financing Request Initiation

Step 4.2 AI Credit Intelligence

Step 4.3 Financier Preferences & Automatic RFQ Broadcast

Step 4.4 Full Disclosure to Financiers

Step 4.5 Bidding & Co-Financing

Step 4.6 Financing Agreement & SGTX Witness Clause

Step 4.7 Financing Fee Collection (0.25% Deducted at Disbursement)

Step 4.8 Repayment & Release

Step 4.9 Financier Historical Data

Step 4.10 DeFi Option

Phase 5: Physical Execution & Multiparty Tracking

Step 5.1 Loading Window Selection

Step 5.2 Booking Confirmation & AI Extraction

Step 5.3 Dynamic Document Requirements

Step 5.4 Granular Provider Status Tracking

Step 5.5 Provider Invoice Upload & AI Comparison

Step 5.6 Barcode Scanning & Predictive Reliability

Step 5.7 Stuck Trade Recovery (Cross-cutting)

Phase 6: Settlement & Payment Orchestration

Step 6.1 Settlement Instruction Generation

Step 6.2 AI PSP Router

Step 6.3 Buyer Approval (Voice, Smart Inbox)

Step 6.4 Payment Execution & Monitoring

Step 6.5 AI-Powered Reconciliation

Step 6.6 Multi-Shipment Settlement

Phase 7: Distressed Cargo Resolution

Step 7.1 Proactive Demurrage Alert

Step 7.2 Declaration of Distressed Cargo

Step 7.3 AI Condition Assessment

Step 7.4 Dynamic AI Pricing Engine

Step 7.5 Triage Dashboard (Three Paths)

Step 7.6 “Check Buyers” Advisory

Step 7.7 Partial Distress & USTN Splitting

Step 7.8 Microcontract & Separate SGTX Fee

Step 7.9 Accelerated Outreach Mode

Step 7.10 AI-Mediated Express Negotiation

Step 7.11 AI-Insurer Integration

Phase 8: Distressed Cargo Outreach

Step 8.1 Access Distressed Listing

Step 8.2 “Check Buyers” (Advisory)

Step 8.3 Select Recipients

Step 8.4 Choose Outreach Mode & Privacy Notice

Step 8.5 AI-Composed Notification

Step 8.6 Send & Track Notifications

Step 8.7 Recipient Response & Microcontract

Phase 9: Global Payment Orchestrator & Fee Collection Infrastructure

Step 9.1 Fee Collection Trigger & Responsibility

Step 9.2 AI-Driven PSP Router

Step 9.3 Dynamic Fee Gross-Up Calculation

Step 9.4 Payment Initiation & Webhook Verification

Step 9.5 PSP Health Monitoring & Automatic Fallback

Step 9.6 OpenBanking Reconciliation

Step 9.7 Voice-Activated Fee Payment Approval

Phase 10: Dispute Resolution & Arbitration

Step 10.1 Filing a Dispute

Step 10.2 AI Triage & Categorisation

Step 10.3 Evidence Package Autocompiler & External Verification

Step 10.4 Predictive Dispute Outcome

Step 10.5 Mediation Log & Multi-Language Support

Step 10.6 AI-Generated Settlement Proposal

Step 10.7 Smart FeeLock Management

Step 10.8 Document Authenticity Check

Step 10.9 Escalation to International Arbitration

Step 10.10 SGTX FEES Dispute Workflow

Step 10.11 QC Dispute Fast-Track

3.6 Stuck Trade Recovery & Draft States

3.7 Negotiation Bot Deadlock Resolution

Part 4: Expanded Microservices Architecture

4.0 Architecture Overview

4.1 Core Service Definitions (47 services, including v6.3 new: inbox-service, tenant-experience-service, workflow-recovery-service, tenant-data-export-service, financier-preference-service, multi-shipment-coordinator, fee-split-orchestrator)

4.2 Zero-Cost Self-Hosted Implementation Matrix

4.3 Inter-Service Communication Map

4.4 Deployment & Scaling on Bare-Metal K3s

4.5 Implementation Checklist

Part 5: Comprehensive Data Model Specifications (Full DDL)

Enums, Identity Layer, Corporate Graph, Governance, Trade Core (including structured container and multi-shipment tables), Packing & Containerisation, Shipments & Milestones, Documents, Financing & Settlement (co-financing, financier preferences), Payment Orchestration, Distressed Cargo, Disputes & Evidence, Logistics Addenda & Container Release, Smart Inbox, Jurisdiction & Ports, Marketplace Partners, Export Jobs, Indexes & RLS

Part 6: User Interfaces & Portals

6.1 Common Shared Components

6.1.1 Smart Inbox (Action-Driven Dashboard)

6.1.2 Trade Command Center (Unified Per-USTN View)

6.1.3 PlainLanguage Governor Decision Panel

6.1.4 Tenant Data Export & Reconciliation Statements

6.1.5 [Removed – Operating Mode Feature Matrix]

6.1.6 Shared Shipments Vault

6.1.7 Accessibility Standards (WCAG 2.2 AA)

6.1.8 Tenant Exit & Data Portability Center

6.1.9 Guided Recovery System

6.1.10 SLA Transparency & Human Escalation Directory

6.1.11 Global UX Rule – The Four Questions

6.1.12 Dual-Mode Toggle (Buyer/Seller Switch)

6.2 Portal-by-Portal Breakdown

6.2.1 Buyer Portal (Structured Container Form, Multi-Shipment Request, Dual-Mode Aware)

6.2.2 Seller Portal (Loading Origin, Manual Logistics Entry, Alternative Ports, Multi-Shipment Response, SGTX Fee Added to Quote)

6.2.3 Logistics Portal (Role-Based Views, Addendum Signing, Container Release Confirmation, Unsubscribe from Anonymous RFQs)

6.2.4 QC Portal (Dynamic Quality Specs, AR Inspection, Override Accountability, AQL Sampling)

6.2.5 Financier Portal (Full Disclosure, Co-Financing, Preference Filters, Collateral Dashboard, Historical Data)

6.2.6 Government Portal (Dynamic Module Configuration, Anonymous Trade Management, Customs API Integration)

6.2.7 Admin Portal (Multisig, Policy Simulation, Natural Language Audit, Configuration Rollback, Tenant Impersonation)

6.2.8 MultiTenant Administration (Employee Invitation, RBAC, Data Scopes, Approval Chains, Tenant Activity Log, Branding, Configuration Versioning)

6.2.9 Marketplace Partner Portal (Rate Limit Gauges, Sandbox, Revenue Share Agreement, Webhook Management)

6.3 Mobile Application (React Native + Expo)

Offline-first, barcode scanning, voice commands, AR inspection, offline queue, multi-shipment aware

6.4 Desktop Application (Tauri v2)

System tray, native notifications, local file printing (ZPL), auto-update, deep links

6.5 Cross-Cutting Features

Self-Sovereign Identity Backup, One-Tap Smart Escrow Claim, Anonymous Market Intelligence Panel (opt-in, differential privacy), Voice Language Matrix & TTS Fallback, Biometric Stress Flag Consent, Non-Marketplace Footer, Role-Specific Value Cards, Timezone & Holiday Adjustment, Emergency Operations Mode, Operating Complexity Score

6.6 Development Setup & ZeroCost Stack

6.7 Implementation Checklist (User Interfaces & Portals)

Part 7: Revenue Model & Fee Structure

7.0 Revenue Philosophy

7.1 SGTX Trade Fee (Dynamic, Commodity- & Country-Pair-Specific, Paid by Seller, Added to Quote)

7.2 Financing Fee (Flat 0.25%, Deducted at Disbursement)

7.3 Marketplace Partner Revenue Share

7.4 AIDriven Fee Forecasting & Rate Simulation

7.5 Distressed Cargo Fee Adjustment

Part 8: Legal, Compliance & Risk Framework

8.0 Legal Foundation

8.1 Multi-Layer Legal Architecture (including Logistics Addendum, SGTX FEES Addendum)

8.2 Compliance Framework (RIA, Sanctions & PEP Screening, AML/KYC, Distressed Cargo Compliance, Regulatory Reporting)

8.3 Risk Framework (10 dimensions)

8.4 Data Privacy & Sovereign Residency

8.5 AI Governance & Explainability

8.6 Contractual Architecture & Smart Clauses

8.7 Dispute Resolution

8.8 Emergency Multisig Override Procedure

Part 9: Hybrid DeFi & Crypto Financing Framework

9.0 Philosophy

9.1 DeFi Eligibility & Dynamic Country Matrix

9.2 Protocol Risk Management

9.3 DeFi PlainLanguage Risk Summary (Mandatory Modal)

9.4 AI-Orchestrated Stablecoin Liquidity Aggregator

9.5 Stablecoin Management & Depeg Detection

9.6 Cross-Chain Interoperability & Bridge Risk Scoring

9.7 Tokenized Trade Assets & Secondary Market

9.8 DeFi-Native Fee Handling via Smart Escrow

9.9 Predictive Liquidation Protection

9.10 Zero-Knowledge Proof of Reserves

9.11 Automated FATF Travel Rule Compliance

Part 10: Global Payment Orchestrator & Settlement Infrastructure

10.0 Non-Custodial Foundation

10.1 Architecture Overview

10.2 AIPowered Dynamic PSP Router

10.3 Trade Principal Settlement

10.4 SGTX Fee Collection (Trade & Financing)

10.5 Multi-Rail Redundancy & Automatic Fallback

10.6 OpenBanking Self-Reconciling Ledger

10.7 Zero-Knowledge Proof of Payment

10.8 Voice-Activated Operations

10.9 Global Regulatory Compliance

Part 11: AI Intelligence Layer – Deep Dive

11.0 AI Philosophy & Governance

11.1 AI Authority Ladder & Three-Tier Routing (Groq, HF local, Ollama)

11.2 AI Infrastructure Stack

11.3 The 7-Layer AI Architecture

11.4 AI Agent Registry (Updated)

11.5 Deep Dive: Key AI Systems (Commodity-Specific Margin Estimator, RIA, Cold-Chain, Digital Twin, Negotiation Bot, Smart Inbox Text Generator, Tenant Message Generator)

11.6 Explainability & Model Drift

11.7 Tenant Health Engine

Part 12: Sovereign Tech Stack & Deployment Topology

12.0 Sovereign by Design

12.1 Technology Stack (Languages, Microservices, Data Stores, Orchestration, AI, Cryptography, Observability)

12.2 Infrastructure & Deployment Topology (Global Sovereign Nodes, Edge/IoT, Load Balancing, Scaling)

12.3 Development Environment

12.4 Cryptography & Key Management

12.5 Backup & Disaster Recovery

12.6 Regional Adaptations & Global Compliance

12.7 Security & Hardening

12.8 Observability & AIDriven Operations

12.9 Implementation Checklist

Part 13: Governor Service – Authoritative Design Specification

13.0 Overview

13.1 Architecture & Core Components

13.2 OPA Policy Engine (including dual-mode, multi-shipment, financing, distressed policies)

13.3 WasmEdge Constitutional Engine

13.4 AI Integration (Three-Tier, with Fallback)

13.5 Tenant Message Generation (v6.3)

13.6 Loom Deterministic Logging

13.7 Configuration

13.8 Non-Negotiable Rules

13.9 Global Dynamic Updates

13.10 Example Decision Record

Part 14: Comprehensive Audit, Observability & Compliance

14.0 Philosophy

14.1 Immutable Audit Architecture (Loom, NATS JetStream, ClickHouse)

14.2 Real-Time Observability Stack (Prometheus, Grafana, Loki, Jaeger)

14.3 Distributed Tracing

14.4 Auditor’s Interface & Compliance Reporting

14.5 Model Drift & AI Audit

14.6 Governance & Verification Gates

Part 15: Zerocost Addons Suite

15.0 Philosophy

15.1 Categorised Module Inventory (62 Addons, Tiers 1–9)

15.2 Implementation Priority Matrix

15.3 Total Cost of Ownership ($0)

15.4 Example Activation Process

Part 16: KYB/KYC & Registration – Aidriven, Jurisdiction-Dynamic

16.0 Purpose & Scope

16.1 KYB/KYC Tiers & Jurisdiction Requirements

16.2 Document Extraction & Verification (Zero-Cost AI Pipeline)

16.3 Biometric Liveness & KYC

16.4 PEP & Sanctions Screening

16.5 Manual Review Fallback & SLA Transparency

16.6 Integration with Onboarding Wizard

16.7 Governance Gates

Part 17: Marketplace API & Revenue Share

17.0 Philosophy

17.1 Architecture & Zero-Cost Stack

17.2 Endpoint Specifications (with rate limits, idempotency, standard errors)

17.3 Revenue Attribution Logic & Auto-Attribution

17.4 Governance Gates

17.5 AIPowered Partner Features

17.6 Implementation Checklist

Part 18: Saved Contacts / Network Feature

18.0 Philosophy

18.1 Core Data Model

18.2 Automatic Contact Saving

18.3 AI-Enriched 360° Trust Portrait

18.4 Predictive Relationship Health Score

18.5 GraphRAG-Powered Indirect Connections

18.6 Smart Labels & Dynamic Grouping

18.7 Organization Memory Layer (Opt-In)

18.8 Voice-Driven Contact Management

18.9 Privacy-Preserving Verifiable Credentials

18.10 Network-Wide Anomaly Alerts

18.11 Portal Integration (Dual-Mode Aware)

18.12 Governance & Privacy Gates

Part 19: Enhanced Commodity & Packing Specifications (Multicommodity)

19.0 Philosophy

19.1 Multi-Commodity Data Representation (Consuming Buyer's Structured Form)

19.2 AI-Driven HS Code Classification & Dual-Use Screening

19.3 Commodity Compatibility & Global Mixing Rules

19.4 Palletisation & Container Loading (ORTools + AI)

19.5 Collaborative Packing Plan Editing

19.6 Container Capacity Heatmap

19.7 Automated Packaging Compliance Checker

19.8 Ecological Packaging Advisor & Carbon Footprint Calculation (CBAM ready)

19.9 Generative AI for Custom Packaging

19.10 Governance Gates

Part 20: Location-Based Trucking & Route Intelligence

20.0 Philosophy

20.1 Core Technology Stack (OSRM, offline, zero-cost)

20.2 AI-Driven Dynamic Route Optimization

20.3 Predictive Dispatching & VRP

20.4 Real-Time Anomaly Detection & Geofencing

20.5 Voice-Guided Turn-by-Turn Navigation (Offline)

20.6 Carbon-Minimising Route Options

20.7 Automated E-Way Bill & Transit Document Generation

20.8 AIPredicted Maintenance Alerts

20.9 Dynamic Loading Dock Scheduling

20.10 Decentralised Proof-of-Location with Consent & Privacy Controls

20.11 Governance Gates

Part 21: Auto-Generated Barcodes for Shipment Tracking

21.0 Philosophy

21.1 Core Technology Stack

21.2 Barcode Generation & Dynamic Content (SSCC18, QR, Verifiable Credential)

21.3 Multi-Modal Pallet Identification (Camera, AI Vision, Voice)

21.4 AR Scan Assistant

21.5 Predictive Scanning Reliability Agent

21.6 Label Print Workflow (ZPL, PDF, Reprint Policy)

21.7 Blockchain-Anchored Barcode Hashes

21.8 Scan-Triggered Milestone Confirmation

21.9 Governance Gates

Part 22: Implementation Roadmap & Scaling Strategy

22.0 Philosophy

22.1 Core Phases (Phase 1 MVP, Phase 2 Scale, Phase 3 Enterprise) – with all v6.3 features

22.2 AIPowered Project Management (Project Oracle)

22.3 Self-Healing CI/CD with Intelligent Rollback

22.4 Predictive Scaling Engine

22.5 Dynamic Geographic Expansion Optimizer

22.6 Zero-Cost Infrastructure Lifecycle Management

22.7 AIGenerated Partnership Proposals

22.8 Continuous Compliance Roadmap Alignment

22.9 Scaling Metrics (Living Dashboard)

22.10 Implementation Checklist

Part 23: Global Country Coverage & Settlement Matrix

23.0 Philosophy

23.1 Core Technology Stack

23.2 Dynamic Coverage Tiers (Full, Standard, Limited, Restricted, Blocked)

23.3 Global Settlement Rail & PSP Matrix (with multi-hop FX)

23.4 Sanctions & Embargoes – Instant Propagation

23.5 Automated Country-Specific Regulatory Reporting

23.6 Distressed Country Factor & Financing Regulations

23.7 Voice-Queried Country Intelligence

23.8 Zero-Cost Country Data Fabric

23.9 Expansion Scoring

23.10 Governance Gates

Part 24: Threat Model & Attack Surface Analysis (STRIDE + MITRE)

24.1 Threat Model Methodology

24.2 STRIDE Analysis Summary (Key Threats & Mitigations)

24.3 MITRE Att&Ck Coverage

24.4 Attack Surface Inventory (v6.3)

24.5 Continuous Controls & Verification

24.6 Residual Risk Acceptance

24.7 Zero-Cost Toolchain

Part 25: SLA & Uptime Guarantees

25.0 Purpose & Scope

25.1 Core Platform SLAs (including Smart Inbox, Tenant Message Generation, Financing Service, Multi-shipment Coordinator)

25.2 Maintenance Windows

25.3 Status Page & Incident Response

25.4 Credit & Compensation Model

25.5 Zero-Cost Monitoring & Enforcement

25.6 Example SLA Incident Flow

Part 26: Glossary & Acronyms

26.1 Core Terms (over 150 terms, including new v6.3 terms: Dual-Mode Toggle, FeeLock, MicroUSTN, Check Buyers, Co-Financing, etc.)

26.2 Acronyms

Part 27: Openapi 3.1 Specification Index

27.0 Overview

27.1 Governor & Platform Endpoints

27.2 Trade & Quote Endpoints (including structured container and multi-shipment)

27.3 Packing & Containerisation Endpoints

27.4 Contract & Fee Endpoints

27.5 Financing Endpoints (co-financing, preferences)

27.6 Shipment & Milestone Endpoints

27.7 Settlement & Payment Orchestrator Endpoints

27.8 Distressed Cargo Endpoints

27.9 Dispute Endpoints

27.10 Network & Saved Contacts Endpoints

27.11 Marketplace Partner Endpoints

27.12 KYB/KYC & Onboarding Endpoints

27.13 Country & Jurisdiction Endpoints

27.14 Admin & Observability Endpoints

27.15 Inbox & Preferences Endpoints

27.16 Tenant Data Export & Statement Endpoints

27.17 Trade Command Center Endpoint

27.18 Dual-Mode & Employee Context Endpoint

Part 28: Appendices

A1: USTN Format Specification (Company-Anchored, No Version)

A2 : Incoterms 2020 Reference Table (Simplified for Cost Rules)

A3 : AI Agent Registry & Authority Mapping (v6.3 – Three-Tier)

A4: Technical Specifications (Key Services)

A5: Distressed Cargo Matrix (Extended, Dynamic)

A6 : AI Provider Fallback Chain (Zero-Cost, No Billing Details)

A7 : Tenant Quick Start Guide (No Operating Modes – Uniform Experience)

A8: Terminal Automation Reference & cd Path Index

Part 29: the Sovereign Trade Operating System – a Manifesto & Living Document

29.1 The Problem: Friction as a Service

29.2 The SGTX Thesis: Invisible Rails, Visible Trust

29.3 Three Unshakable Pillars

29.4 Who SGTX Serves (and Who It Does Not)

29.5 The Zero-Cost Imperative

29.6 The TenantCentric Revolution (v6.3)

29.7 The Governance Compact

29.8 The Future

29.9 A Note to Developers, Auditors, and Future Maintainers

29.10 Closing Words

END OF INDEX

Part 0: Executive Summary & Core Governance Philosophy (v6.3 – Final, Zerocost, Tenantcentric, with All Enhancements)

0.1 Mission Statement

SGTX is a sovereign, AI-governed, non-custodial execution engine for global trade – NOT a marketplace. We provide the infrastructure for organizations to execute cross-border trade with cryptographic certainty, AI-assisted optimization, and zero counterparty risk through non-custodial SGTX FEES protection. Marketplaces integrate via API to leverage SGTX technology, paying a negotiated percentage per successful contract.

Zero-Cost Implementation Note : All integrations use self-hosted OpenAPI 3.1 specs served at /api/v1/openapi.json – no paid API gateways, no SaaS middleware. Partner authentication via ZITADEL (self-hosted, OSS) + Ed25519 JWT signing. Entirely built on a zero-cost, self-hosted stack – open-source software, public APIs, and bare-metal deployment – SGTX carries no licensing fees, no SaaS subscriptions, and requires no billing details whatsoever.

TenantCentric Overlay (v6.3) : Beyond its sovereign architecture, SGTX v6.3 has been purpose-designed for human usability. Every portal now opens to a Smart Inbox – a single, prioritised feed that tells the user exactly what to do next. A Trade Command Center consolidates all information about a trade into one screen. Every compliance block is explained in plain language through a Governor Decision Panel that never exposes technical jargon. A guided Onboarding Wizard and integrated sandbox ensure that new tenants can execute a practice trade within minutes. Trading companies can seamlessly switch between Buyer and Seller dashboards with a dual-mode toggle, while multi-shipment contracts allow per-shipment SGTX FEES collection and flexible amendments without upfront bulk payments. Logistics providers are bound as signatories, and no container is released without SGTX FEES confirmation. All trade documents must include the USTN and relevant GTIDs, ensuring end-to-end traceability.

0.2 Core Governance Philosophy

Organizations are treated as mini-states. Every irreversible action (contract lock, milestone confirmation, settlement) is explicitly authorised by Human + Governor + AI gate. Value is released, never flowed. The platform is non-custodial: it never holds funds, only issues verifiable instructions and locks SGTX FEESs.

Dev Environment Checkpoint : Governor service compiled via cargo build --release on any Linux x64 or macOS. All WASM modules pre-validated in WasmEdge runtime before deployment. No Docker required – native binaries + CRIO runwasi shim.

0.3 Key Principles (G1 to G4)

G1 – Execution Always Gated: Temporal workflow + WASM gate. No irreversible action without it. Enforced by governor_decisions table. Zero-cost: Temporal Core (OSS) self-hosted on K3s; WASM modules compiled with wasmtime compile.

G2 – AI May Block, Never Force: AI returns structured decision object only; never executes. Enforced by ai_inference_records. Zero-cost: vLLM + HF local models + self-hosted Ollama/LocalAI; Rig agents (Rust, OSS) orchestrate.

G3 – Non-Custodial Structural: No fund tables; only SGTX FEESLock + instruction generator. Enforced by SGTX FEES_locks. Zero-cost: NATS JetStream KV (OSS) for lock state; PostgreSQL for audit trail – zero payment processor fees.

G4 – Every Action Attributable: governor_decisions table with cryptographic signature. Event sourcing + audit trails. Enforced by audit_log. Zero-cost: Loom deterministic logging (Rust, OSS) + ClickHouse (OSS) for immutable append-only storage.

0.4 Example Enforcement Flow

An employee attempts to confirm a shipment milestone.

Request → Governor Proxy intercepts → Governor evaluates:

Does the employee have the permission? (OPA)

Is the trade corridor sanctioned? (WasmEdge jurisdiction module)

Is the SGTX FEESLock active? (NATS KV check)

If all checks pass → ALLOW decision, signed cryptographically, logged via Loom. Milestone confirmation proceeds; SGTX FEESLock partially released.

If any check fails → DENY or CONDITIONAL decision with required remediation steps.

0.5 SGTX FEES Supremacy Rule & Container Release Control

Single-shipment contracts : SGTX FEES is due immediately upon contract lock, paid separately by the responsible party(ies) using jurisdiction-aware payment methods.

Multi-shipment contracts : SGTX FEES is collected per-shipment, triggered only after both parties have mutually confirmed readiness for that specific shipment. A new SGTX FEESLock is created per shipment, and the container is NOT released until SGTX issues a formal release confirmation (email/API with a one-time token) after SGTX FEES payment.

Logistics providers are bound as signatories to the trade contract or a dedicated addendum. They must include the USTN and all relevant GTIDs on every document (packing list, invoice, bill of lading, customs declaration).

All payments to SGTX are gross, without deductions. SGTX bank account is located in the USA. The SGTX FEESLock in NATS KV is marked ACTIVE upon payment verification and released as trade milestones are achieved.

Zero-Cost PSP Routing : Payment orchestration uses self-hosted Rust payment-orchestrator service with free-tier PSP SDKs (Stripe Test Mode, Fawry Sandbox, SEPA Test). No live billing details required for dev. Production routing matrix stored in payment_aggregators table – all entries validated against uptime_score ≥ 99% before activation.

Tenant-Facing SGTX FEES Explanation (v6.3) : At the time of contract signing, the SGTX FEES Calculator Preview shows the exact breakdown of the SGTX FEES rate and an interactive slider to adjust the split between buyer and seller. A plain-language message appears: “This small fee secures your trade with cryptographic milestone tracking, AI-assisted logistics, and zero-counterparty-risk settlement. It is paid before contract lock to activate these protections.” For multi-shipment contracts, the message clarifies that SGTX FEES is collected per-shipment.

0.6 USTN Format – Deterministic & Company-Anchored

The USTN (Unique SGTX Transaction Number) is generated at contract lock and incorporates the identities of the involved companies to prevent replay and ensure non-repeatability.

Format : SGTX-{BUYER_GTID_SUFFIX}-{SELLER_GTID_SUFFIX}-{YYYYMMDDHHMMSS}-{RANDOM8}

Components:

SGTX : fixed prefix.

BUYER_GTID_SUFFIX : last 6 alphanumeric characters of the buyer’s GTID (after removing hyphens).

SELLER_GTID_SUFFIX : last 6 alphanumeric characters of the seller’s GTID.

YYYYMMDDHHMMSS : UTC timestamp of contract lock.

RANDOM8 : 8 random alphanumeric characters.

Generation is deterministic and unique. No version suffix is needed because the combination of parties, timestamp, and random ensures uniqueness.

Example: SGTX-1397F3A-456ABC-20260415120000-A1B2C3D4

0.7 Competitive Landscape & Moat (Zero-Cost Stack)

All components are built on free, open-source software, with three AI providers used according to task:

Groq – Fast advisory (A1), Smart Inbox titles, tenant messages, live market suggestions. Limits: 30 req/min, 14,400 req/day. No billing details required.

Hugging Face (self-hosted) – Privacy-critical tasks (A2/A3), document extraction, contract clause analysis, sanctions screening. Unlimited – you run the model on your own server. No external API calls.

Ollama / LocalAI / GPT4All – Fallback for all models. Runs on your server’s CPU/GPU. Unlimited, no external dependencies.

All three require NO billing details, NO credit card, and NO subscription.

Moat Components and Their Zero-Cost Tech Stack

Governor + WASM Constitutional : OPA (Rego) + WasmEdge (OSS) + Rust Axum

SGTX FEESLock Non-Custodial : NATS JetStream KV (OSS) + PostgreSQL 18

AI Authority Ladder (A0–A4): Rig (Rust OSS) + vLLM + HF local models + Ollama/LocalAI

GTID + Saved Contacts : Self-hosted ZITADEL (OSS) + PostgreSQL RLS

Jurisdiction Supremacy : WasmEdge modules + scrapped OFAC/UN/EU lists (free)

Tenant Experience Layer : Smart Inbox (Rust inbox-service), Trade Command Center, Decision Panel (Groq + template fallback)

0.8 Zero-Cost Technology Commitment

Every line of code, every microservice, every AI model, and every deployment artefact is built exclusively on:

Open-source software (Rust, PostgreSQL, NATS, WasmEdge, Temporal, Grafana, Prometheus, etc.)

Free public APIs & data sources (ECB FX rates, OpenStreetMap, satellite data, etc.)

Self-hosted infrastructure (bare-metal K3s clusters, no cloud dependency)

Local AI inference (Hugging Face free tier, vLLM, Rig, self-hosted Ollama/LocalAI)

No billing details are ever required. The platform is fully operational without a single paid license, cloud subscription, or external monetary dependency.

0.9 Implementation Checklist (Linux x64 or macOS – ZeroCost)

Full CD Path – Ready to Run – Zero Manual Input

Set environment variables and validate dependencies. No code blocks – the checklist is textual:

Verify Rust, WasmEdge, OPA, PostgreSQL client, NATS CLI are installed.

Set environment variables : SGTX_DEV_PATH, DATABASE_URL, NATS_URL, HF_TOKEN (optional), GROQ_API_KEY (optional), RUST_BACKTRACE, SQLX_OFFLINE.

Build Governor service with locked dependencies.

Apply database schema (PostgreSQL 18 + pgvector).

Initialize NATS JetStream KV for SGTX FEESLock.

Load OPA policies.

Pre-validate WasmEdge modules.

Initialize Loom logging.

Start Governor service.

Verify health endpoint.

Post-Execution Validation:

All services running, dependencies version-locked.

Zero billing details required, zero manual input.

Governor responding at http://localhost:8080.

OPA policies loaded, WasmEdge modules compiled.

0.10 Part 0 Status

□ Merged, zero-cost annotated, tenant-centric overlay integrated, multi-shipment per-shipment SGTX FEES, container release control, document traceability, dual-mode toggle, and three-tier AI fallback (Groq + HF local + Ollama/LocalAI) – all ready for production on your own server. USTN format updated to include company identifiers and removed version suffix.

END OF PART 0

Part 1: Constitutional & Governance Layer (v6.3 – Merged, Zerocost, Three-Tier AI, Tenantfriendly Decision Explanations)

1.1 Constitutional Layer (ZeroCost WasmEdge + OPA)

The entire platform is governed by WASM-compiled constitutional rules executed in the WasmEdge runtime. Every microservice call, UI action, Temporal workflow step, and database write passes through the Governor Service before execution. No exceptions.

Zero-Cost Implementation:

Constitutional rules stored as .rego (OPA) + .wasm modules in /core/governor/policies/.

WasmEdge installed via package manager (OSS, no billing).

OPA self-hosted via package manager.

All WASM modules precompiled with wasmedge compile for native execution – zero Docker, zero cloud dependency.

Constitutional Module Validation (Automated Script):

Validate that WasmEdge, OPA, and all required modules are present. Precompile constitutional WASM modules idempotently. Check OPA policy syntax offline.

1.2 AI Authority Levels (Three-Tier Zero-Cost Strategy)

The AI Authority Ladder is strictly enforced. No autonomous execution (A5 blocked at WASM level). Three zero-cost inference providers are used depending on latency, privacy needs, and availability. No billing details are ever required.

The Three Providers and Their Roles:

Groq (free tier, no credit card) – Used for fast advisory tasks (A1), Smart Inbox title/description generation, tenant messages, live market suggestions, and other low-latency, non-privacy-critical inferences. Rate limits: 30 requests per minute, 14,400 per day. No billing details required.

Hugging Face (self-hosted, local inference) – Used for privacy-critical tasks (A2/A3) such as document extraction (Donut), contract clause analysis, sanctions screening, and any inference that must not leave the tenant's sovereign node. Unlimited – you run the models on your own server. No external API calls.

Ollama / LocalAI / GPT4All (self-hosted, local inference) – Used as a universal fallback when Groq rate limit is exceeded or Hugging Face models are unavailable. Runs on your server's CPU/GPU. Unlimited, no external dependencies.

AI Authority Matrix with Automated API Selection:

A0 – Observational: Logging only, no influence. Example: analytics, performance monitoring. Recommended API: None (local only). Zero-cost: ClickHouse append-only logs.

A1 – Advisory: Suggestions only, cannot block. Example: educational hints, price recommendations, Smart Inbox titles, tenant messages. Recommended API: Groq (fast, free tier). Fallback: Ollama/LocalAI.

A2 – Constraining: Can block, delay, escalate – never force. Example: compliance prescreen, risk flagging. Recommended API: Hugging Face (local, privacy-preserving). Fallback: Ollama/LocalAI.

A3 – Escalation: Forces human review, cannot decide autonomously. Example: Governor orchestrator final routing, high-risk trade approval. Recommended API: Hybrid (HF for deep analysis + Groq for fast escalation payload). Fallback: Ollama/LocalAI.

A4 – Governance: Enforces execution gates within constitutional bounds. Example: low-risk auto-accept of routine milestones. Recommended API: None (OPA + WasmEdge only). No AI invocation.

A5 – FORBIDDEN: Never permitted. Blocked at WASM compile time. Any A5 attempt rejected pre-execution.

AI Invocation Strategy (Three-Tier):

For A1 (Advisory) : Groq by default. If rate limit exceeded or Groq unavailable, fallback to Ollama/LocalAI with a quantised model (e.g., Llama 3 8B).

For A2 (Constraining) : Hugging Face local inference (Mixtral, Donut, ViT) for compliance-critical checks. If HF model not loaded or server overloaded, fallback to Ollama/LocalAI.

For A3 (Escalation) : Rig agent first calls HF for deep analysis, then Groq for fast formatting. If either fails, fallback to Ollama/LocalAI for both tasks.

All AI calls are logged in ai_inference_records with the provider, model, latency, token count, and fallback events.

Example A2 Decision Object (HF Local Inference, Privacy-Preserving):

{
"decision_id": "dec-00291",
"authority_level": "A2",
"action_requested": "contract.lock",
"verdict": "CONDITIONAL",
"conditions": ["HUMAN_APPROVAL_REQUIRED", "JURISDICTION_REVIEW_PENDING"],
"confidence": 0.74,
"explanation": "Seller jurisdiction (Yemen) is high risk. Bank only + enhanced DD required.",
"loom_hash": "sha256:a4f3b9...",
"recommended_actions": [
"Request additional KYB documentation from seller",
"Route financing through approved bank partner",
"Schedule human review within 24 hours"
],

"_inference_metadata": {
"model": "mistralai/Mixtral-8x7B-Instruct-v0.1",
"api_source": "huggingface_local_vllm",
"fallback_used": false,
"offline_mode": true
}
}

AI API Selection Logic (Automated in Rust):

The router selects the provider based on authority level, privacy requirement, and availability. If the preferred provider fails or rate limit is exceeded, it automatically falls back to the next in chain: Groq → Ollama/LocalAI, HF → Ollama/LocalAI.

1.3 Human Authority Hierarchy (Zero-Cost Identity Stack)

Platform Governance Authority : Sets global constitutional rules, platform-wide policy definition. Verification: Multisig (3/5) + ZITADEL MFA. Zero-cost: ZITADEL self-hosted; Ed25519 keys stored in OS keychain.

Tenant Root : Immutable legal entity (KYC/KYB verified). Full tenant control within jurisdiction. Verification: ZITADEL + Government Registry + biometric liveness. Zero-cost: Donut v3 + PaddleOCR (OSS) for document extraction; registry APIs scraped via curl (free).

Tenant Admins : Invite/remove employees, define internal approval chains. Scoped to tenant; cannot modify constitutional rules. Verification: ZITADEL MFA + role-based permissions. Zero-cost: OPA policies enforce scope.

Employees : Scoped permissions; actions logged and attributable. Verification: ZITADEL + WebAuthn + data scope enforcement. Zero-cost: webauthn-rs crate (OSS); no SMS/email costs.

External Parties : Time-limited scoped access (auditors, consultants). Read-only mostly; auto-expire. Verification: JWT with short TTL (exp claim); no paid session management.

Zero-Cost ZITADEL Setup (Linux x64):

Install ZITADEL from source or package. Initialize config idempotently. Start in background (dev mode, no billing). No credit card required.

1.4 Jurisdiction Supremacy Rule (Dynamic, Zero-Cost Data Sources)

SGTX always applies the strictest applicable rule among:

Country of buyer.

Country of seller.

Country of logistics execution.

Country of financier.

Governing law of contract.

Autoblocked (sanctions): North Korea, Iran, Syria, Cuba, Russia, Belarus.
High-Risk (bank-only + enhanced DD): Iraq, Afghanistan, Yemen, Lebanon, Pakistan.

The platform maintains a dynamic jurisdiction matrix updated from 50+ official sources (UN, OFAC, EU, UK, national lists) – all free and scraped/cached.

Zero-Cost Sanctions Update Pipeline:

Use curl to fetch free XML/JSON feeds from OFAC, UN, EU. Parse and compile to WASM offline. Run via cron.

WasmEdge Jurisdiction Check (Rust):

Check corridor for sanctions level. Return GovernorVerdict::DENY for autoblock, CONDITIONAL for high-risk, ALLOW otherwise.

1.5 Governor Service – Single Point of Truth (Zero-Cost Decision Engine with Tenant-Facing Explanations)

The Governor is the authoritative decision engine. Every action passes through:

OPA policy evaluation.

WasmEdge constitutional gate.

AI consult (if A1–A3) using the three-tier provider strategy.

Ed25519 signing.

Loom deterministic logging.

Tenant-friendly message generation (v6.3).

Decision Flow (Enhanced):

Request → OPA Policy Check → WasmEdge Constitutional Gate → AI (A1-A3, with fallback) → Ed25519 Sign → Generate Tenant Message (if DENY/CONDITIONAL) → Loom Hash → Store in governor_decisions.

Tenant Message Generation (v6.3) : Whenever a verdict is DENY or CONDITIONAL, the Governor immediately calls the AIAgentOrchestrator (A1, using Groq; if Groq unavailable, uses Ollama/LocalAI fallback) to produce a plain-language tenant_message. This JSONB field is stored in the governor_decisions table and contains:

A short, jargon-free summary of why the decision was made.

A structured list of unmet conditions, each with a human-readable label and an action URL.

The message is generated in the user's preferred language
(from employees.preferred_language).

Example Tenant Message (Condition):

{
"summary": "Your contract lock is on hold because the seller's jurisdiction requires enhanced KYC. Our compliance system paused the process to protect both parties.",
"conditions": [
{

"condition_id": "kyc_tier2_seller",
"label": "Seller KYB Tier 2 verification",
"status": "unmet",
"action_url": "/v1/kyc/upgrade?tenant=SGTX-VN-TRD-000456"
},
{
"condition_id": "doc_cert_origin",
"label": "Certificate of Origin from Vietnam",
"status": "unmet",
"action_url": "/v1/documents/upload?trade=..."
}
],
"escalation_hint": "If you believe this is an error, you can request a human review. A compliance officer will respond within 24 hours."
}

This message is what the PlainLanguage Governor Decision Panel (Part 6.1.3) renders in the UI, ensuring tenants never see opaque Loom hashes or OPA rule IDs.

Zero-Cost Stack Mapping for Governor Service:

OPA: Self-hosted Rego engine.

WasmEdge: Native runtime.

Loom: Rust deterministic logger.

Signing : ed25519-dalek crate (OSS). Keys stored in OS Keychain.

Storage: PostgreSQL 18 + pgvector.

Tenant Message Generation : Groq (primary) with Ollama/LocalAI fallback. Non-blocking call; fallback to template if both unavailable.

Automated Decision Evaluation Script (Extended with Tenant Message):

The script evaluates OPA, WasmEdge, AI (with provider fallback), signs with Loom, generates tenant message if needed, and stores in PostgreSQL. All idempotent.

1.5.1 Cryptographic Enforcement Layer (Mandatory)

All Governor decisions must be:

Signed using Ed25519.

Linked to a previous decision hash.

Stored with immutable audit record.

Decision Object Structure : decision_id, verdict, inputs_hash, timestamp, signature.

Execution Rule : No action may proceed unless a valid signed Governor decision is present.

1.5.2 Replay Protection

Each incoming request must include: nonce, timestamp, signature.
Validation: nonce unique, timestamp within allowed window, signature matches payload.
Failure results in immediate rejection.

1.6 ExecutionGate (WasmEdge) – Sandboxed, Zero-Cost

Runs in sandboxed WASM. Blocks if constitutional rules, jurisdiction supremacy, or SGTX FEESLock status fail. Pre-compiled and validated.

1.7 Legal Disclaimer (Zero-Cost, Automatically Injected, Enhanced with Non-Marketplace Messaging)

SGTX is a non-custodial execution platform, not a marketplace. We do NOT hold funds, take title to goods, or broker introductions. We provide the infrastructure for your trades. Headquartered in New Jersey, USA. Users bear 100% compliance responsibility. Max liability = SGTX FEESs paid in last 12 months.

This disclaimer is displayed on every login, contract lock, and SGTX FEES payment screen. Additionally, a persistent footer in every portal reads: "SGTX is a non-custodial execution platform, not a marketplace. We do not hold funds, take title to goods, or broker introductions. We provide the infrastructure for your trades."

PART 1: IMPLEMENTATION CHECKLIST (Linux x64 or macOS – ZeroCost, Full Automation)

Full CD Path – Ready to Run – Zero Manual Input

Validate dependencies : cargo, wasmedge, opa, psql, zitadel (optional).

Set environment variables (SGTX_DEV_PATH, DATABASE_URL, HF_TOKEN optional, GROQ_API_KEY optional, RUST_BACKTRACE, SQLX_OFFLINE).

Validate OPA constitutional policies.

Pre-compile WASM modules idempotently.

Build Governor service with locked dependencies.

Start ZITADEL identity service (self-hosted, zero-cost).

Update jurisdiction sanctions from free sources.

Start Governor service.

Health check and constitutional validation.

AI API connectivity test (Groq optional, HF local optional, Ollama/LocalAI as fallback). No failures if optional APIs unavailable – fallback works.

Post-Execution Validation:

OPA policies loaded and validated.

WasmEdge modules compiled and sandbox-validated.

Governor service responding at http://localhost:8080.

ZITADEL identity service running (dev mode).

Jurisdiction matrix updated from free sources.

AI API connectivity : Groq optional, HF local optional, Ollama/LocalAI available as fallback.

Tenant message generation pathway active (Groq → Ollama fallback chain).

Zero billing details required, zero manual input, zero reinstall.

Part 1 Status: □ Merged, zero-cost enforced, three-tier AI (Groq + HF local + Ollama/LocalAI) with fallback, tenant-friendly decision explanations integrated, ready for production on your own server.

END OF PART 1

Part 2: Identity, Tenants & Internal Authority (v6.3 – Merged, Zerocost, Three-Tier AI, Onboarding Wizard, Organization Graph, Dual-Mode Toggle, Tenant Lifecycle – Fully Expanded – No Operating Modes)

2.1 Global Trade Identity (GTID) – Deterministic, Zero-Cost Generation

Format: SGTX-{COUNTRY_CODE}-{ENTITY_TYPE}-{SEQUENCE}-{CHECKSUM}
Example: SGTX-EG-TRD-002139-7F3A

Components:

SGTX: fixed prefix.

COUNTRY_CODE : ISO 3166-1 alpha2 country code of the tenant's primary jurisdiction.

ENTITY_TYPE : three-letter code: TRD (trader), LOG (logistics), FIN (financier), QC (quality control), GOV (government), REG (regulatory), MP (marketplace partner).

SEQUENCE : zero-padded 6-digit number, unique per (COUNTRY_CODE, ENTITY_TYPE) pair.

CHECKSUM : 4-hexadecimal-digit CRC32-ISO-HDLC of the concatenated fields (first 2 bytes, represented as 4 hex chars).

Generation uses a deterministic CRC-based algorithm. The sequence is stored in PostgreSQL and incremented atomically. Validation function ensures GTID authenticity before any resolution.

2.2 GTID Resolution Service – Zero-Cost, Self-Hosted, with Caching & Validation

Provides a fast endpoint to resolve a GTID into the tenant’s public profile : legal name, jurisdiction, trust scores, KYB status, sanctions clearance, and lifecycle state. All responses respect data privacy. Caching uses Valkey (Redis-compatible). Rate limiting is enforced at the API gateway.

2.3 Tenant Types & Onboarding Flow – Logistics Roles Fully Expanded, Zero-Cost KYB/KYC, Marketplace Partner Added, Private Financier Support

The platform supports multiple tenant types : CORPORATE (buyers/sellers/trading houses), FINANCIAL (institutional banks, DeFi pools, factoring companies, private financiers where legally permitted), LOGISTICS (freight forwarder, shipping line, trucking company, customs clearance broker), QUALITY_CONTROL (inspection agencies, surveyors, lab testing), REGULATORY (customs authorities, port authorities, single window operators), GOVERNMENT (trade ministries, export promotion agencies), MARKETPLACE_PARTNER (external digital trade platforms).

Private Financiers (v6.3) : Non-bank lenders, private credit funds, family offices, peer-to-peer lending platforms are supported. Onboarding checks jurisdiction allowance via a dynamic matrix updated by the Regulatory Intelligence Agent (RIA). Required documentation includes proof of registration or exemption, and a legal basis explanation.

Logistics Roles : Freight Forwarder, Shipping Line, Trucking Company, Customs Broker – each with role-specific fields and verification requirements.

Marketplace Partner Onboarding : Manual onboarding via Admin Portal (multisig approval). After approval, the partner receives an Ed25519 API key and can access the Marketplace Partner Portal.

All KYB/KYC verification uses zero-cost stack : HF Donut + PaddleOCR for document extraction, government registry scraping via curl, and local ML models for sanctions/PEP screening. No external identity service is required.

2.4 Permission System – Action-Based OPA Enforcement, Logistics Subroles, Trader Mode Aware

The permission system is action-based, fine-grained, and enforced in real time by OPA. Every user action corresponds to a specific permission string (e.g., contract.sign). Permissions are assigned to roles, roles to employees. The Governor evaluates every request against the user's permissions, trader mode context (BUY/SELL/DUAL), and tenant lifecycle state. Logistics providers have additional subrole checks (e.g., only a customs broker can submit clearance documents). Data scopes (row-level and field-level) restrict what specific data an employee can see or act upon. All permissions are stored in the database, versioned, and audited.

2.5 Icon Shuffle & Visibility Rules – Zero-Cost UI Logic, Trader Mode Aware

Icon shuffle randomly reorders the positions of key action icons and navigation elements on each login or session, making it harder for an observer to predict a user's next action. The shuffle is deterministic per session (seeded with session ID). Visibility rules dynamically show or hide UI elements based on the employee's permissions, trader mode (BUY/SELL/DUAL), and tenant lifecycle state. All logic is implemented clientside (React) using a compiled OPA WASM policy. For dual-mode trading companies (see 2.8), the role switcher in the header changes the effective trader mode context for the session, which also affects visibility.

2.6 Saved Contacts / Network Feature (v6.3 – Fully Expanded, AI-Enriched, Zero-Cost)

Every interaction between tenants (trade, quote, message, distressed cargo notification) is automatically recorded in the tenant_contacts table. The feature transforms the platform into a relationship intelligence hub. It is not a marketplace – the platform never suggests or ranks contacts for cold outreach. It provides deep, dynamic intelligence on entities a tenant already knows.

Core Data Model : Stores trade count, total value, first/last interaction, is_favorite, is_blocked, trust_snapshot (AI-generated 360° portrait), relationship_health_score (predictive), smart_labels, and indirect connections (via GraphRAG).

Automatic Contact Saving : Contacts are saved automatically on trade request creation, logistics quote acceptance, financing agreement completion, direct message exchange, and distressed cargo notification. No user action required.

AI-Enriched 360° Trust Portrait (Daily Refresh) : Uses internal data (trade history, payment delays, dispute rate) plus external public news sentiment (scraped free RSS) and Groq (or Ollama fallback) to produce concise plain-language portrait.

Predictive Relationship Health Score : LSTM model (local) predicts future strength (0-100) based on trade frequency trend, payment behaviour, responsiveness, dispute trend, and market conditions. Visible only to the tenant.

GraphRAG-Powered Indirect Connections : Traverses corporate graph to discover indirect links (shared logistics providers, common UBOs, sanctions proximity) – displayed in expandable panel.

Voice-Driven Contact Management : Vosk (offline) + HF Mixtral (local) for intent extraction. Examples: “Show details for Mekong Fresh”, “Add Nile Foods to favourites”, “Block any contact from Russia”.

Privacy-Preserving Verifiable Credentials : Using Plonky3 ZK proofs, a contact can prove attributes (e.g., ISO 9001 certified) without revealing full documents. Verified credential appears as a badge.

Network-Wide Anomaly Alerts : Isolation Forest model checks saved contacts daily for sharp deviations (sudden payment delays, negative news sentiment spike, PEP link). Alerts appear in Smart Inbox as advisory.

Portal Integration : The Network tab appears in Buyer, Seller, Logistics, and Financier portals with role-specific views. For dual-mode trading companies, the Network tab shows contacts relevant to the currently active trader mode (Buyer or Seller).

2.7 Tenant Onboarding Wizard & Sandbox (v6.3 – Fully Expanded)

2.7.1 Onboarding Wizard Flow – Step-by-Step

Six sequential steps : Welcome & GTID Confirmation, Organization Details (legal identity), KYB/KYC Tier 1 Verification (document upload), Profile Configuration (trader mode, language, voice/stress preferences), Create First Resource (optional), Enter the Sandbox.

Step 4 – Profile Configuration: Includes trader mode (Buy Only, Sell Only, Buy and Sell – Dual). For dual-mode tenants, the system will later enable the dual-mode toggle in the header after onboarding.

Step 5 – Create First Resource (Role-Specific, Optional): For trading companies (CORPORATE with dual mode), they can add commodities, typical packaging, etc. Skipping is allowed; a reminder banner may appear after going live.

Step 6 – Sandbox Environment: Fully functional, isolated replica with synthetic counterparties, dummy USTNs, guided practice trade, and weekly auto-reset. Tenants can practice before going live.

2.7.2 Sandbox Environment

Activation occurs immediately after Step 5 (or Step 4 if Step 5 is skipped). Data stored in separate PostgreSQL schema (sandbox_*) with RLS. Synthetic USTNs use “SB” prefix. Dummy counterparties pre-provisioned. Guided practice trade walks through entire lifecycle. Reset weekly or manually. Transition to production (“Go Live”) wipes sandbox data and sets lifecycle_state to VERIFIED (or KYB_PENDING if manual review required).

2.7.3 “Skip for Now” & Reminder Banner

Step 5 can be skipped for all tenant types. A reminder banner appears in Smart Inbox after going live, escalating in priority if ignored. No tier-based restrictions.

2.7.4 Draft Auto-Save & Expiry

Debounced auto-save every 30 seconds. Drafts stored in tenant_onboarding_state.step_dataJSONB and versioned in draft_history. Resume on login via Smart Inbox banner. Drafts expire after 14 days (reminders sent at 11 and 13 days). Expired drafts can be restored via Company Admin → Drafts.

2.7.5 Backend Integration & API Endpoints

Endpoints under /v1/onboarding/ for state retrieval, saving, skipping, sandbox reset, and exit-sandbox. All gated by Governor. Sandbox cleaner service runs daily to reset inactive sandboxes.

2.8 Dual-Mode Toggle for Trading Companies (Buyer/Seller Dashboard Switching)

For tenants with type = CORPORATE and default_trader_mode = DUAL, a toggle (Buyer/Seller) appears in the global header. This allows a single employee to switch between Buyer Dashboard and Seller Dashboard contexts within the same session. The toggle does not change permissions – it only filters which UI tabs, data, and actions are presented. The selected mode is stored in employees.active_trader_mode_context (session-based or persisted). Tenant admins can disable the toggle per employee via data_scopes.allow_role_switching = false. The Governor enforces that actions are allowed in the current context; if an employee tries to perform an seller action while in buyer mode, the Decision Panel explains: “You are currently in Buyer mode. Switch to Seller mode to perform this action.”

2.9 Internal Tenant Organization Graph (v6.3 – New)

Enables large enterprises to structure their SGTX presence hierarchically under a single GTID (or parent tenant group).
Tables: tenant_business_units, tenant_departments, tenant_cost_centers, tenant_approval_groups, tenant_approval_policies.

Features:

Delegated administration per business unit.

Approval chains (e.g., contracts above $100k require two department heads).

Cost center tracking for internal reporting.

Cross-tenant group for holding companies (Phase 3 feature, designed from start).

UI : Organization Manager in Company Admin tab (visible to tenant admins). Visual tree view, drag-and-drop to move employees, policy editor, and “Preview as Employee” button.

2.10 Tenant Lifecycle Engine (v6.3 – New)

Unified state machine governing access, feature visibility, compliance actions, and Smart Inbox messaging. States: REGISTERED, ONBOARDING, KYB_PENDING, VERIFIED, LIMITED_MODE, AT_RISK, SUSPENDED, ARCHIVED.

Transition rules and triggers defined in a table. Governor decisions required for certain transitions (e.g., KYB_PENDING → VERIFIED, VERIFIED → AT_RISK). Each state impacts allowed actions (e.g., KYB_PENDING disables real trade creation but allows sandbox). Smart Inbox creates high-priority items on state changes with Groq-generated (or fallback) explanations. Lifecycle history is logged immutably.

PART 2: IMPLEMENTATION CHECKLIST (Linux x64 or macOS – ZeroCost, Full Automation)

Validate dependencies (cargo, psql, redis-cli, opa, wasmedge, zitadel optional).

Set environment variables.

Build identity service (version-locked).

Validate GTID generation module.

Compile OPA policies and WASM modules.

Start cache (Valkey/Redis optional).

Validate trust score model (XGBoost local, training optional).

Start identity service.

Health check and GTID resolution test.

Run network contacts auto-sync script.

Apply onboarding & organization DDL.

Seed sandbox schema.

Apply tenant lifecycle DDL.

Verify dual-mode toggle configuration (add columns active_trader_mode_context and allow_role_switching).

Post-Execution Validation:

GTID generation deterministic, checksum-validated.

OPA policies compiled, logistics subroles enforced.

Identity service responding.

Cache available (optional).

Trust score model loaded.

Network contacts autosync executed.

Onboarding wizard backend ready.

Organization hierarchy tables created.

Tenant lifecycle state machine added.

Dual-mode toggle columns added, data scopes extended.

Zero billing details required, zero manual input.

Part 3, Phase 1: Trade Initiation (Buyer Portal) – Structured Container-Level Request Form (v6.3 – Fully Extended, with Express Mode, AI Container Advisor, Multi-Shipment Schedule Builder, Cloning, and Dual-Mode Toggle Aware)

PHASE 1 GOAL
The buyer creates a precise, multi-container trade request using a structured form with global dropdowns, dependent fields, cloning, and per-container commodity details. The buyer may also request a multi-shipment contract. AI provides advisory suggestions but never overrides manual input. All data is validated by the Governor before submission. The form includes an optional Express Mode (free-text AI parsing) for users who prefer natural language.

PREREQUISITE
The buyer is logged into the Smart Inbox (Part 6.1.1). Their trader mode must be BUY or DUAL (if DUAL, the active context – set by the dual-mode toggle in the header – must be "Buyer"). They navigate to the "New Trade Request" tab in the sidebar.

Step 1.1 – Seller Selection (GTID Entry or Saved Contacts with Advanced Search)

The buyer selects the seller using one of two methods, both accessible from a single, unified input component:

Method A – Direct GTID Entry
The buyer manually types the seller's GTID (e.g., SGTX-VN-TRD-002139-7F3A). The system validates the format in real time and resolves the GTID to the seller's legal name, jurisdiction, trust score, and sanctions status.

Method B – Search from Saved Contacts
The buyer clicks a dropdown icon to open a searchable modal of their saved contacts (from the Network feature). The modal allows:

Search by GTID or company name (fuzzy search, real-time filtering)

Display of each contact's legal name, GTID, trust score, and a "last trade" date

One-click selection

Advanced UI/UX Enhancements (High-End, Professional, Compliant):

Debounced search – As the user types, results appear within 300 milliseconds.

Keyboard shortcuts – Arrow keys to navigate results, Enter to select.

Auto-suggest – If the buyer has previously traded with the seller, the GTID is pre-suggested in the input field (with a "recent" badge).

Trust indicator – A colour-coded badge (green for trust ≥80, amber for 50-79, red for <50) appears next to each search result.

Sanctions clearance icon – A shield icon indicates whether the seller has passed sanctions screening.

Saved contact avatar – If the seller has a company logo (via tenant branding), a small thumbnail is shown.

Accessible labels – ARIA-compliant with screen reader announcements.

Fallback manual entry – If the seller is not in saved contacts, the buyer can still type any valid GTID.

Real-time validation – As the buyer types a GTID, the system verifies checksum and existence without blocking input.

Upon selection (either method), the GTID Resolution Service returns:

Legal name, jurisdiction, trust score, sanctions clearance status.

If the seller is a saved contact, a “Saved Contact” badge appears with a link to the full 360° Trust Portrait.

The buyer may click to view the full Trust Portrait (Groq-generated summary of trade history, payment behaviour, and public news – advisory only).

Governance : The Governor checks that the buyer has permission trade.request.create and that the seller is not blocked, suspended, or from a sanctioned jurisdiction. If the seller is from an autoblocked country, the request is DENY with a plain-language explanation via the Decision Panel.

Step 1.2 – Structured Container & Commodity Entry (Enhanced Form)

This structured form is the primary interface. An optional “Express Mode” toggle allows free-text AI parsing as an alternative for users who prefer natural language, but the structured form is the default for precise trade execution.

1.2.1 – Container Count & Basic Setup

Number of Containers – Number input (min 1, max 50). Default: 1. When changed, the form dynamically generates a tab or accordion for each container (e.g., “Container 1”, “Container 2”, …).

Container Tabs / Accordion – Each container has its own section. Buttons at the bottom of each container:

Clone Container – Copies all data (origin, destination, port, palletized status, pallet size, and all commodities) to a new container. The user can then modify any field, including destination override.

Remove Container – Only shown if number of containers > 1.

UI/UX Enhancements:

Drag-and-drop reordering of container tabs.

Visual progress indicator showing how many containers are fully configured (all required fields filled).

Tooltips explaining each field.

1.2.2 – Per-Container Fields (Global, Dynamic Dropdowns)

Each container section includes the following fields, sourced from live data updated by the Regulatory Intelligence Agent (RIA).

Field	Type	Source / Dependency	Example
Country of Origin	Global dropdown	ISO 3166-1 from jurisdictions tabl e	Vietnam
Destination Country	Global dropdown	ISO 3166-1 from jurisdictions tabl e	Egypt
Port of Discharge	Global dropdown (dependent)	Filtered by Destination Country from portstable (UN/LOCODE + RIA updates)	Alexandria
Palletized?	Toggle (Yes/No)	–	Yes
Pallet Size (if palletized)	Dropdown	International standards: EUR (800×1200 mm), ISO (1000×1200 mm), Custom (free text)	EUR 800×1200 mm
Destination Override	Optional text / dropdown	Defaults to the Destination Country; can be overridden for a specific container (e.g., “Alexandria Free Zone”)	Alexandria Free Zone
Notes (per container)	Text area (optional)	–	“Expedite customs clearance”

UI/UX Enhancements:

Dependent dropdowns – The Port of Discharge dropdown is disabled until a Destination Country is selected. It then loads only ports belonging to that country.

Flag icons – Next to each country dropdown, a small country flag (using a lightweight sprite set) for visual recognition.

Pallet size presets – Dropdown includes EUR, ISO, and a “Custom” option that reveals a free-text field for non-standard dimensions.

Destination Override – Initially hidden; appears when the user clicks “Override destination for this container”. Useful for shipments to special economic zones or specific warehouses.

Inline validation – If a port is not found or is sanctioned, a red warning appears instantly.

1.2.3 – Commodities Within a Container (with AI-Driven Dynamic Product Specification)

Each container can contain one or more commodities.

Starts with one commodity row per container.

Button : “Add another commodity in this container” (unlimited).

Per-commodity fields (base fields):

Field	Type	Source / Dependency	Example
Commodity Type	Global dropdown	Predefined list: Fresh Fruits, Fresh Vegetables, Frozen Fruits, Frozen Vegetables, Grains, Meat, Seafood, Dairy, Chemicals, Machinery, Textiles, Other (with free text)	Fresh Fruits
Product	Combined input (three methods)	Option A: Type HS code (6/10 digits) → autofill product name from HS database (updated by RIA). Option B: Dropdown filtered by Commodity Type (from productstable, synced with WCO). Option C: “Other” free-text.	0805.10 (Oranges) or “Valencia Oranges”
Packaging	Dropdown + free text	Options: Boxes, Mesh bags, Plastic crates, Pallet wrap, Drums, Barrels, Bales, Bins, Carton bags, Jumbo bags, Other (manual entry)	Boxes (10 kg cartons)
Number of Pallets for this commodity	Number input	Integer ≥ 0	22
Net Weight per Unit	Number + unit selector (kg / lb)	User enters; unit can be per box, per bag, etc.	10 kg
Gross Weight per Unit	Number + unit	Auto-filled from net weight + tare factor (e.g., +5%) – user can override manually.	10.5 kg
Notes (per commodity)	Text area (optional)	–	“Stack max 5 layers”

AI-Driven Dynamic Product Specification (A1 – Groq/Ollama, advisory only):

When the buyer selects a product (either by HS code or from the product dropdown), the system automatically calls the Product Specification Agent to generate a dynamic, context-aware specification form tailored to that specific product.

How it works:

Trigger : Buyer selects a product (e.g., “Valencia Oranges” or HS 0805.10).

AI inference : The agent retrieves from a knowledge base (industry standards, WCO data, historical trade patterns) the relevant specification dimensions for that product.

Dynamic form generation : The system replaces the generic “Product Specification” text field with a structured set of input fields specific to the product.

Example for Oranges (Fresh Fruits):

The AI detects the product as “Oranges (Citrus sinensis)” and generates:

Field	Type	Example
Variety	Dropdown	Valencia, Navel, Blood Orange, Mandarin, Other (free text)
Size range	Dual slider or dropdown	72-80 mm, 80-88 mm, 88-96 mm
Colour grade	Dropdown	Bright orange, Orange with green spots, Light orange
Brix (sugar content)	Number input (with unit °Bx)	11.0 – 14.0 (pre-filled typical range)
Defect tolerance	Percentage input	≤2% bruising, 0% mould
Number of pallets per size	Dynamic list	If multiple sizes selected, buyer specifies pallets per size (e.g., 10 pallets of 72-80 mm, 12 pallets of 80-88 mm). System auto-sums to total pallets entered.

Example for Frozen Strawberries:

Field	Type	Example
Variety	Dropdown	Camarosa, Albion, Sweet Charlie, Other
Grade	Dropdown	Grade A (whole, no bruises), Grade B (sliced), Grade C (puree)
Sugar added	Yes/No toggle; if Yes, percentage	10%
Packaging type (frozen)	Dropdown	Polybags, Cartons, Bulk boxes
Temperature requirement	Read-only with override	-18°C (advisory)

Example for Textiles (Cotton Fabric):

Field	Type	Example
Weave type	Dropdown	Plain, Twill, Satin, Jersey
Thread count	Number input	200 TC
Width	Number input (cm or inches)	150 cm
Weight	Number input (gsm)	180 gsm
Colour	Free text or colour picker	White
Roll length	Number input (meters per roll)	100 m
Number of rolls	Number input	50

Implementation details:

The AI generates a JSON schema describing the specification fields. The frontend renders these fields dynamically using a form renderer.

The buyer can accept the AI-suggested fields or override them. All overrides are logged.

If the AI confidence is low (<70%), the system falls back to a generic text area with a note: “We could not determine specifications automatically. Please describe below.”

For multi-commodity containers, each commodity has its own dynamic specification set.

The specification data is stored in trade_requests.specifications JSONB, preserving the structured fields for later use in QC inspection, contract generation, and dispute resolution.

UI/UX Enhancements:

Progress indicator – While the AI is generating specifications, a skeleton loader appears.

Inline edit – All generated fields are editable; changes are highlighted.

Reset to AI – A button to revert to the AI-suggested values if the buyer made changes but wants to reset.

Save as template – The buyer can save the entire specification set as a personal template for future trades of the same product and counterparty.

Compatibility warning – If the buyer selects specifications that conflict with industry norms (e.g., size range too large for standard packaging), a non-blocking advisory banner appears.

Governance (G1U12 – New): The AI specification generation is purely advisory (A1). It never blocks submission. If the AI fails (Groq and Ollama both unavailable), the system falls back to a simple text area with a placeholder: “Enter product specifications (e.g., Grade A, Size 72-80 mm)”. The failure is logged, and a low-priority Smart Inbox alert is sent to the platform admin.

1.2.4 – Cloning & Bulk Operations

To streamline the creation of multiple containers or commodities, the form provides the following advanced operations:

Clone Container

Located at the bottom of each container section.

When clicked, the system creates a new container tab immediately after the current container.

The new container is a perfect copy of the source container, including:

Country of Origin, Destination Country, Port of Discharge

Palletized status and Pallet Size

Destination Override (if any)

All commodities with their specifications, packaging, pallet counts, and notes

After cloning, the user can modify any field independently. The clone operation is logged for audit purposes.

A maximum of 50 containers is enforced.

Bulk Edit (Advanced)

For large shipments (e.g., 10+ containers), a “Bulk Edit” button appears in the toolbar. Clicking it opens a modal that allows the user to:

Apply the same commodity to all containers – Select a commodity from the current container and apply it to every container (adding it to each container's commodity list).

Copy container settings to selected containers – Choose a source container and a range of target containers, then copy origin, destination, port, palletized status, and pallet size (commodities are not copied unless explicitly selected).

Increment container-specific fields – For fields like “Destination Override” that may vary sequentially, the system can generate a pattern (e.g., “Warehouse A1”, “Warehouse A2”, …).

The Bulk Edit modal includes a preview of changes before confirmation. All bulk operations are reversible (undo button appears for 10 seconds after application).

Remove Container

Visible only when number of containers > 1.

Clicking removes the entire container and all its commodities.

A confirmation modal appears : “Are you sure you want to remove Container X? This action cannot be undone.”

The container tab is removed, and remaining tabs are re-numbered.

1.2.5 – Global Notes Bar

At the bottom of the entire structured form, before the submission buttons, a Global Notes section is provided.

Field : Additional Notes (for entire trade) – Text area (max 2000 characters).

Placeholder example : “Seller to provide phytosanitary certificate. Insurance required. Please ensure reefers are pre-cooled to 4°C.”

Character counter – Shows remaining characters (2000 max).

AI assistance (optional, A1) – A small “□ Suggest” button next to the field. When clicked, the AI (Groq, fallback Ollama) analyses the trade details (commodities, origin, destination, incoterm) and suggests common notes, such as:

“Phytosanitary certificate required for fresh fruit export.”

“Bill of lading to be issued in negotiable form.”

“Temperature set point: 4°C for fresh oranges; -18°C for frozen strawberries.”

The buyer can accept, edit, or ignore the suggestion. The AI suggestion is logged but not stored if not accepted.

The global notes are stored in trade_requests.global_notes and displayed to the seller in the quote request view.

1.2.6 – Express Mode (Free-Text AI Parsing, Optional)

For users who prefer natural language over structured forms, a toggle labeled “Use AI Express” is available at the top of the New Trade Request form. When enabled, the structured form is hidden and replaced with a single large text area.

Express Mode Interface:

A text area (minimum height 200px) with placeholder : "Describe your trade in plain English. Example: I need 20,000 kg of Valencia oranges from Vietnam, CFR Alexandria, packed in 10 kg cartons, 22 pallets. Also 5,376 kg of lemons in mesh bags, 14 pallets. Two containers, second container to Port Said."

A microphone icon for voice input (speech-to-text via Vosk offline or Web Speech API, with fallback).

A "Parse" button that triggers the AI Intent Parser (A2, Hugging Face local primary, Ollama fallback).

Processing:

The system sends the raw text to the Intent Parser agent.

The agent extracts : number of containers, per-container origin/destination/port, commodities, HS codes (if inferable), quantities, packaging, pallet counts, incoterm, and multi-shipment schedule.

Extracted data is displayed in a preview panel showing the structured form that would be generated.

For each extracted field, a confidence score (0-100) is shown. Fields with confidence below 80% are highlighted in amber with a note: "Please verify or correct."

The buyer can edit any extracted field directly in the preview (which updates the underlying structured data).

Once confirmed, the system populates the full structured form (behind the scenes) and the buyer proceeds to standard validation and submission.

Fallback and Error Handling:

If the AI cannot parse the text (confidence <50% for critical fields like commodity or quantity), the system displays a message: "We could not fully understand your request. Please try rephrasing or switch back to the structured form."

A "Switch to Structured Form" button is always visible, allowing the buyer to abandon Express Mode and manually enter all fields.

Voice Input Enhancement:

Clicking the microphone opens a voice recording modal. The system uses Vosk (offline) for speech-to-text, then passes the transcribed text to the Intent Parser. This is especially useful for mobile users or those who prefer dictation.

Governance (G1U2, G1U3) : The Express Mode is advisory only. The Governor does not accept a trade request based solely on Express Mode parsing; the buyer must review and confirm the extracted data before submission. The system logs the original raw text, the extracted JSON, and any manual corrections made by the buyer.

1.2.7 – Integration with AI & Governor

The structured form directly populates trade_requests.parsed_specs JSONB with high confidence (no parsing ambiguity).

AI Container Advisor (A1) – After the buyer enters all commodity volumes, the system may display a non-blocking advisory banner generated by Groq (fallback Ollama): "Based on your total volume and commodity type, we suggest 2 × 40' HC reefers. Your

current entry shows 3 containers. Would you like to adjust?"
The buyer can accept (auto-adjusts container count) or ignore. Overrides are logged.

Governor validation – All fields are checked:

Port of discharge must exist in ports for the destination country.

Pallet counts must be positive integers.

Commodity types and HS codes (if provided) are checked against sanctions and dual-use lists.

Draft auto-save – Every 30 seconds (debounced), the form state is saved to the server with status = 'DRAFT'. If the user leaves and returns later, the Smart Inbox shows a "Continue draft trade request" item (priority 30). Drafts expire after 14 days (see Part 3.6) with reminders.

1.2.8 – Example Wireframe (Buyer Portal)

text

```
┌──────────────────────────────────────────────────────────────────────
───┐
│ New Trade Request – Structured Form │
├──────────────────────────────────────────────────────────────────────
───┤
│ Number of Containers: [ 3 ] + │
│ │
│ ┌─ Container 1 ───────────────────────────────────────────────────┐ │
│ │ Country of Origin: [Vietnam ▼] │ │
│ │ Destination Country: [Egypt ▼] │ │
│ │ Port of Discharge: [Alexandria ▼] (filtered by Egypt) │ │
│ │ Palletized: [Yes ●] No ○ Pallet Size: [EUR 800×1200 mm ▼] │ │
│ │ Destination Override (if any): [Alexandria Free Zone] │ │
│ │ │ │
│ │ Commodities: │ │
│ │ ┌─ Commodity 1 ───────────────────────────────────────────────┐ │ │
│ │ │ Commodity Type: [Fresh Fruits ▼] │ │ │
│ │ │ Product: [0805.10 - Oranges] ── or ── [Valencia Oranges ▼] │ │ │
│ │ │ Product Spec: [Grade A] [Size 72-80mm] │ │ │
│ │ │ Packaging: [Boxes (10 kg cartons) ▼] ── or ── [Other: _________ ] │ │ │
│ │ │ Number of Pallets: [22] │ │ │
│ │ │ Notes: [Max 5 layers] │ │ │
│ │ │ [+ Add another commodity in Container 1] │ │ │
│ │ └─────────────────────────────────────────────────────────────┘ │ │
│ │ │ │
```

```
│ │ [Clone Container] [Remove Container] │ │
│
└──────────────────────────────────────────────────────────────┘
│
│ │
│ ┌─ Container 2 ─────────────────────────────────────────────┐ │
│ │ ... (same fields) │ │
│
└──────────────────────────────────────────────────────────────┘
│
│ │
│ Additional Notes (entire trade): │
│ [ ] │
│ │
│ [Save Draft] [Submit Trade Request] │
└───────────────────────────────────────────────────────────────
───┘
```

Step 1.3 – Multi-Shipment Request (Buyer Option)

After completing the container and commodity details, the buyer may toggle "Request multi-shipment contract" (default: off). When enabled, a schedule builder appears below the form.

Each shipment in the schedule includes:

Field	Description
Shipment number	Auto-incremented.
Delivery date	Date picker (future date only).
Port of discharge	Dropdown filtered by destination country (can differ from other shipments; must be valid and not sanctioned).
Number of containers for this shipment	Integer ≥ 1.
Commodities / packaging	By default, inherits from the main container definition. The buyer may click "Edit commodities" to override per shipment (opens a mini-form with same fields as per-container commodity entry).
Actions	Add, clone, or remove shipment lines.

The buyer can define any number of shipments. The schedule is stored in trade_requests.multi_shipment_schedule JSONB.

The total contract value is not calculated until the seller provides logistics costs per shipment (Phase 2).

Global notes for the multi-shipment contract can be added in the existing notes bar.

Governance (G1U12): The Governor validates that:

Each shipment's delivery date is in the future.

Each port of discharge exists and is not sanctioned.

The schedule is not empty when the toggle is on.

If any validation fails, the request is CONDITIONAL with a Decision Panel explanation.

Step 1.4 – AI Container Advisor (Advisory Only)

As described in 1.2.6, after the buyer has entered all data (including multi-shipment schedule, if any), an optional advisory banner may appear. The recommendation is generated by Groq (or Ollama) based on total volume, commodity type, and packaging. The buyer can accept, reject, or ignore. No action is blocked.

Step 1.5 – Marketplace Relationship Detection (Auto-Attribution)

Before submission, the system queries partner_lead_attributions for the most recent marketplace-introduced trade between the same buyer and seller.

If found, a transparency banner is displayed : "This trade will be attributed to [Marketplace Name] because you first connected through them on [date]. The standard revenue share of [X]% will be applied. You may dispute this within 72 hours."

A "Dispute" link opens a form where the buyer can provide a reason.

Disputes are analysed by AI (A2, HF local) and, if necessary, escalated to the Platform Governance Authority (A3).

The dispute window is 72 hours. If no dispute is filed, the attribution becomes final.

Step 1.6 – Governor Prescreen & Submission

When the buyer clicks "Submit Trade Request", the Governor evaluates:

OPA Policy Check

Permission trade.request.create.

Active trader mode context = BUY (or DUAL with Buyer context).

WasmEdge Constitutional Gate

Jurisdiction of buyer, seller, and each port of discharge (including those in the multi-shipment schedule) must not be in the autoblock list.

Each port of discharge must be a valid port for its declared destination country (cross-reference with ports table).

Dual-use goods screening (if an HS code is entered) – if flagged, the verdict becomes CONDITIONAL.

AI Consult (A2 – only if triggered)

For dual-use HS codes, the AI (HF local, fallback Ollama) may add a CONDITIONAL verdict with required documentation.

For incompatible commodity mixing (e.g., apples and kiwis without separators), the AI issues a warning; the buyer must explicitly acknowledge.

Decision Merger & Tenant Message Generation

If all checks pass → ALLOW.

The trade request is created with status = 'PENDING_SELLER_RESPONSE'.

The structured parsed_specs JSONB is stored with all per-container data, plus multi_shipment_schedule if provided.

A Loom hash is generated and stored in governor_decisions.

The seller receives a Smart Inbox item (priority 75) summarizing the request (including multi-shipment schedule if any).

The buyer receives a low-priority confirmation.

If CONDITIONAL or DENY → The Governor generates a plain-language tenant_message (using Groq, fallback Ollama) that lists the specific unmet conditions (e.g., “Port of discharge for container 2 is not valid for Egypt” or “Dual-use goods require an export licence”). The PlainLanguage Governor Decision Panel (Part 6.1.3) opens automatically, and the buyer cannot proceed until the conditions are resolved.

Step 1.7 – Draft Auto-Save (Cross-Cutting)

The structured form autosaves every 30 seconds (debounced). The autosave stores the current state in trade_requests with status = 'DRAFT' and the full parsed_specs JSONB (including any partial multi-shipment schedule). If the buyer leaves the page (e.g., closes the browser), the draft is preserved. Upon next login, the Smart Inbox displays a “Continue draft trade request” item (priority 30). Drafts expire after 14 days of inactivity (reminders at 11 and 13 days). Expired drafts can be restored via Company Admin → Drafts (see Part 2.7.4).

GOVERNANCE GATES (PHASE 1 – UPDATED WITH ALL MODIFICATIONS)

Gate ID	Check	Enforcer	Failure Action
G1U1	Agent mesh session properly initialised (if using Express Mode)	Governor Orchestrator	Block all agent activations
G1U2	Intent classification confidence ≥ 0.85 OR human confirmed (Express Mode)	Intent Parser (A2)	Escalate to human clarification
G1U3	Spec extraction confidence ≥ 0.80 per critical field (Express Mode)	Spec Extractor (A2)	Highlight uncertain fields

G1U4	HS code classification with dual-use check complete	HS Classifier (A2) + WasmEdge	Block if dual-use without licence
G1U5	Jurisdiction prescreen: ALLOW or CONDITIONAL resolved	Compliance PreScreener (A2/A3)	DENY with explicit reason code
G1U6	All agent invocations logged via Loom	Governor Orchestrator	Reject session if logging fails
G1U7	Per-container data consistency (port in destination country, pallet count ≥0, packaging valid)	Governor (WasmEdge)	Highlight error, block submission
G1U8	Marketplace partner attribution recorded (if applicable) – includes auto-detection	Governor (A3)	Revenue attribution required; if disputed, trade held
G1U9	Container type recommendation logged; buyer override logged	Governor (A3)	Override allowed but logged
G1U10	Multi-shipment schedule validation (if toggled): each shipment has future date, valid port, container count ≥1	Governor (WasmEdge)	Block; show which shipment failed
G1U11	All Governor decisions shown to tenant via PlainLanguage Decision Panel if blocked	Governor (A3) + Groq/Ollama	User sees jargon-free explanation and action links; cannot proceed until condition resolved

WORKED EXAMPLE – BUYER USES STRUCTURED FORM WITH MULTI-SHIPMENT REQUEST

Context : Nile Foods (Egypt, trader mode = DUAL, active context Buyer) wants to import 20,000 kg of Valencia oranges from Mekong Fresh (Vietnam) and 5,376 kg of lemons, using two containers, and requests a multi-shipment contract with two shipments.

GTID Entry : Types “SGTX-VN” → autocomplete shows Mekong Fresh. Selects. Trust portrait shows score 88, “Saved Contact – 8 trades”.

Container Setup: Sets Number of Containers = 2.

Container 1 : Origin Vietnam, Destination Egypt, Port of Discharge Alexandria, Palletized Yes, Pallet Size EUR.

Container 2 : Clone Container 1, then change Port of Discharge to Port Said.

Commodities – Container 1:

Add Commodity 1 : Fresh Fruits → HS 0805.10 (Oranges), Grade A, Size 72-80 mm, Packaging: Boxes (10 kg cartons), Number of Pallets: 22, Notes: “Stack max 5 layers”.

Add Commodity 2 : Fresh Fruits → HS 0805.50 (Lemons), Grade B, Size 50-60 mm, Packaging: Mesh bags (8 kg), Number of Pallets: 14.

Container 2 : After cloning, adjust pallet counts (or keep same).

Multi-shipment request : Toggle “Request multi-shipment contract”. Adds two shipments:

Shipment 1 : Delivery date 15 July 2026, Port Alexandria, 1 container, commodities as defined.

Shipment 2 : Delivery date 15 August 2026, Port Port Said, 1 container, same commodities.

Global Notes : “Seller to provide phyto certificate for both products. Insurance required.”

AI Container Advisor : Groq suggests 2×40’ HC reefers – matches user’s entry. No adjustment needed.

Marketplace Attribution : No prior marketplace introduction found – no banner.

Submit : Governor ALLOW. Trade request created with status = PENDING_SELLER_RESPONSE. Seller receives Smart Inbox item: “New multi-shipment trade request from Nile Foods – Oranges + Lemons, 2 containers over 2 shipments, ports Alexandria & Port Said. Accept, modify, or decline.”

Draft Auto-Save : If the user had left mid-form, the draft (including multi-shipment schedule) would have been saved and resumable later.

END OF PHASE 1 (TRADE INITIATION) – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 2: Seller Quote, Packing & Logistics Orchestration (v6.3 – Fully Extended, with EXW Price Entry Integrated into Request View, No Separate EXW Tab)

PHASE 2 GOAL
The seller receives the buyer's structured container-level request (and optional multi-shipment schedule), opens that request in a single unified screen, and from that screen:

Declares the loading origin

Locks EXW prices per commodity (using live market chart and AI-recommended band)

Designs packing based on the buyer's structured data

Determines logistics costs (manually or via RFQ)

Offers alternative delivery ports within the same destination country

May modify the buyer's multi-shipment schedule

Calculates the SGTX platform fee (dynamic, commodity - & country-pair-specific, paid by seller, added to total quote)

Submits the quote package

The buyer sees the total price and the fee amount separately for full transparency. All actions are performed from a single integrated screen – no separate EXW tab.

PREREQUISITE
The seller has accepted the trade request (status = ACCEPTED). The seller's trader mode must be SELL or DUAL (with active context “Seller” if the dual-mode toggle is used). The buyer's parsed_specs (containers, commodities) and optional multi_shipment_schedule are available.

STEP 2.1 – SELLER OPENS THE REQUEST (UNIFIED VIEW)
From the Pending Requests tab in the Seller Dashboard, the seller clicks on a trade request. The system displays a single, scrollable screen that contains:

Read-only section : The buyer's original request – number of containers, per-container origin, destination, port of discharge, palletized status, pallet size, destination override, per-commodity details (type, product, HS code, specification, packaging, number of pallets, notes), global notes, and multi-shipment schedule (if any).

Editable section : Seller's inputs – loading origin, EXW prices, incoterm, logistics costs, alternative ports, packing & containerisation, multi-shipment response, SGTX fee calculation, and submission.

All mandatory fields are clearly marked. The seller never navigates to a separate “EXW Price Lock” tab.

Step 2.2.1 – Loading Origin (Mandatory, with Advanced Geocoding and Route Intelligence)

Before any pricing or logistics steps, the seller must specify the physical origin of the goods. These fields are mandatory and appear at the top of the editable section.

Fields:

Field	Type	Source / Validation	Example
Country of Loading	Global dropdown	ISO 3166-1 from jurisdictions tabl e, filtered by non-sanctioned countries	Vietnam
Port of Loading	Dependent dropdown	Filtered by Country of Loading from portstable (UN/LOCODE + RIA updates)	Saigon (VNSGN)
Alternative Loading Point (optional)	Free text + map picker	Allows seller to specify a warehouse or factory address if different from port	"Mekong Fresh Warehouse, Ben Tre"

Geocoded Coordinates	Auto-filled (read-only)	Nominatim (self-hosted, OSM) converts address to lat/lng for distance calculations	10.8231°N, 106.6297°E

Advanced UI/UX and Zero-Cost Intelligence Features:

Auto-suggest from recent trades – If the seller has previously used a loading origin for the same buyer or commodity, the system pre-fills the country and port with a “Recent” badge. The seller can change it.

Geocoding integration – When the seller enters an alternative loading point (free text), the system calls Nominatim (self-hosted, free) to resolve the address to coordinates. A small map (Leaflet) displays the pin. The seller can drag the pin to correct the location.

Distance to port calculation – Once both the loading point (or port) and the port of loading are known, the system calculates the approximate trucking distance using OSRM (self-hosted, free). This distance is displayed as an advisory: “Estimated trucking distance: 85 km”. This helps the seller estimate local trucking costs (can be used to pre-fill the logistics cost entry).

Port intelligence – Next to the Port of Loading dropdown, a small info icon shows:

Last mile congestion level (from historical AIS data, free)

Average gate waiting time (from carrier reports, anonymised)

Any active disruptions (strike, weather, closure) scraped from free news/RSS via RIA.

Sanctions and compliance check – The Governor (WasmEdge) instantly checks that the country and port are not sanctioned and are compatible with the selected incoterm (e.g., for FOB, the loading port must be in the seller's country). If a violation is detected, the field is highlighted in red, and a Decision Panel explains the issue.

Fallback for offline geocoding – If Nominatim is temporarily unavailable, the system uses a local cached reverse geocoding database (extracted from OSM planet file, refreshed monthly). No external API key required.

Governance (G2U17 – Enhanced): The Governor checks that:

The loading port is not sanctioned and is active in the ports table.

For incoterms FOB, CFR, CIF, CPT, CIP, the port of loading must be in the seller's country (enforced by WasmEdge).

If the seller enters an alternative loading point, the geocoded coordinates are logged but not used for compliance (purely advisory).
Missing or invalid loading origin blocks quote submission; the Decision Panel explains the reason and provides a direct link to correct the field.

Step 2.2 – EXW Price Lock (Fully Dynamic, with per-Ton / per-Kilo / per-Unit Conversion, Weight Unit Toggle, Total Container Price Calculation, AI Fair Price, Live Market Chart, Price Deviation Justification, Post-Lock Price Watch, and Five Advanced Add-Ons)

Overview
The seller locks the EXW price per commodity. The interface is fully dynamic: the seller

can enter the price in tons, kilograms, or the buyer's packaging unit (e.g., per box, per bag). The system instantly calculates equivalent prices in all other units. A global toggle switches all displayed weights between metric tons and pounds. The system also calculates total EXW value per commodity, per container, and for the entire shipment. An AI Fair Price (advisory) based on recent anonymised platform trades is shown alongside a live market chart. If the seller's price deviates significantly above the AI band, a mandatory justification is required. After quote submission, a post-lock price watch monitors market movements and can trigger a price renegotiation.

In addition, five advanced, high-end add-ons are available (all zero-cost, self-hosted). Add-ons 2, 3, and 4 are optional for the seller (configurable in Seller Dashboard → Advanced Pricing Tools). Add-on 5 is controlled exclusively by the Platform Governance Authority (Admin Portal) and can be enabled or disabled per GTID.

A. Live Market Chart & AI Fair Price (Advisory, Zero-Cost, Privacy-Preserving)

Live Market Chart – A 30-day interactive line chart (Chart.js) is displayed for the primary commodity (and optionally for each commodity if multiple). Data is sourced from free public feeds: FAO, USDA, World Bank commodity prices, and, where available, free terminal market data (e.g., CME delayed quotes). The chart is cached daily and updated every 6 hours. A green shaded horizontal band represents the AI-recommended price range (Groq primary, Ollama fallback) based on public indices and seasonality.

AI Fair Price (Advisory) – Next to the chart, a "Fair Price" badge shows a single recommended value (e.g., "Fair price: $1.52 per kg"). This value is computed from anonymised aggregated data of completed SGTX trades for the same (HS6 code, seller country, buyer country) within the last 7 days, weighted by volume, and blended with the public index (70% actual trades, 30% index).

Zero-cost implementation : A nightly PostgreSQL batch query computes volume-weighted average EXW price for each (HS6, seller_country, buyer_country) where trade status = COMPLETED and lock_date > now() – interval '7 days'. The result is cached. If fewer than 5 trades exist, the system falls back to the public index only.

Privacy guarantee : The seller is never told that the price is based on their own or the buyer's past trades. The tooltip only says: "Based on recent market activity and public indices." Individual transaction data is never exposed.

One-click acceptance : A button "Use fair price" pre-fills the EXW price fields with the fair price (converted to the seller's selected entry unit). The seller can then adjust manually.

B. Dynamic Price Input with Smart Unit Conversion (Per Ton / Per Kilo / Per Unit)

For each commodity, the seller sees a compact card with:

Price entry method – Three tabs (or a segmented toggle) labelled:
"Per Ton" | "Per Kilo" | "Per Unit"

"Per Unit" means the buyer's specified packaging unit (e.g., "box", "mesh bag", "carton"). The net weight per unit is taken from the buyer's commodity specification (e.g., 10 kg per box). If the buyer has not provided net weight per unit (rare), the seller is prompted to enter it once; that value is then stored and used for this trade.

Default selection is based on commodity type : per kg for fresh produce, per ton for grains/minerals, per unit for consumer goods.

Price value – A number input with two decimal places. As the user types, the system instantly calculates and displays the equivalent prices in the other two units (read-only, below the input).

Example : Seller enters 1.50perkg→systemshows“1.50perkg→systemshows“1,500.00 per ton” and “$15.00 per box (10 kg box)”.

Currency – Dropdown (USD, EUR, GBP, etc.). Defaults to seller’s preferred currency from their profile.

Conversion logic (zero-cost, client-side with server validation):

1 metric ton = 1,000 kilograms.

1 kilogram = 1,000 grams (not used directly, but consistent).

Per-unit weight : stored as net weight in kg (e.g., 10 kg/box).

Price per unit = Price per kg × net weight per unit.

Price per kg = Price per unit ÷ net weight per unit.

All conversions are performed client-side in JavaScript for instant feedback; upon form submission, the server (Governor) re-validates the conversions to prevent tampering.

C. Global Weight Unit Toggle (Metric Tons / Pounds)

A prominent toggle located at the top of the EXW section (above the first commodity) allows the seller to switch all displayed weights between metric tons (t) and pounds (lb).

Conversion factor : 1 metric ton = 2,204.62 lb (exact). All conversions use this factor.

The toggle affects:

Buyer’s requested quantity (e.g., “20,000 kg” becomes “44,092 lb”).

Net weight per unit (e.g., “10 kg box” becomes “22.05 lb box”).

Price labels (e.g., “1.50perkg”becomes“1.50perkg”becomes“0.68 per lb” – automatically recalculated using the conversion factor).

Important : The toggle is purely for display. The underlying stored values remain in metric (kg, tons) for consistency. When the seller enters a price in “per lb” mode, the system converts it to per kg on the backend. The seller never needs to think about this – the interface feels native.

D. Total EXW Value Calculation (Real-Time, Per Commodity, Per Container, Total Shipment)

As the seller enters EXW prices, the system calculates and displays (read-only):

Total EXW value for this commodity = (price per unit) × (total number of units). Total number of units is derived from the buyer’s total quantity divided by net weight per unit.

Subtotal for the container – Sum of EXW values for all commodities inside that container (displayed at the bottom of each container’s card).

Grand total EXW for all containers – Sum of all container subtotals, displayed prominently at the end of the EXW section.

These totals update in real time and are used in Step 2.7 (SGTX fee calculation). The seller sees them clearly as they adjust prices, helping them understand the financial impact.

E. Price Deviation Justification (Governance G2UA1)

If the entered price (converted to per-kg equivalent) is >30% above the AI band maximum, a red banner appears and a text area is shown. The seller must provide a justification (minimum 20 characters). The justification is stored in seller_quotes.exw_justification and linked to a Governor decision. The quote cannot be submitted until the justification is provided.

If the entered price is >50% below the AI band minimum, an amber advisory warning appears ("Your price is significantly below market average. Are you sure you want to proceed?") but no justification is required. The seller can click "Proceed anyway".

If the price is within the band, no warning or justification is needed.

F. Post-Lock Price Watch (Background, Zero-Cost)

After the seller submits the quote (Step 2.8), a NATS subscription (or a lightweight polling service) monitors daily market price changes for the same commodity and trade corridor using the same free price feeds.

If the market price moves >10% (absolute percentage) from the locked EXW price (converted to per-kg), the seller receives a high-priority Smart Inbox item (priority 85):

"Market price moved +12% since you locked your EXW price at $1.50/kg. Your locked price stands. Would you like to reopen pricing and notify the buyer?"

The seller can click "Reopen Pricing". This creates a Governor decision (G2UA1) and notifies the buyer via Smart Inbox. The buyer can then accept the new price (the seller updates the quote) or continue with the original locked price.

If the seller ignores the alert, no further action is taken. The alert reappears only if the market moves another 5% from the previous alert threshold.

All price watch events are logged for audit.

G. Five Advanced, High-End Critical Add-Ons (Zero-Cost, Self-Hosted)

These add-ons are optional, configurable per seller (except #5 which is admin-controlled). They do not require any paid API or external service.

Add-on 1: Volume-Tiered Pricing

The seller can define up to three price tiers based on total quantity (e.g., 1.50/kgfor0-10,000kg,1.50/kgfor0-10,000kg,1.45/kg for 10,001-20,000 kg, $1.40/kg for >20,000 kg). The system automatically calculates the blended EXW value and displays the effective average price. The buyer sees the tiered structure only if the seller chooses to share it (checkbox).

Implementation : UI with an "Add tier" button; pricing logic stored in seller_quotes.tiered_pricing JSONB. Governor validates that tiers are non-overlapping and increasing in quantity.

Add-on 2 : Dynamic Currency Conversion with Live FX (Advisory)

If the seller selects a currency different from the buyer's requested currency (or from USD), the system shows the approximate converted amount using free ECB daily rates (updated once per day). A note says: "Estimated exchange rate – final rate determined at payment." This helps the seller set prices in their local currency while the buyer sees an equivalent.

The seller can also set a fixed FX rate for the trade (overriding the daily rate), which is then used in the contract. The rate must be within a reasonable band of the market rate (±3%) unless justified.

Implementation : ECB daily rates XML (free). The seller's chosen rate is stored in seller_quotes.fx_rate_override. Governor validates the deviation.

Add-on 3: Historical Price Comparison Tool (Seller-Only)

A small expandable panel shows the seller's own EXW prices for the same commodity and buyer over the last 12 months (if any). This is purely for the seller's reference and never shared. It helps the seller remain consistent.

Implementation : Query seller_quotes for the same seller_tenant_id, product_hs_code, and buyer_tenant_id within the last year; display average, min, max, and a sparkline. No external data.

Add-on 4 : "Lock & Hold" – Temporary Price Lock Without Fees (Time-Limited)

The seller can offer a temporary price lock for a defined period (24 hours, 48 hours, 1 week, etc.) without any fee. This indicates to the buyer that the quoted EXW price is guaranteed only for that lock period. The lock is not a binding contract; it simply means the seller will not change the price during that window. The buyer sees a badge: "Price locked until [date]".

If the buyer accepts the quote before the lock expires, the price holds. If the lock expires, the seller may change the price or extend the lock (a new lock period can be offered).

Implementation : A dropdown in the EXW section: "Lock this price for: Off | 24h | 48h | 1 week | Custom (hours)". The chosen duration is stored in seller_quotes.price_lock_until(timestamp). The buyer's Smart Inbox shows the lock expiry. The Governor does not enforce any financial penalty; it is purely informational. If the seller attempts to change the price after the lock expires but before quote submission, no restriction.

Governance : The lock is advisory. The seller can still revise the price before submission. After submission, the lock information is included in the quote for transparency.

Add-on 5 : Platform-Admin-Controlled "Private Price Floor / Ceiling" (Per GTID)

The Platform Governance Authority (Admin Portal) can set a minimum or maximum allowed EXW price for a specific seller GTID, buyer GTID, or commodity pair. This is used for strategic partnerships, pilot programs, or regulatory compliance. The seller does not see this override; the Governor enforces it silently.

Configuration : Admin Portal → Special Price Controls. The admin selects a seller GTID (or buyer GTID, or both) and a commodity HS code range, then sets a price floor (minimum) and/or price ceiling (maximum). The admin can also enable/disable this control per GTID with a toggle.

Enforcement : When the seller attempts to lock an EXW price, the Governor (WasmEdge) checks the special_price_controls table. If a matching rule exists and the price is outside the allowed range, the Governor returns DENY with a Decision Panel that says: “The price you entered is outside the permitted range for this trade corridor. Please adjust.” The seller is never shown the actual floor/ceiling values (to prevent reverse engineering).

Zero-cost implementation : Simple PostgreSQL table special_price_controls(seller_gtid, buyer_gtid, hs_code_pattern, min_price, max_price, enabled, created_by, approved_by_multisig). Only Platform Governance Authority (multisig) can insert/update rows. Changes are logged in audit_log.

Use case example : A large buyer negotiates a confidential volume discount – the admin sets a ceiling for that buyer-seller pair. Or a seller is temporarily restricted due to compliance issues – a floor is set to prevent dumping.

All add-ons are activated via toggles in the Seller Dashboard → Settings → Advanced Pricing Tools (except #5 which is platform-admin-only). Each add-on, when enabled, adds relevant UI elements to Step 2.2. They are all built on the zero-cost stack (PostgreSQL, Governor, NATS, etc.).

H. Zero-Cost Implementation Summary for Step 2.2

Component	Technology / Data Source	Cost
Live market chart	FAO, USDA, World Bank free feeds; Chart.js	$0
AI Fair Price (advisory)	Internal PostgreSQL batch query (volume-weighted average of last 7 days’ trades) + Groq/Ollama for natural language explanation	$0
Unit conversions	Client-side JavaScript + server validation	$0
Weight unit toggle (metric/pounds)	Client-side factor (1 t = 2204.62 lb)	$0
Total container price calculation	JavaScript on parsed data	$0
Price deviation justification	Governor + Rego policy	$0
Post-lock price watch	NATS subscription + daily price feed polling	$0
Add-on 1: Volume-tiered pricing	UI + JSONB storage	$0
Add-on 2: Dynamic currency conversion	ECB daily rates (free XML)	$0
Add-on 3: Historical price comparison	Internal query on seller’s own data	$0
Add-on 4: Lock & Hold (time-limited, no fee)	Timestamp stored in seller_quotes.price_lock_until	$0
Add-on 5: Admin-controlled price	special_price_controls table + Governor enforcement	$0

floor/ceiling		

I. Governance & Verification

G2UA1 (enhanced) : The Governor validates that the EXW price (converted to per-kg) is positive, that any required justification is provided and meets minimum length (20 characters), and that the derived total EXW values are consistent with the price and quantities.

G2UA2 (new) : If Add-on 5 is active for the seller or buyer, the Governor checks that the price is within the defined floor/ceiling. If outside, verdict DENY with a generic message.

G2UA3 (new) : For Add-on 4 (Lock & Hold), the lock duration is stored but not enforced by Governor; it is purely informational. However, if the seller attempts to change the price after the lock expires but before submission, no restriction. After submission, the lock information is frozen in the quote.

All price entries, justifications, tiered pricing data, FX overrides, and lock periods are stored immutably in seller_quotes and linked to the Governor decision ID. The Loom hash chain ensures that the locked price cannot be repudiated.

J. Example UI Flow (Textual Wireframe for Seller – with Add-ons Enabled)

text

```
┌──────────────────────────────────────────────────────────────
───┐
│ EXW PRICE LOCK – Commodity: Valencia Oranges (HS 0805.10) │
├──────────────────────────────────────────────────────────────
───┤
│ Live Market Chart (last 30 days) AI Fair Price: $1.52/kg [Use] │
│ [[redacted]] Based on recent market activity. │
│ AI band: $1.45 – $1.65/kg │
├──────────────────────────────────────────────────────────────
───┤
│ Advanced Tools: [Volume tiers] [Currency FX] [History] [Lock & Hold] │
├──────────────────────────────────────────────────────────────
───┤
│ Price entry: ○ Per Ton ● Per Kilo ○ Per Unit (10 kg box) │
│ EXW price: [1.50] [USD] per kg │
│ Equivalent: $1,500.00 per ton | $15.00 per box │
│ Lock & Hold: [✓] Lock this price for [48 hours ▼] (no fee) │
├──────────────────────────────────────────────────────────────
───┤
│ Total EXW for this commodity: $30,000.00 │
│ Container subtotal (Container 1): $41,289.60 │
│ Grand total EXW (all containers): $82,579.20 │
```

```
├──────────────────────────────────────────────────────────────────
──┤
│ ⚠ Price is 12% above AI band (max $1.65). Justification required: │
│ [ Organic certification, premium quality, high demand in EU. ] (min 20 chr)│
├──────────────────────────────────────────────────────────────────
──┤
│ [Save EXW Prices] [Back] [Next: Logistics Costs] │
└──────────────────────────────────────────────────────────────────
──┘
```

End of Step 2.2

Step 2.3 – Logistics Cost Entry: Manual or Request Quotes (Fully Expanded, with Incoterm-Aware Dynamic Form, AI Benchmarking, Zero-Cost Route Intelligence, and Advanced RFQ Mode with Pickup Warehouse, Multi-Container Multi-Stop Management, AI Route Optimisation, Batch RFQ, and Professional Options)

Overview

The seller determines all logistics costs required to move the goods from the loading origin to the destination agreed with the buyer. The system provides two modes: Mode A – Manual Entry (the seller enters fixed costs) and Mode B – Request Quotes (RFQ) (the seller sends anonymous or directed requests to logistics providers). All fields adapt dynamically based on the selected incoterm (enforced by WasmEdge). AI benchmarking suggests typical costs using anonymised platform data and free public sources. Zero-cost route intelligence pre-calculates distances and estimated fuel costs using OSRM and OpenStreetMap.

Prerequisites:

The seller has already specified the loading origin (Step 2.2) and selected an incoterm (Step 2.4 – incoterm selection is assumed to be done before this step; if the blueprint order differs, the numbering can be adjusted).

For Mode B, the seller must have confirmed at least one pickup warehouse location (global or per container).

The seller can switch modes at any time before the packing plan is locked. Switching discards unsaved entries with a warning.

MODE A – MANUAL ENTRY (I HAVE FIXED PRICES)

When Mode A is selected, the system displays a list of logistics services. Only services that are mandatory or optional for the chosen incoterm are shown. The WasmEdge incoterms engine validates that mandatory fields are filled before quote submission.

A.1 Incoterm-Aware Dynamic Form

Standard Logistics Services Table (with incoterm applicability):

Service	Applies to Incoterms	Mandatory / Optional	Example Cost
Trucking (origin to port of loading)	All except EXW	Mandatory (if not EXW)	$900
Customs clearance (export)	All except EXW	Mandatory (if not EXW)	$600
THC (terminal handling charge at origin port)	FOB, CFR, CIF, CPT, CIP	Mandatory	$300
Ocean freight (per container or total)	CFR, CIF, CPT, CIP	Mandatory	$2,800 × number of containers
Air freight (per kg)	For air shipments only	Mandatory (if air)	$4.50/kg
International trucking (land border)	DAP, DPU, DDP	Mandatory (if overland)	$1,200
Insurance (optional, can be included)	Any except EXW	Optional	$450
Destination charges (THC, delivery)	DAP, DPU, DDP	Mandatory	$500
Duties (for DDP only)	DDP	Mandatory	15% of value
Other local charges (e.g., documentation, seal)	All except EXW	Optional	$50

Dynamic behavior:

When the seller changes the incoterm (Step 2.4), the list of services refreshes immediately. Previously entered costs for services that are no longer applicable are cleared (with a warning). Costs for services that remain applicable are retained.

Each service row includes:

A description with a tooltip explaining what the service covers.

A cost input (number field) with currency dropdown (default same as EXW currency).

A “Source” badge : “Manual” (or “From RFQ” if previously selected in Mode B then switched).

For trucking and ocean freight, an AI benchmark suggestion (see section A.2).

A.2 AI Cost Benchmarking (Advisory, Zero-Cost, Privacy-Preserving)

For each service (especially trucking, ocean freight, and customs clearance), the system displays an AI-suggested cost range (e.g., “Typical range: 800–800–1,000”). This is derived from:

Anonymised aggregated data of actual logistics costs from completed SGTX trades for the same (origin country, destination country, container type, commodity) within the last 30 days. Computed nightly (volume-weighted average and percentiles).

Public benchmarks (e.g., SIERA for ocean freight, free index data from Drewry, or scraped spot rates) as a fallback when insufficient platform data exists.

Route intelligence – For trucking, OSRM calculates the driving distance between the loading point and the port of loading. Using a configurable cost-per-km (derived from local fuel prices and driver wages, scraped from free government data), the system estimates a base cost and shows a range (±20%).

The benchmark is purely advisory. A small tooltip explains : "Based on recent shipments on this route. Your actual cost may vary."

A.3 Zero-Cost Route Intelligence for Trucking

When the seller enters a loading origin (Step 2.2) and selects a port of loading, the system automatically calls OSRM (self-hosted) to calculate the driving distance and estimated travel time.

The distance is displayed in the trucking service row, along with an estimated fuel cost (using a configurable fuel price per km, updated monthly from free sources). The seller can use this as a baseline for their manual cost entry.

A.4 Validation and Governance for Mode A

G2U18 (enhanced) : The Governor validates that all mandatory services for the selected incoterm have a cost entry. If any mandatory service is missing, the quote submission is blocked, and the Decision Panel lists the missing services with direct links to fill them.

G2U18a (new) : If the seller attempts to submit a quote with a logistics cost that deviates >50% from the AI benchmark range (e.g., 5,000fortruckingwhentherangeis5,000fortruckingwhentherangeis800-$1,000), the Governor returns a CONDITIONAL verdict requiring a justification (e.g., "Express delivery, dedicated truck"). The justification is stored.

MODE B – REQUEST QUOTES (RFQ) WITH PICKUP WAREHOUSE, MULTI-CONTAINER MANAGEMENT, AND ADVANCED PROFESSIONAL OPTIONS

When Mode B is selected, the seller uses a request-for-quote system to obtain competitive bids from logistics providers. The seller must first specify pickup location(s) – optionally per container and with multi-stop routes – then send RFQs, receive quotes, and select the best offer.

B.1 Global Pickup Warehouse Entry (Mandatory)

Field : "Primary Pickup Warehouse / Factory Address" – free text with autocomplete (Nominatim, self-hosted, OSM data).

Geocoding : As the seller types, the system suggests addresses. Upon selection:

The full address is displayed.

A pin appears on an embedded Leaflet map (centered on the coordinates).

The driving distance to the port of loading (from Step 2.2) is calculated via OSRM and shown (e.g., "Distance to Saigon port: 85 km").

The seller can drag the pin to adjust the location manually. The new coordinates are reverse-geocoded to an address.

The global pickup address is stored in seller_quotes.pickup_address and seller_quotes.pickup_coordinates.

B.2 Per-Container Pickup Assignment (For Multiple Containers)

When the trade involves more than one container, the seller may assign a different pickup warehouse to each container. This is critical when goods are stored at different factories.

For each container (as defined by the buyer's container tabs), the seller sees an expandable "Pickup Details" panel.

Options:

Same as global pickup (default) – uses the primary address.

Different pickup location – the seller enters a separate address and pins it on a container-specific mini-map.

When "Different pickup location" is selected:

The system stores the address and coordinates per container in seller_quotes.per_container_pickup JSONB.

The distance to the port of loading is calculated and displayed.

All per-container pickup points are colour-coded on a master map (see B.5).

B.3 Multi-Stop Pickup per Container (Advanced)

For any container, the seller can enable multi-stop pickup (the truck collects goods from multiple warehouses before reaching the port).

A list of stops appears. The seller can add stops (address + geocoding) and optionally specify:

Estimated loading time per stop (minutes).

Priority constraints (e.g., "must pick up stop A before stop B").

The system uses ORTools (VRP solver) to compute the optimal route that minimises total driving distance and respects loading time windows. The optimised route is displayed on the map with numbered pins.

The seller can accept the AI-optimised route or manually reorder stops by dragging the list.

Governance (G2U18h) : The system checks that the route is feasible (no stop is >500 km from the next) and that total estimated driving time does not exceed the loading window (warning only, not blocking).

B.4 Cross-Container Route Optimization (AI Smart Option)

If multiple containers have different pickup locations (and possibly multi-stop), the seller can enable co-optimise entire collection across all containers.

The system solves a Vehicle Routing Problem (VRP) using ORTools:

Inputs : For each container – pickup location(s) (or list of stops), port of loading (same for all containers, or can be different per shipment).

Number of available trucks (default = number of containers, but seller can reduce for consolidation).

Loading time per stop, driver hours, etc.

Output : An optimised dispatch plan showing which truck picks up which containers and the optimal sequence.

The seller can view the plan on the map as coloured routes. They can accept the AI-optimised plan or manually adjust (e.g., force two containers to be picked up by the same truck).

Zero-cost : ORTools (Apache 2.0) runs locally on the seller’s sovereign node.

B.5 Master Map with All Pins

A master map at the top of the RFQ section shows all pickup locations for all containers, each colour-coded by container number.

Clicking a pin opens a tooltip showing container ID, address, and multi-stop sequence (if any).

If a container has a multi-stop route, the route line is drawn between stops, with arrows indicating direction.

For cross-container optimisation, each truck’s route is displayed in a distinct colour.

B.6 Sending RFQs (Batch or Per Container)

Batch RFQ (Option 8) – The seller can create a single RFQ that includes all required trucking movements as a bundle.

The RFQ payload includes:

A summary table of each container’s pickup address(es), port of loading, and estimated distance.

The optimised route (if enabled) as a suggested collection order.

Commodity details, weight, volume, container type, incoterm, loading window.

Providers can quote a single price for the entire bundle or per container.

Alternatively, the seller can send separate RFQs per container (e.g., for different providers).

RFQ Distribution:

Anonymous broadcast – Sent to all eligible logistics providers (based on route, commodity, and provider preferences). The seller’s identity is hidden until a quote is accepted.

Directed – The seller enters specific GTIDs of logistics providers (from saved contacts or search). The system highlights providers used successfully in the past (“Preferred – used 5 times”).

B.7 Advanced Professional Options (Zero-Cost, High-End)

The seller can enable any of the following optional toggles before sending RFQs. All options are built on existing zero-cost components (NATS, OSRM, ORTools, PostgreSQL, Governor).

Option 2 – Multi-Stop Pickup (already covered in B.3)
Option 3 – Special Equipment Requirements
A checklist of special equipment: “Reefer (temperature range)”, “Flatbed”, “Container

chassis”, “Tail lift”, “Air-ride suspension”. The seller checks applicable items. Providers see these requirements and adjust quotes.

Option 4 – Time-Defined Loading Window with Demurrage Protection
The seller specifies a precise loading window (e.g., “Pickup only between 08:00-12:00 on 20 Jun”). If the provider arrives outside the window, demurrage charges may apply. The system can automatically compute an estimated demurrage risk based on historical port congestion (from Phase 5) and display it to the provider.

Option 5 – Quote Validity Period
The seller sets a deadline for providers to submit quotes (e.g., “Quotes must be received within 24 hours”). After the deadline, the RFQ closes automatically. Providers see a countdown timer. This encourages timely responses.

Option 6 – AI “Should-Cost” Model for Quote Comparison
When quotes are received, the system displays an AI-generated “should-cost” range (advisory) based on the pickup location(s), distance, commodity, and current market rates (from anonymised platform data). This helps the seller identify unusually high or low quotes.

If a quote is >30% above the should-cost range, the system flags it with a warning.

If >30% below, it flags potential risk (e.g., provider might be unreliable).
The seller can still select any quote.

Option 7 – Direct Negotiation with Providers via Trade Room
The seller can open a secure chat (Trade Room) with any provider who has submitted a quote to clarify details, ask for a revised quote, or negotiate terms. All messages are logged and can be used as evidence in case of dispute.

Option 8 – Batch RFQ for Multiple Containers (already covered in B.6)

B.8 Receiving and Comparing Quotes

Providers receive RFQs via Smart Inbox and email (co-branded). They submit quotes through their Logistics Portal.

The seller receives notifications when quotes arrive. A “Compare Quotes” panel shows all received quotes per service (or per container, if separate RFQs). Columns include:

Provider name, trust score, quoted price, estimated transit time, special conditions, and a “Select” button.

The seller selects a quote for each service (or for the entire bundle). Once selected, the cost field is populated and locked (the seller can still manually override, but an “Override” flag is logged).

The seller can also request a bundled quote from a freight forwarder (one quote covering multiple services). The bundle is shown as a single line item with a breakdown.

B.9 Provider Visibility and Addendum Requirement

Providers see the exact pickup address(es) on a map, along with distances, special equipment, loading windows, and any optimised routes.

Before contract lock, the selected provider must sign the Logistics Service Addendum (Phase 3). The seller is warned if a provider has not yet signed, but can still select them; the contract will later enforce the signature.

B.10 Governance for Mode B

G2U18c (new) : The seller must confirm at least one pickup warehouse location (global or per container) before any RFQ is sent. Governor validates that pickup_coordinates is not null and that the address resolves to a valid location.

G2U18d (new) : For multi-stop pickups, the system checks route feasibility; if total estimated driving time exceeds the loading window, a warning is shown but RFQ can still be sent.

G2U18e (new) : For directed RFQs, the provider must exist and not be blocked. The Governor rejects RFQs sent to suspended or blocked providers.

G2U18f (new) : The selected quote must be from a provider who has a valid GTID and is not sanctioned. The Governor checks this before allowing the cost to be locked.

G2U18g (new) : All RFQ broadcasts, quote submissions, and selections are logged immutably via Loom.

UI WIREFRAME (MODE B WITH TWO CONTAINERS, MULTI-STOP, AND OPTIONS)

text

```
┌──────────────────────────────────────────────────────────────────
───┐
│ LOGISTICS COST ENTRY [Mode A] [Mode B ●]│
│ Incoterm: CFR │
├──────────────────────────────────────────────────────────────────
───┤
│ MASTER MAP (All pickups) │
│ [Interactive map with pins: Container 1 (blue), Container 2 (green); │
│ route lines; cross-container optimisation on] │
├──────────────────────────────────────────────────────────────────
───┤
│ PICKUP MANAGEMENT │
│ ┌─ Container 1 (Oranges) ──────────────────────────────────────┐ │
│ │ Pickup: ○ Same as global ● Different location ││
│ │ Address: [Mekong Fresh, Ben Tre]  [Distance: 85 km to port] ││
│ │ Multi-stop: [✓] [Add stop] [Optimize route] ││
│ │ Stop 1: Ben Tre (30 min) ││
│ │ Stop 2: My Tho (20 min) ││
│ │ Stop 3: Saigon (15 min) ││
│ │ Optimised route distance: 142 km (saves 18 km) ││
```

```
│
└──────────────────────────────────────────────────────────────────────
┘ │
│ ┌─ Container 2 (Lemons) ───────────────────────────────────────────┐ │
│ │ Pickup: ● Same as global ○ Different location │ │
│ │ Address: [Global pickup: Long An Warehouse] │ │
│ │ Multi-stop: [ ] (single pickup) │ │
│
└──────────────────────────────────────────────────────────────────────
┘ │
│ Cross-container optimisation: [✓] Co-optimise entire collection │
│ AI suggests: Truck A picks up Container 2 then Container 1 (shorter). │
│ [Accept AI plan] [Manual override] │
├──────────────────────────────────────────────────────────────────────
───┤
│ PROFESSIONAL OPTIONS │
│ ☑ Special equipment: [Reefer (-18°C)] [Tail lift] │
│ ☑ Loading window: 14 Jun 08:00-12:00 (demurrage risk: low) │
│ ☑ Quote validity: [48 hours ▼] │
│ ☑ AI should-cost comparison: enabled │
│ ☑ Allow direct negotiation via chat │
├──────────────────────────────────────────────────────────────────────
───┤
│ RFQ DISTRIBUTION │
│ ○ Anonymous broadcast (all eligible logistics providers) │
│ ● Directed: [SGTX-VN-LOG-001 - Fast Trucking] [SGTX-VN-LOG-002 - Ocean Co.│
│ [+ Add from saved contacts] │
├──────────────────────────────────────────────────────────────────────
───┤
│ [Send Batch RFQ (all containers)] [Send per-container RFQ] │
├──────────────────────────────────────────────────────────────────────
───┤
│ QUOTES RECEIVED (0 so far) │
│ (After quotes arrive: comparison table with provider, price, conditions) │
└──────────────────────────────────────────────────────────────────────
───┘
```

ZERO-COST IMPLEMENTATION SUMMARY FOR STEP 2.3

Component	Technology / Data Source	Cost
Dynamic incoterm-aware form	WasmEdge + client-side JS	$0
AI cost benchmarking (platform data)	Nightly PostgreSQL aggregation (volume-weighted percentiles)	$0
Public benchmark fallback	Drewry (free), SIERA, scraped spot rates	$0
Route intelligence (distance, fuel cost)	OSRM (self-hosted), OpenStreetMap, local fuel price DB	$0
Geocoding	Nominatim (self-hosted, OSM)	$0
Map rendering	Leaflet + self-hosted tile server	$0
VRP / route optimisation	ORTools (Apache 2.0) – local Python/Rust	$0
RFQ distribution and quote collection	NATS JetStream, existing notification service	$0
Provider selection from saved contacts	Existing tenant_contacts table	$0
AI should-cost model	Internal query on anonymised completed trades (percentiles) + light regression	$0
Direct negotiation chat	Existing Trade Room (NATS + Yjs)	$0
Governor validations	Rego policies, WasmEdge	$0

No billing details, no external API keys, no cloud services are required. All components are self-hosted and open-source.

End of Step 2.3 (Fully Expanded)

Step 2.4 – Alternative Delivery Ports (Within the Same Destination Country) – Fully Expanded, with AI-Suggested Ports, Cost Impact Analysis, Comparison Table, and Zero-Cost Implementation

Overview

After the primary logistics costs are determined (Step 2.3), the seller may offer alternative ports of delivery within the same destination country as the buyer's original port of discharge. This gives the buyer flexibility to choose a port that may offer lower costs, faster transit, or better inland connectivity. The seller can add any number of alternatives. For multi-shipment contracts, alternatives can be added per shipment.

The system provides AI-suggested alternative ports based on historical cost savings, transit time improvements, and congestion data (all from free sources). Each alternative includes a detailed cost breakdown, transit time, and a comparison with the primary option. The seller can run a separate mini-RFQ for a specific alternative port if logistics costs are unknown.

Prerequisite : The seller has already determined primary logistics costs (Step 2.3) and knows the destination country (from the buyer's request).

A. Adding an Alternative Delivery Port

Button : “+ Add Alternative Delivery Port” at the bottom of the Logistics Cost Entry section (visible after primary costs are at least partially filled).

Clicking opens a new row or modal with the following fields:

Field	Type	Source / Validation	Example
Port of Delivery	Global dropdown	Filtered by the buyer’s destination country from ports table (UN/LOCODE + RIA updates)	Damietta (EGDAM)
Expected Transit Time	Number input (days)	Default suggested by AI (see section B); seller can override	16 days
Additional Logistics Cost	Number (positive, negative, or zero)	Seller can enter manually, or click “Run RFQ for this port” to obtain quotes	-$150
Cost Breakdown	Read-only (auto-calculated)	Primary logistics cost + additional cost	2,000+(−2,000+(−150) = $1,850
Total Delivered Price (before SGTX fee)	Read-only (auto-calculated)	EXW total + total logistics cost (primary + extra)	30,000+30,000+1,850 = $31,850
Reason / Notes	Text area (optional)	Seller explanation	“Lower port charges, good road connectivity”

Governance (G2U19 – Enhanced):

The alternative port must be in the same destination country as the buyer’s original port of discharge (verified by WasmEdge).

The port must not be sanctioned (checked against jurisdictions).

Transit time must be >0 days.

If any alternative port fails validation, the quote submission is blocked, and the Decision Panel lists the invalid ports.

B. AI-Suggested Alternative Ports (Advisory, Zero-Cost)

When the seller focuses on the “Add Alternative” field, the system automatically queries an AI Port Suggester (A1, Groq primary, Ollama fallback) that analyses:

Historical platform data (anonymised) for the same destination country, commodity, and container type: which alternative ports were selected by other sellers in the last 90 days, and what were the average cost savings/deltas.

Public data (free):

Port congestion levels from AIS historical data (scraped from free AIS providers).

Average dwell times from port authority websites (scraped monthly).

Road connectivity scores (from OpenStreetMap).

Distance from primary port – ports within 200 km of the primary port are flagged as “nearby”.

The AI returns up to 3 suggested ports, each with:

Port name, country.

Estimated transit time difference (e.g., “+2 days” or “-1 day”).

Estimated cost difference (e.g., “−150”or“+150”or“+50”).

Confidence score (e.g., “85% confidence based on 12 similar trades”).

A “Use suggestion” button to pre-fill the form.

The suggestions are purely advisory; the seller can ignore them and manually select any valid port.

Zero-cost implementation:

The suggestion engine runs entirely on the seller’s sovereign node. The “historical data” is a nightly aggregated table alternative_port_benchmarks (destination_country, commodity_hs6, container_type, alternative_port, avg_cost_diff_usd, avg_transit_diff_days, sample_count).

Public data is refreshed weekly via RIA scraping (free RSS/HTML).

No external API keys required.

C. Running a Mini-RFQ for an Alternative Port

If the seller does not know the logistics cost for an alternative port, they can click “Run RFQ for this port”. This triggers a simplified RFQ process focused only on the incremental cost difference from the primary port.

Workflow:

Seller selects the alternative port.

The system automatically creates a mini-RFQ for ocean freight and destination charges(trucking from the primary port to the alternative port, or directly from origin to alternative port – depending on the incoterm). The RFQ includes:

Same origin, same commodity, same container type.

New destination port (the alternative).

Reference to the primary quote for comparison.

The RFQ is sent to the same logistics providers who already quoted for the primary route (or to new providers if preferred).

Providers submit incremental quotes (e.g., “Primary cost 2,800,alternativeport2,800,alternativeport2,650”).

The seller receives quotes and selects one. The additional cost (or saving) is automatically populated.

All steps are logged; the selected quote is stored in quote_delivery_options.extra_logistics_cost_details.

Governance (G2U19a – new): The mini-RFQ must be completed before the alternative port can be included in the final quote. The Governor checks that a selected quote exists for the alternative port (unless the seller manually entered the cost).

D. Comparison Table for Buyer (Pre-submission Preview)

The seller sees a comparison table of all delivery options (primary + alternatives) at the bottom of the section, updated in real time. Columns:

Port	Transit Time	Additional Cost	Total Logistics Cost	EXW Total	Total Delivered Price (before SGTX fee)	Actions
Alexandria (primary)	14 days	–	$2,000	$30,000	$32,000	(primary)
Damietta	16 days	-$150	$1,850	$30,000	$31,850	[Edit] [Remove]
Port Said	13 days	+$100	$2,100	$30,000	$32,100	[Edit] [Remove]

The seller can edit or remove any alternative port. The primary port cannot be removed.

E. Multi-Shipment Contracts: Per-Shipment Alternatives

If the trade is a multi-shipment contract (buyer requested multiple future shipments), the seller can add alternative ports per shipment. The UI shows a tab or accordion for each shipment. The process for adding alternatives is identical to single shipment, but the comparison table is displayed per shipment.

Governance (G2U20 – extended): For multi-shipment, each shipment's alternatives are validated independently. If any shipment has an invalid alternative (e.g., port not in destination country), the entire quote submission is blocked, and the Decision Panel specifies which shipment and which alternative.

F. Integration with SGTX Fee Calculation (Step 2.7)

For each delivery option (primary and each alternative), the system calculates:

Total trade value (before fee) = EXW total + total logistics cost (primary + extra).

SGTX fee = dynamic rate × total trade value.

Final quoted price to buyer = total trade value + SGTX fee.

All options are presented to the buyer in the Quote Review & Negotiation tab.

G. UI Wireframe (Textual)

text

```
┌──────────────────────────────────────────────────────────────────────────────
──┐
│ ALTERNATIVE DELIVERY PORTS │
├──────────────────────────────────────────────────────────────────────────────
──┤
│ AI Suggestions: Based on similar trades, consider: │
│ • Damietta – avg saving $150, +2 days transit [Use] │
│ • Port Said – avg +$50, -1 day transit [Use] │
├──────────────────────────────────────────────────────────────────────────────
──┤
│ ┌─ Alternative Port 1 ─────────────────────────────────────────────────┐ │
│ │ Port of Delivery: [Damietta ▼] (Egypt) │ │
│ │ Expected Transit Time: [16] days (AI suggestion: 15-17) │ │
│ │ Additional Logistics Cost: [-150] USD (or [Run RFQ for this port]) │ │
│ │ Cost Breakdown: Primary logistics $2,000 + extra -$150 = $1,850 │ │
│ │ Total Delivered Price (before SGTX fee): $31,850 │ │
│ │ Notes: [Lower port charges, good road connectivity] │ │
│ │ [Save] [Cancel] [Remove] │ │
│
└──────────────────────────────────────────────────────────────────────────────
┘ │
│ [+ Add Alternative Delivery Port] │
├──────────────────────────────────────────────────────────────────────────────
──┤
│ DELIVERY OPTIONS COMPARISON │
│
┌─────────────┬──────────────┬──────────────┬──────────────┬──────────────┐
│
│ │ Port │ Transit Time │ Extra Cost │ Total Log │ Total Deliv │ │
│
├─────────────┼──────────────┼──────────────┼──────────────┼──────────────┤
│
│ │ Alexandria │ 14 days │ – │ $2,000 │ $32,000 │ │
│ │ (primary) │ │ │ │ │ │
│ │ Damietta │ 16 days │ -$150 │ $1,850 │ $31,850 │ │
│ │ Port Said │ 13 days │ +$100 │ $2,100 │ $32,100 │ │
```

```
│
└────────────────┴──────────────────┴──────────────────┴──────────────────┴──────────────────┘
│
├──────────────────────────────────────────────────────────────────────────────────────────────
────┤
│ [Back to Logistics] [Save Draft] [Next: Packing & Containerisation] │
└──────────────────────────────────────────────────────────────────────────────────────────────
────┘
```

H. Zero-Cost Implementation Summary

Component	Technology / Data Source	Cost
Port dropdown (filtered by country)	ports table (UN/LOCODE) + RIA updates	$0
AI-suggested alternative ports	Nightly aggregated table alternative_port_benchmarks + Groq/Ollama for explanation	$0
Mini-RFQ for alternative port	Same RFQ infrastructure as Step 2.3 (NATS, notifications)	$0
Cost breakdown and comparison table	Client-side JavaScript + Governor validation	$0
Transit time & cost difference calculations	Server-side (Rust) + stored in quote_delivery_options	$0
Validation (same country, not sanctioned)	WasmEdge + OPA	$0
Multi-shipment per-shipment alternatives	contract_shipments table + UI tabs	$0

No billing details, no external APIs, no cloud services required.

End of Step 2.4 (Alternative Delivery Ports)

Step 2.5 – Packing & Containerisation

The system consumes the buyer’s structured per-container data from parsed_specs. Seller can:

View and confirm buyer’s palletized status, pallet size, packaging, and number of pallets per commodity.

Calculate palletisation using ORTools solver – optimises cartons per pallet, layers, total pallets.

3D container viewer (Three.js) shows loading plan; collaborative editing (Yjs) for multiple seller employees.

Lock packing plan – once locked, plan becomes immutable. SSCC barcodes generated.

AI Loading Guide (A1, advisory) : Groq (fallback Ollama) generates step-by-step loading guide in seller's language.

Governance Gates (G2U10–G2U14): Feasibility, container legality, weight limits, lot mixing, Loom hash after lock.

Step 2.6 – Seller Responds to Multi-Shipment Request (if Any)

If the buyer requested a multi-shipment contract (Phase 1), the seller sees the proposed schedule in the Quote Submission tab. Options:

Accept as proposed – no changes.

Propose modifications – edit any shipment line (delivery date, port, container count, commodities/packaging) with notes. This becomes a counter-offer.

Reject multi-shipment – propose single shipment instead (or cancel).

Add alternative delivery ports – for any shipment (or per shipment), as in Step 2.4.

Logistics costing per shipment : Because ports and dates may differ, the seller must determine logistics costs for each shipment separately (manual entry or RFQ per shipment). The system calculates a total trade value per shipment (EXW + logistics). The SGTX fee is then calculated per shipment (see Step 2.7). Multi-shipment SGTX FEES is collected per-shipment (not upfront – see payment timing below).

Governance (G2U20) : Governor validates each shipment: future date, valid port, container count ≥1. Invalid shipments block submission.

Step 2.7 – SGTX Platform Fee Calculation (Paid by Seller, Added to Quote, Full Transparency)

The SGTX platform fee is a dynamic, commodity-specific, country-pair-specific percentage. It is paid solely by the seller and added to the total quoted price. Both parties see the fee amount for full transparency.

2.7.1 – Fee Calculation Model (Commodity & Country-Pair Aware)

The fee rate is derived from an estimated profit margin that is specific to the commodity (HS code, perishability, typical margins) and the seller/buyer country pair (currency, logistics costs, tariffs, risk). The model uses hybrid AI:

XGBoost/LightGBM base model – trained on thousands of historical SGTX trades. Features include:

HS6 code + commodity type (sugar, wheat, petroleum, fresh fruit, etc.).

Seller country and buyer country (currency trends, export taxes, port efficiency, regulatory stability).

Logistics costs (from manual entry or RFQ).

Market price indices (FAO, USDA, ECB, etc.).

Seasonality (week/year).

Volume (container count).

Incoterm and shipment type (single- or multi-shipment).

LLM refinement layer (A2, Hugging Face local, fallback Ollama) – reads real-time qualitative signals:

News sentiment about the commodity (e.g., “EU announces higher sugar import quota”).

Geopolitical events affecting the country pair.

Carrier performance data.
The LLM outputs an adjustment factor (±0.1% to ±0.5%) that the Governor applies to the base model’s margin estimate.

Final margin estimate = base XGBoost margin × (1 + LLM adjustment).
Fee rate is then derived from a mapping table (between 0.1% and 2.5%) based on the margin, with the same constitutional bounds.

Real-time self-learning : Every completed trade feeds back into the XGBoost model. If actual margins consistently deviate, the model auto-retrains weekly, and the Governor gates the new weights.

2.7.2 – Fee Calculation Per Delivery Option (and Per Shipment)

For each delivery option (primary port or alternative port), and for each shipment in a multi-shipment contract, the system calculates:

Total trade value (before fee) = EXW total + total logistics costs (as defined by incoterm).

Fee rate = f(estimated margin, commodity, seller/buyer country, market conditions).

SGTX fee amount = Total trade value × Fee rate.

Final quoted price to buyer = Total trade value + SGTX fee.

Example:

Total trade value (EXW + logistics) = $30,000

Fee rate (calculated) = 1.2%

SGTX fee = $360

Final quoted price = $30,360.

2.7.3 – Display to Seller (Quote Submission Tab)

The seller sees a clear breakdown:

text

```
┌──────────────────────────────────────────────────────────────┐
│ Price Breakdown for [Delivery Option / Shipment X] │
├──────────────────────────────────────────────────────────────┤
│ EXW total: $28,000 │
│ Logistics total (trucking, freight, etc.): $1,880 │
│ ──────────────────────────────────────────────────────────── │
│ Total trade value (before SGTX fee): $29,880 │
│ SGTX platform fee (1.2%): $360 │
│ ──────────────────────────────────────────────────────────── │
```

```
│ Final quoted price to buyer: $30,240 │
│ │
│ Payment timing: For single-shipment: due before contract lock. │
│ For multi-shipment: due per shipment, upon each shipment's lock.│
└──────────────────────────────────────────────────────────────┘
```

The seller cannot change the fee rate (constitutional bounds), but they see exactly how it is derived.

2.7.4 – Display to Buyer (Quote Review Tab)

The buyer sees full transparency – the total price and the fee amount separately:

text

```
┌──────────────────────────────────────────────────────────────┐
│ Quoted price (CFR Alexandria): $30,240 │
│ (including SGTX platform fee of $360) │
│ │
│ The seller is responsible for paying the platform fee. │
│ This fee secures cryptographic milestone tracking, AI-assisted │
│ logistics, and zero-counterparty-risk settlement. │
└──────────────────────────────────────────────────────────────┘
```

No slider or split is presented. The buyer simply sees the all-inclusive price and the fee component for transparency.

2.7.5 – Fee Payment Timing (Critical for Multi-Shipment)

Single-shipment contract : The seller pays the SGTX fee upfront before contract lock (Phase 3). The SGTX FEESLock is created and marked ACTIVE only after payment verification.

Multi-shipment contract : The master contract is signed once with no fee paid upfront. For each shipment in the schedule, the following steps occur:

Mutual confirmation – Seller clicks “Ready for Shipment X”, buyer clicks “Confirm Shipment X”. This triggers a per-shipment lock.

Fee payment – The seller must pay the SGTX fee for that specific shipment (calculated on that shipment’s trade value). Payment is processed via PSP.

SGTX FEESLock created – Once payment is verified, a new SGTX FEESLock (or FeeLock) is created for that shipment, and a new USTN is generated for that shipment.

Container release – SGTX sends release confirmation to logistics providers.

Shipment execution – The shipment proceeds through Phases 4-6 independently.

If the seller fails to pay for a shipment, that shipment is not locked and cannot proceed; the master contract remains active for remaining shipments (or may be cancelled per terms).

Governance (G2U22) : The final quoted price must equal Total trade value + SGTX fee. The Governor validates the fee rate within 0.1%–2.5%. If the calculated fee does not match the quote, submission is blocked.

Step 2.8 – Submit Quote Package

The seller clicks "Submit Quote". The Governor performs final checks:

All mandatory fields present : loading origin, EXW price, logistics costs (manual or selected quotes), packing plan locked.

For multi-shipment contracts : schedule fully defined (whether accepted or modified), each shipment validated (G2U20).

Alternative ports (if any) valid (G2U19).

SGTX fee correctly calculated and added to each total price (G2U22).

Fee rate within constitutional bounds.

If all checks pass:

Trade status → QUOTED.

Quote package (including all delivery options, multi-shipment schedule, alternatives, and per-option fees) stored in seller_quotes, quote_delivery_options, and seller_quotes.counter_schedule (for multi-shipment).

The buyer receives a Smart Inbox item (priority 75) with:

Summary of the quote (number of options, multi-shipment if any).

A link to the Quote Review & Negotiation tab where they can see the comparison table with total prices and fee amounts.

The seller receives a confirmation.

If any check fails, the Governor returns CONDITIONAL or DENY with a PlainLanguage Decision Panel explanation (tenant message generated by Groq/Ollama). The seller must correct the issue and resubmit.

GOVERNANCE GATES (PHASE 2 – COMPLETE UPDATED TABLE)

Gate ID	Check	Enforcer	Failure Action
G2U1	Living quote initialised	Governor Orchestrator	Block publication
G2UA1	EXW deviation ≤30% or justified	Price Check (A2)	Block / require justification
G2U2	Route Oracle validation	Route Oracle (A2)	Reject route
G2U3	Carrier sanctions screening	Compliance Validator	Block carrier
G2U3b	Carrier risk score ≥40	Risk Radar (A2)	Block/warn
G2U4	Pricing within market bands (±30%) – advisory	Pricing Dynamics (A2)	Flag for review

G2U5	Contingency route if primary risk >0.3	Risk Radar (A2)	Require contingency
G2U6	Autonegotiation parameters within policy	Governor (A3)	Reject autoaccept
G2U7	Carbon calculation valid	Sustainability Scorer (A1)	Flag correction
G2U8	Disruption monitoring active	Risk Radar (A2)	Degrade to static quote
G2U9	Freeze decision logged (packing plan lock)	Governor (A3)	Cannot proceed
G2U10	Palletisation feasible	Packing Solver (A2)	Require manual adjustment
G2U11	Container type legal for commodity	WasmEdge	Block illegal container
G2U12	Lot mixing allowed by importing country	HF (A2)	Require waiver
G2U13	Container gross weight within limit	ORTools	Reconfigure load
G2U14	Packing plan lock creates Loom hash	Governor (A3)	Cannot lock
G2U15	AI loading instructions generated (advisory)	Groq (A1)	–
G2U16	Logistics bundle optimiser recommendation (advisory)	Bundle Optimiser (A2)	–
G2U17	Loading origin specified	Governor (WasmEdge)	Block quote submission
G2U18	Mandatory logistics costs present (per incoterm)	Governor (WasmEdge)	Block; show missing
G2U19	Alternative delivery port valid (same country, transit time>0)	Governor (WasmEdge)	Block; reject option
G2U20	Multi-shipment schedule valid (each shipment)	Governor (A3)	Block; require correction
G2U21	Price visibility respected (buyer sees fee separately)	Governor (OPA)	Enforced at API level
G2U22	Final quoted price = trade value + SGTX fee (fee within 0.1%–2.5%)	Governor (A3)	Block; recalc required
G2UPACK1	Packing plan USTN required before settlement	Governor (A3)	Hold settlement

WORKED EXAMPLE – SELLER SUBMITS QUOTE WITH SGTX FEE (MULTI-SHIPMENT, ALTERNATIVE PORTS)

Context : Mekong Fresh (Vietnam) receives a multi-shipment request from Nile Foods:

Shipment 1 : 15 July, Alexandria, 1 container (oranges + lemons)

Shipment 2 : 15 August, Port Said, 1 container (same commodities)

Seller actions:

Loading Origin: Vietnam → Saigon.

EXW Price Lock: Oranges 1.50/kg,lemons1.50/kg,lemons2.10/kg.

Logistics – Manual (CFR): Trucking 900,customs900,customs600,
THC 300,oceanfreight300,oceanfreight2,800 per container × 2 = $5,600.

Alternative Port for Shipment 1 : Add Damietta with transit time 16 days, extra cost -$150.

Multi-shipment response : Accept schedule as proposed, but adjust Shipment 2 port to Damietta (same country) and note.

Packing: Validate buyer’s pallet data, lock plan.

SGTX fee calculation (per shipment):

For Shipment 1 (Alexandria primary, Damietta alternative):
Trade value (EXW + logistics) = 30,000.Feerate(commodity-specific,country-pair)=1.230,000.Feerate(commodity-specific,country-pair)=1.2360.
Quoted price = $30,360.

For Shipment 2 (Damietta): similar.

Alternatives also get their own fee (based on their trade value).

Display to seller : Sees breakdown with fee line and payment timing note.

Submit Quote: Governor passes. Status → QUOTED.

Buyer view:

Quote review shows for each option :
“Total 30,360(includesSGTXfee30,360(includesSGTXfee360)”.

Fee payment (later in Phase 3):

Master contract signed (no fee paid).

Shipment 1 mutually confirmed → seller pays $360 → USTN generated → shipment proceeds.

Shipment 2 later: same process.

END OF PHASE 2 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 3: Contracting, Negotiation & SGTX Fee Collection (v6.3 – Fully Extended, with Amendment Flow, Mutual Confirmation, SGTX Witness Clause, Own Contract Upload, Multi-Shipment per-Shipment Fee Payment, Logistics Provider Signatories, Container Release Control, Three-Tier AI, No Operating Modes, No Fee Split)

PHASE 3 GOAL
After the seller has submitted a quote package (Phase 2), the buyer reviews the options (including multi-shipment schedule and alternative ports). The buyer may accept a specific option, negotiate the price, or request amendments to non-price terms. Once both parties mutually confirm an agreement, the contract is assembled with a mandatory SGTX Witness Clause. If the parties wish to use their own contract, they must also sign a separate SGTX FEES Addendum. For single-shipment contracts, the SGTX fee is paid upfront before contract lock; for multi-shipment contracts, the fee is paid per-shipment upon each shipment's lock. All logistics providers must sign a service addendum, and no container is released without SGTX's confirmation after fee payment.

PREREQUISITE
Trade status is QUOTED. The buyer's trader mode must be BUY or DUAL (active context Buyer). The seller's trader mode must be SELL or DUAL (active context Seller).

Step 3.1 – Buyer Reviews Quote Package

The buyer accesses the Quote Review & Negotiation tab. The system displays a comparison table of all delivery options (primary port + alternatives) and, if multi-shipment was requested, a schedule table.

For each delivery option (single shipment or per-shipment in multi-shipment), the buyer sees:

Port of delivery, transit time.

Final quoted price (including SGTX fee).

SGTX platform fee amount (displayed separately for transparency, e.g., "Total 30,240incl.SGTXfee30,240incl.SGTXfee360").

A note: "The seller is responsible for paying the SGTX fee."

The buyer has three choices for each option (or for the entire multi-shipment contract):

Accept – the buyer accepts the quoted price and terms for that option (or the whole schedule). Proceeds to mutual confirmation.

Negotiate – the buyer proposes a different total price (a single number) for one or more options/shipments. The seller may counter.

Amend – the buyer requests changes to non-price terms: weight, packaging, port of discharge, delivery dates, commodity specifications, etc. (all fields that were originally entered in Phase 1 or modified by seller in Phase 2).

Step 3.2.0 – Amendment UI & Workflow

When the buyer clicks “Amend”, a structured amendment form appears (pre-populated with current values). The buyer can change any of the following (per option or per shipment):

Field	Example change
Weight / quantity	from 20,000 kg to 18,000 kg
Packaging	from boxes to mesh bags
Port of discharge	from Alexandria to Port Said (must be valid and same country)
Delivery date (for multi-shipment)	from 15 July to 20 July
Product specification	from Grade A to Grade B
Number of pallets	from 22 to 20

The buyer adds a mandatory note explaining the reason for the amendment. The system validates that the requested changes are commercially reasonable (e.g., weight change does not exceed container capacity, port remains in same destination country). The amendment request is stored in negotiation_amendments with status PENDING.

Governance (G3U1) : Governor checks that amended fields are within allowed ranges. If an amendment would violate a constitutional rule (e.g., changing port to a sanctioned country), the request is DENY with a Decision Panel explanation.

3.2.1 – NEGOTIATION HARMONY: REAL-TIME COUNTER-OFFER PANEL, BOT TRANSPARENCY, AND CHAT-FIRST UI

Purpose
Provide a professional, high-end negotiation interface that mirrors modern trading platforms. Both buyer and seller see a unified, real-time negotiation panel that updates instantly via NATS. The panel includes:

A timeline of all offers and counter-offers (price and non-price amendments).

An AI-assisted negotiation bot (transparent, can be paused or overridden).

Integrated trade room chat for contextual messaging.

Visual diff of changes (price, delivery date, port, etc.).

Prerequisite
A quote has been submitted (Phase 2) and the buyer has started negotiation (Phase 3, Step 3.1). The negotiation session is active.

A. Negotiation Dashboard Layout

The dashboard is displayed in the Trade Command Center under a new tab “Negotiation”. It replaces the older, simpler amendment form.

The dashboard has three vertical columns (desktop) or collapsible sections (mobile):

Left Column (Offer History)	Middle Column (Current Offer & Controls)	Right Column (Trade Room Chat)

Chronological list of all offers and counter-offers. Each entry shows: party, amount (if price negotiation), changed fields (if amendment), timestamp, status (pending, accepted, rejected).	Shows the latest offer from the counterparty. Buttons: Accept, Counter, Reject, Request Info. If AI bot is enabled, shows bot parameters and a “Pause/Override” button.	Full trade room chat (as in TCC) but with negotiation-specific context. Messages can be tagged with offer IDs.

B. Real-Time Updates via NATS

When either party submits a counter-offer, the other party’s dashboard updates instantly. A small notification badge appears in the Smart Inbox and on the TCC tab.

C. Visual Diff of Amendments

When a party proposes amendments (e.g., change port from Alexandria to Damietta, or adjust delivery date), the system shows a side-by-side diff view:

text

```
┌──────────────────────────────┬──────────────────────────────┐
│ Current │ Proposed │
├──────────────────────────────┼──────────────────────────────┤
│ Port: Alexandria │ Port: Damietta │
│ Delivery: 15 Jul 2026 │ Delivery: 20 Jul 2026 │
│ Packaging: Boxes (10 kg) │ Packaging: Mesh bags │
└──────────────────────────────┴──────────────────────────────┘
```

The buyer/seller can accept the diff partially (if multiple changes) or fully.

D. Negotiation Bot Transparency Panel

If either party enables the AI negotiation bot (opt-in, off by default), a panel shows:

Current round (e.g., “Round 2 of 5”).

Last bot message.

Bot parameters (e.g., “Target price : 1.45–1.45–1.55”, “Deadline: 24h”).

A “Pause/Override” button that disables the bot and lets the user take manual control.

All bot decisions are logged and can be reviewed in the audit trail.

Bot logic (advisory only, A1/A2):

The bot can propose counter-offers based on predefined ranges set by the user. It never forces acceptance. The user can see the rationale (“AI suggests $1.48 based on market movement and your historical acceptance rate”).

E. Integrated Trade Room with Offer Linking

In the chat panel, users can type messages and optionally attach them to a specific offer (e.g., “Regarding offer #3, we need the certificate of origin before we accept”).

The chat automatically shows when an offer is accepted or rejected, linking to the offer card.

F. Express Counter-Offer (One-Click)

For price negotiation only, a small widget shows the last offered price and a slider/input field to propose a new price. Clicking "Send Counter" instantly creates a new offer.

For amendments, a structured form (similar to the original amendment UI) appears inline.

G. Timeout and Expiry

The seller can set a validity period for their quote (e.g., 48 hours). The negotiation dashboard shows a countdown timer. If the timer expires, the quote is automatically withdrawn (the buyer can no longer accept).

The buyer can request an extension, which the seller can approve or reject.

H. Zero-Cost Technology Stack for Negotiation Harmony

Component	Technology	Cost
Real-time updates	NATS WebSocket + Zustand	$0
Diff viewer	JavaScript object comparison + React	$0
Chat panel	Existing Trade Room (NATS + Yjs)	$0
Bot transparency panel	Rig agent (Rust) + Groq/Ollama (advisory)	$0
Offer history	PostgreSQL negotiation_sessions table	$0
Timer and expiry	Client-side countdown + server validation	$0

I. UI Wireframe (Buyer Negotiating with Seller)

text

```
┌──────────────────────────────────────────────────────────────
────────────────┐
│ NEGOTIATION – USTN SGTX-EG-20260515-ABC123 [Bot: Paused] │
├──────────────────────────────────────────────────────────────
────────────────┤
│ OFFER HISTORY │ CURRENT OFFER │ TRADE ROOM │
├───────────────────────────────┬────────────────────────────
──┬─────────────┤
│ 14 Jun 09:00 – Buyer proposed $30,240│ Seller counter-offer (14 Jun 11:30) │ [Chat] │
│ (original quote) │ │ Buyer: Can you│
│ 14 Jun 11:30 – Seller counter-offer │ Price: $30,000 (CFR Alexandria) │ reduce price? │
│ $30,000 (pending) │ Delivery: 15 Jul 2026 │ Seller: We can│
│ │ Port: Alexandria │ do $30,000. │
│ │ │ │
│ │ [Accept] [Counter] [Reject] [Chat] │ │
├───────────────────────────────┴────────────────────────────
──┴─────────────┤
```

│ Bot transparency: Not active. │ Timer: Quote expires in 23h 45m │ [Send] │
└──┘

J. Governance

All offers and counter-offers are signed by the respective parties (ZITADEL passkey) and stored immutably.

The Governor validates that any accepted offer does not violate jurisdiction, sanctions, or fee bounds (re-validation).

If a bot is used, its decisions are logged in ai_inference_records.

Integration with Existing Blueprint

This section supplements Step 3.2 (Amendment UI & Workflow) and Step 3.3 (Seller Response). The existing negotiation log and amendment tables are reused; only the UI and real-time features are enhanced.

The Negotiation Bot Transparency Panel is already described in Part 11.5.5; this UI makes it visible to users.

Step 3.3 – Seller Responds to Amendment or Negotiation

The seller receives a Smart Inbox item : "Buyer has requested amendments / price negotiation on trade USTN..." The seller opens the Quote Review & Negotiation tab and sees a side-by-side comparison of original vs. requested values.

For a negotiation (price change):

Seller can : Accept (agrees to new price), Counter (proposes a different price), or Reject (with reason).

Counter-offers follow the same negotiation bot transparency (A2/A3) with Groq/Ollama assistance. The bot shows rounds, last message, and a "Pause/Take Over" button.

For an amendment request:

Seller can : Accept all, Accept some (with notes), Counter-amend (propose different changes), or Reject (with reason).

If the seller accepts only part of the amendment, the buyer can choose to proceed with the partially accepted changes or continue negotiating.

The negotiation/amendment log is stored in negotiation_amendments and negotiation_sessions. Each round increments a round counter (max 5 rounds before deadlock resolution, see Part 3.8).

Governance (G3U2) : All negotiation and amendment messages are signed by the respective parties and logged via Loom.

Step 3.4 – Mutual Confirmation

Once both parties have explicitly agreed on a final set of terms (price, delivery option(s), schedule, and any amendments), each clicks “Confirm Agreement” in their respective portals. The system records a mutual_confirmation timestamp and creates a pre-contract snapshot of all agreed terms. This snapshot is immutable and stored in JSONB.

Governance Gate (G3U3) : The agreement is not considered final until both parties have pressed confirm. The Governor logs a decision with decision_type = “mutual_confirmation”.

Step 3.5 – Contract Assembly with SGTX Witness Clause

The contract is assembled by the Clause Forge agent (HF local, fallback Ollama). The Risk Oracleand Jurisdiction Harmonizer run. The contract includes:

All commercial terms (price, incoterm, delivery options, schedule, amendments).

Mandatory SGTX Witness Clause (automatically inserted, non-removable):

“The parties acknowledge that SGTX Platform (the ‘Platform’) has facilitated the execution of this contract as a non-custodial witness. SGTX is not a party to the underlying trade but provides cryptographic milestone tracking, AI-assisted logistics, and settlement instructions. The Platform’s fee of [X]% of the trade value (calculated as [amount]) is due as follows: for single-shipment contracts, the fee is payable upfront before contract lock; for multi-shipment contracts, the fee is payable per-shipment, each upon mutual confirmation and before that shipment’s USTN is generated. The Platform’s signature below serves as evidence of its role as witness and its right to collect the fee as specified.”

For multi-shipment contracts, the schedule from contract_shipments is incorporated.

The SGTX Governor digitally signs the contract as a witness (a separate signature field, alongside buyer and seller signatures). The contract is stored in contracts with status PENDING_SIGNATURES.

Governance Gate (G3U4) : The Clause Forge output must have confidence ≥0.90 for critical clauses; otherwise, human legal review is required.

Step 3.6 – Upload Own Contract (with SGTX Fees Addendum)

As an alternative to the generated contract, either party may upload their own PDF contract.

Workflow:

In the Contract Signing tab, the user clicks “Upload my own contract” (available only before signatures).

A PDF file is uploaded (max 10 MB).

HF Donut (A2, local) extracts key fields : parties, goods, price, incoterm, governing law, delivery terms, payment terms. Extracted data is displayed for user confirmation. Low-confidence fields are highlighted.

Mandatory : The user must also sign a separate SGTX FEES Addendum – a short document generated automatically that contains the SGTX Witness Clause (same as above) and the fee payment terms. This addendum is non-negotiable.

Both the uploaded contract and the addendum are stored as separate documents, both signed by all parties (buyer, seller, and Governor).

The Governor validates that the uploaded contract does not contain clauses that contradict the mandatory SGTX terms (e.g., no clause attempting to bypass the fee). If a contradiction is found, the Decision Panel appears: “Your uploaded contract contains a clause that conflicts with the SGTX FEES Clause. Please revise or accept the standard SGTX contract.”

If the user refuses to sign the addendum, contract lock is blocked.

Governance Gate (G3U5) : The uploaded contract must pass AI consistency check; any contradiction with mandatory SGTX terms → DENY.

Step 3.7 – Logistics Providers Sign Service Addendum

Before the contract lock can proceed, all logistics providers selected in Phase 2 (ocean freight, trucking, customs broker, etc.) must sign a Logistics Service Addendum. This addendum is automatically generated for each provider and includes:

USTN (to be generated after fee payment) – placeholder.

Obligation not to release the container(s) without SGTX confirmation.

Requirement to include USTN and GTIDs on all documents.

Penalty clause for unauthorised release.

Workflow:

Each provider receives a Smart Inbox item : “Sign logistics addendum for trade USTN ...”

The provider reviews the addendum in the Logistics Portal and clicks “Sign with Passkey”(ZITADEL).

The Governor records the signature in logistics_addenda.

The contract cannot proceed to fee payment until all required providers have signed.

Governance Gate (G3U6) : If any mandatory provider has not signed, the contract lock is blocked, and a Decision Panel explains which provider is missing.

Step 3.8 – SGTX Fee Payment (Single-Shipment Vs. Multi-Shipment)

After mutual confirmation and logistics addenda are signed, the fee payment process begins. The timing differs based on contract type.

3.8.1 – Single-Shipment Contract (Upfront Payment)

The seller receives a high-priority Smart Inbox item : “SGTX fee of $360 for trade USTN ... is due before contract lock.”

The seller navigates to the Pay SGTX Fee screen. The PSP Router (AI) selects the optimal PSP (e.g., Stripe, Payoneer) based on seller's country, fee, and realtime health.

The seller pays the gross amount (including PSP fees).

Once payment is confirmed via PSP webhook, the Governor marks the SGTX FEESLock as ACTIVE and generates the USTN (using the buyer and seller GTID suffixes).

The contract status becomes LOCKED.

The system sends a container release confirmation to the primary logistics provider (see Step 3.9).

3.8.2 – Multi-Shipment Contract (Per-Shipment Payment, No Upfront)

The master contract is signed first without any fee payment.

For each shipment in the schedule, the following steps occur:

Seller clicks "Ready for Shipment X" in the Contract Details → Multi-Shipment Schedule.

Buyer receives a Smart Inbox notification and clicks "Confirm Shipment X".

Mutual confirmation for that shipment triggers a per-shipment lock process.

Seller pays the SGTX fee for that specific shipment (calculated on that shipment's trade value, same rate as quoted).

Upon payment verification, a new SGTX FEESLock is created for that shipment, and a unique USTN is generated for that shipment (based on the same buyer/seller GTID suffixes but with a new timestamp and random).

The contract's contract_shipments row for that shipment is updated with status = LOCKED, ustn = generated USTN, SGTX FEES_lock_id = ...

SGTX sends a container release confirmation for that shipment's containers.

The shipment proceeds through Phases 4-6 independently.

If the seller fails to pay for a shipment, that shipment remains SCHEDULED and cannot proceed; the master contract remains active for remaining shipments (or may be cancelled per terms).

Governance Gate (G3U7) : For multi-shipment, the Governor ensures that the per-shipment fee is paid before that shipment's USTN is generated. The shipment cannot be executed without payment.

3.8.3 – Commission Payment Terms and Automatic Late Fees

The SGTX trade fee (commission) is due according to the following rules, enforced automatically by the Governor and the payment orchestrator.

For single-shipment contracts:

The seller must pay the SGTX trade fee within 7 days of contract lock (i.e., after mutual confirmation and before the USTN is generated).

However, if the container is loaded on the vessel before the 7-day deadline, the fee becomes due within 24 hours of the loading milestone confirmation. Whichever is closer (earlier) applies.

For multi-shipment contracts:

Each shipment's SGTX trade fee follows the same rule independently : due within 7 days of that shipment's lock, or within 24 hours of loading confirmation (if earlier).

Late fee calculation (automatic, zero-cost):

If the seller fails to pay by the due date, a late fee of 0.1% of the unpaid SGTX trade fee is added for each full day of delay. The late fee accrues daily and is capped at 100% of the original SGTX trade fee.

The late fee is calculated at 00:00 UTC each day by a cron job (late-fee-calculator, Rust service) that checks fee_payment_requests where status = 'PENDING' and due_date < NOW(). It updates fee_payment_requests.late_fee_accrued and adds a record to late_fee_events.

The seller receives a Smart Inbox alert (priority 90) on the first day of delay, and daily reminders thereafter.

The late fee is collected at the same time as the original fee (when the seller eventually pays). The PSP split instruction includes both the original fee and the accrued late fee.

Example:

SGTX trade fee = $360. Due date = 20 June.

Seller pays on 23 June (3 days late). Late fee = 360×0.1360×0.11.08. Total due = $361.08.

If the container is loaded before the 7-day deadline:

The loading milestone is confirmed by the shipping line or logistics provider (via barcode scan or manual entry). Upon confirmation, the system automatically:

Calculates the new due date = loading confirmation timestamp + 24 hours.

Updates fee_payment_requests.due_date and sends a Smart Inbox alert to the seller : "Container loaded. SGTX fee of $360 is now due within 24 hours (by 22 June 14:00 UTC)."

If the seller pays within that 24-hour window, no late fee. If not, late fees begin accruing after the 24-hour deadline.

Governance (G3U7 – enhanced):

The Governor checks that the fee payment is completed before the USTN is generated. If the due date passes without payment, the Governor blocks any further milestones (loading, departure, etc.) until payment is made.

For multi-shipment, a delay in one shipment does not block other shipments (each shipment's FeeLock is independent).

Zero-cost implementation:

The late-fee calculator runs on the same cron scheduler as other platform jobs (no external service).

Calculations use Rust decimal arithmetic to avoid floating-point errors.

The 0.1% rate is stored in platform_config table and can be adjusted by multisig if needed.

Step 3.9 – Container Release Confirmation (After Fee Payment)

Immediately after fee payment is verified (for single-shipment or per-shipment), the system automatically sends a release confirmation to the primary logistics provider (the ocean carrier or freight forwarder responsible for the container). This confirmation is sent via:

Email (co-branded : "SGTX via [Seller Name]") containing a one-time release token.

API webhook (if the provider has integrated).

The confirmation includes:

USTN

SGTX FEES payment hash (Loom hash)

One-time release token (UUID, valid 72 hours)

A link: https://sgtx.io/confirm-release?token=...&ustn=...

The logistics provider must acknowledge receipt (click the link or call the API) before the "Container Released" milestone can be confirmed. This acknowledgment is logged in release_confirmations.

Governance Gate (G3U8) : The "Container Loaded" milestone is blocked until the release confirmation has been acknowledged by the provider.

Step 3.10 – Digital Signatures & Contract Lock

After fee payment (and for multi-shipment, after each shipment's fee payment) and release confirmation acknowledged, the signing flow proceeds:

Buyer signs the contract (or the per-shipment addendum for multi-shipment) using ZITADEL passkey.

Seller signs.

SGTX Governor signs as witness (automatic).

The contract's cryptographic_hash is computed, and the status becomes LOCKED. For multi-shipment contracts, the master contract status remains ACTIVE_MASTER, and each locked shipment gets its own row in contract_shipments with status = LOCKED.

Governance Gate (G3U9) : All signatures must be collected, and the SGTX FEESLock (or per-shipment SGTX FEESLock) must be ACTIVE before lock.

Step 3.11 – Post-Lock Actions

USTN (or multiple USTNs for multi-shipment) is stored in shipments and appears on all documents.

Packing plan's ustn foreign key is updated (resolving the nullable column).

Smart Inbox items:

Seller : “Contract locked – USTN generated. Shipment tracking active.”

Buyer : “Contract locked – USTN generated. Awaiting shipment milestones.”

For multi-shipment, each locked shipment triggers its own Smart Inbox items.

GOVERNANCE GATES (PHASE 3 – UPDATED)

Gate ID	Check	Enforcer	Failure Action
G3U1	Amendment request fields within allowed ranges	Governor (WasmEdge)	DENY with Decision Panel
G3U2	All negotiation/amendment messages signed and logged	Governor (A3)	Block progression
G3U3	Mutual confirmation from both parties recorded	Governor (A3)	Cannot proceed to contract
G3U4	Clause Forge confidence ≥0.90 for critical clauses	Clause Forge (A2/A3)	Require human legal review
G3U5	Uploaded contract consistent with SGTX terms	Document Authenticity (A2)	Block; require addendum
G3U6	All mandatory logistics providers have signed addendum	Governor (A3)	Block lock; notify missing
G3U7	SGTX fee paid (single-shipment upfront; multi-shipment per-shipment)	Governor (A3)	Cannot generate USTN
G3U8	Container release confirmation acknowledged by logistics provider	Governor (A3)	Block milestone confirmation
G3U9	All signatures collected, SGTX FEESLock ACTIVE	Governor (A3)	Cannot lock contract
G3U10	Packing plan USTN populated (G2UPACK1)	Governor (A3)	Hold settlement

WORKED EXAMPLE – MULTI-SHIPMENT CONTRACT WITH PER-SHIPMENT FEE

Context : Seller Mekong Fresh and buyer Nile Foods have agreed on a multi-shipment schedule (two shipments). They have mutually confirmed the master contract.

Contract assembly – Clause Forge generates contract with SGTX Witness Clause. Both parties sign the master contract (no fee paid).

Logistics addenda – Trucking company, shipping line, and customs broker each sign via their portals.

First shipment activation – Seller clicks “Ready for Shipment 1”. Buyer confirms.

Per-shipment fee payment – Seller pays $360 via Stripe. Payment verified.

USTN generated for shipment 1 : SGTX-1397F3A-456ABC-20260715120000-A1B2C3D4. SGTX FEESLock ACTIVE.

Container release confirmation sent to shipping line. Provider acknowledges.

Contract lock for shipment 1 (master status remains active). Shipment proceeds.

Second shipment – Two months later, same process repeats. New USTN generated for shipment 2.

Result : No upfront bulk fee, each shipment pays its own fee only when ready.

3.12 – CANCELLATION AND MODIFICATION OF LOCKED CONTRACTS / SHIPMENTS (WITH PARTY-SPECIFIC FEES, CONTAINER NUMBER TRIGGER, AND SCORE IMPACT)

After a contract (single-shipment) or a specific shipment within a multi-shipment contract is locked (USTN generated), the parties may agree to cancel or modify the trade. The platform provides a structured workflow for both scenarios, enforced by the Governor. Cancellation fees and scoring impacts depend on who initiates the cancellation, when, and the reason.

A. Cancellation Fee Matrix (Who Pays What)

Scenario	Cancellation Fee	Payer	Notes
Buyer cancels before container number is assigned (i.e., before booking confirmation upload)	$0	None	No fee; contract voided
Buyer cancels after container number is assigned (booking confirmation uploaded)	$50	Buyer	Buyer must upload booking cancellation as evidence
Seller cancels after contract lock (any time)	$50	Seller	No need for container number trigger
Shipping line cancels or container rolled to another date	$0	None	No fee; platform logs reason
Mutual cancellation (agreed by both parties)	$0	None	Requires mutual confirmation
Force majeure (determined by Platform	$0	None	Requires evidence upload

Governance Authority)			

Important : The $50 cancellation fee is in addition to the SGTX trade fee already paid (which is generally non-refundable unless the cancellation is due to platform error).

B. Workflow for Buyer-Initiated Cancellation After Container Number

The buyer goes to the Trade Command Center → “Request Cancellation”.

The system checks if a container number has been assigned to the shipment (i.e., shipments.container_numbers is not empty, or a booking confirmation document with container number has been uploaded and extracted).

If yes, the buyer is required to upload a booking cancellation document (PDF from the shipping line) as evidence.

The system uses HF Donut to extract the cancellation confirmation number and date. The extracted data is displayed for buyer confirmation.

Once the buyer confirms, the cancellation request is sent to the seller. The seller can Accept or Reject.

If accepted:

The $50 cancellation fee is charged to the buyer (via PSP split).

The contract/shipment status becomes CANCELLED_BY_BUYER.

The buyer’s trust score is reduced (see section D).

The seller receives a Smart Inbox notification.

If the seller rejects (e.g., believes the cancellation is invalid), the dispute process begins (Phase 10). The Platform Governance Authority reviews the evidence.

C. Seller-Initiated Cancellation (After Contract Lock)

Seller clicks “Request Cancellation”. No container number check is required.

The seller must provide a reason (free text, min 20 characters).

The buyer receives the request and can accept or reject.

If accepted:

The $50 cancellation fee is charged to the seller.

The seller’s trust score is reduced.

The contract is cancelled.

If rejected, the trade continues.

D. Shipping Line / Logistics Provider Initiated Cancellation or Rollover

If the shipping line cancels the booking (e.g., vessel omission) or the container is rolled to a later vessel, the logistics provider (or the seller) can upload an official cancellation notice from the carrier.

The system automatically:

Waives any cancellation fee (no charge to buyer or seller).

Records the reason in cancellation_reason (e.g., “Carrier rolled to 5 July”).

Updates the shipment status to ROLLED (new column) and adjusts the ETD/ETA accordingly.

Does not affect the buyer's or seller's trust score.

The shipping line's performance score is updated (see section E).

E. Score Impact (Trust Score and Performance Score)

All cancellations affect the relevant party's score (either the tenant's trust score or the logistics provider's performance score). The exact impact is determined by a combination of automated rules and manual review by the Platform Governance Authority.

For buyers and sellers:

Each cancellation (faulted party) reduces the tenant's trust score by a base amount: 5 points for the first cancellation in 90 days, increasing by 2 points for each subsequent cancellation (capped at 15 points per incident).

The reduction is applied automatically by the trust-score-service when a cancellation is finalised.

The tenant receives a Smart Inbox notification : "Your trust score has been reduced by 5 points due to a cancellation."

After 90 days without cancellation, the score gradually recovers (1 point per month).

For logistics providers (shipping lines, trucking companies):

If a provider cancels or rolls a container (without legitimate force majeure), their performance score (used in carrier risk scoring) is reduced by 3 points per incident.

If the provider can prove force majeure (e.g., port strike, weather), the score is not reduced. The provider must upload evidence.

Platform Administration Review (A3 – Human Escalation):

Any party can dispute the automatic score reduction within 14 days. The dispute goes to the Platform Governance Authority (multisig) for review.

The Authority can:

Uphold the reduction (no change).

Reduce the penalty (e.g., 5 points → 2 points).

Waive the reduction entirely (if cancellation was due to circumstances beyond the party's control).

For logistics providers, they can also adjust the reason code (e.g., "Carrier error" vs "Force majeure").

All decisions are logged immutably.

F. Modification (Changes After Lock) – No Additional Fees Within 7 Days

As previously defined : modifications requested within 7 days of lock incur no extra fees.

If a modification involves changing the container number or the shipping line, the original cancellation rules do not apply; it is treated as an amendment, not a cancellation.

G. Governance Gates (New and Revised)

G3U11 (revised) : Buyer-initiated cancellation after container number assignment requires uploaded booking cancellation document. Governor validates that the document is uploaded and that the extracted cancellation reference is present.

G3U12 (revised) : The $50 cancellation fee is charged to the initiating party (buyer or seller) via PSP split. If the fee payment fails, the cancellation is not finalised, and the trade remains active.

G3U13 (new) : Shipping line cancellations or rollovers require an official notice (PDF) uploaded by the logistics provider or seller. Governor validates that the notice contains a valid carrier reference.

G3U14 (new) : Automatic trust score reduction is applied by trust-score-service upon cancellation finalisation. The reduction is recorded in trust_scores.history JSONB.

G3U15 (new) : Any party can request human review of a score reduction within 14 days. The request is logged as a Governor decision with type score_review_request.

H. UI Example (Buyer Cancellation After Container Number)

text

```
┌─────────────────────────────────────────────────────────────┐
│ REQUEST CANCELLATION – USTN SGTX-EG-20260515-ABC123 │
├─────────────────────────────────────────────────────────────┤
│ ⚠ A container number has been assigned: MAEU8901234 │
│ │
│ To cancel, you must upload the booking cancellation document: │
│ [Choose File] booking_cancel.pdf │
│ │
│ Cancellation fee: $50 (payable by buyer) │
│ │
│ [ ] I understand that my trust score will be reduced. │
│ │
│ [Send Cancellation Request] [Cancel] │
└─────────────────────────────────────────────────────────────┘
```

I. Zero-Cost Implementation

Component	Technology	Cost
Container number detection	Check shipments.container_numbers or document extraction	$0
Booking cancellation document upload & extraction	HF Donut (local)	$0
Cancellation fee collection	Same PSP split as SGTX fee	$0

Trust score reduction	trust-score-service (XGBoost + rules)	$0
Performance score update for logistics providers	carrier-risk-scorer (LightGBM)	$0
Human review workflow	Admin Portal + Governor (multisig)	$0

End of Subsection 2 (Revised – Cancellation and Modification Rules)

END OF PHASE 3 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 4: Universal Trade Finance (v6.3 – Fully Extended, Corrected Final Version, with Full Disclosure, Financier Preferences, Automatic RFQ, Co-Financing, SGTX Witness Clause, Flat 0.25% Financing Fee Deducted at Disbursement, No Fee at Repayment, and Complete Governance)

PHASE 4 GOAL
After a trade (single-shipment) or a specific shipment within a multi-shipment contract is locked, any party (seller or buyer) may request financing. The request triggers an automatic RFQ broadcast to all financiers whose pre-set preferences match the borrower's criteria (countries, amount, settlement type). Financiers see full trade details (buyer/seller names, documents, trust scores, historical financing performance) before bidding – this is the key trust advantage of SGTX. Multiple financiers can co-finance a single request (split the amount). The SGTX financing service fee is a flat 0.25% of the financed amount, deducted from the disbursement via PSP split (non-custodial). SGTX signs as witness on each financing agreement, securing its legal right to collect the fee. The trade execution SGTX fee (dynamic, 0.1%–2.5%) is already paid by the seller at contract lock (Phase 3) and is independent.

PREREQUISITE
A contract (or a specific contract shipment) has status LOCKED and its trade SGTX fee has been paid by the seller. The requester's trader mode must be appropriate (seller for pre-shipment, buyer for post-shipment). Dual-mode toggle affects visibility.

Step 4.1 – Financing Request Initiation

Requester (seller or buyer) navigates to the Financing tab in their portal. A list of eligible locked trades (or specific shipments within multi-shipment contracts) is displayed. The requester selects a trade (or a specific shipment) and clicks "Request Financing".

Request Form (pre-filled from contract data):

Field	Source	Example
USTN (or shipment USTN for multi-shipment)	Contract / contract_shipments	SGTX-...-V1
Total trade value	Contract (EXW + logistics + SGTX fee)	$30,240
Requested amount (P)	User input (default 70% LTV)	$20,000
Financing type	Dropdown: Pre-shipment, Post-shipment, Invoice financing, Structured	Pre-shipment
Tenor (days)	User input	60
Preferred settlement method	Dropdown: Bank transfer, Stablecoin (USDC/USDT), DeFi protocol	Bank transfer
Preferred currency	USD, EUR, GBP, etc.	USD
Collateral type	Dropdown: Goods, Warehouse receipt, Receivables, None	Goods
Special instructions	Text area	“Seller needs working capital before harvest”

The system uses AI (Groq, fallback Ollama) to recommend a maximum LTV (Loan-to-Value) based on commodity perishability, jurisdiction, and historical default rates. The requester can override.

Governance (G4U1) : Only locked trades (or locked shipments) are eligible. The request is stored in financing_requests with status REQUESTED.

Step 4.2 – AI Credit Intelligence (Dynamic, per Request)

Immediately after submission, the Credit Intelligence Agent (A2, HF local, fallback Ollama) computes:

Credit score (0–100)

Default probability (0–100%) using 200+ signals, enhanced with multi-shipment context:

Signal category	Examples
Trade performance	Payment history, dispute frequency, document accuracy from SGTX tables
Corporate intelligence	UBO structure, sanctions proximity, PEP exposure, news sentiment (scraped)
Shipment-specific	Route risk, carrier reliability, perishability, realtime AIS vessel positions
Market intelligence	Commodity price trends (FAO), country risk scores
Behavioural	Negotiation style, responsiveness
Multi-shipment (if applicable)	Previous shipments within the same contract performance

The score and probability are displayed to the requester (advisory). Financiers will see the same later.

Governance (G4U2) : If default probability >15%, the system recommends reducing the facility size or requiring additional collateral (advisory, not blocking).

Step 4.3 – Financier Preferences (Pre-Set by Each Financier)

Each financier (bank or private) defines filters in their Financier Portal → Preferences. These filters determine which RFQs they receive. Stored in financier_preferences table.

Filter categories:

Filter	Description	Example
Accepted borrower countries	List of countries where the borrower (seller or buyer) is located	Egypt, Vietnam, UAE
Minimum trust score of borrower	0–100	≥70
Minimum trade value	Total trade value (EXW + logistics + trade fee)	≥$50,000
Maximum financed amount (per request)	Upper limit on the portion this financier will bid	$500,000
Preferred financing types	Pre-shipment, Post-shipment, Invoice, Structured	Pre-shipment only
Preferred settlement methods	Bank transfer, Stablecoin, DeFi protocol	Bank transfer, Stablecoin
Excluded commodities	HS code ranges or commodity types	Arms, dual-use, perishables
Geographic restrictions	“Accept only from”, “Accept all except”	Accept only UK, EU, USA

Example – Barclays (UK bank):

Accept only from UK, EU, USA.

Trust score ≥75.

Max financed amount £1M.

Settlement bank transfer only.

Example – Private financier (Family Office):

Accept all except sanctioned countries.

Trust score ≥60.

Prefer DeFi stablecoin.

Max $200,000.

Governance (G4U2a) : Preference changes are logged in audit_log. The system filters RFQs in real time – financiers only receive notifications for requests that match their settings.

Step 4.4 – Automatic RFQ Broadcast

The system automatically broadcasts an RFQ to all financiers whose preferences match the borrower's request. The RFQ is delivered via:

Smart Inbox item (priority based on match score).

Email (if financier enabled notifications).

API webhook (if financier integrated).

The RFQ includes:

Anonymised borrower identity (hidden until bid accepted) but all trade details (see Step 4.5) are visible to the financier when they open the RFQ.

Requested amount (P), tenor, financing type, preferred settlement method/currency.

Credit score, default probability, recommended LTV.

Commodity, total trade value, incoterm, ports, vessel (if available), logistics providers.

Governance (G4U3) : RFQ broadcast logged in financing_rfq_log. Financiers can opt out of categories via preferences.

Step 4.5 – Full Disclosure to Financiers (Strength of SGTX)

When a financier opens an RFQ, they see all non-confidential details of the trade and the requesting company before placing a bid. This is the key trust advantage:

Seller and buyer full legal names, GTIDs, jurisdictions, trust scores, KYB status.

Complete trade details : USTN, commodity, HS code, quantity, total trade value, incoterm, ports of loading/discharge, vessel name (if ocean), ETD/ETA, logistics providers.

All documents as read-only PDFs : commercial invoice, packing list, booking confirmation, eBL (if issued), QC reports, insurance certificate.

Historical performance of both counterparties:

Number of previous trades on SGTX, total trade value.

Dispute rate, on-time delivery percentage, average payment delay.

Previous financing history (if the requesting company has requested financing before): number of financing requests, total financed amount, repayment performance (on-time, late, default).

Credit score trend (last 12 months, line chart).

AI credit intelligence : current credit score, default probability, recommended LTV, and a Groq-generated plain-language risk summary.

Governance (G4U3a) : Financiers must have signed a confidentiality agreement (in their MSA) to access full details. Data is presented read-only and cannot be exported.

Step 4.6 – Bidding & Co-Financing

Bidding process:

Financiers submit encrypted bids (blind) via the Financier Portal. Each bid specifies:

Amount they are willing to finance (a portion of P, e.g., 30k,30k,50k).

APR (or all-in cost).

Collateral requirements (must match borrower's request or stricter).

Settlement method (must match borrower's preferred or an acceptable alternative).

Optional note (for negotiation).

Bidding window: configurable by borrower (default 48 hours).

Financiers do not see each other's bids until the window closes (only the total amount of bids received, not individual APRs or identities).

Co-financing : The borrower may accept a combination of bids that sum to P (or less). Each accepted financier enters into a separate financing annex (or a syndicated agreement). The system automatically consolidates accepted bids.

Negotiation : If a financier includes a negotiable note, the borrower's AI negotiation bot (A2/A3) can propose a counter-offer (APR, collateral, etc.). The negotiation bot transparency panel shows rounds, last message, target parameters, and a "Pause/Take Over" button. The borrower can also request extension of the bidding window.

Governance (G4U4, G4U5):

Minimum 2 qualified bids or window extension required.

Blended rate (weighted average) must be within market bands (±50% of benchmark) – flag for usury/compliance review if not.

If any bid uses DeFi, protocol risk score must be ≥60.

Step 4.7 – Financing Agreement & SGTX Witness Clause

After the borrower accepts a set of bids, a financing agreement is assembled. For each financier, an annex is created (or a master syndicated agreement). The agreement includes:

Financed amount (P), APR, tenor, repayment schedule.

Collateral terms.

Mandatory SGTX Financing Witness Clause (automatically inserted, non-removable):

"The parties acknowledge that SGTX Platform (the 'Platform') has facilitated this financing arrangement as a non-custodial witness. The Platform's financing service fee of 0.25% of the financed amount (calculated as [amount]) is payable exclusively by the borrower and shall be deducted from the disbursed principal. The Platform's signature below serves as evidence of its role as witness and its right to collect the fee as specified. The Platform does not guarantee repayment and holds no responsibility for the financier's or borrower's obligations."

For multi-shipment financing, the agreement references the specific shipment(s).

Signatures:

Borrower signs each annex (or master agreement) via ZITADEL passkey.

Each financier signs.

SGTX Governor signs as witness (automatic, Ed25519 signature).

The signed agreement is stored in financing_agreements and financing_annexes. Status becomes ACTIVE (but not yet disbursed).

Governance (G4U6) : The witness clause must be present; otherwise lock is blocked.

Step 4.8 – Financing Fee Collection (Flat 0.25%, Deducted from Disbursement)

Financing fee = 0.25% × Financed amount (flat, no dynamic range).
This fee is separate from the trade SGTX fee already paid by the seller at contract lock. It is collected only once, at disbursement, via PSP split. No fee is collected at repayment.

Collection process (non-custodial, using PSP split):

After the financing agreement is signed, the payment orchestrator creates a disbursement instruction for each financier.

Each financier initiates a payment of their portion P_i to a PSP that supports split payments (e.g., Stripe Connect, Payoneer Split).

SGTX's PSP router instructs the PSP to split the transfer in real time:

P_i – (0.25% × P_i) → borrower's bank account / wallet.

0.25% × P_i → SGTX's bank account (financing fee).

The PSP sends webhooks confirming both legs. The financing SGTX FEESLock (or FeeLock) is marked ACTIVE.

The borrower's received amount is recorded (actual_received = P_i – fee).

Note on who pays:

The financing fee is deducted from the borrower's disbursement. The borrower effectively pays it out of the loan principal. SGTX never holds the full principal.

If the seller requests financing, SGTX already collected the trade fee (e.g., 1%) at contract lock; now collects an additional 0.25% financing fee.

If the buyer requests financing, SGTX collects only the 0.25% financing fee (trade fee already paid by seller).

Governance (G4U7):

The split instruction must be cryptographically signed by each financier before execution.

The borrower sees the net amount they will receive before confirming.

PSP webhooks must be verified (signature, matching amounts).

If the split fails, the entire disbursement is rolled back.

Step 4.9 – Repayment & Release (No SGTX Fee)

The borrower repays the financier(s) directly according to the repayment schedule. The repayment is not split; SGTX does not collect any fee at this stage.

The borrower sends P_i + accrued interest to each financier via PSP (or onchain transfer).

SGTX monitors repayment events via PSP webhooks or onchain events (for DeFi).

Upon full repayment of a financier's portion:

The corresponding financing SGTX FEESLock is released.

The financier's historical record for that borrower is updated (on-time, late, or default).

The borrower's credit score is adjusted (for future financing requests).

The borrower receives a Smart Inbox confirmation.

Governance (G4U8) : Repayment events are immutable, logged in financing_repayments and attached to the financing agreement's Loom chain. No SGTX fee is collected during repayment.

Step 4.10 – Financier-Specific Data: Saved Financing History & Credit Scores

Each financier has a "Financed Companies" tab in their portal (read-only, audit-traced). It stores:

List of all companies (seller or buyer) that the financier has ever financed.

For each company:

Name, GTID, jurisdiction, current trust score.

Total financed amount (sum of all financing agreements).

Repayment performance : number of on-time payments, late payments (average days), defaults.

Credit score trend (line chart over the last 12 months, with annotations for each financing event).

Active financing (outstanding amount, next repayment date, health status).

The financier can refresh credit assessment on demand (triggers a new AI credit intelligence run using the latest trade and repayment data).

Data is updated automatically after each repayment or default event. It is immutable – cannot be edited by the financier.

Governance (G4U9) : Data access is restricted to the financier who owns the history. The platform does not share across financiers.

Step 4.11 – DeFi Option (if Selected by Financier)

If a financier bids with settlement method "DeFi protocol" and the borrower accepts, the following steps occur before disbursement:

DeFi PlainLanguage Risk Summary modal appears (five bullet points, generated by Groq, fallback Ollama). The financier must click “I understand” – recorded in ai_inference_records.

DeFi Protocol AutoSelector (A2) – selects the protocol with the highest risk score (≥60) and sufficient liquidity (e.g., Aave V3 on Polygon, Compound on Base).

Smart Contract Deployment – The defi-service deploys an ERC3525 token (tokenised invoice) and an escrow contract (if needed). The stablecoin loan (USDC, USDT) is deposited into the borrower’s wallet. All gas costs are borne by the financier (or deducted from the principal – configurable).

Health Monitoring – The Stablecoin Monitor (A2) tracks peg deviations using free 0x API / Chainlink feeds:

Deviation >0.5% → warning to both parties via Smart Inbox.

Deviation >2% → freeze new positions; existing positions flagged.

Liquidation Early Warning – The Liquidation Watch Agent (A2, LSTM) predicts health factor for each position every 6 hours. If predicted health factor drops below 1.2, a Smart Inbox alert is sent to the financier and borrower with a Groq-generated recommendation (“Add collateral or repay $X within 24 hours to avoid liquidation”).

Governance (G4U10, G4U11):

The protocol’s smart contract must have passed a static audit (Slither/Aderyn) with no critical findings.

The jurisdiction matrix must allow DeFi for the borrower’s and financier’s countries.

The secondary market pricing model (if tokenized) must be validated before trading is enabled.

GOVERNANCE GATES (PHASE 4 – COMPLETE TABLE)

Gate ID	Check	Enforcer	Failure Action
G4U1	Request only on locked trade/shipment	Governor	Block
G4U2	Default probability <15% or reduced facility	Credit Oracle (A2)	Recommend adjustment (advisory)
G4U2a	Financier preferences validated before RFQ send	Governor	Filter out; no notification
G4U3	RFQ broadcast logged; ≥2 eligible financiers notified	Governor	Retry; escalate if none
G4U3a	Financiers sign confidentiality agreement to view full details	Governor (A3)	Block access to documents
G4U4	Bidding window: ≥2 qualified bids or extension	Liquidity Auctioneer	Extend window

G4U4a	Sum of accepted co-financing bids ≤ requested amount (P)	Governor (A3)	Block; require re-selection
G4U5	Blended rate within market bands (±50% of benchmark)	Risk Dynamics (A2)	Flag for compliance review
G4U5a	DeFi protocol risk score ≥60	DeFi Navigator (A2/A3)	Exclude protocol; fallback to conventional
G4U6	Financing agreement contains SGTX Witness Clause	Governor (A3)	Block signature
G4U7	PSP split instruction signed by financier; net to borrower = P – 0.25%×P	Governor (A3)	Block disbursement; alert
G4U8	Repayment monitored; no SGTX fee collected	Governor (A3)	Log and verify; if fee detected, refund and investigate
G4U9	Financier historical data immutable & audit-traced	Governor (A3)	No alteration; any change triggers alert
G4U10	DeFi protocol smart contract audited (no critical issues)	Compliance Validator (A3)	Block deployment
G4U10a	Jurisdiction allows DeFi for both parties	Governor (WasmEdge)	Block; force conventional financing
G4U11	DeFi Risk Summary acknowledged before any DeFi bid	Governor (A3)	Bid submission blocked
G4U12	Multi-shipment financing: fee per-shipment, deducted per disbursement	Governor (A3)	Block; require separate request for each shipment

WORKED EXAMPLE – SELLER FINANCES SECOND SHIPMENT (MULTI-SHIPMENT)

Context : Mekong Fresh (seller) has a multi-shipment contract with Nile Foods. First shipment already executed. Second shipment (due in 2 months) is locked but not yet activated. Mekong Fresh wants pre-shipment finance for the second shipment.

Request : Seller (trader mode SELL) selects the second shipment (contract_shipment ID). Requests $20,000, 60 days, bank transfer.

Credit Intelligence : Credit score 88, default prob 6.2%. AI recommends LTV 70% ($21,000).

Financier preferences : Barclays (UK) has preferences matching (buyer is in Egypt? Actually seller is Vietnam, but Barclays' filter might include Vietnam – simplified). Barclays receives RFQ.

Full disclosure : Barclays sees seller Mekong Fresh, buyer Nile Foods, all documents (invoice, packing list, B/L), historical trade performance (12 previous trades, 0 disputes), and previous financing (none).

Bidding : Barclays bids 20,000at5.220,000at5.220,000 at 5.8% APR.

Selection: Mekong Fresh accepts Barclays.

Financing agreement & witness : Agreement signed by borrower, Barclays, and SGTX (witness).

Fee collection (disbursement) : P = 20,000,fee=20,000,fee=50. Barclays initiates payment via Stripe Connect. PSP splits: 19,950toMekongFresh,19,950toMekongFresh,50 to SGTX. SGTX FEESLock ACTIVE.

Repayment : Mekong Fresh later repays $20,000 + interest to Barclays via wire transfer. SGTX monitors webhook, updates records. Financing SGTX FEESLock released. Barclays updates Mekong Fresh's history (on-time repayment).

Result : SGTX earned $50 financing fee (plus trade fee already collected at contract lock). No fee collected at repayment.

END OF PART 3, PHASE 4 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 5: Physical Execution & Multiparty Tracking (v6.3 – Fully Extended, with Multi-Shipment Support, Container Release Control, Logistics Provider Signatories, Three-Tier AI, No Operating Modes, QC Override Accountability, Predictive Cold Chain, Batch Barcode Scanning, Stuck Trade Recovery)

PHASE 5 GOAL
After contract lock (single-shipment) or after a specific shipment within a multi-shipment contract has been mutually confirmed and its SGTX fee paid (per-shipment), the shipment moves into physical execution. The platform tracks every milestone – from loading at the seller's warehouse to delivery at the buyer's destination – with AI-powered automation, multisensor consensus, voice-enabled updates, predictive analytics, and autonomous milestone verification. No container is released without SGTX's confirmation (already enforced in Phase 3). All logistics providers are bound by signed addenda. The system integrates barcode scanning, IoT sensors (optional), and real-time alerts via Smart Inbox. For multi-shipment contracts, each shipment executes independently, with its own milestone timeline and SGTX FEESLock.

PREREQUISITE
For single-shipment: Contract status = LOCKED, SGTX FEESLock = ACTIVE, USTN generated.
For multi-shipment: At least one shipment in contract_shipments has status LOCKED (after mutual confirmation and fee payment). The master contract remains ACTIVE_MASTER.

Step 5.1 – Loading Date/Time Window Selection

Portal: Seller & Logistics (Trucking) Portals.

After the contract (or shipment) is locked, the seller and the selected trucking company coordinate a loading window. The system shows the trucking company's availability (from its calendar API or manual input) and suggests optimal slots based on:

Port cutoff times (scraped from terminal websites).

Historical traffic data (from route_correction_factors).

Weather forecasts (OpenWeatherMap free tier).

A simple scheduler UI allows the seller to pick a 4-hour window (e.g., "14 June 2026, 08:00–12:00"). Once confirmed, the loading window is stored in shipments (for single-shipment) or contract_shipments (for multi-shipment).

For multi-shipment, each shipment gets its own loading window.

Smart Inbox : The trucking company dispatcher receives a "Shipment Alert" with the pickup window.

Governance : None beyond permission checks; this step is advisory only.

Step 5.2 – Booking Confirmation Upload & AI Extraction

Portal : Shipping Line Portal → "Booking Confirmation" upload.

Once the shipping line issues a booking confirmation (PDF, email, EDI), the logistics provider or seller uploads it. HF Donut (A2, local) extracts:

Vessel name, IMO number, voyage number.

ETD (estimated departure) and ETA.

Container numbers assigned.

Terminal and port details.

Extracted data is validated against the shipment record (container count, type, temperature set points). If a mismatch (e.g., booking only covers 2 containers instead of 3), the AI (Groq) alerts: "Booking confirmation contains 2 reefers, but the contract requires 3. Please contact the carrier."

Upon successful validation, the system updates shipments (or contract_shipments) with vessel info and ETD.

For multi-shipment, each shipment's booking is separate.

Smart Inbox : All parties receive a "Shipment Alert" when booking is confirmed.

Step 5.3 – Dynamic Document Requirements with AI Validation

Service: document-requirement-service (Rust + Rig + HF).

For every route, commodity, and incoterm combination, the system dynamically generates a required document checklist. For a citrus shipment from Vietnam to Egypt, the list includes:

Commercial Invoice, Packing List, Phytosanitary Certificate, Certificate of Origin, Fumigation Certificate (if required), Export Declaration, Bill of Lading (eBL where available).

AI validation (A2 constraining) : When a responsible party uploads a document, HF Donut extracts key data and compares with the locked contract and packing plan. Any discrepancy >1% flags the document; it cannot be marked “accepted” until reviewed.

Example : Phytosanitary certificate states 20,000 kg but packing list says 17,600 kg → flagged. Groq explains: “The phytosanitary certificate weight does not match the packing list. Please request an amended certificate.”

Governance : If a required document is missing and critical, the loading milestone is blocked. The Decision Panel explains.

Step 5.4 – Granular Provider Status Tracking (Multi-Portal, AI-Enhanced)

This is the heart of Phase 5. Each logistics provider updates the shipment status through their portal or mobile app, often at the pallet level. The system uses multiple sensors and AI to verify and augment these updates.

5.4.1 – Trucking & Pallet Loading (Mobile App)

The SGTX mobile app (React Native, offline-first) features:

Barcode Scanning (ZXingC++) : Scans SSCC18 barcodes on each pallet.

Voice-Activated Updates (A1) : Driver speaks: “Pallet OR005 loaded into container TCNU1234567.” Vosk transcribes offline; HF Mixtral extracts pallet ID and container number. Low confidence triggers confirmation prompt.

AR Overlay (A2 advisory) : Supervisor points phone at container interior; AR.js overlays the loading plan, showing where each pallet should be placed. System detects misplaced pallets via barcode recognition.

Multisensor consensus for “Loaded” milestone : When all pallets for a container are scanned, AI (A2) checks:

Barcode scans per pallet (≥95% coverage).

Optional IoT weight sensor (±5% of packing list weight).

Geofence verification (truck at warehouse).
If consensus ≥95%, milestone is auto-confirmed (A4). Otherwise, human review required.

The confirmed milestone broadcasts via NATS; partial SGTX FEESLock release (e.g., 10%) is triggered.

Smart Inbox : Seller, buyer, and logistics dispatcher receive alerts.

5.4.2 – Port Gate & Customs

Port Gate In : OCR or manual entry updates status to “Gated In”. System cross-references container number with booking data.

Customs Clearance : Broker’s portal shows document checklist. AI validates documents; once all verified, broker submits export declaration. Digital twin simulates clearance time based on historical port congestion.

eBL Interoperability (A2): If carrier uses eBL platform (e.g., TradeLens), HF Donut monitors and extracts bill of lading issuance event – automatically marks “Bill of Lading Issued”.

5.4.3 – Shipping Line & Vessel Tracking

Once vessel departs, AIS receiver (free AISHub) updates vessel position every few hours. Digital twin simulation (A2) uses vessel speed, weather, port congestion to predict ETA deviations. If delay likely, Groq notifies: “MSC Alessia is 8 hours behind schedule due to a storm. Revised ETA Alexandria: 20 June.”

Smart Inbox : Buyer receives “SHIPMENT_ALERT” with updated ETA.

5.4.4 – Predictive Cold Chain Integrity & Quality Forecasting

Data sources : Temperature/humidity sensors (IoT – LoRaWAN, optional; or manual readings).

AI processing : A PyTorch LSTM model (local) ingests realtime temperature series and predicts remaining shelf life at destination. If temperature excursion occurs (e.g., 8°C for >30 min when set point is 5°C), model recalculates. Groq alerts: “Container TCNU5678901 (Lemons) experienced 8°C for 20 min. Shelf life reduced from 30 to 27 days. Recommended action: accelerate customs clearance.”

Smart Inbox : High-priority alert (score 65) to all parties; opens Trade Command Center with temperature graph.

5.4.5 – Automated Dispute Evidence Package

In case of anomaly, an AI agent (A2) automatically compiles evidence:

All milestone timestamps with signatures.

Temperature/humidity logs.

Photos (analysed by HF ViT).

AIS positions.

Weather reports.

Groq summary : “Container TCNU8901234 (Lemons) deviated from temperature set point for 2 hours. Shelf life reduced by 3 days. Root cause: reefer unit maintenance flag.”

The package is stored as dispute_recommendations and can be submitted to insurers or arbitrators.

5.4.6 – AR-Enhanced Container Inspection & QC Override Accountability

For QC inspectors (mobile app):

AR mode recognises SSCC barcode and overlays product, lot, inspection instructions.

HF Vision Transformer (ViT) runs locally, detecting carton damage, mould, etc., highlighting on screen.

Override accountability (v6.3) : If inspector overrides an AI finding, mandatory text reason (≥10 characters) is required. Stored in inspection_logs.inspector_notes with ai_overridden = true and original AI confidence.

The final QC report displays a badge : "⚠ 2 AI findings overridden – see notes."

Smart Inbox : Seller receives alert when QC report is submitted (CONDITIONAL PASS with overrides).

5.4.7 – Autonomous Demurrage/Detention Prediction

XGBoost model (local) predicts probability of demurrage/detention for each container at destination, considering:

Vessel ETA and historical port stay times.

Customs clearance duration (broker performance, document completeness).

Trucking availability.

Free time limits.

If probability >60%, system sends Smart Inbox alert and suggests early booking of priority trucking. Groq explains: "Container TCNU1234567 has 72% risk of 300demurrage.Bookingaguaranteedslotnowcosts300demurrage.Bookingaguaranteedslotnowcosts150 and reduces risk to 20%."

5.4.8 – SGTX FEES Release Automation (Per-Shipment, Already Paid)

The SGTX FEESLock (or per-shipment SGTX FEESLock) was already set to ACTIVE after fee payment. Release conditions are evaluated after each milestone. However, because the SGTX fee is already collected upfront (single-shipment) or per-shipment (multi-shipment), no further SGTX FEES release occurs. The SGTX FEESLock is simply marked FULLY_RELEASED after final delivery confirmation. This is a change: no incremental release; the fee was paid before the shipment started.

Update to original blueprint : SGTX FEESLock release is now purely administrative – it tracks that the shipment has been executed. No partial releases.

Governance : The PlainLanguage Decision Panel appears if a release condition is blocked (e.g., delivery not confirmed). The Decision Panel explains: "SGTX FEES was already paid. The shipment cannot be marked delivered because milestone X is missing."

Step 5.5 – Provider Invoice Upload & AI Comparison

Each logistics provider uploads their invoice (PDF). HF Donut extracts amount, currency, service description and compares with quoted amount (provider_quotes). Discrepancy >1% flagged.

Example : Trucking invoice 320vs.quoted320vs.quoted300 (6.7% deviation). Smart Inbox alert: "Invoice deviation – review before payment approval." Resolution can be manual (seller approves discrepancy).

Step 5.6 – Barcodes in Action (Auto-Generated in Phase 2)

The barcodes (GS1-128 SSCC18 + QR) generated during packing plan lock are now physically applied to pallets. Scanning with SGTX mobile app is the primary method for:

Confirming individual pallet loading (and thus container loading).

QC inspections (linking results to pallet IDs).

Delivery confirmation (scanning at buyer's warehouse).

Each scan creates a shipment_milestones record (with scanned_barcode_id), proving provenance.

Predictive scanning reliability agent (A2, XGBoost) : Monitors scan success rates per batch. If a batch has predicted failure risk >70%, a Smart Inbox alert is sent: "Pallet labels on OR012 to OR022 may fail in outdoor conditions. Reprint recommended."

Step 5.7 – Stuck Trade Recovery (Cross-Cutting)

If any milestone remains unconfirmed beyond its SLA (e.g., "Loading" not confirmed within 24 hours of scheduled window), the workflow-recovery-service triggers escalating reminders:

12h overdue: Smart Inbox reminder to responsible party.

24h overdue: Alert to seller, buyer, and Admin Portal.

48h overdue: Escalate to human mediator (A3).

No autocancellation for physical execution – manual review required. The Shipments Vault shows an alert icon.

For multi-shipment, each shipment's milestones are tracked independently; a delay in one shipment does not block others.

GOVERNANCE GATES (PHASE 5 – UPDATED)

Gate ID	Check	Enforcer	Failure Action
G5U1	Multisource consensus ≥ threshold for automilestone	Milestone Verifier (A2/A3)	Degrade to human verification
G5U2	Byzantine fault tolerance met (<1/3 sources disagree)	Milestone Verifier (A3)	Escalate to investigation
G5U3	Computer vision confidence ≥0.95 for document verification	Vision Verifier (A2)	Require human review
G5U4	Disruption prediction confidence ≥0.70 for autoaction	Disruption Predictor (A2)	Alert only, no autoaction
G5U5	Recovery action cost ≤ tenant autoexecute threshold	Route Optimizer (A2/A3)	Require human approval
G5U6	Smart container	Smart Container	Override with

	decision within safety parameters	Agent (A2)	conservative default
G5U7	SGTX FEES release condition verified by ≥3 independent sources	SGTX FEES Releaser (A2/A3)	Hold release, escalate
G5U8	All sensor data sources logged via Loom for audit	Governor (A3)	Block milestone confirmation
G5U9	Pallet scan events originate from authenticated device	Governor (A3)	Reject scan, log anomaly
G5UA1	Document validation (AI Donut) – mismatch flagged	Doc Validator (A2)	Block document acceptance
G5UA2	Cold chain anomaly detected; shelf life recalculated	Quality Predictor (A2)	Notify responsible parties
G5UA3	Autocompiled dispute evidence package complete	Evidence Agent (A2)	(None – advisory)
G5UA4	AR inspection findings logged	Vision Agent (A2)	(None – advisory)
G5UA5	Demurrage prediction above threshold → notification	Risk Agent (A2)	Advisory only
G5UA6	All blocked actions display PlainLanguage Governor Decision Panel	Governor (A3) + Groq/Ollama	User sees jargon-free explanation; cannot proceed until condition resolved
G5UA7	If QC inspector overrides AI finding, mandatory text reason ≥10 chars required	Application (A2)	Override rejected; reason must be provided
G5UA8	If a milestone is overdue beyond SLA, Stuck Trade Recovery rules activate	Recovery Service (A4)	Autoescalation
G5UA9	For multi-shipment, each shipment's milestones are independent	Governor (A3)	Delay in one shipment does not block others

WORKED EXAMPLE – END-TO-END EXECUTION (MULTI-SHIPMENT, SECOND SHIPMENT)

Context : Mekong Fresh and Nile Foods have a multi-shipment contract. First shipment already delivered. Second shipment (oranges) was locked, SGTX fee paid ($360), and USTN generated.

Loading window selected: 14 June 08:00–12:00.

Booking confirmation uploaded : HF Donut extracts vessel MSC Alessia, ETD 16 June.

Document checklist : Seller uploads phytosanitary certificate – passes AI validation.

Trucking & pallet loading : Driver scans 22 pallets via mobile app. All barcodes verified. Loading milestone auto-confirmed (consensus 100%). Smart Inbox: “Container loaded.”

Port gate & customs : Customs broker submits export declaration via API – cleared.

Vessel departure : AIS updates position. Digital twin predicts ETA on time.

Cold chain: No temperature excursion.

Delivery : Container arrives at Alexandria. Buyer confirms delivery via TCC.

Final milestone : Shipment status = DELIVERED. SGTX FEESLock marked FULLY_RELEASED.

Smart Inbox: Both parties receive “Shipment completed.”

If a temperature excursion had occurred, the AI would have compiled evidence and notified both parties; the buyer could file a dispute with the autocompiled package. The QC inspector’s override (if any) would be flagged.

END OF PHASE 5 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 6: Settlement & Payment Orchestration (v6.3 – Fully Extended, with Multi-Shipment Support, No SGTX Fees Release (Already Paid), Financing Separate, Three-Tier AI, Non-Custodial Splitting, Smart Inbox Integration, Voice Activated Approval, Reconciliation with AI)

PHASE 6 GOAL
After the shipment reaches its destination and the buyer confirms acceptance (or the contractual payment milestone is reached), the platform instructs and verifies the settlement of the trade principal (buyer → seller). The SGTX trade fee has already been collected upfront from the seller at contract lock (Phase 3). No SGTX FEES release occurs here. Separate financing repayments (if any) are handled independently (Phase 4). The platform issues cryptographically signed settlement instructions, monitors external payment rails, reconciles automatically using AI-extracted bank statements or PSP webhooks, and provides full transparency through the Smart Inbox and Tenant Data Export.

For multi-shipment contracts, each shipment settles independently, using its own USTN. The platform never holds funds – only issues splitting instructions for the financing fee (in Phase 4) but for trade settlement, the full principal flows directly from buyer to seller.

PREREQUISITE
Shipment status = DELIVERED (or contractual payment milestone reached). The SGTX

trade fee has already been paid by seller; no further fee collection. For multi-shipment, each shipment settles when its delivery milestone is confirmed.

Step 6.1 – Settlement Instruction Generation (USTN-Linked)

When the delivery milestone is confirmed, the settlement service automatically generates a settlement instruction for the trade principal (the total quoted price the buyer agreed to in Phase 3). This instruction is unique to the USTN (or per shipment USTN for multi-shipment).

Instruction content:

USTN (or shipment USTN)

Buyer GTID, Seller GTID

Amount (total quoted price, including the SGTX trade fee that was already paid – the buyer pays the full price, seller already paid the fee from their side)

Currency (as per contract)

Payment rail (selected by PSP Router – see Step 6.2)

Remittance information : SGTX USTN {USTN} / INV {contract reference}

Governance (G6U1) : The instruction is signed by the Governor (Ed25519) and stored in settlement_instructions with status PENDING. The buyer receives a Smart Inbox item (priority 85) to approve.

Note on multi-shipment : Each shipment's settlement is independent; the buyer approves per shipment after its delivery.

Step 6.2 – AI PSP Router (Payment Rail Selection)

The PSP Router (A2, LightGBM + Groq) selects the optimal payment rail for the buyer → seller transfer based on:

Payer country (buyer), payee country (seller), amount, currency.

Realtime PSP health scores (from psp_health_logs).

Cost (fees + FX spread).

Predicted settlement time.

The router returns the top PSP (e.g., SWIFT, Stripe, Payoneer, local instant payment). Groq generates an explanation for the buyer: "Recommended: SWIFT via Bank of Alexandria. Estimated fees $25, settlement in 2-3 days." The buyer can override (choose another PSP) if permitted by their role.

Governance (G6U2) : The selected PSP must have uptime ≥99% and not be sanctioned in the jurisdictions involved.

Step 6.3 – Buyer Approval (Voice-Activated, Smart Inbox)

The buyer (or their authorised finance user) receives a Smart Inbox item : *"Approve settlement of $30,240 to Mekong Fresh for USTN SGTX-...-V1."* Two approval methods:

Manual: Click "Approve" in the TCC or Smart Inbox.

Voice (A1) : In the mobile app or web, speak: "Approve settlement for USTN SGTX-...-V1." Vosk transcribes offline; HF Mixtral extracts intent; ZITADEL verifies biometrics; Governor signs the instruction.

Once approved, the settlement instruction status becomes AUTHORISED. The payment orchestrator sends the instruction to the selected PSP.

Governance (G6U3) : If the buyer's role requires dual approval (internal approval chain), the instruction is held until all approvers sign.

Step 6.4 – Payment Execution & Monitoring

The PSP processes the transfer. SGTX monitors via:

PSP webhook (cryptographically signed) – preferred.

OpenBanking API (PSD2) – where available, the platform pulls bank statements using HF Donut to extract transaction data.

Manual upload – if no API, the seller can upload a SWIFT confirmation PDF.

Webhook verification : The payment orchestrator verifies the PSP's signature, matches the amount, USTN, and payer/payee. If verified, the settlement status becomes CONFIRMED.

Governance (G6U4) : The PSP webhook must contain the USTN in the reference field; otherwise, AI (HF Donut) attempts to extract it from unstructured text. If USTN not found, the payment is flagged for manual reconciliation.

Step 6.5 – AI-Powered Reconciliation (Self-Reconciling Ledger)

The Reconciliation Engine (A2, HF Donut) continuously matches incoming payment confirmations against open settlement instructions. For each PSP webhook or bank statement line:

Extract amount, currency, value date, reference (USTN pattern).

Compare with settlement_instructions that are AUTHORISED.

If match confidence ≥95% (amount within tolerance, USTN matches, payer/payee consistent), the settlement is autoreconciled. Status → RECONCILED.

If confidence <95% or no match found, a Smart Inbox item is created for the finance team: "Unmatched payment of $X – please review and assign to USTN."

Multi-shipment : Each shipment's USTN is reconciled separately.

Governance (G6U5) : All reconciliation events are logged in audit_log and settlement_reconciliation. The ClickHouse retention is 7 years.

Step 6.6 – Query & Dispute (Voice, Smart Inbox)

Any authorised party can query settlement status via voice : *"What is the status of payment for USTN SGTX-...-V1?"* Groq replies with the current status, ETA, and any issues.

If the seller has not received payment after the expected settlement time, they can raise a settlement dispute via the Disputes tab. This triggers an AI-assisted investigation (A2) – the system compiles evidence (PSP logs, webhook receipts, bank statements) and presents to both parties. Escalates to Platform Governance Authority if unresolved.

Governance (G6U6) : Settlement disputes are handled in Phase 10; no automatic refund is issued.

Step 6.7 – Tenant Data Export & Reconciliation Statements

For every settled trade, the tenant-data-export-service automatically updates the tenant's financial ledger. The buyer and seller can download monthly reconciliation statements (PDF, CSV, JSON) from the Finance section. Each statement includes:

All settled USTNs, amounts, dates, counterparty names.

Base currency conversion (using ECB daily rates).

Ed25519 signature and SHA256 checksum for cryptographic verification.

Governance (G6U7) : Exports are rate-limited and require tenant.export permission.

Step 6.8 – Multi-Shipment Settlement (Independent per Shipment)

If the contract is multi-shipment, each shipment settles independently. The buyer approves and pays for each shipment separately as it is delivered. The master contract remains active until all shipments are settled. The seller's financial records show each shipment's payment separately.

The process for each shipment is identical to single-shipment (Steps 6.1–6.7).

GOVERNANCE GATES (PHASE 6 – UPDATED)

Gate ID	Check	Enforcer	Failure Action
G6U1	Settlement instruction signed and linked to USTN	Governor (A3)	Block instruction creation
G6U2	Selected PSP has country coverage + health score ≥80	PSP Router (A2)	Fallback to secondary PSP
G6U3	Buyer has permission to approve (OPA + internal approval chain)	Governor (OPA)	Hold for additional approvers

G6U4	PSP webhook signature verified, USTN matched	Webhook Verifier (A2)	Reject webhook, log anomaly, flag for manual reconciliation
G6U5	Reconciliation confidence ≥95% for auto-match	SelfReconciler (A2)	Queue for manual review; send Smart Inbox alert
G6U6	Settlement dispute evidence package compiled automatically	Dispute Agent (A2)	Escalate to human mediator if confidence low
G6U7	Monthly reconciliation statement cryptographically signed	TenantDataExportService	Block download if signature not verifiable
G6U8 (v6.3)	No SGTX FEES release – fee already collected; settlement only for principal	Governor (A3)	Block any attempted SGTX FEES deduction
G6U9	For multi-shipment, each shipment settled independently; settlement of one shipment does not affect others	Governor (A3)	Enforced by USTN-level scoping

WORKED EXAMPLE – SINGLE SHIPMENT SETTLEMENT

Context : Nile Foods (buyer, Egypt) has received delivery of oranges from Mekong Fresh (seller, Vietnam). The total quoted price was 30,240,whichincludeda1.230,240,whichincludeda1.2360) already paid by Mekong Fresh at contract lock.

Shipment delivered : Milestone DELIVERED confirmed. Settlement service generates instruction: PAYER: Nile Foods, PAYEE: Mekong Fresh, AMOUNT: $30,240, USTN: SGTX-EG-...-V1.

PSP Router : Recommends SWIFT via Bank of Alexandria (fee $25, 2 days). Buyer approves via voice.

Payment : Bank of Alexandria transfers $30,240 to Mekong Fresh's bank. Webhook received, verified.

Reconciliation : HF Donut extracts USTN from SWIFT MT103 field 70, matches instruction. Status → RECONCILED.

Seller: Receives Smart Inbox: "Payment of $30,240 received."

Buyer : Receives monthly statement showing the settlement. No further fees.

END OF PHASE 6 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 7: Distressed Cargo Resolution (v6.3 – Fully Extended, with Multi-Shipment Support, Triage Dashboard, AI Condition Assessment, Dynamic Pricing, Microcontract & Separate SGTX Fee, Check Buyers Advisory, Privacy Notice, Smart Inbox Integration, Plainlanguage Decision Panel, No Container Release Without Fee

PHASE 7 GOAL
When cargo becomes distressed (physically damaged, abandoned, delayed beyond shelf life, or subject to demurrage), the platform provides a structured resolution workflow. The distressed cargo owner (typically the seller, but could be the buyer or logistics provider) can declare distress, receive AI-assessed condition and pricing, and resolve the lot via sale, compliant disposal, or insurance claim. A microcontract is created for the distressed portion, with a separate SGTX fee(adjusted by country-specific distressed factor). The original trade SGTX fee remains collected (already paid). The Check Buyers advisory feature allows the seller to see potential buyers from their saved contacts before initiating outreach. All actions are governed by the Governor, with full Smart Inbox and Decision Panel transparency.

PREREQUISITE
A shipment (or a specific shipment within a multi-shipment contract) has status IN_EXECUTION or DELIVERED but with damaged goods. The original contract is locked and the SGTX trade fee has already been paid by the seller. The cargo owner has the necessary permission (distressed.declare). For multi-shipment contracts, distress can be declared per shipment without affecting other shipments.

Step 7.1 – Proactive Demurrage & Delay Alert (Pre-Distress)

The system monitors port, customs, and container free-time data via digital twin and disruption predictions. When demurrage risk is detected (e.g., free time expires in 48 hours), a Smart Inbox alert (priority 80) is sent to the buyer, seller, and logistics providers. If no action is taken and the risk materialises, the digital twin flags the shipment as potential distressed, pre-computing possible outcomes.

Example alert : *"Demurrage charges will start in 48h. Estimated charge $50/day. Consider releasing cargo or declaring it distressed."*

Governance: Advisory only – no automatic declaration.

Step 7.2 – Declaration of Distressed Cargo

Portal : Seller Portal (or Buyer/Logistics with permission) → "Distressed Cargo" tab. The user selects a USTN (or a specific shipment USTN for multi-shipment) and clicks "Declare Distressed".

Declaration form:

Select affected pallets/containers (from the packing plan).

Describe the condition (free text, or upload photos).

Choose distress reason : Damage, Quality deterioration, Shelf life expired, Abandonment, Demurrage, Other.

Voice command (A1) : In mobile app, speak: “Pallet LEM003 has mould, mark as distressed.” Vosk transcribes; HF Mixtral extracts purpose.

Governance (G7U1) : Only parties directly involved in the trade can declare. The declaration is logged in distressed_cargo_listings with status PENDING_ASSESSMENT.

Step 7.3 – AI Condition Assessment (A2, HF ViT)

When photos are uploaded as evidence, the HF Vision Transformer (ViT) (local) analyses them:

Detects visual damage (mould, crushing, colour change, moisture stains, insect presence).

Produces a condition score (0–100) per pallet (100 = perfect, 0 = destroyed).

Stores the score in distressed_cargo_listings.condition.

If no photos, the user can manually enter a condition score (subject to review). The AI also estimates remaining shelf life (if perishable) using cold-chain data from Phase 5.

Example output : Condition score 35, tags: “mould detected (92%), crushed cartons (78%)”. The score feeds into dynamic pricing.

Governance (G7U2) : The condition assessment is advisory; the seller can override with a written reason (logged).

Step 7.4 – Dynamic AI Pricing Engine (A2, XGBoost)

The Distressed Pricing Agent (XGBoost, local) estimates a fair market recovery price based on:

Original contract value of the affected goods.

AI condition score.

Remaining shelf life (from cold-chain prediction).

Current commodity price index (FAO, scraped).

Destination demand (scraped local market trends).

Buyer urgency (if Accelerated Outreach is used).

Output : Suggested price range (min-max) and a recommended listing price. Groq generates a business explanation: *“Based on condition score 35, 5 days remaining shelf life, and current demand for citrus at Alexandria, we recommend listing these 1,344 kg of lemons at 0.75–0.75–0.90/kg. Original contract price was $2.10/kg. This price range has a 78% predicted sale rate within 48 hours.”*

The seller can override the price, but the AI's rationale is logged. If the seller attempts to set a price >200% of the suggested range, the Governor issues a CONDITIONAL verdict with a Decision Panel explanation.

Governance (G7U3) : Price override requires justification; unrealistic price blocks listing until resolved.

Step 7.5 – Distressed Cargo Triage Dashboard (Three Paths)

Instead of a raw listing form, the Distressed Cargo tab presents a Triage Dashboard with three clearly labelled paths:

Path A – "Sell Quickly" (max speed recovery).

Sets price expectation to the AI-recommended quick-sale price.

Prepares Accelerated Outreach Mode (Phase 8) but does not send notifications until the seller explicitly opts in and reviews the privacy notice.

Displays a progress dashboard : "Eligible contacts: 5. Outreach ready. [Start Outreach]".

Path B – "Comply with Local Law"

Runs the Jurisdiction Compliance Assistant.

Scans the jurisdictions matrix for distressed sale rules (e.g., mandatory liquidator, CBE notification).

Generates required reports (e.g., CBE notification for Egypt) with one click.

Once compliance items are marked complete, seller can return to Path A or Path C.

Path C – "File Insurance Claim"

Runs the Evidence Package Compiler (Phase 10) with sensor logs, photos, AI condition assessment, and market valuation.

Outputs a downloadable PDF and a Groq-generated claim narrative.

The seller can then still use Path A or B if desired.

Governance : The triage dashboard orchestrates the same underlying agents; no new microservice required.

Step 7.6 – "check Buyers" Advisory (Pre-Outreach, v6.3)

Before initiating any outreach, the seller can click "Check Buyers" (visible in Path A and on the listing page). A modal displays a table of eligible saved contacts (from tenant_contacts where is_blocked = false and relationship includes BUYER or GENERAL). The contacts are filtered and ranked:

Eligibility : Contact must have previously traded in the same commodity category or expressed interest (via buyer_search_requests).

Ranking : Composite score using LightGBM (local) based on past distressed purchase success, trust score, location proximity, response time.

Example display:

Buyer Name	Trust Score	Past Distressed Purchases	Actions
Juice Factory Ltd.	91	3 (100% completion)	Select
Cairo Processors	83	1 (50% completion)	Select

The seller can select potential buyers and then proceed to Accelerated Outreach (Phase 8). No notifications are sent by clicking "Check Buyers" – it is purely advisory. The selected list is carried forward to the outreach step.

Governance (G7U4) : This feature never sends unsolicited messages; it only displays existing contacts. All selections are logged for audit.

Step 7.7 – Partial Distress & USTN Splitting (Microustn)

If only part of a shipment is distressed (e.g., 3 of 22 pallets), the system can split the shipment:

The original USTN is retained for the clean portion.

A new microUSTN is generated for the distressed pallets (format : same as USTN but with a new timestamp and random suffix, and a parent_ustn field linking to the original).

The packing list, commercial invoice, and all documents are automatically updated.

The original trade_request and contracts references are linked to both parts via a parent-child relationship in the database.

Governance (G7U5) : The split requires a Governor decision (A2) to ensure the clean goods remain saleable under the original contract. If buyer consent is required, a Smart Inbox item is sent automatically.

Step 7.8 – Microcontract & Separate SGTX Fee

Once a distressed sale is agreed (via negotiation or direct offer), a microcontract is created under a new contracts row, referencing:

microUSTN

Buyer and seller (distressed cargo buyer)

Agreed price

Special distressed terms

SGTX fee for distressed sale : The original SGTX trade fee was already paid by the seller for the full shipment. For the distressed portion, a new, separate SGTX fee is charged on the distressed sale amount, using the standard dynamic rate (0.1%–2.5%) adjusted by the distressed country factor (from jurisdictions table). The factor accounts for mandatory local costs (e.g., liquidator fees, inspection bonds). Example: If the country factor is 0.85, the effective fee rate becomes standard_rate × 0.85. The fee is paid by the seller (the distressed cargo owner) upfront before the microcontract is locked. The same PSP split mechanism applies.

SGTX FEESLock for distressed sale : A new SGTX FEESLock is created for the distressed microcontract. The original SGTX FEESLock for the main trade is partially released (or held) based on the salvage amount (but since the main fee was already collected, this is just administrative).

Governance (G7U6) : The microcontract must be signed by both parties (buyer and seller) via ZITADEL passkey. The SGTX Governor signs as witness.

Step 7.9 – Accelerated Outreach Mode (v6.3)

If the seller chooses Path A ("Sell Quickly") and selects buyers (via "Check Buyers" or manually), they can initiate Accelerated Outreach (time-boxed, simultaneous notifications). Before any notifications are sent, a mandatory Distressed Cargo Privacy Notice modal appears:

"You are about to notify your selected contacts about this distressed lot. The notification will include: product, quantity, condition score, asking price, and your company name. Only contacts you selected will receive it. Do you want to proceed?"

The seller must explicitly opt in (click "Yes, start outreach"). This action is logged as a Governor decision (distressed_accelerated_outreach_optin). The outreach then proceeds as defined in Phase 8.

Governance (G7U7) : If the seller opts out, no notifications are sent. Accelerated Outreach is per-session opt-in.

Step 7.10 – AI-Mediated Express Negotiation (A2/A3)

Interested buyers can submit offers via a fast-track negotiation channel. Sellers and buyers can opt into AI bots (similar to Phase 3) with accelerated parameters:

Max 3 rounds.

Offer expiry configurable (e.g., 2 hours).

Price must stay within seller's floor and buyer's ceiling.

Bots generate counteroffers via Groq (fallback Ollama).

The Negotiation Bot Transparency Panel shows rounds, parameters, and a "Pause/Override" button. All messages are stored in distressed_cargo_offers and negotiation_sessions.

When an offer is accepted, the microcontract is created (Step 7.8).

Governance (G7U8) : If no agreement reached within expiry, the seller can relist or manually negotiate.

Step 7.11 – AI-Insurer Integration & Claim Support

For Path C ("File Insurance Claim"), the system compiles an evidence package (as in Phase 10) and Groq drafts a claim narrative. The package includes:

Original contract, invoice, packing list.

All milestone logs.

Sensor data (temperature excursions).

Photos with AI condition assessment.

Distressed sale agreement (if sold) or market valuation.

AI-generated claim narrative.

The final package is downloadable as a PDF and can be emailed if SMTP is configured. The seller can submit to their insurer.

Governance (G7U9) : The platform does not submit claims; it only provides the evidence.

Step 7.12 – Governance Gates (Phase 7)

Gate ID	Check	Enforcer	Failure Action
G7U1	Distressed declaration only by authorised party on a locked trade/shipment	Governor (A3)	Block declaration
G7U2	AI condition assessment logged; override requires reason	Vision Agent (A2)	Human verification required
G7U3	Price within AI-suggested range (±200%) or justified	Pricing Validator (A2)	Flag for human review; block if extreme
G7U4	"Check Buyers" only shows existing saved contacts; no notification sent	Governor (OPA)	Block access
G7U5	USTN splitting valid and documents consistent	Governor (A2)	Halt splitting
G7U6	Microcontract locked only after distressed fee paid (factory rate adjusted per jurisdiction)	Governor (A3)	Block lock; show Decision Panel
G7U7	Accelerated Outreach requires explicit privacy notice opt-in	Governor (A3)	Block notification sending
G7U8	Express negotiation rounds logged; bot transparency panel	Negotiation Orchestrator	(Advisory)

	available		
G7U9	Insurance claim evidence package complete (advisory)	Evidence Agent (A2)	None
G7U10 (v6.3)	Before any outreach, privacy notice acknowledged by seller	Governor (A3)	Block notification sending
G7U11	Triage dashboard paths A/B/C are advisory; any irreversible action passes through standard Governor gates	Governor (A3)	As per underlying action
G7U12	All blocked distressed actions display PlainLanguage Governor Decision Panel	Governor (A3) + Groq/Ollama	User sees jargon-free explanation; cannot proceed until condition resolved

WORKED EXAMPLE – DISTRESSED LEMONS WITH "CHECK BUYERS" AND SALE

Context : Mekong Fresh (seller) has a shipment of lemons to Egypt. During transit, a temperature excursion occurs on 3 pallets (LEM003, LEM004, LEM005). Shelf life reduced to 5 days. The buyer declines to accept the damaged pallets.

Declaration : Seller selects USTN, marks pallets LEM003–005 as distressed, uploads photos. AI condition score: 35, mould detected.

Triage Dashboard : Seller clicks "Check Buyers" – sees Juice Factory Ltd. (trust 91, 3 previous distressed purchases) and Cairo Processors (trust 83, 1 purchase). Selects both.

Path A – Sell Quickly: Accepts AI-recommended price $0.80/kg.

Privacy notice : Seller acknowledges. Accelerated Outreach begins (Phase 8).

Negotiation : Juice Factory offers 0.75/kg.Seller'sbotcountersat0.75/kg.Seller'sbotcountersat0.78. Juice Factory accepts.

Microcontract : New microUSTN generated. Distressed SGTX fee (country factor Egypt = 1.0, standard rate 1.03%) = 1.03% of 900=900=9.27. Seller pays $9.27 via PSP. SGTX FEESLock ACTIVE.

Sale completed : Microcontract locked. Juice Factory receives the lemons.

Clean shipment : Remaining good lemons proceed normally. Original SGTX FEESLock unaffected.

END OF PHASE 7 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 8: Distressed Cargo Outreach (v6.3 – Fully Extended, with “check Buyers” Advisory, Privacy Notice, Accelerated Outreach Mode, Co-Branded Notifications, Smart Inbox Integration, Direct Contact Notification, Non-Marketplace, Three-Tier AI)

PHASE 8 GOAL
This is the only mechanism by which a distressed cargo owner can notify potential buyers. It is a direct, permissioned notification system – not a marketplace. The owner may:

Notify selected saved contacts (from tenant_contacts) or specific GTIDs after using the “Check Buyers” advisory (Phase 7).

Use voice commands to select contacts.

Receive AI-generated, personalised notifications (Groq, fallback Ollama) drafted for each recipient.

Choose between Standard Outreach (manual, one-by-one) or Accelerated Outreach Mode(time-boxed, simultaneous, with floor price and real-time ranking).

Accelerated Outreach always requires an explicit opt-in per session and a privacy noticeexplaining exactly what information will be shared and with whom.

All actions are logged, all communication is encrypted, and no unsolicited messages are ever sent. The seller remains in full control.

PREREQUISITE
A distressed cargo listing has been created (Phase 7) with status ACTIVE. The seller has the permission distressed.outreach. For multi-shipment contracts, outreach is per-distressed-microcontract.

Step 8.1 – Access Distressed Listing & Initiate Outreach

Portal : Seller Portal → “Distressed Cargo” tab. After a distressed listing has been created (Phase 7), the owner sees a “Notify Contacts” button within the listing card. Clicking it opens the outreach interface.

Example : Seller Mekong Fresh has a distressed lot of 1,344 kg lemons (microUSTN SGTX-SB-...). AI condition score 35, recommended price $900. Clicking “Notify Contacts” begins targeted outreach.

Governance (G8U1) : Outreach only allowed for cargo flagged as DISTRESSED with an active listing.

Step 8.2 – “check Buyers” Advisory (Pre-Outreach)

Before composing messages, the seller may click “Check Buyers” (same as in Phase 7). This opens a modal showing eligible saved contacts (from tenant_contacts where is_blocked = false and relationship includes BUYER or GENERAL). The list is filtered and ranked:

Eligibility : Contact must have previously traded in the same commodity category or expressed interest (via buyer_search_requests).

Ranking : Composite score using LightGBM (local) based on past distressed purchase success, trust score, location proximity, response time.

Example table:

Buyer Name	Trust Score	Past Distressed Purchases	Actions
Juice Factory Ltd.	91	3 (100% completion)	Select
Cairo Processors	83	1 (50% completion)	Select
Alexandria Trading	76	0	Select

The seller can select one or more buyers. The selected list is carried forward to the outreach step. No notifications are sent from this button – it is purely advisory.

Governance (G8U2) : "Check Buyers" never sends messages; only displays existing contacts. All selections are logged in distressed_contact_checks (audit only).

Step 8.3 – Select Recipients (Saved Contacts or Enter GTID)

After (or without) using "Check Buyers", the seller proceeds to select recipients. The UI presents two options, both on a single screen with clear labelling.

8.3.1 – Notify Saved Contacts (from "Check Buyers" selection or manual selection)

The system displays a list of the owner's saved contacts that are eligible for this distressed lot. The seller can check/uncheck any contact. The list shows:

Legal name, trust score, relationship type (e.g., "Previous trading partner").

"Relevant" badge if the contact has traded this commodity before.

The seller can also use voice: "Select Juice Factory."

The system never preselects contacts – the owner must manually check each recipient. The list is sorted alphabetically (or by relevance if the seller chooses).

8.3.2 – Notify Specific Trader via GTID

The owner can manually enter one or more GTIDs in a text box. The system resolves each GTID in real time (legal name, jurisdiction). The owner must confirm that the GTID is a legitimate business contact (logged).

Governance (G8U3) : The contact must exist in the system and not be blocked. The action is logged but not restricted.

Step 8.4 – Choose Outreach Mode & Privacy Notice

After selecting recipients, the owner chooses between two modes:

A. Standard Outreach

For each selected recipient, a Groq-generated draft notification is displayed (editable).

The owner can edit each message individually or apply a template to all.

Clicking “Send” delivers the messages immediately.

B. Accelerated Outreach Mode (Time-Boxed)

The owner sets an outreach window (2–24 hours) and a floor price (hard minimum, never disclosed to recipients).

The system will simultaneously notify all selected recipients.

Responses are received in real time and ranked by a composite score (response speed + trust score + closeness to asking price).

At the end of the window, Groq recommends the best offer.

Before the notifications are sent, a mandatory privacy notice modal appears:

“You are about to initiate Accelerated Outreach for the distressed lot [description]. The following contacts will be notified simultaneously: [list of names]. The notification will include: the product, quantity, condition score, asking price, and your company name. Recipients will know your identity. Do you want to proceed? [Yes, start outreach] [No, go back]”

The owner must explicitly confirm. This is logged as a Governor decision (distressed_accelerated_outreach_optin).

Governance (G8U4) : Accelerated Outreach is opt-in per session. The system does not automatically activate it, even if used previously. The privacy notice is mandatory.

Step 8.5 – AI-Composed Notification (A1)

For each selected recipient (in either mode), Groq (fallback Ollama) generates a personalised, concise, and factual message in the recipient’s preferred language (if known, else English). The message includes:

Reference to the original USTN and the microUSTN.

Commodity, quantity, condition score (from AI assessment), remaining shelf life.

Asking price (from dynamic pricing engine).

A link to access the secure offer (via the SGTX portal).

Example to Juice Factory Ltd. (English) : “Dear Juice Factory Ltd., we have a distressed lot of 1,344 kg of Eureka lemons from our contract USTN...V1. Condition score 35, 5 days remaining shelf life. Asking price $900 for the lot. Please review this secure offer via your SGTX portal: [link]. – Mekong Fresh”

The AI does not add marketing language; it sticks to facts. The owner can edit the message.

Co-Branded Email Delivery : When SGTX sends the notification via email, the sender name is formatted as “Mekong Fresh via SGTX” <noreply@sgtx.io> to ensure immediate recognition and prevent spam. The subject line includes the truncated microUSTN and commodity.

Step 8.6 – Send & Track Notifications

Once confirmed, notifications are delivered to each recipient's SGTX inbox (NATS-based). Each recipient receives:

Smart Inbox item (priority 75, category NEW_OFFER) : "Distressed lot available from Mekong Fresh – 1,344 kg lemons, condition score 35, $900. [View Offer]"

Push notification (if mobile app installed).

Email alert (co-branded, with unsubscribe link for anonymous RFQs – but distressed outreach is directed, so unsubscribe does not apply).

The owner can see delivery status in their portal : Sent, Delivered (acknowledged by NATS), Read (if recipient opened the notification – tracked with consent). For Accelerated Outreach, the owner sees a live dashboard: number of responses received, ranked best offers, and time remaining.

Governance (G8U5) : All notifications are logged in distressed_contact_notifications with message hash, timestamp, and recipient.

Step 8.7 – Recipient Response & Integration with Phase 7

If a contacted party is interested, they can:

Click the link in the notification → opens the Trade Command Center for that microUSTN, where they can view the full listing, photos, and condition report.

Submit an offer directly through the portal – creates a distressed_cargo_offers record and flows into Phase 7's express negotiation process.

Request more details via the chat panel (Trade Room).

Decline – the owner sees status updated to "Declined."

If the contacted party declines or does not respond, the owner may manually renotify or wait. No automatic escalation or follow-up is sent – the owner is in full control.

Step 8.8 – Post-Acceptance & Microcontract

When an offer is accepted (via Phase 7), the distressed sale proceeds exactly as described in Phase 7 (microcontract creation, distressed SGTX fee, USTN splitting, regulatory compliance). The notification system's role ends at the point of offer acceptance.

Smart Inbox updates:

Seller : The "Outreach Active" item is removed; replaced by "Microcontract pending signature."

Non-accepted responders receive a brief, factual message : "The distressed lot [description] has been sold. Thank you for your interest. – Mekong Fresh (via SGTX)."

GOVERNANCE GATES (PHASE 8 – DISTRESSED OUTREACH)

Gate ID	Check	Enforcer	Failure Action
G8U1	Outreach only for cargo flagged as DISTRESSED with active listing	Governor (A3)	Block notification
G8U2	Recipients must be verified tenants (saved contact or valid GTID)	GTID Resolver	Block invalid GTIDs
G8U3	“Check Buyers” never sends notifications – only advisory	Governor (OPA)	Block any automated send
G8U4	All notification content logged via Loom (who, to whom, when, message hash)	Governor (A3)	Reject send
G8U5	No marketplacelike automatching or open broadcast; only direct notification	OPA policy	Revoke platform permissions if violated
G8U6 (v6.3)	Before Accelerated Outreach, owner must explicitly opt in via privacy notice; session timestamp and contact list are logged	Governor (A3)	Block accelerated send; standard outreach still available
G8U7 (v6.3)	Co-branded emails must clearly identify the sender; impersonation prevention via WasmEdge	Governor (WasmEdge)	Reject email; log anomaly
G8U8 (v6.3)	All outreach actions are surfaced in the sender’s Smart Inbox; if any notification fails, sender receives SHIPMENT_ALERT item	inboxservice (A1)	Informational; no block

WORKED EXAMPLE – FULL OUTREACH FLOW (V6.3)

Context : Mekong Fresh has a distressed lot of 1,344 kg lemons (microUSTN). They want to sell quickly.

Initiate : From Distressed Cargo tab, clicks “Notify Contacts” next to the lemon listing.

Check Buyers : Clicks “Check Buyers” – sees Juice Factory Ltd. (trust 91), Cairo Fruit Processors (trust 83), Alexandria Trading (76). Selects Juice Factory and Cairo Fruit.

Select Recipients : Confirms the two selected contacts appear. Also manually enters GTID of a new contact.

Accelerated Outreach : Chooses Accelerated Outreach Mode, sets window 6 hours, floor price $810. Privacy notice appears listing the three contacts. Reads and clicks “Yes, start outreach.” Logged.

AI Draft: Groq drafts personalised messages. Edits one.

Send & Track : Notifications sent. Dashboard shows “3 sent.” After 1 hour, Juice Factory offers $900; Cairo Fruit views but doesn’t offer; new contact declines.

Ranking : Dashboard ranks Juice Factory’s offer as best (score 94). Groq recommends accepting.

Acceptance : Clicks accept. MicroUSTN generated, microcontract sent for signature. Other two contacts receive close message.

Smart Inbox : Mekong Fresh sees “Microcontract pending signature.” Juice Factory sees “New contract to sign.”

END OF PHASE 8 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 9: Global Payment Orchestrator & Fee Collection Infrastructure (v6.3 – Fully Extended, with Single-Shipment & Multi-Shipment Fee Collection, PSP Routing, Gross-Up Calculation, Webhook Verification, OpenBanking Reconciliation, Smart Inbox Integration, Voice Activated Operations, No Separate “sgtx Fees” Phase)

PHASE 9 GOAL
This phase defines the underlying payment infrastructure that executes the collection of SGTX fees (trade fee and financing fee) when they become due – not a separate user-facing step. The trade SGTX fee is collected upfront at contract lock (single-shipment) or per-shipment(multi-shipment) as described in Phase 3. The financing fee (0.25%) is collected at disbursementas described in Phase 4. Phase 9 provides the PSP routing, dynamic gross-up calculation, webhook verification, and reconciliation for all fee collections, ensuring SGTX receives the exact net amount and that all transactions are cryptographically verifiable. It also includes voice-activated operations for authorised users to approve large fee payments, monitor PSP health, and handle fallbacks. All components are non-custodial – SGTX never holds funds, only issues splitting instructions.

PREREQUISITE
A fee collection event is triggered (Phase 3 for trade fee, Phase 4 for financing fee). The responsible payer (seller for trade fee, borrower for financing fee) has a verified payment method in their portal.

Step 9.1 – Fee Collection Trigger & Responsibility

The fee collection is not initiated by the payer navigating to a separate tab. Instead, it is embedded in the workflow:

Trade fee (single-shipment) : After mutual confirmation and logistics addenda signed, the system automatically creates a fee payment request for the seller. The seller sees a “Pay SGTX fee” screen in the contract signing flow (Phase 3).

Trade fee (multi-shipment) : Before each shipment is locked, the seller receives a per-shipment fee payment request.

Financing fee : Before the loan is disbursed, the borrower receives a fee payment request (deducted from principal – but the PSP split handles it; see Phase 4 Step 4.7). The financing fee is collected via split instruction, but the platform still needs to verify the split.

Responsible party:

Trade fee: Seller (always, as per our modifications).

Financing fee : Borrower (the party requesting financing – seller or buyer). The fee is deducted from the disbursed amount; the borrower effectively pays.

Governance (G9U1) : The fee payment request is stored in fee_payment_requests with status PENDING and linked to the contract or financing agreement.

Step 9.2 – AI-Driven PSP Router (Fee Optimisation)

The PSP Router (A2, LightGBM + Groq) selects the optimal PSP for the fee payer based on:

Payer country (seller’s country for trade fee; borrower’s country for financing fee).

Amount (fee amount).

Currency (USD always – fees are in USD).

Realtime PSP health scores (from psp_health_logs, updated every 30 seconds).

Cost (percentage fee + fixed fee per transaction).

Historical success rate for similar amounts and countries.

The router returns a ranked list of PSPs with estimated gross amounts (including PSP fees). Groq generates a plain-language recommendation displayed to the payer: “Option 1: Stripe (card) – gross 216.90,net216.90,net214.10, instant confirmation. Option 2: Bank transfer (SWIFT) – gross 214.10,fee214.10,fee0 (borne by your bank), but 2-3 days clearing. We recommend Stripe for immediate contract lock.”

The payer selects the preferred method. The selection is stored in fee_payment_requests.psp_selected.

Governance (G9U2) : The selected PSP must have country coverage and realtime health score ≥80. Fallback PSPs are pre-computed.

Step 9.3 – Dynamic Fee Gross-Up Calculation (AI Validated)

The system calculates the gross amount the payer must send so that after PSP fees, the net received equals the exact fee amount due. Formula:

Gross = (Net + Fixed_Fee) / (1 – Percentage_Fee) + Safety_Buffer (safety buffer configurable, default 0.5%).

AI validation (A2, HF Donut) : Before displaying the gross amount, the system fetches the latest fee schedule from the PSP (scraped daily). HF Donut extracts the current percentage and fixed fees from the PSP's PDF or HTML. If the scraped fees differ from the cached values by >0.1%, the block is CONDITIONAL and the payer is asked to confirm manually or wait for fee schedule update.

The calculation is stored in fee_calculations with a Loom hash.

Governance (G9U3) : The gross amount is displayed to the payer; they must confirm before redirect.

Step 9.4 – Payment Initiation & Redirect

The payer is redirected to the PSP's hosted payment page (or uses the PSP SDK).
A payment_attempts record is created with status INITIATED. The payment reference includes the USTN (for trade fee) or financing agreement ID (for financing fee) in the remittance field (e.g., SGTX USTN SGTX-...-V1 or SGTX FIN CF123).

Governance (G9U4) : The PSP webhook endpoint is pre-registered and the signature verification key is stored encrypted.

Step 9.5 – Webhook Verification & SGTX Feeslock Activation

Upon successful payment, the PSP sends a webhook. The payment-orchestrator service:

Verifies the webhook's cryptographic signature (using the PSP's public key stored in payment_aggregators.credentials_encrypted).

Extracts the payment reference and amount.

Checks that the received amount matches the expected gross (within tolerance).

If verified:

Marks the fee_payment_request as PAID.

Updates the payment_attempts status to SUCCEEDED.

For trade fee : Activates the SGTX FEESLock (or per-shipment SGTX FEESLock) to ACTIVE.

For financing fee : Marks the financing SGTX FEESLock as ACTIVE.

Publishes a NATS event fee.payment.confirmed – triggers the next step in Phase 3 (contract lock) or Phase 4 (disbursement).

If verification fails, the attempt is retried (up to 3 times). After 3 failures, a Smart Inbox alert is sent to the payer and the Platform Governance Authority.

Governance (G9U5) : The webhook must contain the USTN or financing agreement ID; otherwise, the payment is flagged for manual reconciliation.

Step 9.6 – PSP Health Monitoring & Automatic Fallback

A lightweight Rust service (psp-health-monitor) pings each active PSP’s health endpoint (or its sandbox) every 30 seconds, recording latency and error rates. A local LSTM model (or simple rule-based) forecasts the likelihood of a PSP being unhealthy in the next hour.

If health score drops below 80 → PSP marked DEGRADED in routing table; a Smart Inbox alert sent to the Platform Governance Authority.

If health score drops below 50 → PSP disabled; any new fee payments are automatically routed to the next best fallback PSP (pre-computed per country).

The fallback logic is a WasmEdge rule, ensuring no single PSP failure halts fee collection.

Governance (G9U6) : The PSP health monitor is self-hosted; no external service required.

Step 9.7 – OpenBanking Reconciliation (Self-Reconciling Ledger)

For PSPs that support PSD2 / OpenBanking APIs, the settlement-service can pull SGTX’s bank statement (with appropriate credentials from the Platform Governance Authority). HF Donut extracts each transaction line and matches it with open fee_payment_requests.

If a match is found (amount, reference, date within tolerance), the fee_payment_request is marked RECONCILED = true.

If an unmatched deposit appears (e.g., a wire from an unknown source), the system alerts the finance team via Smart Inbox.

If a fee payment remains unmatched for >7 days, an escalation is created.

This reconciliation runs daily and is also triggered on demand by the Admin Portal.

Governance (G9U7) : Reconciliation confidence must be ≥95% for auto-match; otherwise, manual review required.

Step 9.8 – Voice-Activated Fee Payment Approval (A1)

For large fee amounts (above a configurable threshold, e.g., $10,000), the system may require a second human authorisation (if the payer’s internal policies require). An authorised signatory can approve via voice:

In the mobile app or web portal, speak : “Approve fee payment for USTN SGTX-...-V1.”

Vosk transcribes offline; HF Mixtral extracts USTN and intent; ZITADEL verifies biometrics.

The Governor signs the payment instruction and releases it to the PSP.

Governance (G9U8) : The voice approval is logged with a Loom hash. If biometric verification fails, the instruction is not sent.

Step 9.9 – Tenant Data Export & Fee Statements

All fee payments (trade fees and financing fees) are included in the tenant's monthly reconciliation statements (Part 6.1.4). For the seller, the trade fee appears as a line item; for the borrower, the financing fee appears as a deduction from the disbursed amount. Statements are cryptographically signed and available for download.

Governance (G9U9) : The statement includes the fee amount, PSP used, date, and USTN/financing agreement ID.

GOVERNANCE GATES (PHASE 9 – SUPPORTING FEE COLLECTION)

Gate ID	Check	Enforcer	Failure Action
G9U1	Fee payment request stored and linked to contract or financing agreement	Governor (A3)	Block fee collection flow
G9U2	PSP selected has country coverage + realtime health score ≥80	PSP Router (A2)	Fallback to secondary PSP
G9U3	Gross-up calculation validated by AI (HF Donut fee schedule)	Fee Calculator (A2)	Recalculate; block if mismatch
G9U4	Payment redirect includes USTN/financing ID in remittance field	Governor (A3)	Reject; log anomaly
G9U5	PSP webhook signature verified; amount matches expected gross	Webhook Verifier (A2)	Reject webhook; retry up to 3 times; escalate on failure
G9U6	PSP health score monitored; automatic fallback if <50	Health Monitor (A2)	Manual override if false positive
G9U7	OpenBanking reconciliation confidence ≥95% for auto-match	SelfReconciler (A2)	Queue for manual review; Smart Inbox alert
G9U8	Voice approval requires ZITADEL biometric verification	Governor (A3)	Reject approval; log failed attempt
G9U9	Monthly fee statement cryptographically signed	TenantDataExportService	Block download if signature invalid

WORKED EXAMPLE – TRADE FEE COLLECTION (SINGLE SHIPMENT)

Context : Mekong Fresh (seller, Vietnam) has agreed a contract with Nile Foods. Trade fee = 360(1.2360(1.230,000). The payment request is embedded in the contract signing flow.

Fee payment request created – Mekong Fresh sees a screen: "Pay SGTX fee of $360 to proceed with contract lock."

PSP Router – Recommends Stripe (gross 370.50,net370.50,net360), or bank transfer ($360 net but slower). Seller selects Stripe.

Gross-up calculation – Stripe fee 2.9% + 0.30→gross=(0.30→gross=(360 + 0.30)/(1–0.029)=0.30)/(1–0.029)=371.03 (with buffer). Seller confirms.

Payment – Redirected to Stripe. Pays $371.03.

Webhook – Verified; SGTX FEESLock set to ACTIVE.

Contract lock – Proceeds. USTN generated.

Reconciliation – Stripe settles $360 to SGTX bank. OpenBanking pulls statement, matches reference. Payment marked RECONCILED.

END OF PHASE 9 – v6.3 WITH ALL MODIFICATIONS

Part 3, Phase 10: Dispute Resolution & Arbitration (v6.3 – Fully Extended, with AI-Mediated Triage, Evidence Package Autocompiler (Including QC Overrides), Predictive Outcome Modelling, Smart Inbox Integration, Plainlanguage Decision Panel, Multi-Shipment & Financing Disputes, SGTX Fees Dispute Workflow, External Verification, Three-Tier AI)

PHASE 10 GOAL
If a party to a trade believes that the executed shipment, payment, or financing has not met the contractual terms – quality defects, late delivery, documentation errors, temperature deviations, payment default, or financing fee miscalculation – they can file a dispute under the USTN (or financing agreement ID). The platform provides a structured, AI-assisted mediation process, with the option to escalate to international arbitration (ICC, DIFC-LCIA). The affected SGTX FEESLock(s) are frozen (but note: trade fee already paid, so freeze is administrative – no refund unless dispute resolves in payer's favour). The system ensures that all evidence is cryptographically verifiable, offers AI-powered outcome prediction, settlement proposal generation, and document authenticity checks. For multi-shipment contracts, disputes can be filed per shipment. For financing disputes (including co-financing), the evidence package includes the financing agreement and repayment records. The QC dispute fast-track automatically flags inspector-overridden AI findings. All steps are gated by the Governor and integrated with Smart Inbox and PlainLanguage Decision Panel.

PREREQUISIT
The trade (or specific shipment, or financing agreement) is locked or settled. The disputing party has the permission dispute.file.

Step 10.1 – Filing a Dispute

Portal : Any party (buyer, seller, logistics provider with standing, financier) → "Disputes" tab.

The filing party navigates to the trade's USTN (or shipment USTN, or financing agreement ID) via the Trade Command Center or Shipments Vault and clicks "File a Dispute". They can type a detailed description, upload evidence, or use voice input.

10.1.1 – Voice-Initiated Filing (A1)
In the mobile app, speak: *"File a dispute for USTN SGTX-EG-...-V1. The oranges in container 2 arrived with mould. I am claiming $2,000 in damages."* Vosk transcribes; HF Mixtral extracts USTN, nature of complaint, affected containers/pallets, claim amount. The form is pre-filled.

10.1.2 – Structured Submission
The disputes record is created:

trade_request_id (or contract_shipment_id for multi-shipment,
or financing_agreement_idfor financing dispute)

filing_party_gtid

dispute_type (QUALITY, DELAY, DOCUMENTATION, PAYMENT, FINANCING_FEE, SETTLEMENT, OTHER)

description, evidence_links, status = FILED

governor_decision_id

Smart Inbox : Filer sees confirmation; counterparty sees alert with dispute type.

Governance (G10U1) : Dispute filing must reference a valid, locked USTN or financing agreement. Governor returns DENY with Decision Panel explanation if not.

Step 10.2 – AI Triage & Categorisation (A2)

The Dispute Triage Agent (HF Mixtral, local) analyses:

Dispute description and evidence.

Contract clauses (or financing agreement).

Uploaded photos (via HF ViT for QC disputes).

Prior dispute history of the parties.

Output:

Refined dispute category.

Severity score (1-5).

Early assessment of mediation likely success.

Recommended resolution path (mediation, arbitration, or settlement).

Example : “Dispute categorised as QUALITY (mould). Severity 3. Based on contract’s force majeure and inspection clauses, mediation is recommended. 65% chance of settlement without arbitration.”

Stored in dispute_recommendations. Both parties see the recommendation in their Smart Inbox.

Governance (G10U2) : Triage is advisory, does not block any action.

Step 10.3 – Evidence Package Autocompiler (v6.3 – with External Verification)

Service : evidence-compiler (Rust). Triggered automatically when a dispute is filed, and can be manually re-run.

10.3.1 – Package Contents (immutable, Loom-hashed)

Category	Data Included	Source
Trade & Contract	USTN, contract clauses, incoterm, governing law, signatures, amendment history	contracts, trade_requests
Shipment Milestones	All milestone confirmations with timestamps, confirmers, method, evidence hashes	shipment_milestones
IoT Sensor Data	Temperature, humidity, shock readings, anomaly flags	iot_sensor_readings
QC Reports & Overrides	Full inspection reports, AI defect detections, inspector overrides with mandatory reasons and original AI confidence	qc_jobs, inspection_logs
Documents	All uploaded and generated documents (invoices, packing lists, certificates) with SHA256 hashes	documents, generated_documents
Communications	Negotiation messages, mediation log, trade room chats	negotiation_sessions, disputes.mediation_log
AI Decision Logs	Governor decisions, AI inference records, SHAP explanations, plainlanguage tenant messages	governor_decisions, ai_inference_records
Financial Data (trade)	SGTX fee payment, settlement instructions, PSP confirmations	fee_payment_requests, settlement_instructions
Financing Data (if financing dispute)	Financing agreement, disbursement records, repayment schedule, fee deductions	financing_agreements, financing_annexes, payment_attempts
QC Override Details	For any dispute involving QC	inspection_logs with ai_overrid

	report: all inspector-overridden AI findings, original AI confidence, inspector's reason, final classification	den = true

Privacy filtering : no raw voice audio, no bank account numbers, no unrelated trades.

10.3.2 – External Verification (v6.3 – New)
The platform provides a public, read-only verification page at https://verify.sgtx.io/evidence/{package_id}?token={verification_token}. The token is generated per package and provided to the disputing parties (and, by court order, to arbitrators). The page performs clientside verification:

Fetches public metadata (loom_hash, platform_signature, compiled_at, dispute_id, package_sha256).

Verifies Ed25519 signature against SGTX public key.

Recomputes hash to confirm integrity.

No trade data is exposed – only hashes and signatures. The full package remains encrypted and can be decrypted only with a key provided to authorised parties.

Governance (G10U3) : The evidence package must be compiled before dispute can proceed to mediation. Missing critical data triggers manual review.

Step 10.4 – Predictive Dispute Outcome (A2)

The system runs a predictive model (XGBoost, local) trained on past SGTX dispute outcomes. Features:

Dispute type, commodity, incoterm, ports.

Party trust scores and dispute history.

Existence of contradictory evidence (e.g., origin QC pass vs. destination damage).

Contractual strength of the complaining clause.

For QC disputes : whether the inspector overrode AI findings and the original AI confidence.

Output:

Estimated probability of filing party winning.

Predicted damage amount (range).

Confidence interval.

Groq presents a neutral, plain-language summary to both parties. The prediction is advisory and stored in dispute_recommendations.

Governance (G10U4) : The prediction does not block any action; it is displayed as “AI-estimated outcome” with a disclaimer.

Step 10.5 – Mediation Log & Multi-Language Support (A1)

The mediation_log JSONB array stores all exchanges. Each message is timestamped and attributed. Groq (fallback Ollama) translates each message into the counterparty's preferred language for understanding (not certified translation). A sentiment analysis model (HF cardiffnlp/twitter-roberta-base-sentiment) monitors messages for hostility; if detected, the system flags the mediator and suggests a cooling-off period (24h pause).

Smart Inbox : Each new mediation message generates a NEGOTIATION item (priority 80). Both parties see the other's messages (translated).

Governance (G10U5) : Mediation log entries are immutable and signed by the author's ZITADEL passkey.

Step 10.6 – AI-Generated Settlement Proposal (A1)

If mediation stalls, the AI (Groq + HF Mixtral) generates a fair settlement proposal based on:

Predictive outcome (Step 10.4).

Contract's liability and liquidated damages clauses.

Undisputed portion (if any, e.g., only 3 of 22 pallets contested).

Industry standards (scraped trade arbitration awards).

Example proposal : "The parties agree to settle as follows: 1) Seller compensates Buyer $1,200 for damaged lemons. 2) Buyer releases seller from further claims under USTN...V1. 3) SGTX fee already paid remains with SGTX. 4) Both parties bear own costs."

The proposal is presented to both parties via Smart Inbox; each can accept, reject, or request modifications. If accepted, the dispute status becomes RESOLVED. The resolution is stored immutably.

Governance (G10U6) : Acceptance requires both parties to sign a settlement addendum (ZITADEL passkey). Governor gate.

Step 10.7 – Smart SGTX Feeslock Management (Freeze & Partial Release)

When a dispute is filed, the Governor freezes the affected SGTX FEESLock(s) (for trade fee) or financing SGTX FEESLock(s) (for financing fee). Because the fee is already paid, freezing prevents any administrative release (e.g., marking the trade as fully settled) until resolution. For undisputed amounts (e.g., 85% of shipment is fine), the AI may propose releasing the unaffected portion administratively. This requires a Governor decision with A3 approval.

For financing disputes, the financing SGTX FEESLock is frozen; no further disbursements or repayments are processed until resolution.

Governance (G10U7) : Partial release requires multisig approval (Platform Governance Authority). Decision Panel explains.

Step 10.8 – Document Authenticity Check (A2)

If a party submits evidence (e.g., a different inspection report) that contradicts previously uploaded documents, the Document Authenticity Agent (HF Donut) cross-checks:

Metadata (dates, author, digital signatures).

Consistency of issuing body with original document.

Any signs of tampering (hash comparison).

If inconsistency detected, AI flags it : “The inspection report submitted by the buyer appears to be from a different surveyor than the original QC booking. Confidence 88%. May require external verification.” This is recorded but does not block mediation; it is presented to the mediator.

Smart Inbox : Both parties see a COMPLIANCE item with the flag.

Step 10.9 – Escalation to International Arbitration

If mediation fails (no settlement after max rounds or parties agree to escalate), the system prepares the case for formal arbitration.

Automated Case Filing (A2):

Retrieves the arbitration request form from a local template (pre-obtained from ICC, DIFC-LCIA, etc.).

Autofills with dispute details, parties, contract references, and a structured claim statement.

Groq drafts the claim narrative in the language required by the arbitration body.

The compiled package (form + evidence) is produced as a PDF, stored in generated_documents.

The filing party’s legal counsel reviews and submits.

Smart Inbox : Both parties receive an informational item: “Arbitration case package ready. Download PDF.” Dispute status → ARBITRATION_PENDING.

Governance (G10U8) : The platform does not submit the case; only prepares documents.

Step 10.10 – SGTX Fees Dispute (SGTX Fee) Workflow (v6.3 – New)

Although the SGTX trade fee is paid upfront, a party may dispute the fee calculation (e.g., incorrect dynamic rate applied, mis-classification of commodity). This is a separate dispute type (FEE_DISPUTE). The workflow:

Tenant clicks “Dispute SGTX Fee” in the Trade Command Center’s financial card.

Pre-filled form with USTN, fee amount, rate, and a reason.

AI (A2, HF local) analyses:

Compares the fee calculation against the contract's parameters and the dynamic formula.

Checks if any adjustment factor was misapplied (e.g., wrong perishability surcharge).

Outputs a recommendation : UPHOLD, ADJUST (partial refund), or REFUND (full).

The dispute is escalated to the Platform Governance Authority (multisig). A new Governor decision with decision_type = "SGTX FEES_dispute" is created.

The Platform Governance Authority reviews the AI analysis and can:

Uphold the fee (no change).

Approve a refund (partial or full).

Request more information.

If a refund is approved, the payment-orchestrator initiates a refund to the original PSP method (or bank transfer). The SGTX FEESLock is adjusted.

Time limit: Must be filed within 90 days of fee payment.

Governance (G10U9) : Any monetary adjustment requires multisig (3/5). Refund is logged immutably.

Step 10.11 – QC Dispute Fast-Track (v6.3)

If the dispute involves a QC report, the evidence package automatically includes a dedicated qc_overrides section. This section lists every AI finding that the inspector overrode, with:

Original AI detection (defect type, confidence %).

Inspector's classification after override.

Inspector's mandatory reason (≥10 characters).

Timestamp and photo hashes.

This data is presented to the mediator/arbitrator upfront, reducing investigation time.

Governance (G10U10) : This is mandatory for any QC-related dispute; no additional gate.

Step 10.12 – Governance Gates (Phase 10 – Complete)

Gate ID	Check	Enforcer	Failure Action
G10U1	Dispute filing references valid USTN / shipment / financing agreement with SGTX FEESLock	Governor (A3)	Reject filing
G10U2	Triage and categorisation logged (advisory)	Dispute Agent (A2)	None
G10U3	Evidence package compiled and Loom-	Evidence Agent	Block escalation until

	hashed before escalation	(A2/A3)	complete
G10U4	Mediation log entries timestamped and attributed	Governor (A3)	Reject message
G10U5	AI-generated settlement proposal (if used) explicitly accepted by both parties	Governor (A3)	Cannot resolve
G10U6	Partial SGTX FEESLock release (if applied) must have Governor approval	Governor (A3)	Hold full amount
G10U7	Document authenticity flags recorded (advisory)	Doc Agent (A2)	None
G10U8	Arbitration package completeness check passed	Arb Agent (A2)	Cannot close dispute
G10U9	SGTX FEES dispute: monetary adjustment requires multisig (3/5)	Governor (A3)	Reject adjustment, log dispute
G10U10	QC dispute fast-track: overridden AI findings flagged in evidence package	Evidence Compiler (A2)	None (informational – ensures transparency)
G10U11 (v6.3)	All blocked actions display PlainLanguage Governor Decision Panel with actionable remediation	Governor (A3) + Groq/Ollama	User sees jargon-free explanation; cannot proceed until condition resolved

WORKED EXAMPLE – QUALITY DISPUTE WITH QC OVERRIDE FAST-TRACK (V6.3)

Context : Nile Foods receives a shipment of lemons. The lemons arrived with mould. Nile Foods files a dispute.

Filing : Nile Foods opens Disputes tab, selects USTN, speaks: “File dispute for mould on lemons, claim $2,000.” Form pre-filled.

Triage & Evidence Compilation : AI categorises QUALITY, severity 3, recommends mediation. Evidence package is compiled. The QC report included: inspector overrode AI finding on pallet OR020 (original AI mould confidence 92% → inspector overrode to “bruising” with reason “Dark spot is surface bruise”). The override is flagged in the package.

Predictive outcome : AI estimates 78% chance for buyer, typical award 1,200–1,200–1,600.

Mediation : Mekong Fresh posts a statement. Groq translates to Arabic for Nile Foods. Sentiment analysis remains neutral. After deadlock, AI proposes settlement: $1,200.

Acceptance : Both parties accept via Smart Inbox. Settlement addendum signed.

SGTX FEES handling : Original SGTX fee already paid. No adjustment. Dispute closed.

QC override transparency : The evidence package clearly shows that the inspector overrode an AI mould detection. The mediator saw this but determined the mould on other pallets was valid.

END OF PART 3, PHASE 10 – v6.3 WITH ALL MODIFICATIONS

Part 4: Expanded Microservices Architecture (v6.3 – Fully Upgraded, with All Modifications: Three-Tier AI, Multi-Shipment, Dual-Mode Toggle, SGTX Fee Paid by Seller, Distressed Enhancements, Financing Co-Financing, Etc.)

4.0 Architecture Overview

SGTX is built as a polyglot, self-hosted, event-driven microservices mesh. Every service is a Rust binary (or Python sidecar) compiled with --release --locked against a version-locked Cargo.lock. All services communicate through the Governor Proxy, which enforces constitutional rules, jurisdiction supremacy, SGTX FEESLock checks, and AI authority gates before any call reaches business logic.

Communication patterns:

Synchronous REST (Axum/Tonic) → immediate actions requiring a Governor verdict.

Asynchronous Messaging (NATS JetStream) → events (milestone updates, RFQ broadcasts, negotiation messages, dispute filings, financing bids, distressed outreach).

Workflow Orchestration (Temporal Core, self-hosted) → long-running stateful processes (contract genesis, financing auctions, distressed cargo resolution, multi-shipment coordination).

IoT Ingestion (MQTT, Mosquitto, self-hosted) → optional real-time sensor data.

All infrastructure runs on bare-metal K3s clusters with CRIO and Firecracker for optional sandboxing. The development environment is any Linux x64 or macOS with native binaries (no Docker, no billing). The architecture now fully supports multi-shipment contracts, dual-mode toggle for trading companies, three-tier AI (Groq + HF local + Ollama), SGTX fee paid solely by seller, co-financing, and all other v6.3 enhancements.

4.1 Core Service Definitions & Governor-Proxied Communication

The table below lists every microservice, its primary technology (all OSS), and its responsibilities in v6.3. New or significantly upgraded services for v6.3 are marked with ★.

#	Service	Technology	Responsibilities
1	governor-service	Rust + Axum +	Policy evaluation,

		WasmEdge + OPA	jurisdiction supremacy, SGTX FEESLock checks, Loom logging. **Every request passes through it. Generates tenant-friendly decision messages (via Groq/Ollama A1) stored in governor_decisions. tenant_message.
2	identity-service	ZITADEL (self-hosted, Apache2.0)	OIDC, SAML, WebAuthn/passkeys, multitenancy, RBAC, GTID generation & resolution. Manages onboarding state (tenant_onboarding_state), dual-mode toggle (employees.active_trader_mode_context), tenant lifecycle (tenants.lifecycle_state), sandbox provisioning API.
3	trade-service	Rust + Axum + Temporal	CRUD on trade requests, phase orchestration, status transitions, USTN generation (now company-anchored). Supports structured per-container request form, multi-shipment schedule storage (trade_requests.multi_shipment_schedule), draft saving, workflow recovery triggers.
4	quote-service	Rust + Axum	EXW price lock, quote assembly, RFQ generation, logistics bundle selection. Feeds live market chart data, manages post-lock price watch, stores alternative delivery ports (quote_delivery_options), supports manual logistics cost entry (seller_quotes.manual_logistics_costs) and

			price visibility toggle.
5	logistics-service	Rust + Axum + Rig	Route optimisation (Route Oracle), carrier matching, RFQ distribution to logistics providers. Handles logistics bundle re-optimisation (diff view), VRP dispatch planner for trucking companies, logistics provider addendum signing, container release confirmation distribution.
6	location-service	Rust + OSRM + Nominatim (self-hosted)	Geocoding, distance/time matrices, trucking route intelligence, GPS trace learning.
7	contract-service	Rust + Axum + Temporal	Contract genesis (Clause Forge, Risk Oracle, Jurisdiction Harmonizer), SGTX FEESLock creation, digital signatures, multi-shipment contract management (contract_shipments). Added SGTX Witness Clause mandatory; supports own contract upload with mandatory addendum; logistics addendum signature collection.
8	finance-service	Rust + Axum + Temporal + XGBoost	Financing requests, blind bidding (NATS KV encrypted), credit intelligence (200+ signals), Monte Carlo simulation, DeFi interface. Enforces co-financing (multiple financiers per request), flat 0.25% financing fee deducted at disbursement, financier preferences filtering, automatic RFQ broadcast, financing witness clause, financier

			historical data storage.
9	shipment-service	Rust + Axum + MQTT	Shipment tracking, milestones, IoT data ingestion, barcode scan events, sensor consensus. Computes estimated margin for sellers (visible in Shipments Vault) and aggregates document readiness scores; supports per-shipment milestones for multi-shipment contracts.
10	settlement-service	Rust + Axum + NATS	Settlement instruction generation, multi-rail verification, FX arbitrage, PSP split reconciliation (for fee collection). Handles trade principal settlement (buyer → seller) – no SGTX FEES release (already collected).
11	payment-orchestrator	Rust + Axum + PSP SDKs (Stripe, Fawry, etc.)	SGTX fee collection (trade fee and financing fee), PSP routing with AI, fee gross-up, webhook verification, PSP health monitoring, automatic fallback, split payment instructions for financing fee deduction.
12	document-service	Rust + Axum + Ceph (or local FS)	Document storage, SHA256 hashing, HF Donut OCR/extraction, document authenticity checks. Mandatory USTN & GTID stamping on all generated documents.
13	barcode-service	Rust	SSCC18, GS1-128, QR code generation per pallet, barcode scanning validation. Includes predictive scanning reliability agent (A2, XGBoost) that warns about potential scan

			failures before they occur; reprint policy enforced (Governor decision required after loading).
14	commodity-service	Rust + Rig	Commodity compatibility checks, HS code classification, container type enforcement, commodity-packing rules. Enhanced with commodity-specific margin estimation for SGTX fee calculation (XGBoost + LLM refinement).
15	kyb-kyc-service	Rust + Axum + Rig	KYB/KYC verification, document scraping from government registries, AML/sanctions screening.
16	trust-score-service	Rust + XGBoost	Dynamic trust scoring (200+ signals), real-time updates, explainability via SHAP. Now includes financing repayment history as a signal.
17	notification-service	Rust + NATS	Email, SMS, push notifications via self-hosted SMTP or free gateways. Supports co-branded email sender name for distressed cargo outreach (e.g., “Seller via SGTX”).
18	audit-service	Rust + ClickHouse + Loom	Immutable audit trail, Loom hash chaining, event sourcing, tamper-evidence verification.
19	ai-agent-orchestrator	Rust + Rig + vLLM (HF local) + Groq + Ollama	Manages all AI agents (A0-A4), three-tier API routing, prompt templating, decision logging, fallback chain (Groq → Ollama, HF → Ollama).
20	marketplace-api-service	Rust + Axum	Partner API endpoints, revenue share attribution, auto-detection of repeated

			direct trades. Implements per-partner rate limiting and feeds the partner portal's usage analytics dashboard.
21	defi-service	Rust + ethers-rs + Polygon + The Graph	Smart contract deployment, stablecoin health monitoring, tokenization, secondary market pricing agent.
22	esg-carbon-service	Rust + LightGBM	ESG scoring, carbon footprint calculation (IMO EEXI), green route recommendation.
23	disruption-service	Rust + LSTM + GDACS/ACLED feeds	Predictive disruption modelling, demurrage/detention prediction, real-time alerts.
24	distressed-service	Rust + Axum + Temporal	Distressed cargo listing, AI condition assessment (HF ViT), dynamic pricing (XGBoost), microcontract creation. Orchestrates the triage dashboard (Paths A/B/C), "Check Buyers" advisory, enforces privacy notice before outreach, handles microUSTN splitting.
25	buyer-search-service	Rust + Axum (only for distressed outreach)	Distressed cargo direct notification to saved contacts / specific GTIDs; no open marketplace. Supports "Check Buyers" advisory query.
26	jurisdiction-service	Rust + WasmEdge	Jurisdiction matrix compilation, sanctions screening, CBDC readiness monitoring, regulatory report generation, distressed country factor calculation.
27	api-gateway	Nginx + Lua + WasmEdge filters	Rate-limiting, WASM-based jurisdictional

			rules, JWT validation, request routing to Governor.
28	fx-oracle-service	Rust + Axum	Multi-source FX rate aggregation (ECB, 0x API), volatility prediction, optimal conversion path finding.
29	corporate-graph-service	Rust + PyTorch Geometric	Ownership graph, UBO detection, fraud cycle detection, graph proximity for distressed buyer outreach.
30	observability-service	Go + OpenTelemetry + VictoriaMetrics	Metrics, distributed tracing, profiling (eBPF), alerting.
31	packing-containerisation-service	Rust + Python (ORTools) via subprocess/gRPC	Palletisation optimisation, 3D container loading, loading instructions generation, SSCC barcode assignment. Supports collaborative editing (Yjs integration) for multi-user packing plan sessions, consumes buyer's structured pallet data.
32	negotiation-orchestrator	Rust + Rig	Automated counter-offer bots for Phase 3 (trade negotiation) and Phase 7 (distressed cargo). Integrates with Groq/Ollama for message generation; provides bot transparency state (round, last message, parameters) to the frontend.
33	voice-gateway	Rust + Vosk (offline)	Speech-to-text for all voice features, intent extraction via HF Mixtral, response generation via Groq/Ollama.
34	evidence-compiler	Rust	Auto-compilation of dispute evidence packs from multiple sources, Loom-hashed output. Now flags QC

			inspector overrides for the dispute fast-track.
35	digital-twin-engine	Rust + Python (Pyro)	Realtime shipment digital twin, “what-if” simulations, predictive ETA, risk forecasting.
36	secondary-market-service	Rust + Polygon	Tokenized asset trading, AI pricing suggestions (advisory).
37	document-requirement-service	Rust + Rig	Dynamic document checklist per trade, AI validation of uploaded docs against contract.
38	qc-service	Rust + Axum	QC booking, inspection point recommendation (AI), AR-enhanced inspection data ingestion. Enforces override accountability: mandatory reason for any AI override, stored in inspection_logs.inspector_notes.
39	network-service	Rust + Axum	Saved contacts management, autosave on interaction, trust portrait generation, relationship health scoring (LSTM). Handles co-branded email delivery for distressed outreach and “Check Buyers” advisory queries.
40	analytics-service	Rust + ClickHouse	Trade analytics, model drift monitoring, platform metrics.
41	inbox-service	Rust + Axum + Groq (free tier) + NATS	Manages the Smart Inbox for all tenants: deterministic urgency scoring (A4), Groq/Ollama title/description generation, snooze/dismiss actions, NATS-driven real-time updates. Subscribes to platform

			events to create/update/delete items per user.
42	tenant-experience-service	Rust + Axum	Orchestrates onboarding wizard (steps, draft saving), sandbox (data isolation, reset), dual-mode toggle context, organisational hierarchy (business units, departments, cost centres). Aggregates data for the Trade Command Center and exposes the /v1/trade/{ustn}/command-center endpoint.
43	workflow-recovery-service	Rust + NATS + Tokio	Monitors SLA timers for stuck trades (e.g., 7 days for PENDING_SELLER_RESPONSE). Publishes reminder and escalation events, and creates Governor decisions for auto-cancellations and per-shipment timeouts. Ensures no trade (or shipment) stalls indefinitely.
44	tenant-data-export-service	Rust + Tera + headless Chrome (optional) + ed25519-dalek	Generates cryptographically signed data exports (CSV/PDF/JSON) and monthly reconciliation statements for tenants (including fee statements). Provides /v1/tenant/export and /v1/tenant/statement endpoints.
★45	financier-preference-service	Rust + Axum	Stores and evaluates financier preferences (country filters, amount limits, settlement types) for automatic RFQ broadcast. Used by finance-service to filter recipients.

★46	multi-shipment-coordinator	Rust + Temporal	Manages per-shipment locking, fee collection, and milestone tracking for multi-shipment contracts. Coordinates independent USTN generation per shipment.
★47	fee-split-orchestrator	Rust + Axum	Handles PSP split instructions for financing fee deduction (0.25%) and for trade fee collection. Ensures atomic split verification.

4.2 Zero-Cost Self-Hosted Implementation Matrix

Service	Key Libraries / Runtimes	External Data Sources (Free)	macOS/Linux Setup
Governor	axum, wasmedge-sdk, opa-wasm, loom, reqwest (for Groq/Ollama)	–	cargo build --release --locked; Groq API key optional
Identity	ZITADEL binary, webauthn-rs	Government registries (scraped)	brew install zitadel or cargo install --git
Packing	ORTools (Python), ort crate	–	pip3 install ortools, cargo build
AI Orchestrator	rig, vllm (HF), reqwest (Groq), ollama client	HF free tier, Groq free tier	pip install vllm ollama, export HF_TOKEN=... GROQ_API_KEY=...
Finance	xgboost, numpy, pandas	ECB rates, FAO indices, AISHub	pip install xgboost
Shipment	mosquitto, paho-mqtt	AISHub, OpenWeatherMap free tier	brew install mosquitto
Settlement	ethers-rs, plonky3	0x API free tier, The Graph	cargo build
Document	donut (transformers), paddleocr	–	pip install transformers
Jurisdiction	wasmedge	OFAC, UN, EU scrapped lists	cargo build --target wasm32-wasi
Voice Gateway	vosk	–	Download Vosk model (~50 MB)
Inbox	axum, reqwest (Groq), nats	–	cargo build

Tenant Experience	axum, serde_json, reqwest	–	cargo build
Workflow Recovery	axum, nats, tokio	–	cargo build
Tenant Data Export	axum, tera, headless chrome (optional), ed25519-dalek	–	cargo build; optional brew install --cask google-chrome for PDF
Financier Preference	axum, postgres	–	cargo build
Multi-shipment Coordinator	axum, temporal	–	cargo build
Fee Split Orchestrator	axum, reqwest	–	cargo build

4.3 Inter-Service Communication Map (Governor-Proxied, Event-Driven)

All synchronous REST calls pass through the Governor Proxy. Asynchronous events are published to NATS JetStream. Examples:

Example 1 – Loading a pallet (Phase 5, v6.3 enhanced):

text

Mobile App → API Gateway → Governor Proxy

→ Governor evaluates OPA policy (shipment.milestone.confirm) + WasmEdge (jurisdiction, SGTX FEESLock)

→ If ALLOW, forwards to shipment-service.

→ shipment-service updates milestone, publishes “pallet.loaded” event to NATS.

→ NATS consumers:

- notification-service → alerts buyer, seller
- inbox-service → creates/updates Smart Inbox items
- multi-shipment-coordinator → updates per-shipment progress
- digital-twin-engine → updates ETA simulation

Example 2 – Financing RFQ broadcast (Phase 4):

text

finance-service receives financing request

→ Calls financier-preference-service to filter eligible financiers

→ Publishes “financing.rfq.broadcast” to NATS with encrypted bids subject

→ Each eligible financier’s inbox-service creates a Smart Inbox item

→ finance-service starts bidding window timer

Example 3 – Multi-shipment shipment activation (Phase 3):

text

Seller clicks “Ready for Shipment X” → API Gateway → Governor

→ Governor checks master contract status, per-shipment fee payment

→ multi-shipment-coordinator workflow activated (Temporal)

→ fee split orchestrator triggers PSP split for fee

→ After payment, generates new USTN, updates contract_shipments

→ Publishes “shipment.locked” event → inbox-service updates buyer

4.4 Deployment & Scaling on Bare-Metal K3s

All services are compiled into static binaries and deployed as OCI containers (without Docker, using CRIO + runwasi). A minimal K3s cluster runs:

WASM microservices inside WasmEdge sandboxes for maximum security (Governor modules, jurisdiction checks).

Native binaries for high-performance services (trade-service, shipment-service, finance-service).

Firecracker microVMs for workloads that need kernel isolation (AI inference, document OCR).

Data residency enforced by deploying sovereign nodes in each region (EU, Egypt, UAE, USA, India, etc.) with strict network policies. The sovereign_nodes table routes requests to the correct regional instance.

New v6.3 services scaling:

inbox-service is stateless; scale horizontally across nodes. NATS JetStream partitions inbox events by employee ID.

finance-service uses NATS KV for encrypted bids; replicas share the same KV bucket.

multi-shipment-coordinator runs as a Temporal worker; multiple workers can handle different contracts.

fee-split-orchestrator is stateless; scales with payment volume.

Example production-scale deployment:

3 replicas of inbox-service per node, fronted by Governor Proxy.

2 replicas of finance-service (NATS KV ensures consistency).

1 singleton workflow-recovery-service per node (to avoid duplicate timers).

4.5 Implementation Checklist (Linux x64 or macOS, ZeroCost)

bash

cd /opt/sgtx

```
# 1. Build all new v6.3 services
for svc in inbox-service tenant-experience-service workflow-recovery-service tenant-data-export-service financier-preference-service multi-shipment-coordinator fee-split-orchestrator; do
cd core/$svc
cargo build --release --locked
cd ../..
done

# 2. Install optional Python packages for ORTools, simulations, and ML
pip3 install ortools pyro-ppl xgboost lightgbm torch transformers vllm

# 3. Download Vosk model (if not already)
cd mobile/models
ls vosk-model-small-en-us-0.15 || curl -L -o vosk-model.zip https://alphacephei.com/vosk/models/vosk-model-small-en-us-0.15.zip && unzip vosk-model.zip

# 4. Check all services health (expected ports 8080-8095)
for port in 8080 8081 8082 8083 8084 8085 8086 8087 8088 8089 8090 8091 8092 8093 8094 8095; do
curl -f http://localhost:$port/health 2>/dev/null || echo "Port $port not active"
done

# 5. Ensure NATS, PostgreSQL, Valkey are running
pg_isready -h localhost -p 5432 -U sgtx_user
nats-server &
valkey-server &
```

Post-execution : All microservices are operational, Governor-gated, and built on zero-cost open-source foundation. The addition of financier preference, multi-shipment coordinator, and fee split orchestrator completes the transition to a fully tenant-centered, action-driven platform with advanced financing and multi-shipment capabilities.

END OF PART 4 – v6.3 WITH ALL MODIFICATIONS

Part 5: Comprehensive Data Model Specifications (v6.3 – Full DDL with All Modifications: Structured Containers, Multi-Shipment, Alternative Ports, Manual Logistics Costs, Dual-Mode Toggle, SGTX Fee by Seller, Financer Preferences, Co-Financing, Distressed Enhancements, Evidence Packages, Etc.)

This part contains the complete physical data model for PostgreSQL 18 with pgvector and TimescaleDB. All v6.3 modifications are integrated. Tables are listed in logical order. SQL syntax is shown in clear blocks, but the blueprint uses sql code blocks – I will present them as plain text with line breaks for readability.

1. ENUMERATED TYPES (ENUMS)

text

CREATE TYPE tenant_type AS ENUM (

'CORPORATE', 'FINANCIAL', 'LOGISTICS', 'QUALITY_CONTROL', 'REGULATORY', 'GOVERNMENT', 'MARKETPLACE_PARTNER'

);

CREATE TYPE logistics_role AS ENUM (

'FREIGHT_FORWARDER', 'SHIPPING_LINE', 'TRUCKING', 'CUSTOMS_BROKER'

);

CREATE TYPE kyb_status AS ENUM (

'PENDING', 'SUBMITTED', 'VERIFIED', 'REJECTED', 'EXPIRED'

);

CREATE TYPE employee_status AS ENUM (

'INVITED', 'PENDING_APPROVAL', 'ACTIVE', 'SUSPENDED', 'DEACTIVATED'

);

```
CREATE TYPE grant_type AS ENUM ('ALLOW', 'DENY');

CREATE TYPE trade_status AS ENUM (
'DRAFT', 'MATCHING', 'PENDING_SELLER_RESPONSE', 'QUOTED',
'NEGOTIATING', 'CONTRACTED', 'FINANCING', 'IN_EXECUTION',
'COMPLETED', 'CANCELLED', 'DISPUTED'
);

CREATE TYPE contract_status AS ENUM (
'DRAFT', 'PENDING_SIGNATURES', 'LOCKED', 'ACTIVE', 'COMPLETED', 'DISPUTED',
'TERMINATED', 'ACTIVE_MASTER'
);

CREATE TYPE shipment_status AS ENUM (
'CREATED', 'GATED_IN', 'LOADED', 'DEPARTED', 'IN_TRANSIT',
'ARRIVED', 'CUSTOMS_IMPORT', 'DELIVERED', 'DISPUTED', 'DISTRESSED'
);

CREATE TYPE SGTX FEES_lock_status AS ENUM (
'ACTIVE', 'PARTIALLY_RELEASED', 'FULLY_RELEASED', 'DISPUTED', 'CANCELLED'
);

CREATE TYPE financing_status AS ENUM (
'REQUESTED', 'BIDDING', 'AWARDED', 'ACTIVE', 'REPAID', 'DEFAULTED'
);

CREATE TYPE governor_verdict AS ENUM (
'ALLOW', 'DENY', 'CONDITIONAL', 'ESCALATE', 'PENDING'
);

CREATE TYPE ai_authority AS ENUM ('A0', 'A1', 'A2', 'A3', 'A4');

CREATE TYPE trader_mode AS ENUM ('BUY', 'SELL', 'DUAL');

CREATE TYPE pallet_status AS ENUM ('PENDING', 'LOADED', 'DAMAGED', 'DELIVERED');
```

CREATE TYPE psp_health_status AS ENUM ('HEALTHY', 'DEGRADED', 'DOWN');

2. IDENTITY LAYER (TENANTS, EMPLOYEES, ROLES, ONBOARDING, DUAL-MODE)

text

```
CREATE TABLE tenants (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
gtid TEXT UNIQUE NOT NULL,
legal_name TEXT NOT NULL,
jurisdiction TEXT NOT NULL,
type tenant_type NOT NULL,
logistics_roles logistics_role[] DEFAULT '{}',
kyb_status kyb_status DEFAULT 'PENDING',
kyb_tier INTEGER DEFAULT 1,
cryptographic_hash TEXT NOT NULL,
public_key_pem TEXT,
certificate_pem TEXT,
immutable_since TIMESTAMPTZ,
risk_score DECIMAL(5,2),
sanctions_cleared BOOLEAN DEFAULT false,
aml_last_screened TIMESTAMPTZ,
default_trader_mode trader_mode DEFAULT 'DUAL',
icon_shuffle_enabled BOOLEAN DEFAULT true,
lifecycle_state TEXT NOT NULL DEFAULT 'REGISTERED', -- REGISTERED, ONBOARDING,
KYB_PENDING, VERIFIED, LIMITED_MODE, AT_RISK, SUSPENDED, ARCHIVED
lifecycle_state_updated_at TIMESTAMPTZ DEFAULT NOW(),
lifecycle_state_reason TEXT,
emergency_mode_active BOOLEAN DEFAULT false,
emergency_mode_reason TEXT,
emergency_mode_activated_at TIMESTAMPTZ,
emergency_mode_activated_by UUID,
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE employees (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
tenant_id UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
email TEXT NOT NULL,
full_name TEXT NOT NULL,
role_id UUID,
kyc_status TEXT DEFAULT 'PENDING',
kyc_tier INTEGER DEFAULT 1,
status employee_status DEFAULT 'INVITED',
invitation_token TEXT,
invitation_expires_at TIMESTAMPTZ,
approved_by UUID REFERENCES employees(id),
mfa_enabled BOOLEAN DEFAULT false,
default_trader_mode trader_mode DEFAULT 'DUAL',
icon_shuffle_permission BOOLEAN DEFAULT true,
preferred_language TEXT DEFAULT 'en',
voice_stress_consent BOOLEAN DEFAULT false,
active_trader_mode_context trader_mode, -- for dual-mode toggle
last_login_at TIMESTAMPTZ,
created_at TIMESTAMPTZ DEFAULT NOW(),
UNIQUE(tenant_id, email)
);

CREATE TABLE roles (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
name TEXT NOT NULL,
permissions TEXT[] NOT NULL,
allowed_trader_modes trader_mode[] DEFAULT ARRAY['BUY','SELL','DUAL'],
created_at TIMESTAMPTZ DEFAULT NOW(),
UNIQUE(tenant_id, name)
);

CREATE TABLE employee_permissions (
employee_id UUID NOT NULL REFERENCES employees(id) ON DELETE CASCADE,
permission TEXT NOT NULL,
grant_type grant_type NOT NULL,
trader_mode_context trader_mode[] DEFAULT ARRAY['BUY','SELL','DUAL'],
```

```
granted_at TIMESTAMPTZ DEFAULT NOW(),
granted_by UUID REFERENCES employees(id),
expires_at TIMESTAMPTZ,
PRIMARY KEY(employee_id, permission, trader_mode_context)
);

CREATE TABLE data_scopes (
employee_id UUID PRIMARY KEY REFERENCES employees(id) ON DELETE CASCADE,
country_access TEXT[],
document_types TEXT[],
max_transaction_value DECIMAL(20,4),
custom_filters JSONB,
trader_mode_filters JSONB,
hidden_cost_components TEXT[], -- e.g., 'freight', 'SGTX FEES'
allow_role_switching BOOLEAN DEFAULT true -- enable/disable dual-mode toggle
);

CREATE TABLE tenant_documents (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
document_type TEXT NOT NULL,
storage_path TEXT NOT NULL,
sha256_hash TEXT NOT NULL,
ai_extracted JSONB,
verified_at TIMESTAMPTZ,
verified_by UUID REFERENCES employees(id),
uploaded_at TIMESTAMPTZ DEFAULT NOW(),
visible_in_modes trader_mode[] DEFAULT ARRAY['BUY','SELL','DUAL']
);

CREATE TABLE trust_scores (
gtid TEXT PRIMARY KEY REFERENCES tenants(gtid) ON DELETE CASCADE,
score DECIMAL(5,2) NOT NULL,
model_version TEXT,
updated_at TIMESTAMPTZ DEFAULT NOW(),
components JSONB,
```

```
buy_mode_score DECIMAL(5,2),
sell_mode_score DECIMAL(5,2)
);

CREATE TABLE trader_mode_sessions (
session_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
employee_id UUID REFERENCES employees(id),
active_mode trader_mode NOT NULL,
shuffle_count INTEGER DEFAULT 0,
last_shuffle_at TIMESTAMPTZ,
created_at TIMESTAMPTZ DEFAULT NOW(),
expires_at TIMESTAMPTZ
);

CREATE TABLE tenant_onboarding_state (
tenant_id UUID PRIMARY KEY REFERENCES tenants(id) ON DELETE CASCADE,
current_step INTEGER NOT NULL DEFAULT 1,
step_data JSONB DEFAULT '{}',
sandbox_active BOOLEAN DEFAULT true,
completed BOOLEAN DEFAULT false,
updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE draft_history (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
entity_type TEXT NOT NULL,
entity_id UUID,
draft_snapshot JSONB NOT NULL,
edited_by UUID REFERENCES employees(id),
edited_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE tenant_lifecycle_history (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
```

```
from_state TEXT NOT NULL,
to_state TEXT NOT NULL,
reason TEXT,
governor_decision_id UUID REFERENCES governor_decisions(decision_id),
changed_by UUID REFERENCES employees(id),
changed_at TIMESTAMPTZ DEFAULT NOW()
);
```

3. CORPORATE GRAPH & ORGANISATION

text

```
CREATE TABLE entity_nodes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
gtid TEXT REFERENCES tenants(gtid),
entity_type TEXT NOT NULL,
external_id TEXT,
properties JSONB
);

CREATE TABLE relationship_edges (
from_id UUID NOT NULL REFERENCES entity_nodes(id) ON DELETE CASCADE,
to_id UUID NOT NULL REFERENCES entity_nodes(id) ON DELETE CASCADE,
relationship TEXT NOT NULL,
weight FLOAT,
properties JSONB,
PRIMARY KEY(from_id, to_id, relationship)
);

CREATE TABLE beneficial_owners (
tenant_id UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
individual_id UUID NOT NULL REFERENCES entity_nodes(id) ON DELETE CASCADE,
ownership_percentage DECIMAL(5,2),
verified_at TIMESTAMPTZ,
ubo_declaration_path TEXT,
PRIMARY KEY(tenant_id, individual_id)
);
```

```
CREATE TABLE tenant_business_units (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
parent_bu_id UUID REFERENCES tenant_business_units(id),
name TEXT NOT NULL,
administrator_employee_id UUID REFERENCES employees(id),
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE tenant_departments (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_unit_id UUID NOT NULL REFERENCES tenant_business_units(id),
name TEXT NOT NULL
);

CREATE TABLE tenant_cost_centers (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_unit_id UUID REFERENCES tenant_business_units(id),
code TEXT NOT NULL,
description TEXT
);

CREATE TABLE tenant_approval_groups (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
name TEXT NOT NULL,
employee_ids UUID[] NOT NULL
);

CREATE TABLE tenant_approval_policies (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
action TEXT NOT NULL,
condition_json JSONB NOT NULL,
required_approvals JSONB NOT NULL,
quorum INTEGER NOT NULL,
```

```
approval_mode TEXT DEFAULT 'parallel',
enabled BOOLEAN DEFAULT true
);
```

4. GOVERNANCE (GOVERNOR DECISIONS, AI INFERENCE, AUDIT)

text

```
CREATE TABLE governor_decisions (
decision_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
decision_type TEXT NOT NULL,
actor_gtid TEXT NOT NULL,
actor_employee_id UUID REFERENCES employees(id),
verdict governor_verdict NOT NULL,
conditions JSONB,
policy_version TEXT NOT NULL,
rule_refs TEXT[],
explainability TEXT,
confidence DECIMAL(5,4),
loom_hash TEXT NOT NULL,
cryptographic_signature TEXT NOT NULL,
tenant_message JSONB,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE audit_log (
id BIGSERIAL,
table_name TEXT NOT NULL,
record_id UUID NOT NULL,
action TEXT NOT NULL,
before JSONB,
after JSONB,
changed_by UUID REFERENCES employees(id),
changed_at TIMESTAMPTZ DEFAULT NOW()
) PARTITION BY RANGE (changed_at);

CREATE TABLE ai_inference_records (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

agent_name TEXT NOT NULL,
authority_level ai_authority NOT NULL,
action_context JSONB NOT NULL,
decision JSONB NOT NULL,
confidence DECIMAL(5,4),
shap_values JSONB,
explanation TEXT,
loom_hash TEXT NOT NULL,
created_at TIMESTAMPTZ DEFAULT NOW()
);

5. TRADE CORE (TRADE REQUESTS WITH STRUCTURED CONTAINER DATA)

text
CREATE EXTENSION IF NOT EXISTS vector;

CREATE TABLE trade_requests (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
buyer_tenant_id UUID NOT NULL REFERENCES tenants(id),
seller_tenant_id UUID REFERENCES tenants(id),
raw_description TEXT,
parsed_specs JSONB NOT NULL, -- structured container/commodity data
specifications JSONB,
embedding vector(1536),
parsing_confidence DECIMAL(5,4),
dual_use_flag BOOLEAN DEFAULT false,
dual_use_category TEXT,
status trade_status DEFAULT 'DRAFT',
match_results JSONB,
assigned_seller_id UUID REFERENCES tenants(id),
marketplace_auto_attributed BOOLEAN DEFAULT false,
multi_shipment_schedule JSONB, -- buyer's multi-shipment proposal (if any)
sla_deadline TIMESTAMPTZ,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
created_by UUID NOT NULL REFERENCES employees(id),
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()

```
);

CREATE INDEX idx_trade_requests_embedding ON trade_requests USING
hnsw(embedding vector_cosine_ops);

CREATE TABLE commodity_compatibility_warnings (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
trade_request_id UUID REFERENCES trade_requests(id),
conflicting_commodities TEXT[],
warning_type TEXT,
message TEXT,
overridden_by UUID REFERENCES employees(id),
override_reason TEXT
);

CREATE TABLE seller_quotes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
trade_request_id UUID NOT NULL REFERENCES trade_requests(id),
seller_tenant_id UUID NOT NULL,
exw_price DECIMAL(20,4) NOT NULL,
exw_currency TEXT NOT NULL,
exw_locked_at TIMESTAMPTZ NOT NULL,
exw_justification TEXT,
incoterm TEXT NOT NULL,
validity_days INTEGER DEFAULT 15,
status TEXT,
manual_logistics_costs JSONB, -- manual entry per incoterm
share_breakdown_with_buyer BOOLEAN DEFAULT false,
loading_country TEXT,
loading_port TEXT,
counter_schedule JSONB, -- seller’s modified multi-shipment schedule
governor_decision_id UUID,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE quote_delivery_options (
```

```
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
seller_quote_id UUID NOT NULL REFERENCES seller_quotes(id),
port_of_delivery TEXT NOT NULL,
transit_days INTEGER NOT NULL,
extra_logistics_cost DECIMAL(20,4) DEFAULT 0,
total_delivered_price DECIMAL(20,4) NOT NULL,
cost_breakdown JSONB,
is_primary BOOLEAN DEFAULT false,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE provider_quotes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
seller_quote_id UUID REFERENCES seller_quotes(id),
provider_gtid TEXT NOT NULL,
provider_type TEXT,
is_bundle BOOLEAN DEFAULT false,
bundle_breakdown JSONB,
total_cost DECIMAL(20,4),
contract_reference TEXT,
is_contract_rate BOOLEAN DEFAULT false,
route_details JSONB,
ai_benchmark_deviation DECIMAL(5,2),
bundle_score DECIMAL,
status TEXT,
governor_decision_id UUID,
submitted_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE quote_selected_services (
seller_quote_id UUID REFERENCES seller_quotes(id),
provider_quote_id UUID REFERENCES provider_quotes(id),
service_type TEXT NOT NULL,
selected_cost DECIMAL(20,4),
PRIMARY KEY(seller_quote_id, service_type)
);
```

```
CREATE TABLE negotiation_sessions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
trade_request_id UUID NOT NULL REFERENCES trade_requests(id),
messages JSONB DEFAULT '[]',
current_offer JSONB,
counter_offers JSONB DEFAULT '[]',
negotiator_config JSONB,
ai_settlement_proposal JSONB,
status TEXT DEFAULT 'ACTIVE',
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
bot_decision_log JSONB DEFAULT '[]',
started_at TIMESTAMPTZ DEFAULT NOW(),
closed_at TIMESTAMPTZ
);

CREATE TABLE negotiation_amendments (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
negotiation_session_id UUID NOT NULL REFERENCES negotiation_sessions(id),
proposed_by_gtid TEXT NOT NULL,
amendments JSONB NOT NULL,
status TEXT DEFAULT 'PENDING',
response_note TEXT,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE contracts (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
trade_request_id UUID NOT NULL REFERENCES trade_requests(id),
contract_type TEXT NOT NULL DEFAULT 'SINGLE_SHIPMENT', -- SINGLE_SHIPMENT or
MASTER_MULTI
incoterm TEXT NOT NULL,
incoterm_rules JSONB NOT NULL,
clauses JSONB NOT NULL,
multi_shipment_schedule JSONB,
governing_law TEXT NOT NULL,
```

dispute_resolution TEXT NOT NULL,

special_requests TEXT,

status contract_status DEFAULT 'DRAFT',

cryptographic_hash TEXT,

buyer_signature TEXT,

seller_signature TEXT,

signed_at TIMESTAMPTZ,

locked_at TIMESTAMPTZ,

SGTX FEES_lock_id UUID,

sgtx_fee_rate DECIMAL(5,4) NOT NULL, -- dynamic rate, paid by seller

sgtx_fee_amount DECIMAL(20,4) NOT NULL,

marketplace_partner_id UUID,

risk_simulation_result JSONB,

quality_specs JSONB DEFAULT '{}',

sla_deadline TIMESTAMPTZ,

SGTX FEES_collection_model TEXT DEFAULT 'UPFRONT', - - UPFRONT or PER_SHIPMENT

governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),

created_at TIMESTAMPTZ DEFAULT NOW(),

updated_at TIMESTAMPTZ DEFAULT NOW()

);

CREATE TABLE contract_shipments (

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

contract_id UUID NOT NULL REFERENCES contracts(id),

shipment_number INTEGER NOT NULL,

scheduled_delivery_date DATE NOT NULL,

port_of_discharge TEXT NOT NULL,

containers_count INTEGER NOT NULL,

total_price DECIMAL(20,4) NOT NULL,

sgtx_fee_amount DECIMAL(20,4), -- per-shipment fee

SGTX FEES_lock_id UUID,

ustn TEXT, -- generated when shipment locked

status TEXT DEFAULT 'SCHEDULED', -- SCHEDULED, LOCKED, SHIPPED, CANCELLED, FAILED_PAYMENT

created_at TIMESTAMPTZ DEFAULT NOW()

);

```
CREATE TABLE fee_payment_requests (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
contract_id UUID REFERENCES contracts(id),
contract_shipment_id UUID REFERENCES contract_shipments(id),
financing_agreement_id UUID REFERENCES financing_agreements(id),
party_type TEXT NOT NULL, -- 'seller' for trade fee, 'borrower' for financing fee
tenant_id UUID NOT NULL,
amount DECIMAL(20,4),
currency TEXT,
payment_method TEXT,
status TEXT DEFAULT 'PENDING',
payment_reference TEXT,
gross_amount DECIMAL(20,4),
psp_fee DECIMAL(20,4),
paid_at TIMESTAMPTZ,
governor_decision_id UUID
);

CREATE TABLE SGTX FEES_locks (
lock_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
contract_id UUID REFERENCES contracts(id),
contract_shipment_id UUID REFERENCES contract_shipments(id),
financing_agreement_id UUID REFERENCES financing_agreements(id),
trade_id UUID REFERENCES trade_requests(id),
fee_rate_pct DECIMAL(5,4) NOT NULL,
fee_usd DECIMAL(20,4) NOT NULL,
currency TEXT DEFAULT 'USD',
status SGTX FEES_lock_status DEFAULT 'ACTIVE',
release_conditions JSONB NOT NULL,
released_pct DECIMAL(5,2) DEFAULT 0,
disputed_amount DECIMAL(20,4),
frozen_amount DECIMAL(20,4),
kv_version BIGINT,
responsible_tenant_id UUID REFERENCES tenants(id),
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
```

```
locked_at TIMESTAMPTZ DEFAULT NOW(),
fully_released_at TIMESTAMPTZ
);
```

6. PACKING & CONTAINERISATION

```text
CREATE TABLE packing_plans (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
ustn TEXT, -- nullable until contract/shipment lock
plan_data JSONB NOT NULL,
loading_guide TEXT,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
locked_at TIMESTAMPTZ DEFAULT NOW()
);

ALTER TABLE packing_plans ADD CONSTRAINT fk_packing_plans_ustn
FOREIGN KEY (ustn) REFERENCES shipments(ustn) DEFERRABLE INITIALLY DEFERRED;

CREATE TABLE pallet_details (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
packing_plan_id UUID NOT NULL REFERENCES packing_plans(id),
container_id UUID REFERENCES shipments(id),
sscc TEXT UNIQUE NOT NULL,
lot_number TEXT,
product_hs_code CHAR(6),
cartons_per_pallet INTEGER NOT NULL,
gross_weight_kg DECIMAL(10,2),
status pallet_status DEFAULT 'PENDING',
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE container_loading_plans (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
packing_plan_id UUID NOT NULL REFERENCES packing_plans(id),
container_id UUID NOT NULL REFERENCES shipments(id),
pallet_ids UUID[] NOT NULL,
```

```
visual_layout JSONB,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

7. SHIPMENTS & MILESTONES

text

```
CREATE TABLE shipments (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
ustn TEXT UNIQUE NOT NULL,
contract_id UUID REFERENCES contracts(id),
contract_shipment_id UUID REFERENCES contract_shipments(id),
contract_sequence_number INTEGER DEFAULT 1,
booking_number TEXT,
loading_date DATE,
loading_time_slot TEXT,
status shipment_status DEFAULT 'CREATED',
current_milestone TEXT,
origin_port TEXT,
destination_port TEXT,
vessel_name TEXT,
imo_number TEXT,
transport_legs JSONB,
ai_predictions JSONB,
carbon_footprint_tons DECIMAL(8,4),
esg_score DECIMAL(5,2),
ais_last_position JSONB,
digital_twin_enabled BOOLEAN DEFAULT false,
shelf_life_remaining_days DECIMAL,
chosen_port_of_delivery TEXT,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE shipment_milestones (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
ustn TEXT NOT NULL REFERENCES shipments(ustn),
milestone TEXT NOT NULL,
confirmed_at TIMESTAMPTZ NOT NULL,
confirmed_by UUID REFERENCES employees(id),
confirmation_method TEXT,
ai_verdict JSONB,
evidence_hash TEXT,
scanned_barcode_id UUID,
sla_deadline TIMESTAMPTZ,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id)
);

CREATE TABLE iot_sensor_readings (
id BIGSERIAL,
ustn TEXT NOT NULL REFERENCES shipments(ustn),
sensor_type TEXT NOT NULL,
value JSONB NOT NULL,
unit TEXT,
anomaly_flag BOOLEAN DEFAULT false,
recorded_at TIMESTAMPTZ NOT NULL
);

CREATE TABLE disruption_predictions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
ustn TEXT NOT NULL REFERENCES shipments(ustn),
prediction_type TEXT NOT NULL,
probability DECIMAL(5,4) NOT NULL,
affected_route TEXT,
predicted_delay_days INTEGER,
data_sources TEXT[],
recommendation TEXT,
ai_model_version TEXT,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE provider_status_updates (
```

```
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
shipment_ustn TEXT REFERENCES shipments(ustn),
submitting_provider_gtid TEXT NOT NULL,
actual_service_provider_gtid TEXT,
milestone TEXT NOT NULL,
status_data JSONB,
confirmed_at TIMESTAMPTZ DEFAULT NOW(),
governor_decision_id UUID
);

CREATE TABLE shipment_barcodes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
shipment_ustn TEXT REFERENCES shipments(ustn),
barcode_type TEXT NOT NULL,
barcode_data TEXT NOT NULL,
pallet_number INTEGER,
sscc TEXT UNIQUE
);
```

8. DOCUMENTS & REQUIREMENTS

text

```
CREATE TABLE shipment_document_requirements (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
shipment_ustn TEXT REFERENCES shipments(ustn),
document_type TEXT NOT NULL,
responsible_party_type TEXT,
responsible_tenant_id UUID,
is_critical BOOLEAN DEFAULT false,
status TEXT DEFAULT 'PENDING',
governor_decision_id UUID
);

CREATE TABLE documents (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
requirement_id UUID REFERENCES shipment_document_requirements(id),
trade_request_id UUID REFERENCES trade_requests(id),
```

```
shipment_ustn TEXT REFERENCES shipments(ustn),
tenant_id UUID NOT NULL REFERENCES tenants(id),
uploaded_by_provider_gtid TEXT,
document_type TEXT NOT NULL,
filename TEXT,
storage_path TEXT NOT NULL,
sha256_hash TEXT NOT NULL,
version INTEGER DEFAULT 1,
previous_version_id UUID REFERENCES documents(id),
uploaded_by UUID NOT NULL REFERENCES employees(id),
metadata JSONB,
ai_extracted JSONB,
verification_status TEXT DEFAULT 'PENDING',
verified_at TIMESTAMPTZ,
governor_decision_id UUID REFERENCES governor_decisions(decision_id),
uploaded_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE generated_documents (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
shipment_ustn TEXT REFERENCES shipments(ustn),
document_type TEXT NOT NULL,
draft_data JSONB NOT NULL,
status TEXT DEFAULT 'DRAFT',
finalized_document_id UUID REFERENCES documents(id),
finalized_at TIMESTAMPTZ
);

CREATE TABLE ebl_capability_matrix (
carrier_id TEXT PRIMARY KEY,
carrier_name TEXT NOT NULL,
supported_platforms TEXT[],
supported_routes JSONB,
last_verified TIMESTAMPTZ,
is_active BOOLEAN DEFAULT true
);
```

```
CREATE TABLE shipment_ebls (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
shipment_ustn TEXT NOT NULL REFERENCES shipments(ustn),
platform TEXT NOT NULL,
platform_reference TEXT NOT NULL,
issue_date DATE,
current_holder_gtid TEXT REFERENCES tenants(gtid),
status TEXT DEFAULT 'ISSUED',
events JSONB DEFAULT '[]',
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

9. FINANCING & SETTLEMENT (CO-FINANCING, FINANCIER PREFERENCES)

text

```
CREATE TABLE financing_requests (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
contract_id UUID NOT NULL REFERENCES contracts(id),
contract_shipment_id UUID REFERENCES contract_shipments(id),
requester_tenant_id UUID NOT NULL REFERENCES tenants(id),
amount DECIMAL(20,4) NOT NULL,
currency TEXT DEFAULT 'USD',
tenor_days INTEGER NOT NULL,
financing_type TEXT NOT NULL,
preferred_settlement_method TEXT, -- 'bank_transfer', 'stablecoin', 'defi'
preferred_currency TEXT DEFAULT 'USD',
collateral JSONB,
credit_intelligence JSONB,
ai_recommended_ltv DECIMAL(5,2),
monte_carlo_default_prob DECIMAL(5,4),
status financing_status DEFAULT 'REQUESTED',
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE financier_preferences (
```

```
financier_tenant_id UUID PRIMARY KEY REFERENCES tenants(id),
accepted_borrower_countries TEXT[], -- ISO codes
min_borrower_trust_score DECIMAL(5,2),
min_trade_value DECIMAL(20,4),
max_financed_amount DECIMAL(20,4),
preferred_financing_types TEXT[],
preferred_settlement_methods TEXT[],
excluded_commodities TEXT[], -- HS chapter or text
geographic_restrictions TEXT[], -- 'accept_only' or 'all_except'
accept_only_countries TEXT[],
all_except_countries TEXT[],
updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE financing_offers (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
financing_request_id UUID NOT NULL REFERENCES financing_requests(id),
financier_tenant_id UUID NOT NULL REFERENCES tenants(id),
amount_offered DECIMAL(20,4) NOT NULL, -- portion of total
effective_apr DECIMAL(6,4) NOT NULL,
all_in_cost DECIMAL(6,4),
collateral_required JSONB,
conditions JSONB,
offer_note TEXT,
settlement_method TEXT, -- 'bank_transfer', 'stablecoin', 'defi'
bid_encrypted BOOLEAN DEFAULT true,
bid_opened_at TIMESTAMPTZ,
status TEXT DEFAULT 'SUBMITTED', -- SUBMITTED, ACCEPTED, REJECTED, COUNTERED
submitted_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE financing_agreements (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
financing_request_id UUID NOT NULL REFERENCES financing_requests(id),
master_agreement BOOLEAN DEFAULT true,
annex_number INTEGER,
```

```
financier_tenant_id UUID NOT NULL,
amount DECIMAL(20,4) NOT NULL,
apr DECIMAL(6,4) NOT NULL,
tenor_days INTEGER NOT NULL,
collateral_terms JSONB,
repayment_schedule JSONB,
sgtx_financing_fee DECIMAL(20,4) NOT NULL, -- 0.25% of amount
sgtx_financing_fee_rate DECIMAL(5,4) DEFAULT 0.0025,
SGTX FEES_lock_id UUID REFERENCES SGTX FEES_locks(lock_id),
defi_contract_address TEXT,
defi_transaction_hash TEXT,
stablecoin TEXT,
cryptographic_hash TEXT NOT NULL,
financier_signature TEXT,
borrower_signature TEXT,
governor_signature TEXT,
status TEXT DEFAULT 'PENDING_SIGNATURES',
disbursed_at TIMESTAMPTZ,
repaid_at TIMESTAMPTZ,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE financing_repayments (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
financing_agreement_id UUID NOT NULL REFERENCES financing_agreements(id),
scheduled_date DATE,
actual_paid_at TIMESTAMPTZ,
principal_paid DECIMAL(20,4),
interest_paid DECIMAL(20,4),
status TEXT DEFAULT 'PENDING', -- PENDING, PAID, LATE, DEFAULTED
late_days INTEGER,
governor_decision_id UUID
);

CREATE TABLE financier_historical_data (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
financier_tenant_id UUID NOT NULL REFERENCES tenants(id),
borrower_gtid TEXT NOT NULL,
total_financed_amount DECIMAL(20,4),
number_of_financings INTEGER,
on_time_repayments INTEGER,
late_repayments INTEGER,
defaults INTEGER,
last_updated TIMESTAMPTZ DEFAULT NOW(),
UNIQUE(financier_tenant_id, borrower_gtid)
);

CREATE TABLE defi_tranche_positions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
financing_agreement_id UUID NOT NULL REFERENCES financing_agreements(id),
protocol_id UUID,
amount_stablecoin DECIMAL(20,8) NOT NULL,
stablecoin TEXT NOT NULL,
chain TEXT DEFAULT 'POLYGON',
contract_address TEXT NOT NULL,
deposit_tx_hash TEXT,
withdrawal_tx_hash TEXT,
depeg_alert_triggered BOOLEAN DEFAULT false,
status TEXT DEFAULT 'ACTIVE',
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE settlement_instructions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
ustn TEXT NOT NULL REFERENCES shipments(ustn),
instruction_type TEXT NOT NULL, -- 'TRADE_PRINCIPAL', 'FEE_SPLIT'
payload JSONB NOT NULL,
status TEXT DEFAULT 'PENDING',
psp_reference TEXT,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

```
CREATE TABLE settlement_confirmations (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
instruction_id UUID NOT NULL REFERENCES settlement_instructions(id),
proof_hash TEXT NOT NULL,
amount_confirmed DECIMAL(20,4),
currency_confirmed TEXT,
fx_rate_applied DECIMAL(16,8),
psp_name TEXT,
webhook_received_at TIMESTAMPTZ,
verified_by_ai BOOLEAN DEFAULT false,
ai_verdict JSONB,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
confirmed_at TIMESTAMPTZ DEFAULT NOW()
);
```

10. PAYMENT ORCHESTRATION & FEE COLLECTION

text

```
CREATE TABLE payment_aggregators (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
name TEXT NOT NULL UNIQUE,
country_codes TEXT[] NOT NULL,
supported_currencies TEXT[],
supports_split BOOLEAN DEFAULT true,
api_endpoint TEXT,
credentials_encrypted JSONB,
uptime_score DECIMAL(5,4),
last_health_check TIMESTAMPTZ,
is_active BOOLEAN DEFAULT true
);

CREATE TABLE payment_attempts (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
fee_payment_request_id UUID NOT NULL REFERENCES fee_payment_requests(id),
aggregator_id UUID NOT NULL REFERENCES payment_aggregators(id),
payer_country TEXT NOT NULL,
```

```
payment_method TEXT NOT NULL,
amount_local DECIMAL(20,4),
currency_local TEXT,
amount_usd DECIMAL(20,4),
fx_rate DECIMAL(16,8),
SGTX FEES_amount_usd DECIMAL(20,4),
psp_fee_usd DECIMAL(20,4),
gross_amount DECIMAL(20,4),
net_amount DECIMAL(20,4),
status TEXT DEFAULT 'INITIATED',
failure_reason TEXT,
retry_count INTEGER DEFAULT 0,
psp_transaction_id TEXT,
split_executed_at TIMESTAMPTZ,
settled_at TIMESTAMPTZ,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE fee_calculations (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
payment_attempt_id UUID REFERENCES payment_attempts(id),
net_fee DECIMAL(20,4),
gross_amount DECIMAL(20,4),
fee_breakdown JSONB,
safety_buffer_applied BOOLEAN DEFAULT true,
governor_decision_id UUID
);

CREATE TABLE psp_health_logs (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
aggregator_id UUID NOT NULL REFERENCES payment_aggregators(id),
health_score DECIMAL(5,2),
latency_ms INTEGER,
checked_at TIMESTAMPTZ DEFAULT NOW()
);
```

11. DISTRESSED CARGO (WITH “CHECK BUYERS”, MICROCONTRACT, SPLITTING)

text

```
CREATE TABLE distressed_cargo_listings (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
original_shipment_ustn TEXT REFERENCES shipments(ustn),
micro_ustn TEXT, -- generated when split
parent_ustn TEXT, -- link to original
seller_tenant_id UUID NOT NULL REFERENCES tenants(id),
product_details JSONB NOT NULL,
quantity DECIMAL(10,2) NOT NULL,
unit TEXT NOT NULL,
current_location TEXT NOT NULL,
condition TEXT NOT NULL,
condition_score INTEGER, -- AI-assessed 0-100
price_expectation DECIMAL(20,4),
price_currency TEXT DEFAULT 'USD',
floor_price DECIMAL(20,4), -- for accelerated outreach
accept_offers_from TEXT[], -- GTIDs
willing_to_ship_onward BOOLEAN DEFAULT true,
listing_expiry TIMESTAMPTZ NOT NULL,
status TEXT DEFAULT 'ACTIVE',
ai_price_recommendation JSONB,
triage_path TEXT, -- 'SELL', 'COMPLY', 'INSURANCE'
privacy_notice_acknowledged BOOLEAN DEFAULT false,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE distressed_cargo_offers (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
listing_id UUID NOT NULL REFERENCES distressed_cargo_listings(id),
buy_tenant_id UUID NOT NULL REFERENCES tenants(id),
amount DECIMAL(20,4) NOT NULL,
currency TEXT DEFAULT 'USD',
message TEXT,
```

```
status TEXT DEFAULT 'PENDING',
ai_risk_flag JSONB,
governor_decision_id UUID NOT NULL REFERENCES governor_decisions(decision_id),
submitted_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE distressed_contact_checks (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
listing_id UUID NOT NULL REFERENCES distressed_cargo_listings(id),
seller_tenant_id UUID NOT NULL,
buyer_gtid TEXT NOT NULL,
match_score INTEGER,
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE distressed_contact_notifications (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
distressed_ustn TEXT NOT NULL,
recipient_gtid TEXT NOT NULL,
message_hash TEXT,
sent_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE micro_contracts (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
parent_contract_id UUID REFERENCES contracts(id),
distressed_listing_id UUID REFERENCES distressed_cargo_listings(id),
micro_ustn TEXT UNIQUE NOT NULL,
buyer_gtid TEXT NOT NULL,
seller_gtid TEXT NOT NULL,
agreed_price DECIMAL(20,4),
sgtx_fee_rate DECIMAL(5,4),
sgtx_fee_amount DECIMAL(20,4),
SGTX FEES_lock_id UUID,
status TEXT DEFAULT 'PENDING_SIGNATURES',
locked_at TIMESTAMPTZ,
```

```
governor_decision_id UUID,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

12. DISPUTES & EVIDENCE

text

```
CREATE TABLE disputes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
trade_request_id UUID REFERENCES trade_requests(id),
contract_shipment_id UUID REFERENCES contract_shipments(id),
financing_agreement_id UUID REFERENCES financing_agreements(id),
filing_party_gtid TEXT NOT NULL REFERENCES tenants(gtid),
dispute_type TEXT NOT NULL, -- QUALITY, DELAY, DOCUMENTATION, PAYMENT,
FINANCING_FEE, FEE_DISPUTE, OTHER
description TEXT,
evidence_links TEXT[],
status TEXT DEFAULT 'FILED',
mediation_log JSONB DEFAULT '[]',
arbitration_case_id TEXT,
resolution_contract_id UUID,
ai_settlement_proposal JSONB,
predicted_outcome JSONB,
governor_decision_id UUID NOT NULL,
filed_at TIMESTAMPTZ DEFAULT NOW(),
resolved_at TIMESTAMPTZ
);

CREATE TABLE dispute_recommendations (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
dispute_id UUID NOT NULL REFERENCES disputes(id),
ai_prediction TEXT,
suggested_settlement JSONB,
confidence DECIMAL(5,4),
shap_values JSONB,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

```
CREATE TABLE evidence_packages (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
dispute_id UUID NOT NULL REFERENCES disputes(id),
package JSONB NOT NULL,
loom_hash TEXT NOT NULL,
verification_token_hash TEXT, -- sha256 of one-time token
compiled_at TIMESTAMPTZ DEFAULT NOW()
);
```

13. LOGISTICS ADDENDA & CONTAINER RELEASE

text

```
CREATE TABLE logistics_addenda (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
contract_id UUID NOT NULL REFERENCES contracts(id),
contract_shipment_id UUID REFERENCES contract_shipments(id),
provider_gtid TEXT NOT NULL,
service_type TEXT NOT NULL,
signed_at TIMESTAMPTZ,
signature_hash TEXT,
addendum_content TEXT,
status TEXT DEFAULT 'PENDING' -- PENDING, SIGNED, REJECTED
);

CREATE TABLE release_confirmations (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
ustn TEXT NOT NULL,
SGTX FEES_lock_id UUID NOT NULL,
release_token UUID NOT NULL UNIQUE,
sent_at TIMESTAMPTZ,
acknowledged_at TIMESTAMPTZ,
acknowledged_by_gtid TEXT,
method TEXT -- EMAIL, API
);
```

14. SMART INBOX & NOTIFICATIONS

```text
CREATE TABLE inbox_items (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
employee_id UUID NOT NULL REFERENCES employees(id) ON DELETE CASCADE,
item_id UUID UNIQUE NOT NULL,
ustn TEXT REFERENCES shipments(ustn) ON DELETE SET NULL,
category TEXT NOT NULL CHECK (category IN
('NEEDS_SIGNATURE','NEEDS_APPROVAL','NEEDS_DOCUMENT','NEEDS_PAYMENT','SHIPME
NT_ALERT','NEW_OFFER','NEGOTIATION','COMPLIANCE','GENERAL')),
priority_score INTEGER NOT NULL CHECK (priority_score BETWEEN 0 AND 100),
title TEXT NOT NULL,
description TEXT,
deadline TIMESTAMPTZ,
action_link TEXT NOT NULL,
snoozed_until TIMESTAMPTZ,
dismissed BOOLEAN DEFAULT FALSE,
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE inbox_history (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
employee_id UUID NOT NULL REFERENCES employees(id) ON DELETE CASCADE,
item_id UUID NOT NULL,
ustn TEXT,
action_taken TEXT,
timestamp TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE inbox_preferences (
employee_id UUID PRIMARY KEY REFERENCES employees(id) ON DELETE CASCADE,
min_score INTEGER DEFAULT 30,
muted_categories TEXT[] DEFAULT '{}',
preferred_language TEXT DEFAULT 'en'
);
```

15.JURISDICTION, COUNTRY, PORTS

text

```
CREATE TABLE jurisdictions (
code TEXT PRIMARY KEY,
name TEXT NOT NULL,
sanctions_level TEXT NOT NULL,
regulatory_body TEXT,
last_updated TIMESTAMPTZ DEFAULT NOW(),
kyc_tier_required INTEGER DEFAULT 2,
registry_api_endpoint TEXT,
customs_api_endpoint TEXT,
cbdc_status TEXT DEFAULT 'NONE',
psp_partners JSONB,
reporting_requirements JSONB,
defi_allowed BOOLEAN DEFAULT false,
defi_restrictions JSONB,
stablecoins_allowed TEXT[],
private_financier_allowed BOOLEAN DEFAULT false,
distressed_sale_allowed TEXT, -- 'YES', 'CONDITIONAL', 'NO'
distressed_country_factor DECIMAL(5,4) DEFAULT 1.0,
distressed_SGTX FEES_factor DECIMAL(5,4) DEFAULT 1.0
);

CREATE TABLE ports (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
unlocode TEXT NOT NULL UNIQUE,
name TEXT NOT NULL,
country_code TEXT NOT NULL REFERENCES jurisdictions(code),
latitude DECIMAL(10,8),
longitude DECIMAL(11,8),
is_active BOOLEAN DEFAULT true
);
```

16. MARKETPLACE PARTNERS & ATTRIBUTION

text

```
CREATE TABLE marketplace_partners (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
partner_name TEXT NOT NULL,
api_key_hash TEXT NOT NULL,
SGTX FEES_split_percent DECIMAL(3,2),
active BOOLEAN DEFAULT true,
api_key_encrypted TEXT,
api_key_created_at TIMESTAMPTZ,
api_key_last_used TIMESTAMPTZ,
ip_whitelist TEXT[]
);

CREATE TABLE partner_rate_limits (
partner_id UUID NOT NULL REFERENCES marketplace_partners(id),
endpoint TEXT NOT NULL,
limit_per_window INTEGER NOT NULL,
window_seconds INTEGER NOT NULL,
updated_at TIMESTAMPTZ DEFAULT NOW(),
PRIMARY KEY (partner_id, endpoint)
);

CREATE TABLE partner_lead_attributions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
trade_request_id UUID REFERENCES trade_requests(id),
marketplace_partner_id UUID REFERENCES marketplace_partners(id),
attribution_type TEXT NOT NULL,
revenue_share_pct DECIMAL(5,4),
governor_decision_id UUID
);

CREATE TABLE webhook_delivery_logs (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
partner_id UUID NOT NULL REFERENCES marketplace_partners(id),
endpoint_url TEXT NOT NULL,
event_type TEXT NOT NULL,
payload JSONB,
response_status INTEGER,
response_body TEXT,
```

```
duration_ms INTEGER,
success BOOLEAN,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

17.EXPORT JOBS & TENANT ACTIVITY

text

```
CREATE TABLE tenant_export_jobs (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
requested_by UUID NOT NULL REFERENCES employees(id),
format TEXT NOT NULL CHECK (format IN ('CSV','JSON','PDF')),
date_range JSONB NOT NULL,
scope TEXT DEFAULT 'ALL',
include_docs BOOLEAN DEFAULT false,
status TEXT DEFAULT 'PENDING',
file_path TEXT,
signature TEXT,
checksum TEXT,
created_at TIMESTAMPTZ DEFAULT NOW(),
completed_at TIMESTAMPTZ
);
```

18.INDEXES & ROW-LEVEL SECURITY (EXAMPLES)

text

```
CREATE INDEX idx_shipments_ustn ON shipments(ustn);
CREATE INDEX idx_shipments_contract_id ON shipments(contract_id);
CREATE INDEX idx_shipments_status ON shipments(status);
CREATE INDEX idx_contract_shipments_contract ON contract_shipments(contract_id);
CREATE INDEX idx_contract_shipments_status ON contract_shipments(status);
CREATE INDEX idx_seller_quotes_trade ON seller_quotes(trade_request_id);
CREATE INDEX idx_quote_delivery_options_quote ON
quote_delivery_options(seller_quote_id);
CREATE INDEX idx_financing_requests_status ON financing_requests(status);
CREATE INDEX idx_financing_offers_request ON financing_offers(financing_request_id);
CREATE INDEX idx_distressed_cargo_listings_active ON distressed_cargo_listings(status)
WHERE status = 'ACTIVE';
```

CREATE INDEX idx_distressed_listings_expiry ON distressed_cargo_listings(listing_expiry);

CREATE INDEX idx_governor_decisions_created ON governor_decisions(created_at DESC);

CREATE INDEX idx_inbox_items_employee_score ON inbox_items(employee_id, priority_score DESC);

CREATE INDEX idx_inbox_items_employee_ustn ON inbox_items(employee_id, ustn);

CREATE INDEX idx_audit_log_tenant_action ON audit_log(changed_by, action, changed_at DESC);

CREATE INDEX idx_tenant_contacts_last_interaction ON tenant_contacts(last_interaction);

CREATE INDEX idx_ports_country ON ports(country_code);

CREATE INDEX idx_jurisdictions_code ON jurisdictions(code);

-- RLS examples (simplified)

ALTER TABLE trade_requests ENABLE ROW LEVEL SECURITY;

CREATE POLICY trade_requests_tenant_isolation ON trade_requests

USING (buyer_tenant_id = current_setting('app.current_tenant_id')::uuid

OR assigned_seller_id = current_setting('app.current_tenant_id')::uuid);

ALTER TABLE inbox_items ENABLE ROW LEVEL SECURITY;

CREATE POLICY inbox_items_employee_isolation ON inbox_items

USING (employee_id = current_setting('app.current_employee_id')::uuid);

END OF PART 5 – v6.3 WITH ALL MODIFICATIONS

PART 6.1 : COMMON SHARED COMPONENTS (v6.3 – FULLY EXTENDED, WITH ALL MODIFICATIONS: DUAL-MODE TOGGLE, SGTX FEE PAID BY SELLER, MULTI-SHIPMENT, FINANCING CO-FINANCING, DISTRESSED ENHANCEMENTS, ETC.)

This part defines the shared user interface components that appear across all portals (Buyer, Seller, Logistics, QC, Financier, Government, Admin, Marketplace Partner). All components are built on the zero-cost stack (Next.js 15, React, Zustand, NATS WebSocket, Tailwind CSS). They are fully integrated with the Governor decision pipeline, the three-tier AI (Groq + HF local + Ollama), and the event-driven backend.

PART 6.1.1 – SMART INBOX (ACTION-DRIVEN DASHBOARD FOR ALL ROLES) – v6.3 FULLY EXPANDED, WITH TRADER PORTAL UNIFICATION, BUYER & SELLER DASHBOARDS, THREE-TIER AI, ALL MODIFICATIONS

6.1.1.0 Purpose & Design Philosophy

The Smart Inbox is the default landing page for every authenticated user in every portal. It replaces traditional dashboards with a single, unified action feed. Every item is anchored to a specific USTN (or shipment USTN, financing agreement ID, or distressed microUSTN). Clicking any item opens the Trade Command Center (or equivalent detail view for financing/distressed). Items are prioritised by a deterministic urgency score (0–100), and each includes a plain-language title and description generated by AI (Groq, fallback Ollama). The inbox respects the dual-mode toggle – a user working within the Trader Portal sees tasks specific to their current mode: when in Buyer mode they see buyer-specific items (contract signing, customs readiness, etc.); switching to Seller mode immediately reveals seller-specific tasks (pending trade requests, EXW deadlines, carrier risk alerts, etc.). The inbox is purely advisory (A1): no irreversible action is taken from the inbox without passing through the Governor.

Core principles restated with v6.3 modifications and the Trader Portal unification:

Action-first, never information-first – users see only what requires immediate attention.

Trade-centric – items link to USTNs, financing agreements, or distressed lots.

Role-adaptive – items are derived from the user's permissions, trader mode (and active dual-mode context: Buyer or Seller), and open obligations.

Prioritised – urgency score computed deterministically (A4) using rules that incorporate deadlines, SLAs, and risk signals. AI (Groq/Ollama) generates the title and description after the score is known.

Dismissable & snoozable – users can snooze (2h, 4h, 8h, 24h) or permanently dismiss items; actions sync across devices via NATS.

Non-marketplace – never suggests unknown counterparties or unsolicited trades.

Multi-shipment aware – items for multi-shipment contracts appear per shipment when that shipment is pending action (e.g., "Confirm Shipment 2", "Pay SGTX fee for Shipment 2").

Financing aware – financiers see bids pending, margin calls, repayment reminders, and co-financing opportunities. Buyers and Sellers who are also borrowers see financing-specific tasks.

Distressed aware – Sellers see outreach responses, buyer offers; Buyers see distressed listings from saved contacts.

Trader Portal dual-mode : When the toggle is set to Buyer, the inbox shows only buyer-centric items; switching to Seller immediately refreshes and shows seller-centric items. The inbox never mixes tasks from both modes unless the user switches modes.

6.1.1.1 Functional Description

6.1.1.1.1 Inbox Item Structure

Each inbox_items record (database) has the following fields, used to render the UI:

item_id: UUID, unique identifier; used for snooze/dismiss.

ustn (or financing_agreement_id, distressed_listing_id): TEXT, primary identifier. For multi-shipment, includes contract_shipment_id.

category: ENUM – NEEDS_SIGNATURE, NEEDS_APPROVAL, NEEDS_DOCUMENT, NEEDS_PAYMENT, SHIPMENT_ALERT, NEW_OFFER, NEGOTIATION, COMPLIANCE, GENERAL. For financiers: BID_OPPORTUNITY, MARGIN_CALL, REPAYMENT_DUE. For Buyers/Sellers: additional context labels are applied internally.

priority_score: INTEGER (0-100), computed urgency.

title: TEXT, short plain-language title, e.g., “Sign contract for USTN … before 14:00 UTC”.

description: TEXT, slightly longer explanation, e.g., “The seller has already paid the SGTX fee. Your signature will lock the contract and generate the USTN.”

deadline: TIMESTAMPTZ, when action is required, adjusted for holidays/weekends per Part 6.5.12.6.

action_link: TEXT, deeplink URL to the relevant screen (e.g., /trade/USTN/contract/sign, /finance/bid/123, /distressed/listing/456).

snoozed_until: TIMESTAMPTZ, if snoozed, when it reappears.

dismissed: BOOLEAN, true if permanently dismissed (kept in history).

6.1.1.1.2 Urgency Scoring Model (Deterministic, A4)

The inbox-service computes a base score using rules and then applies role-specific modifiers. Scores are recalculated whenever the underlying state changes (e.g., deadline approaches, new offer received). The scoring rules are hot-reloadable via OPA policy snippets.

Master scoring table (baseline):

Score 100: Action required by a hard deadline <2h, missing it would irreversibly block the trade (e.g., contract lock expiry, fee payment deadline for multi-shipment shipment)
Score 95: Human approval (e.g., contract signature) required, deadline <24h
Score 90: Counteroffer from counterparty unanswered >24h
Score 85: Critical document missing, vessel ETA <72h
Score 80: Fee payment attempt failed after 3rd retry; manual intervention needed
Score 75: New quote received, validity <1 day
Score 70: Carrier risk score <50
Score 65: Cold-chain temperature excursion unacknowledged >2h
Score 60: Provider invoice discrepancy >5% unresolved >48h
Score 50: Upcoming milestone reminder (e.g., loading window starts in 24h)
Score 40: Financing: bidding window closing in <4h (for financier)
Score 35: Financing: repayment due in 7 days (for borrower)
Score 30: General trade activity (e.g., “Shipment departed”)
Score 20: Distressed: new offer received
Score 10: Completed trade, closed dispute, financed repaid (informational)

Role-specific modifiers (added to baseline):

Buyer (active dual-mode context = BUY):

Contract pending signature >48h → +10.

Customs readiness critical missing → +15.

Seller (active dual-mode context = SELL):

EXW locked >7 days and packing plan not locked → +10.

Alternative delivery port offer created → +5 (advisory).

Logistics:

RFQ open >72h with zero responses → +15.

Container gated but not picked up within 4h of window → +20.

QC:

Inspection report not submitted within 12h of agreed completion → +15.

Financier:

Bidding window <2h and no bid submitted → +10.

Margin call LTV >85% → +20.

Repayment overdue 1 day → +25.

Admin (Platform Governance):

Multisig request pending >24h → +15.

Multi-shipment specific:

For a multi-shipment contract, each pending shipment gets its own item with score based on its individual deadline. The master contract may have a separate “Remaining shipments” informational item (score 20).

Snooze & dismiss : Users can snooze an item for 2h, 4h, 8h, or 24h. The item disappears and reappears after the snooze period unless the underlying urgency score increases by ≥10 points (in which case it reappears immediately). Dismissed items move to history but are not deleted.

Cross-device sync : Snooze/dismiss actions publish a NATS event inbox.item.state_changed. All active client sessions receive the event and update local Zustand store immediately. Conflict resolution: last write wins; losing write logged in inbox_history with action_taken = ’OVERWRITTEN’.

6.1.1.1.3 Smart Inbox UI Layout (Web, Desktop, Mobile)

Web/Desktop (Next.js + Tailwind) : The inbox is a scrollable list grouped by priority band (High: 80–100, Medium: 50–79, Low: 0–49). Each priority band is collapsible.

Example screenshot (textual representation):

```
┌──────────────────────────────────────────────────────────────────────
───┐
│ [GTID Badge] [Notifications] [Dual-Mode Toggle: Buyer ● Seller] [Avatar] │
├──────────────────────────────────────────────────────────────────────
───┤
│ Smart Inbox [Customise] │
├──────────────────────────────────────────────────────────────────────
───┤
│ □ HIGH PRIORITY (2) │
│
┌─────────────────────────────────────────────────────────────────────┐
│
│ │ ⚠ USTN SGTX-EG-...-V1 -- Sign contract before 14:00 UTC (2h 15min) ││
│ │ The seller has paid the SGTX fee. Your signature will lock the trade. ││
│ │ [ Sign Contract ] [ Snooze 2h ] (Deadline: 14:00 UTC) ││
│
```

```
│ ⚠ USTN SGTX-EG-...-V2 -- Phytosanitary certificate missing -- 38h to ETA││
│ Customs clearance will be blocked. ││
│ [ Upload Certificate ] [ Request from Seller ] [ Snooze 4h ] ││

│ □ MEDIUM PRIORITY (3) │

│ □ Shipment 2 of 3 -- Confirm readiness ││
│ Multi-shipment contract #MC001 -- shipment due 15 Aug. ││
│ [ Confirm Shipment ] [ Request Changes ] ││

│ □ Financing request #FR001 -- Bidding window closes in 2h ││
│ Seller Mekong Fresh, $100,000, 60 days. Match score 94. ││
│ [ Submit Bid ] [ View Details ] ││

│ □ LOW PRIORITY (4) │

│ □ USTN SGTX-...-V3 -- Shipment departed ││
│ MSC Alessia has departed Saigon. ETA Alexandria 20 Jun. ││
│ [ Track Vessel ] [ Dismiss ] ││

│ □ AI Summary: "You have 2 high-priority items. Signing the contract for ││
│ V1 will keep your shipment on schedule. The missing certificate for V2 ││
│ must be uploaded soon to avoid customs delays." │
```

Components within the inbox:

Dual-mode toggle (visible only for CORPORATE tenants with default_trader_mode = DUAL and allow_role_switching = true). When toggled between Buyer and Seller, the inbox refreshes to show items relevant to the new mode.

Priority groups – expand/collapse. Default expanded: HIGH only; MEDIUM and LOW collapsed but can be expanded.

AI Summary Card (bottom of inbox) – generated by Groq (fallback Ollama) based on the current items, with a plain-language recommendation. The summary respects the active mode; for a Buyer it might read “Sign the contract for V1, and the missing certificate for V2 needs your attention,” while for a Seller it might read “Your carrier risk alert requires immediate reoptimisation; also respond to the buyer’s counteroffer.”

Customise button – opens a modal where the user can set a minimum priority score threshold (e.g., only show items ≥50) and mute entire categories (e.g., mute GENERAL).

Mobile (React Native) : The inbox is the first tab in the bottom navigation. Items appear as a flat list sorted by score. Swipe left reveals “Snooze” and “Dismiss”. AI summary card is a collapsible banner at the top. Offline mode: cached items stored in WatermelonDB; snooze/dismiss actions queued and synced when online.

6.1.1.1.4 AI Summary Card (Groq/Ollama, A1)

The card is updated after each inbox load or when items are added/removed. The prompt template (sent to Groq, fallback Ollama) incorporates the user’s active mode (Buyer or Seller): “The user is currently in {Buyer/Seller} mode. You have {high_count} high-priority items, {medium_count} medium, {low_count} low. High items: [{titles}]. Summarise in 2 sentences what the user should focus on today, in their preferred language. Do not mention API names or technical terms.” Example for a Buyer: “You have two high-priority items. Signing the contract for V1 will keep your shipment on schedule. The missing certificate for V2 must be uploaded soon to avoid customs delays.” If AI is unavailable, a static template is used: “You have {N} high-priority items. Please attend to them to keep your trades moving.”

6.1.1.1.5 Integration with Workflow (Phases 3, 4, 5, 7, 8, 10)

The inbox is populated by the inbox-service listening to NATS events emitted by other services:

Event: contract.status.changed (to PENDING_SIGNATURES) → Inbox Item: “Sign contract” (Buyer/Seller, whichever is required to sign)
Event: fee.payment.request.created → “Pay SGTX fee” (Seller for trade fee, borrower for financing fee, depending on role)
Event: shipment.milestone.overdue → “Milestone overdue” (responsible party)
Event: financing.rfq.broadcast → “New financing opportunity” (eligible financiers)
Event: financing.bid.submitted → “New bid received” (borrower, which could be a Buyer or Seller)
Event: distressed.listing.created (with “Check Buyers” selection) → “Distressed lot available” (selected buyers)

Event: distressed.offer.submitted → “New offer for distressed lot” (Seller)
Event: dispute.filed → “Dispute filed – review” (counterparty)
Event: document.missing.critical → “Missing document – action required” (responsible party)
Event: multi_shipment.shipment.ready → “Confirm readiness for shipment N” (Buyer or Seller as appropriate)

All event payloads include the employee_id or tenant_id to route the item correctly, respecting the dual-mode context. For the Trader Portal, if the user is in Buyer mode they see only items where they are the buyer; seller-side items are hidden until the toggle switches.

6.1.1.1.6 Governance & Privacy

The inbox is A1 advisory only – it never executes actions. All action buttons link to Governor-gated endpoints.

Inbox items are generated from data the tenant is already authorised to see (enforced by RLS and OPA).

The AI call for title/description is non-blocking; if Groq and Ollama both fail, a template-based fallback is used and logged.

All inbox activity (creation, snooze, dismiss, action taken) is logged in inbox_history (not Loom – these are UI actions, not irreversible trade decisions). However, high-sensitivity actions (e.g., approving a large fee payment) generate a separate Governor decision.

6.1.1.1.7 Configuration & Customisation

Each user can, via the [Customise] button:

Set a minimum priority score (default 0). Items below that threshold are hidden.

Mute entire categories (e.g., GENERAL, SHIPMENT_ALERT). Muted items are not shown.

Choose snooze default durations (customisable).

Reset preferences.

Preferences are stored in inbox_preferences and applied on every inbox load.

6.1.1.2 API Endpoints (used by frontend)

All endpoints are under /v1/inbox and require a valid JWT.

Method: GET /v1/inbox – Returns current list of active items for the authenticated user, sorted by score descending. Accepts query params ?min_score=50 and ?categories=NEEDS_SIGNATURE,NEEDS_DOCUMENT. The response is filtered by the active dual-mode context (Buyer or Seller) automatically.
Method: GET /v1/inbox/history – Returns last 200 items (including dismissed/snoozed) for review.
Method: POST /v1/inbox/{item_id}/snooze – Snoozes an item. Body: {"duration_minutes": 120}.
Method: POST /v1/inbox/{item_id}/dismiss – Permanently dismisses an item (still logged in history).
Method: POST /v1/inbox/preferences – Saves user preferences. Body: {"min_score": 60, "muted_categories": ["GENERAL"]}.

Response format for GET /v1/inbox (simplified):

{
"high_priority": [
{
"item_id": "...",
"ustn": "SGTX-EG-...-V1",
"category": "NEEDS_SIGNATURE",
"priority_score": 100,
"title": "Sign contract before 14:00 UTC",
"description": "Your trade with Mekong Fresh will be cancelled if not signed within 3h.",
"deadline": "2026-06-14T14:00:00Z",
"action_link": "/trade/.../contract/sign",
"snoozed_until": null,
"dismissed": false,
"created_at": "2026-06-14T10:20:00Z"
}
],
"medium_priority": [...],
"low_priority": [...],
"ai_summary": "You have 2 high-priority items..."
}

6.1.1.3 Implementation Notes for Developers

The inbox service is stateless and horizontally scalable; items are stored in PostgreSQL. NATS JetStream is used for event delivery; the service subscribes to all relevant subjects.

The urgency scoring rules are written in Rego (OPA) and can be hot-reloaded. The scoring engine runs inside the Governor proxy to avoid duplication of logic.

The AI title/description generation calls the ai-agent-orchestrator with authority A1, using Groq as primary and Ollama as fallback. The prompt includes the category, score, deadline, relevant trade data (e.g., commodity, counterparty name), and the user’s active mode (Buyer or Seller). The resulting title and description are stored with the item.

For multi-shipment contracts, each shipment gets its own item_id but may share the same ustn (until shipment-specific USTN is generated). The action_link includes the contract_shipment_id to deep-link to the correct shipment.

Dual-mode context : when a user switches between Buyer and Seller via the toggle, the frontend calls POST /v1/employee/switch-context, which returns a new JWT with updated active_trader_mode_context. The inbox subsequently refreshes with the new context, filtering items accordingly. The inbox-service receives the JWT on each request and filters items based on the permission set and active mode.

END OF PART 6.1.1 – SMART INBOX (v6.3 WITH TRADER PORTAL UNIFICATION AND ALL MODIFICATIONS)

PART 6.1.2 – TRADE COMMAND CENTER (UNIFIED PER-USTN VIEW WITH MULTI-SHIPMENT, FINANCING, DISTRESSED SUPPORT) – v6.3 FULLY EXPANDED, WITH TRADER

PORTAL UNIFICATION, BUYER & SELLER PERSPECTIVES, THREE-TIER AI, ALL MODIFICATIONS

6.1.2.0 Purpose & Design Philosophy

The Trade Command Center (TCC) is a single, real-time page that consolidates everything about a specific trade – or a specific shipment within a multi-shipment contract, or a financing agreement, or a distressed micro-contract – into one scrollable, context-rich view. It is accessible from any USTN badge, financing agreement ID, or distressed microUSTN link (from Smart Inbox, Shipments Vault, notifications, or emails). For multi-shipment contracts, the TCC can show the master contract overview and allow drilling into each shipment; for financing agreements, it shows loan details, repayment schedule, and collateral status; for distressed micro-contracts, it shows the distressed sale terms and settlement.

The TCC is fully integrated with the Trader Portal's dual-mode toggle. A user in Buyer mode sees only the buyer's perspective: commercial terms from the buyer's side, pending actions are buyer-only (e.g., approve settlement, upload import documents), and the financial card hides the seller's margin. Switching to Seller mode instantly reveals the seller's perspective: full cost breakdown (EXW, logistics, SGTX fee), estimated margin, and seller-specific pending actions (e.g., confirm shipment readiness, reoptimise logistics). The URL remains the same; the active trader mode context in the JWT determines which view is served. Financiers see a financing-focused view, and logistics providers see only their relevant services.

Core principles:

One page, one entity – all dimensions (trade, shipment, financing, distressed) available via anchored sections and collapsible panels.

Timeline-driven – horizontal timeline showing the entity's journey from initiation to completion. For buyers, the timeline highlights phases from trade request to delivery and settlement; for sellers, from acceptance to settlement.

Action-oriented – prominent pending action panel with a clear call-to-action button, always passing through the Governor.

Privacy-respecting visibility – displayed data filtered by the logged-in tenant's permissions, trader mode context (Buyer or Seller), and financing roles.

Multi-entity aware – supports trade, financing, distressed, and multi-shipment sub-entities.

All data is fetched from a single endpoint (/v1/trade/{ustn}/command-center, or similarly for financing and distressed), which returns aggregated, permission-filtered JSON.

6.1.2.1 Layout & Sections (for a Trade / Shipment)

The TCC is organised into several anchored sections, all on a single page. The exact sections shown vary based on the entity type (trade, financing, distressed) and the user's active mode (Buyer, Seller, Financier, Logistics). Below is the full layout for a standard trade contract, with notes on how each section adapts to different contexts.

A. Persistent Header

Large USTN badge, clickable to copy to clipboard. For multi-shipment, shows the specific shipment's USTN with a label like "Shipment 2 of 3". For financing, shows the financing agreement ID; for distressed, the microUSTN.

Trade Reference ID (contract ID, financing agreement ID, or distressed micro-contract ID).

Global Status Badge (e.g., LOCKED, IN_EXECUTION, DELIVERED, ACTIVE_MASTER for financing, DISTRESS_SALE). Colour-coded: green for completed, amber for in progress, red for blocked/disputed.

Quick Actions : Share Trade Room (generates role-based access link), Export Summary (triggers PDF export of the TCC view), Open in Digital Twin (if available).

Dual-mode notice : For Trader Portal users with DUAL mode, a small text near the status badge indicates the current perspective: "Viewing as Buyer" or "Viewing as Seller". Clicking the text toggles the mode instantly.

B. Trade Progress Timeline (Horizontal)
A horizontal, colour-coded bar displays the classic trade phases: Initiated → Pending Seller Response → Quoted → Negotiating → Contract Signed (fee paid by seller) → In Execution → Delivered → Settled. Each phase is an icon with a label; completed phases are filled with a checkmark, the current phase is highlighted, and future phases are greyed out. For a buyer, the phase "Pending Seller Response" is highlighted until the seller accepts; for a seller, the same phase is "Awaiting Your Response" until they accept. For multi-shipment, each shipment has its own mini-timeline below the master timeline. Clicking a shipment's timeline segment opens that shipment's detailed view within the TCC. For financing: Requested → Bidding → Awarded → Active → Repaid. For distressed: Declared → Assessed → Listed → Outreach → Offer Accepted → Microcontract Locked → Completed.

C. Pending Action Panel
This is the most visually prominent element, positioned directly beneath the timeline. It is a coloured card (amber for standard actions, red for urgent deadlines) that displays:

The single next action the user must take (e.g., "Sign contract for USTN … before 14:00 UTC", "Upload phytosanitary certificate", "Submit your bid for financing request #FR001", "Confirm Shipment 2").

A "Why this matters" sentence generated by Groq, contextualised to the user's role. For a buyer: "Your signature will lock the contract and generate the USTN. The shipment is on a tight schedule." For a seller: "Paying the SGTX fee will activate the contract and allow the buyer to proceed. Your funds will be released only after you confirm the loading."

One or two prominent action buttons (e.g., "Sign Contract", "Pay Fee", "Confirm Shipment") that lead to the relevant form or workflow.

Snooze/Dismiss options, synced with the Smart Inbox item that triggered this action. Snoozing here also snoozes the corresponding inbox item.

For multi-shipment, the panel may show "Confirm Shipment X" for the buyer, or "Pay fee for Shipment X" for the seller, depending on whose action is next.

D. Summary Dashboard (Grid of Cards)
A responsive grid (2-3 columns) of expandable cards, each representing a key dimension

of the trade. Cards are greyed out if the user lacks permission to view them. The content is tailored to the active mode.

D1. Commercial Terms Card

Incoterm (e.g., CFR), along with a tooltip explaining who pays what.

Trade value breakdown.

For a Buyer : Total quoted price (e.g., 30,240),withalineitem“(includesSGTXplatformfeeof30,240),withalineitem“(includesSGTXplatformfeeof360 paid by seller)”. The buyer sees the final amount they must pay to the seller. No further cost details.

For a Seller : EXW price locked, logistics costs (trucking, freight, customs, etc. as per incoterm), SGTX platform fee (added to total), and finally the total quoted price to buyer. Beneath these, the seller sees the estimated margin (total price minus all costs minus fee) highlighted in green or amber.

For multi-shipment, the card shows a summary of prices per shipment, and the seller sees a total margin across all shipments.

D2. Parties Card

Buyer and seller GTIDs, legal names, jurisdictions, trust scores (colour-coded). Each name is clickable to open the contact’s 360° Trust Portrait (if they are in the user’s saved contacts).

For a financier viewing a financing agreement, this card shows the borrower and, if applicable, the counterparty (buyer/seller) details.

Dual-mode aware : A buyer sees the seller’s trust portrait; a seller sees the buyer’s trust portrait.

D3. Shipment Status Card

Vessel name, IMO number, container count, current milestone (e.g., “Departed Saigon, ETA Alexandria 20 Jun”).

A mini-map (Leaflet) showing vessel position (if AIS data available) with a dotted route line.

For multi-shipment, a table of shipments with their individual statuses, ports, and ETAs. Each shipment row is clickable to drill into that shipment’s details.

Cold-chain status : if IoT sensors are active, a temperature gauge shows current vs. set point; any excursion flagged with a red warning.

D4. Documents Card

Number of required, verified, and missing documents.

Expandable checklist of all required documents (phytosanitary certificate, commercial invoice, packing list, etc.). Each document shows status (□ verified, □ pending, □ missing critical) and who is responsible.

For the buyer, documents that the buyer must provide (e.g., import permit) are highlighted. For the seller, export-side documents are highlighted.

Direct upload buttons for the responsible party; request buttons to send a pre-filled message to the counterparty.

D5. Finances Card

Trade principal settlement status : for a buyer, shows the amount owed and whether payment has been authorised; for a seller, shows the amount expected and confirmation of receipt.

SGTX fee status : for both buyer and seller, shows “Paid by seller” with a checkmark. For the seller, the exact fee amount and the date of payment are displayed.

Financing (if any) : for the borrower (buyer or seller), shows financed amount, outstanding balance, next repayment date. For financiers, shows their portion of the loan and collateral status.

For multi-shipment, each shipment’s settlement and fee status is listed separately.

D6. Risk & Compliance Card

Overall trade risk score (0-100) with colour coding.

Jurisdiction alerts (e.g., “Seller is from a high-risk jurisdiction; enhanced KYC completed”).

Sanctions flags (green “Cleared” or red “Blocked”).

Any outstanding compliance items (e.g., missing ESG documentation, CBAM pending).

For the seller, also shows carrier risk scores and any logistical red flags.

D7. Logistics Card

Lists the selected logistics providers (ocean carrier, trucking, customs broker) with their names, service types, and risk scores.

For the seller, provides a “Reoptimise” button to rerun the logistics bundle optimiser if carrier risks have changed.

For the buyer, the card is read-only and shows only the scheduled vessel and estimated transit times.

Container release status : once the SGTX fee is paid and the release confirmed, a green badge “Container Release Confirmed” appears. The logistics provider’s acknowledgment status is shown.

D8. QC Card (if applicable)

Inspection status, inspector name, report link.

AI findings summary : “2 defects found, 1 inspector-overridden”. Clicking shows the detail with the inspector’s mandatory reason.

Overridden AI findings are flagged with an orange badge to draw attention.

D9. Multi-shipment Schedule Card (only for multi-shipment contracts)

A table of all shipments in the schedule with columns : Shipment number, scheduled delivery date, port of discharge, number of containers, total price, SGTX fee, status (SCHEDULED, LOCKED, etc.).

Action buttons per shipment : for buyer, “Confirm Shipment”; for seller, “Pay Fee” or “Mark Ready”. The buttons only appear when the action is due.

A progress bar showing how many shipments have been completed out of the total.

D10. Distressed Information Card (if any part of the trade is distressed)

Shows the distressed microUSTN, affected pallets, condition score, listing price, and outreach status.

For the seller, links to the full distressed triage dashboard. For the buyer, shows offers made or the option to make an offer if the listing was sent to them.

E. Activity Feed (Vertical Timeline)
A vertically scrolling list, chronologically ordered, of every significant event in the entity's lifecycle. Events are drawn from the trade_event_timeline table and include:

"2026-06-10 08:30 – Buyer created trade request"

"2026-06-10 09:15 – Seller accepted request"

"2026-06-11 14:20 – Seller submitted quote (30,240,includingSGTXfee30,240,includingSGTXfee360)"

"2026-06-12 10:00 – Buyer accepted quote"

"2026-06-12 10:05 – Contract signed by seller (SGTX fee paid)"

"2026-06-12 10:06 – Contract locked, USTN generated"

"2026-06-13 08:00 – Loading window confirmed"

"2026-06-14 06:30 – All pallets scanned, container loaded"

etc.
Each entry includes a timestamp, the actor (employee name and tenant), and a link to the associated document or Governor decision. The feed respects permissions: it only shows events the user is authorised to see. For a buyer, commercial cost details are hidden; for a seller, the full breakdown is shown.

F. Collaborative Trade Room (Expandable Panel)
A collapsible panel at the bottom of the TCC that provides real-time chat and a shared task board. Built on NATS for chat and Yjs for the task board, it respects data scopes. For multi-shipment contracts, the trade room is per contract, but messages can be tagged with a shipment number. For financing, a separate "Financier-Borrower" chat is available. The trade room for a buyer and seller shows each other's messages; a logistics provider who is a party to the trade sees only the logistics-related channel. The room includes:

A message input with file attachment support.

Task cards (e.g., "Upload packing list", "Confirm loading date") that can be created and assigned to parties.

An AI assistant that can answer questions about the trade and suggest actions.

6.1.2.1.2 Mobile Adaptation
On screens narrower than 768px, the TCC layout adapts:

The horizontal timeline becomes a vertical step indicator showing only the current phase and next phase.

The Pending Action Panel becomes a full-width banner at the top; swipe to dismiss.

The Summary Cards become a horizontally swipeable carousel.

The Activity Feed is a vertically stacked list with infinite scroll.

The Trade Room is a slide-up bottom sheet.

6.1.2.1.3 Backend Integration

The TCC is powered by the /v1/trade/{ustn}/command-center endpoint (or similar for financing and distressed). This endpoint returns a JSON object containing:

ustn, trade_status, phase_progress (timeline nodes)

pending_action object (action, deadline, why-it-matters, action_link, derived from the Smart Inbox and Governor)

summary_cards object (each key either populated with card data or null if the user lacks permission, respecting dual-mode context)

activity_feed (paginated list of events)

trade_room (chat history and task board state for permitted channels)

permissions_map (compact representation of visible sections)

The frontend subscribes to NATS topics (trade.{ustn}.>) for real-time updates. When an event arrives, the tenant-experience-service sends a partial update via WebSocket, and the frontend patches the relevant panel without a full page reload.

6.1.2.1.4 Cross-Portal Availability & Dual-Mode Awareness

Every portal type (Trader Portal in Buyer or Seller mode, Logistics, QC, Financier, Government, Admin, Marketplace Partner) renders the same TCC component. The single endpoint returns data differently based on the authenticated tenant, employee permissions, and active dual-mode context.

Buyer mode (active = BUY) : Shows commercial terms from the buyer's perspective. Hides seller's margin. Pending actions are buyer's responsibilities (e.g., approve settlement, upload import documents). The "Finances" card shows the total payable amount, not the cost breakdown.

Seller mode (active = SELL) : Shows full cost breakdown (EXW, logistics, SGTX fee) and estimated margin. Pending actions are seller's responsibilities (e.g., confirm shipment readiness, reoptimise logistics). The seller can see the buyer's trust portrait but not the buyer's internal financials.

Financier mode : Shows financing-specific cards (loan details, collateral, repayment schedule). Trade commercial terms are presented in a condensed form.

Logistics provider : Shows only the services they were contracted for, milestones they must confirm, and documents they must upload. No price or margin information.

Government : Shows risk scores, compliance status, and documentation relevant to their jurisdiction.

6.1.2.2 Governance & Decision Panel Integration

If the TCC detects a blocked action (e.g., a user clicks a button but the Governor returns DENY or CONDITIONAL), the PlainLanguage Governor Decision Panel (see 6.1.3) slides open over the TCC, explaining the reason and providing a remediation link. Users are never left wondering why a button is disabled. The panel's content is also stored in the

TCC's response as part of the pending_action object, allowing the frontend to pre-render the explanation.

6.1.2.3 Example Workflow – Buyer Viewing a Multi-Shipment Contract

Buyer mode active : TCC shows master contract overview with a table of 3 shipments. Shipment 1 is LOCKED (fee paid by seller, USTN generated, in execution). Shipment 2 is SCHEDULED with a pending action: "Confirm Shipment 2". Shipment 3 is SCHEDULED with no pending action yet. The pending action panel highlights "Confirm Shipment 2 – deadline 15 Aug". The buyer clicks "Confirm" → Governor checks that the seller has also confirmed readiness. If not, Decision Panel explains: "The seller has not yet marked Shipment 2 as ready. You will be notified when they do."

Same user switches to Seller mode via the header toggle. The TCC instantly reloads with seller perspective. Now the pending action panel shows "Mark Shipment 2 as Ready" and "Pay SGTX fee for Shipment 2" (if not yet paid). The Finances card now shows the seller's margin estimate. The user clicks "Mark Ready" – Governor validates, and the buyer receives an inbox item.

END OF PART 6.1.2 – TRADE COMMAND CENTER (v6.3 WITH TRADER PORTAL UNIFICATION AND ALL MODIFICATIONS)

END OF PART 6.1.2 – TRADE COMMAND CENTER (v6.3 WITH ALL MODIFICATIONS)

PART 6.1.3 – PLAINLANGUAGE GOVERNOR DECISION PANEL (v6.3 – FULLY EXTENDED, WITH ALL MODIFICATIONS: DUAL-MODE TOGGLE, SGTX FEE PAID BY SELLER, MULTI-SHIPMENT, FINANCING, DISTRESSED, ETC.)

PART 6.1.3 – PLAINLANGUAGE GOVERNOR DECISION PANEL – v6.3 FULLY EXPANDED, WITH TRADER PORTAL UNIFICATION, BUYER & SELLER CONTEXTS, THREE-TIER AI, AND ALL MODIFICATIONS

6.1.3.0 Purpose & Design Philosophy

Every Governor verdict that is not ALLOW must be translated instantly into a clear, human-readable message that explains why the action was blocked and what the tenant must do next. The PlainLanguage Governor Decision Panel is the universal, zero-jargon interface between the platform's constitutional enforcement layer and the human user. It replaces the opaque "DENY" or "CONDITIONAL" codes with plain language, structured conditions, and direct call-to-action buttons.

The panel appears automatically whenever a tenant attempts an irreversible action and the Governor returns DENY or CONDITIONAL. It can also be opened on demand via a "What's blocking me?" button found in the Trade Command Center, Smart Inbox, and any workflow screen. For the Trader Portal, the panel is fully aware of the user's current mode (Buyer or Seller). If a user in Buyer mode attempts an action reserved for the Seller, the panel will clearly state that they are in Buyer mode and offer a single-click switch to Seller mode before retrying the action. This eliminates confusion for dual-mode tenants and reduces support tickets.

Core principles reinforced in v6.3:

Zero Jargon – The tenant never sees “Governor”, “OPA”, “WasmEdge”, “Loom”, “A2”. Phrases like “Your trade is on hold because…” or “An additional verification is required…” are used.

Explain, then guide – Three parts: a plain-language summary of what happened, a bullet list of specific unmet conditions each with a status icon and a clickable action, and a clear statement of what happens next once all conditions are met.

Transparent authority – The panel may indicate that a decision was made by “automated compliance rules” or that “human reviewer required”, without revealing AI authority levels.

Realtime updates – If conditions are later resolved (e.g., a document is uploaded), the panel automatically refreshes via NATS events and updates the checklist in real time.

Multi-entity aware – For multi-shipment contracts, the panel is per-shipment; for financing agreements, it covers financing-specific blocks; for distressed cargo, it covers listing or offer blocks.

Dual-mode context – If the user’s active trader mode (Buyer/Seller) is the reason for the block, the panel will inform the user and offer a toggle switch directly inside the panel.

6.1.3.1 Functional Description

6.1.3.1.1 Trigger Points

Automatic on block : Every frontend action that calls a Governor-gated API endpoint is wrapped in a handler. If the response contains a governor_decision_id and the verdict is not ALLOW, the Decision Panel is rendered as a slide-over or modal, depending on the platform (web, desktop, mobile). It is not dismissible until the user acknowledges it by either resolving the conditions or requesting human review.

“What’s blocking me?” button: Available in the Trade Command Center (Risk & Compliance card) when a trade is in a stuck state (e.g., CONDITIONAL verdict unresolved), in the Smart Inbox on any item with a governor_decision_id, and on any workflow screen where a “Next” or “Submit” button is disabled. Clicking it fetches the pending decisions for that entity via the /v1/decisions/{trade_id}/pending endpoint and opens the panel with the most recent blocking decision.

Background unresolved conditions : If a CONDITIONAL verdict has been issued but the tenant has not yet resolved all conditions, the panel can be pinned as a slim banner at the top of the portal, showing a countdown until auto-escalation. The banner can be clicked to reopen the full panel.

6.1.3.1.2 Panel Structure

The panel is divided into five distinct zones, each designed to progressively inform and guide:

A. Header

The header instantly communicates the type of block and the affected entity. It contains:

A colour-coded status badge : “Action Blocked” (red for DENY), “Additional Steps Required” (amber for CONDITIONAL), or “Human Review Needed” (blue for ESCALATE).

The specific action that was attempted, expressed in user language : e.g., “Contract Signing”, “Milestone Confirmation”, “Financing Bid Submission”, “Distressed Listing Publication”.

The entity identifier (e.g., USTN, financing agreement ID, distressed microUSTN) displayed as a clickable badge that, when clicked, opens the relevant Trade Command Center or command-center view in a new tab.

For multi-shipment : the header includes the shipment number and contract ID, e.g., “Action Blocked: Confirm Shipment 2 of MC001”.

For dual-mode context errors : the header may read “Wrong Dashboard Mode”, with the badge in amber.

B. PlainLanguage Explanation

This is the core narrative, a 2-3 sentence summary stored in the governor_decisions.tenant_message JSONB. It is generated by the Tenant Message Generator (A1) at the moment the decision is created. The explanation is personalised to the user’s role and active mode. Examples:

For a DENY due to sanctions on the seller’s jurisdiction : “This trade cannot proceed. The seller’s country (Yemen) is on our autoblock list. SGTX cannot facilitate trades with entities from fully sanctioned jurisdictions. If you believe this is an error, you may request a human review.”

For a CONDITIONAL requiring additional KYC : “Your contract lock has been paused because the seller’s jurisdiction requires enhanced due-diligence documentation. Our compliance system paused the process to protect both parties from regulatory risk. Once the documents below are provided and verified, the trade will automatically advance.”

For a dual-mode mismatch : “You are currently in Buyer mode. This action (submitting a quote) can only be performed from the Seller dashboard. Switch to Seller mode to continue.”

For a multi-shipment per-shipment fee not paid : “Shipment 2 cannot be locked because the SGTX platform fee for this shipment has not been paid. The fee of $360 must be paid by the seller before the shipment can proceed. Please navigate to the fee payment screen to complete this step.”

The explanation never includes raw rule IDs, OPA policies, or confidence percentages unless the user explicitly expands the evidence section.

C. Condition Checklist

This is the actionable core of the panel. It displays a bullet list of specific unmet conditions drawn from the conditions JSONB array and the structured tenant_message.conditions array. Each condition is presented as a card row containing:

A status icon : red cross (□ Unresolved) or green check (□ Resolved).

A human-readable label, e.g., “Upload a valid business registration certificate from the seller”, “Provide a letter of credit from a recognized bank”, “Switch to Seller mode”.

For each condition that can be resolved by the user, a direct action button or link: “Upload Now”, “Contact Seller”, “Switch Mode”, “Pay Fee”. These buttons deeplink to the exact form or worfklow step.

For conditions that must be resolved by another party (e.g., seller KYC), a button to send a pre-filled reminder message or to open the Trade Room for direct communication.

If the condition is informational (e.g., "Sanctions screening passed"), it is shown as resolved.

The checklist is ordered with the user's own pending actions first, then counterparty actions, then completed items. As conditions are resolved, the panel updates in real time via NATS events (if the user remains on the page) or on next page load.

D. Next Step & Escalation

A dedicated area below the checklist explains what will happen after all conditions are met:

"Once these documents are verified (usually within 2 hours), your contract will automatically advance to the signing stage."

"If you believe this decision is incorrect, you can request a human review. A compliance officer will respond within 24 hours."

For dual-mode blocks : "After switching to Seller mode, you will be able to submit your quote immediately."

The "Request Human Review" button triggers an A3 escalation. When clicked:

A new governor_decision entry is created with verdict = ESCALATE and decision_type = 'human_review_request'.

The Admin Portal is notified, and the compliance officer (or designated human reviewer) receives a high-priority Smart Inbox item.

The user sees a confirmation : "Your review request has been submitted. You will receive a notification when a human reviewer has responded."

E. Resolution Timer & Escalation Fallback

If a CONDITIONAL verdict remains unresolved for more than 12 hours, a countdown timer appears at the bottom of the panel: "This hold will be automatically escalated to a human reviewer in 23h 12m if not resolved." When the timer expires, the workflow-recovery-service auto-escalates to A3, creating a new Governor decision and notifying both the Admin Portal and the user.

F. Confidence & Evidence (v6.3 – Enhanced)

If the decision involved an AI model (A2 or A3), the panel displays a badge indicating the confidence level (e.g., "High confidence (94%)") and a toggle to expand an "Evidence sources" section. This section lists, in plain language, the specific data that influenced the decision. For example:

"Sanctions list update from OFAC (2026-06-10) – confidence 98%"

"News article about regulatory change in seller's jurisdiction (2026-06-09) – confidence 78%"

"Carrier risk score drop from 85 to 41 due to labour dispute report – confidence 92%"

The evidence is generated by Groq from the AI inference's SHAP values and metadata, stored in the ai_inference_records.explanation field.

6.1.3.1.3 UI Wireframe (Textual Representation)

```
┌──────────────────────────────────────────────────────────────────────
───┐
│ ⚠ Action Blocked: Contract Signing │
│ USTN: SGTX-EG-2026...-V1 │
├──────────────────────────────────────────────────────────────────────
───┤
│ Why this happened: │
│ The seller's jurisdiction (Yemen) requires enhanced due-diligence │
│ documentation. Our compliance system paused the process to protect both │
│ parties from regulatory risk. │
│ │
│ What's needed: │
│ □ Upload a valid business registration certificate from the seller. │
│ [Upload Now] │
│ □ Provide a letter of credit from a recognized bank. │
│ [Contact Seller] │
│ □ Buyer sanctions check passed. │
│ │
│ Confidence: High (94%) │
│ □ Show evidence ▼ │
│ • UN Sanctions list update (2026-06-10) – confidence 98% │
│ • Seller jurisdiction risk model – confidence 91% │
│ │
│ What happens next: │
│ Once these documents are verified (usually within 2 hours), your contract │
│ will automatically proceed. │
│ │
│ If you believe this decision is incorrect: │
│ [Request Human Review] – a compliance officer will respond within 24 hours. │
│ │
│ This hold will auto-escalate in 23h 12m if unresolved. │
└──────────────────────────────────────────────────────────────────────
───┘
```

6.1.3.2 Backend Integration

Governor Modifications

The Governor service already stores the tenant_message JSONB in the governor_decisions table. The message is generated synchronously but via a non-blocking call to the AI orchestrator, ensuring the decision is persisted even if the AI is temporarily unavailable. The fallback chain is:

Primary : Groq (A1) receives the decision context (action, conditions, jurisdiction, stake-holders, and active dual-mode context) and returns a structured JSON with summary, conditions array, escalation_hint, and optional confidence/evidence.

Fallback: Ollama/LocalAI (same prompt, local inference).

Ultimate fallback : A static English template is used, which includes the decision type, a generic “Additional verification required” summary, and a note: “This message is not available in your preferred language. Please contact support if you need translation assistance.”

The tenant_message is stored alongside the decision and linked to the decision_id. The message is generated in the user’s preferred_language (from employees table), defaulting to English if the language is not supported by the AI provider at invocation time.

API Endpoints:

GET /v1/decisions/{decision_id}/explanation – Returns the full tenant_message for a specific Governor decision, including the condition checklist and evidence. This is called when the user clicks “What’s blocking me?” or when the panel opens.

GET /v1/decisions/{trade_id}/pending – Returns all pending Governor decisions (verdicts not ALLOW) for a given trade, financing agreement, or distressed entity. The panel can list multiple blockages.

POST /v1/decisions/{decision_id}/escalate – Triggers an A3 escalation for the decision, creating a new decision with verdict ESCALATE and decision_type ‘human_review_request’. The body contains an optional reason from the user.

Event Integration:

When a CONDITIONAL verdict is created, the Governor publishes a governor.condition.changed event on NATS with the trade_id and decision_id. The frontend listens for this event and updates the checklist in real time if a condition is resolved by another party. For example, if the seller uploads the missing KYC document, the condition for the buyer’s contract lock automatically flips from “□ Unresolved” to “□ Resolved”, and the panel may auto-close if all conditions are met.

6.1.3.3 Accessibility & Multi-Language

Messages are generated in the tenant’s preferred language (Arabic, Vietnamese, German, English). If the AI cannot produce the language, English is used with a note.

The panel is fully keyboard-navigable : Tab moves between condition rows and action buttons; Enter activates the primary action; Escape closes the panel (if dismissible).

WAI-ARIA roles are applied : the panel is role=”alertdialog” when blocking an action, and live regions announce dynamic updates to screen readers.

Colour is never the sole indicator; status badges also include text and icons.

6.1.3.4 Integration with Multi-Shipment, Financing, Distressed

Multi-shipment : The panel can appear for a specific shipment, e.g., “Cannot lock Shipment 2 because SGTX fee not paid”. The header includes the shipment number and contract ID. Conditions are shipment-specific. Escalation may be to the Platform Governance Authority if the master contract is at risk.

Financing : For a CONDITIONAL verdict when a financier submits a bid (e.g., “DeFi protocol risk score too low”), the panel explains the protocol risk, shows the current score, and suggests alternative protocols. The "Request Human Review" button may escalate to the financier's own compliance team or to the Platform Governance Authority.

Distressed : When a seller attempts to list a distressed lot with an unrealistic price (>200% of AI band), the panel explains: "Your asking price is significantly above the AI-recommended range based on condition and shelf life. Provide a justification to proceed." An inline text field for the justification is shown directly in the panel, and the seller can submit it without leaving the panel.

6.1.3.5 Governance Gate (G1U11)

This component implements the user-facing aspect of the gate : all blocked actions display the panel, and users cannot proceed until conditions are resolved or they request human review. The Governor ensures that the tenant_message is always populated and that any DENY/CONDITIONAL verdict is immediately accompanied by an explanation.

END OF PART 6.1.3 – PLAINLANGUAGE GOVERNOR DECISION PANEL (v6.3 WITH TRADER PORTAL UNIFICATION AND ALL MODIFICATIONS)

PART 6.1.4 – TENANT DATA EXPORT & RECONCILIATION STATEMENTS – v6.3 FULLY EXPANDED, WITH TRADER PORTAL UNIFICATION, BUYER & SELLER PERSPECTIVES, MULTI-SHIPMENT, FINANCING, DISTRESSED, SGTX FEE TRANSPARENCY, CRYPTOGRAPHIC SIGNING, AND ALL MODIFICATIONS

6.1.4.0 Purpose & Design Philosophy

Every tenant must be able to download their complete trade history, financing records, distressed transactions, and financial events in standard, machine-readable formats. This serves accounting reconciliation, regulatory data portability (GDPR, PDPL, etc.), and internal auditing. Exported data is cryptographically signed by the platform for authenticity and integrity.

In the unified Trader Portal, the data export is accessible from both the Buyer dashboard and the Seller dashboard under the Company Admin section. The export logic is the same, but the default filters adapt to the tenant's active mode: when in Buyer mode, the export pre-selects trades where the tenant acted as buyer (buyer); when in Seller mode, it pre-selects trades where the tenant acted as seller (seller). Tenants can override this and export all trades regardless of role. For multi-shipment contracts, exports include per-shipment details; for financing, they include loan disbursements and repayments; for distressed, they include micro-contracts and sale proceeds. SGTX fee transparency is maintained: the trade fee (paid by seller) and the financing fee (deducted from the borrower) are clearly separated in the statements.

Core principles:

Self-service – Tenants generate exports; no support ticket needed.

Cryptographic integrity – Signed with platform's Ed25519 key, includes SHA256 checksum, and a QR code for external verification.

Format choice – CSV (spreadsheets), JSON (API consumption), PDF (formal statements).

Scope control – Filter by date range, USTN, counterparty, trade type (buy/sell), or export all.

Multi-entity support – Trade, financing, distressed, fee payments.

Currency normalisation – All amounts converted to a tenant-selectable base currency using daily ECB rates (with fallback to OpenExchangeRates free tier).

SGTX fee transparency – Separate line item for SGTX fees (trade fee paid by seller, financing fee deducted from borrower). Statements show both the fee amount and the net received/paid.

Dual-mode aware – The export interface respects the active mode (Buyer/Seller) for default filtering, but the user can change.

6.1.4.1 “Export My Data” Button & User Interface

Location : Company Admin → Data & Privacy tab (also accessible from Finance section for monthly statements). Visible to employees with tenant.export permission (typically tenant_admin and finance_user). The button is labelled “Export My Data”.

Dialog options:

Date range : calendar picker with “All time”, “Last 30 days”, “Last 90 days”, “Custom”. Max 12 months per export for performance; for longer periods, the system suggests splitting into multiple exports.

Format : CSV (UTF-8 with BOM), JSON (UTF-8), PDF (formatted statement).

Scope : “All trades”, “Buyer side only” (trades where tenant is buyer), “Seller side only” (trades where tenant is seller), “Financing agreements”, “Distressed transactions”, “Fee payments only”. The default selection is set based on the active dual-mode context: if the user is in Buyer mode, “Buyer side only” is pre-selected; if Seller mode, “Seller side only”.

Include documents checkbox (optional, with a warning about potentially large file size; when checked, ZIP will contain all related documents).

Base currency selection : dropdown with USD, EUR, tenant’s local currency (detected from jurisdiction). Default is the tenant’s saved preference from Company Admin → Finance Settings.

Submit : “Request Export” button creates an export job and returns a job_id.

Processing : The tenant-data-export-service (Rust, Axum) aggregates data from PostgreSQL and ClickHouse, respecting row-level security. It applies currency conversion using ECB daily rates (cached daily) and the OpenExchangeRates free tier as fallback. For PDF generation, it uses Tera templates rendered with headless Chrome to produce a professional layout. The generated file is stored temporarily in encrypted storage, and a record is inserted into tenant_export_jobs with the signed checksum.

Notification : When the job completes, a Smart Inbox item is created (category GENERAL, priority 30) with a direct download link. The link is a presigned URL valid for 24 hours. Tenants can view past export jobs in a table within the Data Export tab, with columns: Job ID, Requested Date, Format, Date Range, Status (PENDING, COMPLETED, FAILED), Download (if completed). Clicking Download retrieves the ZIP.

Cryptographic signature (ZIP package) : The downloadable ZIP contains:

trade_data.csv (or .json or .pdf) according to the selected format.

signature.sig – Ed25519 signature of the data file, generated with the platform's private key.

checksum.sha256 – plaintext SHA256 hash of the data file.

README.txt – instructions on how to verify the signature and checksum using the SGTX public key (published at /v1/platform/public-key). The README also includes a verification URL: https://verify.sgtx.io where users can drag-and-drop the ZIP to verify.

6.1.4.2 Export Format Specifications (v6.3 – Enhanced)

CSV Schema (trade_data.csv) – extended for multi-shipment, financing, distressed, fees, and Buyer/Seller context:

Columns (in order):

USTN / Entity ID

Entity Type (Trade, Financing, Distressed, Fee)

Shipment Number (for multi-shipment, empty for single or non-shipment entities)

Date (event date : contract lock, settlement, disbursement, distressed sale)

Role (BUYER or SELLER – indicates whether the tenant was buyer or seller for trades, borrower for financing)

Counterparty GTID

Counterparty Legal Name

Incoterm (for trades)

Commodity (HS code and description)

Quantity (with unit)

EXW Value (USD, shown only for seller side; for buyer side this may be omitted or shown as "—")

Logistics Cost (USD, for seller side)

SGTX Trade Fee (USD, paid by seller)

SGTX Fee Payer (always "Seller" for trade fee)

Financed Amount (USD, if any)

Financing Fee (USD, 0.25% of financed amount)

Disbursement Received (USD, net amount borrower received after fee deduction)

Repayment Made (USD, total repaid including interest)

Distressed Sale Amount (USD)

Net Settlement (USD, final amount received or paid)

Status (SETTLED, ACTIVE, etc.)

Original Amount (in original currency if different from base)

Original Currency

FX Rate (ECB rate used for conversion)

Base Currency (as selected by tenant)

All monetary values are in the chosen base currency unless indicated. For multi-shipment, each shipment appears as a separate row, with the master USTN and shipment number.

JSON Schema (export.json):

The JSON export is structured as a nested object to allow programmatic consumption:


```
{
"export_metadata": {
"tenant_gtid": "SGTX-VN-TRD-002",
"tenant_legal_name": "Mekong Fresh Co.",
"exported_at": "2026-06-15T10:30:00Z",
"date_range": {"from": "2026-01-01", "to": "2026-06-01"},
"base_currency": "USD",
"role_filter": "SELLER",
"signature": "ed25519:base64...",
"checksum_sha256": "a1b2c3..."
},
"trades": [
{
"ustn": "SGTX-EG-...-V1",
"role": "SELLER",
"counterparty_gtid": "SGTX-EG-TRD-001",
"counterparty_name": "Nile Foods",
"incoterm": "CFR",
"commodity": {"hs_code": "0805.10", "description": "Valencia Oranges"},
"quantity": {"value": 20000, "unit": "kg"},
"exw_value_usd": 28000.00,
"logistics_cost_usd": 2000.00,
"sgtx_trade_fee_usd": 360.00,
"sgtx_fee_payer": "SELLER",
"settlement": {
"net_received_usd": 30360.00,
"status": "SETTLED",
"settled_at": "2026-06-15"
},
"shipments": [
{
"shipment_number": 1,
"ustn": "SGTX-EG-...-V1-S1",
"contracted_price_usd": 30360.00,
"sgtx_fee_usd": 360.00,
"status": "DELIVERED"
}
]
}
],
```

"financing_agreements": [
{
"agreement_id": "FA-001",
"role": "BORROWER",
"financed_amount_usd": 20000.00,
"financing_fee_usd": 50.00,
"net_disbursed_usd": 19950.00,
"apr": 5.2,
"tenor_days": 60,
"repayments": [
{"date": "2026-07-15", "principal_usd": 10000.00, "interest_usd": 260.00, "status":
"PAID"}
]
}
],
"distressed_transactions": [
{
"micro_ustn": "SGTX-...-D1",
"role": "SELLER",
"sale_amount_usd": 900.00,
"sgtx_fee_usd": 9.27,
"fee_payer": "SELLER",
"buyer_gtid": "SGTX-EG-TRD-003",
"buyer_name": "Juice Factory Ltd.",
"date": "2026-06-20"
}
],
"fee_summary": {
"total_sgtx_fees_paid": 419.27,
"breakdown": {
"trade_fees": 360.00,
"financing_fees": 50.00,
"distressed_fees": 9.27
}
}
}

The JSON schema explicitly includes the role field (BUYER or SELLER) for each transaction, enabling tenants to differentiate their buying and selling activities.

PDF Layout (Monthly Reconciliation Statement):

The PDF statement is designed as a formal, printable document with these sections:

Header : SGTX logo, tenant legal name, GTID, statement period (month/year), base currency, and a “Role” indicator (e.g., “Viewing as: Seller”).

Trade Summary Table : Columns – Date, Entity Type, USTN / ID, Description (including counterparty and commodity), Original Amount (in original currency), Original Currency, Base Amount (converted), FX Rate. Rows are grouped by trade, with multi-shipment shipments indented.

Financing Summary Table : Columns – Date, Agreement ID, Financed Amount, Fee Deducted (0.25%), Net Received, Repayments (Principal + Interest).

Distressed Sales Table : Columns – Date, MicroUSTN, Original USTN, Buyer, Sale Amount, SGTX Fee.

Fee Summary Box : “Total SGTX fees paid in [Month]: X(Trade:X(Trade:Y, Financing: Z,Distressed:Z,Distressed:W)”.

Footer : “This statement is cryptographically signed. Verify at verify.sgtx.io” and a QR code linking to the verification page for this specific statement. The last page includes the Ed25519 signature, SHA256 checksum, and verifying instructions.

6.1.4.3 Monthly Reconciliation Statement (Finance Section)

The Finance section in Company Admin contains a “Monthly Statements” tab, visible to employees with finance.view permission. The tab displays an interactive web view of the current month’s statement, with options to switch months and download as PDF. The web view is built with the same data as the PDF, using charts for fee breakdown and currency distribution.

Example web view (simplified):

Tenant : Mekong Fresh Co. (SGTX-VN-TRD-002) – Base currency: USD – Viewing as: SELLER – June 2026

Date	Entity Type	USTN / ID	Description	Original Amount	Original Currency	Base Amount	FX Rate
2026-06-10	Trade Fee	SGTX-EG-...-V1	SGTX trade fee paid (seller)	360.00	USD	360.00	1.0000
2026-06-15	Settlement	SGTX-EG-...-V1	Settlement received from buyer (Nile Foods)	47,950.30	USD	47,950.30	1.0000
2026-06-20	Financing	FIN-001	Loan disbursement (net after fee)	19,950.00	USD	19,950.00	1.0000
2026-06-20	Financing Fee	FIN-001	SGTX financing fee deducted (0.25%)	50.00	USD	50.00	1.0000
2026-06-25	Distressed	DST-001	Distressed sale proceeds (lemons, microcont	900.00	USD	900.00	1.0000

			ract)				
2026-06-25	Distressed Fee	DST-001	SGTX distressed fee paid (seller)	9.27	USD	9.27	1.0000

Net Cash Flow : 68,381.03TotalSGTXfeespaidinJune:68,381.03TotalSGTXfeespaidinJune:419.27 (Trade: 360.00,Financing:360.00,Financing:50.00, Distressed: $9.27)

If the user had switched to Buyer mode, the statement would instead show buyer-side transactions (payments made to sellers, import-side fees if any, etc.). The role filter is prominent and can be changed interactively.

6.1.4.4 Currency Normalisation for Multi-Currency Statements

Display currency choice : The tenant selects a base currency in Company Admin → Finance Settings → Statement Currency. Options: USD, EUR, GBP, tenant’s local currency (auto-detected). This selection applies to all employees within the tenant.

Exchange rate source : ECB reference rates (free daily XML feed, fetched at 16:00 CET) are used as the primary source. For currencies not covered by ECB, the system falls back to OpenExchangeRates free tier (limited to 1,000 requests/month, sufficient for cached daily rates). If both fail, the statement shows “Rate unavailable – manual conversion required” for that line item, and the tenant can enter a manual rate via the web interface, which is stored per tenant per currency pair and used for future calculations.

On-statement presentation : Each transaction line shows two amount columns – Original Amount (in the currency of the transaction) and Base Amount (converted to the chosen base currency). For transactions already in the base currency, the Original Amount column may be omitted or show the same value with a note “(base currency)”. The FX rate column displays the rate applied.

Multi-hop conversions : If a transaction involved a currency pair not directly available (e.g., VND to USD), the system uses the ECB rate for VND→EUR and then EUR→USD, displaying the composite rate in the FX column.

6.1.4.5 Accounting Webhook (for ERP Integration)

Tenants wanting real-time ERP integration can register a webhook endpoint. The “Integrations” tab under Company Admin allows an admin to set up to 5 webhooks. Configuration includes:

URL (HTTPS required).

Events to subscribe : settlement.completed, fee.paid, financing.disbursed, distressed.sale.completed.

Secret for HMAC-SHA256 signature (auto-generated, shown once).

Optional: include documents (boolean).

When a matching event occurs, the tenant-data-export-service POSTs a JSON payload (similar to the export JSON for that single entity) to the registered URL. The payload includes an X-SGTX-Signature header. The service expects a 200 OK response within 10 seconds; it retries up to 3 times with exponential backoff. Delivery logs are visible in the Integrations tab. This enables direct synchronization with accounting systems.

6.1.4.6 Backend Implementation & Endpoints

Endpoints:

POST /v1/tenant/export – Creates an export job. Request body: {“format”: “CSV”, “date_range”: {“from”: “2026-01-01”, “to”: “2026-06-01”}, “scope”: “SELLER”, “include_documents”: false, “base_currency”: “USD”}. Returns {“job_id”: “...”}. Scope can be “ALL”, “BUYER”, “SELLER”, “FINANCING”, “DISTRESSED”, “FEES”.

GET /v1/tenant/export/{job_id} – Returns job status (PENDING, PROCESSING, COMPLETED, FAILED) and, if completed, a presigned download URL (valid 24h), checksum, signature, and expiry.

GET /v1/tenant/statement?month=2026-06&format=pdf&role=SELLER – Returns the monthly reconciliation statement as JSON (default) or PDF. The role parameter filters to buyer or seller transactions; if omitted, shows all.

POST /v1/tenant/webhook/register – Registers an accounting webhook.

GET /v1/tenant/webhook/logs – Retrieves delivery logs for the last 30 days.

Database table (tenant_export_jobs) : stores job metadata, file path (in encrypted object store), checksum, signature, status, created_at, completed_at.

Security : Export endpoints are governed by OPA and RLS. Only data belonging to the authenticated tenant is ever included. The platform’s Ed25519 private key, used for signing exports, is stored in SoftHSMv2 and accessed only by the tenant-data-export-service.

6.1.4.7 Worked Example – Seller Downloads Monthly Statement with Multi-Currency and Financing

Context : Mekong Fresh (Vietnam, base currency = USD) has one trade (USD), one financing disbursement (USD), and one distressed sale (USD) in June. The finance manager, while in the Seller dashboard, goes to Company Admin → Monthly Statements. The statement shows all three lines, with role “SELLER” annotated. The manager downloads the PDF, verifies the signature via the public key, and sends it to the external auditor.

If the same manager switched to Buyer mode, the statement would be empty (no buy-side transactions in that month). The manager can change the role filter to “ALL” to see everything.

END OF PART 6.1.4 – TENANT DATA EXPORT & RECONCILIATION STATEMENTS (v6.3 WITH TRADER PORTAL UNIFICATION AND ALL MODIFICATIONS)

END OF PART 6.1.4 – TENANT DATA EXPORT & RECONCILIATION STATEMENTS (v6.3 WITH ALL MODIFICATIONS)

PART 6.1.5 – [REMOVED: OPERATING MODE FEATURE MATRIX]

Note : The Tenant Operating Modes (SIMPLE / ADVANCED / ENTERPRISE) have been removed from the blueprint as per earlier modifications. Therefore, Part 6.1.5

(original “Operating Mode Feature Matrix”) is omitted. All tenants now have uniform access to all features, with visibility and permissions controlled solely by trader_mode (BUY/SELL/DUAL), dual-mode toggle context, data scopes, and RBAC roles. No progressive feature disclosure based on “modes” exists. The numbering of subsequent parts is adjusted accordingly.

PART 6.1.6 – SHARED SHIPMENTS VAULT (COMMON COMPONENT FOR ALL PORTALS) – v6.3 FULLY EXPANDED, WITH TRADER PORTAL UNIFICATION, BUYER & SELLER MODES, MULTI-SHIPMENT SUPPORT, THREE-TIER AI, AND ALL MODIFICATIONS

6.1.6.0 Purpose & Design Philosophy

The Shared Shipments Vault is a universal, reusable component that displays a paginated, filterable, sortable table of shipments – identified by USTN – across every SGTX portal that requires shipment visibility. It is not a standalone page; it is embedded into the Buyer (now Buyer) and Seller (now Seller) dashboards of the Trader Portal, as well as into the Logistics, QC, Financier, Government, and Admin portals. The same codebase renders the vault differently based on the authenticated user’s tenant type, permissions, data scopes, and, critically for the Trader Portal, the active dual-mode context (Buyer or Seller).

Design principle : One component, many configurations. Developers build the vault once as a React component (with a companion Rust aggregation service), and each portal instantiates it with a configuration object that determines which columns are visible, what default filters apply, which row actions are available, and which additional slide-out tabs are rendered. The vault is fully multi-shipment aware: for contracts with multiple shipments, each shipment appears as a separate row, with the master USTN shown as a parent reference, and columns can be drilled into. The vault is also real-time: when a milestone is confirmed, a document is uploaded, or a vessel position changes, the corresponding row updates without a full page reload, thanks to NATS WebSocket subscriptions.

For the Trader Portal, the vault replaces the former separate “Inbound” and “Outbound” tables. When the dual-mode toggle is set to Buyer, the vault automatically filters to shipments where the tenant is the buyer, and shows buyer-specific columns like “Customs Readiness” and “Documents Missing”. When set to Seller, it filters to outbound shipments and shows seller-specific columns like “Estimated Margin” and “Logistics Quotes Status”. Users can override the automatic filter and view all shipments regardless of role, but the default respects the active mode. The toggle instantly switches the vault’s context without navigating to a different URL.

6.1.6.1 Core UI Layout – Fully Expanded

The Shared Shipments Vault occupies the full width of its parent container, which is typically the main content area of the Trader Portal (in either Buyer or Seller mode), the Logistics Portal, the QC Portal, the Financier Portal, the Government Portal, or the Admin Portal. Its layout is divided into four distinct horizontal bands, vertically stacked, plus a slide-out panel that overlays from the right edge. All UI elements are rendered using Tailwind CSS, Radix UI primitives, and custom React components from the shared component library. The vault is designed to be the primary working surface for logistics coordinators, trade managers, finance users, and government officials who need to monitor and act upon tens or hundreds of shipments simultaneously.

Global State Bar

At the very top of the vault, a thin horizontal bar displays summary statistics derived from the currently active filters. This bar is not interactive but provides at-a-glance context. It shows, from left to right:

Total shipments in view (e.g., “Showing 48 shipments”).

Active mode indicator – for the Trader Portal, it displays “Buyer View” or “Seller View” in a small pill that matches the colour of the dual-mode toggle. For other portals, it shows the portal name (e.g., “Logistics View – Freight Forwarder”).

Quick summary of pending actions – e.g., “3 require document upload”, “2 quotes expiring soon”. These are derived from the Smart Inbox items linked to the visible shipments and are computed via a lightweight API call that runs in parallel with the main data fetch.

A “Last updated” timestamp with a subtle pulsing dot when the NATS WebSocket connection is live, or a static clock when in polling fallback.

Active / Archive Toggle

Immediately below the global state bar, a segmented control labelled “Active” and “Archived” acts as the primary view filter. This toggle is present in all portals but has slightly different semantics per role:

In the Buyer dashboard, Active means shipments where the status is one of CREATED, GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED, or CUSTOMS_IMPORT. Archived includes DELIVERED, COMPLETED, CANCELLED, and DISPUTED.

In the Seller dashboard, Active includes the same statuses as Buyer minus CUSTOMS_IMPORT (since that is an import step) and plus any shipment awaiting seller action. Archived is the same.

In Logistics and QC portals, Active means any job not yet fully completed; Archived means the job is closed (delivered, cancelled, or dispute resolved).

In the Financier portal, Active means loans that are outstanding; Archived means repaid or defaulted.

The toggle is implemented as two large buttons side by side, with the active selection highlighted in the brand primary colour and a badge count inside. For example, “Active (3)” and “Archived (12)”. Clicking “Archived” triggers a new API call with the archived flag, and the table is replaced with the filtered results. The user’s choice is persisted in local state and also, if the user is logged in across multiple devices, broadcast via a NATS user-preference event so that other sessions reflect the same choice. The toggle can be overridden by deep-link parameters (e.g., ?archive=true) for shared URLs.

Toolbar

The main toolbar sits below the Active/Archive toggle and spans the full width. It is a flex container (wrapping on smaller screens) that houses all filtering, search, and view-mode controls. It is designed to be dense but clear, with every element having a visible label or tooltip.

Search Bar : A text input with a magnifying glass icon on the left. As the user types, a debounce of 300 milliseconds triggers a server-side full-text search. The search examines USTN fields, container numbers, vessel names, buyer and seller legal names, and, for multi-shipment contracts, the master contract USTN. The search is implemented on the backend using PostgreSQL tsvector columns, with a ranking function that prioritises exact USTN matches. Pressing Enter or clicking the search icon immediately executes the search (bypassing the debounce). A small “x” button on the right clears the search and restores the previous filter set.

Filter Chips : A horizontal row of compact, multi-select dropdown buttons, each representing a filter dimension. The chips automatically collapse into a “More Filters” dropdown on mobile widths.

Status Chip : Opens a panel with checkboxes grouped into Pre-shipment, In Transit, and Post-shipment, each with the relevant status values. The user can select any combination. A “Select All” / “Clear All” link is provided at the bottom. The chip label reflects the number of selected statuses (e.g., “Status (3)”).

Commodity Chip : Opens a searchable tree list of HS chapters (2-digit) and their sub-headings (4-digit). Selecting a chapter automatically includes all its sub-headings. The label shows the HS code if a specific one is selected, or “All Commodities” otherwise.

Port Chip : Opens two sub-inputs – one for origin port and one for destination port. Each provides an autocomplete field that queries the global ports database (UN/LOCODE). The chip label is “Ports” unless both are filled, in which case it shows e.g., “SGN → ALEX”.

Risk Score Chip (Government and Admin only) : Opens a dual-handle range slider from 0 to 100, with preset buttons for “Low (0-25)”, “Medium (26-70)”, “High (71-100)”. The current range is displayed on the chip.

Date Chip : Opens a quick-select panel with preset date ranges (“Last 7 days”, “Last 30 days”, “Next 7 days”) and a custom date range picker using the Radix Calendar component. The selected range is reflected on the chip as “Jun 1 – Jun 15” or the preset name.

Saved Filters Chip : A dropdown listing all the user’s saved filter sets. Selecting one applies all the saved criteria and updates the search bar, chips, and sort order. A “Save current filters” button at the bottom of the dropdown opens a modal to name the set.

All filter chips are additive; applying a new filter does not remove the previous ones. The active filters are visually indicated by a small dot on the chip and a count badge. Clicking the “x” on a chip removes that specific filter. A “Clear All Filters” link appears at the end of the chip list when any filter is active.

View Toggle : Two icon buttons, “Table” and “Map”, placed on the rightmost side of the toolbar. The Table view is the default and shows the shipment grid. The Map view replaces the table with a full-height Leaflet map that displays vessel positions for shipments with status IN (DEPARTED, IN_TRANSIT, ARRIVED). The map is only enabled when the portal’s configuration permits it (config.mapEnabled = true). When the map is active, the table is hidden, and a “Reset Map View” button appears to re-centre the map on all visible vessels.

Export Button : A download icon with the label “Export”. Clicking it triggers a POST request to /v1/shipments-vault/export with the current filter parameters. A small toast notification appears while the CSV is being generated (for small exports, instant; for large, a job ID is returned). When ready, the browser triggers a download of the CSV file. The user is not navigated away from the vault.

Refresh Button : A circular arrow icon. When clicked, it forces a full re-fetch of the shipment data from the API, bypassing any client-side cache. The button spins during the request. The timestamp in the global state bar updates after the refresh. This is useful when the user wants to ensure they are seeing the absolute latest state, for example after a bulk upload.

Main Data Table

The table occupies the central portion of the vault and is built using a custom React virtualised table component (based on @tanstack/react-table) to handle thousands of rows smoothly. It features:

Column Headers : Each column header is a clickable button that cycles through three states: default sort (unsorted), ascending (arrow up), and descending (arrow down). The currently sorted column is highlighted with a subtle background colour. The sort order is communicated to the backend via the sort and order query parameters. Certain columns (e.g., “Quick Actions”) are not sortable.

Column Reordering and Resizing : Users can drag a column header to a new position; the new order is saved in user_preferences and persists across sessions. Columns can also be resized by dragging the right edge of the header; a minimum width is enforced per column type.

Row Selection : Each row has a checkbox at the far left. The checkbox in the header selects or deselects all currently loaded rows. Selected rows are highlighted in a light blue. A bulk action toolbar appears above the table when at least one row is selected, offering actions applicable to multiple shipments: “Batch Document Request”, “Assign to User” (for logistics), “Export Selected”, “Add to Watchlist”. The specific actions depend on the user’s role and permissions.

Row Hover and Click : Hovering over a row changes its background to a slightly darker shade. Clicking anywhere on the row (except on action buttons or checkboxes) opens the slide-out detail panel for that shipment. The clicked row is then marked as “active” with a left border indicator.

Infinite Scroll / Pagination : By default, the table uses traditional pagination with page numbers and a rows-per-page selector at the bottom. However, for users who prefer continuous scrolling, there is a “Load more” button at the bottom of each page, and an optional setting (in the Customise menu) to enable infinite scroll, where the next page is automatically appended as the user scrolls down.

Empty State : When no shipments match the current filters, the table area displays a friendly illustration and text: “No shipments found. Try adjusting your filters or checking the Archived view.” A “Clear All Filters” button is provided directly in the empty state.

Slide-Out Detail Panel

The slide-out panel is triggered by clicking a shipment row or by clicking a “Details” icon button in the “Quick Actions” column. It slides in from the right edge over the vault, pushing the remaining content to the left slightly on desktop, or taking over the full screen on mobile.

The panel has a fixed header with:

The USTN (or shipment identifier) as a large, clickable badge that opens the full Trade Command Center in a new browser tab.

The current status badge.

Quick summary: counterparty name, commodity, incoterm, ETA.

Close button (X) and a “Pop out” icon that opens the current panel content as a modal for detached viewing.

The body of the panel is divided into four tabs:

Summary Tab : Displays the most important information in a slim card layout: parties, ports, vessel, container list (with scan status per pallet), and the pending action panel for that specific shipment. If the shipment is part of a multi-shipment contract, a mini-progress bar shows the status of all shipments.

Documents Tab : A checklist of all required documents, each with a status icon and, for the responsible party, an “Upload” button. For documents that have been uploaded, a “View” link opens the document in a PDF viewer or image lightbox. AI verification results are shown with confidence badges. A “Request All Missing” button sends a batch message to the counterparty.

Containers Tab : A visual representation of the container loading plan (if the packing plan is locked). It shows a simplified 3D or 2D block diagram of pallet positions, with each pallet coloured by scan status (scanned, pending, damaged). Clicking a pallet shows its SSCC, product, lot, and weight. For multi-shipment, a dropdown selects the specific container within the shipment.

Timeline Tab : A vertical stepper listing every milestone, from trade initiation to the current state. Each step shows the timestamp, the confirmer, the confirmation method (barcode, voice, manual), and a link to the Governor decision. This tab is scrollable and can contain dozens of entries for long-running shipments.

The panel can be closed by clicking the X, clicking outside the panel (on desktop), or pressing the Escape key.

Map View (When Enabled)

When the view toggle is set to “Map”, the table area is replaced by a Leaflet map that fills the available height. The map tiles are served from a self-hosted OpenStreetMap tile server, ensuring zero cost.

Vessel Markers : Each visible shipment that has a known vessel position (from AIS data) is displayed as a ship icon rotated according to the vessel’s heading. Clicking the marker shows a tooltip with the USTN, vessel name, current speed, ETA, and distance to destination. A line is drawn from the vessel’s current position to the destination port, dashed if the vessel is not yet departed, solid if underway.

Port Markers : For shipments where the vessel is at port or not yet assigned, a port marker is shown with the USTN and status.

Clustering : When multiple vessels are in close proximity, the markers cluster into a numbered circle, which expands on click.

The map view respects the same filters as the table view; only shipments matching the current filter criteria are displayed on the map. The map is not available for the Archived view since archived shipments are no longer in transit.

Pagination Controls

At the bottom of the table (or below the map), a pagination bar provides:

“Rows per page” dropdown: 10, 25, 50, 100.

Page number buttons, with ellipsis for large page counts.

“Previous” and “Next” buttons.

A text summary: “Showing 1-25 of 187 shipments”.

For infinite scroll mode, the pagination bar is replaced by a “Loading more shipments…” spinner that appears as the user scrolls near the bottom.

Save and Load Filter Sets

A small star icon next to the “Saved Filters” dropdown allows users to save the current combination of search term, filter chips, sort column, and sort order as a named set. A modal asks for a name and an optional description. Saved filter sets are immediately available in the dropdown and can be deleted or renamed from a “Manage Filters” sub-panel. These sets are synchronised across the user’s sessions via a NATS user-preference event, so a set created on the desktop app is available on the mobile app. The platform does not limit the number of saved sets.

Responsive Adaptation

On viewports narrower than 768 pixels, the layout adapts significantly:

The global state bar condenses to just the total count and a single-line active mode indicator.

The toolbar wraps into two lines : search bar on the first line, filter chips and view toggle on the second. Chips that overflow are placed inside a “More” dropdown.

The table transforms into a card list. Each card represents one shipment and shows a stack of the most important fields: USTN, status, vessel, ETA, and a “Details” expandable section that reveals the additional columns. Row actions become swipeable (left to reveal actions).

The slide-out panel takes over the full screen and becomes a modal, with a back button at the top to return to the card list.

The map view is replaced by a simplified map occupying the top half of the screen, with the card list below.

All interactive elements meet minimum touch target sizes of 44×44 pixels. The vault is tested on real mobile devices using BrowserStack or similar, ensuring that truck drivers and warehouse staff can quickly check shipment statuses on their phones.

Dual-Mode Awareness in the Vault

For the Trader Portal specifically, the vault's toolbar includes a small, persistent indicator of the current mode: "Buyer View" or "Seller View", rendered as a tag next to the Active/Archive toggle. This indicator is not a control itself but reinforces which context the user is in. When the user flips the dual-mode toggle in the global header, the vault performs a full refresh: it calls the /v1/shipments-vault endpoint with the updated mode parameter (BUYER or SELLER), resets the filters to the appropriate defaults, and re-renders with the correct columns. This transition is animated with a quick fade to indicate the context change. The slide-out panel, if open for a row that is no longer valid in the new mode, automatically closes.

END OF SUB-SECTION 6.1.6.1 – CORE UI LAYOUT

6.1.6.2 Common Columns (Available to All Portals) – Fully Expanded

The Shared Shipments Vault defines a baseline set of columns that appear in every instantiation of the component, regardless of whether the user is a Buyer, a Seller, a logistics provider, a QC inspector, a financier, a government official, or a platform administrator. These columns provide the essential identifying, tracking, and temporal information that any stakeholder needs to locate and understand a shipment at a glance. Each column is rendered as a dedicated React component that adapts its display to the available data, the user's permissions, the active dual-mode context (Buyer or Seller), and the real-time event stream. The columns described below are universally available; portal-specific columns are defined in the per-portal configuration sections (6.1.6.3.1 through 6.1.6.3.7) and are simply added to this common set.

USTN / Shipment ID

This is the primary identifier for every row and the most prominent column. The column displays the full USTN string, which for single-shipment trades follows the standard SGTX-{BUYER_SUFFIX}-{SELLER_SUFFIX}-{TIMESTAMP}-{RANDOM8} format. For a Buyer viewing an inbound shipment, the buyer suffix is their own company's GTID suffix, which is subtly highlighted in the brand colour. For a Seller viewing an outbound shipment, the seller suffix receives the same treatment. The USTN is rendered as a clickable badge with a slight shadow and the SGTX primary colour border. Clicking the badge navigates the user to the Trade Command Center for that specific trade, opening it in the same tab by default, or in a new tab if the Ctrl or Command key is held.

For multi-shipment master contracts, each shipment appears as a separate row. In these rows, the USTN column shows the shipment-specific USTN (e.g., SGTX-EG-...-V1-S2) with a superscript label reading "Shipment 2". Directly below the shipment USTN, in a smaller font and prefixed by a subtle chain icon, a secondary link labeled "Master: SGTX-EG-...-V1" is displayed. Clicking this secondary link opens the Trade Command Center view of the master contract, which displays the full schedule and aggregated information. The way this master link behaves is dual-mode aware: when the user is in Buyer mode, the master contract TCC shows the buyer's perspective; when in Seller mode, it shows the seller's perspective.

The USTN column also includes a small copy-to-clipboard icon that appears on hover. Clicking the icon copies the full USTN (or shipment USTN) to the system clipboard and briefly displays a “Copied!” tooltip. For government auditors and administrators, an additional shield icon is shown when the shipment’s milestones have been blockchain-anchored (the optional high-integrity mode that anchors Loom hashes on Polygon). Clicking this shield opens a small modal that displays the Polygon transaction hash and a link to the block explorer, providing an immutable external proof of the shipment’s milestone chain. The USTN column header is sortable; clicking it sorts by the USTN string alphabetically. For multi-shipment rows, the sort takes the master USTN into account, so all shipments belonging to the same master contract naturally group together when sorted ascending or descending.

Status

The status column displays the current lifecycle stage of the shipment as a colour-coded pill badge. The standard status values and their associated colours are: CREATED (slate grey), GATED_IN (sky blue), LOADED (teal), DEPARTED (indigo), IN_TRANSIT (light blue), ARRIVED (amber), CUSTOMS_IMPORT (violet), DELIVERED (emerald green), DISPUTED (red with a warning triangle), DISTRESSED (dark red with a broken container icon), and CANCELLED (dark grey with a horizontal line through the text). Each badge also includes a small icon to the left of the text—for example, a truck for DEPARTED, a ship for IN_TRANSIT, a warehouse for ARRIVED—ensuring that colour is not the sole differentiator and that the column meets WCAG 2.2 Level AA requirements.

Hovering over or focusing on the status badge opens a rich tooltip that provides significantly more context than the label alone. The tooltip contains: the exact UTC timestamp of the last status change; the party who confirmed the last milestone (e.g., “Confirmed by Fast Trucking Co., Ahmed S., 14 Jun 2026 08:42 UTC”); and a one-sentence, plain-language explanation of the current stage, generated by the AI (Groq primary, Ollama fallback) in the user’s preferred language. For example, a shipment with status CUSTOMS_IMPORT might show the tooltip: “Arrived at Alexandria port. Customs clearance initiated. Phytosanitary certificate verified; import permit is still pending from the buyer’s broker.” A shipment with status IN_TRANSIT might show: “MSC Alessia is south of Crete, traveling at 18 knots. On schedule. ETA Alexandria 20 Jun 06:00 UTC. No cold-chain anomalies detected.”

The status column is deeply integrated with the real-time event system. When the shipment’s status changes—a milestone is confirmed, a document is verified, a vessel departs—a NATS event (shipment.status.changed.{ustn}) is published. The vault’s client-side NATS listener receives this event and updates the badge in the table instantly. The update is accompanied by a brief pulse animation on the badge to draw the user’s attention. If the user has the table filtered by status and the new status no longer matches the active filter, the row fades out smoothly over approximately half a second, providing a visual cue that the shipment has moved out of the current view.

For the Trader Portal, the status column has slightly different visual weight depending on the active mode. In Buyer mode, the CUSTOMS_IMPORT status badge is rendered with a subtle pulsing border, indicating that the buyer likely has an action to take—uploading an import permit, paying customs duties, or arranging last-mile delivery. In Seller mode, the

DEPARTED and IN_TRANSIT statuses have a slightly more prominent appearance because the seller's remaining actions during sea transit are minimal, but they still need to monitor progress and be alerted to any delays. The status column header is sortable; the sort order uses a predefined ordinal mapping that reflects the chronological progression of a trade (CREATED first, DELIVERED last, with DISPUTED and CANCELLED placed at the very end). This ensures that a sort by status always presents shipments in a logical, lifecycle-ordered manner.

Vessel / Booking

This column displays the identity of the transport. Its content changes dynamically based on the shipment's stage. When a booking has been confirmed and a vessel is assigned, the column shows the vessel name (e.g., "MSC Alessia") in bold, with the IMO number displayed in a smaller, lighter font directly beneath. A small ship icon precedes the vessel name, and if the vessel is currently at sea and AIS data is available, the icon is rendered in blue and the vessel name is a clickable link that opens a mini-map overlay showing the vessel's current position, heading, speed, and the last AIS update timestamp.

If a booking is confirmed but the specific vessel has not yet been named or assigned (common in the early stages), the column shows the booking reference number prefixed by a document icon. A tooltip explains: "Booking reference #XYZ12345. Vessel assignment expected within 24 hours." If neither booking nor vessel information is available—for example, for a shipment that has been created but not yet accepted by the seller—the column displays a dash and the tooltip reads "No booking data yet."

For multi-shipment contracts, the vessel and booking information is specific to each shipment row. If two shipments in the same master contract are carried on different vessels, they will correctly display the respective vessel names. The vessel name in the column links to the Trade Command Center's shipment status card, which includes the full vessel details, the AIS track, and any disruption predictions. The column header is sortable; when sorted, shipments with a vessel name appear before those with only a booking reference, and both appear before those with no transport data. Within the same group, sorting is alphabetical.

Origin / Destination

This column presents the trade's origin and destination as a compact, visually clear pair. It consists of two port codes (UN/LOCODE) arranged vertically: the origin port on top, a downward-pointing arrow in the middle, and the destination port below. Next to each port code, a small flag icon of the corresponding country is rendered, using a lightweight flag sprite set. For example, a shipment from Saigon, Vietnam, to Alexandria, Egypt, would display "SGN □□" above a downwards arrow and "ALEX □□" below.

Clicking on either the origin or destination port code opens a tooltip that provides the full port name, the country, a mini informational card about the port. For the destination port, the tooltip additionally shows the expected remaining transit time (calculated from the AI's ETA prediction minus the current time) and any known port congestion alerts. If the digital twin or disruption service has flagged an issue at the destination (e.g., "Strike expected at Alexandria port on estimated arrival date"), a small warning triangle appears next to the port code, and the tooltip turns amber to draw attention.

For multi-shipment contracts where different shipments have different ports of discharge, each row correctly reflects its own origin and destination. The Origin / Destination column is also the basis for the map toggle: when the user switches to the map view, the origin and destination ports are used to draw the route lines and position the vessel markers. The column header is not sortable in the traditional sense because it contains two distinct values; however, a custom sort implementation allows sorting by origin port code alphabetically, then by destination port code as a secondary criterion.

ETD / ETA

The Estimated Departure and Estimated Arrival column is a dual-line display similar to Origin / Destination. The top line shows “ETD:” followed by the date and time in the user’s local timezone (converted from the stored UTC timestamp). The bottom line shows “ETA:” with the same format. If the AI digital twin has computed a confidence interval for the ETA, the time is appended with a small “±Xh” badge. For example, “ETA: 20 Jun 2026 06:00 ±3h”. If the AI predicts a delay that would cause the ETA to slip beyond the contractual delivery date or beyond a free-time deadline, the ETA text and badge turn amber, and a small delay warning icon appears next to the time. Clicking the delay warning icon opens a tooltip generated by Groq that explains the reason for the predicted delay: “A storm system in the Mediterranean is expected to add 18 hours to the transit time. The captain has adjusted course to minimise impact. Revised ETA confidence is 75%.”

A small information icon next to the ETD allows the user to see the port cutoff times and the loading window, when available. For shipments in the early stages, the ETD may be preceded by “Est.” to indicate that it is an estimate based on the booking confirmation rather than a confirmed departure. Once the vessel departs, the ETD is replaced by the actual departure time, with a small green checkmark, and the text changes to “Departed:” to distinguish it from the estimated value.

For multi-shipment contracts, each shipment row shows its own ETD and ETA. The column is fully sortable; sorting by ETD is the default sort order for the vault when it first loads. When the map view is active, the ETD and ETA are used to interpolate vessel positions along the route, providing a smooth animation of the ship icon between ports.

Last Updated

This column shows a human-readable relative timestamp, such as “2 hours ago”, “15 minutes ago”, or “1 day ago”. The absolute UTC timestamp is available in a tooltip on hover and is also copied to the clipboard if the user clicks on the timestamp text. The relative time is computed client-side using a lightweight library that updates every minute, so the values stay fresh even when the user has the page open for an extended period without manual refresh.

The Last Updated column reflects the most recent change to the shipment’s state—a milestone confirmation, a document upload, a status change, a new disruption prediction, or even a counterparty message in the trade room. It is directly derived from the shipments.updated_at field in the database. When a new real-time event arrives via NATS, the value is immediately updated without a full row re-render, ensuring the user always sees the freshest activity timestamp.

For the purpose of sorting and filtering, the underlying value is the full UTC timestamp. This column is often used manually by logistics coordinators and trade managers to quickly identify shipments that have been sitting without updates—a likely indicator of a stuck trade. When a shipment's last updated timestamp exceeds the expected update interval (e.g., no milestone confirmation for 24 hours despite the vessel being at sea), the row is lightly shaded with a warning tint, and the "Last Updated" text is displayed in amber to draw attention.

Quick Actions (Common)

On the far right of the table, a "Quick Actions" column appears as a series of small icon buttons. These are not toggled by a configuration prop but are automatically enabled based on the user's permissions and the shipment's current status. The common Quick Actions available across all portals are:

View TCC (eye icon) : Always available. Opens the Trade Command Center for the specific USTN in a new tab.

Request Document (paperclip with a plus icon) : Visible if the user has the document.request permission and the shipment has missing required documents. Clicking it opens a pre-filled message draft addressed to the responsible counterparty, requesting the specific missing document.

Upload Document (upload arrow icon) : Visible if the user's tenant is the responsible party for at least one missing or pending document. Clicking it opens the document upload form pre-scoped to the shipment.

Add to Watchlist (star icon) : Always visible. Toggles the shipment's watched status for the user, which influences the Smart Inbox and the Shipments Vault's "Watchlist" filter.

All Quick Action icons have accessible labels (aria-label) and tooltips. They do not trigger actions that directly modify trade state without an explicit Governor decision; the "View TCC" and "Add to Watchlist" actions are purely read-only or preference changes, while "Request Document" and "Upload Document" lead to forms that themselves pass through the Governor.

Interaction Between Common Columns and the Dual-Mode Toggle

The common columns described above are always present regardless of whether the user is in Buyer or Seller mode within the Trader Portal. However, the data and emphasis within those columns subtly shift:

The USTN column, as noted, highlights the user's own GTID suffix in the brand colour, reinforcing their role.

The Status column emphasises buyer-relevant statuses (CUSTOMS_IMPORT) or seller-relevant statuses (LOADED) with a pulsing border depending on the active mode.

The Origin / Destination column does not change, but the context of whether the user is sending or receiving the goods is communicated by the entire dashboard layout.

The ETD / ETA column may prioritise the ETA in Buyer mode (since buyers care most about arrival) and the ETD in Seller mode (since sellers care most about departure and proof of loading). This is a minor visual change—the more critical time is shown in a slightly larger font.

When the user flips the toggle, the vault performs a full data re-fetch with the updated mode parameter, and the common columns re-render with the new context. The transition is seamless and does not interrupt any open slide-out panel; if a panel was open for a shipment that is no longer visible in the new mode, the panel closes gracefully.

END OF SUB-SECTION 6.1.6.2 – COMMON COLUMNS

6.1.6.3 Role-Specific Columns & Configurations

6.1.6.3.1 Trader Portal – Buyer Dashboard (Buyer Mode) – Role-Specific Columns & Configuration

When the Shared Shipments Vault is rendered inside the unified Trader Portal and the user's active dual-mode context is Buyer (active_trader_mode_context = BUY), the vault immediately adapts to show inbound shipments—those where the authenticated tenant is the buyer. All the common columns described in 6.1.6.2 remain fully present, but they are supplemented by a set of buyer-specific columns that surface information critical for an importing organisation: who is sending the goods, whether the import documentation is ready, how many documents the buyer still must provide, and a precise, AI-enriched arrival estimate with delay warnings. The vault's default filters, row actions, and slide-out panel tabs are also tailored to the Buyer perspective, ensuring the user never wastes time looking at outbound shipments or seller-side cost breakdowns while in Buyer mode.

The Buyer Dashboard's vault configuration is automatically applied when the user's JWT contains the claim active_trader_mode_context: "BUY". If the tenant has only the BUY trader mode, the toggle is hidden and the Buyer configuration is permanent. For dual-mode tenants, switching the toggle from Seller to Buyer instantly calls the backend with mode=BUYER, re-fetches the shipment list, and re-renders the vault with the columns and behaviours described below. The transition is seamless and preserves the user's scroll position, active filters, and selected rows.

Additional Columns for Buyer Dashboard

The following columns are appended to the common column set whenever the vault operates in Buyer mode. They appear immediately after the basic tracking columns (USTN, Status, Vessel, Origin/Destination, ETD/ETA) and before the "Last Updated" and "Quick Actions" columns. Their order can be customised by the tenant admin from the "Customise Columns" settings, but the default order reflects the natural workflow of an buyer: who, when, and what do I need to do.

Seller Name (Seller Identity)

This column displays the legal name of the exporting counterparty—the company that is shipping the goods to the buyer. The name is rendered as a link styled in the secondary text colour, and clicking it opens the seller's 360° Trust Portrait if the seller is a saved contact of the buying tenant. If the seller is not yet in the buyer's network, a smaller "Add to Network" icon appears next to the name, allowing the buyer to manually save the contact directly from the vault, without navigating away.

Beneath the seller's legal name, two small badges may appear. The first is a trust score badge (numeric, colour-coded green/amber/red according to the score ranges), representing the current platform-wide trust score of that seller. The second badge, appearing only if the seller is a “Saved Contact” and if the AI has computed a relationship health score, shows a simple indicator: a small heart icon with the health score (0-100) and a trend arrow (up, flat, down). Hovering over the health score shows a Groq-generated plain-language summary of the relationship’s trajectory, such as “Strong and growing – 8 trades completed, 0 disputes.”

The Seller Name column is sortable alphabetically and is also searchable via the global search bar. When a user types a seller’s name in the search, matches are highlighted across all rows.

Customs Readiness

The Customs Readiness column provides an instant visual summary of the import clearance documentation status. It is represented as a traffic-light indicator comprising three stacked lights, each corresponding to a category of required import documents.

The top light represents “Buyer-side documents” : those the buyer must provide, such as an import permit, tax identification forms, or a foreign exchange allocation certificate. It is green if all buyer-side documents are verified or uploaded, amber if some are pending but not yet critically late, and red if one or more critical documents are missing and the vessel’s ETA is within 72 hours (configurable threshold).

The middle light represents “Seller-side documents” : those the seller must provide, such as the commercial invoice, packing list, and certificate of origin. It works identically but reflects the counterparty’s compliance.

The bottom light represents “Third-party documents” : those provided by logistics providers, government agencies, or inspection bodies, such as the phytosanitary certificate, fumigation certificate, or pre-shipment inspection report. If any of these are missing or flagged by AI, the light reflects that.

Each light is not just a colour; it contains a small icon—a checkmark for green, a clock for amber, a warning triangle for red. Hovering over any light opens a detailed tooltip listing the exact documents, their current statuses, who is responsible, and a direct action link: “Upload import permit”, “Request from seller”, or “Message broker”. The tooltip for the top light is tailored to the buyer; the others are informational. The overall readiness is summarised by a single icon next to the column header: green shield, amber hourglass, or red exclamation mark.

This column is updated in real time through NATS. When a document is uploaded and verified, the corresponding light changes from red to amber to green, and a brief highlight animation confirms the update. The column header is sortable, with a custom ordinal that places fully-ready shipments first (green), then those with minor issues (amber), then those with critical missing documents (red). This helps customs specialists prioritise their work.

Documents Missing Count (Buyer’s Responsibility)

This column displays a simple numeric badge indicating how many required documents for which the buyer’s own organisation is the responsible party are currently missing or in a non-verified state. The count is computed dynamically from the shipment_document_requirements table, filtered by responsible_party_type = ’BUYER’ (or buyer) and status not in ('VERIFIED'). The badge is circular, colour-coded: grey if zero, amber if 1-2 documents are missing, red if 3 or more are missing, or if any critical document (is_critical = true) is among the missing ones.

Clicking the badge opens the slide-out panel's Documents tab, pre-filtered to "Buyer's Pending". This immediately takes the user to the exact documents they need to address, eliminating any search. A small tooltip on hover reads, for example, "2 documents needed: Import Permit, Tax ID Declaration. Both required before customs clearance."

For buyers in heavily regulated corridors, this column is one of the most used features of the vault. Combined with the Customs Readiness column, it allows an import manager to scan a list of dozens of inbound shipments and quickly identify the ones where their own action is overdue.

Estimated Arrival (with Delay Prediction)

This column builds on the common ETD/ETA column but is enhanced specifically for the Buyer perspective. It displays the ETA in a larger, bolder format, because the arrival time is the single most important temporal metric for an buyer. The ETA is shown in the user's local timezone with a confidence band extracted from the AI digital twin. A small ship icon with a dotted line appears next to the arrival time. If the AI digital twin or disruption service predicts a delay, the arrival time is rendered in amber, with the probable delay duration appended in parentheses: "ETA 20 Jun 08:00 (+14h delay)". A warning triangle icon appears, and clicking it opens a Groq-generated explanation of the delay cause, impact on demurrage, and recommended actions. For example: "Storm in Eastern Mediterranean is adding 14 hours to transit. Your free storage period at Alexandria begins on 21 Jun. No demurrage risk yet, but consider booking priority trucking to minimise port storage."

The column also shows a small countdown timer underneath the ETA if the vessel is within 72 hours of arrival. The timer is in days and hours, counts down in real time, and turns red when the remaining time is less than 24 hours. This gives the buyer a visceral sense of urgency to finalise customs documents and arrange inland logistics. The Estimated Arrival column is sortable by the absolute ETA value; when sorted, delayed shipments are not automatically grouped separately, but the amber colour makes them visually prominent.

Default Filters for Buyer Dashboard

When the vault first loads in Buyer mode, it applies a set of default filters that focus the user's attention on active inbound shipments. The status filter is pre-set to include only statuses that represent goods in motion or at the border: GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED, and CUSTOMS_IMPORT. Statuses like CREATED, CANCELLED, and DISPUTED are excluded by default, though the user can add them via the status chip. The Active/Archive toggle at the top of the vault respects this default: Active shows the above statuses; Archived shows DELIVERED, COMPLETED, and any CANCELLED or DISPUTED shipments.

The commodity filter is not pre-set, as buyers may deal with multiple products. The date range filter defaults to "Next 7 days" for ETA, showing only shipments expected to arrive within a week. This is the most actionable time window for an buyer. The risk score filter is not applicable in Buyer mode.

These defaults can be overridden by the user; the changes are saved in the user's preferences and re-applied on the next visit. A "Reset to Buyer Defaults" button appears in the filter toolbar if any filter deviates from the standard.

Row Actions for Buyer Dashboard

Along with the common Quick Actions (View TCC, Request Document, Upload Document, Add to Watchlist), the Buyer dashboard adds the following role-specific row actions:

Track Shipment (map pin icon) : Opens the map view with the specific vessel centered and the route highlighted. This action is available for shipments in IN_TRANSIT or ARRIVED status.

Request Documents from Seller (message icon with paperclip) : Sends a pre-filled, AI-generated message to the seller's trade room, listing the specific missing documents and their criticality. The message is written in the counterparty's preferred language if known.

View Seller Profile (person icon) : Opens the counterparty's 360° Trust Portrait in a new panel, including trade history, delivery performance, dispute rate, and relationship health score.

Initiate Customs Clearance (shield with checkmark icon, visible only if the shipment has status CUSTOMS_IMPORT and the buyer has customs clearance permission): Triggers the customs clearance workflow from the vault, allowing the buyer to validate the AI-generated customs declaration and submit it to the national system (if integrated) or download it for manual submission.

All row actions are rendered as small icon buttons in a dedicated column or as part of the Quick Actions group, depending on screen width and configuration. Each action has an accessible label and a tooltip.

Slide-out Panel Extra Tabs for Buyer Dashboard

When a shipment row is clicked in Buyer mode, the slide-out detail panel includes an additional tab beyond the standard Summary, Documents, Containers, and Timeline:

Customs Readiness tab : This tab presents a detailed, step-by-step checklist of every import document required for the specific trade corridor and commodity. Each document row shows: document name, responsible party (Buyer, Seller, or Third-party), status (with verification confidence), AI-extracted summary if already uploaded, and direct action buttons (Upload, Request, or View). The tab also includes a "Submit Customs Declaration" button at the bottom, enabled only when all critical documents are verified. This tab consolidates the customs workflow into a single view accessible directly from the vault, eliminating the need to navigate to the separate Customs Readiness tab in the portal sidebar for quick checks.

Every element of the Buyer dashboard's vault—from the columns to the filters to the slide-out tab—is designed to answer the four universal UX questions. The Seller Name tells the user who is sending the goods. The Customs Readiness column signals what is happening with documentation. The Documents Missing Count tells the user exactly what they need to do, and the Estimated Arrival column tells them what happens next and when. When a document is missing, the number and the red light provide the blocker.

Together, these columns transform the vault from a passive tracking grid into an active, task-oriented workspace that keeps the buyer in control of every inbound shipment.

END OF SUB-SECTION 6.1.6.3.1 – TRADER PORTAL – BUYER DASHBOARD COLUMNS

6.1.6.3.2 Trader Portal – Seller Dashboard (Seller Mode) – Role-Specific Columns & Configuration

When the Shared Shipments Vault is rendered inside the unified Trader Portal and the user's active dual-mode context is Seller (active_trader_mode_context = SELL), the vault adapts to show outbound shipments—those where the authenticated tenant is the seller. All common columns remain, but they are supplemented by seller-specific columns that surface information vital for the exporting organisation: who is buying the goods, what the projected profit margin looks like, the loading status of each container, and the progress of logistics quotations. The vault's default filters, row actions, and slide-out panel tabs are reconfigured to match the seller's workflow, ensuring the user sees only outbound shipments and seller-relevant actions without needing to navigate to a different URL or portal.

The Seller Dashboard configuration is activated automatically based on the JWT claim active_trader_mode_context: "SELL". For tenants with only SELL trader mode, this configuration is permanent. For dual-mode tenants, toggling from Buyer to Seller immediately sends a request to the backend with mode=SELLER, refreshing the vault with the columns, filters, and actions described below. The switch is instantaneous and preserves the user's active selections.

Additional Columns for Seller Dashboard

The following columns appear when the vault operates in Seller mode. They are placed after the core tracking columns (USTN, Status, Vessel, Origin/Destination, ETD/ETA) and before "Last Updated" and "Quick Actions". The default order follows the seller's natural priorities: first, who is the customer; second, what is the financial outcome; third, is the cargo moving; fourth, are the logistics costs finalised. All columns are customisable via the "Customise Columns" settings.

Buyer Name

This column displays the legal name of the importing counterparty—the buyer. The name is rendered as a clickable link, and selecting it opens the buyer's 360° Trust Portrait if that buyer is a saved contact. For new counterparties, a small "Add to Network" icon is available.

Beneath the buyer's name, two small badges may appear : a trust score badge (colour-coded green for ≥80, amber for 50-79, red for <50) reflecting the platform's assessment of the buyer's reliability, and a relationship health badge (heart icon with a numerical score and a trend arrow) that appears only for saved contacts. Hovering over the health badge shows a Groq-generated summary such as "Stable relationship – 12 trades, average payment delay 2 days". The column is sortable alphabetically and searchable.

Estimated Margin

This column provides the seller with a real-time snapshot of the projected profit for each shipment. It is calculated as the EXW contract value minus the sum of all confirmed logistics invoices minus the SGTX trade fee (which the seller has already paid at contract lock). The result is displayed as a monetary amount in the trade's currency, with a colour-coded background: green if the actual margin is within 2% of the estimated margin at the time of quoting, amber if it is 2% to 5% lower, red if it is more than 5% lower or if the margin has turned negative, and grey if no logistics invoices have been uploaded yet, indicating that the margin cannot yet be accurately computed.

Hovering over the margin value opens a detailed tooltip that breaks down the calculation: EXW value, total freight paid so far, total trucking paid so far, total customs brokerage paid so far, the SGTX fee that was already paid, and the resulting net margin. This tooltip also displays the original estimated margin from the quote, allowing the seller to compare planned versus actual performance. For multi-shipment contracts, the margin is calculated per shipment; the tooltip also shows an aggregated master contract margin if the user hovers over the master contract row.

This column is visible only to employees with permission to view financial data; for warehouse operators or junior staff, the column is hidden or shows "Confidential" based on the hidden_cost_components data scope. The column is sortable, with green shipments typically placed first and red last.

Loading Status

The Loading Status column indicates how far the physical preparation of the shipment has progressed. It uses an icon-and-text combination: "Pending" (an empty container icon with a clock) means no pallets have been scanned; "Partial" (a half-filled container icon) appears when some but not all pallets for the associated container(s) have been scanned as loaded; "Fully Loaded" (a filled container icon with a green checkmark) appears when every pallet for every container in that shipment has been successfully scanned and the loading milestone has been confirmed.

For multi-shipment contracts, the loading status is per-shipment. The status updates in real time as barcode scans are received from the mobile app. If the loading window for a container has passed and the status is still "Pending" or "Partial", the icon gains a red exclamation mark, and the text turns amber to indicate a potential delay. Hovering displays the exact number of pallets loaded out of the total, the window deadline, and a link to the Barcode Print tab or to the packing plan viewer. The column is sortable, with fully loaded shipments appearing first by default.

Logistics Quotes Status

This column provides a quick summary of the logistics procurement process for the shipment. It shows, in a compact format, the number of quotes received versus the number of required services. For example, a display of "2/3" with a checkmark indicates that two out of three logistics services (e.g., ocean freight and customs clearance) have selected quotes, but one (e.g., trucking) still needs a decision. If all required quotes are selected, the display shows a green "Complete" badge. If an RFQ has been sent but no responses have been received, the display shows a clock icon and the text "Awaiting responses".

Hovering over the status opens a tooltip listing each required logistics service (as defined by the incoterm and the seller's manual entry or RFQ choices), the status of each (Quote Received, Quote Selected, RFQ Sent, Not Yet Requested), and, if a quote has been accepted, the provider's name and cost. The tooltip also provides a direct "ReOptimise" button if the carrier risk scores have changed significantly since the original selection. This column allows the seller to identify shipments where logistics are completely arranged versus those that still need attention without leaving the vault. It is sortable, with fully-quoted shipments first.

Default Filters for Seller Dashboard

Upon initial load in Seller mode, the vault applies a set of filters that focus on outbound shipments actively in motion or awaiting preparation. The status filter is pre-set to include CREATED, GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED, and CUSTOMS_IMPORT (though CUSTOMS_IMPORT is rare for outbound, it may appear for shipments where the seller assists with import clearance at destination). Cancelled and disputed shipments are excluded from the Active view but appear in the Archive. The Active/Archive toggle follows the same semantics: Active shows the above statuses; Archived includes DELIVERED, COMPLETED, CANCELLED, and DISPUTED.

The date range filter defaults to "Next 14 days" for ETD, as sellers are most concerned with upcoming departures. The risk score filter is not applied by default. These defaults can be customised, and a "Reset to Seller Defaults" button restores them.

Row Actions for Seller Dashboard

In addition to the common Quick Actions, the Seller dashboard provides the following role-specific row actions:

View Quote (document icon with a dollar sign) : Opens the quote details for that shipment, including the EXW price, the logistics cost breakdown, the SGTX fee applied, and the total price quoted to the buyer. Only available if the quote has been submitted.

ReOptimise (refresh icon with a truck) : Triggers a re-evaluation of the logistics bundle for that shipment using the latest carrier risk scores and available quotes. The result is displayed in a diff view, comparing the existing selected providers with the new AI recommendation, and the seller can accept or dismiss the change. This action is visible only when the logistics bundle has already been selected and the shipment is not yet in transit.

Print Barcodes (printer icon) : Opens the Barcode Print tab pre-scoped to the pallets of this shipment, allowing the seller to generate and print SSCC labels immediately.

Confirm Shipment (checkmark icon, visible for multi-shipment contracts where the shipment is in SCHEDULED status and the seller has not yet marked it as ready): Triggers the "Ready for Shipment" confirmation, which notifies the buyer and starts the per-shipment locking process.

Mark as Loaded (truck icon, visible if the shipment is in LOADED status but the milestone has not yet been confirmed due to missing barcode scans): Provides a manual override to confirm loading, which requires a Governor decision with a mandatory reason.

All row actions are presented as icon buttons with tooltips and accessible labels.

Slide-out Panel Extra Tabs for Seller Dashboard

When a shipment row is clicked in Seller mode, the slide-out detail panel includes an additional tab beyond the standard Summary, Documents, Containers, and Timeline:

Cost Breakdown tab : This tab is visible only to users with permission to view financial data and is never shown to warehouse operators or junior staff with hidden_cost_components scopes. It displays a comprehensive, line-by-line breakdown of the shipment's finances from the seller's perspective. Sections include: EXW price locked (per commodity, with the date locked and the AI-recommended range for reference); Logistics costs (each service category showing the provider's name, quoted cost, accepted cost, and any invoice discrepancy flagged); SGTX trade fee (the dynamic percentage and amount, with a small "i" icon explaining the factors that contributed to the rate, such as commodity margin, country pair, and any adjustment factors); and the resulting estimated margin. For multi-shipment contracts, the tab displays per-shipment costs and a master contract summary. The tab also includes a small chart (sparkline) showing the cumulative cost versus the original quote, updated in real time as logistics invoices are uploaded and verified. This allows the seller to monitor cost overruns directly from the vault.

Every element of the Seller dashboard's vault is designed to answer the four universal UX questions for the seller. The Buyer Name tells the seller who their customer is. The Estimated Margin column signals the financial health of the shipment. The Loading Status and Logistics Quotes Status tell the seller exactly what needs attention: finalising a logistics quote, completing the loading, or both. The Cost Breakdown tab answers the detailed "what is happening with my costs" question. When a margin turns red or a loading window is missed, the vault immediately makes the blocker visible, empowering the seller to act before small issues escalate into disputes or financial losses.

END OF SUB-SECTION 6.1.6.3.2 – TRADER PORTAL – SELLER DASHBOARD COLUMNS

6.1.6.3.3 Logistics Portal – Freight Forwarder, Shipping Line, Trucking Company, Customs Broker – Role-Specific Columns & Configuration

The Logistics Portal serves multiple distinct roles, each with its own operational focus. The Shared Shipments Vault adapts to whichever logistics role the authenticated employee holds: FREIGHT_FORWARDER, SHIPPING_LINE, TRUCKING, or CUSTOMS_BROKER. If an employee holds more than one role (e.g., a company operating both forwarding and trucking), a role-selector dropdown appears at the top of the vault, allowing the user to switch between their assigned perspectives. The dual-mode toggle is not applicable to logistics tenants, as they do not act as buyers or sellers.

The vault's baseline behaviour for all logistics roles is to show only those shipments in which the authenticated tenant has a direct involvement—either an accepted quote (via quote_selected_services) or an assigned document responsibility (via shipment_document_requirements.responsible_tenant_id). This strict filtering ensures that forwarders, carriers, truckers, and brokers never see trades outside their contracted jobs. The columns, filters, and actions described below are layered on top of the common columns (USTN, Status, Vessel, Origin/Destination, ETD/ETA, Last Updated) that were defined in 6.1.6.2.

Common Logistics Columns (All Roles)

The following two columns appear for every logistics user regardless of their specific role. They provide essential context about the service they are providing and its commercial status.

Service Type

This column displays the specific logistics service that the provider is delivering for the shipment. It uses a compact text-and-icon label: “Ocean Freight” (ship icon), “Trucking” (truck icon), “Customs Brokerage” (document with stamp icon), or in the case of a forwarder who has bundled multiple services, “Forwarding – Ocean + Trucking” (a combination icon). The label is derived from the service_type field in the provider_quotes and quote_selected_services tables. If the provider is involved in multiple services for the same shipment (e.g., both trucking and customs), the column shows the primary service or, for forwarders, a unified “Forwarding” label. For freight forwarders who have subcontracted, a small “(sub)” badge appears alongside the service type if the work is being performed by a partner from their network.

My Quote Status

This column shows whether the provider’s quote for this shipment has been accepted, is pending, or was rejected. The status is displayed as a coloured badge: “Accepted” (green checkmark), “Pending” (amber clock), “Rejected” (red cross). If the provider submitted a bundled quote (forwarder), the badge reflects the overall acceptance status of the bundle. Hovering over the badge reveals the date the quote was submitted, the amount quoted, and if rejected, the reason provided by the seller (e.g., “Carrier risk score too low”). If the quote is still pending, the tooltip includes a countdown to the RFQ deadline and a “Modify Quote” link. This column is critical for logistics sales teams to track their pipeline and follow up on outstanding proposals.

Role-Specific Columns and Behaviours

Freight Forwarder

Freight forwarders see the following additional column when they have an accepted quote or are coordinating a shipment.

Bundle Breakdown – This column provides a compact, inline summary of the services bundled under the forwarder’s quote. For example, “Ocean: 2,800,Trucking:2,800,Trucking:900, Customs: $600”. Each line item shows the service type, the cost the forwarder quoted, and a small indicator if the forwarder has subcontracted that service to a partner from their network (a link icon). Hovering expands the full detail of each sub-service, including the subcontractor’s name, their quote reference, and the forwarder’s margin on that sub-service (visible only to the forwarder). If the forwarder has not yet finalised the subcontracting, a “Pending” badge appears next to that line item. This column helps forwarders track the profitability and completeness of their consolidated offerings directly from the vault.

Row Actions for Freight Forwarder – In addition to the common actions, the forwarder can trigger “Update Milestone” for any milestone they are responsible for (e.g., “Confirm Customs Clearance”, “Gate In at Terminal”), “Upload Document” (e.g., House Bill of Lading, Forwarder’s Cargo Receipt), and “ReOptimise” if the logistics bundle needs to be re-evaluated.

Slide-out Extra Tab – The “My Tasks” tab lists all pending milestones and document uploads specifically assigned to the forwarder for that shipment, with direct action buttons.

Shipping Line

For employees of a shipping line, the vault displays columns and actions tailored to vessel operations and booking management.

Booking Status – This column indicates the status of the booking confirmation for the shipment. It shows one of: “Pending” (the seller has requested a booking, but the line has not yet issued a confirmation), “Confirmed” (booking details have been provided and AI-extracted), “Rejected” (the line declined the booking, with the reason displayed in a tooltip). Next to the status, the container numbers assigned are listed in a small, scrollable list. If the eBL has been issued, a green “eBL” badge is appended, with the issuance date and the platform (e.g., TradeLens) visible on hover.

Webhook/API Health (Background Indicator) – While not a column, the toolbar in Shipping Line mode includes a small status light indicating the health of the line’s integration (API, email polling, or manual). If the API is healthy, a green dot appears; if degraded (e.g., 2 consecutive failed attempts), an amber dot; if disabled, a red dot and a prompt to check the Integration Profile.

Row Actions for Shipping Line – “Issue Booking Confirmation” (opens a form to manually confirm a booking if the API/email channel is not being used), “Upload eBL” (if the line’s eBL platform does not send automatic webhooks), “Update Milestone” (for vessel departure, arrival, etc.), and “View TCC” (filtered to the line’s services).

Slide-out Extra Tab – “Booking Details” tab shows the full booking information: vessel, voyage, ETD, container details, terminal, and the line’s internal reference. It also displays the history of API calls and emails related to this shipment.

Trucking Company

Trucking companies use the vault to manage pickups, deliveries, and route execution.

Pickup / Delivery Window – This column displays the scheduled time window for the trucking service. For example, “Pickup: 14 Jun 08:00-12:00” or “Delivery: 20 Jun 06:00-10:00”. Next to the window, a small icon indicates whether the window is upcoming (blue clock), active (green truck), overdue (red clock), or completed (green check). If the actual arrival time deviates from the window, the deviation is shown in parentheses (e.g., “+35 min”). Hovering over the window shows the full route, the driver’s name, and the current GPS-based ETA if the driver is en route.

Route Progress – For active trips, this column shows a progress bar and a percentage based on the distance covered versus the total planned route. The progress is derived from GPS traces from the mobile app. A red bar indicates a significant delay or a geofence breach. Clicking the progress bar opens the map view centred on the truck.

Row Actions for Trucking – “Start Trip” (confirms departure), “Confirm Delivery” (completes the milestone), “Upload Invoice”, “View Route”, and “Message Dispatcher”.

Slide-out Extra Tab – “Route Details” tab shows the full OSRM-computed route, turn-by-turn instructions, and the live GPS trace (if available) overlayed on the map.

Customs Broker

Customs brokers interact with the vault primarily to verify documents, clear shipments, and submit declarations.

Clearance Status – This column shows the customs clearance progress for the shipment. The values are “Pending” (documents not yet submitted), “In Review” (declaration submitted to customs authority, awaiting response), “Cleared” (authority has approved), and “Held” (authority has flagged or rejected). If the status is “Held”, a tooltip displays the reason from the customs response and a link to the relevant document for amendment.

Documents Verified – A numeric badge showing how many documents the broker has reviewed out of the total assigned to them. For example, “3/5 Verified”. If any AI-flagged discrepancy remains unresolved, a warning triangle appears next to the count, and the tooltip lists the flagged documents and the discrepancies.

Row Actions for Customs Broker – “Verify Document” (opens the document verification queue filtered to that shipment), “Submit Declaration” (if the country has a customs API connector, sends the declaration; otherwise, generates a PDF for manual submission), “Override AI Flag” (allows the broker to accept a flagged document after providing a mandatory reason), and “View TCC”.

Slide-out Extra Tab – “Customs Declaration” tab shows the fully populated declaration form (based on AI extraction of all trade documents), the broker’s submission history, and any response from the customs authority. The tab includes a “Submit to Customs” button that is enabled only when all required documents are verified.

Common Logistics Row Actions (All Roles)
Beyond the role-specific actions, every logistics user can access the following common actions, enforced by permissions:

View TCC (eye icon) – Opens the Trade Command Center filtered to show only the services and data relevant to that logistics provider.

Upload Document (upload icon) – Available for any document where the provider is the responsible party.

Update Milestone (flag icon) – For milestones that the provider must confirm (e.g., “Truck Departed”, “Container Gated In”).

Message Seller/Forwarder (chat icon) – Opens the Trade Room for that shipment, scoped to the logistics channel.

Sign Addendum (pen icon, visible when a logistics addendum is pending signature) – Opens the addendum for review and signing via ZITADEL passkey. This action is highlighted with a red badge until completed.

Acknowledge Release (key icon, visible after SGTX sends a container release confirmation) – Allows the provider to acknowledge the release token, which is required before the container can be loaded.

Real-Time Updates Specific to Logistics
The vault in the Logistics Portal subscribes to NATS subjects that are particularly relevant to service providers. Events such as shipment.milestone.confirmed, document.uploaded, and provider.quote.accepted trigger immediate row updates. For trucking companies, location pings from the mobile app update the Route Progress bar in real time. When a new RFQ is broadcast anonymously and the provider’s preferences match, the vault’s

toolbar displays a non-intrusive notification badge: “New RFQ received” with a link to the RFQ Inbox tab.

Unsubscribe Mechanism Integration
The toolbar of the Logistics Portal vault includes a subtle, persistent indicator of the provider’s anonymous RFQ opt-in status. When the provider has opted out, a small grey badge “Anonymous RFQs: Opted Out” is shown next to the filters, with a link to Company Admin → Notification Preferences to change the setting. This ensures the provider is never left wondering why they are not receiving certain broadcast opportunities. The feature is also referenced in the Warm-Up Program emails, which are co-branded and respect the opt-out choice.

Governance and Decision Panel Integration
Whenever a logistics action is blocked—for example, attempting to confirm a milestone before the container release confirmation has been acknowledged—the PlainLanguage Governor Decision Panel slides open directly within the vault overlay. The panel explains the reason in plain language (“The container release confirmation has not yet been acknowledged. Please check your email or notifications for the release link and acknowledge it before confirming this milestone.”) and provides a direct action button.

This configuration ensures that the Shared Shipments Vault, when instantiated within the Logistics Portal, becomes a fully integrated, role-aware operational console that covers the end-to-end needs of the logistics provider without exposing commercial data outside their scope.

END OF SUB-SECTION 6.1.6.3.3 – LOGISTICS PORTAL – ROLE-SPECIFIC COLUMNS

6.1.6.3.4 QC Portal – Role-Specific Columns & Configuration

When the Shared Shipments Vault is instantiated inside the QC Portal, it is configured exclusively for quality control inspectors, laboratory analysts, and surveyors—employees of tenants with the type QUALITY_CONTROL. The dual-mode toggle does not apply because QC tenants have a single role context. The vault immediately filters to show only those shipments for which the QC provider has an active or completed inspection job. Every common column remains available, but a set of QC-specific columns, row actions, and slide-out tabs is layered on top, giving the inspector a complete operational view without exposing any commercial values such as prices, margins, or SGTX fees.

Additional Columns for QC Portal

Inspection Status
This column is the primary operational indicator for QC personnel. It displays the current lifecycle stage of the inspection job using a colour-coded badge with an icon. The values are: ASSIGNED (grey briefcase) – the seller has booked the QC provider but the job has not yet been accepted; ACCEPTED (blue check) – the QC provider has confirmed the job; IN_PROGRESS (amber magnifying glass) – the inspector has started on-site work, with a progress percentage showing how many mandatory pallets have been examined; COMPLETED (green badge with a report icon) – the inspection report has been digitally signed and submitted; DISPUTED (red warning triangle) – the QC report is being challenged by a party.

For jobs in the IN_PROGRESS state, a small progress bar sits directly beneath the badge, showing the fraction of pallets inspected out of the total required by the AQL sampling plan. Clicking the status badge opens the full inspection job page, pre-loaded with the dynamic quality specifications, the container and seal numbers, and the AI-recommended inspection points.

Defect Count

This column provides an instant numeric summary of the defects found during the inspection. It shows a circular badge with the total number of defects, colour-coded based on severity: green if all defects are within the allowed tolerance per the contract specifications, amber if any defect exceeds the minor threshold, and red if any defect exceeds the critical threshold (e.g., mould on a zero-tolerance product). If no defects were found, the badge displays a green “0” with a checkmark icon.

Hovering over the badge opens a tooltip that categorises the defects by type (e.g., “Mould: 2, Bruising: 1, Carton damage: 0”) and indicates, for each type, whether the count is within the contractual acceptable quality limit (AQL). If any AI-detected finding was overridden by the inspector, the tooltip includes an orange note: “⚠ 1 AI finding overridden – see report notes.” This column allows QC managers to quickly triage which shipments need immediate attention or dispute preparation.

Report Due

The Report Due column displays a colour-coded deadline for when the final signed inspection report must be submitted. For jobs in ACCEPTED or IN_PROGRESS status, it shows the agreed deadline (normally 24 to 48 hours after the inspection window, configurable per QC provider contract). The date is displayed in the user’s local timezone. As the deadline approaches, the colour transitions: more than 6 hours away is neutral, between 6 and 2 hours turns amber, and less than 2 hours turns red with a pulsing animation. If the report has not been submitted and the deadline passes, the column shows “OVERDUE” in bold red, and a high-priority Smart Inbox item is automatically generated for the inspector.

For COMPLETED jobs, this column instead shows the actual date and time of submission, with a green checkmark. Clicking the date opens the signed report PDF in a modal viewer.

Default Filters for QC Portal

Upon loading, the vault applies a filter set that focuses the QC professional on their immediate workload. The status filter is preset to include jobs with inspection status ASSIGNED and IN_PROGRESS. This keeps the inspector’s view limited to work that is either awaiting their acceptance or currently being performed. Jobs that are ACCEPTED but not yet started also appear if the inspection window is within the next 48 hours.

The commodity filter is not pre-set but can be used to view only perishable goods or specific HS chapters. The date range filter defaults to “Next 7 days” for the inspection due date. The search bar respects the USTN, the seller’s name, and the product description. These defaults can be customised, and a “Reset to QC Defaults” button restores the factory settings.

Row Actions for QC Portal

In addition to the common Quick Actions (View TCC, etc.), the QC portal provides specialised row actions that directly drive the inspection workflow:

Accept Job (handshake icon, visible only for jobs with status ASSIGNED) : Confirms that the QC provider will carry out the inspection. The status changes to ACCEPTED and the seller is notified.

Start Inspection (camera icon, visible for ACCEPTED or IN_PROGRESS jobs) : Launches the mobile-optimised inspection flow (or opens the web-based checklist if on desktop). This is the primary action for the inspector when they arrive on-site.

Upload Report/Continue Inspection (document icon with a plus) : If the inspection is partially done, the button says “Continue Inspection”; if all mandatory pallets have been examined and the report draft is ready, it says “Finalise Report”. This action opens the report drafting interface with the AI-generated findings, defect log, and mandatory override reasons, ready for the inspector to review, edit, and sign with ZITADEL passkey.

Override AI Finding (toggle icon, visible only if inspection is in progress) : Allows the inspector to manually reject an AI-detected defect by providing a mandatory reason (minimum 10 characters). This is also available from within the inspection flow, but the row action gives quick access to the recent AI findings for that shipment.

View Report (magnifier icon, visible for COMPLETED or DISPUTED jobs) : Opens the finalised, digitally signed report in a PDF viewer with Loom hash verification data.

Slide-out Panel Extra Tabs for QC Portal

When a shipment row is clicked in QC mode, the slide-out panel includes two additional tabs that are essential to the inspector’s work.

Inspection Checklist Tab
This tab presents the quality inspector with a structured, step-by-step list of every mandatory check required for the specific commodity and contract. The checklist is populated from the dynamic quality_specs JSONB that was extracted from the contract during Phase 3. It is ordered by priority: first, the high-priority pallets that the AI has flagged (e.g., “OR-019 – end of lot, historically higher defect rate – Priority: HIGH”); next, the random samples required by the AQL table; and finally, any optional additional pallets that the seller requested. Each checklist item shows the pallet ID (SSCC), the product description, the expected quality specification (size, colour, brix, defect allowances), and a status indicator: a red circle for not yet inspected, a green checkmark for inspected with all findings, or an amber exclamation for inspected with one or more defects found. The checklist also includes the mandatory container and seal number fields, with live OCR validation against the booking data. The inspector can check off items directly from this tab if the mobile app is not being used.

QC Override Log Tab
This tab is visible only after an inspection has been completed and contains at least one instance where the inspector overrode an AI detection. It presents a table listing every override event: the pallet ID, the AI’s original detection (defect type, confidence percentage), the inspector’s final classification, the mandatory reason the inspector provided (minimum 10 characters), and the timestamp. For example: “OR-020 – AI: Mould (88%) → Inspector: Bruising – Reason: ‘Dark spot is surface bruise, no mould under

magnifier' – 2026-06-14 09:23 UTC". This tab is automatically included in the dispute evidence package if the QC report is later contested, and it is crucial for transparency. For the inspector, it serves as an audit log of their professional judgment.

Real-Time Updates Specific to QC

The QC vault subscribes to NATS subjects related to inspection job assignments and milestone updates. When a new inspection job is assigned to the QC provider, a row appears in the vault (if it matches the filters) and is highlighted with a brief animation. When pallet scans come in from the mobile app, the Inspection Status progress bar and Defect Count update in real time, allowing QC managers at the office to monitor on-site progress. If a dispute is filed against a QC report, the status immediately changes to DISPUTED, and a red border appears around the row. Smart Inbox items are generated in parallel.

Governance and Decision Panel Integration

QC inspectors are bound by strict override accountability. If an inspector attempts to finalise a report without completing the mandatory AQL sample, or if they try to override an AI finding without providing a 10-character reason, the Governor blocks the action, and the PlainLanguage Governor Decision Panel appears. The panel explains the exact requirement: "You must inspect all 5 randomly selected pallets before submitting the report." or "A reason of at least 10 characters is required to override the AI mould detection. Please describe why the AI finding is incorrect." The inspector cannot proceed until the conditions are resolved.

Additionally, if the report submission is blocked because the packing plan's USTN is not yet available (e.g., the contract has not been locked), the Decision Panel explains this dependency and provides a link to the Trade Command Center. All QC-related Governor decisions are logged immutably with Loom hashes and linked to the inspection job.

This configuration transforms the Shared Shipments Vault into a complete quality assurance workspace, tightly integrated with the dynamic contract specifications, mobile AR inspection, and the platform's constitutional governance.

END OF SUB-SECTION 6.1.6.3.4 – QC PORTAL – ROLE-SPECIFIC COLUMNS

6.1.6.3.5 Financier Portal – Role-Specific Columns & Configuration

When the Shared Shipments Vault is rendered inside the Financier Portal, it is configured for employees of tenants with type FINANCIAL—institutional banks, private credit funds, DeFi pools, and other lending entities. The dual-mode toggle is not applicable; financiers have a single role context. The vault's purpose from the financier's perspective is to monitor the shipments that are linked to their active financing agreements, track the performance of those loans, and respond rapidly to changes in collateral value or repayment schedules. The vault therefore shows only shipments where the financier is a party to an active or recently repaid financing agreement—it does not show the broader universe of platform trades. All common columns defined in 6.1.6.2 are fully present, supplemented by a set of financier-specific columns that surface the critical metrics: how much was lent, how much is still outstanding, the current loan-to-value ratio, the next repayment date, and the overall health of the loan. Row actions, slide-out tabs, and real-time alerts are all tailored to the credit and risk management workflow.

Additional Columns for Financier Portal

Financed Amount
This column displays the principal amount that the financier has committed to the specific shipment under a financing agreement. For a single-financier loan, it is the full loan principal. For co-financing arrangements where multiple financiers share the same request, the column shows this specific financier’s portion (the annex amount). The amount is shown in the loan’s currency (typically USD) with two decimal places, and it is formatted with the currency symbol. Beneath the principal, a smaller line indicates the portion of the total financing request that this represents—for example, “60,000of60,000of100,000 (60%)”. This provides immediate context about the financier’s share of the risk, which is particularly important when multiple lenders are involved. The column is sortable by amount, allowing a credit officer to see the largest exposures at the top.

Outstanding Balance
Closely tied to the Financed Amount, this column reflects the current remaining debt—the principal plus accrued interest that has not yet been repaid. For loans with a repayment schedule, it shows the outstanding principal plus any interest that has accrued since the last payment. For bullet repayment loans (interest and principal at maturity), it shows the full principal plus accrued interest. The amount is colour-coded: green if the loan is performing and on schedule, amber if the loan is approaching maturity within 30 days without a repayment plan, and red if the loan is overdue or in default. A small sparkline chart next to the amount shows the outstanding balance over time (past 30 days), providing a visual indication of whether the balance is decreasing as expected. Hovering over the sparkline reveals a tooltip with the exact repayment schedule and which instalments have been paid. The column is sortable; by default, loans with the highest outstanding balance appear first.

LTV (Loan-to-Value)
This column is present only for financing agreements that are collateralised by goods, warehouse receipts, or other tangible assets. It displays the current Loan-to-Value ratio as a percentage, calculated as the outstanding balance divided by the current market value of the collateral. The LTV is a critical risk metric: a rising LTV indicates that the collateral is losing value relative to the debt. The percentage is rendered inside a compact horizontal bar: green for LTV ≤ 70%, amber for 70% < LTV ≤ 85%, and red for LTV > 85%. If the LTV exceeds 85%, the bar pulses gently to draw attention.

Hovering over or clicking the LTV bar opens the full Collateral Monitoring Dashboard as a modal overlay, with five components: a circular LTV gauge chart showing the exact ratio and the current collateral value; a 30-day LTV history line chart with horizontal threshold lines at 70% and 85% and annotations for events like collateral top-ups or commodity price drops; a Liquidation Risk Meter (a linear gauge predicting the probability of liquidation within 7 days using a local LSTM model); a table of all margin calls issued for this loan; and an interactive “Simulate Price Drop” slider that allows the financier to drag a hypothetical collateral value decline up to 50% and see the resulting LTV, whether a margin call would be triggered, and the additional collateral required to restore 70% LTV. This simulation is purely client-side and does not affect the actual loan, but it helps the financier prepare for downside scenarios.

Next Repayment

This column shows the date of the next scheduled repayment instalment (principal and/or interest) for the loan. The date is displayed in the user's local timezone, with the day of the week. If the repayment is due within 7 days, the date is highlighted in amber; if it is past due, it turns red with the text "Overdue by X days". A small calendar icon next to the date allows the financier to add the repayment date to their external calendar via an .ics download. Hovering over the date shows the amount due and the payment method (e.g., "$5,200 via SWIFT to Barclays Account"). The column is sortable, with overdue dates appearing at the very top, followed by upcoming dates, then loans without a scheduled repayment (e.g., bullet loans). This column is essential for cash flow forecasting and for ensuring that follow-up actions are taken on delinquent borrowers.

Repayment Health

This column provides a concise, colour-coded summary of the borrower's repayment behaviour for the specific loan and, more broadly, across all their financing agreements with that financier. It is represented as a simple badge: "On Track" (green checkmark) for loans where all repayments have been made on time; "Watch" (amber eye) for loans where one or two payments are late but within a grace period; "Delinquent" (red exclamation) for loans where payments are more than 30 days overdue; and "Defaulted" (dark red with a broken circle) for loans that have been declared in default. If the financier has historical data on the borrower from the Financed Companies tab, a small secondary badge below the primary health badge shows the borrower's overall repayment track record—for example, "Borrower: 12 on-time, 1 late". This provides the financier with a quick credit history reference without leaving the vault. The column is sortable, with the worst-performing loans naturally sorted to the top to prioritise action.

Margin Call Alert

This column is a binary indicator : a red warning triangle appears if the loan's LTV has crossed the financier's margin call threshold (configurable, default 75%) and a margin call has not yet been issued or the outstanding call has not been resolved. If no margin call is active, the cell is blank. A tooltip on the warning triangle shows the date the threshold was breached, the current LTV, and a direct "Issue Margin Call" button. If a margin call has been issued but not yet resolved, the button reads "Pending Resolution", and the tooltip includes the deadline given to the borrower and the amount of additional collateral required. This column ensures that no margin call ever goes unnoticed, even when the financier is managing dozens of loans.

Default Filters for Financier Portal

The vault applies a strict initial filter that selects only those shipments linked to an active financing agreement where the authenticated financier's tenant is a party. This is enforced server-side via the financier-preference-service and the financing-agreement tables; the financier cannot accidentally view shipments outside their portfolio. Within this filtered set, the Active/Archive toggle is configured as follows:

Active view : shipments where the financing agreement has status ACTIVE, DISBURSED, or BIDDING (if the financier has submitted a bid that is not yet awarded).

Archived view : shipments where the agreement status is REPAID, DEFAULTED, or TERMINATED.

The status filter is automatically scoped to the financing agreement's status, not the shipment's logistics status, though the common status column still reflects the trade milestone. The commodity filter is available for financiers who want to monitor their exposure by HS chapter, and the date range filter defaults to "All Active" to give a full portfolio view. These defaults can be saved and customised.

Row Actions for Financier Portal

Beyond the common Quick Actions, the Financier portal provides the following role-specific row actions, all governed by the OPA-enforced RBAC:

View Collateral Dashboard (gauge icon) : Opens the full Collateral Monitoring Dashboard for the selected loan in a modal overlay, as described under the LTV column. This is the primary risk management tool.

Issue Margin Call (exclamation triangle icon, visible when LTV exceeds the threshold and no active margin call exists): Opens a pre-filled form that allows the financier to specify the required additional collateral amount (default: amount needed to restore 70% LTV), the deadline, and a message to the borrower. Once issued, a Governor decision is created, and the borrower receives a high-priority Smart Inbox item.

View Financing Agreement (document icon) : Opens the full financing agreement PDF, including the SGTX Witness Clause, the repayment schedule, the collateral terms, and the digital signatures.

Record Manual Repayment (money icon, for cases where the repayment is not automatically detected via PSP webhook): Allows the financier to enter a repayment amount and date, which triggers an AI reconciliation attempt and updates the outstanding balance.

Refresh Credit Intelligence (brain icon) : Triggers a new AI credit intelligence run for the borrower, pulling in the latest trade data, repayment history, and public news. The updated credit score and default probability are displayed in a small modal and recorded.

View TCC (eye icon, as common) : Opens the Trade Command Center filtered to the financier's view, showing the loan details, the underlying trade's progress, and documents.

Slide-out Panel Extra Tabs for Financier Portal

When a shipment row is clicked in the Financier vault, the slide-out panel includes two additional tabs beyond the standard Summary, Documents, Containers, and Timeline:

Repayment Schedule Tab : This tab presents a detailed table of every scheduled repayment for the loan, whether it is an instalment or a bullet payment. Columns include the due date, the principal portion, the interest portion, the total amount due, the status (PAID, PENDING, LATE, DEFAULTED), the actual payment date, and a link to the PSP transaction or manual record. A visual progress bar at the top shows the total percentage of the loan that has been repaid. For co-financing, each financier sees only their own portion's schedule, but a note indicates the borrower's total repayment obligation across all financiers. The tab also includes early repayment calculator: the financier can enter a hypothetical early repayment amount and see the impact on remaining interest and the final maturity date.

Borrower Profile Tab : This tab provides a consolidated, read-only view of the borrower's financial behaviour and identity, drawn from the Financed Companies tab and the platform's trust-score service. It includes the borrower's GTID, legal name, jurisdiction, KYB tier, and overall trust score. A line chart shows the borrower's credit score trend over the last 12 months, annotated with each financing event (disbursement, repayment, default). A summary table lists all historical financing agreements between this financier and the borrower, including the total amount financed, the number of agreements, on-time repayments, late repayments, and defaults. There is also a "Refresh Credit Score" button to run the AI credit intelligence on demand. This tab equips the financier with a full 360-degree view without navigating to another part of the portal.

Real-Time Updates Specific to Financier

The Financier vault subscribes to NATS subjects that are specifically relevant to credit and collateral management. Events such as financing.repayment.received, collateral.value.changed (triggered when a commodity price feed updates the market value of goods), and financing.margin_call.issued cause immediate updates to the affected rows. The Outstanding Balance, LTV, and Repayment Health columns refresh in real time. If the LTV crosses the 85% threshold while the vault is open, the entire row is highlighted with a red border for five seconds, and a native browser notification (if permitted) is shown, regardless of whether a margin call has been issued yet. This ensures that credit officers monitoring the vault can react within seconds to adverse market moves.

When a new financing request that matches the financier's preferences is broadcast via the automatic RFQ system, the vault's toolbar displays a non-intrusive notification badge: "New Opportunity: $XXX,XXX" with a direct link to the Financing Opportunities tab. This bridges the vault's portfolio-monitoring function with the active deal-origination workflow.

Governance and Decision Panel Integration

Financier actions are subject to the same constitutional and compliance governance as every other role. If a financier attempts to issue a margin call that exceeds the contractual limits, or if the jurisdiction matrix has been updated to block new financing activities in the borrower's country, the Governor returns a DENY or CONDITIONAL verdict. The PlainLanguage Governor Decision Panel slides open and explains, for example: "The margin call amount requested exceeds the maximum allowed under the financing agreement. Maximum additional collateral that can be demanded is $12,000. Adjust the amount and retry." or "The borrower's jurisdiction (Country X) has been downgraded to Restricted. No new margin calls can be issued until the compliance review is completed." This ensures that credit decisions are always aligned with the platform's enforced rules.

All financier-specific actions—viewing the collateral dashboard, issuing a margin call, recording a manual repayment—are logged immutably via Loom and linked to the relevant financing agreement and Governor decision. The Financed Companies tab is also subject to data privacy rules; a financier can only view their own historical data, never that of another financier.

This configuration makes the Shared Shipments Vault a powerful, real-time credit portfolio management console, fully integrated with the platform's financing lifecycle, risk models, and governance framework.

END OF SUB-SECTION 6.1.6.3.5 – FINANCIER PORTAL – ROLE-SPECIFIC COLUMNS

6.1.6.3.6 Government Portal – Customs, Port Authority, Trade Ministry, Regulatory Agencies – Role-Specific Columns & Configuration

When the Shared Shipments Vault is instantiated inside the Government Portal, it serves as the primary operational interface for officials from customs, port authorities, trade ministries, sanitary and phytosanitary agencies, and other regulatory bodies. Access is limited to employees of tenants with the type GOVERNMENT or REGULATORY, onboarded through a manual diplomatic channel governed by the Platform Governance Authority. The dual-mode toggle does not apply to government tenants, as they have a single, jurisdiction-scoped role context. The vault is configured to show only those shipments that fall within the official's jurisdiction—origin, destination, or transit country matches the government tenant's country code—enforced by PostgreSQL Row-Level Security and OPA policies. Within this jurisdictional scope, a set of specialised columns, row actions, and slide-out tabs is appended to the common columns, giving the official a real-time, dynamic view of trade flows, compliance status, and clearance workflows without exposing commercial values, SGTX fee details, or counterparty financial data.

The Government Portal's vault is unique in that its exact feature set depends on the agency's dynamically configured modules. A customs agency that has enabled "auto_clearance" and "api_integration" sees a different set of columns and actions than a port authority that only uses "live_trade_monitor". The vault reads the government_profiles.module_config JSONB to conditionally render columns, actions, and tabs. The baseline configuration described below assumes all modules are enabled, as is typical for a full-featured customs authority like Egypt Customs, and then notes how the display simplifies when certain modules are turned off.

Additional Columns for Government Portal

Risk Score
This column displays the SGTX AI-computed risk score for each shipment, a number between 0 and 100 that quantifies the overall risk associated with the trade. The score is computed by the platform's counterparty risk models, sanctions screening, commodity-specific risk factors, and route-specific disruption predictions. It is presented as a horizontally filled bar, colour-coded according to the thresholds configured in the agency's risk_scoring module. In the default configuration, green represents scores 0-25 (low risk, eligible for auto-clearance), amber represents 26-70 (medium risk, manual inspection recommended), and red represents 71-100 (high risk, mandatory manual inspection). The exact thresholds are overridable by the government admin via the dynamic module settings.

Hovering over the risk score bar reveals a detailed tooltip that breaks down the contributing factors: counterparty trust scores, jurisdiction risk level, commodity HS code classification, carrier performance history, and any recent sanctions or news flags. The tooltip is generated by Groq in the officer's preferred language, for example: "Risk Score 22 – Low. Buyer and seller are both VERIFIED with trust scores above 85. No sanctions flags. Cargo is citrus, a standard low-risk commodity. Carrier has a 92% on-time delivery record." For high-risk shipments, the tooltip explicitly flags the concerning factor: "Risk Score 72 – High. Seller jurisdiction (Country Y) has been downgraded to Restricted. Enhanced due diligence documentation not yet fully verified. Recommend manual inspection before clearance."

The column is sortable, and the agency can set an auto-clearance threshold in its configuration: shipments with a score below that threshold are automatically cleared without any human action, and the officer sees a log entry indicating the auto-clearance. The Risk Score column is the primary triage tool for customs officials managing high volumes.

Clearance Status

This column shows the customs clearance progress for the shipment within the government's jurisdiction. The values and their associated badges are: PENDING – the shipment has arrived or is approaching but clearance has not yet been initiated; IN REVIEW – a clearance workflow has been started (manually or by auto-clearance logic) and is pending either document verification, risk assessment, or multi-agency approval; CLEARED – all required steps have been completed and the goods are authorised for release; HELD – the shipment has been stopped by an official or by an automated rule due to missing documents, high risk score, or a sanctions flag, and cannot proceed until the hold is lifted; REJECTED – clearance has been definitively denied with a required reason, and the seller/buyer have been notified.

If the agency has enabled the api_integration module and connected to a national customs system (e.g., Nafeza for Egypt, VNACCS for Vietnam), the clearance status also shows the external declaration reference number, such as "CLEARED – NFZ12345". The officer can click the reference to view the response from the national system. The column is sortable, with shipments needing action (IN REVIEW, HELD) appearing at the top by default.

Declaration Reference

This column is visible only if the api_integration module is enabled and an external customs system connector is active. It displays the reference number assigned by the national single-window system to the customs declaration for this shipment. The reference is a clickable link that, when selected, opens the external system's status page (if the officer has SSO or a valid session) or displays a modal with the full declaration data as submitted by SGTX. If the declaration has been auto-submitted but the external system has not yet responded, the reference shows "Submitted – awaiting response" with a clock icon. If the connector has failed and the declaration could not be submitted, the column shows "Connector error – manual submission required" in red, and the officer can download a pre-filled PDF declaration from the slide-out panel's Customs Declaration tab. For jurisdictions without API integration but with the permit_issuance module active, this column may instead show the digital permit number.

Anomaly Flags

This column provides a visual summary of any AI-detected anomalies or discrepancies in the shipment's documentation, data, or behaviour. It is a cumulative indicator based on the output of several AI agents: the document validation agent (A2) that flags weight mismatches, invoice discrepancies, and missing certifications; the sanctions rescreener that detects late-appearing PEP links; and the route anomaly detector that identifies deviations from the planned logistics corridor.

The column displays a circular badge with the total number of active flags, coloured amber for informational warnings and red for critical issues that require immediate human review. Hovering over the badge opens a list of each flag, with a one-sentence Groq-generated explanation and a direct link to the relevant document or dashboard section. For example: "1. Phytosanitary certificate weight mismatch – declared 20,000 kg, packing list shows 17,600 kg. Confidence: 96%. [Review Document] 2. Vessel deviated from planned route by 12 nautical miles near a high-risk area. [View Route]". For shipments with no anomalies, the cell is blank. This column allows customs officers to quickly identify shipments where AI has already highlighted a problem, focusing their attention where it is most needed.

Integration Badge

This column is present when the government tenant has at least one active API connector. It shows small icons representing the external systems with which data has been synchronised for this shipment: a customs system icon (e.g., a building with a flag), a phytosanitary system icon (a leaf), a port community system icon (an anchor), etc. Each icon is coloured green if the synchronisation was successful, amber if it was attempted but with warnings, or red if it failed. Clicking an icon opens the integration connector log for that specific shipment, showing the full API request and response. This column is crucial for IT-savvy officers who need to troubleshoot integration failures quickly without leaving the vault.

Default Filters for Government Portal

The vault applies a jurisdiction-scoped filter automatically and irrevocably : only shipments where origin_country, destination_country, or any transit_country equals the government tenant's assigned country code are included. This is enforced server-side by PostgreSQL RLS and the Governor Proxy.

Within this jurisdictional set, the status filter defaults to showing all shipments that are currently active or at the border: ARRIVED, CUSTOMS_IMPORT, IN_TRANSIT (if the vessel is within 48 hours of arrival), and any shipment with a pending clearance action (HELD, IN REVIEW). Shipments that have been CLEARED or DELIVERED are moved to the Archived view, though they can be accessed by toggling the Archive.

The risk score filter can be used to show only high-risk shipments (useful for risk-based inspection), and the date range filter defaults to "Arriving in the next 7 days". The commodity filter is available for agencies that only deal with specific goods (e.g., Agriculture Ministry filtering to HS chapters 06-14). These defaults are saved per user and can be reset to agency defaults.

Row Actions for Government Portal

The following row actions are available, displayed as icon buttons, and are gated by the permissions assigned to the officer's role and the dynamic module configuration:

Approve Clearance (green shield with checkmark) : This action is available when the clearance status is IN REVIEW and all required conditions (document verification, risk assessment, and multi-agency approvals if applicable) are met. Clicking it changes the status to CLEARED, creates a Governor decision (decision_type = ’customs_clearance’), and, if the api_integration module is active, automatically pushes the approved status to the connected national system. The officer is prompted for a confirmation: “Approve clearance for USTN...? This will release the goods and notify the buyer and seller.”

Hold (red hand icon) : Places the shipment on hold. A mandatory reason must be provided (free text, minimum 20 characters). This action creates a Governor decision, stops any auto-clearance process, and notifies the buyer, seller, and logistics providers via Smart Inbox. The officer can later lift the hold.

Reject Clearance (red X icon) : Permanently rejects the clearance request. A detailed reason is mandatory. The buyer and seller are notified, and the shipment’s status is updated. The owner must re-submit corrected documents and request a new clearance.

View Documents (folder icon) : Opens the Document Verification tab of the slide-out panel, filtered to the documents that are pending or flagged by AI. This is the primary action for document review officers.

Submit to Customs System (upload icon, visible if api_integration is active and the declaration has not yet been submitted): Manually triggers the API call to submit the pre-filled customs declaration to the national system.

Issue Permit (stamp icon, visible if permit_issuance module is active and the shipment requires a government-issued permit): Opens the permit generation form, where the officer reviews the auto-filled application and digitally issues the permit.

View TCC (eye icon, common) : Opens the Trade Command Center with full shipment details, filtered to the government’s perspective (commercial data hidden, compliance data prominent).

Slide-out Panel Extra Tabs for Government Portal

When a shipment row is clicked, the slide-out panel provides several specialised tabs, depending on which modules are enabled.

Compliance Tab (always present)
This tab gives the officer a complete view of the shipment’s regulatory compliance status. It lists every document required for import/export in the officer’s jurisdiction, with AI verification results. Sanctions flags, PEP matches, and dual-use goods classifications are prominently displayed. A “Missing Documents” section highlights what is still needed, who is responsible, and provides one-click “Request Document” buttons that send a pre-filled message to the counterparty. The tab also shows the overall risk score breakdown and, if the multi_agency_workflow module is active, the status of all other agencies that must approve the shipment.

Multi-Agency Workflow Tab (if multi_agency_workflow module is enabled)
For shipments that require sequential or parallel approvals from multiple government bodies (e.g., Customs, Ministry of Agriculture, Ministry of Health), this tab renders a visual workflow stepper. Each step shows the agency name, the approval status (PENDING, APPROVED, REJECTED), the official who acted, the timestamp, and any comments. The officer can approve or reject their own agency’s step directly from this tab. A progress percentage indicates how many agencies have signed off. If the officer’s agency is not

the current active step, the interface is read-only. This tab is critical for coordinating complex regulatory approvals without separate communication channels.

Anonymous Trade Detail Tab (if anonymous_trade module is enabled and the shipment is an anonymous trade)
For shipments where the buyer and seller identities are protected (government-to-government or defence-related), this tab shows the redacted trade data that the officer is authorised to see. Party names are replaced with “Government Agency – [Country]”. The officer can view the HS code, weight, and required permits, but not the commercial invoice or the names of the counterparties unless explicitly authorised. A “Request Full Access” button initiates a multisig-governed declassification procedure, which, if approved by the Platform Governance Authority, temporarily grants the officer read access to the real identities for the duration of the review session. This access is logged, time-limited, and audited.

Customs Declaration Tab (if customs_declaration module is enabled)
This tab displays the fully populated customs declaration form, auto-filled by the AI using extracted trade data, invoices, packing lists, and certificates. The officer can review every field, edit if necessary, and then click “Approve & Submit” to send the declaration to the integrated national customs API. If no API is available, a “Download PDF” button generates the official declaration form for manual submission. A history log shows all previous submissions and responses.

Integration Log Tab (if api_integration module is enabled)
Provides a technical view of all API calls made to external systems for this shipment, including timestamps, request payloads, HTTP status codes, response bodies, and any error messages. This tab is intended for IT support staff or technically proficient officers troubleshooting connector issues.

Real-Time Updates Specific to Government

The Government vault subscribes to NATS events that are particularly relevant to border and regulatory processes. When a vessel’s AIS position indicates that it has entered the jurisdiction’s waters or is approaching the port, the vault highlights the row. New document uploads by the buyer or seller trigger the Document Verification queue to refresh, and any AI-detected anomalies cause the Anomaly Flags column to update immediately. Multi-agency approval events from other agencies are also streamed in real time, so an officer waiting for, say, the Agriculture Ministry’s clearance sees the status change without refreshing.

When the RIA detects a new sanction or regulatory change affecting the jurisdiction, the vault’s toolbar displays an alert banner: “New regulatory update affecting [Country X] shipments. Click to review impacted shipments.” The affected shipments are automatically filtered, and the officer can take immediate action to hold or review them.

Governance and Decision Panel Integration

Government actions are subject to the full Governor gating. If an officer attempts to approve a clearance that is blocked by a missing mandatory document or a multi-agency approval that has not yet been granted, the PlainLanguage Governor Decision Panel slides open and explains: “Clearance cannot be approved because the sanitary certificate from the Ministry of Health has not been verified. Please wait for the Health Ministry’s approval or contact their liaison.” The panel provides a direct link to the Multi-Agency Workflow tab. Similarly, if a new embargo is detected, any attempt to clear a shipment involving the sanctioned country is met with a DENY verdict and an explanation: “This shipment involves Country Y, which has been added to the autoblock list as of [date]. Clearance is permanently denied.”

All clearance decisions, holds, permit issuances, and multi-agency approvals are logged immutably as Governor decisions with Loom hashes, establishing a complete, court-admissible audit trail for every regulatory action taken on the platform.

This configuration makes the Shared Shipments Vault a sovereign, dynamic, and fully-integrated border management console that adapts to each agency’s mandate and module configuration, all within the zero-cost, self-hosted infrastructure of the Government Portal.

END OF SUB-SECTION 6.1.6.3.6 – GOVERNMENT PORTAL – ROLE-SPECIFIC COLUMNS

6.1.6.3.7 Admin Portal – Platform Governance Authority – Role-Specific Columns & Configuration

When the Shared Shipments Vault is rendered inside the Admin Portal, it is configured exclusively for the members of the Platform Governance Authority – the predefined 3-of-5 multisig that holds ultimate decision-making power over the entire SGTX platform. This is not the tenant-level Company Admin; it is the platform’s own constitutional oversight body. The vault therefore has the broadest possible scope – every shipment, every USTN, every financing agreement, and every distressed micro-contract across all tenants is visible in principle – but this visibility is constrained by the zero-knowledge governance model. By default, the vault shows only redacted, aggregated, or system-level indicators. Full, unredacted visibility into any specific trade requires a multisig-approved break-glass procedure that is logged, time-limited, and notified to the affected tenants. This dual-layer design preserves the platform’s promise of privacy and sovereignty while equipping the multisig with the operational intelligence it needs to detect systemic risks, investigate abuse, and ensure constitutional integrity.

In addition to the privacy-preserving vault columns described below, the Admin Portal includes two governance capabilities that are unique to the Platform Authority: the ability to set special SGTX FEES rates for agreed pairs of GTIDs (a privileged form of fee customization), and an integrated AI-powered Customer Care Chatbot that can impersonate tenants (with PIN-based approval) to resolve issues automatically, escalating to a human support agent only when necessary. These capabilities are tightly governed, fully logged, and are accessed from dedicated tabs within the Admin Portal, though their triggers may appear in the vault’s row actions.

Zero-Knowledge Visibility Model for the Vault

In day-to-day operation, the Admin vault applies a restrictive data privacy filter enforced by the Governor Proxy. This filter blocks access to:

Full legal names of buyers and sellers (only country codes and GTID prefixes are shown)

Exact commodity descriptions (only 2-digit HS chapter, e.g., “08 – Fruit”, not “0805.10 Valencia Oranges”)

Port details (only country codes for origin and destination)

Trade values, fee amounts, and finance principal amounts

Counterparty trust scores (only a system-level risk flag)

Identities of logistics providers (only the service type and a system-generated risk score)

What remains visible is exactly what the Platform Authority needs for systemic oversight:

Anonymised USTN (e.g., “SGTX-EG-...-V1”)

Trade corridor (country pair)

Commodity category

Platform risk score

Governance state flags (dispute, overdue milestone, fee payment pending, stuck trade, sanctions alert)

SLA compliance indicators (e.g., “3 of 5 milestones confirmed on time”)

Aggregated system metrics (not per-trade, but available in the toolbar)

When a specific investigation is required – for example, a tenant reports a persistent fee calculation error, or a dispute pattern is detected in a particular corridor – any multisig member can initiate a “Request Full Audit View” action from the vault. This action requires the approval of at least two additional multisig members (i.e., a 3-of-5 quorum). Once approved, the admin who initiated the request is granted a time-limited, fully-logged, read-only session (maximum 15 minutes) that reveals all redacted fields for the specific USTN or entity. A persistent red banner displays “FULL AUDIT MODE – SESSION LOGGED”. Every page view and API call during this session is recorded as a Governor decision with decision_type = ’admin_audit_view’, including the identities of all approving multisig members and the precise data accessed. After the session expires, the admin automatically reverts to the redacted view, and an automated email notification is sent to the tenant’s registered admin contact: “The Platform Governance Authority conducted a compliance review of your trade [USTN] on [date/time]. No action was taken.” This process balances transparency and accountability.

Common Columns for Admin Portal Vault

The following columns appear in the Admin’s day-to-day vault. They are built on top of the anonymised common columns (truncated USTN, status, corridor).

USTN Mask – The USTN is truncated to its first segment: “SGTX-EG-...” (or “SGTX-ANON-...” for anonymous trades). It is not clickable in day-to-day mode, preventing accidental access. Only during an active audit session does it become a full clickable link to the Trade Command Center.

Commodity Category – Two-digit HS chapter (e.g., “08”) or a broad category label (e.g., “Perishables”, “Electronics”). This is sufficient to detect sector-specific risk patterns.

Corridor – Origin country code → Destination country code (e.g., “VN→EG”). No port details are shown.

Platform Risk Score – Full 0-100 value, colour-coded. This is essential for governance; high-risk corridors need no identification of the specific parties to trigger a review.

Governance Flags – A set of compact icons that summarise the operational status of the trade from the platform's perspective. Icons include:

Active Dispute (red gavel)

Overdue Milestone (amber clock with exclamation)

Fee Payment Pending (dollar sign with hourglass)

Stuck Trade (anchor with a red cross)

Sanctions Alert (shield with a warning triangle)

Multi-shipment Anomaly (branching arrow with a yellow dot)
Each icon's tooltip provides a one-sentence, jargon-free explanation generated by Groq. For example: "Shipment 2 of 3 is 48 hours past its loading window. Escalation reminder sent to seller."

Platform Health Metrics (Aggregated) – Not a column per row, but a live summary bar above the table shows: total active trades, percentage of milestones within SLA, fee collection success rate, and current dispute ratio. This is always visible and is drawn from ClickHouse aggregations.

Break-Glass Indicator – A small, discreet dot next to each row. Green means "no access requested". If an audit session is active for that row, the dot turns red and a tooltip reads "Under active audit by [multisig member]".

Row Actions in Admin Portal Vault

In addition to the anonymised common actions, the Admin vault offers:

Request Full Audit View (shield with a key icon) : Initiates the multisig approval flow for break-glass access to the full trade details. The requesting admin provides a mandatory reason (free text). The request is broadcast to all other multisig members via Smart Inbox. Once the quorum approves, the view unredacts, and a Loom-logged Governor decision is created.

Force Milestone (hammer icon, visible only during an active audit session and if an emergency override is justified): Allows the Platform Authority to manually confirm or adjust a milestone. This action requires the same multisig approval as the audit view, plus an additional confirmation step with a reason. It is intended solely for extreme cases (e.g., a bug preventing legitimate confirmation). The action is permanently logged and can be reviewed by any tenant who is affected.

Initiate Special Rate (percentage icon) : This action is available on any row (in day-to-day or audit mode) and opens the Special SGTX FEES Rate management interface, pre-filled with the buyer and seller GTIDs from the selected trade. This enables the multisig to create a customised fee rate for that specific GTID pair. The process is described in the next section.

View Decision Log (scroll icon) : Opens the full Loom-chained Governor decision log for the trade (only the sequence of verdicts, not the underlying data, unless in audit mode).

Special SGTX FEES Rate Management

The Platform Governance Authority can, via a dedicated tab in the Admin Portal or from the vault's row action, define a special SGTX FEES rate that overrides the dynamically computed SGTX trade fee for a specific pair of GTIDs – for example, "SGTX-VN-TRD-002" (a particular seller) and "SGTX-EG-TRD-001" (a particular buyer). This capability is intended for strategic partnerships, pilot programs, or long-term volume agreements that fall outside the standard commodity- and country-pair model. It is not a backdoor for arbitrary discounts; every special rate is bound by the same constitutional floor (0.1%) and ceiling (2.5%) and must be approved by the multisig.

Workflow to create a special rate:

An admin initiates the action from the Admin Portal's "Special Rates" tab, or by using the "Initiate Special Rate" action on a vault row. The interface prompts for:

Seller GTID (pre-filled if from vault)

Buyer GTID (pre-filled if from vault)

New fee rate (percentage, within 0.1%–2.5%)

Effective date range (start and optional end)

Reason (mandatory, free text, e.g., "Strategic partnership – high volume corridor")

The proposal is submitted as a multisig request. All other multisig members receive a Smart Inbox item: "Special rate proposal: [Seller] → [Buyer] at X%. Review and approve or reject."

If the quorum (3-of-5) approves, the Governor records a decision of type 'special_rate_override' and inserts a row into the special_SGTX FEES_rates table. This table includes the buyer_gtid, seller_gtid, rate, effective_from, effective_to, and the Governor decision ID.

When a trade is initiated between those two GTIDs, the Fee Calculator checks the special_SGTX FEES_rates table before falling back to the dynamic model. If a matching active special rate exists, it uses that rate. The seller still pays the fee, and the buyer still sees the total price with the fee component. The AI Margin Estimator is informed of the override so that the LLM refinement does not incorrectly adjust it.

Special rates are visible in the Admin Portal's "Special Rates" manager, which lists all active and expired overrides with their justification and approval history. Tenants cannot see this list. When a special rate expires, the dynamic model resumes automatically.

All special rate actions are logged immutably and can be audited via the Configuration History tab.

AI-Powered Customer Care Chatbot

The Admin Portal includes a Customer Care tab that hosts a conversational AI agent (authority level A1 for advisory, A3 for actions requiring impersonation). This chatbot is the first line of support for tenant inquiries. It is capable of diagnosing problems, retrieving relevant trade data, and even performing limited actions on behalf of the tenant – but only with the tenant's explicit, PIN-based consent. The bot never overrides the tenant's autonomy; every sensitive action is approved by the tenant before execution.

Tenant-Initiated Chat Flow:

Access : From any portal, a tenant clicks the "Help" or "Chat with Support" button, which opens a chat panel powered by the customer-care-service (Rust, Axum, integrated with Groq for natural language and HF Mixtral for intent extraction).

Initial Triage : The AI bot greets the tenant and asks them to describe their issue. It parses the query using natural language understanding. If the question is about a specific USTN, the bot can reference that trade's data only after the tenant provides the USTN – it does not have access to all tenant trades by default.

Automated Resolution : For common, non-sensitive issues ("I can't find my missing document", "Why is my contract blocked?", "How do I pay the SGTX fee?"), the bot provides direct, accurate answers by querying the Governor's tenant_message, the document checklist, or the fee payment status – all through the standard, permission-scoped API. It can also generate pre-filled links to the exact form.

Action with PIN Approval : If the bot identifies that it can resolve the issue by performing an action on the tenant's behalf (e.g., "I need to reset my barcode printer settings", "Please send a reminder to the seller for the missing phytosanitary certificate"), it proposes the action: "I can send a pre-filled reminder to the seller for the missing certificate. This will be logged. Do you want me to proceed?" If the tenant agrees, the bot requests the tenant's 6-digit support PIN. This PIN is set by the tenant admin in Company Admin → Support Settings and is never stored in plaintext; it is verified via a one-time challenge. Once the tenant enters the correct PIN, the bot performs the action by temporarily impersonating the tenant's employee account. This impersonation is:

Narrowly scoped – only the specific API call(s) needed for the task.

Fully logged as a Governor decision with decision_type = 'customer_care_bot_impersonation', including the action, the tenant's GTID, the USTN, and the timestamp.

Notified to the tenant via Smart Inbox : "Customer Care Bot sent a document request on your behalf at [time]."

Escalation to Human : If the bot fails to resolve the issue after three attempts, or if the tenant explicitly types "talk to a human" or "phone support", the bot immediately offers to escalate. It asks the tenant to confirm their preferred contact number (from the tenant's profile) and schedules a callback. The escalation ticket appears in the human agent's queue within the Admin Portal's "Human Support Queue" tab (accessible only to designated support personnel, who are distinct from the Platform Governance Authority). The human agent then contacts the tenant by phone, using a self-hosted VoIP bridge (e.g., Asterisk or a simple WebRTC session) that encrypts the call and does not rely on third-party telephony APIs. The call is not recorded by default; if recording is required for compliance, the tenant is informed and must consent.

Human Agent Interface : Support agents have a dedicated dashboard that shows queued tickets, the tenant's identity, the USTN (if applicable), the chatbot's transcript, and any actions the bot already performed. The agent can continue the conversation via the same chat interface (text) or, once on the phone, they have full context. Agents are bound by the same break-glass access rules: they can only see full trade details if the tenant provides explicit permission or if the multisig approves a temporary audit view for support purposes.

Zero-Cost Technology Stack for the Chatbot:

Chat API and intent handling : Rust Axum + Rig (agent framework)

NLP : HF Mixtral (local) for intent extraction; Groq (free tier) for response generation and tenant-friendly explanations

PIN verification : bcrypt hash stored in employees.support_pin_hash; verified via a challenge-response

Impersonation : ZITADEL session override with scope-limited JWT

Logging: Loom for all bot-initiated actions

VoIP : Self-hosted Asterisk or WebRTC (using open-source Janus Gateway) – no external telephony cost

Integration with the Rest of the Blueprint

The addition of the special SGTX FEES rates requires a new database table (special_SGTX FEES_rates) in Part 5 and a reference in Part 7 (Revenue Model). The customer care chatbot requires the customer-care-service in Part 4 and a new tab "Customer Care" in Part 6.2.7. The zero-knowledge vault columns and the break-glass procedure must be reflected in Part 13 (Governor policies for admin audit access) and Part 14 (audit logs). These changes are consistent with the v6.3 philosophy and can be incorporated without altering the platform's core constitution.

END OF SUB-SECTION 6.1.6.3.7 – ADMIN PORTAL – PLATFORM GOVERNANCE AUTHORITY – ROLE-SPECIFIC COLUMNS & CONFIGURATION

6.1.6.4 Common Row Actions Across All Portals (Permission-Based)

This sub-section defines the set of row-level action buttons that appear in every instantiation of the Shared Shipments Vault, regardless of the user's portal or role. These actions are rendered as small icon buttons within the "Quick Actions" column (or a dedicated actions column) and are visible or hidden based on two factors: the authenticated employee's permissions (as defined by their role and data scopes) and the current state of the shipment. A truck driver, for example, will see "Confirm Milestone" but never "Issue Margin Call"; a buyer in the Trader Portal will see "Request Document" but not "ReOptimise". The actions listed here are common across all portals; the portal-specific actions (e.g., "Start Inspection" for QC, "Accept Job" for logistics) are described in their respective sub-sections (6.1.6.3.1 through 6.1.6.3.7).

Every action button has a tooltip that displays its name and, if the action is currently unavailable due to state or permission, a brief explanation. For example, a "Confirm Milestone" button may be greyed out with the tooltip: "You cannot confirm this milestone because the container release confirmation has not yet been acknowledged." All actions that modify trade state pass through the Governor and are subject to the constitutional rules; if a Governor verdict is DENY or CONDITIONAL, the PlainLanguage Governor Decision Panel opens instead of executing the action.

View Trade Command Center

Icon: eye

Permission : none (basic visibility of the shipment is required, which is implied by the vault's row being present)

Description : Opens the full Trade Command Center for the selected USTN in a new browser tab. For multi-shipment contracts, the TCC opens for the specific shipment if a shipment USTN is available, or for the master contract if the row represents the master. This action is always available and serves as the primary navigation method from the vault into the detailed trade view.

Portals: All.

Request Document

Icon: paperclip with a plus sign

Permission: document.request

Description : Initiates a pre-filled message to the counterparty (or the responsible logistics provider) requesting a specific missing document. The system presents a modal listing all currently missing required documents, with checkboxes to select which ones to request. Upon confirmation, a Groq-generated message is posted in the trade room, in the recipient's preferred language, and a new Smart Inbox item is created for the responsible party. This action is only available when at least one required document has a status other than "Verified" and the responsible party is not the requesting user's own tenant. For the buyer, this typically means requesting seller-side documents; for the seller, requesting buyer-side or third-party documents.

Portals : Trader Portal (both Buyer and Seller dashboards), Logistics Portal, Government Portal.

Upload Document

Icon: upward arrow with a tray

Permission: document.upload

Description : Opens the document upload form pre-filtered to the selected shipment's document requirements. The user can drag-and-drop a file or select one from their device. The file is immediately scanned by the AI document pipeline (HF Donut + PaddleOCR) and cross-referenced against the contract requirements. If the AI verification passes with confidence ≥85%, the document is marked as "Verified" and the shipment's readiness indicator updates. If confidence is lower, the document is flagged for manual review. This action is only available when the current user's tenant is listed as the responsible party for at least one pending document.

Portals : All, applicable when the user's tenant is the responsible party.

Confirm Milestone

Icon: flag

Permission: shipment.milestone.confirm

Description : Allows the user to manually confirm a logistics milestone for which they are responsible. The modal lists the next milestone(s) pending their confirmation. The user selects a milestone, optionally provides a note, and signs the confirmation with their ZITADEL passkey. The Governor then validates that the container release (if applicable) has been acknowledged by the logistics provider and that the FeeLock is active. If the milestone is the final one in a container, the container's loaded/delivered status is updated and the SGTX FEESLock release conditions are evaluated. This action is also available via voice command on the mobile app. It is critical for logistics providers updating pickup, departure, and delivery events, and for sellers/buyer confirming loading completion.

Portals : Trader Portal (Buyer and Seller dashboards for milestones they are obligated to confirm), Logistics Portal (all roles).

Add to Watchlist

Icon: star

Permission: none

Description : Toggles the "watched" status of the shipment for the current user. A watched shipment appears in a dedicated "Watchlist" filter within the vault and may generate lower-threshold Smart Inbox items (e.g., a general update that would normally be priority 20 might be bumped to priority 40 if the shipment is watched). This is a purely personal preference and does not affect other users. The star icon is filled when watched, empty when not.

Portals: All.

Export Shipment Data

Icon: download arrow

Permission : tenant.export (or a scoped permission for export)

Description : Triggers a micro-export of the selected shipment's summary data as a CSV or PDF. The export includes the USTN, status, milestone timeline, document checklist status, and, for users with finance permissions, the trade value and fee summary (buyer sees total price; seller sees cost breakdown). The file is generated server-side and downloaded immediately. For government auditors, the export is cryptographically signed and includes Loom hashes. This action is always visible but may be greyed out if the user lacks export permissions.

Portals: All, subject to export permissions.

ReOptimise Logistics Bundle

Icon: refresh arrows with a truck

Permission: logistics.bundle.reoptimise

Description : Available only in the Seller dashboard of the Trader Portal. This action triggers a re-evaluation of the logistics bundle (carrier selection) for the shipment. The system fetches updated carrier risk scores, available quotes, and route data, and presents a side-by-side comparison (diff view) of the current selected providers versus the AI-recommended optimal bundle. The seller can then accept the new recommendation, which updates the selected quotes, or dismiss it. This action is visible only when the shipment is not yet in transit and a logistics bundle has already been selected. It is a key cost-optimisation tool for sellers.

Portals: Trader Portal – Seller Dashboard only.

Issue Margin Call

Icon: exclamation triangle with a dollar sign

Permission: financing.margin_call

Description : Available only in the Financier Portal for shipments linked to active financing agreements. This action opens the margin call form, pre-filled with the current LTV, the required additional collateral to restore a safe threshold, and a deadline. The financier can adjust the amount and message, then submit the margin call. The Governor checks that the LTV exceeds the configured threshold and that a margin call is not already active. Upon submission, the borrower receives a high-priority Smart Inbox item, and the loan's margin call log is updated.

Portals: Financier Portal only.

Approve Clearance / Hold / Reject

Icon : green shield (Approve), red hand (Hold), red X (Reject)

Permission: customs.clearance.approve (or .hold, .reject)

Description : Government portal actions for customs clearance. "Approve Clearance" releases the goods and pushes the status to the integrated customs API if available. "Hold" pauses the clearance process with a mandatory reason. "Reject" definitively denies clearance, requiring the buyer/seller to correct documents and reapply. All actions are logged as Governor decisions and are subject to the agency's dynamic module configuration.

Portals : Government Portal only, visible when the relevant customs modules are enabled.

Request Full Audit View (Admin)

Icon: shield with a key

Permission : admin.audit_view (restricted to members of the Platform Governance Authority multisig group)

Description : Available only in the Admin Portal. This action initiates the break-glass procedure to view the full, unredacted details of a shipment. The requesting admin must provide a reason. The request is broadcast to all other multisig members; at least two additional approvals are required. Once approved, the admin's session for that trade becomes "Full Audit Mode" for 15 minutes, revealing legal names, exact commodities, ports, values, and all documents. A red banner is shown, and every page view is logged. After expiry, the view reverts to redacted, and the tenant is automatically notified. This action is the only way the Platform Governance Authority can access private trade data operationally.

Portals: Admin Portal only.

Force Milestone (Admin)

Icon: hammer

Permission: admin.force_milestone (multisig-gated)

Description : A rare emergency override available only in the Admin Portal during an active Full Audit Mode session. It allows the Platform Governance Authority to manually confirm or adjust a shipment milestone that is stuck due to a verified platform bug or an exceptional circumstance beyond the tenants' control. The action requires the same multisig quorum as the audit view, plus an additional confirmation with a detailed justification. It is permanently logged and can be reviewed by any affected tenant.

Portals: Admin Portal only.

Initiate Special Rate (Admin)

Icon: percentage symbol

Permission: admin.special_rate (multisig-gated)

Description : Available in the Admin Portal, this action opens the Special SGTX FEES Rate proposal form, pre-filled with the buyer and seller GTIDs from the selected trade. It allows the Platform Governance Authority to propose a custom trade fee rate for that GTID pair, subject to constitutional bounds (0.1%–2.5%) and multisig approval. This action is described in full in 6.1.6.3.7.

Portals: Admin Portal only.

Chat with Customer Care Bot (Initiate Support)

Icon: speech bubble with a question mark

Permission: none (all users)

Description : This action opens the AI-powered Customer Care Chatbot panel directly from the vault, with the context of the selected shipment pre-loaded (USTN, current status, and recent document/milestone history). The bot will use this context to provide more accurate, faster support without requiring the tenant to explain which trade they are referring to. The bot can diagnose issues, retrieve trade data, and, with the tenant's explicit PIN-based approval, perform certain support actions on their behalf. If the bot cannot resolve the issue, it will offer to escalate to a human support agent, who will call the tenant's registered phone number using the platform's self-hosted VoIP. This action is the front-line entry point to the customer care system described in 6.1.6.3.7.

Portals : All portals (Buyer, Seller, Logistics, QC, Financier, Government, Admin). In the Admin Portal, it connects to a separate internal support queue for platform-level issues.

Governance and Permissions Enforcement

Every row action button is rendered based on a client-side evaluation of the user's permission set (embedded in the JWT) and the shipment's state, but the definitive enforcement is server-side. When an action button is clicked, the frontend sends an API request to the appropriate service endpoint. The Governor Proxy intercepts this request, evaluates the OPA policies (which include the user's permissions, the active dual-mode context, the shipment's status, and any jurisdictional constraints), and either allows the action, denies it with a plain-language explanation, or returns a CONDITIONAL verdict with remediation steps. This ensures that even if a button is erroneously rendered due to a client-side bug, no unauthorised action can be performed.

For the Trader Portal's dual-mode toggle, the row actions are aware of the active context. A user in Buyer mode will never see "ReOptimise" (seller-only) even if they have the technical permission; the OPA policy for that action requires active_trader_mode_context = SELL. Similarly, a user in Seller mode will not see "Approve Clearance" (buyer/government action). This contributes to a clean, uncluttered interface.

This complete set of common row actions, combined with the portal-specific actions in 6.1.6.3.1–6.1.6.3.7, transforms the Shared Shipments Vault from a passive data table into an active command centre that aligns exactly with each user's responsibilities and permissions.

END OF SUB-SECTION 6.1.6.4 – COMMON ROW ACTIONS ACROSS ALL PORTALS (PERMISSION-BASED)

6.1.6.5 Global Filters, Search, and Saved Filter Sets (All Portals)

The Shared Shipments Vault provides a consistent, powerful, and intuitive filtering and search system that works identically across all portals—whether the user is a Buyer, Seller, logistics provider, QC inspector, financier, government official, or platform administrator. The filtering engine is server-side, powered by PostgreSQL full-text search and indexed query parameters, ensuring that even vaults containing hundreds of thousands of shipments respond within milliseconds. The filter UI is rendered as part of the vault's toolbar and is designed to be discoverable, keyboard-accessible, and easily resettable. Every user can create, save, and reload personalised filter sets that persist across sessions and devices. The system respects the user's active dual-mode context (Buyer or Seller) to pre-select sensible defaults, but the user is always free to override them.

Search Bar

The search bar is the most prominent element in the toolbar. It accepts free-form text and performs a real-time, server-side search across multiple indexed fields. The search is implemented using PostgreSQL's tsvector columns, which maintain pre-computed lexemes for the following fields: shipments.ustn, shipments.booking_number, shipments.vessel_name, tenants.legal_name (for buyer and seller), and trade_requests.raw_description (for any legacy free-text trade descriptions). The search is case-insensitive, ignores diacritics, and supports partial matches. For example, typing "Mek" will return all shipments involving Mekong Fresh as either the buyer or seller.

A debounce of 300 milliseconds prevents excessive server load while the user is typing. However, pressing Enter immediately triggers the search, bypassing the debounce. The search bar also includes a small "x" button to clear the current search term and restore the previous filter state. When a search term is active, the vault's result set is narrowed to rows that match, and the matching text in the visible columns (e.g., a vessel name or counterparty name) is highlighted with a subtle background colour. The search term is automatically saved as part of any saved filter set.

Filter Chips

Immediately below the search bar (or wrapping to the next line on smaller viewports), a row of compact, multi-select filter chips provides access to the most common filtering dimensions. Each chip, when clicked, opens a dropdown panel that contains checkboxes, a slider, or a date picker, depending on the dimension. The chosen values are displayed as a summary on the chip itself (e.g., “Status (3)”, “Ports: SGN→ALEX”). An active chip has a small coloured dot and a count badge. The chips are additive: selecting values in one chip does not reset the others. The filter is applied as soon as the user confirms the selection (by clicking “Apply” in the dropdown or simply by closing it, depending on the chip type).

The following standard filter chips are available in every portal:

Status Chip : Opens a panel with status values grouped into logical categories: Pre-shipment (CREATED, GATED_IN, LOADED), In Transit (DEPARTED, IN_TRANSIT), At Destination (ARRIVED, CUSTOMS_IMPORT), and Closed (DELIVERED, DISPUTED, CANCELLED, DISTRESSED). The user can select any combination of statuses. A “Select All” / “Clear All” link is provided. The number of selected statuses is reflected on the chip badge. For the Buyer dashboard, the default pre-selection includes inbound-relevant statuses; for the Seller dashboard, outbound-relevant statuses. The user can override these by simply checking or unchecking boxes.

Commodity Chip : Opens a searchable, hierarchical tree list of HS chapters (2-digit codes) and their 4-digit sub-headings. Selecting a chapter automatically includes all its sub-headings. The label shows the HS code if a specific commodity is selected, or “All Commodities” otherwise. This chip is useful for a trader dealing primarily in one product category to quickly filter the vault.

Port Chip : Opens two separate autocomplete fields: one for the origin port and one for the destination port. Each field queries the global ports table using the UN/LOCODE. The user can fill in one or both. The chip label updates to reflect the selection, e.g., “Origin: SGN” or “SGN → ALEX”. This chip is essential for logistics providers and government officials monitoring specific trade corridors.

Risk Score Chip (Government and Admin portals only) : Displays a dual-handle range slider (0 to 100). The user can drag the lower and upper bounds to define a range, or click preset buttons: “Low (0-25)”, “Medium (26-70)”, “High (71-100)”. The chip label shows the current range. This chip enables risk-based inspection for customs officers and allows platform administrators to focus on high-risk segments.

Date Chip : Opens a panel with quick-select presets (“Last 7 days”, “Last 30 days”, “Next 7 days”, “Next 30 days”) and a custom date range picker (using a calendar widget). The filter is applied to the ETD and ETA columns: a shipment is included if its ETD or ETA falls within the selected range. The chip label shows the chosen range, e.g., “Jun 1 – Jun 15”. The default for the Buyer dashboard is often “Next 7 days” (ETA focus); for the Seller dashboard, “Next 14 days” (ETD focus). This is customisable.

Saved Filters Chip : A dropdown listing all of the user's previously saved filter sets. Selecting one instantly applies all the saved criteria—search term, statuses, commodity, ports, date range, risk score—and updates all other chips accordingly. The dropdown also contains a "Save current filters" option and a "Manage Filters" link that opens a modal where the user can rename or delete saved sets.

All filter chips can be dismissed individually by clicking the small "x" on the chip. A "Clear All Filters" link appears at the end of the chip row whenever any filter is active, restoring the vault to its default state for the user's role. The active filters are also reflected in the URL query string (e.g., ?status=LOADED,DEPARTED&search=Mekong&commodity=08), enabling deep linking and browser back/forward navigation.

Active / Archive Toggle and Its Interaction with Filters

The segmented toggle at the top of the vault—Active / Archived—acts as a pre-filter that is orthogonal to the chip-based filters. When "Active" is selected, the vault shows shipments with statuses defined as active for the user's portal (e.g., for a Buyer, all statuses up to DELIVERED). When "Archived" is selected, the vault shows shipments that are completed, cancelled, or disputed. The status chip automatically updates to reflect the active/archive selection: if the user toggles to "Archived", the status chip switches to the archived statuses (DELIVERED, COMPLETED, CANCELLED, DISPUTED) and any previously selected active statuses are cleared. The user can still manually add an active status in the status chip while in "Archived" mode, but the toggle provides a convenient single-click view switcher.

Dual-Mode Context and Default Presets

For the Trader Portal, the vault's initial filter state depends on the active dual-mode toggle context. In Buyer mode, the vault defaults to:

Status : GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED, CUSTOMS_IMPORT

Date: Next 7 days (ETA)

Active/Archive: Active

In Seller mode, the defaults are:

Status : CREATED, GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED

Date: Next 14 days (ETD)

Active/Archive: Active

These defaults are only applied when the vault is first loaded in that session, or when the user explicitly clicks "Reset to [Buyer/Seller] Defaults". The user's subsequent filter changes are remembered via the saved filter set mechanism and are reapplied unless the user resets them.

Saved Filter Sets

Any combination of active search term, filter chips, sort column, and sort order can be saved as a named filter set. The "Save current filters" option in the Saved Filters chip prompts the user for a name (e.g., "My Citrus Shipments – Egypt"). The set is stored in the user_preferences.filter_sets JSONB column and is immediately available in the dropdown. Saved filter sets are synchronised across the user's devices via a NATS user-preference event, so a set created on the desktop app appears on the mobile app. The platform does not limit the number of saved sets per user.

When a saved filter set is loaded, the vault re-fetchces data with all the stored criteria. If the set includes a status filter that conflicts with the current Active/Archive toggle, the toggle automatically switches to include those statuses (with a brief notification). Saved sets can also be shared as a URL: clicking "Copy Link" in the Saved Filters dropdown copies a full URL with all query parameters, which can be sent to a colleague. When the colleague opens the link, their vault is populated with the same filters, but their own permissions and data scopes still apply, so they may see fewer rows.

Performance Considerations

Behind the scenes, the vault's filter and search queries are served by the /v1/shipments-vaultendpoint, which translates the URL query parameters into optimised SQL queries. Frequently accessed filter combinations (e.g., the default Buyer active filter) are cached in Valkey with a short TTL (30 seconds) to provide sub-100-millisecond responses for common requests. The cache is invalidated immediately when any shipment in the result set is updated (via NATS-driven cache busting). For complex full-text searches, PostgreSQL's GIN indexes on the tsvector columns ensure that even searches across millions of rows complete within tens of milliseconds. The frontend employs optimistic UI: after a filter change, the table is cleared and a subtle loading skeleton is shown, replaced by the new data as soon as the API response arrives.

Accessibility and Keyboard Navigation

The entire filter bar is keyboard-accessible. The Tab key moves focus through the search bar and each filter chip; Enter opens the chip's dropdown; arrow keys navigate within the dropdown; Space toggles checkboxes; Escape closes the dropdown and returns focus to the chip. The search bar supports standard shortcuts (Ctrl+F focuses it). Screen readers announce the number of active filters and the result count whenever a filter is applied, using an aria-live region below the toolbar. All filter chips have clear aria-label attributes describing their current state.

END OF SUB-SECTION 6.1.6.5 – GLOBAL FILTERS, SEARCH, AND SAVED FILTER SETS (ALL PORTALS)

6.1.6.6 Real-Time Updates (All Portals)

The Shared Shipments Vault is not a static snapshot. It is a live, event-driven component that reflects the current state of every shipment without requiring the user to manually refresh the page. The real-time update mechanism is built on NATS JetStream, the platform's asynchronous messaging backbone. Every state-changing event—a milestone confirmation, a document upload, a status transition, a new disruption prediction, a financing repayment, a container release acknowledgment—is published by the responsible microservice to a dedicated NATS subject. The vault's client-side WebSocket connection subscribes to these subjects for all visible shipments and updates the relevant table row or detail panel immediately.

The following subjects drive the vault's real-time behaviour:

shipment.status.changed.{ustn} – fires when the shipment's overall status changes (e.g., from LOADED to DEPARTED). The vault updates the Status badge with a brief pulse animation and, if the new status no longer matches the active filters, the row fades out after one second.

shipment.milestone.confirmed.{ustn} – fires when any milestone is confirmed. The vault updates the Loading Status, the milestone timeline in the slide-out panel, and recalculates the Compliance Flags if an overdue milestone is resolved.

document.uploaded.{ustn} – fires when a required document is uploaded and AI-verified. The vault updates the Documents Missing count, the Customs Readiness or Documents Readiness indicator, and the document checklist in the slide-out panel.

contract.status.changed.{trade_request_id} – fires when a contract is locked, disputed, or terminated. The vault refreshes the commercial status context.

fee.payment.confirmed – fires when a fee payment is verified. The vault updates the Fee Payment Pending flag and the SGTX FEESLock status (for Admin view).

financing.repayment.received, financing.margin_call.issued, collateral.value.changed– fires for financiers, updating the Outstanding Balance, LTV, and Repayment Health columns.

distressed.listing.created, distressed.offer.submitted – updates the Distressed Information card for the relevant shipment.

multi_shipment.shipment.locked – when a shipment within a multi-shipment contract is locked, the vault updates that shipment’s row and the master contract’s aggregated progress bar.

container.release.acknowledged – for logistics providers, the vault updates the milestone accessibility and the Release Confirmation badge.

When a relevant event is received, the vault’s frontend listener performs an optimistic update: it applies the changes directly to the local Zustand store without a full page reload. Only the specific fields referenced in the event payload are patched. The updated row is briefly highlighted in a pale colour to draw the user’s attention. If the event indicates that the row should be removed from the current filtered view (e.g., a status change makes it no longer match the selected statuses), the row is removed with a smooth fade-out animation, and a small toast notification appears at the bottom of the vault: “Shipment SGTX-... has been delivered and moved to Archived.”

In the Trader Portal, the real-time updates respect the dual-mode context. When the toggle is set to Buyer, the vault subscribes only to events for shipments where the authenticated tenant is the buyer. Switching to Seller immediately resubscribes to the seller’s shipments. This prevents unnecessary network traffic and ensures that no cross-role data leaks into the UI.

If the WebSocket connection is lost (e.g., network interruption), the vault automatically retries with exponential backoff and displays a non-intrusive yellow banner at the top of the table: “Live updates paused. Retrying in 10 seconds…” During this time, the vault continues to serve stale data from the client-side cache. Once the connection is re-established, the banner disappears, and a full re-fetch is triggered to synchronise any missed events. For offline-first scenarios (mobile app or desktop app with intermittent connectivity), the same NATS-based replenishment occurs via the background sync engine.

6.1.6.7 Accessibility and Responsive Design (All Portals)

The Shared Shipments Vault is built to meet WCAG 2.2 Level AA standards in every portal. All interactive elements—buttons, filter chips, table rows, and slide-out panel controls—are fully operable by keyboard. The tab order follows the visual layout: first the toolbar controls (search, filter chips, toggles), then the table header and rows, and finally the pagination bar. A “Skip to main content” link at the top of the vault allows keyboard users to bypass the toolbar and jump directly to the first shipment row.

Within the data table, the Arrow Up and Arrow Down keys move the focus ring between rows. Pressing Enter on a focused row opens the slide-out detail panel. When the panel is open, focus is trapped inside it until the user presses Escape to close it. Every column header is a button that announces its sort state (ascending, descending, or unsorted) to screen readers via aria-sort. The row selection checkboxes have an accessible label derived from the USTN, e.g., “Select shipment SGTX-EG-...-V1”.

All status badges, risk scores, and colour-coded indicators include a text label and an icon, ensuring that colour is never the sole means of conveying information. The urgency colours (red, amber, green) pass a minimum contrast ratio of 4.5:1 against the background. Charts and sparklines used in the LTV or Margin columns have an alternative text description accessible to screen readers, and a data table view is available as a fallback for the LTV history chart.

Responsive design is handled through Tailwind CSS breakpoints. On viewports narrower than 768 pixels (tablet and mobile), the vault’s layout changes significantly:

The toolbar wraps into two lines, and excess filter chips collapse into a “More Filters” dropdown.

The table transforms into a card list, where each shipment is a card showing the most critical fields: USTN, status, vessel, ETA, and the two most role-relevant additional columns (e.g., Customs Readiness and Documents Missing for Buyer; Estimated Margin and Loading Status for Seller). A “Show details” expandable section reveals the remaining columns.

The slide-out panel becomes a full-screen modal with a back button.

Touch targets for action buttons and filter chips are enlarged to at least 44×44 pixels.

The Active/Archive toggle and view toggles remain fixed at the top for quick access.

On very small screens (mobile phones, <480px), the column customisation is automatically simplified, and the user can swipe through a carousel of shipment cards. The vault is tested on real iOS and Android devices using the SGTX mobile app’s WebView, ensuring parity with the desktop experience.

6.1.6.8 Configuration API and Developer Integration

The Shared Shipments Vault is designed to be instantiated from any SGTX portal by passing a configuration object of type ShipmentsVaultConfig. This object, defined in TypeScript, allows developers to specify which columns are visible, which row actions are available, what the default filters should be, and any extra tabs for the slide-out detail panel. The vault component itself is a reusable React component located in packages/shared-ui/src/components/ShipmentsVault.

The configuration interface is as follows:

typescript

```
interface ShipmentsVaultConfig {
visibleColumns: Array<
'ustn' | 'status' | 'vessel' | 'originDest' | 'etdEta' |
'sellerName' | 'buyerName' | 'customsReadiness' | 'documentsMissing' |
'estimatedMargin' | 'loadingStatus' | 'logisticsQuotes' |
'serviceType' | 'myQuoteStatus' | 'inspectionStatus' | 'defectCount' |
'financedAmount' | 'outstandingBalance' | 'ltv' | 'repaymentHealth' |
'riskScore' | 'clearanceStatus' | 'anomalyFlags' | 'tenantName'
>;
columnRenderers?: Record<string, (value: any, row: any) => ReactNode>;
defaultFilters?: {
status?: string[];
dateRange?: [Date, Date];
riskScoreMin?: number;
riskScoreMax?: number;
commodity?: string;
mode?: 'BUYER' | 'SELLER';
};
rowActions: Array<
'viewTcc' | 'requestDocument' | 'uploadDocument' | 'confirmMilestone' |
'reoptimise' | 'marginCall' | 'approveClearance' | 'holdClearance' | 'rejectClearance' |
'startInspection' | 'acceptJob' | 'updateMilestone' | 'signAddendum' | 'acknowledgeRelease' |
'requestAuditView' | 'forceMilestone' | 'initiateSpecialRate' | 'chatSupport'
>;
extraTabs?: Array<{ id: string; label: string; component: React.ComponentType<{ ustn: string }> }>;
mapEnabled: boolean;
refreshInterval?: number;
}
```

Each portal imports the vault and passes its specific config. For example, the Trader Portal in Buyer mode passes defaultFilters.mode: 'BUYER', visibleColumns with buyer-specific fields, and extraTabs with the Customs Readiness tab. When the user toggles to

Seller mode, the portal re-renders the vault with a new config
where mode is 'SELLER' and the columns and tabs switch accordingly. The vault component handles the transition by internally caching the previous data for a few seconds to avoid a flicker while the new API call is in flight.

Developers can also extend the vault with custom column renderers and additional slide-out tabs without modifying the core component. The columnRenderers prop allows overriding the default rendering of any column; for example, the Financier portal might provide a custom renderer for the LTV column that displays a more detailed gauge. Extra tabs are passed as React components that receive the ustn and the full shipment object as props, enabling fully isolated, portal-specific functionality inside the vault.

The configuration is type-checked at build time, ensuring that only valid column names, action names, and filter structures are used. The vault's runtime validates the config and falls back to safe defaults if an invalid combination is detected, logging a warning to the browser console in development mode.

6.1.6.9 Backend Endpoints for the Shared Shipments Vault

The vault is powered by two primary API endpoints, served by the tenant-experience-service(Rust, Axum). Both endpoints require a valid JWT and respect the user's permissions, data scopes, and active dual-mode context.

GET /v1/shipments-vault

Returns a paginated, filtered, sorted list of shipment summaries for the authenticated user. The endpoint accepts the following query parameters:

page (integer, default 1) – Page number.

per_page (integer, default 25, max 100) – Number of rows per page.

sort (string) – Column to sort by. Accepts any column from the visible set,
e.g., ustn, status, etd, eta, created_at, updated_at.

order (string, asc or desc) – Sort order.

status (string, comma-separated) – Filter by shipment statuses, e.g., LOADED,DEPARTED.

date_from and date_to (ISO 8601 date strings) – Filter by ETD/ETA range.

search (string) – Full-text search term.

commodity (string) – Filter by 2-digit HS chapter, e.g., 08.

risk_min and risk_max (integers) – Risk score range (only available for government and admin roles; ignored for others).

mode (string, buyer or seller) – Overrides the active dual-mode context for the query. This is automatically set by the frontend when the toggle is switched. If not provided, the default is derived from the JWT's active_trader_mode_context claim.

The response is a JSON object:

json

{

"data": [

{
"ustn": "SGTX-EG-...-V1",
"status": "LOADED",
"vessel": "MSC Alessia",
"origin": "SGN",
"destination": "ALEX",
"etd": "2026-06-14T08:00:00Z",
"eta": "2026-06-20T06:00:00Z",
"last_updated": "2026-06-13T10:30:00Z",
"columns": {
"seller_name": "Mekong Fresh",
"customs_readiness": "amber",
"documents_missing": 2,
...
}
}
],
"meta": {
"current_page": 1,
"per_page": 25,
"total_pages": 8,
"total_count": 187,
"active_mode": "BUYER"
}
}

The columns object contains the role-specific fields. The values present in columns depend on the user's permissions and the visibleColumns requested by the frontend (the frontend sends a lightweight hint in a custom header, but the server always enforces access). If the user lacks permission for a column, its value is null.

The service uses PostgreSQL's ROW_NUMBER() for pagination and GIN indexes for full-text search. Frequently used filter combinations are cached in Valkey with a 30-second TTL. When a shipment update event is received by the inbox-service or the shipment-service, a targeted cache invalidation keyed by USTN forces a refresh of the affected pages on the next request.

POST /v1/shipments-vault/export

Triggers a synchronous export of the current filtered list as a CSV file. The request body contains the same filter parameters as the GET endpoint. For small result sets (up to 10,000 rows), the CSV is generated in memory and returned immediately as a downloadable file with a Content-Disposition header. For larger sets, the service returns a 202 Accepted response with a job_id, and the export is processed asynchronously by the tenant-data-export-service. The user receives a Smart Inbox notification when the export is ready, with a presigned download URL valid for 1 hour. The exported CSV includes the columns visible to the user, respecting all data scopes and redaction rules.

Performance Characteristics

The endpoint is designed to serve hundreds of concurrent requests with sub-100-millisecond latency for cached queries. The query planner uses partial indexes scoped by tenant_id to ensure row-level security is enforced efficiently. For the Admin Portal's platform-wide view (all tenants), the query runs against a ClickHouse materialised view to avoid overloading the operational PostgreSQL instance, and results are cached for 5 minutes.

Both endpoints are fully documented in the OpenAPI 3.1 specification (Part 27 of the blueprint) and are available in the platform's interactive API explorer.

6.1.6.10 Example Usage in the Trader Portal (Buyer and Seller)

This worked example demonstrates how the same underlying Shared Shipments Vault component adapts to two different roles within the Trader Portal, using the dual-mode toggle. The example follows Nile Foods, a dual-mode trading company tenant, as two employees use the vault in the course of a typical week.

Scenario – Buyer Mode

Ahmed, an import manager at Nile Foods, logs into the Trader Portal. The dual-mode toggle in the header is set to "Buyer." The Smart Inbox is his landing page, but he clicks the "Shipments" tab to open the Shared Shipments Vault.

Default View : The vault loads in Active mode, showing three inbound shipments. The columns include USTN, Status, Vessel, Origin/Destination, ETD/ETA, Seller Name, Customs Readiness, and Documents Missing. The status filter is pre-set to show only shipments at sea or arriving soon.

Quick Triage : Ahmed notices that one shipment—oranges from Mekong Fresh—has a red "Customs Readiness" light and a Documents Missing badge showing "2". He hovers over the badge and sees that he needs to upload the import permit and a tax declaration.

Action : He clicks the "Upload Document" button on that row, uploads the two documents, and the Customs Readiness light immediately turns amber (the phytosanitary certificate from the seller is still pending). The Documents Missing badge drops to "0".

Real-Time Update : Ten minutes later, the seller uploads the phytosanitary certificate. Ahmed sees the row highlight briefly; the Customs Readiness light turns green, and a toast notification appears: "Shipment SGTX-EG-...-V1: all documents verified."

Archive : Later in the month, the shipment is delivered. Ahmed switches to the "Archived" toggle and finds the completed shipment in the list, with a green "DELIVERED" badge. He clicks the USTN to open the Trade Command Center for a final review.

Scenario – Seller Mode

Fatima, the export manager, arrives and switches the header toggle to “Seller.” The vault immediately refreshes with the Seller configuration.

Default View : The vault now shows four outbound shipments: two of textiles to Italy, one of dates to Germany, and one of frozen vegetables to the UK. The columns include Buyer Name, Estimated Margin, Loading Status, and Logistics Quotes Status.

Monitoring Margins : Fatima scans the Estimated Margin column. The dates shipment shows an amber margin—down 3% from the estimate. She hovers over it and sees that the trucking invoice was higher than expected. She clicks the “Cost Breakdown” extra tab in the slide-out panel to review the details.

Logistics Action : The frozen vegetables shipment shows “Logistics Quotes: 2/3” with a clock icon. Fatima clicks “ReOptimise” to see if a better carrier is now available. The diff view shows that Carrier Y is now significantly cheaper than the previously selected Carrier X, with a better risk score. She accepts the change, and the Logistics Quotes status updates to “Complete”.

Loading Status : The dates shipment shows “Partial” loading. She clicks the row and opens the Containers tab in the slide-out panel, where she can see that 18 of 22 pallets have been scanned. She messages the warehouse operator via the trade room.

Multi-Shipment : The textiles to Italy are part of a multi-shipment contract. The master contract row shows a progress bar (1 of 3 shipments locked). Fatima clicks the master USTN link and sees the full schedule in the TCC. She then confirms the second shipment, and the master row’s progress bar updates in real time.

This example illustrates how the vault unifies the operational experience for both buying and selling workflows without requiring separate portal logins or UI duplication. The same component, through configuration, serves two radically different use cases seamlessly.

END OF SUB-SECTIONS 6.1.6.6 THROUGH 6.1.6.10 – SHARED SHIPMENTS VAULT (v6.3 WITH TRADER PORTAL UNIFICATION AND ALL MODIFICATIONS)

PART 6.1.7 – ACCESSIBILITY STANDARDS (WCAG 2.2 AA) – v6.3 FULLY EXTENDED, WITH ALL MODIFICATIONS

6.1.7.0 Purpose

This section defines the minimum accessibility requirements for all SGTX portals, mobile applications, and desktop applications. Compliance with WCAG 2.2 Level AA is mandatory for every user interface component. These standards ensure that users with visual, auditory, motor, or cognitive impairments can effectively use the platform, including the dual-mode toggle, Smart Inbox, Trade Command Center, decision panels, and all workflow screens for trade, financing, and distressed cargo.

Scope

All frontend code (web, mobile, desktop) must adhere to these standards. Third-party components (e.g., Chart.js, Three.js) must be configured or wrapped to meet the requirements. Automated testing tools (axe-core, Lighthouse) are run in CI; manual testing is performed quarterly. No part of the platform may exclude users based on disability.

6.1.7.1 Keyboard Navigation

Focus Order

All interactive elements (buttons, links, form fields, tabs, modal controls) must be reachable and operable using the Tab key in a logical order (left to right, top to bottom).

Focus indicators must be clearly visible : a 2-pixel solid outline in a high-contrast colour (minimum 3:1 contrast against background). The default browser outline may be used but must not be suppressed without a replacement.

Skip to content link : The first keyboard-focusable element on every page must be a “Skip to main content” link that jumps past the header navigation.

Focus Trapping

When a modal (e.g., Decision Panel, DeFi Risk Summary, SGTX FEES Dispute modal) opens, focus must be moved to the modal container.

Focus must not leave the modal while it is open. The Tab key cycles within the modal, and the Escape key closes the modal and returns focus to the element that opened it.

The same rule applies to slideout panels (e.g., Shipments Vault detail panel).

Keyboard Shortcuts (Optional but Recommended)

Global shortcuts can be enabled in user preferences (off by default):

Ctrl + K (or Cmd + K on Mac) : focus search bar in Smart Inbox.

Ctrl + Shift + I (or Cmd + Shift + I) : open AI Assistant panel.

Ctrl + Shift + D (or Cmd + Shift + D): open Company Admin.

Esc: close any modal or slideout panel.

For dual-mode toggle, Ctrl + Shift + M switches between Buyer and Seller modes (announced by screen reader).

6.1.7.2 Screen Reader Support (ARIA)

ARIA Roles and Attributes

Use native HTML elements where possible (button, a, input, select). Only when custom components are necessary, apply appropriate ARIA roles (button, dialog, tablist, tab, etc.).

All interactive elements must have an accessible name. For icons without visible text, use aria-label (e.g., “Close modal”, “Download statement”, “Switch to Seller mode”).

Live regions : For dynamic content updates (Smart Inbox new item, Trade Command Center real-time milestone update, dual-mode toggle context change, financing bid arrival), use aria-live="polite" or role="status" so screen readers announce changes without interrupting the user.

Headings Structure

Each page must have exactly one H1 (the page title). Subsequent headings (H2, H3) must follow a logical hierarchy without skipping levels.

The Smart Inbox uses H2 for section titles ("High Priority", "Medium Priority"). The AI Summary Card is wrapped in a <section> with aria-label="AI Summary".

Forms and Validation

Every form field must have an associated <label> (visible or using aria-label if the label is not visible but context is clear).

Validation errors must be announced by screen readers. Use aria-describedby to link the error message to the field. The error summary must be focusable (or announced automatically via live region).

6.1.7.3 Colour Contrast and Visual Design

Colour Contrast Minimums

Normal text (smaller than 18pt or 14pt bold) : contrast ratio at least 4.5:1.

Large text (18pt or 14pt bold and larger) : contrast ratio at least 3:1.

UI components (buttons, borders, focus indicators) : contrast ratio at least 3:1 against adjacent colours.

The default Tailwind CSS palette has been audited; custom colours must be tested using a tool like Contrast Checker. The following combinations are prohibited: light grey on white, dark grey on black, red on green, green on red.

Colour as Information

Colour may not be the sole method of conveying information. For example:

Status badges (green/amber/red) must also include an icon or text label (e.g., "Low risk").

Charts (Cash Position, LTV history) must have patterns or labels; a screen-reader accessible data table must be available as an alternative.

The capacity heatmap (Three.js) must provide a text-based alternative : a table of density percentages per container zone, accessible via a "Show data table" button.

6.1.7.4 Audio and Voice Features

Voice Commands

Voice commands are an optional feature. They do not replace keyboard navigation. All voice-activated actions must also be possible via standard input (mouse, keyboard, touch).

When voice recognition fails or the system falls back to English for unsupported languages, a visible error message must appear: "Voice command not recognised. You can type your request instead." The message must be announced by screen readers.

Audio Alerts

Important notifications (high-priority Smart Inbox items, margin calls, demurrage alerts) may use a short beep or chime, but this must be accompanied by a visible alert and, for screen reader users, a live region announcement.

Users must be able to mute audio alerts via Company Admin → Notification Preferences.

6.1.7.5 AR and 3D Content (Three.js / AR.js)

Fallbacks for 3D Container Viewer

The 3D container viewer (Three.js) must provide a text-based representation of the loading plan. A “Show text layout” button below the canvas generates a table listing pallet positions, product, lot, and weight.

Users can navigate this table with keyboard and screen reader. The table is also exported when the user downloads the packing plan as PDF.

Augmented Reality (AR) Overlay (QC Portal)

AR is a supplementary feature. The inspector must be able to complete the inspection without AR using the standard camera view and manual defect logging.

When AR mode is active, the same information (pallet ID, defect highlights) must be available in a non-AR panel (e.g., a sidebar list). The screen reader should announce: “AR mode active. Use the sidebar to view inspection points.”

6.1.7.6 Mobile Application (React Native)

Touch Targets

Minimum touch target size : 44x44 points (iOS) or 48x48dp (Android). All buttons, icons, and interactive cards must meet this size.

The barcode scanner’s “Scan” button, voice command button, and snooze/dismiss swipe actions must be large enough and provide sufficient spacing to avoid accidental taps.

Screen Reader Support (iOS VoiceOver, Android TalkBack)

All native components (View, TouchableOpacity, TextInput) must have accessible labels. Use accessibilityLabel and accessibilityHint where appropriate.

Custom components (e.g., the AR overlay) must include an accessibilityLabel (e.g., “AR inspection view. Double-tap to start camera.”).

When the offline queue syncs or a scan succeeds, a live region announcement (accessibilityLiveRegion="polite") should inform the user.

6.1.7.7 Automated Testing and Compliance Monitoring

CI Pipeline

axe-core (Jest + axe) runs on all React component unit tests. Violations fail the build.

Lighthouse CI runs on a representative set of pages (Smart Inbox, Trade Command Center, Buyer Portal, Seller Portal, Financier Portal). Score targets: Accessibility ≥ 95.

For mobile, use Detox with axe-devtools-react-native or equivalent.

Quarterly Manual Audits

A checklist based on WCAG 2.2 Level A and AA is executed by a QA engineer.

Any violation is logged as a bug with severity High and must be fixed within two sprints.

User Preferences

All users can set their preferred high-contrast theme (grayscale or inverted) via Company Admin → Accessibility.

Font size can be increased to 125% and 150% without breaking layout (responsive design using rem units).

6.1.7.8 Exception Handling

Temporary Exceptions

The 3D container viewer's heatmap overlay is supported only with a text fallback – no exception.

Voice command limitations for unsupported languages are documented; the fallback (display text only) is acceptable.

No permanent exceptions are granted. Any component that cannot meet WCAG 2.2 AA must be redesigned before release.

END OF PART 6.1.7 – ACCESSIBILITY STANDARDS (v6.3 WITH ALL MODIFICATIONS)

PART 6.1.8 – TENANT EXIT & DATA PORTABILITY CENTER (v6.3 – FULLY EXTENDED, WITH MULTI-SHIPMENT, FINANCING, DISTRESSED DATA PORTABILITY)

6.1.8.0 Purpose

Enterprises and regulated organizations require a clear, auditable, and self-service mechanism to leave the platform. The Tenant Exit & Data Portability Center provides a structured workflow for tenants who wish to close their account, ensuring that all data is exported, legal obligations are satisfied, and the tenant's exit is logged immutably. The process respects multi-shipment contracts, active financing agreements, and distressed transactions, preventing exit until all obligations are resolved or transferred.

This section defines the exit workflow, the portability of data (including trade, financing, and distressed records), and the transition to the ARCHIVED or DELETED lifecycle states (see Part 2.10). It is available to tenant admins from the Company Admin → Exit Center tab.

All actions are gated by the Governor, and every step is logged in the audit trail.

6.1.8.1 Exit Workflow Overview

The tenant exit process consists of five steps, each requiring explicit confirmation. The tenant may abort at any step before final confirmation.

Flow:

Initiate Exit Request – Tenant admin clicks "Request Account Closure".

Pre-Exit Checks – Governor verifies open disputes, active contracts, unsettled fees (SGTX trade fees are already paid, but financing repayments may be outstanding), legal holds.

Generate Export Package – System compiles all tenant data (trades, multi-shipment shipments, financing agreements, distressed transactions, documents, audit logs, financial summaries) into a signed ZIP (same format as Part 6.1.4, extended with financing and distressed data).

Closure Confirmation – Tenant admin reviews the export package and signs a closure confirmation (ZITADEL passkey).

Finalise – Tenant lifecycle state becomes ARCHIVED. Data retained according to jurisdiction. After mandatory retention period (default 7 years), tenant may request deletion to transition to DELETED.

If any pre-exit check fails, the exit request is blocked, and the Decision Panel explains the reason (e.g., “Open dispute #DSP001 must be resolved before exit” or “Financing agreement #FA001 has an outstanding balance of $5,000 – repay or transfer before exit”).

6.1.8.2 Pre-Exit Checks (Governor-Enforced)

Check	Description	Failure Action
Open disputes	Any dispute with status not RESOLVED or CLOSED	Block exit; list dispute IDs
Active contracts	Single-shipment contract with status LOCKED or ACTIVE and not COMPLETED or TERMINATED	Block exit; require termination first
Multi-shipment master contract	Master contract with any shipment not yet COMPLETED or CANCELLED	Block exit; list pending shipments
Active financing agreements	Financing agreement with status ACTIVE or DISBURSED and outstanding balance >0	Block exit; require repayment or transfer
Unsettled SGTX fees	Any fee payment request with status PENDING (unpaid)	Block exit; require payment (though trade fees paid upfront, financing fees may be pending)
Legal hold	Tenant flagged with legal_hold = true (compliance or court order)	Block exit; notify compliance officer
Distressed cargo listings	Any active distressed listing not yet CLOSED	Block exit; require closure or transfer

The tenant can resolve these issues before retrying exit, or request a compliance override (requires multisig approval). For financing agreements, the tenant may request to transfer the agreement to another legal entity (see Transfer of Ownership, 6.1.8.6).

6.1.8.3 Export Package (Data Portability)

The exit export package includes all data that the tenant is entitled to under GDPR / PDPL / local law. It is generated by the same tenant-data-export-service as standard exports (Part 6.1.4), but with a complete scope (all trades, all documents, all audit logs, all financial events, financing, distressed). The package is signed with the platform's Ed25519 key.

Contents:

trade_data.csv – all trades (including cancelled, disputed, multi-shipment separated by row).

financing_data.csv – all financing agreements (requests, offers, agreements, repayments, fees).

distressed_data.csv – all distressed listings, micro-contracts, offers, sales.

documents/ folder – all uploaded and generated documents (PDFs, images).

audit_log.csv – full audit trail for the tenant (see Part 6.2.8.6).

financial_summary.csv – all SGTX fee payments, settlement receipts, financing disbursements, distressed sale proceeds.

contacts.csv – saved contacts (GTIDs, relationship history, trust snapshots).

lifecycle_history.csv – state transitions from tenant_lifecycle_history.

governor_decisions.json – all Governor decisions affecting the tenant (anonymised for other parties).

signature.sig, checksum.sha256, README.txt – for verification.

The export is generated asynchronously. The tenant receives a Smart Inbox item when ready (priority 90). The download link expires after 7 days.

Retention after exit : The export package is stored on the platform for 30 days after exit, then deleted. Tenants must download and keep their own copy.

6.1.8.4 Closure Confirmation & Signature

After the export package is downloaded, the tenant admin must sign a closure confirmation. The modal displays:

"I confirm that I have downloaded all my data, resolved all open obligations (including financing repayments and dispute resolutions), and understand that after closure I will not be able to access the platform. My data will be retained for 7 years as required by law, after which it may be deleted. I may request permanent deletion earlier only if permitted by applicable law."

The admin signs with ZITADEL passkey (WebAuthn). The signature is stored in tenant_lifecycle_history for the ARCHIVED transition.

6.1.8.5 Post-Exit State & Retention

State	Retention Period	Access	Restoration

ARCHIVED	7 years (configurable per jurisdiction)	No login; data retained for compliance and audit	Platform Governance Authority can restore via multisig (rare)
DELETED	After retention period, tenant requests deletion	Data irreversibly anonymised or deleted (except where law requires retention)	Not possible

During the ARCHIVED period, the tenant's GTID is reserved and cannot be reused. The tenant's legal name is retained in audit logs for traceability.

The tenant receives a final Smart Inbox confirmation : "Your account has been closed and is now archived. Your data will be retained until [date]. A permanent deletion request can be submitted after that date."

6.1.8.6 Transfer of Ownership (Alternative to Exit)

For tenants who want to leave but transfer active contracts, financing agreements, or distressed listings to another legal entity (e.g., during acquisition), the platform provides a Transfer Ownershipworkflow instead of full exit, accessible from the same tab.

Transfer workflow:

Tenant admin initiates "Transfer Ownership".

Enters the destination tenant's GTID.

Governor checks that the destination tenant is VERIFIED and has compatible jurisdiction (for financing, also checks financier approvals).

Both parties sign a transfer agreement (ZITADEL passkey).

All active contracts (including multi-shipment schedules), open disputes (with counterparty consent), financing agreements (with financier consent), and distressed listings are reassigned to the destination tenant.

Original tenant's lifecycle state becomes ARCHIVED after transfer completion.

Transfer is logged as a Governor decision and requires no manual approval unless compliance flags are raised.

6.1.8.7 Data Deletion Request (Post-Archival)

After the mandatory retention period (7 years), the tenant (or authorised representative) may request permanent deletion. The request:

Must be submitted via email to privacy@sgtx.io or via a form in the (now inaccessible) tenant portal using a special one-time link provided after exit.

Requires identity verification (notarised document or court order).

Once approved, the Platform Governance Authority (multisig) initiates deletion. All personal data is anonymised or removed. Trade records are aggregated and stripped of identifiers.

Deletion is logged in the audit_log with the approval decision ID.

The tenant receives a confirmation letter (PDF) signed by the platform.

6.1.8.8 Zero-Cost Implementation

Component	Technology	Cost
Exit workflow	Governor + tenant-data-export-service	$0
Export package	Same as Part 6.1.4	$0
Closure signature	ZITADEL passkey	$0
State transition	tenant_lifecycle_history + OPA	$0
Transfer ownership	Governor + contract reassignment logic	$0

No external identity verification or data escrow service is required.

END OF PART 6.1.8 – TENANT EXIT & DATA PORTABILITY CENTER (v6.3 WITH ALL MODIFICATIONS)

PART 6.1.9 – GUIDED RECOVERY SYSTEM (CONFUSION RECOVERY LAYER) – v6.3 FULLY EXTENDED, WITH MULTI-SHIPMENT, FINANCING, DISTRESSED, DUAL-MODE AWARE

6.1.9.0 Purpose

Users can become lost or confused during complex workflows – repeatedly clicking the same button, abandoning forms, navigating backwards, or encountering repeated Governor denials. The Guided Recovery System detects these behavioural signals and proactively offers assistance before the user gives up. The system is advisory only (A1) – it never blocks or overrides user actions. It appears as a non-intrusive help panel that can be dismissed. All recovery events are logged in user_friction_events for platform improvement.

The recovery system is aware of:

Dual-mode toggle – if a user is in Buyer mode but the required action is seller-side, the system suggests switching mode.

Multi-shipment contracts – if a user is stuck on a particular shipment, the recovery offers to jump to that shipment's TCC.

Financing – if a financier hasn't submitted a bid, the system reminds them of the bidding window.

Distressed – if a seller hasn't responded to offers, the system prompts review.

6.1.9.1 Friction Detection Signals

The system monitors the following behavioural patterns in real time (clientside and serverside):

Signal	Detection Method	Threshold	Example
Repeated form submission	Same POST to same endpoint without field	3 attempts within 60 seconds	User clicks "Submit Quote" three times

	changes		without changing anything
Page abandonment mid-workflow	User navigates away from a partially filled form without saving	Form open >5 minutes, then navigation to unrelated page	User starts new trade request, then clicks to Shipments Vault before submitting
Rapid back-and-forth navigation	Sequence of back and forward between two pages	4 transitions within 30 seconds	User goes between Quote Review and Contract Signing repeatedly
Governor denial repetition	Same decision_type denied with same conditions	2 denials within 10 minutes	User tries to lock packing plan twice, same weight limit error
Inactivity on a pending action	Smart Inbox high-priority item open but no action taken	Item viewed 3 times over 2 hours without action	User opens “Sign Contract” item three times but never signs
Hover on disabled button (Enterprise only)	User hovers over a disabled UI element multiple times	3 hovers within 30 seconds	User hovers over greyed-out “Submit” button because document missing
Dual-mode mismatch	User attempts an action that requires opposite trader mode	Detected when action permission fails due to active mode	Seller tries to sign contract while in Buyer mode

When any threshold is crossed, the system logs a user_friction_event and triggers the Guided Recovery modal (or inline panel). For dual-mode mismatch, the recovery suggests switching mode and provides a direct toggle.

6.1.9.2 Recovery Offer Types

Depending on the detected friction, the system offers one or more of the following recovery actions:

Offer Type	Description	Example UI
Explain this step	Shows plain-language explanation of the current workflow step, why it is needed, and what data is required	“You are trying to lock the packing plan. The system requires all pallet weights to be entered and total weight to be within container limits. You are missing weight for pallet OR019.”
Continue where you stopped	Restores the last saved draft of the current workflow (from draft_history)	“You have a saved draft from 2 hours ago. Resume?”
Show an example	Displays a mock example of the current form filled correctly (synthetic data)	“Here is an example of a completed EXW price lock for oranges.”
Guided walkthrough	Launches a step-by-step overlay (similar to onboarding wizard) for the current workflow	Starts a 3-step guide: “Step 1: Enter EXW price. Step 2: Confirm. Step 3: Lock.”

Switch trader mode(dual-mode)	Recognises that the user is in the wrong mode (Buyer vs Seller) and offers to switch	“You are currently in Buyer mode, but this action requires Seller mode. Switch now?”
Jump to pending shipment(multi-shipment)	If a user is stuck on a multi-shipment contract, jumps to the next pending shipment	“Shipment 2 of 3 is pending confirmation. Go to Shipment 2?”
Contact support	Opens a pre-filled chat with the AI Assistant or a support ticket form	“Still stuck? Contact support. We will respond within 2 hours.”

The offer is displayed in a small, non-modal panel at the bottom right of the screen (dismissible). If the user dismisses the same offer three times for the same workflow, the system stops showing offers for that workflow type for 7 days.

6.1.9.3 Governor-Triggered Recovery

When a Governor decision returns CONDITIONAL or DENY, the PlainLanguage Decision Panel already explains the reason. However, the Guided Recovery System adds an extra “Show me how to fix this” button inside the panel. Clicking it launches a contextual walkthrough that guides the user through the exact remediation steps (e.g., “Upload the missing document for seller KYB Tier 2”).

6.1.9.4 Smart Inbox Integration

If a high-priority Smart Inbox item remains unactioned for more than 24 hours (viewed but not acted upon), the system automatically creates a follow-up item with a lower priority (60) that asks: “You haven’t signed the contract yet. Need help? [Explain] [Guided walkthrough] [Contact support]”. This reduces abandonment without being annoying.

6.1.9.5 Database Schema (Additions to Part 5)

sql

```
CREATE TABLE user_friction_events (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(id),
employee_id UUID NOT NULL REFERENCES employees(id),
ustn TEXT,
financing_agreement_id UUID,
distressed_listing_id UUID,
signal_type TEXT NOT NULL,
details JSONB,
recovery_offered TEXT[],
recovery_accepted TEXT,
dismissed BOOLEAN DEFAULT false,
```

created_at TIMESTAMPTZ DEFAULT NOW()

);

6.1.9.6 Example Workflow – Repeated Form Submission (Dual-Mode Mismatch)

Scenario : Seller Mekong Fresh tries to lock the EXW price three times in one minute without changing the price, but they are in Buyer mode (dual-mode toggle set incorrectly).

After the third identical submission, the frontend detects the pattern.

A small panel appears : "You have tried to lock the EXW price three times without change. This action requires Seller mode, but you are currently in Buyer mode. Switch to Seller mode to continue."

The panel includes a button : Switch to Seller mode. Clicking it toggles the dual-mode context and reloads the page.

After switching, the user successfully locks the price.

The friction event is logged.

6.1.9.7 Zero-Cost Implementation

Component	Technology	Cost
Friction detection	Frontend event listeners + backend API (Rust)	$0
Recovery offers	Predefined templates + Groq/Ollama (for explanations)	$0 (free tier)
Guided walkthrough	Overlay + step tracking (same as onboarding wizard)	$0
Data storage	PostgreSQL	$0

No external user analytics or session replay service is required.

END OF PART 6.1.9 – GUIDED RECOVERY SYSTEM (v6.3 WITH ALL MODIFICATIONS)

6.1.10 Service Level Visibility & Human Support Directory (All Portals)

Purpose

When a tenant action requires human intervention – for example, a KYB manual review, a dispute mediation, a compliance officer sign-off, a financing bid approval, or a government customs clearance – the tenant must know precisely what to expect and who is handling their case. This section defines the universal mechanism by which SGTX provides real-time transparency about expected response times, current queue positions, and the responsible support teams or individuals. It integrates directly with the Smart Inbox, the PlainLanguage Governor Decision Panel, and the new Customer Care Chatbot, ensuring that tenants are never left in the dark.

The information is surfaced through three channels:

Smart Inbox Items – Every pending human action displays estimated SLA, queue position, and responsible party directly in the inbox item.

Decision Panel – When an action is blocked by a CONDITIONAL verdict awaiting human review, the panel includes the SLA details.

Customer Care Chatbot – The chatbot provides on-demand SLA checks and can escalate to a human agent if the SLA is breached or the tenant requests it.

The Human Escalation Directory – a read-only page in Company Admin – lists all relevant contacts for the tenant's jurisdiction and active workflows, including compliance officers, dispute mediators, and the Platform Governance Authority.

All SLA data is derived from the platform's workflow-recovery-service, which monitors every pending human task, and from the jurisdictions table, which defines standard response times per action type and per country.

Display Rules for Pending Human Actions

For every action that is not automatically processed (i.e., it requires a human reviewer – a compliance officer, a mediator, a credit committee, etc.), the Smart Inbox item and the Decision Panel must display:

Estimated Response Time – in hours or days, based on the defined SLA for that action type and the current workload of the responsible team. The estimate is continuously updated by an AI model (LightGBM, A2) that considers the number of pending items, historical throughput, and real-time staffing signals (if available).

Current Queue Position – the number of pending items of the same type that are ahead of this item, scoped to the same jurisdiction or team. For example, "You are number 5 of 12 KYB reviews pending for Egypt."

Responsible Party – a human-readable label identifying the team or role handling the review, e.g., "Compliance Team – Egypt", "Dispute Mediator – ICC Paris", "Financing Approval Committee", "Platform Governance Authority". For direct phone support following a chatbot escalation, the actual agent's name may be displayed once assigned.

Escalation Path – a clear statement of what happens if the response time is exceeded, e.g., "If not resolved within 48 hours, this will be automatically escalated to the Platform Governance Authority." The escalation contact is also listed.

If the estimated response time is exceeded, the system automatically generates a follow-up Smart Inbox item (priority 85) with an apology and an updated ETA. For multi-shipment contracts, each shipment's pending action carries its own SLA; a delay in one shipment does not affect others. For financing agreements, margin calls and repayment disputes have their own SLAs. For distressed cargo, listing approval SLAs are shown where a human review is triggered.

Standard SLAs Per Action Type

The following SLAs are defined in the platform's constitutional policies and may be overridden per jurisdiction via the jurisdictions table. All durations are in business hours (as defined by the responsible party's timezone and local holidays, adjustable via 6.5.10).

KYB manual review (Tier 1–2 documents with low confidence): 48 hours

KYB manual review (Tier 3+ enhanced due diligence): 72 hours

KYC biometric liveness manual review: 24 hours

Dispute mediation initial response: 72 hours

Compliance officer review (sanctions/PEP flag, legal hold) : 24 hours

Financing bid manual approval (if automatic RFQ fails) : 48 hours

Multi-shipment shipment readiness confirmation (manual overrides) : 24 hours

Distressed listing approval (if manual review triggered) : 12 hours

Government customs clearance (manual) : variable, based on jurisdiction’s customs_review_sla(default 24 hours)

Platform Governance Authority multisig requests : 72 hours (standard), 24 hours (urgent)

These SLAs are used by the workflow-recovery-service to trigger reminders and auto-escalations.

Data Sources and Calculation

Queue Position is calculated by the inbox-service by counting the number of pending items of the same decision_type where verdict = CONDITIONAL and status != RESOLVED, ordered by created_at timestamp, and filtered by the relevant jurisdiction or team assignment. For the Customer Care Chatbot’s escalations, the human support queue is separate and managed by the customer-care-service.

Responsible Party is derived from the jurisdictions table (for compliance teams, which are assigned per country), the dispute mediator assignment (from the dispute’s governing law), the financing agreement’s credit committee, or the Platform Governance Authority’s internal roles.

SLA Performance History is tracked in ClickHouse. The sla-monitor service feeds actual resolution times into a continuous learning model that refines the AI’s estimated response times.

Example – Smart Inbox Item with SLA Transparency

text

 HIGH – KYB manual review pending (priority 85)

Your account verification requires manual review.

Estimated response: 48 hours. Your position in queue: 3.

Responsible: Compliance Team – Egypt.

If not resolved within 48 hours, escalate to Platform Governance Authority.

[View status] [Contact support via chatbot]

The “Contact support via chatbot” button opens the Customer Care Chatbot with context pre-filled.

Example – Decision Panel with SLA for Financing

text

⚠ Action Blocked: Financing Bid Approval

The bid requires manual review because the borrower's credit score dropped below the auto-approval threshold.

Estimated response: 24 hours (Financing Approval Committee).

Queue position: 1.

Responsible: SGTX Credit Committee.

If not resolved within 24 hours, the bid will be automatically escalated to the Platform Governance Authority.

[Request Human Review] [Ask Chatbot]

Human Escalation Directory (Tenant-Facing)

Each tenant admin has access to a read-only Human Support Directory page within Company Admin. This page lists all human escalation contacts relevant to the tenant's active workflows and jurisdiction:

Compliance Contact – The designated compliance officer or team for the tenant's jurisdiction. Includes anonymised team name (e.g., "SGTX Compliance – MENA"), email, and SLA response time.

Dispute Mediator – If a dispute is active, the assigned mediator's name and organisation (e.g., "ICC Paris"), along with the case reference and expected response time.

Platform Governance Authority – The contact email for the multisig (e.g., governance@sgtx.io), for tenant-wide escalations when all other channels are exhausted.

Credit Committee (if tenant uses financing) – The contact for financing-specific escalations, drawn from the financing agreement's designated approval body.

Approved Liquidators / Insurers (for distressed cargo) – Per jurisdiction, a list of pre-vetted liquidators or insurance contacts that the tenant may need for distressed sales.

The directory is automatically generated from the jurisdictions and active workflow data. Tenants cannot modify it, but they can request an update through the Customer Care Chatbot or by emailing the Platform Governance Authority.

Integration with the Customer Care Chatbot

The Customer Care Chatbot, accessible from any portal, serves as the first point of contact for SLA-related inquiries. A tenant can ask:

"What is the status of my KYB review?"

"How long until my dispute is assigned to a mediator?"

"Escalate my stuck trade to the Platform Governance Authority."

The chatbot will respond with the current SLA estimate, queue position, and responsible party, drawn from the same data sources. If the response time has been exceeded, the chatbot will proactively offer to escalate: "Your KYB review has exceeded the 48-hour SLA. Would you like me to escalate this to the Platform Governance Authority? I'll need your support PIN to proceed." With the tenant's PIN confirmation, the bot creates a

Governor decision (decision_type = 'human_escalation_request'), which triggers an immediate notification to the appropriate human team and logs the event.

If the tenant requests to speak directly with a human agent, the chatbot collects the tenant's preferred phone number and initiates a callback via the platform's self-hosted VoIP infrastructure. The human agent then has access to the full escalation context and can take over resolution.

Governance Gates

G-SLA1 : Every pending human action must display SLA information in the Smart Inbox or Decision Panel. The Governor (A4) ensures that any item created with a required human review includes the SLA metadata.

G-SLA2 : If the SLA is breached, the workflow-recovery-service automatically creates an escalation Governor decision and notifies the Platform Governance Authority via the Admin Portal's Smart Inbox.

G-SLA3 : Queue positions must be recalculated every 6 hours (or on demand when a user queries). The inbox-service uses last known positions if the data source is temporarily unavailable.

G-SLA4 : Any escalation via the Customer Care Chatbot requires the tenant's PIN and creates a Loom-logged decision. The chat transcript is attached to the escalation for the human agent.

END OF SUB-SECTION 6.1.10 – SERVICE LEVEL VISIBILITY & HUMAN SUPPORT DIRECTORY (v6.3 WITH ALL MODIFICATIONS)

6.1.11 Universal UX Principle – The Four Questions Every Screen Must Answer (All Portals)

Constitutional Requirement

Every screen, modal, panel, and dashboard in every SGTX portal must instantly answer four fundamental questions for the user, without requiring them to click, hover, or search. This is a constitutional user experience rule enforced by design reviews, automated UI audits, and the platform's QA process. No screen may display raw data without an immediately understandable, actionable meaning.

This principle applies identically to the Trader Portal (whether the user is in Buyer or Seller mode), the Logistics Portal, the QC Portal, the Financier Portal, the Government Portal, the Admin Portal, the Marketplace Partner Portal, and both the mobile and desktop applications. It is the single most important design test: if a user cannot answer all four questions within three seconds of looking at a screen, that screen is considered defective.

The Four Questions

What is happening? – The immediate, unambiguous state of the entity being viewed. This could be a trade, a shipment, a financing agreement, a dispute, or a user's own account status. It is not a technical label but a human-readable phrase.

What do I need to do? – The single most important action the user should take next. If multiple actions are available, the most urgent or critical one is highlighted. If no action is required at this moment, the user is explicitly told that.

What is blocking me? – Any obstacle that prevents the user from making progress. This could be a missing document, a Governor hold, an unmet condition, a pending counterparty action, or a dual-mode mismatch. If nothing is blocking, the answer is simply “No blockers.”

What happens next? – The immediate consequence of taking the suggested action, or the next automated step the system will take if no action is required. This gives the user confidence that the process is understood and under control.

These four questions must be answered in the order they are asked, and each answer must be visible within the user’s primary focal area on the page—typically the top half, or immediately below any persistent navigation header.

How the Four Questions Are Answered in Every Portal Component

The following table maps the platform’s shared UI components to the specific elements that answer each of the four questions.

Component	What is happening?	What do I need to do?	What is blocking me?	What happens next?
Smart Inbox	Each item’s description (first sentence) summarises the current state of the trade or task. The priority band immediately signals urgency.	The item’s title states the action (e.g., “Sign contract”, “Upload phytosanitary certificate”). The primary action button is prominent.	The description may include the reason for urgency, e.g., “Vessel arrives in 38h.” If the action is impossible, the item notes the blocker. The snooze/dismiss options are secondary.	The description often includes the outcome, e.g., “Trade will be automatically cancelled if not signed within 3h.” The AI Summary Card reinforces this.
Trade Command Center	The Persistent Header with the USTN badge, status badge, and the timelineprovide immediate state. The status badge is colour-coded and large.	The Pending Action Panel (prominently coloured card) states exactly what to do and provides a large action button. For multi-shipment, it may say “Confirm Shipment 2.”	The Pending Action Panel also explains why the action is needed. If a Governor hold exists, the Decision Panel is shown. The “What’s blocking me?” button is available.	The Pending Action Panel includes a “Why this matters” sentence. The timeline’s next phase is highlighted.
PlainLanguage Governor Decision Panel	The Header (“Action Blocked: Contract Signing”) and the PlainLanguage Explanation (“Wh	The Condition Checklist provides specific action buttons for each unmet condition. The “Request Human	The Condition Checklist lists each unresolved condition with a red cross and a direct action link.	The “What happens next” section at the bottom describes the outcome once conditions are

	y this happened") directly answer what is happening.	Review" button is the fallback.		met, and the auto-escalation timer provides a deadline.
Shared Shipments Vault	Each row's Status badge, USTN, and role-specific columns (e.g., "Customs Readiness" for buyer) provide a snapshot. The Active/Archive toggle contextualises.	Row actions provide the primary action button, e.g., "Upload Document", "Confirm Milestone". The most critical action is always visible.	Missing documents, overdue milestones, and red status badges show blockers. Hovering reveals details. Dual-mode mismatch is surfaced if the user is in the wrong mode.	The ETA/ETD column and the Loading Status column show the next milestone. The slide-out panel's Timeline tab provides a full step-by-step view.
Forms (New Trade Request, Quote Submission, etc.)	The page header or form title immediately states the goal (e.g., "Create New Trade Request"). Any inline status messages update the user on auto-save or AI processing.	The primary submit button ("Submit Trade Request", "Pay Fee") is always prominent and is labelled with the action verb.	Validation errors appear inline next to the specific field and also in a summary at the top. The "Why is this required?" tooltip explains mandatory fields.	A helper text below the submit button explains the immediate effect, e.g., "After submission, the seller will be notified and your request will appear in your Shipments Vault."
Dual-Mode Toggle (Header)	The toggle itself displays the current mode ("Buyer" or "Seller"). All UI elements adapt to reflect that context.	The toggle invites the user to switch mode if an action is blocked. The Decision Panel can also offer a direct switch button.	If an action is attempted in the wrong mode, the Decision Panel explains: "You are currently in Buyer mode. Switch to Seller mode to perform this action."	After switching, the portal immediately refreshes, and the previously blocked action becomes available.
Multi-Shipment Schedule (within TCC)	Each shipment row shows its status, delivery date, and port. The master contract shows an aggregated progress bar.	Action buttons per shipment: "Confirm Shipment", "Pay Fee", "Mark Ready". The next pending action is highlighted.	If a shipment cannot be locked because a fee is unpaid, the blocker is shown inline. The decision panel explains.	The progress bar shows overall contract completion. The Timeline section shows the phased delivery plan.
Customer Care Chatbot	The chat panel's initial message summarises the user's current query and relevant trade	The bot proposes specific solutions: "I can send a reminder. Do you want me to proceed?"	If the bot cannot resolve the issue, it states what is missing and offers to escalate.	After an action, the bot confirms what was done and what the user can expect next, e.g., "The

	context.			document request has been sent. The seller will be notified.”

Automated Testing and Enforcement

The CI pipeline includes a dedicated UX Audit job that scans each screen’s rendered output and checks for the presence of elements that answer the four questions. The audit fails if:

A screen has no clear primary action button (Q2).

A screen displays an error message without explaining the underlying blocker (Q3).

A screen’s title is generic (e.g., “Trade Data”) instead of describing the current state (Q1).

A screen’s primary action does not have an adjacent helper text explaining the immediate consequence (Q4).

Manual QA checklists for every new feature explicitly require verifying the four questions. During design reviews, the designer must annotate mockups with how each question is answered. This is not optional; it is a condition of merging any frontend code into the main branch.

Examples

Example 1 – Contract Signing Screen (Buyer Perspective)

What is happening? “Contract ready for signature. The seller has paid the SGTX fee. All parties have agreed to the terms.”

What do I need to do? “Sign contract” (large primary button).

What is blocking me? “No blockers. You can sign now.”

What happens next? “After signing, the contract will lock, the USTN will be generated, and the shipment will move to the execution phase.”

Example 2 – Stuck Shipment in the Vault (Seller Perspective)

What is happening? “Shipment 2 of 3 is stuck at ‘Loading’ for 48 hours.”

What do I need to do? “Contact your warehouse operator or force the milestone (requires justification).”

What is blocking me? “Pallet OR020 has not been scanned. The loading window has passed.”

What happens next? “If the milestone is not confirmed within 12 hours, the shipment will be escalated to the Platform Governance Authority.”

Example 3 – Customer Care Chatbot (Any Portal)

What is happening? “You have asked why your contract is blocked.”

What do I need to do? “Upload the Seller’s KYB Tier 2 document. I can send a request to the seller if you confirm.”

What is blocking me? “The Governor has placed a CONDITIONAL hold because the seller’s jurisdiction requires enhanced due diligence.”

What happens next? “If you provide your support PIN, I will automatically request the document from the seller. Otherwise, you can upload it manually here.”

Exception Handling

Temporary loading states and error screens are exempt from the four-question rule but must still provide a meaningful message. A loading screen must display “Loading trade data – please wait” (answering Q1). An error screen must display “An unexpected error occurred. Please try again or contact support” (answering Q1 and Q2 via a “Retry” or “Contact Support” button). Any screen that cannot answer any of the four questions must not be shown to the user; instead, a fallback screen with a “Contact Support” button is rendered.

END OF SUB-SECTION 6.1.11 – UNIVERSAL UX PRINCIPLE – THE FOUR QUESTIONS EVERY SCREEN MUST ANSWER (v6.3 WITH ALL MODIFICATIONS)

6.1.12 Buyer/Seller Mode Toggle for the Unified Trader Portal (Formerly Dual-Mode Toggle)

Purpose

Trading companies that both import and export goods need a single, unified workspace. The Buyer/Seller Mode Toggle, positioned in the global header of the Trader Portal, enables employees of tenants with default_trader_mode = DUAL to switch instantly between the Buyer Dashboard(for inbound, import-side activities) and the Seller Dashboard (for outbound, export-side activities). The toggle does not alter the user’s permissions; it filters the visible navigation tabs, data, Smart Inbox items, Shipments Vault columns, Trade Command Center financial cards, and available actions to match the selected context. This eliminates the need for separate Buyer and Seller portals, reduces cognitive load, and ensures that a single employee can manage both sides of their company’s trade without logging out or switching accounts.

UI Placement and Appearance

The toggle is located in the global header, to the right of the notifications bell and to the left of the user avatar. It is rendered as a segmented button with two clearly labelled segments: Buyer and Seller. The active mode is highlighted with the brand’s primary colour and a subtle underline, while the inactive mode remains neutral. The segments are wide enough to accommodate the translated labels in all supported languages (minimum width 80px each). A small animated transition occurs when the user switches modes—the highlight slides smoothly from one segment to the other.

The toggle is visible only when all of the following conditions are met:

The authenticated tenant has type = CORPORATE.

The tenant’s default_trader_mode is DUAL.

For the specific employee, data_scopes.allow_role_switching is true (tenant admins can disable the toggle for employees who should only work in one role, such as a finance user who only manages payables).

The employee's assigned role includes allowed_trader_modes that contain both BUY and SELL.

For users who only have BUY or only SELL, the toggle is hidden, and the corresponding dashboard is displayed permanently.

The toggle is also accessible via keyboard shortcut (Ctrl + Shift + M or Cmd + Shift + M on macOS) and via voice command ("Switch to Buyer mode" or "Switch to Seller mode"). When activated by voice, the VoiceCommandButton provides audible confirmation.

Behavior on Toggle

When the user clicks the toggle to switch modes, the frontend immediately calls the dedicated endpoint POST /v1/employee/switch-context with the body {"active_trader_mode_context": "BUY"} or "SELL". The identity-service updates the user's session and returns a new JWT containing the updated active_trader_mode_context claim. The portal then performs a soft reload—the page does not fully refresh, but all data-fetching hooks re-execute with the new context, and the UI re-renders accordingly.

Specifically, the following changes occur instantly:

Smart Inbox – The inbox refreshes, showing only items relevant to the new mode. In Buyer mode, items include contract signing, customs readiness, and counter-offers from sellers. In Seller mode, items include pending trade requests, EXW deadlines, carrier risk alerts, and logistics quote statuses. The AI Summary Card also updates.

Shipments Vault – The vault's configuration swaps between the Buyer and Seller column sets, default filters, and row actions. The data is re-fetched with the appropriate mode parameter.

Trade Command Center – For any open USTN, the financial cards and pending actions adapt: the Buyer sees the total payable amount and buyer-side actions; the Seller sees the EXW breakdown, logistics costs, SGTX fee, estimated margin, and seller-side actions.

Side Navigation – The sidebar tabs change. In Buyer mode, the navigation shows "New Trade Request", "Quote Review & Negotiation", "Contract Signing", "Customs Readiness", "Shipments", "Distressed Cargo" (as buyer), "Disputes", "Saved Contacts", "Financing" (if applicable), and "Company Admin". In Seller mode, the navigation shows "Pending Requests", "EXW Price Lock", "Containerisation & Packing", "Logistics Builder", "Quote Submission", "QC Booking", "Document Finalisation", "Barcode Print", "Shipments", "Distressed Cargo & Notifications" (as seller), "Disputes", "Saved Contacts", "Cash Position", and "Company Admin".

Network Tab – In Buyer mode, the Network tab shows sellers, logistics providers, financiers, and QC providers. In Seller mode, it shows buyers, logistics providers, QC providers, and financiers. The trust portraits and relationship health scores remain consistent.

The toggle can be switched at any time, even mid-workflow. If a user is in the middle of filling out a form (e.g., creating a new trade request in Buyer mode) and accidentally switches to Seller mode, the form state is preserved in the browser's local storage for a short period, and upon switching back, the user receives a prompt: "You have an unsaved draft in Buyer mode. Resume?” However, the best practice is to complete the current action before switching.

Governor Enforcement

The dual-mode toggle is a UI convenience, but the enforcement is server-side. Every API request includes the JWT claim active_trader_mode_context. The Governor’s OPA policies evaluate whether the requested action is permitted in the current context. For example:

contract.sign is allowed in both contexts (both parties must sign the contract), but the context determines which signature is being applied.

seller_quote.submit is allowed only when active_trader_mode_context = SELL.

buyer_quote.accept is allowed only when active_trader_mode_context = BUY.

trade.request.create (initiating a new trade request) is allowed only in BUY mode.

logistics.bundle.reoptimise is allowed only in SELL mode.

If a user attempts an action that is not allowed in their current mode—for example, an employee in Buyer mode tries to submit an EXW price lock—the Governor returns a DENY verdict with a tenant_message that is designed for this specific scenario: “You are currently in Buyer mode. This action (submitting an EXW price) can only be performed in Seller mode. Switch to Seller mode to continue.” The PlainLanguage Governor Decision Panel displays this message, and the panel itself includes a prominent “Switch to Seller Mode” button that, when clicked, triggers the toggle swap and then automatically retries the action. This provides a seamless recovery path for users who inadvertently attempt actions in the wrong context.

Database and API

The employees table stores two relevant fields:

default_trader_mode (BUY, SELL, or DUAL) – the employee’s default context, set during onboarding or by a tenant admin. This is used when the employee first logs in if no active context is set.

active_trader_mode_context (BUY or SELL, nullable) – the currently selected mode for this session. If null, the system falls back to default_trader_mode. The frontend sends this value in the switch-context request, and the backend updates the session and issues a new JWT with that claim.

The data_scopes table includes the boolean column allow_role_switching. When false, the toggle is hidden for that employee, and they are locked into their default_trader_mode. This is typically used for finance clerks, warehouse operators, or junior staff who should only see one side of the business.

The API endpoint POST /v1/employee/switch-context accepts the JSON body {"active_trader_mode_context": "BUY"} and returns a new JWT. The endpoint is gated by the Governor to ensure the employee has allow_role_switching = true and the requested mode is a subset of their allowed_trader_modes.

Worked Example – Dual-Mode Trading Company Employee

Nile Foods, a large Egyptian trading company, has default_trader_mode = DUAL. Ahmed, a senior trade manager, has default_trader_mode = DUAL and allow_role_switching = true.

Ahmed logs in. The portal defaults to the last active mode from his previous session, which was Buyer. He sees his inbound shipments, including a container of Vietnamese oranges arriving next week. The Smart Inbox shows a high-priority item: “Sign contract for USTN SGTX-EG-...-V1 before 14:00 UTC.” He signs the contract, and the USTN is generated.

An hour later, Ahmed needs to respond to a buyer’s counter-offer on an outbound shipment of Egyptian dates to Germany. He looks at the toggle in the header, currently showing “Buyer” as active. He clicks “Seller.” The portal instantly switches: the Smart Inbox now shows “Pending Responses: Buyer Nile Foods submitted a counter-offer 26 hours ago.” The sidebar navigation now includes “Pending Requests,” “EXW Price Lock,” and “Logistics Builder.” He navigates to the pending request, adjusts the EXW price, and submits the counter-offer.

Later that day, Ahmed is reviewing a shipment in the Trade Command Center while in Seller mode. He realises he needs to check the customs readiness of the inbound oranges. He flips the toggle back to Buyer. The TCC for the inbound shipment updates immediately, showing the buyer’s perspective. He sees that the phytosanitary certificate is still missing and uses the “Request from Seller” button to send a reminder.

Throughout this day, Ahmed never logs out. The toggle allows him to fluidly move between the two sides of his company’s operations, with the system always ensuring he sees the right data, the right actions, and the right compliance explanations.

Accessibility

The toggle is fully keyboard-accessible. It is a single tab stop; the left and right arrow keys move between the Buyer and Seller segments; Enter or Space activates the switch. The active segment has aria-pressed="true". The voice command functionality is available in all languages supported by Vosk, with fallback to English parsing. The toggle’s state change is announced by screen readers via an aria-live region: “Switched to Buyer mode” or “Switched to Seller mode.”

END OF SUB-SECTION 6.1.12 – BUYER/SELLER MODE TOGGLE FOR THE UNIFIED TRADER PORTAL (v6.3 WITH ALL MODIFICATIONS)

6.1.13 – MY SHIPMENTS TABLE (UNIFIED, ROLE-AWARE, REAL-TIME SHIPMENT LISTING WITH ADVANCED FILTERING, ORIGINAL VS ACTUAL DATES, CLICKABLE ROLE-BASED DOCUMENTS, AND DUAL-MODE TRADER SUPPORT)

6.1.13.0 Purpose & Design Philosophy

The “My Shipments” table is the single source of truth for every shipment a user can access. It replaces fragmented shipment lists and provides a professional, high-density operational view. The table is:

Role-aware and dual-mode aware – A corporate trader sees inbound or outbound shipments depending on the Buyer/Seller toggle. Logistics, QC, financiers, and government officials see only what their role permits.

Real-time – Updates via NATS when milestones change, documents are uploaded, or statuses update.

Highly filterable and searchable – By status, date range, commodity, port, vessel, container number, and more.

Exportable – CSV or PDF of the current filtered view (cryptographically signed).

Actionable – Click any USTN to open the Trade Command Center. Inline document access, notes, and quick actions.

Compliant – Document downloads are strictly controlled by role and row-level security.

Additional high-end, professional enhancements (zero-cost) included in this specification:

Predictive delay warnings – AI estimates probability of delay based on historical performance and current conditions, shown as a coloured badge.

Custom saved views – Users can save filter, sort, and column configurations as named views.

Bulk actions – Select multiple shipments to batch-request documents or add notes.

Document expiry alerts – If a certificate (e.g., phytosanitary) is expiring soon, a warning icon appears.

Offline cache – The table caches last-seen data in IndexedDB for offline access on desktop/mobile.

All features are built on the zero-cost, self-hosted stack (PostgreSQL, NATS, React, TanStack Table, Leaflet for maps, etc.). No billing details ever required.

6.1.13.1 Core Data Model Additions

To support the enhanced table, the following columns and tables are added to Part 5:

sql

```
-- Add actual departure and arrival timestamps to shipments
ALTER TABLE shipments ADD COLUMN actual_departure TIMESTAMPTZ;
ALTER TABLE shipments ADD COLUMN actual_arrival TIMESTAMPTZ;

-- Add container numbers array (if not already present)
ALTER TABLE shipments ADD COLUMN container_numbers TEXT[];

-- User-specific notes per shipment
CREATE TABLE user_shipment_notes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
shipment_ustn TEXT NOT NULL REFERENCES shipments(ustn) ON DELETE CASCADE,
```

user_employee_id UUID NOT NULL REFERENCES employees(id) ON DELETE CASCADE,

note_text TEXT NOT NULL,

created_at TIMESTAMPTZ DEFAULT NOW(),

updated_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(shipment_ustn, user_employee_id)

);

-- Document role visibility (can also be stored in documents.visible_to_roles array)

ALTER TABLE documents ADD COLUMN visible_to_roles TEXT[] DEFAULT ARRAY['BUYER','SELLER','FINANCIER'];

ALTER TABLE documents ADD COLUMN visible_to_tenant_ids UUID[]; - - for logistics own uploads

-- User saved views for the table

CREATE TABLE user_saved_views (

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

user_employee_id UUID NOT NULL REFERENCES employees(id) ON DELETE CASCADE,

view_name TEXT NOT NULL,

filters JSONB NOT NULL,

sort_column TEXT,

sort_order TEXT,

visible_columns TEXT[],

created_at TIMESTAMPTZ DEFAULT NOW()

);

6.1.13.2 Column Definitions (Complete List)

The table includes the following columns. Columns with (role-specific) appear only for certain roles. All columns are sortable unless noted.

Column Name	Description	Data Source	Role Availability
USTN	Unique shipment identifier. Clickable link to Trade Command Center.	shipments.ustn	All
Shipment Number (multi-shipment)	e.g., “2 of 3”. Links to master contract.	contract_shipments.shipment_number	All
Contract Number	Master contract reference.	contracts.reference_number	All
Status	Current milestone with colour-coded	shipments.status +	All

	badge and icon. Shows delay warning if applicable.	computed delay	
Commodity	Primary commodity (HS code + product name).	trade_requests.parsed_specs	All
Quantity	Total quantity (weight or units) with unit.	Aggregated from commodities	All
Origin (POL)	Port of loading (UN/LOCODE + name).	shipments.origin_port	All
Destination (POD)	Port of discharge.	shipments.destination_port	All
Number of Containers	How many containers.	shipments.container_count	All
Container Numbers	Comma-separated list, clickable to copy.	shipments.container_numbers	All
ETD (Original)	Originally estimated departure date.	shipments.etd	All
Actual Departure	Actual departure date (if vessel sailed). Displayed only if differs from ETD.	shipments.actual_departure	All
ETA (Original)	Originally estimated arrival date.	shipments.eta	All
Actual Arrival	Actual arrival date (if delivered). Displayed only if differs from ETA.	shipments.actual_arrival	All
Delay Indicator	“On time”, “+X days”, “Delayed”. Colour-coded.	Computed from actual vs original	All
Predictive Delay Warning	AI probability of future delay (e.g., “70% chance of delay”). Advisory.	AI predictor (LSTM)	Buyer, Seller, Logistics
Payment Status	Unpaid, Paid, Overdue, Disputed.	settlement_instructions.status	Buyer, Seller, Financier
Documents	Clickable badge with count of accessible documents. Opens modal with download links.	documents with RLS	All (permissions enforced)
Document Expiry Alerts	Warning icon if a certificate expires within 7 days.	documents.expiry_date	Buyer, Seller, Government
Import / Export	“Import” (buyer) or “Export” (seller).	Derived from user role vs trade	Buyer, Seller
Buying From (Buyer mode)	Seller’s legal name and GTID.	tenants.legal_name	Buyer mode

Selling To (Seller mode)	Buyer’s legal name and GTID.	tenants.legal_name	Seller mode
Logistics Provider	Primary carrier or forwarder.	quote_selected_services	Buyer, Seller
Risk Score	AI-computed risk (0-100) for compliance or credit. Colour-coded.	shipments.risk_score	Government, Financier, Admin
Notes	User-private free-text notes (editable inline).	user_shipment_notes	All (personal)
Last Updated	Timestamp of last status change or document upload.	shipments.updated_at	All

Role-specific additional columns (e.g., for logistics, QC, financier) are defined in the same way as the Shared Shipments Vault (Part 6.1.6.3) and are automatically appended when the user belongs to that role.

6.1.13.3 UI Layout and Interaction

Toolbar (above the table)

Global Search – Free-text search across USTN, container numbers, vessel name, buyer/seller name, and commodity.

Filter Bar – Collapsible row of filter chips:

Status – Multi-select dropdown (e.g., Loaded, Departed, Arrived, Delivered).

Date Range – Presets: “Next 7 days”, “Next 30 days”, “Last 7 days”, “Custom”. Applied to ETD/ETA.

Commodity – Type-ahead HS code or product name.

Port – Origin or destination port (UN/LOCODE).

Risk Score (government/financier) – Slider range.

Delay Status – “On time”, “Delayed”, “Severely delayed”.

Saved Views – Dropdown to save, load, rename, or delete current filter/sort/column configuration. Stored per user.

Column Visibility – Button to show/hide columns. Preferences saved per user and per mode (Buyer/Seller).

Export – CSV or PDF of current filtered view (async with notification).

Refresh – Manual refresh (real-time updates also via NATS).

Main Table

Virtual scrolling (react-window) for smooth performance with thousands of rows.

Clickable rows – Clicking anywhere on a row opens the Trade Command Center for that USTN in a new tab (configurable to same tab).

Inline notes – Click the Notes column to edit directly (textarea saves on blur).

Bulk selection – Checkboxes on each row; bulk action bar appears (e.g., “Request Documents”, “Add Tag”, “Export Selected”).

Row actions (icons on the right) :

View TCC

Upload Document (if user is responsible)

Message Counterparty (opens Trade Room)

Add to Watchlist

✏ Edit Notes

Expandable row details (optional) – Click a “+” icon to see a mini-timeline of the last 5 milestones.

Document Modal (clicking the Documents badge)

Lists all documents the user is allowed to see, with file name, type, upload date, and download icon.

For buyer/seller : “Download All” button (ZIP of all accessible documents).

For documents with expiry dates (certificates), shows “Expires in X days”.

Modal respects role-based visibility (enforced server-side).

Pagination

Default 25 rows per page. Options: 10, 25, 50, 100.

Alternative infinite scroll mode (user preference).

Empty State

“No shipments match your filters. Try adjusting or clear filters.”

6.1.13.4 Role-Specific Default Views (Dual-Mode Aware)

User Role / Mode	Default Filters	Default Columns (visible)
Trader – Buyer mode	Status = active inbound (CREATED, GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED, CUSTOMS_IMPORT). Date range = next 7 days (ETA).	USTN, Status, Commodity, POL, POD, ETD, ETA, Delay Indicator, Containers, Documents, Buying From, Payment Status, Notes
Trader – Seller mode	Status = active outbound (CREATED, GATED_IN, LOADED, DEPARTED, IN_TRANSIT, ARRIVED). Date range = next 14 days (ETD).	USTN, Status, Commodity, POL, POD, ETD, ETA, Delay Indicator, Containers, Documents, Selling To, Payment Status, Notes
Logistics	Shipments with an accepted quote; status = active (any).	USTN, Status, Service Type, Pickup Window, Route Progress, Documents (own uploads), Notes
QC	Assigned inspection jobs; status = active.	USTN, Status, Commodity, Inspection Status, Defect Count, Report Due, Documents (QC reports)
Financier	Shipments linked to active financing agreements.	USTN, Status, Financed Amount, Outstanding Balance,

		LTV, Next Repayment, Repayment Health, Risk Score
Government	Shipments where origin/destination/transit = jurisdiction.	USTN, Status, Risk Score, Clearance Status, Declaration Reference, Anomaly Flags, Documents (compliance)
Admin	All shipments (anonymised by default).	USTN Mask, Status, Corridor, Commodity Category, Platform Risk Score, Governance Flags

Dual-mode toggle effect : When the user switches from Buyer to Seller (and vice versa), the table automatically reloads with the saved preferences for that mode. The toggle is only visible for CORPORATE tenants with default_trader_mode = DUAL (or BUY/SELL). Other roles never see the toggle.

6.1.13.5 Original vs Actual Dates & Delay Handling

The table distinguishes between planned and actual dates.

ETD / ETA (Original) – Always shown.

Actual Departure / Actual Arrival – Shown only if they exist and differ from the original. The difference (delay) is displayed inline, e.g., “+2 days”.

Delay Indicator Column – Summarises the overall delay status:

Green checkmark: On time or early.

Amber clock: 1-3 days late.

Red warning: >3 days late.

Predictive Delay Warning (AI, advisory) – For shipments not yet completed, an XGBoost model (trained on historical AIS, weather, port congestion) predicts probability of delay >1 day. Displayed as “70% delay risk” with a tooltip. This model runs locally, updated daily.

Tooltip on delay – Clicking the delay indicator shows a Groq-generated explanation: “Delayed due to port strike in Alexandria (estimated +4 days).”

Data sources:

Original dates from shipments.etd and shipments.eta.

Actual dates from shipment_milestones where milestone = ‘DEPARTED’ (actual_departure) and milestone = ‘DELIVERED’ (actual_arrival). The system automatically populates shipments.actual_departure and actual_arrival when those milestones are confirmed.

6.1.13.6 Role-Based Document Access – Detailed Enforcement

Document download and visibility are strictly controlled:

Buyer and Seller (trader modes) : Can see all documents related to the trade (invoice, packing list, BL, certificates, QC reports, insurance, etc.). No distinction between buyer and seller – both see the full set.

Logistics Provider : Can only see documents that they themselves uploaded (e.g., booking confirmation, delivery note, invoice). They cannot see commercial invoices or QC reports.

QC Provider : Can only see their own inspection reports and attached evidence.

Government Official : Can see only compliance-related documents: certificates (phytosanitary, origin, fumigation), customs declaration, permits, and any document flagged as required by the RIA. They do not see commercial invoice amounts or internal logistics costs.

Financier : Full disclosure – all trade documents (as per Phase 4).

Admin : Can see all documents but requires break-glass or multisig approval for full access (anonymised by default).

Technical enforcement:

Each document record has a visible_to_roles array and optionally visible_to_tenant_ids. When a user requests a document, the Governor checks:

User's role and tenant ID.

For logistics : also checks if the user's tenant matches the uploaded_by_tenant_id (or if the document was explicitly shared).

For government : checks that the document type is in the compliance_document_types list (maintained by RIA).

Download URLs are presigned and expire after 1 hour. All download attempts are logged.

Document expiry alerts:

If a certificate (e.g., phytosanitary) has an expiry date stored in documents.metadata.expiry_date and it is within 7 days, a small ⚠ icon appears next to the document badge. Hover shows "Expires on 25 Jun 2026".

6.1.13.7 Real-Time Updates via NATS

The table subscribes to the following NATS subjects:

shipment.status.changed.{ustn} – updates status, delay indicator, and last updated.

shipment.milestone.confirmed.{ustn} – updates actual departure/arrival, delay.

document.uploaded.{ustn} – updates document count and expiry alerts.

payment.status.changed.{ustn} – updates payment status.

shipment.risk_score.updated – updates risk score (government/financier).

When an event arrives, the affected row is highlighted with a brief amber flash and the data is patched. No full page reload.

6.1.13.8 Bulk Actions & Saved Views

Bulk actions (available when at least one row is selected via checkbox):

Request Documents – Sends a pre-filled message to counterparties for missing docs on all selected shipments.

Add Note – Adds the same private note to all selected shipments.

Export Selected – Exports only the selected rows (CSV or PDF).

Add to Watchlist – Watches all selected shipments.

Saved views:

Users can save the current combination of filters, sort column, sort order, and visible columns as a named view (e.g., “My Delayed Citrus Shipments”).

Views are stored in user_saved_views and synchronised across devices via NATS.

A dropdown in the toolbar allows loading, renaming, or deleting views.

6.1.13.9 Offline Cache (Desktop & Mobile)

The table caches the last loaded data in IndexedDB (web) or WatermelonDB (mobile). When offline, the user can still view the cached shipments, filter locally, and see stale data with a “Last updated: [time] (offline)” banner. Queued actions (e.g., adding notes) are stored and synced when connectivity returns.

6.1.13.10 Performance and Accessibility

Virtual scrolling for rows (TanStack Table + react-window).

Debounced search (300ms).

Keyboard navigation – Tab, arrow keys, Enter to open TCC.

Screen reader support – ARIA labels for all interactive elements.

High contrast mode – Follows system preferences or user setting.

6.1.13.11 Integration with Other Blueprint Parts

Part 5 – Adds the tables and columns listed in 6.1.13.1.

Part 6.1.1 (Smart Inbox) – Clicking a shipment-related inbox item highlights the corresponding row and scrolls it into view.

Part 6.1.6 (Shared Shipments Vault) – The My Shipments table uses the same aggregation service but with a different UI configuration.

Part 6.2.1 (Trader Portal) – The “Shipments” tab now renders the My Shipments table.

Part 6.1.4 (Data Export) – Exports from this table use the same signed export service.

6.1.13.12 UI Wireframe (Buyer Mode, with Delayed Shipment and Document Modal)

text

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ MY SHIPMENTS (Buyer) [Saved Views ▼] [Export] [⚙]│
├──────────────────────────────────────────────────────────────────────────────┤
│ Search: [________________] Status: [Active ▼] Date: [Next 7 days ▼] [More filters] │
├──────┬─────────┬──────────┬──────────┬──────────┬──────────┬──────────┬───────┬────────┤
│ ☐ │ USTN │ Status │ Commodity│ POL │ POD │ ETD/ETA │ Delay │ Docs │
├──────┼─────────┼──────────┼──────────┼──────────┼──────────┼──────────┼───────┼────────┤
│ ☐ │ SGTX... │ □ Arrived│ Oranges │ Saigon │ Alex │ ETD:14Jun│ +4 days │ □ 5 │
│ │ │ (delayed)│ │ │ │ Act:16Jun│ │ (click) │
│ │ │ │ │ │ │ ETA:20Jun│ │ │
│ │ │ │ │ │ │ Act:24Jun│ │ │
│ Buying From: Mekong Fresh (SGTX-VN-TRD-002) Containers: MAEU8901234, MAEU8901235 │
│ Payment: Unpaid (due 25 Jun) Notes: [Awaiting customs clearance] ✏ │
│ Actions: [View TCC] [Message Seller] [Add to Watchlist] │
└──────────────────────────────────────────────────────────────────────────────┘
```

[Document Modal when clicking Docs badge]

```
┌─────────────────────────────────────────────────────────────┐
│ Documents for USTN SGTX-EG-20260515-ABC123 [Download All] │
├─────────────────────────────────────────────────────────────┤
│ □ Commercial Invoice.pdf (uploaded 10 Jun) [Download] │
│ □ Packing List.pdf (uploaded 10 Jun) [Download] │
│ □ Bill of Lading.pdf (uploaded 12 Jun) [Download] │
│ □ Phytosanitary Certificate (expires 30 Jun) ⚠ [Download] │
│ □ Certificate of Origin.pdf (uploaded 9 Jun) [Download] │
└─────────────────────────────────────────────────────────────┘
```

6.1.13.13 Zero-Cost Technology Stack

Component	Technology	Cost
Table & virtual scrolling	TanStack Table, react-window	$0
Real-time updates	NATS WebSocket + Zustand	$0
Filtering & sorting	Server-side PostgreSQL + client-side URL state	$0
Document modal & downloads	Governor + presigned URLs	$0
Role-based visibility	OPA + RLS	$0
Export (CSV/PDF)	tenant-data-export-service	$0
Predictive delay warnings	LSTM model (local) + AIS data	$0
Saved views	user_saved_views table + NATS sync	$0
Offline cache	IndexedDB (web) / WatermelonDB (mobile)	$0
Map integration (optional)	Leaflet + OSM tiles	$0

No billing details, no external APIs, no cloud dependencies. All components are self-hosted and open-source.

6.1.13.14 Implementation Checklist

Add database tables and columns (Part 5 DDL).

Extend tenant-experience-service to support the new aggregation and filtering.

Build React component MyShipmentsTable with all subcomponents.

Implement role-based column visibility using OPA policies.

Add NATS subscriptions for real-time updates.

Implement document modal with permission checks.

Add offline caching (IndexedDB / WatermelonDB).

Integrate into Trader Portal (replace or supplement existing shipments tab).

Test with dual-mode toggle (Buyer/Seller) for corporate tenants.

Ensure no other roles see the toggle.

Deploy and verify with real shipment data.

End of Part 6.1 – Common Shared Components (v6.3 with all modifications). Next: Part 6.2 – Portal-by-Portal Breakdown.

PART 6.2.1 – TRADER PORTAL (BUYER & SELLER DASHBOARDS) – v6.3 FULLY EXTENDED, WITH UNIFIED PORTAL, DUAL-MODE TOGGLE, STRUCTURED CONTAINER FORM, MULTI-SHIPMENT, LOADING ORIGIN, MANUAL LOGISTICS, ALTERNATIVE PORTS, SGTX FEE ADDED BY SELLER, AND ALL MODIFICATIONS

6.2.1.0 Access & Purpose

Access: Employees of tenants with type = CORPORATE and any trader mode (BUY, SELL, or DUAL).
The single Trader Portal replaces the formerly separate Buyer and Seller Portals.
For tenants with default_trader_mode = DUAL and employees with allow_role_switching = true, the Buyer/Seller Mode Toggle (Part 6.1.12) appears in the global header. This toggle switches the entire portal between the Buyer Dashboard (for import/ procurement activities) and the Seller Dashboard (for export/ sales activities).
For tenants with only BUY or only SELL, the toggle is hidden and the appropriate dashboard is displayed permanently.

Purpose:

Buyer Dashboard – Initiate trade requests using a structured per-container form, request multi-shipment contracts, review and compare quotes (including alternative delivery ports), negotiate or amend terms, sign contracts (SGTX fee already paid by the seller), track inbound shipments (active and archived), manage documents and customs readiness, view distressed cargo from trusted counterparties, file disputes, access buyer-centric financing, and manage company settings.

Seller Dashboard – Receive incoming trade requests (including multi-shipment proposals), lock EXW prices with AI market intelligence, design packing based on the buyer's structured data, determine logistics costs (manually or via RFQ), offer alternative delivery ports, calculate the SGTX trade fee (paid by seller, added to the total quote), respond to multi-shipment schedules, submit quotes with full price transparency, manage outbound shipments, handle distressed cargo as a seller (including "Check Buyers" and Accelerated Outreach), file disputes, and manage company settings.
Both dashboards share the same session and URL; the toggle instantly reconfigures the portal without re-authentication. All portal features are uniformly available to every verified tenant – no operating modes exist; visibility is controlled solely by RBAC, data scopes, and the active dashboard mode.

6.2.1.1 Landing Page – Smart Inbox (Buyer or Seller Context)

The Smart Inbox (Part 6.1.1) is the default landing for all users. Its content is entirely determined by the active toggle state.

In Buyer mode, the inbox surfaces incoming shipment alerts, pending contract signatures, unanswered counter-offers from sellers, customs readiness deadlines, multi-shipment confirmations, and financing opportunities where the tenant is the borrower.
In Seller mode, the inbox surfaces pending trade requests, EXW lock deadlines, carrier risk alerts, logistics quote expirations, incoming counter-offers from buyers, QC report submissions, and financing opportunities where the tenant is the borrower.
The AI Summary Card (Groq/Ollama) adapts its language: “You have two high-priority items: sign the contract for the inbound citrus shipment, and upload the missing phytosanitary certificate” (Buyer) vs. “Your carrier risk alert requires immediate reoptimisation. Also, respond to Nile Foods’ counter-offer before it expires” (Seller).

6.2.1.2 Navigation & Tabs – Buyer Dashboard

When the toggle is set to Buyer, the left sidebar presents a task-based workflow order:

Order	Tab Name	Description	Relevant Phases
1	Inbox	Smart Inbox – default landing	All
2	Shipments (Active / Archive)	Shared Shipments Vault with toggle; inbound shipments	5,6,7
3	New Trade Request	Structured container form, GTID entry, multi-shipment request option	1
4	Quote Review & Negotiation	Compare seller’s delivery options, landed cost breakdown, AI negotiation bot	3
5	Contract Signing	Review final contract, sign with ZITADEL passkey (SGTX fee already paid by seller, shown as informational line)	3
6	Customs Readiness	Document checklist for import clearance, one-click requests	5.3
7	Distressed Cargo	View listings from saved contacts, make offers as buyer	7/8
8	Disputes	File dispute, mediation log, AI settlement proposals	10
9	Saved Contacts	Seller profiles with performance metrics, trust portraits, relationship health	18
10	Financing (if enabled)	Request financing, view bids, manage	4

		active loans	
11	Company Admin	Invite employees, RBAC, data scopes, approval chains, branding, exit center	6.2.8

6.2.1.3 Navigation & Tabs – Seller Dashboard

When the toggle is set to Seller, the sidebar presents the natural export workflow order:

Order	Tab Name	Description	Relevant Phases
1	Inbox	Smart Inbox – default landing	All
2	Shipments (Active / Archive)	Shared Shipments Vault with toggle; outbound shipments, estimated margin column	5,6,7
3	Pending Requests	Accept/decline incoming trade requests (including multi-shipment proposals)	1→2
4	EXW Price Lock	AI-recommended price range, live market chart, post-lock price watch	2.1
5	Containerisation & Packing	Palletisation solver (ORTools), 3D container loading, collaborative editing, loading guide	2.2
6	Logistics Builder	RFQ distribution, AI bundle optimiser, manual cost entry, alternative delivery ports	2.3, 2.4
7	Quote Submission	Assemble total delivered price (EXW+logistics+SGTX fee), submit to buyer	2.7, 2.8
8	QC Booking	Select QC provider, AI-recommended inspection points	2.4
9	Document Finalisation	Sign packing list & invoice with passkey, translate automatically	3.5
10	Barcode Print	Generate ZPL/PDF label sheets with dynamic content	2.2→5
11	Distressed Cargo & Notifications	Declare distressed cargo, triage	7/8

		dashboard, “Check Buyers”, Accelerated Outreach with privacy notice	
12	Disputes	File/respond to disputes, AI-compiled evidence, mediation log	10
13	Saved Contacts	Enriched profiles of buyers, logistics, QC providers	18
14	Cash Position	90-day rolling cash forecast, AI-generated summary, detailed ledger	All
15	Company Admin	Employee management, data scopes, approval chains, branding, exit center	6.2.8

6.2.1.4 Shared Components Across Both Dashboards

VoiceCommandButton – examples: “Show my inbound shipments arriving next week” (Buyer), “Accept the lowest logistics quote for USTN …” (Seller), “Switch to Seller mode”.

AIAssistantPanel – answers natural-language queries scoped to the active mode.

PlainLanguage Governor Decision Panel – appears automatically when an action is blocked; if the block is due to a dual-mode mismatch, the panel offers to switch mode.

Customer Care Chatbot – accessible from any tab; can impersonate the user (with PIN) to perform support actions; escalates to human support if unresolved.

6.2.1.5 Detailed Tab-by-Tab Specifications

(Only tabs that are mode-specific or have significant v6.3 enhancements are detailed; common administrative tabs reference Part 6.2.8.)

New Trade Request Tab (Buyer Side, Phase 1)

This is the primary interface for creating a new trade request. It replaces the older free-text form. An optional “Express Mode” toggle allows free-text AI parsing as a fallback, but the structured form is the default for precision.

Step 1 – GTID Entry & Resolution: The buyer manually enters the seller’s GTID. Autocomplete resolves to legal name, jurisdiction, trust score, and sanctions status. If the seller is a saved contact, a badge appears.

Step 2 – Structured Container & Commodity Entry: Number of Containers (1–50). Dynamic tabs per container with fields for Country of Origin, Destination Country, Port of Discharge (dependent dropdown), Palletized? (toggle), Pallet Size (EUR/ISO/Custom), Destination Override, and container-level notes. Inside each container, the buyer adds commodities (Commodity Type, Product via HS lookup or dropdown, Product Specification, Packaging, Number of Pallets, per-commodity notes). “Clone Container” copies all fields. Global notes bar at the bottom.

Step 3 – Multi-Shipment Request (Optional): Toggle “Request multi-shipment contract”. A schedule builder appears with shipment number, delivery date, port (can differ), container count, and commodities (inherited from main, with per-shipment override). Stored in trade_requests.multi_shipment_schedule.

Step 4 – AI Container Advisor (Advisory): After commodity entry, Groq (fallback Ollama) suggests container type and count. Banner: “Based on your volume, we suggest 2 × 40′ HC reefers. Adjust?” Buyer can accept or ignore.

Step 5 – Marketplace Relationship Detection: System checks partner_lead_attributions. If a prior marketplace introduction exists, a banner explains attribution and revenue share, with a 72-hour dispute link.

Step 6 – Governor PreScreen & Submission: All fields validated. If ALLOW, trade request is created with status PENDING_SELLER_RESPONSE. If DENY/CONDITIONAL, the Decision Panel appears.

Draft Auto-Save : Autosave every 30 seconds; resumable from Smart Inbox (“Continue draft trade request”). Drafts expire after 14 days.

Quote Review & Negotiation Tab (Buyer Side)

When the seller submits a quote (which may include multiple delivery options and/or a multi-shipment schedule), the buyer sees a comparison table.

Single-shipment quotes : Columns: Port of Delivery, Transit Time, Total Price (including SGTX fee, displayed as “Total includes SGTX fee of $X”), Actions. The buyer can Accept, Negotiate (propose a different total price), or Amend (request changes to non-price terms such as weight, packaging, port, etc.). Expandable “Landed Cost Breakdown” panel shows EXW, logistics, SGTX fee (informational), and total.

Multi-shipment quotes : A schedule table per shipment with same options. The buyer can accept the whole schedule, negotiate per-shipment, or amend.

Negotiation Bot Transparency Panel is displayed when bot is active.

Quote Submission Tab (Seller Side, Phase 2)

The seller assembles the quote with full cost breakdown.

Loading Origin (mandatory) : Country of Loading, Port of Loading.

EXW Price Lock : Live market chart, AI-recommended band, price deviation rules.

Logistics Builder : Toggle between “I have fixed prices” (manual entry per incoterm, WasmEdge validated) and “Request quotes” (RFQ to logistics providers). Alternative delivery ports can be added.

SGTX Fee Calculation : Dynamic rate (0.1%–2.5%, commodity- and country-pair-specific) applied to total trade value (EXW + logistics). Fee is paid by the seller and added to the total quoted price. The seller sees a clear breakdown: EXW + logistics = X;SGTXfee(YX;SGTXfee(YY; Total quoted to buyer = $Z.

Submit Quote : Governor validates all fields. Status → QUOTED. The buyer receives a Smart Inbox item.

Contract Signing Tab (Both Sides, Phase 3)

After mutual confirmation, the contract is assembled with the mandatory SGTX Witness Clause. For single-shipment, the seller pays the fee before lock. For multi-shipment, the master contract is signed upfront with no fee; each shipment's fee is paid when that shipment is locked. The buyer sees "SGTX platform fee: $X (paid by seller)" as an informational line. Both parties sign with ZITADEL passkey. The Governor signs as witness.

Customs Readiness Tab (Buyer Side)

Dynamic checklist of required import documents with traffic-light status (verified, pending, missing). One-click upload or request buttons.

Distressed Cargo & Notifications Tab (Seller Side)

Declare distressed : Select pallets, describe condition, upload photos. AI condition assessment (HF ViT) and dynamic pricing (XGBoost) run.

Triage Dashboard : Three paths – "Sell Quickly", "Comply with Local Law", "File Insurance Claim".

"Check Buyers" advisory: Ranked list of eligible saved contacts. No notifications sent.

Accelerated Outreach : Time-boxed, simultaneous notifications. Mandatory privacy notice opt-in.

Microcontract creation : New microUSTN, separate SGTX fee (adjusted by country distress factor).

6.2.1.6 Worked Example – Dual-Mode User Journey

Nile Foods (Egypt, dual-mode) wants to import oranges from Mekong Fresh (Vietnam) and export dates to Germany.

Login : Toggle set to Buyer. Smart Inbox shows inbound alerts. Creates new trade request for oranges using the structured form, with a multi-shipment schedule (2 shipments).

Mekong Fresh submits a quote with two delivery options. Buyer reviews in Quote Review tab, accepts Alexandria option.

Contract signed (SGTX fee shown as paid by seller). USTN generated. Shipment tracking begins.

Later, the buyer sees a Smart Inbox item : "Distressed lot of lemons available from Mekong Fresh." Makes an offer, acquires micro-contract.

Switch to Seller : Toggle flips to Seller. Portal now shows outbound shipments. Receives a request from a German buyer for dates.

Seller locks EXW price using the live chart, designs packing, manually enters logistics (CFR), calculates SGTX fee (1.1%), and submits the quote. Buyer accepts.

Seller pays the fee via PSP. Contract locks, USTN generated. Shipment executes.

Switch back to Buyer to approve settlement for the oranges shipment. Settlement paid. Shipment archived.

End-of-month : Seller dashboard's Cash Position tab shows net cash flow from both export and import activities (since the company is dual-mode). The finance manager downloads the monthly reconciliation statement with role filters.

END OF PART 6.2.1 – TRADER PORTAL (BUYER & SELLER DASHBOARDS) – v6.3 WITH ALL MODIFICATIONS

PART 6.2.2 – SELLER DASHBOARD – IN-DEPTH TAB SPECIFICATIONS (TRADER PORTAL) – v6.3 FULLY EXTENDED, WITH ALL MODIFICATIONS

This part provides the detailed functional specification for each tab of the Seller Dashboard within the unified Trader Portal. It assumes the user has already activated the seller context via the Buyer/Seller Mode Toggle (Part 6.1.12) and that the overall portal structure and shared components are as described in Part 6.2.1. All features described here are available to every verified tenant with a trader mode of SELL or DUAL; no operating modes exist. Visibility of individual fields and actions is governed by the employee's RBAC permissions, data scopes, and the hidden_cost_componentsscope.

6.2.2.1 Pending Requests Tab

Lists all incoming trade requests with status PENDING_SELLER_RESPONSE. Each row displays the buyer's GTID, legal name, trust score, commodity summary (HS code and product), requested incoterm, and a "Multi-shipment" badge if the buyer requested a multi-shipment contract. The seller can:

Accept – moves to the EXW Price Lock tab and creates an seller_quote record. The buyer is notified via Smart Inbox.

Decline – with an optional reason. The buyer is notified.

For multi-shipment requests, the seller sees the proposed schedule (the buyer's multi_shipment_schedule) and must either accept the schedule as proposed, propose modifications (counter-schedule), or reject the entire multi-shipment request.

6.2.2.2 EXW Price Lock Tab

After accepting a request, the seller must lock the EXW price before proceeding.

Loading Origin (mandatory) : Country of Loading and Port of Loading, selected from global dropdowns (sourced from jurisdictions and ports). The Governor (G2U17) validates that the loading port is not sanctioned and is compatible with the incoterm (e.g., for FOB it must be in the seller's country). EXW lock is disabled until this is set.

Live Market Chart : A 30-day line chart (Chart.js) for the primary commodity and trade corridor, using free commodity price feeds (FAO, USDA, etc.). A green shaded band shows the AI-recommended price range (Groq, fallback Ollama). The seller enters an EXW price per commodity.

Price Deviation Rule (G2UA1) : If the entered price deviates more than 30 % above the AI band, the Governor returns a CONDITIONAL verdict and requires a written justification (A2, HF local). Prices below the band are allowed without justification.

Post-Lock Price Watch : After lock, a NATS subscription monitors daily market price changes. If the market moves more than 10 % from the locked price, the seller receives a Smart Inbox item: "Market price moved +12 % – your locked price stands. Reopen pricing?” The seller can dismiss or reopen, which triggers a Governor decision and notifies the buyer.

6.2.2.3 Containerisation & Packing Tab

Consumes the buyer’s structured per-container data from trade_requests.parsed_specs. The seller can:

View the buyer’s palletised status, pallet size, packaging, and number of pallets per commodity.

Run the palletisation calculator (ORTools CP-SAT solver) to optimise cartons per layer, layers per pallet, and total pallets. The solver uses carton dimensions (mapped from the packaging field), pallet type, max stacking height (from commodity notes or regulations), and container internal dimensions. Output: cartons per pallet, total pallets, and a 3D loading plan (pallet positions within container).

View the 3D container viewer (Three.js) with collaborative editing (Yjs + WebRTC) for multiple employees. Each user’s cursor is visible, and permissions limit what they can edit (warehouse operators see only pallet positions).

Toggle the capacity heatmap overlay (opt-in, differential privacy ε=0.1) that shows historical loading density for the same commodity and route. High-density zones green, underutilised red.

Lock packing plan – once locked, the plan becomes immutable, SSCC barcodes are generated, and a Loom hash is recorded (G2U9, G2U14). A Groq-generated loading guide in the seller’s preferred language is stored in packing_plans.loading_guide.

AI Loading Guide (A1, advisory) : Groq (fallback Ollama) produces a step-by-step loading guide covering container precooling, pallet order, air gaps, and seal verification.

6.2.2.4 Logistics Builder Tab

The seller determines logistics costs using one of two modes, toggled by a segmented control:

Mode A – I have fixed prices (manual entry): A form that adapts based on the selected incoterm (validated by WasmEdge). Mandatory fields per incoterm are enforced; missing required fields block progression. Available services: trucking, customs clearance, THC, ocean/air freight, insurance, destination charges, duties. For EXW all logistics fields are disabled; for FOB sea freight, destination charges, and insurance are disabled; for CFR/CIF ocean freight and origin charges are mandatory; for DDP destination charges and duties are mandatory. Manual entries are stored in seller_quotes.manual_logistics_costs.

Mode B – Request quotes (RFQ): The seller sends anonymous or directed RFQs to logistics providers. The system distributes the RFQ, providers respond, and the seller selects the best quotes. Selected costs become the logistics cost for the quote.

Alternative Delivery Ports : The seller can add alternative ports within the same destination country. Each alternative includes port (dropdown filtered by destination country), expected transit time (AI may suggest), and additional logistics cost (can be positive or negative). The system calculates total delivered price for each alternative. The Governor validates that the alternative port is in the same country, not sanctioned, and transit time > 0 (G2U19).

ReOptimise Diff View : After an initial logistics bundle is selected, the seller can refresh carrier risk scores. The system presents a side-by-side comparison of the current selection versus the new AI-recommended bundle, and the seller can accept or dismiss the change.

6.2.2.5 Quote Submission Tab

The seller assembles the final quote package.

Summary : Primary delivery option plus any alternatives. For multi-shipment, a table of shipments with per-shipment prices.

SGTX fee calculation : The system computes the dynamic trade fee (0.1 %–2.5 %, commodity- and country-pair-specific) based on the EXW + logistics total. The fee is paid by the seller and added to the total quoted price. The seller sees a breakdown: EXW + logistics = X;SGTXfee(Y X;SGTXfee(Y Y; Total quoted to buyer = $Z.

Display to buyer : The buyer sees the total price and a separate line “(includes SGTX platform fee of $Y)”.

Submit Quote : The Governor validates all fields (G2U22). If passed, trade status → QUOTED, the buyer receives a Smart Inbox item, and the seller receives a confirmation.

6.2.2.6 QC Booking Tab

The seller selects a QC provider for pre-shipment inspection. The system recommends inspection points based on historical defect data and contract quality specs. The seller can view QC reports once submitted.

6.2.2.7 Document Finalisation Tab

The seller digitally signs the packing list and commercial invoice using ZITADEL passkey. Documents can be automatically translated into the buyer’s preferred language via Groq (with disclaimers that it is not a certified translation). Signed documents are stored immutably and visible to the buyer.

6.2.2.8 Barcode Print Tab

The seller prints SSCC-18 and QR labels for each pallet.

Label generation : PDF sheets (Tera + headless Chrome) or direct ZPL for Zebra printers. The seller selects a template (Standard, Customs-Ready, Consignee) and language. AI (LightGBM) optimises label placement on sheets.

Reprint policy : Before loading, reprints are free and logged. After a pallet is scanned as LOADED, a “Request Reprint” button opens a modal requiring a mandatory reason (≥ 10 chars). A Governor decision (barcode.reprint.post_loading) is created; if ALLOW, the reprint proceeds (same SSCC, enhanced contrast). The Decision Panel explains any denial.

6.2.2.9 Distressed Cargo & Notifications Tab

The seller manages distressed goods.

Declare distressed : Select pallets via the packing plan, describe condition, upload photos. AI condition assessment (HF ViT) generates a score (0-100) and tags (e.g., “mould detected 92 %”). Dynamic pricing (XGBoost) suggests a recovery price.

Triage Dashboard : Three paths – “Sell Quickly” (Accelerated Outreach), “Comply with Local Law” (generates jurisdiction-specific reports/ notifications), “File Insurance Claim” (compiles evidence package).

“Check Buyers” advisory: Ranked list of eligible saved contacts, based on past distressed purchase success, trust score, and proximity. No notifications are sent from this button. Selected contacts can be carried forward to outreach.

Accelerated Outreach : The seller sets a time window (2-24 h) and a floor price. Before any notification is sent, a mandatory privacy notice appears listing the contacts and the shared information. The seller must explicitly opt in (logged as a Governor decision).

Microcontract creation : When an offer is accepted, a new microUSTN is generated, a microcontract with a separate SGTX fee (adjusted by the country-specific distressed factor) is created, and a new SGTX FEESLock is activated.

6.2.2.10 Disputes Tab

The seller can file or respond to disputes. The AI-compiled evidence package (including QC overrides flagged) is available. Mediation log supports multi-language translation. AI-generated settlement proposals can be accepted. SGTX FEES disputes (SGTX fee calculation) are routed to the Platform Governance Authority.

6.2.2.11 Saved Contacts Tab

Displays enriched profiles of buyers, logistics partners, QC providers, and financiers that the seller has interacted with. Each contact includes performance dashboards (order frequency, negotiation speed, payment behaviour, dispute rate, financing history), a 360° Trust Portrait (Groq-generated summary using internal trade data and public news), predictive relationship health score (LSTM), indirect connections via GraphRAG, and smart labels. Voice commands are supported.

6.2.2.12 Cash Position Tab

A financial cockpit for the seller.

90-day rolling cash forecast: Expected inflows from settlements, expected outflows for logistics and already-paid SGTX fees. AI-generated plain-language summary.

Detailed ledger : Every cash event (trade settlement received, SGTX fee paid, logistics invoice paid, financing disbursement net, financing repayment) with USTN, date, amount, and status. For multi-shipment, each shipment’s cash flows shown separately.

Currency normalisation : All amounts converted to the seller’s chosen base currency using ECB daily rates. Multi-currency transactions show original amount, FX rate, and base amount.

6.2.2.13 Company Admin Tab (Seller-Side Configuration)

While fully described in Part 6.2.8, the seller dashboard’s Company Admin tab offers role templates specific to sellers (trade_manager, warehouse_operator, finance_user, viewer, customs_specialist). Data

scopes can hide cost components (hidden_cost_components) from specific employees (e.g., a warehouse operator never sees EXW price or freight costs).
The allow_role_switching flag in data scopes controls whether the Buyer/Seller toggle is available per employee.

END OF PART 6.2.2 – SELLER DASHBOARD – IN-DEPTH TAB SPECIFICATIONS (TRADER PORTAL) – v6.3 WITH ALL MODIFICATIONS

PART 6.2.3 – LOGISTICS PORTAL – ROLE-BASED VIEWS (FREIGHT FORWARDER, SHIPPING LINE, TRUCKING COMPANY, CUSTOMS BROKER) – v6.3 FULLY EXTENDED, WITH UNSUBSCRIBE MECHANISM, WARM-UP PROGRAM, CONTAINER RELEASE CONFIRMATION, LOGISTICS ADDENDUM SIGNING, CUSTOMER CARE CHATBOT, AND ALL MODIFICATIONS

6.2.3.0 Access & Purpose

Access : Employees of tenants with type = LOGISTICS and at least one assigned logistics_role(FREIGHT_FORWARDER, SHIPPING_LINE, TRUCKING, CUSTOMS_BROKER). The Buyer/Seller Mode Toggle is not applicable; logistics tenants operate within a single, role-focused context. If an employee holds multiple logistics roles, a role selector dropdown within the portal allows them to switch between their assigned perspectives.

Purpose : Provide a unified but role-adaptive workspace where logistics service providers respond to requests for quotes (RFQs), submit bundled quotes, receive booking requests, update shipment milestones, verify documents, sign logistics addenda, receive container release confirmations, monitor their own performance, manage their notification preferences, and access support. The portal is strictly not a marketplace – it only surfaces requests that have been explicitly directed to the provider by a seller or forwarder, or anonymous broadcasts for which the provider is eligible based on their route network and performance. Providers are never shown unsolicited business opportunities from unknown parties.

All logistics providers are subject to the platform's constitutional governance. They must sign a Logistics Service Addendum before any contract they are involved in can be locked (Phase 3). Container release is controlled by SGTX after the seller's trade fee payment, and the provider must acknowledge the release before loading can commence. The portal integrates the Smart Inbox, Shared Shipments Vault (configured per role), PlainLanguage Governor Decision Panel, and the Customer Care Chatbot. It also includes the Logistics Provider Warm-Up Program and a prominent, one-click unsubscribe mechanism for anonymous RFQs.

6.2.3.1 Common Elements Across All Logistics Roles

6.2.3.1.1 Landing Page – Smart Inbox

Every logistics user lands on the Smart Inbox (Part 6.1.1). It aggregates all open service requests, pending milestones, document verification tasks, performance alerts, addendum signing requests, and container release confirmations into a single prioritised feed. Common logistics inbox items include:

 High – RFQ response window closing in 2 h (score 95)

 High – Logistics addendum pending signature; contract lock requires your signature (score 90)

 High – Container release confirmation received; acknowledge to proceed (score 85)

 Medium – Customs clearance documents flagged by AI – 3 discrepancies need resolution (score 75)

 Medium – Booking confirmation pending from shipping line; seller waiting (score 70)

 Low – Upcoming pickup window in 12 h, dispatch plan available (score 50)

6.2.3.1.2 Shipments Tab – Shared Shipments Vault (Logistics Configuration)

The Shipments tab renders the Shared Shipments Vault (Part 6.1.6) with a filter preset that limits visible rows to only those shipments where the authenticated provider has an accepted quote or an assigned document responsibility. For multi-shipment contracts, each shipment appears as a separate row. The vault's columns, row actions, and slide-out tabs are configured according to the user's active logistics role (detailed in 6.1.6.3.3). Common columns like USTN, Status, Vessel, Origin/Destination, ETD/ETA, and Last Updated are always present. Role-specific columns (Service Type, My Quote Status, Booking Status, Pickup Window, Clearance Status, etc.) appear automatically.

Clicking any USTN opens the Trade Command Center with a logistics-filtered view – only the services, documents, and milestones relevant to that provider are displayed. Commercial values and counterparty financial data are never shown.

6.2.3.1.3 Request Types and Visibility

When a logistics provider receives a request (RFQ, booking, document verification, addendum signing), it arrives in their Smart Inbox and appears under their role-specific tab. Visibility depends on the sourcing mode chosen by the seller:

Directed (FULL visibility) : The seller explicitly entered the provider's GTID. The provider sees the seller's full legal name, GTID, complete trade details (commodity, volume, ports, contract ID if any), and any special instructions.

Broadcast / Anonymous (ANONYMOUS visibility) : The seller sent an anonymised RFQ to all eligible providers. The provider sees only: commodity HS chapter, total volume/weight, origin/destination country, loading window, and a match score (0–100). The seller's identity is hidden until the quote is accepted. A tooltip (Groq-generated) explains the match score: "You are seeing this because your route network covers Vietnam–Egypt and you have a strong on-time performance rating (92 %)."

6.2.3.1.4 Logistics Addendum Signing (Mandatory)

Before any contract can be locked (Phase 3), every logistics provider selected for the trade must sign a Logistics Service Addendum. This addendum is generated automatically and contains:

The USTN (populated after fee payment).

An obligation not to release the container(s) without SGTX's formal release confirmation.

A requirement to include the USTN and all relevant GTIDs on every document (packing list, invoice, bill of lading, customs declaration).

A penalty clause for unauthorised release.

The provider receives a high-priority Smart Inbox item : “Sign logistics addendum for trade USTN…”. They review the addendum PDF and sign using their ZITADEL passkey. The signature is immutably recorded in logistics_addenda. The contract cannot proceed to fee payment until all required providers have signed. If a provider is missing, the Governor blocks the lock and the seller’s Decision Panel explains which provider is outstanding.

6.2.3.1.5 Container Release Confirmation (After Fee Payment)

After the seller pays the SGTX trade fee (Phase 3), the system automatically sends a Container Release Confirmation to the primary logistics provider (the ocean carrier or the freight forwarder responsible for the container). The confirmation is delivered via:

Email (co-branded : “Seller Name via SGTX” noreply@sgtx.io) containing a one-time release token (UUID, valid 72 h).

API webhook (if the provider has integrated their system).

The provider must acknowledge the confirmation by clicking the link or calling the API. This acknowledgment is recorded in release_confirmations. The “Container Loaded” milestone is blocked by the Governor (G3U8) until acknowledgment is received. The provider’s Smart Inbox shows a priority 85 item: “Container release confirmation received – click to acknowledge.”

6.2.3.1.6 Logistics Provider Warm-Up Program & Unsubscribe Mechanism

Co-branded Notification Emails : When SGTX sends an anonymous broadcast RFQ email, the sender name is formatted as “Seller via SGTX” noreply@sgtx.io. The email body clearly states: “You are receiving this because you are registered on SGTX as a logistics provider serving [route]. An anonymous seller has requested a quote. Your identity will be revealed only if they accept your quote.”

Unsubscribe / Opt-Out : Every co-branded email includes an “Unsubscribe from anonymous RFQs” link in the footer. Clicking it sets tenants.anonymous_rfq_opt_out = true and displays a confirmation: “You have unsubscribed from anonymous RFQs. You will still receive directed requests from known sellers. You can re-enable this at any time in Company Admin → Notification Preferences.”

Within the portal, providers manage this preference under Company Admin → Notification Preferences: Receive anonymous RFQs (default ON), Receive match score explanations (ON), Receive warm-up program emails (ON). Opt-out providers are immediately excluded from all future anonymous broadcast RFQs but continue to receive directed requests. The seller is never notified of the opt-out. The Performance dashboard displays a small indicator: “Anonymous RFQs: Opted out. [Change]”.

Reengagement Emails : If a provider has not interacted with the platform for 30 days, a reengagement email summarises the number of missed opportunities (anonymised). If the provider has opted out of anonymous RFQs, this email is not sent.

6.2.3.1.7 Access to Customer Care Chatbot

A Chat with Support button, represented by a speech bubble icon, is always available in the Logistics Portal’s toolbar. The Customer Care Chatbot (described in full in 6.1.6.3.7) can assist logistics providers with common issues – understanding a match score, troubleshooting a booking integration, resetting a label printer configuration – using its standard PIN-based impersonation for actions on the provider's behalf. If the issue requires human intervention, the bot escalates to the human support queue and arranges a callback.

6.2.3.2 Freight Forwarder Portal

Access : Employees with logistics_roles containing FREIGHT_FORWARDER.

Tabs : Inbox, Shipments (Shared Shipments Vault), RFQ Inbox, Bundled Quotes, Partner Network, Performance, Company Admin.

RFQ Inbox & Quote Submission : Each RFQ row displays the service type, commodity, volume, route, deadline, and match score. An AI Quote Assistant panel alongside suggests a price range based on historical quotes for the same lane and a competitor benchmark, with a "Prefill" button. Forwarders can submit quotes for individual services or create bundled quotes (e.g., ocean + trucking) using subcontractors from their Partner Network.

Partner Network : A private, forwarder-managed directory of trucking companies and customs brokers they regularly work with. Each entry includes the partner's GTID, legal name, trust score, serviceable routes, and historical performance. The forwarder can check "Also offer trucking via subcontractor" and select a partner; the system automatically includes the partner's cost in the bundled quote.

6.2.3.3 Shipping Line Portal & Unified Integration

Access : Employees with logistics_roles containing SHIPPING_LINE.

Integration Channels (dynamic evaluation):

API (preferred) : SGTX calls POST /bookings on the line's endpoint. Health checks every 5 min; three consecutive failures degrade to email.

Email (fallback) : Structured email sent to the line's booking desk. SGTX polls the IMAP inbox and uses HF Donut to extract booking data; confidence ≥ 80 % auto-confirms.

Manual Portal (last resort) : The booking request appears in the line's SGTX portal inbox for manual processing.

Portal Tabs : Inbox, Shipments, Booking Requests (with status: PENDING, PROCESSED, FAILED), Webhook/API Log (last 100 calls, health checks), Health Dashboard, Integration Profile Editor, Performance Dashboard, Company Admin.

eBL Issuance: If the line supports eBL, the system automatically marks "Bill of Lading Issued" when the eBL platform sends a webhook or when HF Donut detects the issuance event.

6.2.3.4 Trucking Company Portal

Access: Employees with logistics_roles containing TRUCKING.

Tabs : Inbox, Shipments, Dispatch Planner, Quote Submission, Mobile App Integration, Performance Dashboard, Company Admin.

Dispatch Planner : Aggregates all pending pickups and dropoffs within the region, runs an ORTools VRP solver to minimise empty miles, respects driver hours and time windows. Output: optimised driver assignment and route map (OSRM). Dispatcher can drag-and-drop to adjust; routes are pushed to drivers' mobile apps. Geofence alerts notify the dispatcher if a driver deviates.

Quote Submission : For each transport request, the provider sees pickup and delivery locations, container type, weight, loading window, match score, and a quote form with an AI-suggested price based on route length and corridor benchmarks.

6.2.3.5 Customs Broker Portal

Access : Employees with logistics_roles containing CUSTOMS_BROKER.

Tabs : Inbox, Shipments, Document Verification Queue, Customs Declaration, Quote Submission, Performance Dashboard, Company Admin.

Document Verification Queue : Each row shows USTN, document type, AI verification status (PASSED / FLAGGED / PENDING), AI-extracted fields, and discrepancies. Actions: Accept (override AI, mark verified), Reject (send back to uploader with note), Submit to Customs (if destination country has API integration). Flagged discrepancies require a broker note.

Customs Declaration Automation : For countries with integrated customs APIs (e.g., Egypt Nafeza, Vietnam VNACCS), the broker clicks "Submit to Customs" on a verified document set. The system auto-fills the official declaration form, signs with the broker's credentials (encrypted), calls the customs API, and stores the declaration reference. If no API exists, a pre-filled PDF is generated for manual submission.

6.2.3.6 Provider Performance Self-Service Dashboard

Every logistics provider has a Performance tab displaying:

On-time delivery percentage (30/60/90 days) per trade lane.

Dispute rate (percentage of jobs where the provider was a party to a dispute).

Risk score history (line chart with annotation events, e.g., "Labour dispute – score dropped").

Invoice accuracy (percentage of invoices that matched the quoted amount).

Anonymous benchmark against other providers in the same category and region (differential privacy, $\varepsilon = 0.1$).

Anonymous RFQ opt-out status indicator.

Groq-generated performance summary (e.g., "Your on-time performance has improved to 89 %. You are in the top quartile for document accuracy.").

6.2.3.7 Worked Example – Freight Forwarder End-to-End

Global Freight Solutions (Vietnam, FREIGHT_FORWARDER) receives an anonymous broadcast for an Egypt-bound citrus shipment.

Smart Inbox : "Ocean freight: 3 × 40′ HC Reefer, Saigon→Alexandria. Match score 91. Respond within 12 h." The forwarder clicks "Respond".

RFQ Inbox : AI Quote Assistant suggests 8,200.TheforwarderdecidestobundletruckingviatheirPartnerNetworksubcontractor,FastTruckingCo.Totalcombinedquote:Ocean8,200.Theforwarderdecidestobundletruckin gviatheirPartnerNetworksubcontractor,FastTruckingCo.Totalcombinedquote:Ocean8,200 + Trucking 900=900=9,100. Submitted.

Quote accepted : Seller's details (Mekong Fresh) become visible. Full shipping instructions appear.

Addendum signing : Smart Inbox item: "Sign logistics addendum for USTN…" The forwarder reviews and signs with passkey.

Shipments Tab : The shipment appears in the vault with "My Quote Status: Accepted". Milestones are tracked.

Container release : After the seller pays the SGTX fee, the forwarder receives an email with the release token, acknowledges it, and the loading milestone becomes available.

Performance Tab : After trade completion, the forwarder sees updated on-time delivery statistics and benchmark position.

6.2.3.8 Governance & Permissions

All logistics providers only see data for their own contracted jobs, enforced by RLS and OPA. The Shared Shipments Vault and TCC filter accordingly.

The PlainLanguage Governor Decision Panel appears automatically if a provider attempts an action blocked by the Governor (e.g., confirming a milestone before acknowledging container release).

The opt-out from anonymous RFQs is a user preference, not a trade action, and does not create a Governor decision, but it is logged in audit_log for compliance.

6.2.3.9 Zero-Cost Technology Stack Summary

Portal frontend: Next.js 15, Tailwind, Radix UI

Real-time updates: NATS WebSocket

AI Quote Assistant: Groq (free tier) + LightGBM (local)

Dispatch Planner (VRP): ORTools (Python)

Document extraction: HF Donut (local)

Email polling & processing : lettre + imap + self-hosted Postfix

Customs API integration: REST client (Rust reqwest)

Mobile app (drivers): React Native + Expo + OSRM (offline)

Performance benchmarks: OpenDP (differential privacy)

Governance: WasmEdge, OPA, Loom

Co-branded emails (Warm-Up) : notification-service with Sender header

Unsubscribe mechanism: PostgreSQL + audit log

Customer Care Chatbot : Rust Axum + Rig + Groq + ZITADEL impersonation

No billing details required. All infrastructure runs on the platform's sovereign K3s nodes.

END OF PART 6.2.3 – LOGISTICS PORTAL – ROLE-BASED VIEWS (v6.3 WITH ALL MODIFICATIONS)

PART 6.2.4 : QC PORTAL – ROLE-BASED INSPECTION MANAGEMENT (v6.3 FULLY EXTENDED, WITH DYNAMIC QUALITY SPECS, AR INSPECTION, OVERRIDE ACCOUNTABILITY, AQL SAMPLING, QC DISPUTE FAST-TRACK, SMART INBOX INTEGRATION, AND ALL MODIFICATIONS)

6.2.4.0 Access & Purpose

Access: Employees of tenants with type = QUALITY_CONTROL.

Purpose : Receive inspection bookings, access dynamic quality specifications extracted from trade contracts, perform on-site inspections (using mobile AR, voice, barcode scanning), record container/seal numbers, capture defects with AI vision assistance, and submit signed inspection reports that may influence shipment milestones – all within a single workspace that never exposes commercial values. The portal is not a marketplace; it handles jobs only from sellers who have explicitly selected the QC provider. The portal integrates the Smart Inbox, Shared Shipments Vault, PlainLanguage Governor Decision Panel, and enforces override accountability (mandatory reason for AI overrides) and QC dispute fast-track (overrides automatically flagged in evidence packages).

Dual-mode toggle : Not applicable (QC tenants have a single role).

6.2.4.1 Landing Page – Smart Inbox

After login, the QC inspector lands on the Smart Inbox (Part 6.1.1). It surfaces time-sensitive inspection jobs, overdue reports, and any disputes involving the QC provider.

QC-specific inbox items (examples):

Priority	Example Item	Score
□ **High**	USTN SGTX-...-V1 – Inspection report deadline is in 2h. Report not yet submitted.	95
□ **High**	USTN SGTX-...-V2 – Seller disputed QC report; review AI-compiled evidence package within 48h.	85
□ **Medium**	USTN SGTX-...-V3 – New inspection job assigned (oranges, 22 pallets). Accept or decline.	70
□ **Medium**	USTN SGTX-...-V4 – Inspection photos contain 3 AI-detected defects. Review and confirm or override.	65
□ **Low**	USTN SGTX-...-V5 – Inspection completed yesterday; report signed and archived.	10

AI Summary Card (Groq/Ollama) : “Your most urgent task is the inspection report for the orange shipment in Alexandria – submit it now to avoid blocking the loading milestone. One new job is waiting for acceptance.”

Clicking any USTN badge opens the Trade Command Center filtered to inspection data; clicking action buttons deeplinks to the exact inspection form.

6.2.4.2 Navigation & Tabs

Tab Name	Description	Relevant Phases
Inbox	Smart Inbox (default landing)	All
Shipments	Shared Shipments Vault, filtered to shipments where the QC provider has an inspection job	5, 10
My Schedule	Calendar view of all upcoming and past inspections with job details	2.4, 5
Inspection Queue	List of assigned inspection jobs with status, deadline, quick actions (“Start Inspection”, “View Report”)	2.4, 5
Performance	Self-service performance dashboard (turnaround time, dispute rate, defect detection accuracy)	All
Company Admin	Invite employees, assign roles, update serviceable commodities/regions	6.2.8

The AIAssistantPanel and VoiceCommandButton are available everywhere; voice commands work online and offline.

6.2.4.3 Dynamic Quality Specifications (From Contract)

During Phase 3 contract lock, the Clause Forge Agent (HF Mixtral) extracts product-specific quality specifications from the contract’s clauses and stores them as JSONB in contracts.quality_specs. These specs are automatically transmitted to the QC provider when an inspection job is created, with no manual data entry.

Example quality specs for a citrus shipment:

json

```
{
"product": "Valencia Oranges",
"hs_code": "0805.10",
"lot": "VN-2405-A",
"packaging": "10 kg cartons, 40 per EUR pallet",
"specifications": {
"size": "72–80 mm diameter",
"colour": "bright orange, no green",
```

"brix_min": 11.0,
"defects_allowed": {
"mould": "0%",
"cracks": "<2%",
"bruising": "<3%",
"insect_damage": "0%"
},
"foreign_matter": "none",
"temperature_range": { "min": 4, "max": 8, "unit": "°C" },
"shelf_life_days_min": 21
},
"inspection_points": [
{ "pallet_id": "OR-001", "priority": "random_sample" },
{ "pallet_id": "OR-019", "priority": "high", "reason": "end of lot, historically higher defect rate" },
{ "pallet_id": "OR-020", "priority": "high", "reason": "end of lot" }
],
"mandatory_checks": ["container_number", "seal_number", "temperature_logger", "pallet_stacking"]
}

The QC provider sees these specs in the Inspection Job Details page. Commercial values (price, SGTX fee) are never shown; only quality and logistics data.

6.2.4.4 Inspection Job Lifecycle

Step 1 – Booking Confirmation & Assignment

When an seller books a QC provider (Phase 2.4), the system creates a qc_jobs record with status = 'ASSIGNED'. A Smart Inbox item and an email are sent. The provider can accept (status → ACCEPTED) or decline (with reason). If declined, the seller is notified to select another QC provider.

Step 2 – Inspector Views Job Details (Web)

From the Inspection Queue tab, the inspector opens a job to see:

Seller & buyer names (for context, hidden if not needed per policy).

Full product specifications (as above), with critical thresholds highlighted in red.

Priority inspection points (colour-coded : red = high priority, blue = random sample).

Container and seal number fields (to be filled during inspection).

Special instructions from seller.

"Start Inspection" button (opens the mobile-optimised inspection flow, or launches the mobile app).

Step 3 – On-Site Inspection (Mobile App)

The inspector uses the SGTX mobile app (React Native + Expo) for field operations. The app works fully offline.

A. Container & Seal Verification

The inspector scans the container's barcode (or enters manually); the system compares with the expected container number from the booking. Mismatch is flagged.

The inspector enters the seal number and takes a photo. The seal number is stored; if broken later, it becomes dispute evidence.

AI (HF ViT) can attempt to read the seal number from the photo as backup.

B. Pallet-Level Inspection with AR Overlay

The inspector points the phone at a pallet. The app recognises the SSCC barcode and overlays:

Product name, lot, expected quality specs.

Priority indicator (red/blue).

The inspector can take photos; the AI (HF ViT) runs locally and automatically highlights potential defects with red bounding boxes and labels (mould, bruising, carton damage).

The inspector can record defects via voice : "Pallet OR019, mould on 3 cartons, severity medium." Vosk transcribes offline; HF Mixtral extracts structured data.

C. Defect Logging & AI Override Accountability (v6.3 – Enhanced)

Each AI-detected defect is listed. The inspector can tap "Confirm" (accepts AI finding) or "Override" (rejects AI finding).

If the inspector overrides an AI finding, a mandatory text reason (minimum 10 characters) is required. The reason is stored in inspection_logs.inspector_notes alongside ai_overridden = true and the original AI confidence score (original_ai_confidence).

UI enforcement : When the inspector taps "Override", a text field appears with a live character counter. The "Submit" button remains disabled until ≥10 characters are entered. Placeholder: "Explain why the AI detection was incorrect (min. 10 characters). E.g., 'Dark spot is surface bruise, not mould'."

All overrides are tracked; the final report displays a badge : "⚠ 2 AI findings overridden – see notes."

D. Sampling Plan Compliance – AQL Calculation (ISO 28591, General Inspection Level II)

The app enforces AQL-based random sampling based on lot size. The following table is used:

Lot Size (pallets)	Sample Size (pallets)	Acceptance Quality Limit (AQL)
2 – 8	2	0.65 (critical) / 1.5 (major) /

		4.0 (minor)
9 – 15	3	Same
16 – 25	5	Same
26 – 50	8	Same
51 – 90	13	Same
91 – 150	20	Same
151 – 280	32	Same
281 – 500	50	Same
> 500	80	Same

The random selection algorithm is deterministic (seeded with USTN + inspector ID) so that any inspector auditing the same lot would select the same pallets.

The inspector must inspect the randomly selected pallets before the report can be submitted.

After mandatory random samples, the app prompts for the high-priority AI-recommended pallets; they can be inspected or skipped (logged).

Override : A QC admin can increase the sample size (but not decrease below the ISO requirement) via the web portal, with a mandatory reason: “Client requested additional inspection”.

E. Final Checks & Seal Reconfirmation

Before closing, the app asks to reconfirm the seal number; GPS location and timestamp are recorded.

Step 4 – Report Generation (AI-Assisted)

After all mandatory pallets are inspected, the system autogenerates a draft inspection report using Groq (A1, fallback Ollama) based on collected data:

Draft report example:

text

QC Inspection Report – USTN SGTX-EG-20260415120000-ABC12345-V1

Inspector: John Nguyen (SGS)

Date: 14 Jun 2026

Container: MAEU8901234

Seal: SGTX-SEAL-001 – Intact

Sampling : 5 random pallets (OR003, OR007, OR011, OR015, OR019) + 1 high-priority (OR020)

Defects found:

| Pallet | Defect | Severity | Qty Affected | Within Spec? | Photo |

|--------|--------|----------|--------------|--------------|-------|

| OR-019 | Mould | Medium | 3 cartons | No (0% allowed) | img_123 |

| OR-020 | Bruising | Low | 2 cartons | Yes (<3%) | img_124 |

Verdict: FAIL (mould on OR019 exceeds contract limit)

Recommendation: Replace affected cartons before loading.

AI Overrides: 1 override recorded – see notes.

Signed: [Digital Signature]

The inspector reviews, can edit text, and digitally signs with ZITADEL passkey. The signed report is stored in documents with a Loom hash.

Step 5 – Report Submission & Milestone Impact

qc_jobs status → COMPLETED.

If verdict is FAIL and seller enabled “block on fail”, the loading milestone for that container is automatically set to BLOCKED_QC_FAIL. Seller must manually override (Governor A3) or request reinspection.

If verdict is PASS or CONDITIONAL, the loading milestone can proceed.

Smart Inbox items for the seller are generated accordingly.

6.2.4.5 QC Dispute Fast-Track (v6.3)

If the buyer or seller disputes the QC report within 48 hours of receipt (via their Disputes tab), the system automatically:

Flags all inspector-overridden AI findings in the evidence package.

Attaches the original AI detection confidence scores and the inspector’s mandatory override reasons.

Renders a comparison table : “AI said mould (confidence 92%). Inspector overrode to ‘bruising’. Reason: ‘dark spot looked like bruise under warehouse lighting’.”

This data is available for the mediator/arbitrator, promoting transparency without slowing the initial inspection.

6.2.4.6 QC Portal – Web Interface (Management & Review)

Inspection Queue Tab : A table of all assigned jobs with columns: USTN, product, pallets, location, deadline (colour-coded), status (ASSIGNED, ACCEPTED, IN_PROGRESS, COMPLETED, DISPUTED). Quick actions: “Start Inspection” (opens mobile-friendly view), “View Report”, “Message Seller”.

My Schedule Tab : A full-calendar view (FullCalendar.js, opensource) showing all inspection jobs with date, time, location, and status. Push notifications are sent 48h and 24h before the earliest inspection window (if vessel ETA is known).

Inspection Job Details Page (Web) : When an inspector opens a job on the web for planning or review, they see the same quality specifications, pallet summary, photo gallery, and report draft (editable). A “Submit Report” button is only enabled when all mandatory pallets have inspection records.

Performance Tab (Self-Service Dashboard):

Average turnaround time (assignment to report submission).

Dispute rate (inspections later disputed).

Defect detection confirmation rate (AI findings confirmed vs. overrode; and how often overrides were upheld in disputes).

Benchmarks against other QC providers (anonymised, OpenDP).

Groq-generated summary : “Your average turnaround is 2.3 days, faster than regional average (3.1 days). Defect detection accuracy is 94%. Override rate is 3% with 100% of overrides validated during disputes.”

Settings – Serviceable Commodities & Regions:
The QC provider can update their own profile (logged, requires admin role): commodity HS code ranges, serviceable countries and ports, inspection methods (in-person, remote video, lab analysis), certifications (ISO 17025, GLP). These fields are used by the seller’s sourcing engine.

6.2.4.7 Governance & Permissions

QC portal users can only view shipments where their tenant has a qc_jobs record. The Shared Shipments Vault enforces this.

All AI findings and inspector overrides are logged in inspection_logs and ai_inference_records. The override reason is mandatory and auditable.

The final QC report is signed with ZITADEL passkey and Loom-hashed, ensuring non-repudiation.

The QC provider cannot view commercial values or SGTX fee details.

The Decision Explanation Panel appears if a report submission or milestone confirmation is blocked by the Governor.

6.2.4.8 Worked Example – End-to-End QC Inspection (v6.3)

Context : Seller Mekong Fresh books SGS for a preshipment inspection of oranges.

Smart Inbox : SGS inspector sees: “New inspection job for USTN SGTX-EG-...-V1 (oranges, 22 pallets). Deadline: within 48h of vessel arrival.” Accepts job.

Inspection Queue : Opens job details. Quality specs show: mould 0%, brix ≥11.0. Inspection points: random sample 5 pallets (OR003, OR007, OR011, OR015, OR019) + high-priority OR020.

On-site (mobile) : Scans container MAEU8901234 – matches. Enters seal number. Takes photo.

Pallet inspection : App randomly selects 5 pallets per AQL table. Scans each. AI detects mould on OR019 with bounding boxes. Inspector confirms. No other defects. On OR020, AI detects a dark spot (confidence 88%) labelled as “mould”. Inspector examines closely, overrides to “bruising (minor)”. Mandatory reason field appears; inspector types: “Dark spot is surface bruise, no mould visible under magnifier.” System logs override with original confidence 88%.

Report draft : Groq writes: “OR019: mould on 3 cartons – FAIL. OR020: bruising (inspector override from mould) – within spec. Verdict: CONDITIONAL PASS (mould limited to one pallet).” Report badge shows “⚠ 1 AI finding overridden – see notes.”

Inspector signs with passkey. Report stored. Loading milestone proceeds (verdict not FAIL). Smart Inbox for seller: “QC report submitted – CONDITIONAL PASS with 1 override.”

Later dispute : Buyer disputes quality of OR019 mould. Dispute fast-track automatically includes the original AI mould detection (confidence 92%) and the inspector’s confirmation; no override error. Dispute resolved with evidence.

6.2.4.9 Zero-Cost Technology Stack

Component	Technology	Cost
Mobile app (offline)	React Native + Expo + WatermelonDB	$0
AR defect detection	HF ViT (Core ML / TensorFlow Lite)	$0 (local)
Voice transcription	Vosk (offline)	$0
Report drafting	Groq (free tier) or local LLM fallback	$0
Sampling plan (AQL)	Custom logic + ISO 28591 tables	$0
Barcode scanning	ZXingC++	$0
Calendar view	FullCalendar.js (MIT)	$0
Performance benchmarks	OpenDP (differential privacy)	$0
Document storage & signing	PostgreSQL + ZITADEL	$0

No billing details required.

END OF PART 6.2.4 – QC PORTAL (v6.3 WITH ALL MODIFICATIONS)

PART 6.2.5 : FINANCIER PORTAL (v6.3 FULLY EXTENDED, WITH FULL DISCLOSURE, CO-FINANCING, FINANCIER PREFERENCES, AUTOMATIC RFQ BROADCAST, FLAT 0.25% FINANCING FEE DEDUCTED FROM DISBURSEMENT, SGTX WITNESS CLAUSE, SMART INBOX INTEGRATION, DUAL-MODE NOT APPLICABLE, AND ALL MODIFICATIONS)

PART 6.2.5.0 – FINANCIER PORTAL – ACCESS & PURPOSE (v6.3 FULLY EXTENDED, WITH ALL MODIFICATIONS)

Access

The Financier Portal is the dedicated workspace for employees of tenants with type = FINANCIAL. This includes institutional banks, private credit funds, DeFi liquidity pools, family offices, factoring companies, and any other entity that provides trade finance or supply chain finance through the SGTX platform. The portal is not role-switched via the Buyer/Seller Mode Toggle; financiers have a single, undivided context. Their permissions, views, and actions are entirely scoped to their function as lenders, investors, or credit insurers.

Financiers onboard through the same KYB/KYC pipeline as other tenant types (Part 16), with the addition of a private financier eligibility check: non-bank lenders must pass a dynamic jurisdiction matrix that confirms private lending is permitted in their country and in the borrower's country. Required documentation includes proof of registration or regulatory exemption, and a legal basis explanation. Once verified, the financier gains access to a portal that provides a complete, real-time view of financing opportunities, active loans, collateral positions, and historical performance data, all governed by the platform's constitutional rules and integrated with the Smart Inbox, Trade Command Center, PlainLanguage Governor Decision Panel, and Customer Care Chatbot.

Purpose

The Financier Portal transforms the traditional, opaque trade finance workflow into a transparent, data-rich, and automated experience. It is designed to allow financiers to:

Discover financing requests that match their risk appetite automatically. Every time a borrower—whether a buyer or a seller—requests financing against a locked trade or a shipment within a multi-shipment contract, the platform broadcasts a Request for Quote (RFQ) to all financiers whose pre-set preferences (accepted countries, minimum trust score, minimum trade value, maximum financed amount, preferred settlement methods, excluded commodities) match the request. Financiers do not need to search or browse; the right opportunities come to them via their Smart Inbox and the Financing Opportunities tab.

See full trade details before placing a bid. This is the key trust advantage of SGTX. When a financier opens an RFQ, they see all non-confidential details of the underlying trade: seller and buyer full legal names, GTIDs, jurisdictions, trust scores, the complete contract and shipment data (USTN, commodity, HS code, quantity, incoterm, ports, vessel), all trade documents as read-only PDFs, and the historical performance of both counterparties—including previous trade history, dispute rate, on-time delivery, and, critically, the requesting party's previous financing history (number of loans, total financed, repayment punctuality). This enables informed credit decisions without relying on third-party credit bureaus.

Submit encrypted, blind bids, including co-financing portions. A financier can bid on the entire requested amount or a portion of it. The bid is encrypted with the borrower's public key and stored in NATS JetStream; even SGTX cannot read it until the borrower decrypts it after the bidding window closes. Co-financing allows multiple financiers to share the risk on a single request, with the platform automatically consolidating accepted bids and managing separate annexes.

Manage active loans with real-time collateral monitoring. For collateralised financing, the portal provides a comprehensive Collateral Monitoring Dashboard showing the current Loan-to-Value (LTV) ratio, a 30-day LTV trend, liquidation risk prediction, and an interactive simulator to test the impact of commodity price drops. Proactive alerts (via Smart Inbox) notify the financier when LTV crosses predefined thresholds, and a streamlined margin call workflow allows the financier to issue calls directly from the portal.

Track repayments and maintain private, immutable historical data. The repayment schedule for each loan is displayed, and repayments are automatically reconciled from PSP webhooks or manual inputs. Every repayment event updates the financier’s private historical record on that borrower, which contributes to future credit decisions. This data is visible only to the financier who owns it.

Participate in DeFi-based financing with mandatory risk disclosure. For financiers who choose to use DeFi protocols (Aave, Compound, etc.), the portal provides a DeFi Protocol Risk Oracle that continuously scores protocols, a stablecoin de-peg monitor, a liquidation early warning system, and a mandatory DeFi PlainLanguage Risk Summary modal that must be acknowledged before any DeFi bid. The platform deploys smart contracts as a facilitator but never takes custody of crypto assets.

Access a secondary market for tokenised trade assets. Financiers can list or purchase tokenised invoices (ERC-3525) with AI-suggested fair values, enabling liquidity without waiting for maturity.

Generate regulatory reports and monitor portfolio compliance. The portal auto-generates jurisdiction-specific reports (e.g., FinCEN SAR, ECB statistical return, CBE monthly remittance) and provides an AI-generated portfolio summary with exposure breakdowns.

Access support instantly. The Customer Care Chatbot is available from every screen; it can assist with understanding risk scores, navigating the co-financing flow, or troubleshooting repayment reconciliation. It escalates to a human support agent if needed.

All actions in the Financier Portal are governed by the same constitutional rules as the rest of SGTX. The platform collects its financing fee (0.25 % of the financed amount) automatically via a non-custodial PSP split at the moment of disbursement; the financier never pays SGTX directly. Every bid, agreement, disbursement, and margin call is immutably logged via Loom and linked to a Governor decision.

The portal respects the platform’s zero-cost, self-hosted, open-source philosophy : all analytics, risk models, and communication run on the sovereign node infrastructure; no external credit scoring or data-broker services are required.

END OF PART 6.2.5.0 – FINANCIER PORTAL – ACCESS & PURPOSE (v6.3 WITH ALL MODIFICATIONS)

6.2.5.1 Financier Portal – Landing Page – Smart Inbox (Action-Driven Feed for Lenders)

After login, every financier lands on the Smart Inbox (Part 6.1.1). The inbox is the financier’s command centre, aggregating all time-sensitive financing actions, risk alerts, repayment deadlines, and new opportunities into a single, prioritised feed. The inbox is purely advisory (A1); every item is linked to a specific financing agreement ID, USTN, or distressed microUSTN, and clicking any item opens the relevant detail view—usually the Trade Command Center for the underlying trade, the Financing Opportunities tab, or the Collateral Monitoring Dashboard.

The financier’s inbox respects the same deterministic urgency scoring model as all other portals, but with additional role-specific score modifiers. The AI Summary Card (Groq, fallback Ollama) generates a plain-language daily briefing tailored to the financier’s portfolio.

Financier-Specific Inbox Items

The following examples illustrate the types of items a financier sees, their typical priority scores, and the actions they link to.

Priority	Example Item	Score	Action Link
▯ **High**	Financing request #FR001 – Bidding window closes in 2 h. Request matches your risk appetite (match score 94). No bid submitted yet.	95	Opens the Financing Opportunities tab, pre-loaded with the request details and the bid submission form.
▯ **High**	Active agreement #FA001 – Collateral LTV reached 82 % (threshold 75 %). Margin call required.	90	Opens the Collateral Monitoring Dashboard for that loan, with the “Issue Margin Call” form pre-filled.
▯ **Medium**	Borrower credit score for USTN SGTX-...-V2 dropped from 88 to 71 (news: strike at buyer’s warehouse). Review exposure.	75	Opens the Trade Command Center for that USTN, showing the borrower’s updated credit intelligence.
▯ **Medium**	Financing repayment for USTN SGTX-...-V3 is due in 5 days. Outstanding: $5,200.	70	Opens the Repayment Schedule tab for that agreement.
▯ **Low**	DeFi protocol Aave V3 risk score unchanged at 94. No action.	30	Informational only; opens the DeFi Protocol Comparison tab if clicked.
▯ **Low**	New financing request from seller Mekong Fresh – $30k, 60 days, citrus. Match score 88.	50	Opens the Financing Opportunities tab.

Additional items may include : co-financing invitation received, margin call resolved by borrower, repayment overdue, new DeFi opportunity, secondary market listing matched to a held token, regulatory report due, and portfolio concentration warning.

Role-Specific Urgency Score Modifiers

The base urgency score for any item is computed using the universal rules (Part 6.1.1), then the following financier-specific modifiers are applied:

Bidding window < 2 h and no bid submitted → +10 – ensures opportunities are not missed.

Margin call LTV > 85 % → +20 – elevates critical collateral risk above other medium-priority tasks.

Repayment overdue 1 day → +25 – escalates potential defaults immediately.

Co-financing invitation from a partner with trust score < 70 → +5 – adds slight caution.

DeFi protocol risk score drops below 60 while position is active → +15 – forces attention to potential protocol failure.

Regulatory report due within 7 days → +10 – prevents compliance lapses.

AI Summary Card

The summary card is refreshed on every inbox load and reads something like:

“One bidding window closes soon – submit your bid to stay competitive. A margin call on an existing loan requires immediate attention. Also, a borrower’s credit score has dropped; review exposure. Your portfolio has 3 active loans totalling $120,000; all repayments are current except one overdue by 2 days.”

The prompt sent to Groq incorporates the financier’s active portfolio metrics (total exposure, weighted average APR, number of margin calls outstanding, upcoming repayment total) and the language is purely business-focused, never mentioning API names or technical limits.

Integration with Workflow

The Smart Inbox for financiers is populated by the inbox-service subscription to NATS events:

financing.rfq.broadcast → creates a new opportunity item for each matching financier.

financing.bid.submitted → notifies the borrower (but for the financier, a status change on their own bid may appear as a low-priority “Bid submitted” confirmation).

financing.margin_call.issued → creates a high-priority item for the borrower and a confirmation item for the financier.

collateral.value.changed → if the LTV crosses a threshold, creates an alert item.

financing.repayment.received → updates the repayment schedule and may generate a low-priority “Repayment received” confirmation.

borrower.credit.score.changed → generates a medium-priority item if the score drops by more than 10 points.

All items are scoped to the financier’s own agreements and preferences; the inbox never shows another financier’s bids or portfolio data.

Governance & Privacy

The inbox is A1 advisory only; action buttons link to Governor-gated endpoints.

The AI-generated summary does not include individually identifiable borrower data beyond what is already in the items; it uses aggregated language.

All inbox activity is logged in inbox_history for audit purposes.

END OF PART 6.2.5.1 – FINANCIER PORTAL – LANDING PAGE – SMART INBOX (v6.3 WITH ALL MODIFICATIONS)

6.2.5.2 Financier Portal – Navigation & Tabs (Overview of All Financier Workspaces)

After landing on the Smart Inbox, the financier accesses a focused set of tabs via the left sidebar. Each tab corresponds to a distinct phase of the lending lifecycle, from discovering opportunities to managing active risk and reviewing historical performance. The exact set of tabs is determined by the financier's permissions and enabled features (e.g., the DeFi tab appears only if the financier has opted into DeFi and the jurisdiction matrix allows it). All tabs are integrated with the Smart Inbox for real-time alerts, the Trade Command Center for deep dives into specific trades, the PlainLanguage Governor Decision Panel for blocks, and the Customer Care Chatbot for support.

The following tabs are presented in the order that matches a natural financing workflow.

19. Inbox
    The default landing page, as detailed in 6.2.5.1. All time-sensitive actions—bidding window expirations, margin call alerts, repayment reminders, credit score changes—appear here. The financier can act directly from the inbox (e.g., "Submit Bid", "Issue Margin Call") or click to navigate to the relevant tab with pre-loaded context.
20. Financing Opportunities
    This tab lists all open financing requests that match the financier's pre-set preferences (countries, trust score, trade value, financing type, settlement method, excluded commodities). Each row is a summary card showing the USTN (or shipment USTN), commodity, amount, tenor, AI-computed credit score, default probability, recommended LTV, and a match score (0–100) indicating how well the request aligns with the financier's appetite. A "View Details" button opens the Full Disclosure page, where the financier sees the buyer and seller legal names, all trade documents, historical performance, and AI credit intelligence—before placing a bid. From this page, the financier can submit an encrypted, blind bid (including a portion for co-financing) or initiate a negotiation.

Filters allow the financier to narrow the list by country, commodity, amount, or urgency (bidding window closing). This tab is the primary deal-origination workspace.

21. My Bids & Agreements
    This tab has two sub-views, toggled by a segmented control:

Active Bids : Lists all bids the financier has submitted but that are still pending (bidding window open). Each bid shows the request ID, USTN, amount offered, APR, current leading bid (anonymised), and time remaining. The financier can increase or withdraw a bid. Alerts appear if they are outbid.

Won Agreements (Active Loans) : Displays every financing agreement where the financier is a party and the loan is active (status ACTIVE, DISBURSED, or in repayment). Columns include: USTN, borrower name, financed amount, outstanding balance, next repayment date (colour-coded), repayment health, collateral status, and a margin call alert badge. Clicking a row opens the full agreement view, which includes the Collateral Monitoring Dashboard (LTV gauge, history chart, margin call log, price drop simulator), repayment schedule, and the original agreement PDF. The financier can issue margin calls, view the Trade Command Center, or access the chat.

22. DeFi (If Enabled)
    Visible only if the financier has activated DeFi capabilities in their Company Admin and the jurisdiction matrix allows DeFi for their country. This tab contains three sub-sections:

Protocol Comparison : A dynamically updated table of DeFi protocols (Aave V3, Compound, etc.) with risk scores, TVL, latest audit status, recent incidents, and estimated APY. An AI recommendation (Groq) suggests the safest protocol for the current market.

Stablecoin Health : Monitors USDC, USDT, DAI peg status. Deviation > 0.5 % triggers a warning; > 2 % freezes new positions.

My DeFi Positions : For each active DeFi loan, shows protocol, chain, amount, stablecoin, health factor, and a liquidation early warning (LSTM prediction). A "Liquidation Risk" gauge is displayed.

The mandatory DeFi PlainLanguage Risk Summary modal appears before any DeFi bid is submitted, and the acknowledgment is logged.

23. Secondary Market
    For tokenised trade assets (ERC-3525), this tab allows the financier to view their own listings and available purchase opportunities. The financier can list a tokenised invoice for sale with an AI-suggested price, or browse anonymised secondary market listings with AI-estimated fair values. Purchases are executed atomically on-chain via the defi-service. A 0.1 % platform SGTX FEES applies to secondary sales (not to be confused with the financing fee, which is already collected at disbursement).
24. Portfolio & Compliance
    A high-level oversight tab that gives the financier a holistic view of their lending book and regulatory standing.

AI-Generated Portfolio Summary (Groq) : A narrative like "Your portfolio has 3 active loans totalling $120,000. Weighted average APR 5.2 %. No margin calls pending. One loan is exposed to citrus (risk score 88)."

Exposure Breakdown : Doughnut chart by commodity, region, and risk band.

Regulatory Reports : Auto-generated, jurisdiction-specific reports (e.g., FinCEN SAR, ECB statistical, CBE monthly) ready for download. Reports are compiled using Tera templates and can be auto-emailed if SMTP is configured.

Risk Appetite Validation : AI continuously compares actual portfolio concentrations against the financier's stated preferences and warns of over-concentration (e.g., "40 % of your portfolio is in citrus from Vietnam. Consider diversifying.").

25. Financed Companies
    A private, read-only, audit-traced directory of every borrower the financier has ever financed. For each company, it stores: GTID, legal name, jurisdiction, current trust score, total financed amount, number of financings, on-time/late/default repayment counts, a credit score trend line (12 months) with event annotations, and details of any active financing. The financier can refresh the credit assessment on demand,

triggering a new AI credit intelligence run. This tab is the financier's institutional memory; data is never shared across financiers.

26. Company Admin
    Standard tenant administration. Financier admins can invite employees and assign roles: credit_analyst (can view opportunities and submit bids), trader (DeFi), compliance (view portfolio and reports), viewer (read-only). They can set the Natural Language Risk Appetiteprofile (e.g., "I prefer short-term trade finance for agricultural commodities below $50k"), configure financier preferences (country filters, min trust score, max amount, etc.), define exposure limits, manage notification settings, and access the Tenant Activity Log and Exit Center (Part 6.2.8).

All tabs are connected : a bid submitted in "Financing Opportunities" moves to "Active Bids" and, once awarded, to "Won Agreements." A repayment event updates both "Won Agreements" and the "Financed Companies" record. The Smart Inbox ties them together, ensuring the financier never misses a critical action.

END OF PART 6.2.5.2 – FINANCIER PORTAL – NAVIGATION & TABS (v6.3 WITH ALL MODIFICATIONS)

6.2.5.3 Financier Portal – Financing Opportunities Tab – Automatic RFQ Broadcast, Full Disclosure, and Bid Submission

This tab is the primary deal-origination workspace for financiers. It replaces the traditional, relationship-driven process of finding trade finance deals with an automated, data-rich, and fully transparent pipeline. Every financing request that matches the financier's pre-set preferences is automatically delivered here, ranked by a match score, and accompanied by complete trade details and AI-generated credit intelligence. The financier can then submit an encrypted, blind bid, optionally specifying a portion for co-financing, directly from the same interface.

6.2.5.3.1 Opportunity List (Automatic RFQ Inbox)

When a borrower—a buyer or a seller—creates a financing request against a locked trade or a locked shipment (Phase 4.1), the finance-service immediately evaluates the request against every active financier's preferences. Those preferences, defined in the Financier Portal's Company Admin tab (see 6.2.5.10), include accepted borrower countries, minimum trust score, minimum trade value, maximum financed amount per request, preferred financing types and settlement methods, and excluded commodities. The system also respects geographic restrictions (e.g., "accept only borrowers from the EU" or "accept all except sanctioned countries").

Only financiers whose preferences match the request receive the RFQ. The RFQ is delivered simultaneously via three channels:

A Smart Inbox item (priority based on match score and bidding window urgency) that links directly to the request details.

An email notification (co-branded, with the sender "SGTX Trade Finance" if the borrower is anonymous, or "Borrower Name via SGTX" if the borrower has consented to identity disclosure before bid acceptance—this is configurable per financier).

An API webhook (if the financier has integrated their own loan origination system; see Part 17 for webhook specifications).

The RFQ broadcast is fully logged in financing_rfq_log, and financiers can opt out of specific categories (e.g., all pre-shipment finance, or all requests involving perishable goods) via their preference settings without affecting their eligibility for other requests.

Request List Interface

The Financing Opportunities tab displays all open RFQs as a filterable, sortable table. Each row presents a summary card with the following columns:

USTN (or Shipment USTN) : The identifier of the underlying trade or shipment. Clicking it opens the Trade Command Center filtered to the financier's view, where they can see all trade milestones and documents in the context of the financing request. For multi-shipment contracts, the specific shipment USTN is shown.

Commodity, Amount, Tenor, Financing Type : A concise summary, e.g., "Oranges, $100,000, 60 days, Pre-shipment". The commodity is described at the HS-2 level in the list view to give a quick overview; the full HS code and product name are revealed on the detail page.

Credit Score (0–100): The AI-computed credit score for the borrower, colour-coded (green ≥80, amber 50–79, red <50).

Default Probability : The Monte Carlo-simulated probability of default, expressed as a percentage with a small confidence interval.

AI-Recommended LTV : The maximum Loan-to-Value ratio recommended by the Credit Intelligence Agent (A2), with a tooltip explaining the factors (commodity perishability, jurisdiction, historical recovery rates).

Match Score (0–100): A composite score representing how well this request aligns with the financier's stated preferences and historical success patterns. A higher score indicates a stronger fit. The score is calculated using a LightGBM model that weighs preference alignment, corridor familiarity, and the financier's past bidding and repayment outcomes on similar requests. A "Why am I seeing this?" tooltip, generated by Groq (fallback Ollama), provides a plain-language explanation: "Your route network covers Vietnam–Egypt, and you have a 94% acceptance rate on citrus pre-shipment finance. Your match score of 91 reflects strong alignment."

Deadline : The bidding window expiry time, colour-coded (red if < 4 h, amber < 24 h). A countdown timer appears for requests closing within 2 hours.

Action : A prominent "View Details" button, plus a quick "Submit Bid" shortcut that opens the bid form directly with pre-filled fields.

The financier can filter the list by borrower country, commodity, financing type, amount range, or minimum match score. By default, the list is sorted by match score descending, but the financier can sort by deadline, amount, or credit score. The list updates in real time as new RFQs arrive or as bidding windows close.

6.2.5.3.2 Request Details Page – Full Disclosure (The SGTX Trust Advantage)

When the financier clicks "View Details" on any RFQ, they are taken to a dedicated page that reveals all non-confidential trade and counterparty data. This is the core differentiator of SGTX trade finance: financiers do not rely on opaque credit scores or third-party reports; they see the actual trade, the actual documents, and the actual historical performance before committing capital.

The page is divided into sections, each backed by read-only data, and the financier must have signed the platform's confidentiality agreement (in their Master Service Agreement) to access this level of detail. No data can be exported or copied from this view; screenshot prevention is enforced client-side as an additional deterrent.

Trade Overview:

USTN, commodity with full HS code and product description, total trade value, incoterm, ports of loading and discharge, vessel name (if assigned), ETD/ETA.

The SGTX trade fee that the seller has already paid (shown as "SGTX fee : $X – paid by seller on [date]" for full transparency, though this does not affect the financier's decision).

Counterparty Details:

Buyer and seller full legal names, GTIDs, jurisdictions, KYB tier, trust scores (with trend sparkline over the last 12 months).

A "Sanctions Cleared" badge (green) or a red flag if any party is under review.

Documents (Read-Only):

Commercial invoice, packing list, booking confirmation, bill of lading (or eBL), phytosanitary certificate, certificate of origin, QC report (if available), insurance certificate. All are displayed as embedded PDFs or images with a "View" button. The documents are the originals uploaded by the trading parties and verified by the platform's AI pipeline; their SHA256 hashes are displayed for integrity verification.

Historical Performance:

For both the buyer and the seller : total number of previous trades on SGTX, total trade value, dispute rate, on-time delivery percentage, and average payment delay.

For the requesting party (the borrower) : previous financing history—total number of financing requests, total amount financed, and a breakdown of repayment performance (number of on-time payments, late payments with average days, and any defaults). This data is drawn from the financier_historical_data table and is visible only to the financiers who were involved in those past financings; the platform never shares one financier's data with another.

AI Credit Intelligence:

A plain-language risk summary generated by Groq (fallback Ollama) that synthesises the credit score, default probability, key risk factors, and a recommendation. Example: "Mekong Fresh has a strong repayment track record (8 on-time, 0 late). The underlying trade is a citrus shipment on a well-established corridor with a trusted buyer. Recommended LTV 70 %. Overall risk: Low."

The full credit score trend line for the borrowing party over the last 12 months, annotated with each past financing event (disbursement, repayment).

The Monte Carlo simulation results, showing the distribution of expected loss and the confidence interval.

This full disclosure dramatically reduces the information asymmetry that plagues traditional trade finance. It allows smaller, specialised financiers to compete with large banks on an equal footing, and it enables rapid, data-driven credit decisions without calling for additional paperwork.

6.2.5.3.3 Bid Submission Form (Encrypted, Blind, Co-Financing Enabled)

From the Request Details page, the financier clicks “Submit Bid” to open the bid form. The form is pre-filled with the request’s parameters and allows the financier to specify:

Amount Offered : The portion of the total requested amount (P) that the financier is willing to finance. This can be the full amount or any smaller portion, enabling co-financing. The minimum tranche size is configurable per financier (default $10,000). The form displays the remaining unfinanced amount if other financiers have already committed through co-financing, but individual bids are hidden until the window closes.

APR (All-In Cost) : The annual percentage rate, inclusive of all fees and interest. The system benchmarks this against current market rates for the corridor and commodity, flagging the financier if their proposed APR deviates by more than ±50 % from the benchmark (usury/compliance review triggered).

Collateral Requirements : The financier specifies what collateral they require, which must match or be stricter than the borrower’s originally stated collateral type. For example, if the borrower offered “Goods”, the financier may require “Goods + Warehouse Receipt”.

Settlement Method : Must match one of the borrower’s accepted methods or an acceptable alternative. Options include bank transfer, stablecoin (USDC/USDT), or a specific DeFi protocol.

Conditions: Free text for any special conditions (optional).

Note to Borrower : A message that will be passed to the borrower’s AI negotiation bot if the borrower has enabled automated negotiation. This can be used to signal flexibility on terms.

Encryption and Blind Bidding : The bid is encrypted with the borrower’s public key (Ed25519) on the client side before transmission. The encrypted bid is stored in NATS JetStream KV. Even the platform cannot read the bid until the borrower decrypts it after the bidding window closes. This ensures a fair, blind auction where financiers compete on terms without knowing each other’s offers.

DeFi Option : If the financier selects a DeFi protocol as the settlement method, a mandatory DeFi PlainLanguage Risk Summary modal appears before the bid can be submitted. This modal presents exactly five bullet points in the financier’s preferred language (generated by Groq, fallback Ollama):

What stablecoins are (if used).

What ‘health factor’ means.

What happens if collateral drops in value.

SGTX does NOT guarantee DeFi protocol safety.

Past performance is not indicative of future results.

The financier must click “I understand” to proceed. This acknowledgment is stored in employees.defi_risk_acknowledged_at and logged immutably.

Submission : Upon submission, the bid is stored in financing_offers with status SUBMITTED. The financier’s Active Bids list updates, and a confirmation appears. If the financier has enabled notifications, they will be alerted if they are later outbid. The bidding window timer continues; the financier can modify or withdraw their bid at any time before the window closes.

All bid activity is governed by the standard SGTX compliance gates : sanctions checks on the financier’s jurisdiction, protocol risk scores for DeFi, and rate reasonableness checks. If a bid is rejected by the Governor, the PlainLanguage Governor Decision Panel immediately explains the reason and provides remediation steps.

END OF PART 6.2.5.3 – FINANCIER PORTAL – FINANCING OPPORTUNITIES TAB (v6.3 WITH ALL MODIFICATIONS)

6.2.5.4 Financier Portal – My Bids & Active Loans (Bid Management, Collateral Monitoring, and Repayment Tracking)

This tab is the financier’s operational centre for managing bids that are still pending and for overseeing loans that have been awarded and are currently active. It is divided into two sub-views, toggled by a segmented control at the top: Active Bids and Won Agreements. Each sub-view provides a dedicated, real-time workspace that connects bid activity directly to the underlying trades and the financing lifecycle.

6.2.5.4.1 Active Bids (Submitted, Not Yet Awarded)

This sub-view lists every bid the financier has submitted but which has not yet been accepted or rejected by the borrower. It is designed to give the financier full control over their outstanding offers in a competitive, time-sensitive auction environment.

Table Columns:

Financing Request ID : The unique identifier of the request. Clickable; opens the original RFQ details page (read-only) so the financier can re-review the trade and credit information.

USTN (or Shipment USTN) : The trade or shipment the financing is linked to. Clicking it opens the Trade Command Center in a new tab, filtered to the financier’s perspective.

Amount Offered : The portion of the total requested amount that the financier bid, displayed with the currency. If the bid is for co-financing, a small badge shows the percentage of the total request: “60,000(6060,000(60100,000)”.

APR (All-In Cost) : The annual percentage rate submitted, along with a small indicator showing whether it is above, below, or at the current leading APR (if the borrower’s auction reveals anonymised leader information). The indicator is a coloured arrow: green if the bid is currently the lowest APR, amber if it is within 1% of the lowest, red if it is significantly higher.

Leading Bid (Anonymised) : For auctions where the borrower has enabled visibility of the leading terms, the lowest APR among all bids is shown here. The amount and the financier's identity are never revealed. The column reads something like "Leading APR: 4.8%" or "Not available until window closes" depending on the auction settings.

Time Remaining : A countdown timer to the bidding window closure. Colour shifts from neutral to amber (< 4 h) to red (< 1 h). Once the window closes, the status updates automatically.

Status : The current state of the bid: SUBMITTED, WITHDRAWN, OUTBID (if the borrower's system notifies financiers that a lower APR has been submitted), or EXPIRED. If the borrower has accepted a combination of bids for co-financing and this bid was not selected, the status will reflect REJECTED.

Actions:

Increase Bid (pencil icon) : Opens the bid modification form. The financier can adjust the amount offered or the APR (lower it). The original bid remains in the history; a new version of the bid is submitted, replacing the previous one. All versions are logged.

Withdraw (trash icon) : Removes the bid from the auction. A confirmation modal is shown. The withdrawal is recorded in financing_offers with status WITHDRAWN. A withdrawn bid cannot be reinstated.

Real-time Updates : The Active Bids table subscribes to NATS subjects like financing.bid.submitted (to update the leading APR if the borrower shares that information) and financing.bidding_window.closed. When the financier is outbid, a Smart Inbox item is immediately generated: "You have been outbid on financing request #FR001. New leading APR: 4.8%. You have 6 h remaining to improve your bid." The financier can then decide to increase their bid (lower the APR) directly from that Smart Inbox item, which deep-links to the bid modification form.

If the bidding window closes and the financier's bid is the winning one (or part of a winning combination), the row automatically moves to the "Won Agreements" sub-view and a celebration toast appears. Conversely, if the bid is not selected, the row moves to an archived history table visible via a "Past Bids" link at the bottom of the Active Bids view.

6.2.5.4.2 Won Agreements (Active Loans) – With Collateral Monitoring Dashboard

This sub-view displays every financing agreement where the financier is a lender and the loan is in an active state (ACTIVE, DISBURSED, or in repayment). It is the core credit monitoring and risk management workspace. Each row represents an active loan and provides at-a-glance health metrics, with a deep-dive available through an expandable collateral monitoring row or a dedicated slide-out panel.

Table Columns:

USTN (or Shipment USTN) : Clickable; opens the Trade Command Center with the financing view overlaid, showing the loan details and the trade's progress side by side.

Borrower Name : The legal name of the borrowing party. Clicking it opens the borrower's full profile from the "Financed Companies" tab, providing context on their overall relationship and repayment history.

Financed Amount : The principal amount (P) that the financier committed. For co-financed deals, this shows the financier's portion, with a small label like "Your share: $60,000".

Outstanding Balance : The current remaining debt – principal plus any accrued interest not yet paid. Updated in real time as repayments are received. Colour-coded: green if on schedule, amber if approaching maturity, red if overdue. A sparkline chart next to the amount shows the balance decreasing over the last 30 days.

Next Repayment Date : The due date of the next scheduled instalment (or the bullet maturity date). Displayed in the financier's local timezone. If overdue, the date turns red and shows "Overdue by X days". A small calendar icon allows the financier to export the date to their external calendar.

Repayment Health : A summary badge: "On Track" (green checkmark), "Watch" (amber eye – minor delays), "Delinquent" (red exclamation), "Defaulted" (dark red broken circle). Hovering shows the number of on-time vs. late payments for this specific loan.

Collateral Status : A link or icon that opens the Collateral Monitoring Dashboard for that loan. The icon is coloured green ($\text{LTV} \leq 70\%$), amber ($70\,\% < \text{LTV} \leq 85\%$), or red ($\text{LTV} > 85\%$) to provide instant visual warning.

Margin Call Alert : A red warning triangle badge appears if a margin call has been issued but not yet resolved, or if the LTV has crossed the financier's margin call threshold and no call has been issued yet. Hovering reveals the current LTV and a direct "Issue Margin Call" button.

Actions:

View Collateral Dashboard (gauge icon) : Opens the full dashboard in a modal.

Issue Margin Call (exclamation icon) : Enabled only when LTV exceeds the threshold and no active call exists.

Record Manual Repayment (money icon) : For cases where repayment is made outside the automatic PSP webhook (e.g., a wire transfer that was not automatically reconciled). The financier enters the amount and date; the system attempts AI reconciliation and updates the ledger.

View Financing Agreement (document icon) : Opens the signed PDF agreement with the SGTX Witness Clause.

Refresh Credit Intelligence (brain icon) : Triggers a new AI credit assessment for the borrower, incorporating the latest data.

Collateral Monitoring Dashboard (Modal Overlay)

This is the most sophisticated risk management tool in the Financier Portal. It is accessed by clicking the Collateral Status icon or the "View Collateral Dashboard" action. The dashboard is rendered as a responsive, five-section modal that can also be popped out into a full-screen view for intense monitoring sessions.

A. LTV Gauge Chart
A circular gauge displays the current Loan-to-Value ratio. The gauge is divided into three colour bands: green 0-70%, amber 70-85%, red 85-100%. The needle points to the exact

percentage, which is also displayed numerically. Below the gauge, the collateral value and the outstanding loan value are shown in human-readable format, e.g., “Collateral: 85,000(marketprice),Loan:85,000(marketprice),Loan:60,000 → LTV 70.6%”. The market price of the collateral is derived from the platform’s commodity price feeds (FAO, USDA, etc.) for goods, or from real-time oracles for stablecoin collateral.

B. LTV History Line Chart
A 30-day line chart plots the LTV daily, with horizontal threshold lines at 70% and 85%. Event annotations are placed on the chart: “Collateral added +$10,000 on 15 Jun”, “Market price drop −5% on 20 Jun”, “Margin call issued 22 Jun”, “Margin call resolved 25 Jun”. Hovering over any point shows the date, LTV, and collateral value. This provides a clear narrative of the loan’s risk trajectory.

C. Liquidation Risk Meter
A linear gauge from 0-100% predicts the probability of liquidation within the next 7 days, using an LSTM model trained on historical SGTX lending data, commodity volatility, and DeFi protocol health. The meter is colour-coded (green 0-20%, amber 20-50%, red 50-100%). A tooltip generated by Groq explains the factors driving the current prediction: “Your livestock collateral is exposed to high feed-price volatility (+15% this week), elevating liquidation risk to 34%. Consider a proactive margin call.”

D. Margin Call Log
A table of all margin calls that have been issued for this loan, including: date, LTV at issuance, amount of additional collateral required, deadline, status (PENDING, RESOLVED, EXPIRED), and the borrower’s response. This is a complete audit trail for each credit action.

E. “Simulate Price Drop” Interactive Slider
A slider that allows the financier to drag a hypothetical collateral value down from 0% to 50% decline. As the slider moves, the LTV gauge, collateral value, and new LTV are updated in real time. A text output shows: “If the market price drops by 10%, LTV would be 78.4%. A margin call would be triggered if you had not issued one. To restore 70% LTV, an additional $8,400 in collateral would be required.” This simulation is entirely client-side and does not affect the actual loan. It is invaluable for “what-if” planning and pre-emptive risk mitigation.

Proactive Alerts : The financier’s Smart Inbox receives automatic notifications when LTV crosses 70% (priority 60, advisory), crosses 85% (priority 85, action required), or when a margin call is issued, resolved, or expires. The notification includes a one-click link to the Collateral Monitoring Dashboard.

Margin Call Management Workflow
When the financier clicks “Issue Margin Call” from the dashboard or the row action, a modal opens pre-filled with:

Current LTV and the threshold.

Recommended additional collateral amount (the amount needed to bring LTV back to 70%).

The collateral type (e.g., goods, stablecoin) and the address or warehouse receipt reference.

A deadline (default 48 hours for conventional, 24 hours for volatile commodities).

A free-text message to the borrower.

The financier can adjust the amount and deadline, then submit. The Governor verifies that a margin call is permitted under the financing agreement and the jurisdictional rules, and that the LTV indeed exceeds the threshold. Upon approval, a high-priority Smart Inbox item is sent to the borrower. If the borrower adds collateral or repays part of the loan, the margin call is automatically resolved; if the deadline passes without action, an escalation alert is sent to the financier and the Platform Governance Authority, and the loan is flagged for potential enforcement.

Real-time Updates in Won Agreements
The table subscribes to NATS events : financing.repayment.received, collateral.value.changed, financing.margin_call.issued, financing.margin_call.resolved. Columns like Outstanding Balance, Repayment Health, and Collateral Status update instantly. The LTV gauge in the modal also updates live if the modal is open. If a repayment is missed and the loan becomes delinquent, the entire row is briefly highlighted in red to draw the financier's immediate attention.

Integration with the Customer Care Chatbot
From any row in the Won Agreements table, the financier can click the universal "Chat with Support" icon to open the Customer Care Chatbot with the specific loan context pre-loaded. The chatbot can assist with questions like "Why did this LTV jump suddenly?" (by querying the market price feed and event log) or "What are my options if the borrower doesn't respond to the margin call?" (by outlining the platform's dispute and distressed cargo procedures). If the chatbot cannot resolve the issue, it offers to escalate to a human support agent, who will call the financier's registered phone number.

All data in this tab is strictly scoped to the authenticated financier; the Governor enforces that no cross-lender information leakage occurs, even for co-financed loans where multiple financiers share the same underlying trade—they each see only their own portion and aggregated anonymised data about the other tranches.

END OF PART 6.2.5.4 – FINANCIER PORTAL – MY BIDS & ACTIVE LOANS (v6.3 WITH ALL MODIFICATIONS)

6.2.5.5 DeFi Financing Tab – Protocol Comparison, Stablecoin Health, and Active DeFi Positions (If Enabled)

The DeFi tab appears in the Financier Portal only when the financier has explicitly enabled DeFi capabilities in their Company Admin preferences and when the dynamic jurisdiction matrix confirms that both the financier's country and the borrower's country permit decentralised finance for trade credit. It is not a speculative trading interface; it is a risk-aware, fully monitored gateway that allows financiers to allocate a portion of their lending portfolio to DeFi protocols while remaining within the platform's constitutional governance.

The tab is divided into three sections : a protocol comparison table, a stablecoin health monitor, and a view of the financier's own active DeFi positions. All data is sourced from free, publicly available on-chain feeds (The Graph, 0x API, Chainlink free price feeds) and processed locally on the sovereign node.

Protocol Comparison Table

The top section displays a dynamically updated table of all DeFi protocols that SGTX has assessed and approved for use. Each protocol (e.g., Aave V3 on Polygon, Compound on Ethereum Mainnet, etc.) appears with:

Risk Score (0–100): Computed by the DeFi Protocol Risk Oracle (A2) from audit findings (HF Donut parsing PDF audit reports from GitHub), total value locked (TVL) trends, exploit history, governance activity, and oracle health. Scores are colour-coded: ≥85 (green, low risk), 60–84 (amber, acceptable), 40–59 (red, elevated risk – new positions blocked), <40 (critical, protocol suspended).

TVL and Liquidity : Current total value locked, normalised for the protocol's market cap, and specifically the available liquidity for the stablecoins supported (USDC, USDT, DAI). A low liquidity warning flag appears if the protocol cannot currently absorb the financier's typical loan size without significant slippage.

Latest Audit Status : The date of the most recent audit, the auditor, and a summary of findings (green if no critical issues, amber if remediated, red if open). A link to the full audit report (IPFS) is provided.

Recent Incidents : Any exploit events in the last 12 months, drawn from scraped news and on-chain data, summarised with a severity tag.

Estimated APY : The current supply APY for the stablecoin, updated every 15 minutes from The Graph.

AI Recommendation (Groq, fallback Ollama) : A one-sentence recommendation: "Aave V3 offers the highest safety with acceptable yield. Compound has slightly higher yield but elevated risk due to a recent governance change."

The financier can click any protocol row to see a detailed risk breakdown and a "What you should know" Groq-generated summary tailored to non-technical lenders.

Stablecoin Health Widget

This section monitors the peg status of every stablecoin allowed on the platform. The list is dynamic, maintained by the Stablecoin Monitor Agent (A2). Initially it covers USDC, USDT, and DAI. Each stablecoin shows:

Current peg deviation : Displayed as a percentage (e.g., USDT: -0.51%). Colour-coded green (≤0.5%), amber (>0.5%), red (>2%).

Peg history chart : A 7-day line chart with a horizontal reference line at 1.00.

Action status : "Stable" (green), "Warning – no new positions allowed" (amber), or "Paused – all new lending frozen" (red). When deviation exceeds 2%, the Governor (A2) automatically freezes new DeFi positions in that stablecoin across the entire platform and escalates to the Platform Governance Authority.

My DeFi Positions

For the financier, this table lists every active DeFi loan they have provided through SGTX. Columns include:

Agreement ID, USTN, and Borrower: Identifying the loan.

Protocol and Chain: e.g., “Aave V3 – Polygon”.

Amount (Stablecoin): The amount lent in USDC or USDT.

Current Health Factor : The health factor of the borrower’s position (if the protocol reports it on-chain). Displayed with a gauge and colour coding. A tooltip explains: “Health factor below 1 means the position is at risk of liquidation.”

Liquidation Early Warning (LSTM) : A predicted probability that the health factor will fall below 1.2 within the next 24 hours. Green (low risk), amber (moderate), red (high). If a warning is raised, the financier receives a Smart Inbox alert with Groq-generated advice.

Actions : “Add Collateral” (not available to the financier directly; it sends a pre-filled message to the borrower), “Withdraw & Repay” (initiates an on-chain transaction from the borrower’s wallet, requiring the borrower’s own key), and “View On-Chain” (opens the Polygon/Etherscan transaction).

DeFi PlainLanguage Risk Summary (Mandatory)
Before the financier can submit any new DeFi bid or accept a DeFi-based counter-offer, the mandatory five-bullet risk summary modal appears (as described in 6.2.5.3.3). The financier must click “I understand”. This acknowledgment is stored in employees.defi_risk_acknowledged_at and logged.

Governance : All DeFi activities are governed by Phase 4 gates (G4U5a through G4U5k). The protocol must have a risk score ≥60 for new positions, the jurisdiction must allow DeFi, the smart contract must have passed static analysis (Slither/Aderyn) with no critical issues, and the DeFi Risk Summary must be acknowledged. The Governor enforces these rules; any violation blocks the bid with a plain-language Decision Panel explanation.

6.2.5.6 Secondary Market Tab – Tokenised Trade Assets

The Secondary Market tab provides a venue for financiers to sell or purchase tokenised trade finance assets, increasing liquidity and enabling portfolio rebalancing without waiting for loan maturity. It is an optional feature, enabled per financier in Company Admin. All transactions occur on-chain (Polygon) via self-deployed ERC-3525 tokens; SGTX takes a 0.1% facilitation SGTX FEES on secondary sales, collected via smart contract at the moment of atomic swap.

My Listed Tokens : The financier sees any of their tokenised invoices that they have listed for sale. For each listed token, they see the financing agreement ID, the USTN, the remaining principal, the AI-estimated fair value (updated continuously based on trade progress, commodity prices, and borrower credit score), the asking price they set, and the listing expiry. They can modify the price or cancel the listing.

Available Tokens (Market Listings) : A table of anonymised tokenised invoices listed by other financiers. The displaying data is deliberately limited to protect privacy: the USTN is shown as “SGTX-...-V1”, the commodity category (2-digit HS chapter), the trade progress (e.g., “60% complete, goods at sea”), the remaining tenor, the AI-estimated fair value (expressed as a percentage of face value), and the asking price. The financier can click “Analyse” to see a Groq-generated valuation explanation: “Invoice for USTN...V1 is 60% complete, expected repayment in 30 days. Based on current orange prices and buyer credit score, fair value is 0.985 USDC per 1.0 face value. The seller is asking 0.99 USDC." The financier then submits a purchase offer.

If the seller accepts, the defi-service executes an atomic swap on Polygon : the buyer's stablecoin is transferred to the seller, the ERC-3525 token representing the invoice is transferred to the buyer, and the 0.1% SGTX FEES is split (0.05% to SGTX via smart contract to SGTX's fee wallet). All transactions are logged and reconcile via on-chain monitors.

Governance : Secondary market trades are subject to sanctions screening and jurisdictional restrictions. The secondary market pricing model must be validated by the Platform Governance Authority before trading can be enabled for a specific token.

6.2.5.7 Portfolio & Compliance Tab – Aggregated Risk Management and Regulatory Reporting

The Portfolio & Compliance tab provides the financier with a high-level, holistic view of their entire lending book and their regulatory obligations. It is the tab where the financier goes to understand their aggregate risk exposure, generate required reports for central banks or financial intelligence units, and ensure that their portfolio remains aligned with their stated risk appetite.

AI-Generated Portfolio Summary : At the top, Groq (fallback Ollama) generates a daily plain-language summary of the financier's portfolio, updated each morning or on demand. Example: "Your portfolio has 3 active loans totalling $120,000. All repayments are current except one loan where payment is overdue by 2 days. Your weighted average APR is 5.2%. No margin calls are pending. One loan is exposed to citrus from Vietnam (risk score 88). Your overall portfolio health score is 92/100."

Exposure Breakdown : A responsive doughnut chart (Chart.js) breaks down the portfolio by commodity (HS chapter), by borrower country, and by risk band (low/medium/high). Hovering over any segment reveals the specific amount and percentage. The chart is interactive; clicking a segment filters the underlying data table to show only those loans.

Risk Appetite Validation (A2, advisory) : An AI agent continuously compares the actual portfolio composition against the financier's stated risk appetite (set up in Company Admin as a natural-language statement parsed by HF Mixtral). If a drift is detected—for example, the financier stated a preference for "agricultural commodities below 50k"butcurrentlyhas4050k"butcurrentlyhas40100k—the system issues a Smart Inbox advisory and displays a warning here: "Your current portfolio has 40% exposure to electronics, exceeding your stated preference for agricultural commodities. Consider rebalancing."

Regulatory Reports : A dedicated section automatically generates jurisdiction-specific regulatory reports. The platform's Reporting Agent (A2) compiles the necessary data from settled trades, financing disbursements, repayments, and distressed sales. Reports are populated into official templates using HF Donut for PDF-based forms and Tera for XML/HTML. Examples include:

FinCEN SAR (USA) : Suspicious Activity Report, if any transaction patterns triggered AML alerts.

ECB Statistical Return (EU) : Aggregate cross-border trade finance flows.

CBE Monthly Remittance Report (Egypt) : Form 10, detailing all foreign currency transactions settled in the month.

Local VAT/GST reporting : If required by the financier’s jurisdiction.

Each report is generated as a draft, displayed with a “Review” button, and the financier can download it as a signed PDF or have it auto-emailed to their compliance officer if SMTP is configured. The generation is logged immutably.

Continuous Compliance Alerts : If the Compliance Agent (A2) detects an upcoming KYB expiration for any borrower, a new sanctions designation affecting a borrower’s country, or a breach of the financier’s own exposure limits, a prominent alert banner appears at the top of this tab and a Smart Inbox item is created.

6.2.5.8 Financed Companies Tab – Historical Performance Data on Borrowers

This tab is the financier’s institutional memory. It stores a private, read-only, audit-traced record of every company that has ever received financing from this financier through the SGTX platform. It is a rich, data-driven alternative to the traditional “credit file” and is automatically updated after every disbursement and repayment.

Company List : A searchable table lists all financed companies. Columns include: borrower’s GTID, legal name, jurisdiction, current trust score, total financed amount across all agreements, number of financings, and a summary of repayment performance (e.g., “12 on-time, 1 late, 0 defaults”). Clicking a row opens the detailed company profile.

Company Profile:

Credit Score Trend : A 12-month line chart with annotations for each financing event (disbursement, repayment, default if any). This gives the financier an immediate visual sense of whether the borrower’s creditworthiness is improving or deteriorating.

Financing History Table : A list of all agreements between this financier and the borrower, with columns: agreement ID, date, amount, APR, tenor, status, and repayment outcome. Each row links to the full agreement details in the Won Agreements tab.

Active Financing (if any) : Details of any current outstanding loan with this borrower, pulled from the active loans table.

Refresh Credit Assessment : A button that triggers a fresh AI credit intelligence run for this borrower, incorporating the latest trade data, repayment history, and public news sentiment. The new score is compared to the previous score and the difference is highlighted.

Data Privacy : This data is strictly private to the financier. It is never shared with other financiers, nor with the borrower (though the borrower can see their own public trust score). The data is encrypted at rest and access is gated by RLS and the Governor. The financier cannot modify any historical data; it is append-only.

6.2.5.9 Bid Negotiation & Bot Transparency Panel

When a borrower has enabled AI-assisted negotiation on their financing request, and the financier includes an optional “Note to Borrower” in their bid, the borrower’s negotiation bot (A2/A3) may respond with a counter-offer. This section describes how the financier interacts with that automated negotiation.

Smart Inbox Notification : The financier receives a medium-priority Smart Inbox item: “Borrower counter-offered: lower APR by 0.3% – respond within 24 h.” Clicking it opens the bid in the Active Bids view, with the counter-offer details displayed.

Negotiation Bot Transparency Panel : If the financier has also enabled their own auto-negotiation bot (configurable in Company Admin), the system will display a transparency panel that shows:

The current round (e.g., “Round 2 of 3”).

The last message from the borrower’s bot.

The parameters the financier’s bot is working within (minimum acceptable APR, maximum acceptable collateral requirement, etc., as set by the financier).

A “Pause/Override” button that allows the financier to take manual control at any time.

The financier can accept the counter-offer, reject it, or propose new terms manually. All messages are encrypted and logged in negotiation_sessions. The bot cannot force acceptance; it can only propose. The Governor ensures no binding agreement is reached without explicit human signature (ZITADEL passkey) from both parties.

6.2.5.10 Company Admin Tab – Financier-Specific Configuration

This tab contains the standard multitenant administration functions described in Part 6.2.8, with additional financier-specific settings.

Role Templates:

credit_analyst – Can view opportunities, submit bids, and see their own bids.

trader – Manages DeFi positions and secondary market activity.

compliance – Views portfolio, generates regulatory reports, and sees the Financed Companies tab.

viewer – Read-only access to the portfolio and opportunities, no bid/transaction capability.

Natural Language Risk Appetite : A free-text field where the financier describes their investment appetite in plain language, e.g., “I prefer short-term pre-shipment finance for agricultural commodities below $50,000, with borrowers from Egypt, Vietnam, or Kenya, and I am willing to accept up to 70% LTV.” This text is parsed by HF Mixtral and automatically translated into the structured financier_preferencesfilters that drive the automatic RFQ broadcast. The financier can review the interpreted filters and adjust them manually.

Preferences Configuration : A detailed form mirroring the financier_preferences table: accepted borrower countries (multi-select or “All except”), minimum trust score, minimum trade value, maximum financed amount per request, preferred financing types (pre-shipment, post-shipment, etc.), preferred settlement methods (bank transfer, stablecoin, DeFi), excluded commodities (HS code ranges), and geographic restrictions. Changes are logged and take effect immediately for new RFQ broadcasts.

Exposure Limits : The admin can set per-commodity and per-country exposure caps. If the portfolio exceeds a cap, the system blocks new bids in that category and alerts the financier.

Notification Preferences : Control over email and push notifications for new RFQs, bid status changes, margin call alerts, and repayment reminders. The financier can also toggle the “Receive match score explanations” and “Receive warm-up program emails” options, similar to the logistics Warm-Up program.

6.2.5.11 Governance & Decision Explanation Panel (Financier Context)

Every action the financier takes—submitting a bid, issuing a margin call, acknowledging a DeFi risk summary—is gated by the Governor. If the Governor returns a DENY or CONDITIONAL verdict, the PlainLanguage Governor Decision Panel (Part 6.1.3) slides open and explains the block in plain language.

Common scenarios include:

Bid rejected due to sanctions : “The borrower’s jurisdiction (Country Y) has been added to the autoblock list as of [date]. Financing requests from this jurisdiction are not permitted.”

Margin call blocked by contract : “The margin call amount requested exceeds the maximum allowed under the financing agreement. Maximum additional collateral that can be demanded is $12,000. Adjust the amount and retry.”

DeFi protocol risk too low : “The selected DeFi protocol has a risk score of 55, which is below the permitted threshold of 60. Please select a protocol with a higher risk score or switch to conventional settlement.”

Co-financing sum mismatch : “The total amount of accepted co-financing bids (85,000)exceedstheborrower’srequestedamount(85,000)exceedstheborrower’srequested amount(80,000). Adjust your acceptance.”

The panel always provides a direct link to the relevant form or documentation, and the “Request Human Review” button triggers an A3 escalation to the Platform Governance Authority or the financier’s own compliance officer if configured.

All financing-related Governor decisions are permanently logged with Loom hashes and visible in the Admin Portal’s audit trail. The Customer Care Chatbot can also be invoked from the Decision Panel to get immediate assistance.

6.2.5.12 Worked Example – Bank Finances Pre-Shipment Loan (Co-Financing)

This example follows a conventional bank, "Bank X" (UK), as it finances a pre-shipment loan for Mekong Fresh (Vietnam seller) against a shipment of oranges to Nile Foods (Egypt). The loan is co-financed with a DeFi pool.

Opportunity Discovery : Bank X's credit analyst logs into the Financier Portal. The Smart Inbox shows: "New financing opportunity – $100,000, 60 days, citrus. Match score 94." The analyst opens the Financing Opportunities tab, clicks "View Details".

Full Disclosure : The Request Details page reveals: seller Mekong Fresh (trust 91, 8 previous trades, 0 disputes), buyer Nile Foods (trust 88), all trade documents (invoice, packing list, B/L), credit score 89, default probability 4.2 %. AI recommends LTV 70 %. Based on this data, the analyst decides to proceed.

Bid Submission : The analyst submits an encrypted bid: 60,000at4.8 60,000at4.8 40,000 at 5.2 % APR via Aave V3 (the DeFi Pool Y had to acknowledge the DeFi Risk Summary before their bid was accepted).

Borrower Evaluation : Mekong Fresh sees both bids (after window closes) and accepts the combination: Bank X 60kat4.8 60kat4.8 40k at 5.2 %. Co-financing total $100,000.

Agreement & Witness : The finance-service assembles a master agreement with two annexes. The SGTX Financing Witness Clause is automatically inserted. The financing fee is 0.25 % × 100,000=100,000=250. All parties sign via ZITADEL passkey; the Governor signs as witness.

Disbursement : Bank X sends 60,000tothePSP(StripeConnect).DeFiPoolYsends60,000tothePSP(StripeConnect).DeFiPoolYsends40,000 USDC to the smart contract, which forwards it to the PSP. The PSP splits: 99,750toMekongFresh,99,750toMekongFresh,250 to SGTX. Mekong Fresh receives net 99,750.Therepaymentobligationis99,750.Therepaymentobligationis100,000 + interest.

Monitoring : Bank X monitors the loan via the Won Agreements tab. The Collateral Monitoring Dashboard shows LTV stable at 65 % (the goods are worth more than expected). No margin call needed.

Repayment : Two months later, Mekong Fresh repays $100,000 + interest to Bank X and DeFi Pool Y via wire and on-chain transfer, respectively. SGTX's reconciliation engine detects both payments. The financing agreement status becomes REPAID. Bank X's "Financed Companies" tab automatically updates Mekong Fresh's profile with a new on-time repayment. The relationship is strengthened for future deals.

Throughout the process, Bank X's interactions are governed by the platform, logged immutably, and supported by the Smart Inbox and Customer Care Chatbot as needed. The entire financing lifecycle is completed without a single phone call, paper document, or manual spreadsheet.

END OF PART 6.2.5 – FINANCIER PORTAL (v6.3 WITH ALL MODIFICATIONS)

PART 6.2.6 : GOVERNMENT PORTAL (CUSTOMS, PORT AUTHORITY, TRADE MINISTRY, REGULATORY AGENCIES) – v6.3 FULLY EXTENDED, WITH DYNAMIC MODULE CONFIGURATION, ANONYMOUS TRADE MANAGEMENT, CUSTOMS API INTEGRATION, MULTI-AGENCY WORKFLOW, LIVE TRADE MONITOR, SMART INBOX INTEGRATION, AND ALL MODIFICATIONS

6.2.6.0 Access & Purpose

Access : Employees of tenants with type = GOVERNMENT or REGULATORY. Onboarding is via a manual diplomatic channel (A3 escalation by the Platform Governance Authority). Each jurisdiction can create multiple sub-tenants (e.g., Egypt Customs, Egypt Ministry of Agriculture, Egypt Port Authority), each with its own login, roles, and dynamically configurable modules.

Purpose : Provide a sovereign digital workspace where government officials monitor trade flows through their jurisdiction, verify and approve documents, issue permits, manage customs clearances, participate in multi-agency approval workflows, conduct audits, and handle highly confidential anonymous trades – all within a zero-cost, self-hosted environment that seamlessly integrates with their existing national systems (Nafeza, VNACCS, TradeNet, etc.). The portal is fully integrated with the Smart Inbox, Shared Shipments Vault, Trade Command Center, and PlainLanguage Governor Decision Panel. The dual-mode toggle is not applicable.

6.2.6.1 Dynamic Government Onboarding & Module Configuration (v6.3)

Government tenants can configure their portal via a dynamic module wizard that appears after agency type selection (customs, agriculture, port authority, etc.). The configuration is stored in government_profiles.module_config (JSONB) and enforced by OPA policies and frontend conditional rendering.

Available Modules (Full Catalogue):

Module Name	Description	Dependencies	Configurable Parameters
document_verification	Review AI-verified trade documents, override, and stamp	None	• Require broker override for AI-flagged discrepancies (yes/no) • Auto-accept low-risk documents (confidence >95%) (yes/no) • Notify seller when document rejected (yes/no)
risk_scoring	View SGTX AI-computed risk scores (0-100), set autoclearance thresholds	None	• Autoclear if risk score < slider (0-100, default 25) • Manual inspection if score > slider (0-100, default 70) • Colour thresholds (green ≤25, amber

			26-70, red >70)
auto_clearance	Automatically approve low-risk shipments without manual intervention	risk_scoring enabled	• Apply to all low-risk shipments (yes/no) • Notify officer on autoclearance (yes/no)
multi_agency_workflow	Sequential or parallel approvals with other ministries	None	• Ministries involved (multiselect from list of government tenants in same jurisdiction) • Approval mode: sequential / parallel / anyone • Timeout (days) after which auto-escalation occurs
anonymous_trade	Handle government-to-government or sensitive commodity trades with identity redaction	None	• Require multisig (Platform Governance) for declassification (yes/no) • Automatically redact all non-government parties (yes/no) • Log access to real identities (yes/no)
api_integration	Connect to national single window, customs API, or other system	None	• Endpoint URL (editable) • Authentication type (none / API key / mutual TLS / OAuth2) • API key / certificate (stored encrypted) • Test connection button
permit_issuance	Issue digital export/import permits linked to USTNs	None	• Permit validity period (days, default 30) • Auto-expire permits after validity (yes/no) • Require countersignature from another agency (yes/no)
customs_declaration	Auto-fill and submit customs declarations via API	api_integration	• Declaration format (XML / EDIFACT / JSON) • Auto-submit on document verification (yes/no) • Receive

			confirmation reference (store in shipment record)
live_trade_monitor	Realtime dashboard of shipments in jurisdiction	None	• Refresh interval (seconds, default 60) • Show vessel positions on map (yes/no) • Filter by risk score, port, commodity (preset filters)
audit_trail	Access full Loom-chained audit logs for all shipments	None	• Require reason for each audit view (yes/no) • Export audit logs as signed PDF (yes/no)

Configuration Wizard UI (Textual Wireframe):

Step 1: Agency & Jurisdiction (prefilled)

text

┌──
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│ Configure Egypt Customs Step 1 of 3 │

├──
───┤

│ Agency: Customs │

│ Jurisdiction: Egypt (EG) │

│ Verified: □ Diplomatic channel approved │

│ Contact email: customs@example.gov.eg (editable) │

│ │

│ [Next] │

└──
───┘

Step 2: Module Toggle & Configuration (Core Step)

text

┌──
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│ Enable Modules & Set Rules Step 2 of 3 │

├──
───┤

│

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│

```
│ │ □ Document Verification [Configure] │ │
│ │ Review AI-verified trade documents, override, and stamp. │ │
│
└──────────────────────────────────────────────────────────────┘
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│
│ │ □ Risk Scoring [Configure] │ │
│ │ View SGTX risk scores, set autoclearance thresholds. │ │
│ │ └─ Autoclear if risk score < [ 25 ] (slider) │ │
│ │ └─ Manual inspection if score > [ 70 ] (slider) │ │
│
└──────────────────────────────────────────────────────────────┘
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│
│ │ □ AutoClearance (requires Risk Scoring) [Configure] │ │
│ │ Automatically approve low-risk shipments. │ │
│ │ └─ Notify officer on autoclearance? [Yes] │ │
│
└──────────────────────────────────────────────────────────────┘
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│
│ │ □ Multi-Agency Workflow [Configure] │ │
│ │ Sequential or parallel approvals with other ministries. │ │
│ │ └─ Ministries involved: [Agriculture, Health] (multiselect) │ │
│ │ └─ Approval mode: [Sequential ▼] │ │
│
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│
│ │ ☐ Anonymous Trade Management [Configure] │ │
│ │ Handle government-to-government trades with redaction. │ │
│ │ └─ Require multisig for declassification? [Yes] │ │
```

```
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│
│ │ □ API Integration [Configure] │ │
│ │ Connect to Nafeza (Egypt Single Window). │ │
│ │ └─ Endpoint: [https://nafeza.gov.eg/api/v1] (editable) │ │
│ │ └─ API Key: [••••••••] (masked) [Test Connection] │ │
│ │ └─ Status: □ Connected (200 OK) │ │
│
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│
│ │
│ [Back] [Next] │
└───────────────────────────────────────────────────────────────
───┘
```

Step 3: Review & Activate

text

```
┌───────────────────────────────────────────────────────────────
───┐
│ Review & Activate Step 3 of 3 │
├───────────────────────────────────────────────────────────────
───┤
│ You have enabled the following modules: │
│ • Document Verification (auto-accept low-risk) │
│ • Risk Scoring (autoclear <25, manual >70) │
│ • AutoClearance (notify officer on autoclear) │
│ • Multi-Agency Workflow (Agriculture, Health – sequential) │
│ • API Integration (Nafeza – connection successful) │
│ │
│ These settings can be changed later from the Integrations tab. │
│ │
│ [Back] [Activate Profile] │
└───────────────────────────────────────────────────────────────
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```

After activation, the portal's navigation tabs are dynamically rendered based on enabled modules. Configuration changes are logged in audit_log with decision type government_config_update.

6.2.6.2 Landing Page – Smart Inbox

Government users land on the Smart Inbox (Part 6.1.1). It aggregates time-sensitive tasks and alerts specific to their agency and jurisdiction.

Government-specific inbox items (examples):

Priority	Example Item	Score
□ **High**	USTN SGTX-EG-...-V1 – Clearance hold: Phytosanitary certificate missing. Vessel arrived 2h ago – demurrage risk.	95
□ **High**	USTN SGTX-ANON-...-V9 – Anonymous trade pending multi-agency approval (Defence). Your step: "Ministry Approval".	90
□ **Medium**	USTN SGTX-EG-...-V2 – AI risk score 68, exceeds autoclearance threshold. Manual inspection recommended.	75
□ **Medium**	Nafeza integration: Connector health degraded – 3 failed declaration submissions in last hour. Check API.	80
□ **Low**	Monthly CBE report due in 5 days. Auto-generated draft ready for review.	50
□ **Low**	USTN SGTX-EG-...-V3 – Auto-cleared. No action required.	20

AI Summary Card (Groq/Ollama) : "One vessel is waiting for a missing certificate. An anonymous trade requires your approval to proceed. The Nafeza connector needs attention."

6.2.6.3 Navigation & Tabs (Dynamic per Enabled Modules)

Example for Egypt Customs with all modules enabled:

Tab Name	Description
Inbox	Smart Inbox (default landing)
Live Trade Monitor	Shared Shipments Vault filtered to jurisdiction, with clearance status and integration badges
Clearance Workflow	Review shipments, AI-generated clearance recommendations, approve/hold/reject,

	override risk scores
Document Verification	Verify uploaded trade documents; AI-flagged discrepancies highlighted
Multi-Agency Workflow	View pending approvals from other ministries, approve your agency's step
Anonymous Trade Management	Manage government-to-government or high-confidentiality trades with identity redaction
Integrations	Configure and monitor connectors to national systems (Nafeza, VNACCS, etc.)
Permit Issuance	Issue digital export/import permits linked to USTNs
Audits & Reports	View full audit trails, generate regulatory reports
Company Admin	Invite officials, assign roles, set jurisdictional scopes

6.2.6.4 Live Trade Monitor Tab (Shared Shipments Vault)

Renders the Shared Shipments Vault (Part 6.1.6) with a filter preset to the government's jurisdiction (e.g., all shipments where destination, origin, or transit country equals "EG").

Additional columns visible to government users:

Risk Score – SGTX AI-computed risk (0-100), colour-coded.

Document Readiness – traffic light (green=all docs verified, amber=some pending, red=critical missing).

Declaration Status – if customs integration active (e.g., Nafeza), shows status of auto-submitted declaration (e.g., "Submitted – Accepted", "Pending", "Failed").

Integration Badge – small icon indicating whether the shipment's data has been successfully pushed to the national system.

Clicking any USTN opens the Trade Command Center with full shipment details. For anonymous trades (see 6.2.6.7), the USTN is displayed as SGTX-ANON-... and only authorised officials can see the real identities.

Filters : risk score range, commodity, port, vessel, document completeness. "Archived" view includes cleared, delivered, and distressed shipments.

6.2.6.5 Clearance Workflow Tab

The core operational tab for customs. Displays a queue of shipments that have arrived at the border or are approaching the port.

AI Clearance Recommendation Card (per shipment):

text

USTN: SGTX-EG-20260415120000-ABC12345-V1

Vessel: MSC Alessia – Arrived Alexandria, 10 Jul 2026

Risk Score: 22 (Low)

Documents: 5/5 verified

Recommendation : AUTOCLEAR (eligible under current autoclearance threshold of 30)

[Approve Clearance] [Override → Manual Review] [Hold for Inspection]

If auto_clearance module enabled and risk score ≤ threshold, the system automatically marks the shipment as “Cleared” and pushes status to seller/buyer. The officer sees a log entry.

If risk score above threshold, the officer must manually review. Options : Approve Clearance(release), Hold (pause, wait for missing document or physical inspection), Reject (block with required reason).

Every decision is logged as a Governor decision (decision_type = 'customs_clearance') with officer’s GTID and Loom hash.

Integration with national systems : If the customs API connector is active, clicking “Approve Clearance” automatically sends the final declaration status to the national system and stores the reference. If the connector fails, a manual fallback PDF is generated.

6.2.6.6 Document Verification Tab

Lists all documents required for shipments in the jurisdiction, focusing on those pending or flagged by AI.

USTN	Document Type	Responsible Party	AI Verification Status	Extracted Fields & Discrepancies	Action
...	Phytosanitary Certificate	Seller	FLAGGED	Weight mismatch (20,000 vs 17,600 kg). Confidence 96%.	[Accept] [Reject] [Request Amendment]

Actions : Accept (override AI, mark verified), Reject (send back with note), Request Amendment(pre-filled message). All actions logged.

6.2.6.7 Multi-Agency Workflow Tab

For shipments requiring approvals from multiple government bodies (e.g., Customs + Agriculture + Health). The RIA determines which agencies must approve based on HS code, commodity, and destination country rules. A workflow instance is created in agency_approval_workflows.

Each step has a responsible agency, a deadline, and a status.

The Smart Inbox of each agency shows pending steps.

Approvals/rejections are recorded with official’s GTID and timestamp.

Once all steps approved, the shipment automatically moves to “Cleared” (or next step).

For anonymous trades, each agency sees only the minimum data needed (e.g., HS code, weight, phytosanitary status). Party identities are redacted unless the agency is explicitly authorised.

Example display for sequential workflow:

text

Shipment USTN: SGTX-EG-...-V1

Current step : Agriculture Ministry (PHYTO approval) – deadline 15 Jun 2026

Previous: Customs – □ approved

Next: Health Ministry – pending

[Approve] [Reject] [Add Comments]

6.2.6.8 Anonymous Trade Management Tab

Handles government-to-government and highly sensitive commodity trades where buyer/seller identities are protected.

Anonymous Trade Lifecycle:

Initiation : A government tenant creates a trade request with is_anonymous = true. Counterparty must also be a government tenant with anonymous_trade_enabled = true. SGTX generates an anonymous USTN (e.g., SGTX-ANON-20260415120000-ABC12345-V1). The real USTN is stored in anonymous_trade_mappings and visible only to the two governments and the Platform Governance Authority.

Document Redaction : For all logistics providers, carriers, and non-government parties, consignee/shipper names are replaced with “Government Agency – [Country]”. Invoices and packing lists are automatically redacted.

Multi-Agency Approvals : Each approving agency sees only the data it needs. For example, Ministry of Agriculture sees “Live animals, 50 head, from Country X to Country Y” without knowing the specific ministries involved.

Delivery & Closure : Trade executes normally. All redacted documents carry a watermark “ANONYMOUS TRADE”. Full audit log encrypted.

Anonymous Trade Management Tab UI:

Anonymous USTN	Real USTN (internal, clickable)	Counterparty	Commodity	Status	Actions
SGTX-ANON-...	SGTX-EG-...-V1 (revealed only to authorised)	Government GTID	HS Code (redacted)	IN_EXECUTION	[View Full Details] [Revoke Anonymity] [Audit Log]

Declassification Procedure (v6.3) : If a government decides an anonymous trade no longer requires secrecy, an authorised officer clicks “Revoke Anonymity”. This creates a Governor decision that requires multisig approval (3/5) from the Platform Governance Authority. If approved, the anonymous_trade_mappings record is updated with revoked_at and a reason. The real USTN becomes visible to standard government monitors. Historic redacted documents are not retroactively changed. Both governments and the Platform Governance Authority receive notifications.

6.2.6.9 Integrations Tab (Dynamic per Country)

Allows government admin to manage connectors to national systems. The RIA continuously updates a library of available connectors based on scraped APIs, EDIFACT standards, open data.

Connector examples:

Customs Declaration API (Nafeza, VNACCS, ICS2, ACE)

Phytosanitary Certificate Verification (TRACES)

Certificate of Origin Verification

Vessel Arrival Notification (AISHub)

Permit Issuance (Return API)

Single Window Integration (TradeNet)

Sanctions List Sync

Setup:

Admin selects a connector from the library.

Enters credentials (API key, client certificate, SFTP address).

Clicks “Test Connection”. If successful, the integration is activated and appears as a badge in Live Trade Monitor.

All connector calls are logged in integration_connector_logs.

Example – Egypt Customs with Nafeza:
When a shipment arrives and documents are verified, the system auto-populates the XML declaration, signs it with the broker’s certificate (or customs officer’s), and submits to https://nafeza.gov.eg/api/v1/declaration. Response (reference number, status) is stored. Officer sees “Declaration NFZ12345 submitted – Accepted.” If Nafeza is down, the connector health degrades and Smart Inbox alerts the officer.

6.2.6.10 Permit Issuance Tab

For agencies like Agriculture or Health that issue import/export permits. Officials review applications (auto-filled with trade data), check AI-verified documents, and click “Issue Permit”. The system generates a digitally signed permit (PDF), stores it in documents, and updates the shipment’s document checklist. If api_integration is active, the permit is simultaneously pushed to the national system.

6.2.6.11 Audits & Reports Tab

Government audit teams can:

Search the full Governor decision log for their jurisdiction.

View Loom-chained audit trail of any USTN.

Verify data integrity by clicking “Verify Loom Chain”, which recalculates hashes clientside.

Generate jurisdictional regulatory reports (e.g., CBE monthly remittance, trade statistics) with one click; Groq drafts the narrative.

All audit access is logged and subject to the same permissions.

6.2.6.12 Decision Explanation Panel & Recovery

When a government officer attempts an action that is blocked (e.g., “Clearance blocked due to missing Sanitary Certificate”), the Decision Explanation Panel (Part 6.1.3) shows:

The exact regulation or rule.

The missing document or condition.

A direct link to request the document from the responsible party.

If a shipment gets stuck in a blocked state for >48h, the Smart Inbox escalates the item.

6.2.6.13 Worked Example – Egypt Customs Clearance (Standard)

Context : Vessel MSC Alessia arrives Alexandria. USTN SGTX-EG-...-V1 has risk score 22, documents complete.

Smart Inbox : “Vessel arrived – clearance recommendation: AUTOCLEAR.” (score 80)

AutoClearance enabled : system automatically processes clearance, pushes status to Nafeza, and stores reference NFZ12345.

Officer sees : “Auto-cleared – Nafeza reference NFZ12345.” No manual action.

Seller and buyer receive notifications.

6.2.6.14 Worked Example – Anonymous Defence Trade

Context : Egypt Ministry of Defence creates an anonymous trade for spare parts to Ghana Ministry of Defence.

USTN SGTX-ANON-20260415120000-XYZ12345-V1 generated.

Logistics providers see : “Origin: Alexandria, Destination: Tema. Consignee: Government Agency – Ghana.” No further details.

Multi-Agency : Egypt Customs sees a pending step: “Anonymous trade USTN-ANON... – HS 8710 (military parts) – Ministry of Defence approval attached. Customs approval required.” Officer approves.

Ghana Customs sees only the HS code and that consignee is their own government. They approve.

Both defence ministries see full details. Trade completes. Audit trail shows who accessed real identities.

6.2.6.15 Governance & Permissions

Government tenants only see data for their own jurisdiction (enforced by RLS and OPA).

anonymous_trade_mappings table access-controlled; only employees of involved governments and Platform Governance Authority can query real USTN.

Declassification of an anonymous trade requires multisig approval (3/5).

All integration connectors use encrypted credentials and mutual TLS. Failed API calls are retried and logged.

6.2.6.16 Zero-Cost Technology Stack

Component	Technology	Cost
Dynamic module configuration	PostgreSQL JSONB + RIA updates	$0
Connector library	Rust microservices, open protocols	$0
Customs API integration	REST client, mutual TLS (self-signed)	$0
EDIFACT/EDI parsing	edifact crate (Rust)	$0
Anonymous trade redaction	PostgreSQL RLS + Governor logging	$0
Regulatory report generation	Tera + headless Chrome + Groq	$0
Voice commands	Vosk + HF Mixtral + Groq	$0

No billing details required. Each government deploys connectors on their own sovereign node or SGTX hosts them as part of the sovereign node.

END OF PART 6.2.6 – GOVERNMENT PORTAL (v6.3 WITH ALL MODIFICATIONS)

PART 6.2.7 – ADMIN PORTAL – PLATFORM GOVERNANCE AUTHORITY (MULTISIG OVERSIGHT, SYSTEM CONFIGURATION, TENANT SUPPORT, CUSTOMER CARE CHATBOT, SPECIAL RATE MANAGEMENT) – v6.3 FULLY EXTENDED, WITH ALL MODIFICATIONS

6.2.7.0 Access & Purpose

Access : Only members of the Platform Governance Authority—the predefined 3-of-5 multisig group that holds ultimate decision-making authority over the SGTX platform—may access the Admin Portal. Authentication requires ZITADEL passkey (WebAuthn) with mandatory biometric verification. Every irreversible action (policy changes, jurisdiction matrix updates, PSP fallback triggers, tenant impersonation) requires multisig approval collected asynchronously through the portal's approval queue. No single admin can unilaterally alter the platform's constitution.

Purpose : The Admin Portal is the platform's own operating system. It provides a single, secure control room for system-level administration, including:

Real-time monitoring of platform health and global trade flows (with zero-knowledge aggregate views).

Editing and simulating OPA constitutional policies and WasmEdge modules.

Managing the global jurisdiction matrix, sanctions rules, and DeFi permissions.

Onboarding and managing marketplace partners, including revenue share agreements.

Overseeing platform incidents and generating post-mortems.

Providing tenant support via tenant impersonation (read-only, logged, multisig-approved) and the Customer Care Chatbot (with PIN-based tenant consent).

Defining special SGTX FEES rates for specific GTID pairs (within constitutional bounds).

Version-controlled configuration rollback across all platform settings.

All admin actions are governed by the same Governor they define; no admin action bypasses constitutional rules. Every action is logged immutably with Loom and visible in the audit trail.

6.2.7.1 Landing Page – Smart Inbox (Admin Context)

After biometric login, the admin lands on the Smart Inbox (Part 6.1.1). The admin inbox aggregates pending multisig approvals, system alerts, policy drift warnings, tenant impersonation requests, and AI-generated operational recommendations. It is the sole entry point for all time-sensitive governance actions.

Typical Admin Inbox Items:

 High – Multisig request #MS042: Approve new jurisdiction matrix entry for Nigeria (pending 29 h). Score 100.

 High – PSP health alert: Fawry degraded (score 42). SGTX FEES collection from Egypt automatically switched to Payoneer. Review fallback. Score 95.

 Medium – Policy drift detected: Credit scoring model accuracy dropped to 76 % (threshold 80 %). Retraining triggered; review new model before promotion. Score 80.

 Medium – Incident #INC20260614001: Governor 5-minute outage. Automated post-mortem draft ready for review. Score 75.

 Low – AI policy tuner suggests Rego update: “Reduce max SGTX FEES rate for distressed citrus to 1.8 %.” Simulate impact. Score 60.

 Low – New marketplace partner onboarding request from TradeBridge. AI risk score: 22 (Low). Ready for auto-approval. Score 50.

The AI Summary Card (Groq/Ollama) provides a concise daily briefing, e.g., “One multisig request has been pending for over a day. The Fawry fallback activated successfully; no immediate revenue impact. A draft post-mortem for the Governor incident is ready for your review.” Clicking any item deeplinks to the exact admin tab and form.

6.2.7.2 Navigation & Tabs

The left sidebar presents the following tabs, visible only to authenticated multisig members:

Order	Tab Name	Description
1	Inbox	Smart Inbox (default landing)
2	Platform Health	Realtime dashboards, predictive node health, PSP

		status, operational insights
3	Constitutional Policies	Edit OPA Rego policies and WASM modules with AI impact simulation
4	Governor Log & Audit	Natural-language searchable decision explorer; Loom chain integrity verification
5	Jurisdiction Matrix	Edit global jurisdiction rules, sanctions levels, DeFi/stablecoin permissions
6	PSP Manager	Monitor PSP health, manage fallback chains, add newly discovered payment rails
7	Special Rate Manager	Define and review special SGTX FEES rates for specific GTID pairs
8	Incidents	Incident lifecycle management, automated post-mortem generation
9	Marketplace Partners	Onboard new marketplace partners, manage revenue split agreements, review disputes
10	Tenant Management	View all tenants, impersonate for support (readonly, logged), manage impersonation logs
11	Customer Care Hub	Oversee the AI chatbot, monitor support queues, handle escalations
12	Configuration History	Version-controlled manifest of all platform configuration, with diff view and rollback
13	Internal Admin	Manage multisig members, internal roles, and access keys

All tabs are accessible only with full multisig membership. The PlainLanguage Governor Decision Panel triggers automatically if any admin action is blocked by the constitution.

6.2.7.3 Tab-by-Tab Detailed Specifications

6.2.7.3.1 Platform Health Tab (Zero-Knowledge System Oversight)

This is the command centre for real-time operational awareness. It renders four collapsible zones, all data sourced from Prometheus, ClickHouse, and NATS.

Real-Time Metrics : Active trades count (all USTNs with status CONTRACTED, FINANCING, IN_EXECUTION, and multi-shipment per-shipment counters), 24-hour GMV, fee collection rate (trade fees and financing fees separately), dispute rate, Governor decision latency (p95), node health grid (CPU, memory, disk per sovereign node, colour-coded). No individual trade values or tenant identities are shown; everything is aggregated.

Predictive Health (A2, LSTM) : Line charts forecasting disk usage, latency, PSP error rates over next 7 days. Example alert: "Cairo node disk usage projected to reach 91 % in 5 days. Recommended maintenance window: 20260620 02:00-04:00 EET." A "Schedule Maintenance" button creates a maintenance_windows record.

PSP Health & Fallback Monitor : Table of all connected PSPs with current health score, last check, latency, failure rate (1h), status (HEALTHY, DEGRADED, DOWN). Admin can manually force fallback ("Mark as DOWN") with required reason, or view the current fallback chain per country. Fallback changes require multisig approval.

AI-Generated Operational Summary (Groq, refreshed daily) : "Yesterday, the platform processed 34 trades totaling $1.2 M GMV. PSP Fawry had a 12-minute degradation; fallback to Payoneer worked with zero fee loss. Node Cairo CPU averaged 62 %. No incidents."

6.2.7.3.2 Constitutional Policies Tab (AI-Assisted Impact Simulation)

Allows the multisig to edit the platform's Rego policies and recompile WASM constitutional modules. The interface includes:

Policy Editor : Monaco code editor (VSCode-style) with syntax highlighting for Rego. All current policies are listed in a file tree (permissions.rego, contract.rego, constitutional_rules.rego, etc.).

Impact Simulation : Admin proposes a change and clicks "Simulate Impact". The system compiles the proposed Rego to a temporary WASM module, replays a random sample of 10,000 historical Governor decisions through current vs. proposed policies, and computes affected decisions, estimated fee revenue impact (based on affected trade values), predicted dispute rate change (XGBoost model), and constitutional violations introduced. Results are displayed in a summary card with a Groq-generated recommendation.

Submit for Approval : Once simulated, the admin can submit the change for multisig approval. All other multisig members receive Smart Inbox items. Upon 3-of-5 signatures, the new Rego is deployed, the WASM module is recompiled, the Governor hot-reloads it, and the change is versioned in Configuration History.

6.2.7.3.3 Governor Log & Audit Tab (Natural Language Query)

Immutable decision explorer with natural-language querying powered by Groq converting plain English to ClickHouse SQL. Admins can ask: "Show all decisions where fee release was blocked in the last week," or "List trades where the buyer was from Egypt and a dispute was filed." Results are paginated and exportable to CSV. Clicking a row shows the full decision JSON, including the tenant_message displayed to the user. A "Verify Loom Chain" button recalculates all hashes from genesis to the selected decision; any mismatch triggers a P0 incident.

6.2.7.3.4 Jurisdiction Matrix Editor Tab

Editable table of all 195+ jurisdictions with fields : country code, name, sanctions level, DeFi allowed, distressed sale allowed, reporting requirements, etc. Includes:

Conflict detection rules (WasmEdge) : e.g., "A country with sanctions_level = 'Autoblock' cannot have defi_allowed = true." Conflicts block saving with a red banner.

Revert to RIA Value : Button to restore the latest automatically scraped regulatory value, if the RIA has proposed an update.

Mass Update : “Apply RIA Recommendations” processes all pending RIA suggestions in a batch diff view before multisig approval.

6.2.7.3.5 PSP Manager Tab

Monitor and manage all payment service providers. Table with name, countries served, health score, latency, failure rate, status. Actions: “Force Fallback”, “Restore”, “View Fallback Chain”. Fallback Chain Editor per country allows reordering priority; changes require multisig approval. An “Add New Rail” form, manually or from RIA discovery, allows admins to integrate new PSPs; the system tests connectivity and, if successful, adds it to the live routing table.

6.2.7.3.6 Special Rate Manager Tab

Provides the interface for the Platform Governance Authority to define special SGTX FEES ratesfor specific pairs of GTIDs. This overrides the dynamically computed trade fee rate for trades between those two counterparties. The tab lists all active and expired special rates with justification, rate, validity period, and approval status.

Workflow to create a special rate:

Admin clicks “Propose Special Rate”. A form prompts for Seller GTID, Buyer GTID, new fee rate (0.1%–2.5%), effective date range, and a mandatory reason (free text, e.g., “Strategic volume agreement for citrus corridor”).

The proposal is submitted as a multisig request. All other multisig members receive a Smart Inbox item: “Special rate proposal: [Seller] → [Buyer] at X%. Review.”

Upon 3-of-5 approval, the Governor records a special_rate_override decision and inserts a row into the special_SGTX FEES_rates table. The Fee Calculator then uses this fixed rate for any new trade between those GTIDs during the effective period.

Special rates can be edited or revoked before expiry via the same multisig process. Expired rates are automatically removed from the active list.

All actions are permanently logged and visible in Configuration History.

The tab also displays a quick summary of all active special rates, including the parties, rate, and justification. The Admin Portal’s Shared Shipments Vault (see 6.1.6.3.7) also allows initiation of a special rate directly from a trade row using the “Initiate Special Rate” action.

6.2.7.3.7 Incidents Tab

Manages the lifecycle of platform incidents, with AI-assisted post-mortem drafting.

Incident Queue : List of open/resolved incidents, sorted by severity (CRITICAL, HIGH, MEDIUM, LOW). Each row shows incident ID, title, start time, duration, status.

Automated Evidence Collection : When an incident is created (via Prometheus Alertmanager webhook), a Rust agent gathers logs (Loki), metrics (Prometheus), traces (Jaeger), and correlates with recent config changes. This is attached to the incident.

Draft Post-Mortem (Groq, A1) : AI generates a draft including title, date, impact, root cause analysis, resolution steps, data loss assessment, and recommendations. Admin can edit inline.

Publish : Admin clicks “Publish to Status Page” (status.sgtx.io). The incident is stored permanently in incidents.

6.2.7.3.8 Marketplace Partners Tab

Onboard and manage marketplace partners that connect via the Marketplace API.

Partner List : Table with partner name, jurisdiction, current revenue split, API key status, active/inactive toggle.

Onboard New Partner : Admin enters partner legal name, jurisdiction, contact email, proposed revenue split (e.g., 0.5% partner / 0.8% SGTX). AI (Clause Forge) drafts a custom revenue-share agreement (English and partner’s local language). Admin reviews; sends secure signing link to partner. Partner signs via ZITADEL passkey. Upon signature, an Ed25519 API key is auto-generated.

Disputed Attributions : When a partner disputes an auto-attributed trade, a dispute record appears here. Admin reviews AI recommendations (from Phase 10) and approves/denies the dispute with multisig approval. Revenue adjustment handled by settlement service.

6.2.7.3.9 Tenant Management & Impersonation Tab

Provides oversight and support tools for all tenants.

Tenant Search : Search by GTID, legal name, jurisdiction, type. Clicking a tenant shows their KYB tier, trust score, active trades (including multi-shipment contracts), financing agreements, distressed listings, and a link to their audit trail. This view respects the zero-knowledge model: by default, trade values, counterparty names, and sensitive commercial data are redacted. Only aggregate counts and statuses are shown. Full details require a break-glass audit view (see below).

Impersonation (Support Mode) : Admin clicks “Impersonate” next to a tenant. A modal requires a mandatory reason and triggers a multisig approval request. Once 3-of-5 approvals are collected, the admin is granted a time-limited (max 30 min), read-only session inside that tenant’s portal, with a persistent red banner “IMPERSONATING [Tenant Name] – Readonly.” Every page view is logged as a Governor decision (decision_type = ‘admin_impersonation’) and the tenant is automatically notified after the session ends.

Break-Glass Audit View : For investigating specific trades, an admin can request full un-redacted access to a trade’s data. This follows the same multisig approval as impersonation and is logged with the same rigour. The session is limited to 15 minutes and only for the specific USTN.

Impersonation Log : Subtabs shows all impersonation sessions with admin, tenant, reason, time, and pages viewed.

6.2.7.3.10 Customer Care Hub

This tab provides the oversight interface for the platform’s AI-powered customer support system and the human escalation queue.

Chatbot Performance Dashboard : Real-time metrics on the Customer Care Chatbot: number of conversations handled, resolution rate (percentage of inquiries resolved without human escalation), average conversation duration, and a breakdown of topics (e.g., “Document issues 40%, Login problems 25%, Fee questions 15%”). This helps the Platform Governance Authority assess the bot’s effectiveness and identify systemic issues.

Human Support Queue : A table of all tickets that have been escalated from the chatbot to a human agent, plus any direct support calls. Each row shows: ticket ID, tenant GTID and company name, USTN (if applicable), the reason for escalation, the current status (QUEUED, IN_PROGRESS, RESOLVED, CLOSED), the assigned human agent, and the SLA timer. Admin can reassign tickets or force-close them if needed.

Chat Transcripts : A searchable log of all chatbot interactions, stored with the tenant’s consent. Transcripts are anonymised for platform analytics but identifiable by tenant for audit and dispute resolution. The admin can review conversations where the bot failed to resolve the issue to improve the bot’s training data.

Escalation Rules Configuration : Define under what conditions the bot should automatically escalate (e.g., after three failed attempts, or when a specific keyword like “lawsuit” or “legal action” is mentioned). The admin can also configure the maximum wait time in the human queue before a second-level escalation to the Platform Governance Authority itself.

Integration with the Chatbot’s PIN Mechanism : The admin can see aggregated statistics on PIN-based actions (how many impersonations the bot performed, success rate) but can never see the tenant’s actual PIN, which is stored as a bcrypt hash.

6.2.7.3.11 Configuration History Tab (Version Control & Rollback)

All platform-wide configuration is treated as a versioned manifest. This tab provides a Git-like history and rollback capability for OPA policy files, WASM binary hashes, jurisdiction matrix snapshots, PSP lists and routing rules, AI model version references, special SGTX FEES rates, and government connector configurations.

Version Timeline : Auto-incremented versions, each with timestamp, admin who initiated, list of changed components, and a “Diff” button showing side-by-side comparison with previous version.

Rollback : Admin selects a previous version and clicks “Rollback to this version”. This creates a new version (incremented) that is a copy of the target, ensuring full audit. The system restores all configuration items from the snapshot, recompiles WASM modules, and hot-reloads the Governor. All multisig members are notified.

6.2.7.3.12 Internal Admin Tab

Manages the Platform Governance Authority’s own membership and internal roles.

Multisig Members : List of current members (employee GTIDs), their public keys, status (active/suspended). Adding or removing a member requires 3-of-5 approval of the remaining members.

Internal Roles : platform_admin (full access), auditor (read-only logs and config), support(tenant impersonation and incident viewing, no policy editing).

API Keys for Platform Services : Manage internal API keys used by platform services for inter-service authentication.

6.2.7.4 Governance & Decision Explanation Panel

Every admin action is subject to the same Governor as any tenant. If a multisig member attempts to set a SGTX FEES rate above 2.5 %, force a milestone without justification, or alter a jurisdiction rule that violates a constitutional conflict, the PlainLanguage Governor Decision Panel opens with a clear explanation, the specific rule violated, and the proper procedure (e.g., “Submit a policy amendment via the Constitutional Policies tab with impact simulation”). If a multisig request is stuck (e.g., not enough approvals collected), the Smart Inbox escalates after 48 h.

6.2.7.5 Worked Example – Admin Responds to PSP Degradation and Proposes a Special Rate

Context : The multisig is composed of five members. On a typical morning, Member A logs into the Admin Portal.

Smart Inbox : “PSP health alert: Fawry degraded (score 42). SGTX FEES collection from Egypt automatically switched to Payoneer. Review.”

PSP Manager Tab : Member A sees Fawry’s health chart showing a spike in failures. The fallback to Payoneer is active and working. Member A acknowledges the alert and posts a comment that the operations team will investigate the root cause. No further action needed.

Later : Member B receives a notification that a new special rate proposal has been submitted by Member C: a rate of 0.8 % for all trades between SGTX-VN-TRD-002 (a large Vietnamese seller) and SGTX-EG-TRD-001 (their largest Egyptian buyer) for a 12-month period, with a justification “High-volume corridor strategic partnership.”

Special Rate Manager Tab : Member B opens the proposal, reviews the justification, and checks the counterparties’ trust scores and trade volume. Satisfied, they click “Approve.” Member D also approves. The quorum is reached (3-of-5); the Governor records the special rate and updates the fee calculation engine. The new rate takes effect immediately for any new trades between those GTIDs.

Configuration History : The activation of the special rate appears as a new version (Version 127). Member B can see the diff, who approved, and when.

Throughout the day, the Customer Care Hub shows a steady flow of chatbot interactions; a few have escalated, and the human support team is handling them. The Platform Health dashboard shows all nodes green.

6.2.7.6 Zero-Cost Technology Stack

Policy simulation: Replay historical decisions + XGBoost

Natural language audit: Groq (free tier) + ClickHouse

Predictive health dashboard : LSTM (PyTorch local) + Prometheus

Conflict detection: WasmEdge rules

Incident post-mortem: Groq + Loki/Jaeger/OpenTelemetry

Partner onboarding automation : Clause Forge (HF Mixtral) + Groq

Tenant impersonation: ZITADEL session override + OPA

Configuration version control : PostgreSQL snapshots + Governor logging

Customer Care Chatbot : Rust Axum + Rig + Groq + ZITADEL impersonation

Special rate engine : Governor-enforced fee calculation override

VoIP for human support : Self-hosted Janus Gateway (WebRTC) or Asterisk

No billing details required. All infrastructure is self-hosted.

END OF PART 6.2.7 – ADMIN PORTAL – PLATFORM GOVERNANCE AUTHORITY (v6.3 WITH ALL MODIFICATIONS)

PART 6.2.8 : MULTITENANT ADMINISTRATION PER PORTAL (v6.3 – UNIFIED & EXPANDED, WITH DUAL-MODE TOGGLE SETTINGS, DATA SCOPES, TENANT ACTIVITY LOG, CONFIGURATION VERSIONING & ROLLBACK, TENANT BRANDING (FREE WHITELABEL), AND ALL MODIFICATIONS)

6.2.8.0 Purpose

Every SGTX portal – Buyer, Seller, Logistics, QC, Financier, Government, Admin, and Marketplace Partner – shares a common tenant administration framework. This part defines the universal patterns for:

Employee invitation and lifecycle

Role-based access control (RBAC) with portal-specific templates

Data scopes that limit what an employee can see (including cost hiding for sellers and dual-mode toggle disabling)

Internal approval chains for high-value actions

Tenant Activity Log (self-service audit trail)

Support impersonation (readonly) by the Platform Governance Authority (Part 6.2.7)

Tenant Branding (whitelabel, free for all tenants)

Tenant Configuration Versioning & Rollback (Enterprise mode)

Individual portal sections (6.2.1–6.2.7) contain a brief Company Admin tab description and then reference this part for full technical specification. Marketplace partners have a separate admin in Part 6.2.9.

All administration actions are gated by the Governor and logged immutably via Loom. No tenant admin can modify platform-wide policies – only their own tenant's employees, roles, scopes, branding, and approval chains.

6.2.8.1 Unified Employee Invitation Flow

Every portal uses the same invitation and onboarding process, powered by ZITADEL (self-hosted).

Step-by-step:

Tenant Admin enters email – From the "Company Admin" tab, the admin enters the new employee's email address and selects a role template (see 6.2.8.2).

An employees row is created with status = INVITED and an invitation_token (expires in 72 hours).

Invitation email sent – Email contains a ZITADEL registration link, including the tenant's legal name.

Employee sets up account – Employee clicks link, creates ZITADEL account, registers a WebAuthn passkey (biometric or security key), and accepts the platform's Terms of Service.

Admin approval – Tenant admin sees employee with status PENDING_APPROVAL. They can Approve (status → ACTIVE) or Reject (with reason). Auto-approval can be enabled if the employee's email domain matches the tenant's verified domain.

Login & role assignment – Employee logs in; ZITADEL issues a JWT containing role_id, tenant_id, permissions, data scopes, and (for dual-mode tenants) the active_trader_mode_context and allow_role_switching.

Expiry and reinvite : Invitation expires after 72h; admin can click "Reinvite" to generate a new token. Bulk invitation via CSV upload supported. Deactivation: admin can set employee to SUSPENDED or DEACTIVATED.

6.2.8.2 Role Templates & Permissions (Portal-Specific)

Each portal type has predefined role templates. Tenant admins can assign these templates or create custom roles by selecting individual permissions from the full action list. Permissions are evaluated via OPA (permissions.rego).

6.2.8.2.1 Trader Portal (Buyer / Seller) – already defined.
6.2.8.2.2 Logistics Portal – already defined.
6.2.8.2.3 QC Portal – already defined.
6.2.8.2.4 Financier Portal – already defined.
6.2.8.2.5 Government Portal – already defined.
6.2.8.2.6 Admin Portal – already defined.

Note on dual-mode toggle : For tenants with default_trader_mode = DUAL and employees with allow_role_switching = true (in data scopes), the header toggle is visible. Tenant admins can disable the toggle per employee via data_scopes.allow_role_switching = false.

6.2.8.3 Data Scopes (Extended for Cost Hiding & Dual-Mode Toggle)

Beyond role-based permissions, tenant admins can apply data scopes to any employee to further restrict what data they can see or act upon. Stored in data_scopes table (JSONB custom_filtersand structured columns).

Common scope dimensions (re-stated with additions):

Scope	Description	Example
Country	Restrict to specific jurisdictions	Employee can only view trades involving Egypt.
Commodity (HS Chapter)	Restrict to specific product	Only view trades for HS

	categories	chapters 08 (fruit) and 09 (coffee).
Max Transaction Value	Ceiling on trade value employee can approve/view	Junior trade manager cannot approve quotes > $50,000.
Document Type Visibility	Hide certain document types	Driver cannot view commercial invoices.
Trader Mode (BUY/SELL)	Restrict to buyer or seller context	Finance user in a dual-mode company only sees seller-side settlements.
Service Type (Logistics)	Restrict to a specific service	Dispatcher only sees trucking jobs.
Financing Type	Limit to conventional or DeFi	Trainee trader can only bid on conventional loans.
Hidden Cost Components (Seller)	Hide specific cost elements from employee (e.g., freight, SGTX FEES, SGTX fee)	Warehouse operator sees no pricing; junior trade manager sees EXW but not freight.
Allow Role Switching	Enable/disable dual-mode toggle for this employee	Finance user kept in Buyer mode only; toggle hidden.

Enforcement : Data scopes are injected into the JWT as claims and enforced by OPA at API gateway and PostgreSQL RLS. Frontend uses scopes to hide UI elements (defence in depth).

6.2.8.4 Internal Approval Chains (v6.3 – Expanded)

Tenant admins can define internal approval chains for high-value or sensitive actions within their own tenant. Separate from platform's constitutional governance (multisig). Actions
gated: contract.sign, quote.submit, payment.approve, employee.invite, trade.cancel, financing.bid.submit (and custom extensions).

Policy definition (stored in tenant_approval_policies):

sql

-- Example policy: contract.sign over $100k requires 2 trade managers and 1 finance approver

INSERT INTO tenant_approval_policies (tenant_id, action, condition_json, required_approvals, quorum, approval_mode)

VALUES (..., 'contract.sign', '{"trade.value" : {">": 100000}}',

'[{"role": "trade_manager", "count": 2}, {"group_id": "finance-approvers", "count": 1}]',

3, 'parallel');

Approval Workflow:

Employee attempts gated action. Governor checks policy; if applicable, does not execute immediately.

Creates pending_approval record; sends Smart Inbox items to each approver (category NEEDS_APPROVAL, priority based on SLA).

Approvers see request with details and buttons : Approve, Reject, Request Changes.

Once quorum reached, Governor automatically executes the original action and notifies requestor.

If any approver rejects, action is permanently blocked; requestor receives Smart Inbox rejection reason.

Approval Graph Visualization (Enterprise mode) : Interactive graph (React Flow / D3) showing requestor, action, approvers (pending, approved, rejected), conditions, and estimated completion time based on historical approval times. Available from Trade Command Center and Company Admin.

SLA & Escalation : Each policy can have an SLA (hours). If approver does not act within SLA: reminder sent; after 2× SLA, escalation – manager override or Platform Governance Authority notification.

6.2.8.5 Support Impersonation (by Platform Governance Authority)

As described in Part 6.2.7.3.8, Platform Governance Authority members with the support role can temporarily impersonate a tenant employee for debugging, under strict controls:

Requires multisig approval (3-of-5)

Session is readonly (all write buttons disabled)

Session expires after 30 minutes

Red banner displayed : "IMPERSONATING [Tenant Name] – Readonly."

Every page view logged with decision_type = 'admin_impersonation'

Tenant notified after session via email.

Tenant admins cannot opt out of impersonation; it is a condition of the Master Service Agreement for support purposes. They can view all impersonation sessions involving their tenant in their own activity log.

6.2.8.6 Tenant Activity Log (v6.3 – Fully Expanded, Cryptographically Signed Exports)

Tenant admins can view a complete, immutable log of all employee actions within their tenant for internal audits, compliance, and security monitoring.

Access : Company Admin → Tenant Activity Log tab. Visible only to users with tenant_activity_log.view permission (typically tenant_admin and auditor roles). Readonly – no deletion or modification.

Log Contents : Timestamp, employee name, employee GTID, action (permission string or human-readable), USTN (or financing agreement ID, distressed listing ID), hashed IP address, Governor decision ID (clickable), details JSONB.

Filtering & Search : Date range, employee, action type, USTN, decision ID. Advanced natural-language search supported (Groq, fallback to structured).

Export (Signed) : Admin can export filtered log as CSV, JSON, or PDF. Exports are cryptographically signed with Ed25519, include SHA256 checksum, verification instructions, and QR code. Revocation list available for security incidents.

Retention : 7 years (compliance). After 90 days, logs moved to ClickHouse.

6.2.8.7 Tenant Configuration Versioning & Rollback (Enterprise Mode)

For enterprise tenants, all configuration changes (roles, permissions, data scopes, approval chains, notification preferences, branding) are versioned to allow audit, rollback, and recovery from misconfiguration.

What is versioned : Full snapshot of tenant's configuration state at each change. Stored in tenant_config_history with auto-incremented version number, timestamp, admin who made change, change reason.

UI – Configuration History (Company Admin → Configuration History):

Table of past versions (version, date, admin, change reason). View Diff (side-by-side JSON diff). Rollback to this version button.

Rollback workflow : Admin selects previous version, clicks "Rollback", confirms with reason. System creates a new version (incremented) that is a copy of the target, applies the snapshot (replaces current roles, permissions, etc.), logs the rollback as a Governor decision. Smart Inbox notification to all tenant admins.

Retention : 7 years; older versions moved to ClickHouse after 90 days; rollback still possible but with higher latency.

6.2.8.8 Tenant Branding (Whitepaper – Free for All Tenants, No Tier Restrictions)

All tenants can customise the look and feel of their portal (logo, colours, portal name, tab labels, footer text, email sender name). No additional fees, no upgrade required.

Customisable Elements:

Logo (PNG/SVG, max 500KB) – falls back to SGTX logo if missing/invalid.

Favicon (ICO/PNG, max 100KB)

Primary colour (hex, WCAG contrast validated)

Secondary colour (hex)

Portal name (max 30 characters) – appears in browser tab title and header.

Navigation tab labels (custom per tab, max 20 characters) – resettable individually.

Footer text – must retain the non-marketplace disclaimer (appended automatically).

Email sender name – appears as "From" name in emails; platform appends legal disclaimer. Actual domain remains noreply@sgtx.io.

Login page background (optional image or solid colour, max 2MB JPG/PNG)

Configuration UI : Company Admin → Branding. Live preview (iframe) shows changes before saving. Accessibility colour contrast validation built in.

Storage & Delivery : Branding JSONB stored in tenants.branding. Assets stored in tenant's sovereign node object store. Identity service injects branding into JWT as custom claim. Frontend applies CSS custom properties (--brand-primary, --brand-secondary), updates <title> and favicon, and applies text replacements via React context. Fallback to SGTX defaults if any asset fails.

Security : Only tenant admins can modify branding. Changes logged in audit_log. Platform Governance Authority can revert branding if it violates platform policies (e.g., impersonating SGTX, removing required legal text). Uploaded assets scanned for malware using ClamAV (free, self-hosted).

6.2.8.9 Example – Tenant Admin Uses Activity Log and Rollback

Context : Nile Foods discovers a suspicious trade request created from an unknown commodity. Admin investigates via Tenant Activity Log, filters by trade.request.create last 7 days, finds entry from a former employee whose account was not deactivated. Admin immediately deactivates the former employee. Exports log as signed CSV for internal security report.

Later, an accidental permission change : Admin mistakenly removes contract.sign from all trade managers. Within minutes, contracts cannot be signed. Admin goes to Configuration History, sees version 42 (current) vs 41 (correct). Clicks "Rollback to version 41", enters reason, confirms. Within seconds, permissions restored. Smart Inbox notifies all tenant admins.

6.2.8.10 Zero-Cost Implementation

Component	Technology	Cost
Employee invitation	ZITADEL (self-hosted)	$0
RBAC + OPA	OPA + Rego	$0
Data scopes	PostgreSQL RLS + JWT claims	$0
Approval chains	Governor + OPA + workflow-recovery-service	$0
Tenant Activity Log	ClickHouse + tenant-data-export-service	$0
Configuration versioning	PostgreSQL snapshots + Governor logging	$0
Tenant branding	PostgreSQL JSONB + CSS custom properties	$0
Impersonation	ZITADEL session override + Governor logging	$0

No external identity provider, audit, or branding service required.

END OF PART 6.2.8 – MULTITENANT ADMINISTRATION PER PORTAL (v6.3 WITH ALL MODIFICATIONS)

PART 6.2.9 : MARKETPLACE PARTNER PORTAL (v6.3 – FULLY EXTENDED, WITH API RATE LIMIT GAUGES, USAGE ANALYTICS, SANDBOX, REVENUE SHARE AGREEMENT MANAGEMENT, WEBHOOK MANAGEMENT, LEAD MANAGEMENT, AND ALL MODIFICATIONS)

6.2.9.0 Access & Purpose

Access : Employees of registered marketplace partners (e.g., TradeBridge, Alibaba, other digital trade platforms). Tenant type: MARKETPLACE_PARTNER (new enum value). Authentication: ZITADEL + API key (Ed25519) for API calls. Portal login uses ZITADEL passkey.

Purpose : Self-service portal for partners to manage their integration, view leads (intents), monitor revenue, configure webhooks, dispute attributions, access a test sandbox, and monitor their API rate limits. The portal complements the Marketplace API (Part 17) by providing a user interface for non-technical partner staff. All actions are logged and gated by the Governor. The portal offers the same Smart Inbox landing page and role-based admin controls as other portal types.

Dual-mode toggle : Not applicable (marketplace partners have a single role).

6.2.9.1 Partner Dashboard – Key Metrics & AI Recommendations

The dashboard is the partner's command centre, showing real-time performance metrics and AI-generated insights.

Key metrics (from ClickHouse + PostgreSQL):

Total leads (intents) submitted – last 7, 30, 90 days, displayed as a line chart.

Conversion rate (intents → attributed trades) – percentage and trend arrow.

Total SGTX FEES earned (USD, split between partner and SGTX) – breakdown by month.

Average trade value (USD) – across attributed trades.

Top performing corridors – origin-destination pairs, ranked by GMV.

Recent attributed trades – list of USTNs with date, trade value, SGTX FEES amount.

AI Summary Card (Groq, A1 – fallback Ollama):
Refreshed on each dashboard load: "Your conversion rate increased to 18% this month (up from 14% last month). Vietnam-Egypt citrus corridor is your best performer (8 trades, 12,000SGTX FEES).ConsiderfocusingmarketingeffortsthereYourtotalSGTX FEESearnedthisquarteris12,000SGTX FEES).Considerfocusingmarketingeffortsthere.YourtotalSGTX FEESearnedthisquarteris45,200, which is 12% above target."

Voice commands (supported via VoiceCommandButton):

"Show my conversion rate."

"List top performing corridors."

"What is my total SGTX FEES this month?"

6.2.9.2 Leads Management (Intent Inbox)

Purpose : Review all intents submitted via the API, their status, and take action.

Table columns:

Intent ID	Raw Text	Parsed Specs (summary)	Viability Score	Status	Created At	Action
int9876	“20,000 kg oranges, CFR Alexandria”	oranges, 20,000 kg, CFR, Egypt	87	ACCEPTED	2026-06-10	View
int9877	“5,000 kg apples to Iran”	apples, 5,000 kg, Iran	12	REJECTED	2026-06-11	View
int9878	“lemons, Vietnam, need QC”	lemons, quantity missing	55	CONDITIONAL	2026-06-12	Resolve

Status meanings:

ACCEPTED – Lead is viable; partner can proceed to redirect buyer to SGTX for full trade creation.

REJECTED – Lead is not viable (sanctions, jurisdiction block, unsupported commodity). Reason shown in a tooltip.

CONDITIONAL – Lead has low confidence; partner must collect more information (e.g., quantity) before retrying.

Actions : View (opens modal with complete intent response, parsed specs, recommendations, next action, and Groq-generated explanation if rejected/conditional). Resolve – for conditional intents, allows partner to edit raw text or fill missing fields and resubmit (creates new intent via API).

Filtering & search : by status, date range, commodity, viability score range.

6.2.9.3 Webhook Management & Event Logs

Purpose : Configure webhook endpoints to receive real-time events (trade locked, SGTX FEES settled, dispute filed). Test and monitor delivery.

UI – Webhook Endpoints:

List of registered endpoints (URL, events subscribed, status (active/inactive), created date).

Add Endpoint – form: URL, select events via checkboxes (trade.locked, SGTX FEES.settled, dispute.filed, lead.rejected), optional secret rotation policy.

Edit – change URL or events, regenerate secret.

Delete – remove endpoint.

Test – sends a test.ping event to the endpoint and shows delivery result (success/failure, response body, latency).

Event Delivery Logs:

Table showing recent deliveries : timestamp, event type, endpoint URL, HTTP status, duration, error message (if any).

Retry button for failed deliveries (manually resend the event).

Download logs as CSV for debugging.

Security : Webhook signatures are verified using the partner's Ed25519 public key (stored in marketplace_partners). The portal displays the current public key and allows rotation. A note explains: "Your public key is used by SGTX to sign all webhook payloads; you should use your private key to verify them on your server. Rotating the key invalidates the old one after a 24-hour grace period."

6.2.9.4 Revenue Attribution Dispute Management

Purpose : Allow partners to dispute automatically attributed trades (e.g., "This trade should not be attributed to me") and track resolution.

Dispute Workflow:

Partner sees an attributed trade in the "Attributed Trades" list (on dashboard) that they believe is incorrect.

They click "Dispute" – a modal opens asking for reason (free text) and optional evidence upload (PDF, image).

The dispute is created in the disputes table (type = REVENUE_ATTRIBUTION) and assigned a unique ID. The Platform Governance Authority (multisig) receives a notification.

The AI (HF Mixtral) analyses the dispute and the partner's history, generating a recommendation visible to the Platform Governance Authority in the Admin Portal.

The partner can track the dispute status in the "Disputed Attributions" tab:

Dispute ID	USTN	Attributed Revenue Share	Reason	Status	AI Recommendation
disp001	SGTX-EG-...	$389.28 (0.8%)	"This trade originated from a different marketplace."	IN_REVIEW	"Likely misattribution – recommend reversal."

Once resolved (by multisig), the partner sees the outcome (attribution upheld, reversed, or adjusted). If reversed, the revenue share is removed from the partner's account and the trade is reattributed (or marked as unattributed). The partner receives a Smart Inbox notification.

6.2.9.5 API Key Management & Usage Analytics

Purpose : Self-service management of API keys, monitoring of API usage, and real-time rate limit visibility.

UI – API Keys:

Display current API key (masked, e.g., sgtx_live_••••••••), creation date, last used date.

Generate New Key – old key is invalidated after a 24-hour grace period (both keys work during grace). Partner must update their systems.

Revoke Now – immediately invalidates the key (use with caution).

Usage Analytics (v6.3 enhanced):

Chart (Chart.js) showing API requests per endpoint over time (last 7 days, 30 days).

Breakdown : intent/analyze, trade/initiate, suppliers/match, analytics.

Error rate (4xx, 5xx) over time.

Rate limit status – a gauge chart showing current utilisation vs. limit for each endpoint (e.g., "intent/analyze: 850/1000 requests this hour (85%)"). If a limit is exceeded, the gauge turns red and a tooltip shows the time until reset.

Rate limit increase request : A "Request Increase" button opens a form that submits a limit change request to the Platform Governance Authority. The request is logged and the partner can track its status.

Rate Limit Configuration (v6.3) : Default per-partner rate limits are defined in partner_rate_limits. The partner admin can view their current limits but cannot modify them directly. Limits are visible in a read-only table with columns: Endpoint, Limit, Window (seconds), Current Usage. When a rate limit is hit, the API returns HTTP 429 with a Retry-After header and a Groq-generated message: "You've reached your hourly limit for intent analysis. Your limit resets at 14:00 UTC. Please wait or contact your account manager to request an increase."

Security : Partners can whitelist IP addresses (CIDR ranges) – requests from non-whitelisted IPs are rejected with HTTP 403. The portal shows the current whitelist and allows editing.

6.2.9.6 Integration Guide & Test Sandbox

Purpose : Provide a self-service sandbox environment for partners to test their integration before going live.

Sandbox Endpoints (separate from production):

https://sandbox.sgtx.io/v1/partner/intent/analyze

https://sandbox.sgtx.io/v1/partner/trade/initiate

https://sandbox.sgtx.io/v1/partner/webhook/register (for test webhooks)

API key obtained from Sandbox tab (not production key).

UI – Sandbox Tab:

Test Intent Form : Partner enters raw text (e.g., “I want 10,000 kg of bananas to Germany”) and clicks “Analyze”. The system returns the same response structure as production, but uses synthetic data (market prices, risk scores) generated by Groq to simulate real market conditions. No real trades are created.

Predefined Scenarios : Dropdown to test edge cases: “Sanctions block”, “Low viability score”, “Conditional (missing fields)”. Selecting a scenario prefills raw text and shows expected outcome.

Simulate Trade Initiation : After receiving an intent_id, partner can click “Simulate Trade” to create a synthetic attributed trade (for testing webhooks). This creates a partner_sandbox_leadrecord and sends a test webhook to the partner’s registered sandbox webhook endpoint.

Test Webhook Endpoint : Partner enters a test webhook URL (e.g., https://myapp.com/sgtxtest). The system sends a test.ping event and shows delivery result.

Sandbox Dashboard : Shows simulated leads, attributed trades, and SGTX FEES amounts (reset daily). Partners can use this to demo the platform internally or train their integration team.

Data Isolation : Sandbox data is stored in a separate PostgreSQL schema (sandbox_* tables) and is completely isolated from production. No real trades are affected. All sandbox data is automatically purged every 24h.

6.2.9.7 Revenue Share Agreement Management

Purpose : View, download, and propose amendments to the revenue share agreement between the partner and SGTX.

UI – Agreement Tab:

Current Agreement:

Revenue split: e.g., “Partner: 0.5%, SGTX: 0.8%”

Effective date: 2026-01-01 to 2027-01-01 (if term-limited)

Download signed PDF (from documents table)

Amendment Request:

Partner proposes a new split (e.g., 0.6% partner, 0.7% SGTX) and provides justification (free text).

AI (Groq) generates a performance-based analysis : “Based on your 30% increase in lead volume, a 0.6% split would increase your revenue by $12,000 annually while maintaining SGTX’s margin. Recommendation: Accept.”

The request is sent to the Platform Governance Authority (multisig). The partner can track status (PENDING, APPROVED, REJECTED) in the same tab.

If approved, a new agreement is generated (Clause Forge + Groq), signed via ZITADEL by both parties, and stored immutably.

If rejected, the reason is displayed.

Expiry Notifications : Partner receives portal notifications and emails 90, 60, and 30 days before the agreement expires (if term-limited). These appear as Smart Inbox items.

6.2.9.8 Data Model Additions (to be added to Part 5)

sql

```
-- New tenant type
ALTER TYPE tenant_type ADD VALUE 'MARKETPLACE_PARTNER';

-- Marketplace partner profile (extends marketplace_partners table)
ALTER TABLE marketplace_partners ADD COLUMN api_key_encrypted TEXT;
ALTER TABLE marketplace_partners ADD COLUMN api_key_created_at TIMESTAMPTZ;
ALTER TABLE marketplace_partners ADD COLUMN api_key_last_used TIMESTAMPTZ;
ALTER TABLE marketplace_partners ADD COLUMN ip_whitelist TEXT[];
ALTER TABLE marketplace_partners ADD COLUMN sandbox_enabled BOOLEAN DEFAULT
true;

-- Webhook delivery logs
CREATE TABLE webhook_delivery_logs (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
partner_id UUID NOT NULL REFERENCES marketplace_partners(id),
endpoint_url TEXT NOT NULL,
event_type TEXT NOT NULL,
payload JSONB,
response_status INTEGER,
response_body TEXT,
duration_ms INTEGER,
success BOOLEAN,
created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Partner sandbox leads (reset daily)
CREATE TABLE partner_sandbox_leads (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
partner_id UUID NOT NULL REFERENCES marketplace_partners(id),
raw_text TEXT,
parsed_specs JSONB,
```

viability_score INTEGER,

created_at TIMESTAMPTZ DEFAULT NOW(),

expires_at TIMESTAMPTZ DEFAULT NOW() + INTERVAL '7 days'

);

-- Rate limits (already defined in Part 5 as partner_rate_limits)

6.2.9.9 Zero-Cost Technology Stack (Marketplace Partner Portal)

Feature	Technology	Cost
Dashboard & metrics	ClickHouse + PostgreSQL + Chart.js	$0
Lead management	PostgreSQL + Groq (for synthetic data)	$0
Webhook management & logs	NATS + reqwest	$0
Dispute management	Existing disputes table + A2 AI	$0
API key management	pwhash + encrypted storage	$0
Sandbox environment	Separate subdomain + synthetic data generator	$0
Agreement management	Clause Forge (HF Mixtral) + Groq + ZITADEL	$0
Rate limit monitoring	partner_rate_limits table + gauge chart (Chart.js)	$0

No billing details required. Sandbox uses the same infrastructure as production but with a flag to route to synthetic data.

6.2.9.10 Example Workflow – Partner Onboards, Tests, Goes Live

Onboarding : SGTX admin uses Admin Portal's "Marketplace Partner Onboarding" to invite "TradeBridge". TradeBridge receives email, sets up ZITADEL account, logs into Partner Portal. Smart Inbox shows onboarding task: "Complete your profile."

Sandbox testing : TradeBridge developer navigates to Sandbox tab, generates sandbox API key. Submits test intent: "10,000 kg bananas to Germany". Returns intent_id sbx001 with viability score 92. Simulates trade initiation and sees test webhook delivered to staging endpoint. Sandbox dashboard shows simulated leads and SGTX FEES.

Production setup : Developer goes to API Keys tab, generates production API key, copies it, updates integration. Whitelists server IP. Verifies key with production intent/analyze dummy request.

Live leads : TradeBridge sales team sees live leads in Leads tab. Monitors conversion via dashboard. One lead flagged CONDITIONAL (missing quantity). Sales manager clicks "Resolve", fills missing quantity, resubmits → ACCEPTED.

Attribution dispute : A trade auto-attributed but TradeBridge believes it belongs to another partner. Partner manager clicks “Dispute”, uploads evidence. Platform Governance Authority reviews AI recommendation (suggests misattribution) and reverses attribution. TradeBridge receives Smart Inbox notification.

Performance review & agreement amendment : After six months, dashboard shows conversion rate increased to 22%. AI recommends higher revenue split. Partner manager proposes new split via Agreement tab. Platform Governance Authority approves, new agreement signed, stored.

Rate limit management : Sales team sees 429 error. Check API Usage gauge: intent/analyze at 1000/1000. Clicks “Request Increase”. Admin Portal receives request; after multisig approval, limit raised. Gauge updates.

END OF PART 6.2.9 – MARKETPLACE PARTNER PORTAL (v6.3 WITH ALL MODIFICATIONS)

PART 6.3 : MOBILE APPLICATION (REACT NATIVE + EXPO) – v6.3 FULLY EXTENDED, WITH OFFLINE-FIRST ARCHITECTURE, THREE-TIER AI, VOICE COMMANDS, BARCODE SCANNING, AR INSPECTION, DUAL-MODE TOGGLE CONTEXT, MULTI-SHIPMENT SUPPORT, AND ALL MODIFICATIONS

6.3.0 Purpose & Design Philosophy

The SGTX mobile application is the field-operational extension of the platform, designed for drivers, warehouse staff, and QC inspectors who work in environments where connectivity may be intermittent or absent. It is an offline-first tool built with React Native and Expo, leveraging local databases (WatermelonDB), offline AI models (Vosk for speech-to-text, ONNX Runtime for defect detection), and a resilient sync engine that queues actions when offline and replays them when connectivity returns.

The mobile app is not a scaled-down version of the web portal; it is a task-focused interface that surfaces only the jobs assigned to the authenticated worker. It does not display commercial values, counterparty negotiation details, or any data outside the worker’s scope. It is governed by the same OPA policies and ZITADEL authentication as the web portals, and every milestone confirmation, barcode scan, and inspection report is cryptographically signed and passed through the Governor. The app respects the dual-mode toggle context (a user with dual-mode may act as buyer or seller, but field workers typically have a single role). Multi-shipment contracts are supported: each shipment appears as a separate job with its own milestones and barcode sets.

Core Principles:

Offline-first – All critical functions (scanning, voice commands, defect logging) work without internet. Data is queued locally and synced when a connection is available.

Role-adaptive – The home screen shows only the worker’s assigned tasks: a driver sees pickups and deliveries, a warehouse worker sees loading jobs, a QC inspector sees inspection assignments. Dual-mode toggle is hidden for field workers (only one role per device).

Hands-free operation – Voice commands (Vosk offline, HF Mixtral intent extraction) allow workers to confirm milestones, log defects, and navigate without touching the screen.

Privacy-preserving – Raw audio and images never leave the device unless explicitly uploaded as evidence. Biometric stress analysis is optional, off by default, and requires explicit consent.

Non-marketplace – The app never suggests counterparties or business opportunities; it is a pure execution tool.

Multi-shipment aware – For multi-shipment contracts, each shipment appears as a separate job line; the app knows which shipment's barcodes are being scanned and which milestone to update.

6.3.1 Architecture & Technology Stack (Zero-Cost, Offline-First)

Layer	Technology	Purpose
Framework	React Native 0.76 + Expo	Cross-platform mobile shell (iOS/Android)
State Management	Zustand	Lightweight stores per session
Local Database	WatermelonDB	Offline persistence of tasks, shipments, scan queues
Sync Engine	Custom (Rust bridge via WASM or native module)	Queues actions offline, replays via NATS when online
Authentication	ZITADEL WebAuthn (passkey)	Biometric (fingerprint/face) login, JWT issuance
Barcode Scanning	ZXingC++ (via expo-camera)	SSCC18, GS1-128, QR code recognition
Voice / Speech-to-Text	Vosk (offline models bundled)	Transcribes commands and milestone updates offline
On-Device NLP	HF Mixtral (quantised, ONNX) or DistilBERT	Intent extraction from voice transcripts
Computer Vision	HF ViT (Core ML / TensorFlow Lite)	Pallet defect detection, container number reading
Augmented Reality	AR.js + expo-camera	Overlays inspection instructions, defect highlights
Offline AI Assistant	ONNX Runtime + quantised Llama 3.2 1B	Answers basic queries about cached trade data when offline
Biometric Stress	pyAudioAnalysis / librosa port to WASM (optional)	Advisory stress detection on voice samples (off by default)
Push Notifications	Expo Notifications + NATS WebSocket	Realtime alerts when online
Offline Maps	OSRM (self-hosted) + OpenStreetMap tiles	Turn-by-turn navigation with pre-cached tiles
Multi-shipment Support	WatermelonDB tables linked to contract_shipment_id	Each shipment tracked separately

All processing runs on the device; no cloud APIs required for core functions.

6.3.2 Key Modules (Detailed, with v6.3 Enhancements)

6.3.2.1 Barcode Scanner (Offline, Multi-shipment Aware)

Powered by ZXingC++ compiled for mobile.

Supports GS1-128 (linear) and QR codes.

Continuous scanning mode : camera view active; successful decode triggers haptic confirmation (Expo Haptics) and automatically advances to next expected pallet in the job list.

Offline validation : Scanned SSCCs are checked against locally cached pallet_details (synced when job was assigned). If the SSCC is not found locally, the scan is queued and validated upon reconnection.

Multi-shipment : The app knows which shipment's pallets are being loaded (from job assignment). Scanning a pallet that belongs to a different shipment shows a warning and does not confirm the milestone.

Each scan creates a shipment_milestones record (with scanned_barcode_id) and is signed with the device's ZITADEL certificate.

6.3.2.2 Voice Input & Voice-Driven Workflows (Offline, Cross-Language)

Offline STT : Vosk models for English, Arabic, Vietnamese, German are bundled. User downloads their language on first launch. Transcription runs on-device; no audio leaves the device.

Intent Extraction : Transcribed text passed to local HF Mixtral (or DistilBERT) that extracts intent and parameters. Example: "Pallet OR005 loaded into container TCNU1234567" → {action: "milestone.confirm", pallet_id: "OR-005", container: "TCNU1234567"}.

Voice Commands Catalogue (offline):

"Load pallet [SSCC] into [container]"

"Mark pallet [SSCC] as damaged – reason [text]"

"Show my next pickup"

"Container [number] fully loaded"

"Start inspection for USTN [ustn]" (or for multi-shipment: "Start inspection for shipment [shipment number] of [contract ID]")

"Override defect on pallet [SSCC] – reason [text]"

Online Voice Queries (when connected) : The voice gateway (Rust service) handles complex queries with Groq: "What's the status of USTN ...?" generates spoken summary.

Hands-free confirmation : After voice command recognised, app speaks back confirmation (device TTS) and waits for "yes" or "no" before executing.

6.3.2.3 AR Inspection Mode (QC, Offline, with Override Accountability)

Available only to QC inspectors with an active inspection job (including multi-shipment jobs).

Uses AR.js and device camera to overlay loading plan onto live container view. Each pallet position marked with coloured bounding box (red = high-priority inspection, blue = random sample, green = already inspected).

On-Device Defect Detection : HF ViT (Core ML on iOS, TensorFlow Lite on Android) runs locally on each captured photo. Detects mould, bruising, carton damage, foreign material, drawing bounding boxes in real time.

Override Accountability (v6.3) : If inspector disagrees with AI detection, they tap “Override”. App immediately prompts for mandatory text reason (minimum 10 characters). Reason, original AI confidence score, and override flag stored locally and synced later. Inspector cannot submit report until all mandatory pallets inspected and overrides justified.

AR mode also assists reading container and seal numbers : camera capture of printed numbers, local OCR (PaddleOCR or ONNX) extracts them for verification against booking data.

6.3.2.4 Offline Queue & Sync Engine (Resilient)

All state-changing actions (milestone confirmations, barcode scans, inspection report submissions) are first written to local WatermelonDB with synced = false flag.

Background sync process listens for connectivity. When online, replays queued actions in chronological order, each sent to the appropriate service via Governor-proxied API.

Conflicts (e.g., same pallet scanned by another device) resolved by server; app surfaces discrepancy to user.

Offline queue status visible as badge on home screen : “3 actions pending sync”.

Multi-shipment sync : Each action is tagged with contract_shipment_id to ensure correct routing.

6.3.2.5 Mobile Smart Inbox & Shipments Vault (Simplified, Offline)

Home screen is a simplified Smart Inbox (see Part 6.1.1). Displays worker’s assigned tasks as flat list of cards, sorted by urgency (e.g., “Pickup window starts in 2h”, “Inspection overdue by 1h”).

Each card shows : USTN (or shipment identifier), task description, deadline, prominent action button (e.g., “Start Loading”, “Begin Inspection”).

Swipe left reveals “Snooze” and “Dismiss” (synced with web Smart Inbox when online).

Mobile Shipments Vault tab shows all shipments the worker is involved in, filtered to their role. Drilldown views show document checklist, milestone timeline, container details. All data cached for offline viewing. For multi-shipment, each shipment listed separately with its own status.

6.3.2.6 Offline Maps & Turn-by-Turn Navigation (Trucking)

OSRM routes pre-downloaded for the worker's region (driver selects region on first login). Turn-by-turn navigation works offline.

Voice prompts via device TTS.

Route deviation alerts (geofence) are queued and sent when online.

6.3.3 Role-Specific Workflows (Detailed Examples with v6.3)

6.3.3.1 Truck Driver Flow (Multi-shipment Aware)

Context : Driver Ahmed assigned to pick up two containers for two different shipments of a multi-shipment contract (Shipment 1: oranges to Alexandria, Shipment 2: lemons to Port Said). The app shows two separate jobs.

Login : Fingerprint (ZITADEL). App loads assigned jobs from local cache (synced earlier).

Home Screen : Two cards: "Pickup: Shipment 1 – Oranges, Alexandria. Window 08:00-12:00" and "Pickup: Shipment 2 – Lemons, Port Said. Window 13:00-17:00". Taps first job → "Navigate" (offline maps).

At Warehouse – Loading (Shipment 1): Taps "Start Loading". Camera scanner activates. Scans each pallet's SSCC. Each scan beeps, pallet ID appears with green check. Voice command: "Pallet OR006 loaded" (confirmation spoken back). After all pallets scanned, app auto-detects completion, prompts "Confirm container loaded?" Ahmed confirms. Action queued (offline) → SGTX FEESLock release (partial) queued.

Sync : After leaving warehouse, enters cellular coverage; queued milestones sync. Smart Inbox items for seller/buyer updated in real time.

Anomaly : During transit, geofence deviation alert (offline detection) – app records deviation, prompts driver for reason (recorded locally). When online, alert sent to dispatcher.

6.3.3.2 Warehouse Worker Flow (Label Printing, AR Loading Assist)

Prints SSCC labels via Bluetooth to Zebra printer (or generates PDF for office printer).

Uses AR mode to verify pallet placement according to loading plan (overlay shows designated spots). If pallet misplaced, red highlight appears.

Offline; all scans and placement confirmations queued.

6.3.3.3 QC Inspector Flow (with Override Accountability, Multi-shipment)

Context : Inspector Nguyen assigned to inspect Shipment 1 (oranges) at port. Job details include quality specs, random sample pallets (AQL), high-priority pallets (OR019, OR020). Multi-shipment: separate job for Shipment 2.

Login: Jobs appear. Selects inspection for Shipment 1.

Job load : Offline – quality specs, pallet list, inspection points cached.

On-site – Container & Seal: Scans container barcode (matches). Enters seal number, takes photo.

Pallet inspection (AR) : Points phone at OR019. AR overlay shows product, lot, high-priority flag. AI detects mould (92% confidence). Inspector confirms defect – logs.

Override (OR020) : AI detects dark spot as mould (88%). Inspector examines, overrides to bruising. Mandatory reason field appears; types "Dark spot is surface bruise, no mould under magnifier." Override stored locally with original AI confidence.

Sampling compliance : App tracks mandatory random pallets (5) and high-priority pallets (2). Progress bar. Cannot submit report until all completed.

Report draft (offline) : Groq model (or local template) drafts report. Inspector reviews, signs with passkey (offline signature stored, synced later). After sync, QC report submitted; loading milestone proceeds (if not FAIL). Override flagged in report badge.

6.3.4 Offline Capabilities & Data Caching

WatermelonDB stores : assigned jobs, shipment details, document checklists, pallet information, barcode reference data, multi-shipment contract structures.

Sync schedule : on login, every 5 minutes when online, and on demand when user taps "Sync Now".

Offline Smart Inbox : caches last 50 items with priority scores and action links.

Voice models (Vosk) and quantised NLP model bundled; no network needed.

Offline AI Assistant (Llama 3.2 1B via ONNX) : answers queries about cached data: "How many pallets left to load for Shipment 1?" – queries local DB, generates text response.

Offline maps tiles pre-cached for worker's region.

6.3.5 Security & Privacy

Authentication : ZITADEL passkeys (fingerprint/face). JWT tokens short-lived, refreshed silently.

Data at Rest : WatermelonDB encrypted using device native encryption (iOS Data Protection / Android Keystore). App requires biometric authentication to open after backgrounding.

Data in Transit : All API calls use TLS 1.3 with certificate pinning to SGTX API gateway.

Voice & Photo Privacy : Raw audio and photos processed on-device. Only extracted intents (text) and defect annotations (structured JSON) transmitted. Original audio and full-resolution images never uploaded unless user explicitly attaches as evidence.

Biometric Stress Flag : Off by default. Requires explicit consent (onboarding wizard or Company Admin → Privacy). Analysis runs on-device; only advisory flag (boolean + confidence score) sent to server, visible only to tenant admin. Never blocks actions.

6.3.6 Zero-Cost Technology Stack

Component	Technology	Cost
Mobile Framework	React Native + Expo	$0
Local Database	WatermelonDB	$0
Barcode Scanning	ZXingC++	$0
Speech-to-Text	Vosk (offline models)	$0
On-Device NLP	HF Mixtral / DistilBERT (quantised ONNX)	$0
Computer Vision	HF ViT (Core ML / TFLite)	$0
AR Overlay	AR.js	$0
Offline AI Assistant	ONNX Runtime + Llama 3.2 1B	$0
Maps & Navigation	OSRM (self-hosted) + OpenStreetMap tiles	$0
Biometric Auth	ZITADEL WebAuthn	$0
Encryption	Device native + TLS	$0

No billing details required. All voice, vision, and AI processing runs locally on the device.

6.3.7 Implementation Checklist (macOS / Linux Host for Build)

bash

```
cd /path/to/sgtx/mobile

# 1. Install dependencies
npm install

# 2. Prebuild for iOS and Android
npx expo prebuild

# 3. Download Vosk models (English, Arabic, Vietnamese, German) into assets/vosk/
# (See script in mobile/scripts/download_vosk.sh)

# 4. Convert HF models to ONNX for on-device inference (optional)
python3 scripts/convert_to_onnx.py --model mixtral --output assets/models/

# 5. Build for iOS (requires macOS with Xcode)
npx expo run:ios

# 6. Build for Android
```

npx expo run:android

7. Test offline sync: disable network, perform scans, re-enable, verify replay.

6.3.8 Example – Driver Offline, Multi-shipment

Driver Ahmed has two pending shipments (Shipment A and Shipment B) for a multi-shipment contract. He loads Shipment A pallets in a warehouse with no signal. All scans queued. After leaving, signal returns; queued milestones sync, each correctly attributed to the correct shipment USTN. The SGTX FEESLock for Shipment A partially releases. Shipment B remains unaffected.

END OF PART 6.3 – MOBILE APPLICATION (v6.3 WITH ALL MODIFICATIONS)

PART 6.4 : DESKTOP APPLICATION (TAURI v2) – v6.3 FULLY EXTENDED, WITH SYSTEM TRAY, NATIVE NOTIFICATIONS, OFFLINE CACHING, DIRECT PRINTING (ZPL), AUTO-UPDATE, DUAL-MODE TOGGLE, AND ALL MODIFICATIONS

6.4.0 Purpose & Design Philosophy

The SGTX Desktop Application wraps the Next.js 15 web portal inside a lightweight native shell powered by Tauri v2. It is designed for logistics coordinators, trade managers, finance users, and warehouse supervisors who work primarily from a desktop or laptop. The desktop app provides a dedicated, always-on experience that leverages the full capabilities of the host operating system: persistent system-tray notifications, local file system access for printing and document storage, automatic background updates, and tighter integration with OS-level features (global shortcuts, native menus, deeplink handling). It is built on the same zero-cost, open-source philosophy as the rest of the platform. No additional billing, no proprietary components.

The app respects the dual-mode toggle (trading companies can switch between Buyer and Seller views), supports multi-shipment workflows, integrates with Smart Inbox real-time updates, and provides direct ZPL printing for barcode labels.

6.4.1 Architecture & Technology Stack

Layer	Technology	Purpose
Native Shell	Tauri v2 (Rust backend)	Window management, system tray, file I/O, printing, auto-updater, IPC
Frontend	Next.js 15 (React, TypeScript) – same build artifact as web portal	All portal UI: Smart Inbox, Trade Command Center, Shipments Vault, etc.
Rendering Engine	OS-native WebView (WebKit on macOS, WebView2 on Windows, WebKitGTK on Linux)	Displays the Next.js application; no embedded Chromium

State Management	Zustand (identical to web)	Scoped stores per user session
Realtime Updates	NATS WebSocket client (compiled to WASM)	Live milestones, RFQ updates, Smart Inbox item creation
Authentication	ZITADEL OIDC + PKCE, WebAuthn (passkey via platform authenticator)	Login and token refresh
Secure Storage	OS keychain (macOS Keychain, Windows Credential Manager, Linux Secret Service) via Tauri plugin tauri-plugin-secure-store	Stores refresh tokens and API credentials
Auto-Updater	Tauri built-in updater with self-hosted JSON endpoint and Ed25519 signature verification	Downloads and applies updates from https://updates.sgtx.io
File System Access	Tauri fs plugin (Rust) with permission-scoped directories	Reading/writing PDFs, CSVs, and label sheets to user-selected folders
Printing	Tauri shell plugin (calls lp/lpr on Unix, raw socket to printer IP on Windows/macOS for ZPL)	Direct printing to label and document printers
System Tray	Tauri tray-icon plugin	Background presence with context menu and notification badges
Notifications	Tauri notification plugin (native OS notifications)	Push alerts for high-priority Smart Inbox items
Deep Links	Custom URI scheme sgtx://registered with the OS	Clicking a notification or external link opens the correct portal screen

All components are open-source and self-hosted.

6.4.2 Key Features (Expanded with v6.3)

6.4.2.1 System Tray & Native Notifications

When the user closes the main window, the application minimises to the system tray (menu bar on macOS, notification area on Windows, system tray on Linux). The tray icon provides:

Right-click context menu:

“Open Dashboard” – restores main window to Smart Inbox.

“Active Shipments” – submenu of the 5 most critical shipments (loaded from local cache, updated via NATS).

“Quick Actions” – voice command activation (opens VoiceCommandButton).

“Switch Mode” – if dual-mode tenant, toggle between Buyer and Seller without opening main window.

“Quit” – fully exits.

Badge count overlay : Displays number of high-priority (score ≥80) Smart Inbox items. Updates in real time via NATS WebSocket.

Native OS notifications : For high-priority items (demurrage warning, carrier risk alert, contract pending signature) the app sends a native notification. Clicking it restores main window and navigates directly to the relevant Trade Command Center (or financing/distressed view). For multi-shipment, includes shipment number.

Example : Logistics coordinator Maria sees a popup: "⚠ Container TCNU1234567 – demurrage risk in 6h. Estimated charges $300." She clicks, window restores to TCC for that USTN, showing demurrage prediction and "Arrange Pickup" button.

6.4.2.2 Local File System for Printing & Document Storage

Through Tauri's fs plugin, the desktop app enables:

Print barcode labels directly : From Seller Portal → Barcode Print tab, user clicks "Print". Frontend sends SSCC data to Tauri backend (via invoke), which constructs ZPL (Zebra Programming Language) and sends directly to configured printer IP/port (raw TCP, port 9100). For standard laser printers, a PDF is rendered clientside and sent to OS print spooler (lp on macOS/Linux, lpr on Windows). User can also save PDF to local folder.

Save signed documents : Finalised packing list, commercial invoice, or SGTX addendum can be saved to local disk via native save dialog.

Bulk export of shipment data : "Export My Data" uses desktop app to write large CSV/JSON exports directly to file system, with background progress indicator in tray.

6.4.2.3 Auto-Update with Signature Verification

The app checks a self-hosted update server (https://updates.sgtx.io) at startup and every 4 hours. The update manifest (latest.json) contains version, download URL (platform-specific binary), Ed25519 signature of binary hash, and release notes (Groq-generated). When new version available:

Downloads binary in background.

Verifies Ed25519 signature against platform's public key (hardcoded).

Prompts user : "A new version of SGTX Desktop (v6.3.1) is available. Update now or on next restart?"

On confirmation, applies update and relaunches.

Updates are mandatory after grace period (default 7 days). Ensures all clients stay current.

6.4.2.4 Offline Capabilities & Caching

Same service worker as web portal enables offline access. Additionally, the Tauri backend can cache larger datasets (e.g., full list of active shipments, Smart Inbox items) in local SQLite (via Tauri sql plugin or frontend IndexedDB). When offline, the app can:

Display Smart Inbox with latest synced items.

Open Trade Command Center for cached USTNs and view milestones, documents, activity feeds.

Queue actions (document uploads, signature attempts) for replay when online.

6.4.2.5 Deep Links & Custom URI Scheme

The app registers sgtx:// URI scheme. External applications (email, ERP) can open specific screens: e.g., sgtx://trade/SGTX-EG-...-V1 brings app to foreground and navigates to TCC for that USTN; sgtx://shipment/{shipment_id} for multi-shipment; sgtx://financing/{agreement_id} for financing. Links are generated in exported CSV reports for seamless integration.

6.4.2.6 Multi-Window Support (Optional)

Power users can open additional windows. Each window is an independent instance of the Next.js SPA, sharing the same authentication session and real-time NATS connection (managed by Rust backend). Right-click a USTN badge → “Open in New Window” keeps that trade’s Command Center open while working in main window.

6.4.3 Security & Sandboxing

Frontend JavaScript runs inside WebView sandbox; no direct file system, shell, or OS API access. Privileged operations must be invoked via Tauri commands, implemented in Rust, with permissions whitelisted in tauri.conf.json.

Content Security Policy (CSP) enforced by WebView, preventing XSS.

Auto-updater only accepts binaries signed with platform’s Ed25519 key.

Token storage uses OS keychain, not localStorage.

6.4.4 Worked Example – Logistics Coordinator (Desktop)

Morning : Maria boots laptop; SGTX app starts automatically (login item). Tray icon badge shows “3”. At 10:15, native notification: “⚠ Carrier risk alert: Fast Trucking Co. score dropped to 41. Run reoptimisation.” Clicks notification → main window opens to Logistics Builder tab for that trade, with “ReOptimise” button highlighted. She clicks ReOptimise, diff view shows alternative provider, switches. Closes window (app returns to tray). Later, she prints barcode labels: opens Seller Portal, selects container, clicks “Print”. App sends ZPL directly to warehouse Zebra printer. Labels print instantly. End of day, app auto-updates silently. Next morning, update applied on launch.

6.4.5 Zero-Cost Technology Stack

Component	Technology	Cost
Native Shell	Tauri v2 (Rust, MIT)	$0
Frontend	Next.js 15 (same build as web)	$0
WebView	OS-native (WebKit, WebView2, WebKitGTK)	$0 (included with OS)
Auto-Updater	Self-hosted HTTP server + Ed25519 signatures	$0
Printing (ZPL)	Raw TCP socket to printer IP (Rust std::net)	$0
Printing (PDF)	OS print spooler via lp/lpr or	$0

	Win32 API	
Keychain Storage	Tauri secure-store plugin	$0
System Tray & Notifications	Tauri tray-icon and notification plugins	$0
Build & Signing	pnpm tauri build + self-hosted signing	$0

No billing details required. Desktop app distributed via direct download from SGTX portal and auto-updates from same infrastructure.

6.4.6 Implementation Checklist (macOS/Linux – Build Host)

```bash
cd /path/to/sgtx/portals

# 1. Install Tauri CLI and dependencies
pnpm install
pnpm add -D @tauri-apps/cli@v2

# 2. Initialise Tauri (if not already)
pnpm tauri init

# 3. Configure capabilities (file system, notifications, printing) in
src-tauri/capabilities/default.json

# 4. Build Next.js frontend
pnpm run build

# 5. Build Tauri desktop app
pnpm tauri build # generates .dmg, .msi, .AppImage in src-tauri/target/release/bundle

# 6. Sign binaries with Ed25519 for auto-updater
cd src-tauri
cargo run --release -- sign --key /path/to/update_private.key

# 7. Deploy to update server
scp ./target/release/bundle/macos/SGTX.app.tar.gz updates.sgtx.io:/var/www/updates/
# Update latest.json manifest
```

END OF PART 6.4 – DESKTOP APPLICATION (v6.3 WITH ALL MODIFICATIONS)

PART 6.5 : CROSSCUTTING FEATURES (ALL PORTALS) – v6.3 FULLY EXTENDED, WITH DUAL-MODE TOGGLE, VOICE LANGUAGE MATRIX, EMERGENCY OPERATIONS MODE, OPERATING COMPLEXITY SCORE, AND ALL MODIFICATIONS

6.5.0 Purpose & Scope

The following features are available in every SGTX portal – web, desktop, and mobile – and are built on zero-cost, open-source technology. They transcend any single phase or tenant role, providing universal capabilities that enhance security, collaboration, intelligence, and user autonomy. All components are self-hosted, require no billing details, and respect the platform's non-marketplace constitutional principles.

This section includes features that were previously scattered across the blueprint, now consolidated and updated with v6.3 modifications: Self-Sovereign Identity Backup, One-Tap Smart Escrow Claim, Anonymous Market Intelligence Panel (opt-in, differential privacy), Voice Language Matrix & TTS Fallback, Biometric Stress Flag Consent, PlainLanguage Governor Decision Panel (already covered in 6.1.3, referenced), Tenant Data Export (covered in 6.1.4), Smart Inbox & Trade Command Center (covered in 6.1.1, 6.1.2), Non-Marketplace Footer, Role-Specific Value Cards, Timezone Handling & Holiday Adjustment, Emergency Operations Mode, and Operating Complexity Score.

6.5.1 Self-Sovereign Identity Backup (Shamir Secret Sharing)

Tenants can split their master account recovery key (or any critical secret) into multiple shares using Shamir Secret Sharing (via the sharks Rust crate, compiled to WASM for browser, native for mobile). Scheme: t-of-n (e.g., 3-of-5). Each share is encrypted with the public key of a trusted saved contact (another GTID) and transmitted via encrypted NATS message. No single party – not even SGTX – can reconstruct the secret without a quorum.

Example : Nile Foods tenant admin creates a 3-of-5 backup of the company's recovery phrase. Shares sent to three trusted trade partners and two internal directors. If admin loses access, they initiate recovery via voice: "Begin recovery for Nile Foods." System prompts required shareholders to approve via their portals; once quorum met, shares recombined client-side, secret restored. Process logged via Loom.

6.5.2 One-Tap Smart Escrow Claim (ColdChain Disputes)

When a shipment arrives and the Trade Command Center or Dispute Evidence Viewer detects a qualifying cold-chain deviation (based on IoT sensor data and contract clauses), a single button appears: "Initiate Smart Escrow Release." Tapping triggers:

Compiles evidence package (temperature logs, AI-assessed quality impact, contract terms).

Generates ZKproof (Plonky3) that deviation occurred, without exposing raw sensor data.

Submits conditional release instruction to DeFi smart escrow (if applicable) or conventional SGTX FEESLock release logic.

Logs action with Governor decision (A3 escalation if conditions ambiguous).

User confirms with ZITADEL passkey. No manual evidence gathering.

6.5.3 Anonymous Market Intelligence Panel (v6.3 – Expanded, Opt-In, Differential Privacy)

Provides tenants with aggregated, anonymised market statistics – average freight rates per corridor, typical financing APR ranges, on-time delivery percentages – computed using OpenDP differential privacy (ε = 0.1). No individual trade or tenant data can be reconstructed.

Opt-in only : Tenants must explicitly enable in Company Admin → Privacy & Data → Market Intelligence. The setting applies to entire tenant. Opt-in modal explains data usage and privacy guarantee. Preview with synthetic data available before opting in.

Data contribution : When opted in, the tenant's completed trades are included in nightly aggregation job. Fields extracted: commodity (HS chapter), origin, destination, freight rate, SGTX FEES split (no longer split, but fee rate), financing APR, on-time delivery. Identifiers stripped, differential privacy applied.

Panel content (live, after opt-in):

Average ocean freight rate (Vietnam → Egypt) : 8,200basedon34anonymisedshipments,±8,200basedon34anonymisedshipments,±300 confidence interval.

Typical SGTX fee (seller paid): 1.03% (range 0.8%–1.5%).

Average financing APR (preshipment, agricultural, 60-90 days) : 5.2% (±0.4%).

On-time delivery percentage (logistics providers in corridor) : 89% (median).

Corridor volume trend: +12% month-over-month.

Filters : By commodity (HS chapter), region, time period. Cell counts below minimum threshold (10) show "Insufficient data".

Opt-out : Immediate; panel hidden, data excluded from future aggregates. Historical aggregates unchanged (by design of differential privacy). Logged.

Zero-cost: OpenDP, ClickHouse, Groq for explanations.

6.5.4 Voice Language Matrix & TTS Fallback (v6.3 – Updated)

Offline Vosk model support:

Language	Vosk Model Bundled	Groq Multilingual	Offline Commands	Offline Summaries
English	□ (small-en)	□	□	□
Arabic	□ (vosk-ar)	□	□	□
Vietnamese	□ (custom, bundled)	□	□	□
German	□ (vosk-de)	□	□	□

Other (e.g., Thai, Hindi)	□ Not bundled	□ (response in user’s preferred language)	Partial (English command set)	□

If user’s language not in offline list, VoiceCommandButton shows “□ English only” badge. User can speak commands but processed in English; responses generated in user’s preferred language via Groq.

TTS Fallback:

For supported languages (English, Arabic, Vietnamese, German) : use device TTS (online or offline).

For other languages : Groq returns text only; app shows microphone icon with note: “Spoken responses not available in [language]. Text shown below.”

Tenant admins can disable voice output for unsupported languages via Company Admin → Voice Settings.

Cross-language UX : When command not recognised, app displays translated error message (Groq) and suggests typing or retrying in English.

Voice command catalogue (English examples, with translations for supported languages) includes: “Show my active shipments”, “Accept the lowest quote for USTN ...”, “File a dispute for mould on lemons”, “Load pallet OR005 into container TCNU1234567”, “Override defect on pallet OR020 – reason: surface bruise”.

6.5.5 Biometric Stress Flag Consent (v6.3 – Off by Default, Explicit Consent Required)

Feature off by default for every employee. Enabled only after explicit consent via onboarding wizard or Company Admin → Privacy Settings. Consent screen:

“Voice Stress Analysis Consent: SGTX can optionally analyse the tone of your voice during high-value actions (large payments, contract signatures) to flag elevated stress. Analysis runs entirely on your device; no raw audio stored or transmitted. Result is advisory note visible only to your tenant admin. This feature never blocks any action.”

I understand and consent.

I do not consent.

Consent stored in employees.voice_stress_consent. Withdrawal immediately disables feature and anonymises previously collected flags. Admins cannot force-enable.

6.5.6 PlainLanguage Governor Decision Panel (already specified in 6.1.3) – referenced.

6.5.7 Tenant Data Export & Reconciliation Statements (already specified in 6.1.4) – referenced.

6.5.8 Smart Inbox & Trade Command Center (already specified in 6.1.1, 6.1.2) – referenced.

6.5.9 Non-Marketplace Footer & Role-Specific Value Cards

Every portal page includes persistent footer:

“SGTX is a non-custodial execution platform, not a marketplace. We do not hold funds, take title to goods, or broker introductions. We provide the infrastructure for your trades.”

Role-specific value cards (dismissible) appear on dashboard:

Buyer : “Execute trades with cryptographic certainty – no counterparty risk, AI-optimised logistics, zero custody.”

Seller : “Lock EXW prices with AI market intelligence; source logistics via competitive RFQ; SGTX fee paid by seller, added to quote.”

Logistics : “Receive qualified RFQs from sellers you know. Submit quotes, track performance, and get paid upon verified milestones.”

QC : “Receive inspection bookings with dynamic quality specs. Use AI vision to detect defects and sign reports.”

Financier : “Access pre-vetted trade finance opportunities with AI credit intelligence. Co-financing, full disclosure, flat 0.25% financing fee.”

Government : “Monitor trade flows, verify documents, integrate with national systems – sovereign and secure.”

Admin : “Govern the platform with multisig security, simulate policy changes, and manage configuration.”

6.5.10 Timezone Handling & Holiday/Weekend Adjustment for Urgency Scoring (v6.3)

Storage : All timestamps in PostgreSQL as TIMESTAMPTZ (UTC). Deadlines stored in UTC.

Display : Convert to user’s local timezone (precedence: employee’s timezone → browser/device → fallback Europe/London). IANA database via chrono-tz (Rust) / moment-timezone (JS).

Holiday/Weekend Adjustment for Urgency Scoring : For deadlines that cannot be performed on non-business days (e.g., government document submission, bank transfer), the effective deadline is shifted to the next business day at 09:00 local time. Business day definition: Monday–Friday excluding public holidays in the primary jurisdiction (from public_holidays table, updated by RIA). For cross-jurisdiction, stricter rule applies.

Examples where adjustment applies : contract signature, government document submission, bank transfer. Examples where not applied: demurrage charges (accrue daily), automated milestone confirmation, distressed offer expiry.

Tenant override (Enterprise mode) : can disable adjustment per trade during negotiation (checkbox “Apply holiday adjustment to this deadline” default checked). Override logged as Governor decision.

6.5.11 Emergency Operations Mode (v6.3 – New)

External events (sanctions updates, political instability, port closures) may trigger a simplified, risk-aware mode where only critical actions are available. EOM is triggered automatically by RIA when certain risk thresholds crossed, or manually by tenant admin (or Platform Governance Authority).

Trigger conditions : new sanctions against country involved in active trade; jurisdiction tier downgrade to Restricted or Blocked; major port closure affecting active trades; manual activation.

Interface simplification : portal reduces to only: Smart Inbox (priority only), Shipments Vault (readonly for active, can update existing milestones), Export My Data, Company Admin (limited), Exit workflow. New Trade Request, Quote Review, Financing requests, new bids, etc., hidden or disabled. Persistent red banner: “⚠ EMERGENCY OPERATIONS MODE – Limited functionality. Reason: [reason]. Contact support if error.”

Allowed actions : view existing trades, confirm existing milestones, upload required documents, pay outstanding fees, export data, initiate tenant exit.

Exit from EOM : automatic when condition resolved + 24h stability; manual by tenant admin (if risks resolved); manual by Platform Governance Authority. Smart Inbox notification on exit.

Database : tenants.emergency_mode_active, reason, activated_at, activated_by. Table emergency_mode_events.

6.5.12 Operating Complexity Score (v6.3 – New)

Advisory metric (A1) warning tenant admins when configuration complexity exceeds recommended levels for tenant size and activity. Score range 0–100 (lower is better). Computed daily from components:

Component	Weight	Measure	Optimal Range
Number of custom roles	15%	Count beyond default templates	0-3
Number of approval chains	20%	Count of active policies	0-2
Data scopes per employee	15%	Average number of scope rules per employee	0-1

Active employee count	10%	Normalised	<10 employees (lower tolerance)
Monthly trade volume	10%	Actual trades per month	High volume justifies complexity
Custom permission overrides	20%	Number of non-default assignments	0-5
Conflicting policies	10%	Overlapping/ contradictory policies	0

Score colour : Green 0-30 (optimal), Amber 31-60 (moderate – review), Red 61-100 (high – simplify). Displayed in Company Admin → Overview and as Smart Inbox recommendation when score enters Amber or Red (priority 75). Recommendation includes Groq-generated summary and suggestions. Dismissible; reappears if score remains high for 7 days.

END OF PART 6.5 – CROSSCUTTING FEATURES (v6.3 WITH ALL MODIFICATIONS)

Part 7: Revenue Model & Fee Structure (v6.3 – Aidriven, Tenant-Transparent, Zerocost, with SGTX Fee Paid by Seller, No Split, Dynamic Commodity & Country-Pair Margin Model, Flat 0.25% Financing Fee)

7.0 Revenue Philosophy & Non-Custodial Foundation

SGTX earns revenue only on successfully executed trades and financing agreements – never on listings, advertisements, or access. The platform is structurally non-custodial; it never holds trade principal or loan funds. Revenue consists of two separate, independent fees:

SGTX Trade Fee – calculated dynamically as a percentage of the total trade value (EXW + logistics costs, as defined by the incoterm). This fee is paid solely by the seller and is added to the quoted price that the buyer sees. The fee is collected upfront before contract lock for single-shipment contracts, or per-shipment for multi-shipment contracts (each shipment's fee paid before that shipment is locked). The fee rate is bounded by the constitution (0.1% – 2.5%) and is derived from an AI-estimated profit margin that is commodity-specific and country-pair-specific.

SGTX Financing Fee – a flat 0.25% of the financed amount. This fee is paid by the borrower (the party requesting financing) and is deducted from the disbursed principal via PSP split. The financier never pays SGTX. The fee is collected at the moment of loan disbursement (for each financing request; for multi-shipment financing, per-shipment as each shipment's financing is activated). This fee is separate from the trade fee and applies regardless of whether the borrower is the seller or buyer.

All calculations are AI-governed, real-time, and fully explainable. No central authority manually sets rates. The Governor enforces all fee-related rules.

7.1 SGTX Trade Fee – Core Formula (Commodity- & Country-Pair-Specific)

The final trade fee rate is a function of the estimated trade profit margin, which itself is computed per commodity (HS code, perishability, etc.) and per seller/buyer country pair.

text
final_rate = clamp(0.1%, 2.5%, base_rate + adjustments)

7.1.1 Profit Margin Estimation (AI – XGBoost + LLM Refinement)

The Margin Estimator Agent (XGBoost, retrained weekly on completed SGTX trades) uses the following features:

Commodity HS6 code, commodity type (sugar, wheat, fresh fruit, etc.), perishability class.

Seller country (seller's jurisdiction) – currency trends, export taxes, port efficiency, regulatory stability.

Buyer country (buyer's jurisdiction) – import tariffs, sanitary rules, port congestion, payment reliability.

Logistics costs (from manual entry or selected quotes).

Market price indices (FAO, ECB, USDA, scraped daily).

Seasonality (week/year).

Volume (container count, total weight).

Incoterm and shipment type (single-shipment or multi-shipment).

A second layer (LLM – HF local or Ollama) ingests qualitative signals: news sentiment about the commodity, geopolitical events affecting the country pair, carrier performance data. The LLM outputs an adjustment factor (±0.1% to ±0.5%) that the Governor applies to the base margin estimate.

Example : Oranges from Vietnam to Egypt: XGBoost predicts margin 18.6%. LLM notes a recent strike at a major Vietnamese port and adds +0.3% risk adjustment → effective margin 18.9%. Base rate per mapping table (margin → fee) = 0.90% plus country boost, seasonality, etc., final rate 1.03%.

7.1.2 Adjustment Factors (All Computed Locally, No External APIs)

Factor	Source	Example
Country boost	country_commodity_factors table (LightGBM)	Vietnam-Egypt citrus: +0.05%
Seasonality	seasonal_adjustments table	Oranges in December: +0.10%
Geopolitical risk	Realtime news sentiment (HF NLP) + official risk ratings	Stable corridor: –0.02%
Volume discount	Cumulative trade volume between parties over 12 months	Repeat customer: –0.10%
Perishability surcharge	HS code match + perishable list	Reefer: +0.10%
Anomaly correction	Detected outlier trade patterns	None
Distressed country factor (if distressed sale)	jurisdictions.distressed_country_factor	UAE (liquidator fee): 0.85

All adjustments are bounded; final rate clamped.

7.1.3 Collection Timing

Single-shipment contract : Seller pays the trade fee upfront before contract lock. The fee is added to the quoted price (buyer sees the total, but fee is paid by seller).

Multi-shipment contract : Master contract signed with no upfront fee. For each shipment, after mutual confirmation of readiness, the seller pays the fee for that specific shipment (calculated on that shipment's trade value). Only then is the shipment locked and its USTN generated.

Governance : All fee calculations are logged in fee_calculations and SGTX FEES_calculations(renamed to fee_calculations). The Governor validates that the final quoted price equals EXW + logistics + SGTX fee. The fee rate must be within constitutional bounds.

7.2 Financing Fee (Flat 0.25%)

text
Financing_fee = 0.25% × Financed_amount (minimum $10)

This fee is separate from the trade fee. It applies whenever a party (seller or buyer) requests financing against a locked trade or shipment.

Collection mechanism (non-custodial PSP split):

Financier(s) disburse the full principal to a PSP that supports split payments.

SGTX's payment orchestrator instructs the PSP to split the transfer:

Principal – fee → borrower's account.

fee → SGTX's account.

The borrower's repayment obligation remains the full principal plus interest.

If the financing is co-financed by multiple financiers, each financier's disbursement is split independently, and SGTX collects the fee proportionally (total fee = 0.25% × total principal).

Timing : For single-shipment financing, fee deducted at disbursement. For multi-shipment financing, each shipment's financing has its own fee deducted when that shipment's loan is disbursed.

Governance : The split instruction must be cryptographically signed by the financier. The PSP webhook verification confirms both legs. The financing agreement includes a mandatory SGTX witness clause securing the fee.

7.3 Marketplace Partner Revenue Share

When a trade originates from a registered marketplace partner via the Marketplace API, SGTX shares a portion of the trade fee (not the financing fee) with that partner. The split is negotiated during partner onboarding and stored in marketplace_partners.SGTX FEES_split_percent(renamed to revenue_share_percent). The attribution is automatic (first-touch) and visible to both buyer and seller with a dispute window. For multi-shipment, attribution applies to the master contract and each shipment's fee is shared proportionally.

7.4 AIDriven Fee Forecasting & Rate Simulation (Admin Tools)

The Admin Portal includes a Revenue Intelligence Dashboard (multisig only) with:

Predictive revenue forecasting (LSTM) for trade fees and financing fees.

Rate simulation sandbox – test impact of changing the fee mapping table or adjustments before deployment.

Anomaly & fraud highlights – unusual fee patterns.

All tools use open-source libraries and local ClickHouse data.

7.5 Distressed Cargo Fee Adjustment

For distressed cargo sales, a new microcontract is created with a separate SGTX fee calculated on the distressed sale amount. The standard trade fee rate is multiplied by the distressed country factor (from jurisdictions.distressed_SGTX FEES_factor) to account for mandatory local costs (e.g., liquidator fees). Example: UAE factor 0.85 reduces the fee from 1.03% to 0.8755% on the distressed amount. This fee is paid by the seller of the distressed lot (the original cargo owner) at microcontract lock.

7.6 Example – Trade Fee Calculation (Single Shipment)

Trade : Oranges, Vietnam → Egypt. EXW 28,000,logistics28,000,logistics2,000, total trade value 30,000.Marginestimator→18.6SGTXtradefee=30,000.Marginestimator→18.6SGTXtradefee=30,000 × 1.03% = 309.00?Let′srecalc:Actually1.03309.00?Let′srecalc:Actually1.03309, not 360.That'sfine.Feepaidbyseller,addedtoquote.Buyerseestotal360.That'sfine.Feepaidbyseller,addedtoquote.Buyerseestotal30,309. Seller pays $309 before contract lock.

**Financing (seller requests 20,000):∗∗Financingfee=20,000):∗∗Financingfee=50. Deducted from disbursement: borrower receives 19,950,SGTXreceives19,950,SGTXreceives50. Repayment $20,000 + interest.

END OF PART 7 – v6.3 WITH ALL MODIFICATIONS

Part 8: Legal, Compliance & Risk Framework (v6.3 – Fully Upgraded, Dynamic, Global, Zerocost, with All Modifications: Removed Operating Modes, Three-Tier AI, SGTX Fee Paid by Seller, Multi-Shipment, Financing Co-Financing, Distressed Enhancements, Etc.)

8.0 Legal Foundation & Governance-Integrated Law

SGTX is a non-custodial execution platform, domiciled in New Jersey, USA. It never holds funds, only issues cryptographically signed instructions and locks fees via non-custodial SGTX FEESLock (FeeLock). Every legal instrument, compliance rule, and risk control is embedded directly into the platform's operational governance, enforced by the Governor, and continuously updated by AI agents that monitor regulatory changes across all 195+ recognised jurisdictions. The entire framework is self-executing code (WasmEdge + OPA) and self-updating documentation (Groq + HF local), ensuring the platform remains compliant with global laws at all times, without manual legal review, and at zero cost.

8.1 Multi-Layer Legal Architecture

Every trade and financing agreement executed on SGTX is underpinned by a pyramid of legal documents that stack from the platform constitution down to trade-specific addendums. All documents are dynamically templated, versioned, and signed on-chain or via ZITADEL passkeys.

Document Hierarchy (simplified, with v6.3 additions):

Platform Constitution (Governor WASM rules + OPA policies) – immutable except by multisig (3/5). Defines fee bounds (0.1%–2.5%), AI authority levels (A0-A4, A5 forbidden), jurisdiction supremacy, and dispute resolution.

Master Service Agreement (MSA) – primary contract between each tenant and SGTX. Declares SGTX a non-custodial facilitator. Tenants agree to all annexes.

Annexes & Addendums (dynamically appended per trade):

AI & Automation Addendum – discloses AI levels (A0-A4), explains that AI may block but never force, and obliges platform to explain AI decisions upon request.

Voice & Biometric Data Addendum – governs offline voice processing (Vosk), biometric stress flag (off by default, explicit consent required). No raw audio leaves device.

Digital Twin & IoT Addendum – clarifies that predictive ETA, shelf-life estimates, and digital twin simulations are non-binding estimates.

Packing Plan Addendum – incorporates locked packing plan as irrevocable annex.

Distressed Cargo Addendum – describes AI condition assessment, dynamic country-law SGTX FEES factor, and compliance obligations.

eBL / Smart Document Addendum – covers electronic bills of lading and AI-verified documents.

Payment & Stablecoin Addendum – discloses non-custodial nature of FeeLock, PSP splitting, and risks of stablecoin settlement.

Financing Addendum – included in financing agreements, covering the 0.25% financing fee, PSP split, and SGTX witness clause.

Logistics Service Addendum – signed by each selected logistics provider, binding them not to release containers without SGTX confirmation, and requiring USTN/GTIDs on all documents.

SGTX FEES Addendum – mandatory separate document when parties upload their own contract; includes witness clause and fee terms.

Terms of Service (ToS) – clickthrough agreement binding employees to RBAC rules, data privacy, and platform usage policies.

Trade-Specific Smart Clauses – generated by Clause Forge, may include self-executing conditions (Solidity pseudocode) for DeFi escrows.

All addenda are dynamically generated by Groq (fallback Ollama) and verified by HF Mixtral for legal consistency.

8.2 Compliance Framework – Global, Dynamic, Self-Updating

The compliance engine is a living regulatory brain that continuously scans every jurisdiction, extracts new rules, and integrates them into the platform's enforcement logic without human intervention.

8.2.1 The Regulatory Intelligence Agent (RIA)

Rust/Rig service consuming free RSS feeds, official gazette APIs, and central bank websites from 200+ countries.

Uses HF NLP (DistilBERT) to detect regulatory changes (new sanctions, changed VAT, new mandatory documents, licensing, reporting obligations).

HF Donut extracts full text from PDFs; Groq summarises changes in plain language and local language.

Proposes updates to jurisdictions table; non-critical changes auto-approved (A4), substantive changes escalate to multisig (A3).

Once approved, rules take effect immediately; all active trades in that jurisdiction reevaluated.

8.2.2 Sanctions & PEP Screening (Worldwide, Real-Time)

Data sources : OFAC SDN, UN Consolidated List, EU Sanctions List, UK Sanctions List, 300+ national lists (scraped daily). PEP lists from open government portals and Wikidata (AI-verified).

Screening : multilingual sentence-transformer embedding (HF all-MiniLM-L6-v2) compared against sanction/PEP names using pgvector cosine similarity (threshold 0.92). Potential matches flagged for human review.

GraphRAG (corporategraphservice) detects indirect sanctions exposure (e.g., UBO linked to sanctioned entity through shell companies). High-risk findings block trade initiation.

Continuous rescreening every 6 hours; new sanctions trigger immediate suspension.

8.2.3 AML/KYC Automation

KYB/KYC tiered (T1-T4) and jurisdiction-dynamic. Required tier stored in jurisdictions.kyc_tier_required.

Document extraction : HF Donut + PaddleOCR (local). Cross-checked against free government registries (scraped).

Biometric liveness: ZITADEL passkey + WebAuthn.

SAR (Suspicious Activity Report) auto-drafting : AI (A2) monitors trade and payment patterns (sudden volume spikes, circular trades, payments to high-risk jurisdictions). Groq drafts SAR in format required by relevant FIU (FinCEN, EMLCU, etc.). Compliance officer reviews; stored as signed document.

8.2.4 Distressed Cargo Compliance (Dynamic Country Law)

The jurisdictions table includes distressed_sale_allowed (YES/CONDITIONAL/NO), distressed_country_factor, and distressed_SGTX FEES_factor.

RIA updates these from local laws. When a distressed listing is created, the Governor applies the country factor to the SGTX fee. Conditional requirements (e.g., licensed liquidator) are enforced via WasmEdge rules.

8.2.5 Regulatory Reporting Automation

The Reporting Agent (A2) compiles data from settled trades (including multi-shipment per-shipment settlements, financing disbursements/repayments, distressed sales).

Groq populates jurisdiction-specific forms (e.g., Egypt CBE monthly remittance, EU ECB statistical report, FinCEN SAR).

Reports are stored in generated_documents and can be auto-emailed if SMTP configured.

8.3 Risk Framework – Living & Phase-Integrated

SGTX categorises risk into dimensions, each with automated detection, response, and mitigation. The risk matrix is a living JSON document stored in the database and continuously updated by the risk-monitor (Rig agent).

#	Risk	Detection	Response	Mitigation
1	Counterparty default	Trust scores (XGBoost), credit intelligence (200+ signals), Monte Carlo simulation	Freeze FeeLock, alert financier, trigger distressed cargo resolution	Voluntary security deposit, escrow, insurance
2	Sanctions violation	Realtime embedding comparison, GraphRAG proximity scan	Block trade initiation (G1U5), suspend tenant, autodraft SAR	Continuous screening, jurisdiction matrix updates
3	Document fraud	HF Donut cross-check against original templates, metadata analysis, hash comparison	Reject document, flag for human audit (G5U3), escalate to dispute	Mandatory original uploads, cryptographic chaining
4	Payment default / FX crisis	FX volatility monitoring (LSTM), PSP health, Monte Carlo liquidity predictions	Hold settlement (G9U2), suggest alternative rail, freeze FeeLock	Multi-rail redundancy, stablecoin options, escrow
5	Force majeure	Disruption prediction (LSTM), digital twin simulation, news sentiment	Activate contingency route (G2U5), notify parties, adjust deadlines	Insurance, dual-route planning
6	Cold-chain failure	IoT sensor anomaly detection (LSTM), shelf-life prediction	Alert parties, declare distressed if threshold breached, autocompile insurance evidence	Reefer monitoring, QC inspection at origin
7	AI hallucination / misclassification	Confidence threshold (G1U2, G3UA1) and human-in-the-loop for irreversible actions	Block action, escalate to human (A3), log with SHAP explanation	A0-A4 authority ladder, no A5, explainability
8	Voice data privacy	On-device processing only, raw audio never	(Preventive) – no breach possible	Legal addendum, zero-cloud

		leaves device		architecture
9	AR false positive in QC	Human override required for all AR-generated defect flags	Inspector must confirm or reject; system logs decision	Continuous model retraining on confirmed vs. rejected flags
10	DeFi protocol exploit	On-chain monitoring (The Graph), daily audit report parsing (HF Donut)	Freeze DeFi position, prevent new allocations, alert financiers (G4U5)	Only audited protocols, TVL limit, stablecoin de-peg protection

Each risk is monitored by specific agents and reported on the Admin Portal's risk dashboard, with trends and predictive alerts.

8.4 Data Privacy & Sovereign Residency (Worldwide GDPR-Class Protection)

SGTX treats data sovereignty as a constitutional right. The sovereign_nodes table maps each country to a specific K3s cluster where data must reside. All data operations are routed to the appropriate node. The RIA updates data residency rules in real time based on new regulations.

European Union (GDPR) : Data stored on EU-based sovereign nodes. Right-to-erasure requests executed by Governor after ensuring no open obligations. Data export (Part 6.1.4) provides portability.

Egypt (PDPL) : Data on Egyptian node, with logging for regulatory audits. Stress flag consent mandatory.

India (DPDP Act) : Consent management integrated into onboarding.

Voice & AR data : Processed entirely on device; raw audio and images never transmitted unless explicitly uploaded. Facial blurring runs locally.

Biometric Stress Flag : Off by default; explicit consent required; consent can be withdrawn.

Differential Privacy (OpenDP) : For any opt-in aggregated analytics (market intelligence panel), strict $\varepsilon=0.1$ ensures no individual trade or tenant can be reconstructed.

8.5 AI Governance & Explainability (Legal Binding)

All AI actions are logged with model, authority level, confidence, SHAP values, and a plain-language explanation (Groq). This satisfies the platform's legal obligation to provide explainability under the AI & Automation Addendum.

The A5 prohibition is hardcoded at WASM level; any attempt to execute an autonomous action is rejected and logged as a constitutional violation.

Users can request human review of any AI-generated flag through Dispute or Mediation channels. Platform must provide full AI decision log within 48 hours (contractual SLA).

The PlainLanguage Governor Decision Panel (Part 6.1.3) ensures tenants always receive a jargon-free explanation of any blocking decision.

The AI Policy Tuner (A2) continuously reviews AI decisions and proposes Rego policy adjustments if systemic bias or drift is detected.

8.6 Contractual Architecture & Smart Clauses

Governing Law : Jurisdiction Harmonizer agent selects appropriate governing law and arbitration forum.

Incorporation of Trade Instruments : UCP 600 (documentary credits), URDG 758 (demand guarantees), eUCP are natively supported by Clause Forge.

Digital Signatures : ZITADEL passkeys (WebAuthn) provide eIDAS, ESIGN, and similar compliance.

Smart Clauses (Optional) : Clause Forge generates Solidity pseudocode for self-executing conditions (e.g., automatic payment release upon delivery). Human-audited before deployment; execution limited to specific DeFi escrow or FeeLock instructions.

8.7 Dispute Resolution – Evidentiary Weight & AI Assistance

The platform does not replace courts or arbitration; it provides an irrefutable, tamper-evident evidentiary record that parties can use in any forum.

Loom-chained audit trails : Every milestone, document, and AI decision is hashed and chained.

Autocompiled evidence packages : In a dispute, evidence-compiler gathers sensor logs, AI summaries, image annotations, communications, and QC overrides (flagged) into a single cryptographically sealed package.

AI Settlement Proposal : The system suggests a fair settlement based on contract terms, similar past cases, and predictive outcome modelling (non-binding).

Arbitration Integration : Automated case filing (ICC, DIFC-LCIA) – forms filled by Groq, ready for legal counsel.

8.8 Emergency Multisig Override Procedure (v6.3)

For situations where two or more multisig members become unavailable simultaneously, an emergency override procedure allows the remaining members to act after a waiting period (24h standard, 4h for critical actions) and with at least 2 approvals from remaining members. All overrides are logged, posted to public status page, and subject to post-emergency review. Replacement of unavailable members follows a separate procedure.

8.9 Zero-Cost Enablers

All legal, compliance, and risk components use free, open-source technology:

Component	Technology	Cost
Regulatory scraping & NLU	HF Donut, finetuned DistilBERT, Groq (summaries)	$0
Sanctions screening	OFAC/UN/EU free lists, sentence-transformers (HF), GraphRAGRS	$0
Document verification	HF Donut, PaddleOCR	$0
SAR drafting / reporting	Groq, Tera templates	$0
Risk simulation	Python numpy/scipy, local LSTM	$0
Differential privacy	OpenDP	$0
Sovereign data residency	K3s + PostgreSQL, bare-metal	$0
Governance enforcement	WasmEdge, OPA, Loom	$0

No billing details required.

END OF PART 8 – v6.3 WITH ALL MODIFICATIONS

Part 9: Hybrid DeFi & Crypto Financing Framework (v6.3 – Fully Upgraded, Tenantcentric, with Dynamic Country Matrix, Protocol Risk Oracle, Stablecoin Monitor, Liquidation Early Warning, Tokenized Trade Assets, Zk Proofs of Reserves, and All Modifications)

9.0 Philosophy & Non-Custodial Extension into DeFi

SGTX never holds trade principal or loan funds. The same principle applies to its hybrid financing layer. The platform instructs and verifies – it never stores, lends, or borrows crypto assets for any party. All DeFi interactions are initiated and controlled by the tenants themselves, using smart contracts that are either self-deployed or deployed by SGTX's defi-service strictly as a facilitator. The platform collects only its financing fee (0.25%) via PSP split at disbursement, never through on-chain transfers.

The framework is globally dynamic : the eligibility of a trade for DeFi financing changes in real time based on the jurisdiction matrix (continuously updated by the Regulatory Intelligence Agent from Part 8), the risk scores of protocols, and the live status of stablecoins. No single country rule is hardcoded; every rule is scraped, interpreted, and enforced by AI agents. In v6.3, the financier experience includes mandatory DeFi PlainLanguage Risk Summary modal, Negotiation Bot Transparency Panel for financing counteroffers, and full integration with Smart Inbox and Decision Panel. All modifications from earlier (co-financing, financier preferences, automatic RFQ broadcast, flat 0.25% fee deducted at disbursement) are fully integrated here.

9.1 DeFi Eligibility & Dynamic Country Matrix

Every financing request (Phase 4) is first evaluated for DeFi eligibility. The decision is governed by a dynamic, AI-maintained country matrix that classifies every jurisdiction into three tiers, updated automatically as laws change.

9.1.1 Country Classification (Self-Updating)

The jurisdictions table stores defi_allowed (BOOLEAN), defi_restrictions (JSONB), and stablecoins_allowed (TEXT[]). These fields are populated and updated by the RIA, which continuously scrapes official gazettes, central bank announcements, and financial regulator statements.

Example classifications (dynamic, as of current AI knowledge – simplified):

Full DeFi allowed : Singapore, Switzerland, UAE (ADGM/DIFC), Hong Kong.

Restricted (MiCA compliance, licensed operators) : European Union, United Kingdom, Japan, Australia.

Prohibited: Egypt, India (except CBDC), China, Russia.

Dynamic updates : when a country changes its stance (e.g., Egypt allows stablecoin settlement in a special economic zone), the RIA picks up the news, HF NLP extracts new rules, Groq summarises, and the Governor (A4 for non-critical) or multisig (A3 for major) approves the update. Within minutes, all pending financing requests are reevaluated.

9.1.2 DeFi Eligibility Gate (Phase 4 Gate G4U5a)
Before any DeFi option is presented to a tenant, the Governor verifies:

Both the financier's and the borrower's jurisdictions allow DeFi for the transaction type.

The stablecoin to be used is legal in both jurisdictions (from stablecoins_allowed list).

All required licences or registrations (if known) are flagged to the tenant (advisory).

If a jurisdiction is classified as Prohibited, DeFi options are hidden; the financing request proceeds through conventional rails only. The Decision Panel explains if a user attempts to select DeFi.

9.2 Protocol Risk Management (Dynamic, Real-Time)

SGTX does not endorse any protocol; it assesses and monitors them using a DeFi Protocol Risk Oracle (Rust agent, A2) that produces a continuously updated risk score (0-100). Scores are stored in defi_protocols.risk_score.

9.2.1 Components of Risk Score (All Zero-Cost)

Audit findings : HF Donut parses audit reports (PDF) from GitHub/IPFS; critical findings → score penalty.

TVL and liquidity: From The Graph (free tier), normalised.

Exploit history: Scraped news + on-chain events.

Governance activity : Recent proposals; high-risk changes (e.g., parameter tweaks) reduce score temporarily.

Oracle health : Chainlink feed deviation, fallback mechanisms.

9.2.2 Scoring Thresholds

≥85 (Low risk) – all DeFi operations allowed.

60–84 (Acceptable) – warning shown to user; must acknowledge.

40–59 (Elevated risk) – new positions blocked (A2 Constraining). Existing positions flagged for review.

<40 (Critical risk) – protocol suspended; existing debt must be withdrawn or refinanced within grace period (default 7 days).

9.2.3 Automated Smart Contract Auditor (A2 – Static Analysis)
Before deploying any financing-related smart contract (tokenization, escrow), defi-

service runs opensource static analysis (Slither for Solidity, Aderyn for Rust/ink!). Any critical finding blocks deployment. A manual review by the Platform Governance Authority can override.

9.3 DeFi PlainLanguage Risk Summary (v6.3 – Mandatory Modal)

Appears before any user (financier or borrower) submits a bid, accepts a bid, or initiates a DeFi-settled financing agreement. Contains exactly five bullet points generated by Groq (fallback Ollama) in the user's preferred language:

What stablecoins are (if used) : "You will be lending/borrowing in USDC, a digital dollar that aims to keep a 1:1 value with the US dollar."

What 'health factor' means : "Your loan's health factor measures its safety. If it drops below 1, collateral could be sold to repay the loan, and you might lose part of your capital."

What happens if collateral (goods or crypto) drops in value : "If the value of the goods securing the loan falls sharply, the borrower may need to add more collateral, or the loan could be partially liquidated."

SGTX does NOT guarantee DeFi protocol safety : "SGTX checks that the protocol has passed audits, but we cannot guarantee it will never be hacked. You are responsible for the risks of the specific protocol you choose."

Past performance is not indicative of future results : "A protocol with a clean history can still encounter problems. No returns are guaranteed."

The user must click "I understand" to proceed. Acknowledgement is stored in employees.defi_risk_acknowledged_at, acknowledged_protocols, acknowledged_stablecoins. The modal reappears if 90 days have passed, or if the user selects a new protocol or stablecoin combination.

9.4 AI-Orchestrated Stablecoin Liquidity Aggregator

When a financing request is accepted and the parties choose stablecoin settlement, the platform can split the principal across multiple protocols to minimise concentration risk, if the financier opts in.

A linear programming solver (ORTools, free) allocates the loan amount across the top N protocols (default N=2) based on risk scores, APY, and TVL.

Objective : maximise total yield while keeping weighted average risk score ≥85.

Groq explains the allocation : "Allocating 60% to Aave (APY 3.2%, risk 94) and 40% to Compound (APY 3.8%, risk 88). Blended APY 3.5%."

The financier can override the allocation.

9.5 Stablecoin Management & Global Compliance

Only stablecoins that are globally compliant can be used. The list is dynamic, maintained by the Stablecoin Monitor Agent (A2). Initially: USDC, USDT, DAI (where permitted). Additional stablecoins are added if they meet criteria: fully backed, regular attestations, not sanctioned, market cap ≥$5B.

Depeg Detection & Response:

Fetch prices from free oracles (Chainlink free feeds, 0x API) every minute.

Deviation >0.5% → warning to all active position holders (A1).

Deviation >2% → Governor freezes new stablecoin positions (A2), escalates to Platform Governance Authority (A3). Existing loans can be repaid but no new loans opened until peg restored.

Groq drafts notification : “USDT is currently at $0.98, a 2.1% deviation. All new USDT loans paused until peg is restored.”

9.6 Cross-Chain Interoperability with AI Bridge Risk Scoring

If a trade requires assets to move across chains (e.g., Polygon → Ethereum for yield), the Bridge Risk Agent (A2) monitors bridge contracts using The Graph (free) for TVL, withdrawal delays, and past exploits. Assigns bridge risk score (0-100). Score <70 triggers warning; <50 blocks the cross-chain transfer. Groq recommends alternatives.

9.7 Tokenized Trade Assets & Secondary Market

When a financing agreement is tokenized (ERC3525 for invoices), the platform provides an AI valuation engine for secondary market pricing.

The engine continuously updates fair value based on trade progress (milestones), commodity price changes (FAO), discount rate models.

Groq produces a valuation explanation : “Invoice token for USTN...V1 is 60% complete, expected repayment in 30 days. Based on current orange prices and buyer credit score, fair value is 0.985 USDC per 1.0 face value.”

The price is stored in secondary_market_prices; platform never executes trades, only provides information.

9.8 DeFi-Native Fee Handling via Smart Escrow

For DeFi-settled trades, the FeeLock (SGTX FEESLock) can be represented as a smart contract escrow on Polygon. The escrow holds the SGTX fee amount in stablecoin and releases it to SGTX’s wallet only when all release conditions are met (verified by an SGTX oracle that signs milestone confirmations). The oracle is a Rust service that watches the SGTX FEES_locks table and posts signed attestations. Gas costs minimal. If a dispute occurs, the escrow can be frozen.

9.9 Predictive Liquidation Protection (A2)

The Liquidation Watch Agent (LSTM) monitors health factor of each DeFi position in real time.

If predicted health factor falls below 1.2 in the next 24 hours, sends early warning via NATS to both borrower and financier (Smart Inbox priority 85).

Groq advises : *"Your ETH collateral has high predicted volatility (+15%). Your health factor may drop below 1.0 in 18 hours. Consider repaying $5,000 or adding 0.5 ETH to avoid liquidation."*

Advisory only; no automatic closure.

9.10 Zero-Knowledge Proof of Stablecoin Reserves (A2/A3)

Financiers can optionally generate a ZKproof (using Plonky3) that they hold sufficient stablecoin reserves to back their financing offer, without revealing exact balance or wallet address. The proof is generated client-side (browser or mobile) and verified by the Governor before the bid is accepted. Increases trust and privacy, especially for large trades.

9.11 Automated FATF Travel Rule Compliance

For any stablecoin transfer that settles a trade or finances a loan, the platform automatically embeds the GTID of both sender and receiver as a JSON memo within the transaction. Format: {"sender":"SGTX-EG-TRD-000123","receiver":"SGTX-VN-TRD-000456","ustn":"SGTX-EG-...-V1"}. Before broadcasting, the AI (WasmEdge) verifies the memo is present and matches the settlement instruction.

9.12 Governance Gates (Phase 4/9, DeFi-Specific – Updated)

All original gates (G4U5a–G4U5k) are preserved and updated for v6.3. They include jurisdiction checks, stablecoin permissions, protocol risk score, smart contract audit, bridge risk, ZKproof verification, FATF memo, stablecoin de-peg, liquidation early warning, and mandatory DeFi risk summary acknowledgement.

9.13 Zero-Cost Tech Stack (DeFi)

Component	Technology	Cost
DeFi protocol monitoring	The Graph (free tier), HF NLP, RSS scraping	$0
Smart contract auditing	Slither / Aderyn (OSS)	$0
Stablecoin price feeds	Chainlink free feeds, 0x API (free)	$0
Liquidity aggregator	ORTools (Apache2.0)	$0

ZKproofs	Plonky3 (Apache2.0/MIT)	$0
Tokenization	Self-deployed contracts on Polygon (gas ≈ $0.01–0.10)	≈$0
Oracle for smart escrow	Rust service (self-built)	$0
Voice monitoring	Vosk (offline) + Groq (free)	$0

No billing details required. Gas costs are borne by the tenants, not the platform.

9.14 Example End-to-End DeFi Financing Flow (v6.3 Enhanced)

Seller Mekong Fresh requests $100,000 preshipment finance for oranges (Phase 4). Credit score 89, default prob 4.2%.

Financier "DeFi Pool Asia" (Singapore) receives RFQ (automatic broadcast). Opens details: sees full trade data, seller/buyer names, trust scores, documents. Match score 94.

Financier clicks "Bid" with DeFi option. DeFi PlainLanguage Risk Summary modal appears – five bullet points. Clicks "I understand".

DeFi Tab shows comparison : Aave V3 (risk 94, APY 3.2%) vs Compound (risk 88, APY 3.8%). Selects Aave V3.

Submits encrypted bid : $100,000 at 4.5% APR via Aave V3. Bidding window closes; seller accepts.

Financing agreement assembled with SGTX witness clause. Financing fee = 250(0.25250(0.25100,000). All parties sign.

Disbursement : DeFi Pool sends 100,000USDCtoPSP.PSPsplits:100,000USDCtoPSP.PSPsplits:99,750 to Mekong Fresh, 250toSGTX.MekongFreshreceivesnet250toSGTX.MekongFreshreceivesnet99,750.

Smart contract deployed on Polygon (ERC3525 token). Loan active.

During transit, health factor drops; Liquidation Watch Agent sends early warning. Mekong Fresh adds more collateral. Health factor recovers.

Repayment : Mekong Fresh repays $100,000 + interest to DeFi Pool. SGTX monitors on-chain event. Financing SGTX FEESLock released. Financier's historical record updated.

END OF PART 9 – v6.3 WITH ALL MODIFICATIONS

Part 10: Global Payment Orchestrator & Settlement Infrastructure (v6.3 – Fully Upgraded, with Non-Custodial PSP Splits, Trade Principal Settlement, Financing Fee Deduction, Multi-Rail Redundancy, AI Reconciliation, Smart Inbox Integration, Voice Activated Approvals, and All Modifications)

10.0 Non-Custodial Foundation & Scope

SGTX never touches trade principal or loan funds. The payment orchestrator issues instructions, monitors execution, and verifies completion – it never holds funds. All money flows directly between banks, PSPs, or blockchains of the trading parties and financiers. The system's role is to:

Select the optimal payment rail for each payment (trade principal from buyer to seller, SGTX trade fee from seller, SGTX financing fee from borrower via split).

Provide multi-rail redundancy and automatic fallback.

Reconcile all settlements in real time using AI-extracted bank statements, PSP webhooks, and blockchain events.

Ensure global compliance by dynamically adapting to jurisdiction-specific rules (updated by RIA).

Every component is self-hosted, open-source, and requires zero billing. The orchestrator is integrated with the Smart Inbox (settlement approvals, payment confirmations, reconciliation discrepancies appear as actionable items), the PlainLanguage Governor Decision Panel (if any payment is blocked), and the Tenant Data Export (all settlement events feed into signed statements). No SGTX FEES release occurs here – trade fees are already collected upfront in Phase 3, financing fees at disbursement in Phase 4.

10.1 Architecture Overview

Two primary flows:

Trade principal settlement (buyer → seller) – triggered after delivery confirmation (Phase 6). The full quoted amount (including the SGTX fee component) is transferred from buyer to seller. The seller has already paid the SGTX fee separately; no deduction from this transfer.

SGTX fee collection (trade fee and financing fee) – handled via PSP split instructions (non-custodial). For trade fee: seller pays gross amount; PSP splits net to SGTX. For financing fee: borrower pays gross amount via split from disbursement.

The orchestrator uses the same PSP Router, FX Arbitrage Hunter, Reconciliation Engine, and Voice Gateway as originally designed, now enhanced with multi-shipment support and financing awareness.

10.2 AIPowered Dynamic PSP Router (Unchanged in core, enhanced)

The PSP Router (A2, LightGBM + Groq) selects the best payment service provider for each payment (trade principal, trade fee, financing fee) based on:

Payer country, payee country, amount, currency.

Realtime PSP health scores (updated every 30 seconds).

Cost (fees + FX margin).

Historical success rates.

Returns rank with Groq explanation displayed to the payer. Fallback PSPs pre-computed; automatic failover if primary fails. Multi-currency conversions use FX Arbitrage Hunter (LSTM + 0x API/ECB). For multi-shipment, each shipment's settlement and fee payments are independent.

10.3 Trade Principal Settlement (Buyer → Seller)

Trigger : Shipment milestone DELIVERED confirmed (or contractual payment milestone). For multi-shipment, each shipment settles independently.

Workflow:

Settlement service generates instruction for the total quoted price (buyer's obligation). Instruction includes USTN (or shipment USTN), amount, buyer GTID, seller GTID, remittance info.

PSP Router recommends rail; buyer approves via Smart Inbox (priority 85) or voice ("Approve settlement for USTN…").

Payment instruction sent to PSP. SGTX monitors via webhook.

Upon confirmation, settlement status becomes SETTLED. Buyer and seller receive Smart Inbox notifications.

Reconciliation engine matches PSP webhook to open instruction; updates tenant's financial ledger. Monthly statements reflect the settlement.

Governance (G6U1–G6U7): Instruction signed, PSP health ≥80, approval required (with internal approval chains possible), webhook signature verified, reconciliation confidence ≥95% for auto-match.

10.4 SGTX Trade Fee Collection (Seller Pays, Added to Quote)

The trade fee is collected before contract lock (single-shipment) or before each shipment lock(multi-shipment). The flow is embedded in Phase 3 (not a separate user step). The orchestrator handles:

Gross-up calculation (PSP fee + safety buffer) using AI-validated fee schedules (HF Donut).

Redirect to PSP payment page.

Webhook verification → FeeLock (SGTX FEESLock) marked ACTIVE.

Reconciliation with SGTX bank account (OpenBanking or manual upload).

10.5 Financing Fee Collection (Borrower Pays via PSP Split)

As defined in Phase 4, the financing fee (0.25% of financed amount) is deducted from the disbursed principal using a PSP split instruction. The orchestrator:

Receives disbursement instruction from finance service after financing agreement signed.

For each financier, creates a split instruction : principal – fee to borrower, fee to SGTX.

Verifies PSP webhooks for both legs.

Marks financing SGTX FEESLock ACTIVE. The borrower’s net received amount is recorded.

If the split fails, the entire disbursement is rolled back; the financier is notified and can retry.

10.6 Multi-Rail Redundancy & Automatic Fallback

The psp-health-monitor (Rust) pings each PSP’s health endpoint every 30 seconds. LSTM model predicts failure. If health score <80 → degraded; <50 → disabled. When a payment attempt on the primary rail fails (webhook timeout, PSP rejection), the router instantly selects the next best PSP from the real-time ranking and retries automatically (up to 3 times, with configurable delay). The payer receives a Smart Inbox notification: “Stripe payment failed. Retrying via Payoneer – estimated fee increase $1.20. Cancel within 30 seconds if desired.” The fallback is logged as a Governor decision.

10.7 OpenBanking Self-Reconciling Ledger

For PSPs / banks that support PSD2 / OpenBanking APIs, the settlement-service pulls SGTX’s bank statement periodically. HF Donut extracts each transaction, matches against open fee_payment_requests and settlement_instructions using amount, reference, date. If confidence ≥95%, auto-reconciled. Unmatched items trigger Smart Inbox alert for manual review. Reconciliation runs daily and can be triggered on demand from Admin Portal.

For tenants who consent, the same OpenBanking can reconcile their own payments (not required for core operation).

10.8 Zero-Knowledge Proof of Payment (v6.3 optional)

For privacy-sensitive trades, a payer can generate a ZKproof (Plonky3) that a specific settlement instruction was fulfilled, without revealing amount, counterparty, or rail. The proof is constructed using the payment receipt hash and the settlement instruction hash. Verified by counterparty's node, purely through SGTX network, no external chain cost. The platform stores the proof; underlying data remains encrypted. When used, the settlement confirmation in TCC shows a badge "□ Privacy-Preserved Settlement" with proof hash.

10.9 Voice-Activated Settlement & Fee Payment Operations (A1)

Authorised users can approve or query payments by voice. Examples:

"Approve settlement for USTN SGTX-...-V1." – Governor signs instruction after ZITADEL biometric.

"What is the status of payment for USTN ...?" – Groq replies with status, ETA, PSP.

"Retry the fee payment for contract #C001." – Triggers fallback.

Works offline for basic queries when cached status available.

10.10 Adaptive SGTX FEES/Fee Optimization (already covered in 7.4 – Admin tools)

10.11 Global Regulatory Compliance for Payment Routing

Every payment is checked against the dynamic jurisdiction matrix (updated by RIA). Sanctions check: payer, payee, beneficiary banks, intermediate corridors screened in real time. If a new embargo or capital control is detected, affected rails disabled instantly. Example: new US sanction on a specific Egyptian bank → payment orchestrator blocks any pending settlement involving that bank and suggests alternative route. The payer sees Decision Panel explanation.

10.12 Governance Gates (Phase 9 & 6 – Consolidated)

Gate ID	Check	Failure Action
G9U1	PSP selected has country coverage + health ≥80	Fallback to secondary PSP
G9U2	FX rate within 1% of contract locked rate	Hold payment, alert treasury
G9U3	Fee amount matches contract (±0.1%)	Block settlement, escalate

G9U4	PSP webhook signature verified, USTN matched	Reject webhook, log anomaly
G9U5	Jurisdiction allows split settlement method	Block method, suggest alternative
G9U6	Gross-up calculation validated by AI	Recalculate
G10U1	Multi-rail fallback executed within 30s	Escalate to manual
G10U2	OpenBanking reconciliation confidence ≥95%	Queue for manual review
G10U3	ZKproof verified (if used)	Reject settlement
G10U4	All blocked settlements display Decision Panel	User sees explanation and remediation

10.13 Zero-Cost Tech Stack (Payment Orchestrator)

Component	Technology	Cost
PSP routing & adaptive learning	LightGBM / Bandit, Groq explanations	$0
Rail discovery	HF NLP, Donut, Groq	$0
FX prediction & arbitrage	LSTM (PyTorch), 0x API, ECB rates	$0
Voice commands	Vosk, HF Mixtral, Groq	$0
OpenBanking reconciliation	HF Donut, free PSD2 sandbox APIs	$0
ZKproofs	Plonky3	$0
Messaging	NATS JetStream	$0
Governance	WasmEdge, OPA, Loom	$0

No billing details required.

10.14 Example Settlement Flow (Multi-Shipment, Including Financing Fee Split)

Context : Mekong Fresh (seller) and Nile Foods (buyer) have a multi-shipment contract. Shipment 1 delivered. Buyer approves settlement of 30,240(includesSGTXfeecomponent).Selleralreadypaidtradefee30,240(includesSGTXfee component).Selleralreadypaidtradefee360 upfront. No further fee.

Delivery milestone confirmed. Settlement instruction generated. Buyer receives Smart Inbox item (priority 85). Approves via voice.

PSP Router selects SWIFT via Bank of Alexandria. Buyer's bank transfers $30,240. Webhook verified. Status SETTLED. Seller receives Smart Inbox confirmation.

Meanwhile, seller had requested financing for Shipment 1 earlier. Financing agreement: 20,000at5.220,000at5.250 deducted at disbursement via PSP split: 19,950toseller,19,950toseller,50 to SGTX.

Seller repays $20,000 + interest to financier directly. SGTX monitors via webhook, updates financier’s historical record.

Monthly statement for Mekong Fresh shows : trade settlement received 30,240,financingdisbursementnet30,240,financingdisbursementnet19,950, financing fee 50,tradefeealreadypaid50,tradefeealreadypaid360 (separate entry). All reconciled.

END OF PART 10 – v6.3 WITH ALL MODIFICATIONS

Part 11: AI Intelligence Layer – Deep Dive (v6.3 – Fully Upgraded, Global, Zerocost, Tenant-Transparent, with Three-Tier AI, Commodity-Specific Margin Estimation, Natural Language Explanations, Smart Inbox Text Generation, and All Modifications)

11.0 AI Philosophy & Governance

SGTX's AI is not an autonomous actor – it is a constellation of specialised agents that observe, advise, constrain, and escalate. No AI can execute a trade, move funds, or lock a contract without explicit human authorisation. This is enforced by the AI Authority Ladder (A0–A4), with A5 permanently forbidden at the WASM constitutional level.

All AI components are self-hosted, zero-cost, and globally dynamic. They use a three-tier strategy:

Groq (free tier, no credit card) – fast advisory (A1), Smart Inbox titles/descriptions, tenant messages, live market charts, daily summaries. Limits: 30 req/min, 14,400 req/day.

Hugging Face (self-hosted, local inference) – privacy-critical tasks (A2/A3): document extraction (Donut), contract clause analysis, sanctions screening, dual-use classification, commodity-specific margin estimation. Unlimited (models run on your own server).

Ollama / LocalAI / GPT4All (self-hosted, local inference) – universal fallback when Groq rate limit exceeded or HF models unavailable. Runs on server's CPU/GPU. Unlimited, no external dependencies.

Every decision is logged immutably via Loom and linked to a Governor verdict. Explainability is provided via SHAP values and plain-language summaries (Groq/Ollama). The intelligence layer adapts to the entire planet in real time – new regulations, new shipping routes, new fraud patterns – without requiring human retraining or manual updates.

11.1 AI Authority Ladder & Three-Tier API Routing

The ladder is the platform's non-negotiable constitution for artificial intelligence.

Level	Name	Capability	Example Use Cases	Preferred API	Fallback

A0	Observational	Logging only, no influence	Performance monitoring, trend analysis	None (local)	–
A1	Advisory	Suggestions only, cannot block	Price recommendations, Smart Inbox titles, tenant messages, loading guides	Groq	Ollama/ LocalAI
A2	Constraining	Can block, delay, flag, escalate – never force	Compliance prescreen, risk flagging, document verification, cold-chain anomaly alert	Hugging Face (local)	Ollama/ LocalAI
A3	Escalation	Forces human review, cannot decide autonomously	Final Governor routing, high-risk trade approval, negotiation deadlock resolution	Hybrid (HF for deep analysis + Groq for fast output)	Ollama/ LocalAI
A4	Governance	Auto-executes within constitutional bounds	Routine milestone confirmation, low-risk rail additions, autoreconciliation	None (OPA + WasmEdge)	–
A5	FORBIDDEN	No autonomous execution	–	Blocked at WASM compile time	–

Multi-Provider Fallback Chain:

For A1 : Try Groq → if rate limited or error → Ollama/LocalAI.

For A2 : Try HF local → if model not loaded or server overloaded → Ollama/LocalAI.

For A3 : Try HF for deep analysis → if fails, Ollama; then Groq for formatting → if fails, Ollama.

All calls logged in ai_inference_records with provider, model, latency, token count, and fallback status.

11.2 AI Infrastructure Stack (Zero-Cost, Self-Hosted)

Component	Technology	Purpose
Inference Servers	vLLM (Apache2.0) for local LLMs; ONNX Runtime for edge models	Serve Mixtral, Donut, ViT, small offline LLMs

Model Repository	Hugging Face free tier (cached locally)	Download models once; all inference local
Agent Framework	Rig (Rust, OSS)	Every AI agent is a Rig agent with defined authority and lifecycle
Machine Learning	XGBoost, LightGBM, PyTorch, scikit-learn (all local)	Credit scoring, margin estimation, FX prediction, risk simulation
NLP / Sentiment	Finetuned DistilBERT, FinBERT (HF local)	Sanctions screening, news sentiment, regulatory change detection
Document AI	Donut (local), PaddleOCR	Extract data from invoices, certificates, SWIFT confirmations
Computer Vision	ViT (local), Transformers.js (WebGPU)	Pallet damage detection, AR inspection overlay
Speech-to-Text	Vosk (offline)	Voice commands, offline voice-filled forms
Fast Generation	Groq API (free tier)	Instant summaries, market insights, loading guides, Smart Inbox titles, tenant messages
Simulation	Python numpy/scipy, Pyro	Monte Carlo risk simulation, digital twin
Vector Search	pgvector (PostgreSQL)	Semantic search, sanctions fuzzy matching, RAG
Governance & Logging	WasmEdge, OPA, Loom	All AI decisions enforced and logged immutably

No billing details required.

11.3 The 7-Layer AI Architecture (Recap with v6.3 Enhancements)

The AI layer is organised into seven layers, from raw data ingestion to final governance-gated action.

Layer 1 : Data Ingestion & Preprocessing – Free sources: AISHub, OpenWeatherMap, ECB, FAO, RSS news, OFAC/UN/EU lists, central bank sites, PSP webhooks, IoT sensors. HF Donut extracts PDF data; HF NLP classifies news sentiment.

Layer 2 : Feature Engineering & Embeddings – Commodity embeddings (HS code + trade lane) using sentence-transformers (512dim). Tenant embeddings for sanctions screening. Document embeddings for semantic search. Stored in pgvector.

Layer 3 : Risk Modeling & Predictive Analytics – Counterparty risk (XGBoost, 200+ signals). Geopolitical risk (news sentiment). Cold-chain integrity (LSTM). Disruption risk (multimodal). DeFi protocol risk (Bayesian). PSP/currency risk (LightGBM + LSTM). Commodity-specific margin estimation (XGBoost + LLM refinement) – see 11.5.

Layer 4 : Graph Intelligence & Network Analytics – Corporate Graph (PyTorch Geometric) for UBO detection, sanctions proximity. Trade network graph for circular obligations, distressed buyer discovery. Knowledge Graph RAG for legal/compliance queries.

Layer 5 : Prediction & Simulation – Digital Twin engine (Pyro). Monte Carlo simulation (10,000 scenarios). Packing & containerisation (ORTools + AI). Voice intent parsing (Vosk + HF Mixtral). Smart Inbox urgency scoring (deterministic A4, title/description via Groq/Ollama A1).

Layer 6 : Optimization & Autonomous Actions (Bounded) – Logistics bundle optimiser (neural CF). PSP router (reinforcement learning). Autoreconciliation (HF Donut + rule engine). Autonomous milestone confirmation (A4, multisource consensus ≥95%).

Layer 7 : Governance AI Interface – Governor orchestration with WasmEdge. Model drift monitor. Explainability engine (SHAP + Groq translation). Tenant-facing AI explainability (PlainLanguage Decision Panel).

11.4 AI Agent Registry (Complete, v6.3 Updated)

All agents from the original blueprint are retained, with the following modifications:

Intent Parser, Container Advisor, Margin Estimator, Clause Checker, Negotiation Bot, Credit Intelligence, DeFi Risk Oracle, etc., now use three-tier AI (Groq primary with Ollama fallback for A1; HF local with Ollama fallback for A2; hybrid for A3).

Smart Inbox Text Generator (A1) – new: generates title and description from structured context using Groq (fallback Ollama).

Tenant Message Generator (A1) – new: generates plain-language decision explanations stored in governor_decisions.tenant_message.

Commodity-Specific Margin Estimator (A2) – XGBoost model refined by LLM (HF local or Ollama) using commodity type, country pair, market news.

All agents run under ai-agent-orchestrator and are version-locked.

11.5 Deep Dive: Key AI Systems (v6.3 Enhanced with Commodity-Specific Margin)

11.5.1 Commodity-Specific & Country-Pair Margin Estimator

Base XGBoost model trained on thousands of historical SGTX trades. Features : HS6 code, commodity type (sugar, wheat, fresh fruit, etc.), seller country, buyer country, logistics costs, market price indices, seasonality, volume, incoterm.

LLM refinement layer (HF local, fallback Ollama) reads real-time qualitative signals: news sentiment about the commodity (e.g., “EU announces higher sugar import quota”), geopolitical events affecting the country pair, carrier performance data. Outputs an adjustment factor (±0.1% to ±0.5%).

Final margin estimate used for SGTX trade fee calculation (Part 7). Output is stored and can be explained to the seller via the Trade Command Center: *“Margin estimated at

18.6% based on current market prices and historical data. A recent port strike added +0.3% risk adjustment."*

11.5.2 Regulatory Intelligence Agent (RIA) – The Global Brain – unchanged, but now also updates distressed country factors, financing regulations, and DeFi eligibility.

11.5.3 Cold-Chain Quality Predictor – unchanged (LSTM on IoT sensor data). Now integrates with Smart Inbox for proactive alerts and One-Tap Smart Escrow Claim.

11.5.4 Digital Twin with "What-If" Simulation – unchanged.

11.5.5 AI-Mediated Negotiation (Phase 3 & 7) – unchanged, with bot transparency panel.

11.5.6 Voice-Activated Operations – unchanged, with offline Vosk and HF Mixtral.

11.5.7 Smart Inbox Urgency Scoring & Title Generation (v6.3) – deterministic rule engine (A4) computes priority score. Then calls Groq (or Ollama) to generate title and description. Fallback template if AI unavailable.

11.5.8 Tenant-Friendly Decision Explanations (v6.3) – Governor calls Groq (or Ollama) to produce structured tenant_message JSON for any DENY/CONDITIONAL verdict. Includes plain-language summary, condition checklist, action URLs, and confidence/evidence if available.

11.6 Explainability & Model Drift

Every AI decision (A1–A3) logged with SHAP values and Groq-generated plain-language explanation.

Model Drift Monitor compares predicted vs actual outcomes; if performance drops below threshold, triggers automatic retraining or policy suggestion to Admin Portal.

Users can request full explainability report from AIAssistantPanel for any AI-driven action that affected their trade.

11.7 Tenant Health Engine (v6.3) – Already described in Part 11.11 of original blueprint, now integrated with three-tier AI for summary generation. It computes a health score (0-100) for each tenant based on KYB completeness, trade success rate, dispute frequency, document responsiveness, milestone delay rate, inbox action responsiveness, unresolved Governor conditions, and onboarding completion. Displayed in Company Admin → Health dashboard with Groq-generated recommendations.

11.8 Deployment, Governance & Model Versioning

All models stored in /models/, versioned in MLmodel files.

WasmEdge containers sandbox AI execution for A2/A3.

OPA policies define which agents can interact with which data and at what authority level.

Loom logging provides immutable replay of every AI inference.

ai-agent-orchestrator manages agent lifecycle, health checks, and automatic restart.

11.9 Example AI Workflow: From Trade Initiation to Settlement (v6.3)

Phase 1: Buyer speaks request; Vosk transcribes, HF extracts specs; Container Advisor (Groq) recommends reefers. Smart Inbox title generated by Groq: “New trade request created – awaiting seller response.”

Phase 2: EXW price lock – live market chart (Groq). AI suggests price band. Seller locks price. Margin Estimator (XGBoost + LLM) computes 18.6% margin → fee rate 1.03%.

Phase 3: Negotiation Bot (Groq + HF) runs 3 rounds; deadlock resolved by HFsuggested compromise. Tenant Message Generator produces explanation if block occurs.

Phase 4: Credit Intelligence (XGBoost) approves financing; DeFi Risk Oracle (HF) confirms protocol safety. DeFi Risk Summary modal (Groq) presented.

Phase 5: Cold-chain anomaly detected; LSTM predicts shelf-life reduction; Groq alerts via Smart Inbox. QC inspector overrides AI defect with mandatory reason.

Phase 6: Settlement instruction generated; buyer approves via voice; PSP Router selects rail; HF Donut reconciles bank statement.

Phase 7: Distressed cargo – AI condition assessment (HF ViT), pricing (XGBoost), triage dashboard with Groq explanations.

Phase 10: Dispute filed; Evidence Compiler flags QC override; AI predicts outcome; mediates with Groq-translated messages.

Every step logged, every AI decision explained, every fee accounted for – all at zero cost.

END OF PART 11 – v6.3 WITH ALL MODIFICATIONS

Part 12: Sovereign Tech Stack & Deployment Topology (v6.3 – Fully Upgraded, Global, Zerocost, with All Modifications: Three-Tier AI, Multi-Shipment, Dual-Mode Toggle, SGTX Fee Paid by Seller, Financer Preferences, Etc.)

12.0 Sovereign by Design

SGTX is engineered to run entirely on infrastructure controlled by its operators. No cloud provider, no proprietary platform, no licensing fees. Every component from the kernel to the AI model is open-source and can be deployed on bare-metal servers, virtual private servers, or even a single developer machine. This guarantees that the platform can be operated in any jurisdiction, even those with strict data sovereignty laws, without external dependencies.

The deployment topology is globally distributed and self-healing. AI agents continuously monitor every node, predict failures, and orchestrate failovers without human intervention. Sovereign nodes automatically adapt to local regulations, and new regions can be added with a single command. The stack fully supports all v6.3 modifications: multi-shipment contracts, dual-mode toggle, three-tier AI, structured container forms, financier preferences, co-financing, and distressed cargo enhancements.

12.1 Technology Stack (Zero-Cost, Open-Source)

All versions are locked; no automatic updates without testing.

12.1.1 Core Languages & Runtimes

Technology	Version	Purpose
Rust	1.88 (stable)	All microservices, Governor, agents, tooling (including new v6.3 services: inbox-service, tenant-experience-service, workflow-recovery-service, tenant-data-export-service, financier-preference-service, multi-shipment-coordinator, fee-split-orchestrator)
TypeScript	5.8	React-based portals (Next.js 15)
Python	3.12	Machine learning (XGBoost, LightGBM, PyTorch,

		Transformers, ORTools), simulation

12.1.2 Microservices & Communication

Component	Version	Purpose
Axum (Rust)	0.8	HTTP REST framework for all services
Temporal	1.26 (OSS)	Workflow engine for long-running processes (contract genesis, financing auctions, multi-shipment coordination)
NATS	2.11 (OSS)	Asynchronous messaging (JetStream), KV store for FeeLock (SGTX FEESLock), event bus. v6.3: inbox-service, tenant-experience-service, workflow-recovery-service rely on NATS for real-time updates
MQTT	Mosquitto 2.0	IoT sensor data ingestion (optional)

12.1.3 Data Stores

Component	Version	Purpose
PostgreSQL	18	Primary relational database, supports pgvector, TimescaleDB, RLS. v6.3: new tables for inbox, onboarding, organisation graph, data export, financier preferences, multi-shipment, distressed tracking, etc.
pgvector	0.8	Vector embeddings for sanctions screening, semantic search
TimescaleDB	2.18	Hypertables for timeseries IoT sensor data
ClickHouse	24.12	Immutable audit log, analytics, metrics warehousing, tenant activity log
Valkey (Redis fork)	8.0	Caching (optional)

All databases run on ZFS or ext4 with full-disk encryption (LUKS/dm-crypt, free).

12.1.4 Orchestration & Sandboxing

Component	Version	Purpose
K3s	1.31	Lightweight Kubernetes for production clusters (bare-metal)
CRIO	1.32	Container runtime (no Docker)

runwasi	0.7	WASM container runtime for WasmEdge-based services
WasmEdge	0.14	Secure sandbox for constitutional rules, jurisdiction gates, AI agents
Firecracker	1.10	MicroVM for AI inference and document processing (optional isolation)
Cilium	1.17	eBPF-based networking, observability, and security policies

Development on Linux x64 or macOS uses native binaries and Podman; no Docker required.

12.1.5 AI & ML Frameworks (All Local/Free)

Component	Version	Purpose
vLLM	0.10	Serve Hugging Face models locally (Mixtral, etc.) – optional
Transformers (Hugging Face)	4.52	Finetuned models for NLP, vision, document extraction
ONNX Runtime	1.21	Edge AI inference (mobile, browser)
XGBoost	3.0	Credit scoring, margin estimation, SGTX FEES optimization
LightGBM	4.6	PSP routing, disruption prediction
PyTorch	2.7	LSTM for cold chain, digital twin
Rig	0.6 (Rust)	AI agent framework
Vosk	0.3	Offline speech-to-text
Ollama	latest	Local LLM fallback (runs on CPU/GPU)

Free tier Groq API is used for non-privacy-critical, low-latency natural language generation; Ollama is the fallback.

12.1.6 Cryptography

Algorithm	Purpose
Ed25519	Digital signatures for Governor decisions, contract signatures, data exports
SHA256	Document hashing, Loom hash chains, integrity verification
Dilithium3 / Kyber1024	Post-quantum key exchange and signatures (for long-term secrets)
Plonky3	Zero-knowledge proofs for private settlements,

	reserve proofs
TLS 1.3	All inter-service and client-server communication
AES256GCM	Backup encryption, secret storage
LUKS/dm-crypt	Full-disk encryption on all nodes

Key material stored in each node's HSM (software-based via softhsm2, free) or OS keychain.

12.1.7 Observability & Monitoring

Component	Purpose
Prometheus + Grafana	Metrics dashboards, alerting
VictoriaMetrics (optional)	Long-term metrics storage (ClickHouse can also serve)
Loki	Log aggregation
Jaeger	Distributed tracing (OpenTelemetry)
Parca + eBPF	Continuous profiling
Prometheus Alertmanager	PagerDuty-style notifications (self-hosted webhooks)

All observability stack self-hosted; no SaaS monitoring.

12.2 Infrastructure & Deployment Topology

12.2.1 Global Sovereign Node Architecture

Traffic is routed via Anycast / GeoDNS to the nearest sovereign node. The Governor Proxy (WasmEdge) ensures data residency by enforcing routing based on sovereign_nodes table. Each sovereign node region (e.g., EU – Frankfurt, MENA – Cairo, Asia-Pacific – Singapore) runs a full K3s cluster with PostgreSQL, Valkey, and object store. Cross-region replication is asynchronous via NATS JetStream streams and WAL shipping for PostgreSQL. New regions are provisioned automatically by a CLI tool reading YAML specs.

12.2.2 Edge & IoT Integration

Low-cost LoRaWAN sensors (optional, ~$50 one-off) can be attached to containers for real-time temperature, humidity, and shock monitoring. Data ingested via Mosquitto MQTT brokers at the nearest sovereign node. Edge processing (anomaly detection) runs on a local Raspberry Pi or directly on the mobile app (offline LSTM via ONNX).

12.2.3 Global Load Balancing & Failover

Anycast DNS (free via cloudflared or self-hosted CoreDNS) distributes initial requests. Governor Proxy maintains a global health map of all nodes (updated via NATS heartbeats). If a node becomes unhealthy, traffic is automatically rerouted to the next closest region. AI (LightGBM) predicts node failures based on resource metrics and triggers controlled failover with near-zero downtime.

12.2.4 Dynamic Scaling

Predictive AutoScaler (LSTM) forecasts load per node based on historical trade volumes and scheduled tasks (e.g., end-of-month reporting). Automatically adjusts pod replicas in K3s using kubectl scale. v6.3 services (inbox-service, finance-service, multi-shipment-coordinator) are stateless or use NATS KV for state, allowing horizontal scaling.

12.3 Development Environment (Linux x64 or macOS)

All services run as native processes or in lightweight Podman containers. Dependencies are version-locked (Cargo.lock, pnpm-lock.yaml, uv.lock). The canonical development path is /opt/sgtx (or any directory). Scripts are idempotent. AI models are downloaded once into /models/. No billing details required.

12.4 Cryptography & Key Management

Tenant Keys : Ed25519 keypairs generated client-side. Private keys encrypted with AES256GCM and stored in the tenant's sovereign node. Public keys broadcast as needed.

Platform Keys : Platform Governance Authority uses a 3-of-5 Shamir Secret Sharing scheme for the master signing key. Shares stored in hardware tokens (YubiKey) or software emulation (softhsm2).

Certificate Authority : Internal CA based on stepca (free, OSS) issues X.509 certificates for all services. Short-lived certificates automatically renewed via ACME.

Quantum-Resistant Readiness : Dilithium3 and Kyber1024 used for data that must remain confidential for >10 years (contract archives, identity records).

12.5 Backup & Disaster Recovery

RPO : ≤5 minutes (continuous WAL shipping and NATS JetStream mirroring).

RTO: ≤30 minutes (automated failover to next region).

Backup Strategy : PostgreSQL continuous WAL archiving + daily pg_dump to another sovereign node; NATS JetStream mirrors across ≥2 regions; ClickHouse replicated merge tree. All backups encrypted with AES256GCM.

Restore Testing : backup_validator Rust script restores latest backup into a sandbox K3s cluster monthly, runs integrity checks, and reports via Groq.

Georedundancy : Minimum two regions for production; critical services (Governor, identity) run active-active.

12.6 Regional Adaptations & Global Compliance (Dynamic)

The topology adapts in real time:

Data Residency : enforced by sovereign_nodes table and jurisdiction-routing WasmEdge module.

Regional Compliance Modules : WasmEdge plugins for local encryption standards (Egypt PDPL, India DPDP Act) and logging requirements.

China-Specific : Airgapped node with OCI containers reviewed by local compliance team.

EU (GDPR / CBAM) : Right-to-erasure process triggers cascade deletion on EU node, logged immutably. CBAM reporting via esg-carbon-service.

USA (FedRAMP / OFAC) : FIPS-140-3 compliant cryptography (via OpenSSL FIPS module, free from Red Hat).

Automatic updates : RIA monitors regional regulations and updates WasmEdge compliance plugins.

12.7 Security & Hardening (Zero-Cost)

Immutable infrastructure : K3s worker nodes read-only after boot.

Network policies : Cilium enforces microsegmentation per OPA policies.

Intrusion detection : ids-agent (Rust) monitors system logs and network traffic using free Suricata rules. Alerts to Prometheus.

Vulnerability scanning : trivy scans all container images before deployment; critical vulnerabilities block rollout.

Chaos engineering : chaos-orchestrator (Go, OSS) randomly terminates pods and nodes to test resilience; Groq generates summary.

12.8 Observability & AIDriven Operations

Metrics : sgtx_governor_decisions_total, sgtx_fee_lock_released_usd, per-phase latency, PSP success rates, sgtx_inbox_items_created_total, sgtx_tenant_exports_completed_total, etc.

Dashboards: Grafana dashboards auto-deployed from JSON.

AIOps : Rig agent (A2) analyses Prometheus metrics and logs to predict incidents, creates Jira ticket (self-hosted plane or taiga), preemptively scales or cleans logs.

Distributed tracing : Every request carries trace_id, governor_decision_id, tenant_id; Jaeger visualises lifecycle.

12.9 Implementation Checklist (Linux x64 or macOS)

(Full checklist script provided – no coding block, but textual summary): Validate dependencies (cargo, wasmedge, opa, psql, nats, etc.). Build all services (including v6.3 services). Initialise database. Start infrastructure (PostgreSQL, NATS, Valkey). Start Governor and all microservices. Seed jurisdictions. Verify health endpoints.

END OF PART 12 – v6.3 WITH ALL MODIFICATIONS

Part 13: Governor Service – Authoritative Design Specification (v6.3 – Fully Upgraded, Zerocost, Tenantfriendly Decision Explanations, Three-Tier AI, Multi-Shipment, Financing, Distressed, Dual-Mode Context, and All Modifications)

13.0 Overview

The Governor is the single path through which every irreversible action must pass. It is the software embodiment of the platform’s constitution. Without a Governor decision (ALLOW, DENY, CONDITIONAL, ESCALATE), no contract can be locked, no milestone confirmed, no fee collected, no document accepted, and no payment settled.

The service is a Rust binary built with Axum, backed by three enforcement engines:

OPA (Open Policy Agent) – for policy evaluation (RBAC, business rules, dual-mode context).

WasmEdge – for high-integrity constitutional checks, jurisdiction supremacy, and fee bounds.

Loom – for deterministic, immutable logging and cryptographic chaining.

In v6.3, the Governor integrates directly with the AI Authority Ladder using three-tier AI (Groq, HF local, Ollama). It generates tenant_message JSONB for any DENY/CONDITIONAL verdict (PlainLanguage Decision Panel). It handles new decision types: fee.pay, financing.bid.submit, distressed.microcontract.lock, logistics.addendum.sign, multi_shipment.shipment.lock, SGTX FEES.dispute, and fee.split.confirm. The Governor also enforces dual-mode context (Buyer/Seller) and multi-shipment per-shipment locks.

13.1 Architecture & Core Components

text
External Request
↓
API Gateway (Nginx)
↓
Governor Proxy (Axum intercept)
↓
┌─────┼─────┐
↓ ↓ ↓

OPA WasmEdge AI Router (Rig)
(Rego) (WASM) │
↓ ↓
Decision Merger
↓
Generate Tenant Message (if not ALLOW)
↓
Sign & Loom Log
↓
governor_decisions (PostgreSQL)
↓
NATS Event (optional)

All request fields are validated : actor GTID, action, resource, dual-mode context (active_trader_mode_context from JWT), multi-shipment shipment ID if applicable.

13.2 OPA Policy Engine

Policies written in Rego cover every domain, with v6.3 additions:

permissions.rego – RBAC, dual-mode context checks, data scopes, cost hiding.

fee.rego – validates fee calculations (trade fee rate within 0.1%–2.5%, financing fee 0.25%).

financing.rego – financier preference filtering, co-financing sum ≤ requested amount, split instruction approval.

distressed.rego – distressed listing conditions, price reasonableness, privacy notice acknowledgment.

multiship.rego – multi-shipment per-shipment lock conditions, fee payment per shipment.

logistics.rego – logistics addendum signing, container release confirmation.

dual_mode.rego – allows actions only if active_trader_mode_context matches required mode (BUY/SELL).

Policies are hot-reloaded after multisig approval; AI-assisted simulation runs in Admin Portal before deployment.

13.3 WasmEdge Constitutional Engine

Compiled WASM modules that enforce non-negotiable rules:

constitutional_rules.wasm – fee bounds (0.1%–2.5%), A5 prohibition, multisig requirements.

jurisdiction_matrix.wasm – strictest jurisdiction rule, updated hourly by RIA.

incoterms_engine.wasm – incoterm validation, logistics cost requirements (FOB no sea freight, etc.).

fee_gate.wasm – validates gross-up calculation and split instructions.

distressed_country_gate.wasm – applies country factor for distressed fees.

dual_mode_gate.wasm – checks that action is allowed in current dual-mode context (Buyer/Seller).

All modules are pre-compiled, sandboxed (no network/filesystem), and signed.

13.4 AI Integration (Three-Tier, with Fallback)

The Governor consults AI only after OPA and WasmEdge allow, and only for actions configured to require AI input. The ai-agent-orchestrator routes:

A1 tasks (advisory) → Groq, fallback Ollama.

A2 tasks (constraining, privacy-critical) → HF local, fallback Ollama.

A3 tasks (escalation) → HF local for deep analysis + Groq for formatting, fallback Ollama.

AI outputs (verdict, conditions, confidence, evidence) are merged into the final Governor decision. The AI cannot convert a DENY into an ALLOW; it can only add CONDITIONAL conditions or escalate to human.

13.5 Tenant Message Generation (v6.3)

Whenever the final verdict is DENY or CONDITIONAL, the Governor calls the Tenant Message Generator (A1) with the decision context. The generator uses Groq (primary) or Ollama (fallback) to produce a structured tenant_message JSONB:

summary – short plain-language explanation.

conditions – array of {condition_id, label, status, action_url}.

escalation_hint – optional.

confidence – if AI involved.

evidence_sources – optional array.

The message is stored in governor_decisions and rendered by the PlainLanguage Governor Decision Panel (Part 6.1.3). If Groq and Ollama both fail, a static template fallback is used with a notice.

13.6 Loom Deterministic Logging

Each decision's loom_hash = SHA256(previous_loom_hash || decision_json || cryptographic_signature). The chain's genesis block is the hash of the constitutional rules WASM binary. An hourly background job verifies the chain; any mismatch triggers a P0 incident.

13.7 Configuration (governor.toml)

Key settings : OPA policy directory, Wasm module directory, AI provider endpoints and fallback order, Groq API key (optional), HF token (optional), Ollama URL, Loom log path, signing key path, tenant message template directory, cache TTL, and timeouts.

13.8 Non-Negotiable Rules (Hardcoded)

governor_decision_id NOT NULL in all tables that record state changes.

Axum middleware injects app.current_tenant_id, app.current_employee_id, app.active_trader_mode_context for RLS.

AI A5 prohibition at WASM level; any attempt to execute A5 triggers DENY and logs a constitutional violation.

Tenant message generation step cannot be bypassed for non-ALLOW verdicts.

13.9 Global Dynamic Updates

The Governor subscribes to NATS topics for policy changes (Rego updates, WASM module updates, AI model version changes). All updates are logged, versioned, and require multisig approval for substantive changes.

13.10 Example Decision Record (v6.3)

json

{
"decision_id": "d4e5f6...",
"decision_type": "contract.lock",
"actor_gtid": "SGTX-EG-TRD-000123",
"actor_employee_id": "emp-001",
"active_trader_mode_context": "BUY",
"verdict": "CONDITIONAL",
"conditions": [{"condition_id": "kyc_tier2_seller", "label": "Seller KYB Tier 2"}],
"policy_version": "v6.3.1",
"rule_refs": ["contract.rego:23", "constitutional_rules.wasm:0x12ab"],
"explainability": "Seller jurisdiction high risk; enhanced KYC required.",
"confidence": 1.0,
"loom_hash": "sha256:a1b2c3...",
"cryptographic_signature": "...",
"tenant_message": {
"summary": "Your contract lock is on hold because the seller's jurisdiction requires enhanced KYC.",
"conditions": [{"condition_id": "kyc_tier2_seller", "label": "Seller KYB Tier 2 verification", "status": "unmet", "action_url": "/v1/kyc/upgrade?tenant=..."}],
"escalation_hint": "If you believe this is an error, request a human review."
},

"created_at": "2026-06-14T08:30:00Z"

}

END OF PART 13 – v6.3 WITH ALL MODIFICATIONS

Part 14: Comprehensive Audit, Observability & Compliance (v6.3 – Fully Upgraded, Global, Zerocost, with All Modifications: Multi-Shipment, Financing, Distressed, Tenant Activity Log, Smart Inbox History, Cryptographic Verification)

14.0 Philosophy

Every action on the SGTX platform must leave a cryptographically verifiable, immutable trace. Observability is not a luxury – it is a constitutional requirement. This part defines:

Audit – how every state change, Governor decision, and AI inference is recorded immutably and can be replayed.

Observability – how the platform is monitored, measured, and debugged in real time, entirely with self-hosted open-source tools.

Compliance – how audit trails automatically satisfy regulatory reporting and demonstrate continuous control.

All components are free, open-source, and require no billing details. The observability stack is designed to work globally across sovereign nodes, and AI agents constantly watch for anomalies, drift, and potential compliance issues, often resolving them before a human notices. In v6.3, the audit trail captures tenant-facing decision messages (tenant_message in governor_decisions), Smart Inbox activity (item creation, snooze, dismiss, action taken), financing and distressed events, and multi-shipment per-shipment milestones.

14.1 Immutable Audit Architecture

14.1.1 Loom – Deterministic Hash Chain

Loom is a purpose-built Rust library that creates a sequential, cryptographically linked log of every Governor decision and every significant database write.

Every governor_decisions row includes a loom_hash calculated as SHA256(previous_loom_hash || decision_json || cryptographic_signature). The decision_json includes the tenant_message field, ensuring plain-language explanations are also immutably chained.

The hash chain is anchored at the genesis block (hash of the compiled constitutional WASM modules).

Append-only : no update or deletion possible. Any attempt to alter a past entry breaks the chain and is immediately detectable.

14.1.2 Event Sourcing via NATS JetStream

All state-changing events (milestone confirmed, document uploaded, fee paid, financing disbursement, distressed sale, etc.) are published as NATS JetStream messages with a unique sequence number, timestamp, and subject. These streams are:

Persistent – stored on disk with replication across at least two sovereign nodes.

Replayable – any consumer can replay the entire event history for a trade, financing agreement, or distressed lot to reconstruct its exact state. This includes events that trigger Smart Inbox item creation (trade.status.changed, document.uploaded, financing.bid.submitted, distressed.offer.received), enabling the inbox-service to rebuild the inbox from scratch if needed.

Tamper-evident – each message is hashed, and the stream's integrity is periodically verified.

14.1.3 ClickHouse for Long-Term Retention

While PostgreSQL holds the operational audit tables (audit_log, governor_decisions, ai_inference_records, inbox_history), a dedicated ClickHouse cluster stores these same records in a columnar format optimised for analytics and compliance queries. Data is retained for a minimum of 7 years (configurable per jurisdiction). ClickHouse's ReplicatedMergeTree engine ensures data replication across sovereign nodes without cloud costs.

14.1.4 Tamper-Evidence & Verification

A background job (audit-chain-verifier) runs every hour:

Recalculates every Loom hash from the genesis block forward and compares it with the stored value.

Verifies that all audit_log entries have a corresponding governor_decision_id.

Checks that no database row has been modified without a corresponding audit_log entry (by comparing checksums of critical tables against a separate Merkle tree stored in ClickHouse).
If any verification fails, Groq drafts an immediate incident report; the alert is sent via Prometheus Alertmanager to the Admin Portal's Smart Inbox as a critical item.

14.2 Real-Time Observability Stack

All monitoring tools are self-hosted, open-source, and connected by OpenTelemetry for unified telemetry.

Component	Purpose
Prometheus	Scrapes metrics from every microservice, NATS, PostgreSQL, K3s nodes. v6.3 new metrics: sgtx_inbox_items_created_total, sgtx_inbox_snoozed_total, sgtx_tenant_exports_com

	pleted_total, sgtx_tenant_message_generation _seconds, sgtx_financing_bids_submitted_total, sgtx_distressed_listings_created_total.
VictoriaMetrics (optional)	Long-term, high-cardinality metrics storage.
Grafana	Dashboards for platform health, trade volumes, PSP success rates, AI drift, compliance status, Smart Inbox engagement (items per tenant, average time-to-action), Tenant Experience dashboard (onboarding completion rates, dual-mode toggle usage).
Loki	Aggregates structured and unstructured logs from all services.
Jaeger (with OpenTelemetry)	Distributed tracing across the mesh.
Parca + eBPF	Continuous profiling of CPU, memory, and I/O without code instrumentation.
Prometheus Alertmanager	Routes alerts to the SGTX mobile app (push notification), email, or self-hosted incident management webhook.

14.2.1 Metrics Pipeline

Every microservice exposes a /metrics endpoint in Prometheus exposition format. Key metrics include:

sgtx_governor_decisions_total{verdict=...}

sgtx_governor_decision_latency_seconds

sgtx_fee_lock_released_usd (trade fees and financing fees)

sgtx_trade_requests_created_total

sgtx_shipment_milestones_confirmed_total

sgtx_ai_inference_duration_seconds{model, authority, provider}

sgtx_psp_attempts_total{psp, status}

sgtx_inbox_items_created_total{category, priority}

sgtx_inbox_items_snoozed_total

sgtx_tenant_exports_completed_total{format}

sgtx_tenant_message_generation_seconds{language, fallback_used}

sgtx_financing_requests_created_total

sgtx_distressed_microcontracts_locked_total

Business-level dashboards in Grafana allow real-time tracking of revenue, active shipments, dispute rates, AI accuracy per model, tenant engagement, and dual-mode toggle usage.

14.2.2 AIDriven Observability (AIOps)

A dedicated Observability Agent (Rig, A2) continuously analyses metrics and logs:

Anomaly detection – using a local Isolation Forest model to flag unusual patterns (e.g., sudden drop in trade completions on a certain corridor, spike in Smart Inbox dismissals indicating alert fatigue, unusual financing bid withdrawal patterns).

Predictive Alerting – forecasts resource exhaustion (disk, memory) and preemptively triggers scaling or cleanup.

Root-cause suggestion – when an incident occurs, the agent correlates events across traces, logs, and metrics, and uses Groq to propose a likely root cause: “Governor latency spike at 14:05 correlates with a large number of concurrent multi-shipment locks after a trade fair. No systemic issue. Consider scaling governor pods during peak hours.”

14.3 Distributed Tracing

Every API request carries a traceparent header (W3C Trace Context). The Governor adds governor_decision_id, tenant_id, and active_trader_mode_context as span attributes. This allows developers and auditors to:

Trace a single user click from the browser, through the API gateway, Governor, microservices, database queries, NATS events, and even the AI inference calls.

Replay the exact path and timing of any trade operation, financing disbursement, or distressed sale, including the Smart Inbox item creation and tenant message generation.

Identify bottlenecks or failures with service maps automatically generated by Jaeger.

All tracing data is stored in ClickHouse; a representative sample is retained for 7 years.

14.4 Auditor’s Interface & Compliance Reporting

14.4.1 Audit Explorer (Admin Portal)

A dedicated page in the Admin Portal allows a compliance officer to:

Search by USTN, financing agreement ID, distressed listing ID, GTID, date range, or action type. Natural-language queries are supported via Groq (e.g., “Show all decisions where fee release was blocked in the last week”) and converted to ClickHouse SQL.

View a timeline of every Governor decision, the request payload, the resulting state change (before/after), and the plain-language tenant_message that was shown to the user.

Verify the Loom chain integrity for any trade, financing, or distressed transaction with a single click.

Export the full evidence package for a trade as a tamper-evident PDF, digitally signed by the Governor.

14.4.2 Automated Regulatory Reports

Using the audit data, Groq compiles periodic reports required by various regulators:

Egypt CBE monthly remittance form – populated with all settled trades involving Egyptian parties, including financing fee deductions.

EU ECB statistical report – trade values, financing, cross-border flows, distressed sales.

FATF suspicious transaction reports – if any SARs were drafted.

CBAM reports – carbon footprint data from carbon_footprint_calculations.

The reports are exported as PDFs (via Tera templates + headless Chrome) and can be auto-emailed if SMTP is configured. All report generation is logged immutably.

14.4.3 Continuous Compliance Monitoring

A Compliance Agent (A2) runs continuously across all sovereign nodes:

Checks that all tenants with KYB expiration within 30 days are flagged for renewal; the tenant's Smart Inbox receives a COMPLIANCE item.

Verifies that no active trade, financing agreement, or distressed listing involves a sanctioned entity (rescreens every 6 hours using latest sanctions lists).

Ensures all sovereign nodes meet data residency requirements (e.g., EU trades only processed on EU nodes). Any violation triggers an immediate alert and blocks further processing.

14.5 Model Drift & AI Audit

Every AI model is audited for accuracy and fairness:

The Model Drift Monitor (A2) compares predictions (credit scores, ETA, margin estimates) against actual outcomes weekly. If drift exceeds a threshold, it automatically triggers retraining or suggests a policy change.

Every AI inference is logged with SHAP values and a Groq-generated plain-language explanation. This log is available to tenants upon request, ensuring transparency and regulatory compliance for automated decision-making. The same Groq-generated explanations are used in the PlainLanguage Governor Decision Panel when an AI agent blocks or escalates an action.

If an AI model's performance degrades, the Governor can automatically reduce that model's authority level (e.g., from A2 to A1) until it is retrained.

14.6 Zero-Cost Tech Stack Recap

Component	Cost
Loom (Rust)	$0
NATS JetStream	$0
ClickHouse (OSS)	$0
Prometheus, Grafana, Loki, Jaeger, Parca	$0
OpenTelemetry	$0
Groq (for report drafting, summaries, tenant message generation)	$0 (free tier)
Hugging Face models (for log anomaly detection)	$0 (local)

No billing details required.

14.7 Governance & Verification Gates

Gate ID	Check	Enforcer	Failure Action
GA1	Loom chain integrity verified every hour	Audit Chain Verifier (A4)	Immediate PagerDuty alert; freeze new decisions
GA2	All audit_log entries have a corresponding governor_decision_id	DB consistency check (A4)	Quarantine orphaned record; alert
GA3	Model drift within acceptable bounds for all active models	Model Drift Monitor (A2)	Reduce model authority; trigger retraining
GA4	No active trade, financing, or distressed involves a sanctioned entity	Sanctions Rescreener (A2)	Suspend entity; draft SAR
GA5	All sovereign nodes meet data residency policies	Node Compliance Agent (A2)	Reroute traffic; alert
GA6	Regulatory reports generated and filed for required periods	Reporting Agent (A2)	Block new trades in that jurisdiction until resolved
GA7	Smart Inbox history immutable and audit-traced	inbox_history table (append-only)	Alert if modification attempted

14.8 Implementation Checklist (macOS/Linux – ZeroCost)

Summary of steps:

Start observability stack (Prometheus, Grafana, Loki, Jaeger).

Build audit service.

Seed ClickHouse with audit schema.

Verify Loom chain (first run).

Ensure inbox_history is being recorded (simple SQL count).

Configure alerting rules for SLA breaches.

END OF PART 14 – v6.3 WITH ALL MODIFICATIONS

Part 15: Zerocost Addons Suite (v6.3 – Fully Upgraded, Global, Zerocost, with All Modifications: Three-Tier AI, Multi-Shipment, Financing, Distressed, Dual-Mode Toggle, SGTX Fee Paid by Seller, and All Prior Addons Extended)

15.0 Philosophy & Zero-Cost Commitment

The SGTX AddOns Suite is a collection of 62 advanced capabilities that extend the platform into areas of federated learning, graph intelligence, privacy-preserving computation, digital twins, chaos engineering, carbon accounting, and more. Every addon is designed to run on the same zero-cost, self-hosted infrastructure, using exclusively open-source libraries and free public data. None of these modules require a paid license, a cloud subscription, or a credit card. They are pre-integrated into the platform architecture and can be activated independently, allowing operators to tailor the platform's intelligence and resilience to their needs without increasing operational costs. Activation is performed via a single CLI command or a toggle in the Admin Portal (Configuration History records the activation as a versioned change). Groq (A1) generates a notification summarising the activation and any dependencies.

All addons have been reviewed for compatibility with v6.3 modifications : multi-shipment contracts, dual-mode toggle, three-tier AI (Groq + HF local + Ollama), SGTX fee paid by seller, financier preferences, co-financing, distressed cargo enhancements, and the removal of operating modes. Where relevant, each addon description notes how it interacts with these features.

15.1 Categorised Module Inventory (62 Modules, v6.3 Enhanced)

The suite is organised into nine tiers, from foundational AI improvements to advanced autonomous infrastructure. Each module is briefly described, along with its zero-cost technology stack and a concrete example of its use. Modules marked with ★ are new or significantly enhanced for v6.3.

Tier 1: Advanced AI & Machine Learning (Addons 1–6)

Federated Learning Network – Multiple sovereign nodes collaboratively train a shared fraud-detection model without sharing raw data. Uses OpenFL (Apache2.0), PyTorch. Example: three regional nodes (EU, MENA, APAC) train an anomaly detection model on their own trade data; only encrypted model updates are exchanged, resulting in a global model that identifies money-laundering patterns without violating data sovereignty. v6.3 enhancement: now also supports federated training of commodity-specific margin estimation models across nodes, improving fee accuracy for new corridors.

Graph Neural Network Risk Engine – Detects hidden sanctions links and complex fraud patterns by learning over the corporate ownership graph. Uses PyTorch Geometric, GraphRAGRS. Example: the engine discovers that a shell company in a low-risk jurisdiction is indirectly controlled by a sanctioned entity through a chain of four holding companies; the Governor automatically blocks the trade and drafts a SAR. v6.3: extended to multi-shipment contracts; can detect risk patterns across shipments of the same master contract.

Causal Inference Engine – Determines root causes of trade failures or disputes, not just correlations. Uses DoWhy (MIT), EconML (MIT). Example: after a spike of citrus mould disputes, the engine identifies that delays caused by a specific carrier increase the probability of mould claims by 40%, prompting the platform to recommend alternative carriers. v6.3: now analyses multi-shipment delays and financing defaults causally.

Local LLM Operations – Finetuned Llama 3 or Mistral models run entirely on-premise for contract analysis, negotiation, and document drafting. Uses vLLM, HF Transformers (local). Example: a tenant drafts a bespoke force majeure clause using a locally hosted LLM; the model ensures the language complies with Egyptian and Vietnamese law without any data leaving the sovereign node. v6.3: the local LLM also serves as fallback for Smart Inbox title generation and tenant messages when Groq rate limit is exceeded.

TimeSeries Foundation Models – Pretrained models (LagLlama, Apache2.0) for cold-chain forecasting, FX, and disruption prediction, adaptable to new corridors with minimal data. Uses PyTorch. Example: a new trade lane between Kenya and India has sparse sensor data; the model leverages patterns learned from similar tropical routes to predict shelf-life within 5% accuracy. v6.3: enhanced with multi-shipment schedule data for better ETA predictions.

RAG Knowledge System – Retrieval-augmented generation over all platform trade data, legal documents, and regulations for instant Q&A. Uses pgvector, Groq/Ollama, LangChainRs. Example: an buyer asks, “What documents are needed for importing medical devices into Germany?” The system retrieves the latest EU MDR requirements from its regulatory knowledge base and generates a checklist. v6.3: now includes financing agreements and distressed cargo regulations.

Tier 2: Document & Media Intelligence (Addons 7–12)

Document Intelligence Pipeline – Converts any uploaded PDF, scan, or image into structured, verified trade data. Uses HF Donut, PaddleOCR, Tesseract (Apache2.0). v6.3: enhanced to extract multi-shipment schedule references and financing fee terms.

Satellite Analytics – Uses free ESA Sentinel-2 imagery to monitor port congestion, crop yields, and infrastructure. Uses SentinelHub (free tier), HF ViT. Example: detects an unusual queue of vessels outside Alexandria port, adjusts ETA predictions for all inbound shipments by +12h and notifies buyers via Smart Inbox. v6.3: now integrates with distressed cargo detection (e.g., visible damage at port).

Video Analytics – Analyses CCTV or drone footage to verify loading, container condition, and security. Uses OpenCV (Apache2.0), PyTorch Video. Example: a drone flyover confirms that pallets are loaded in the correct sequence; a discrepancy triggers a real-time alert. v6.3: adds overlay for multi-shipment container identification.

Audio Forensics – Verifies voice commands and detects stress, coercion, or impersonation (privacy-preserving, on-device). Uses pyAudioAnalysis, Librosa. v6.3: integrated with biometric stress flag consent workflow.

Handwriting & Signature Verification – Compares uploaded signatures and handwritten declarations against known samples. Uses Siamese Networks (local TF/Keras). v6.3: now used for QC inspector override justification verification.

Multi-Language Real-Time Translation – Translates contracts, packing lists, and mediation messages instantly using local models. Uses No Language Left Behind (NLLB, MIT), LibreTranslate (AGPL). v6.3: extended to translate tenant messages in Decision Panel.

Tier 3: Simulation & Digital Twins (Addons 13–18)

Digital Twin Simulator – Full supply-chain digital twin, from factory floor to final delivery, with real-time sensor inputs. Uses Pyro, custom Rust simulation engine. v6.3: now supports multi-shipment contracts; each shipment has its own twin, and the master contract twin aggregates.

Agent-Based Market Simulation – Models thousands of virtual traders to forecast commodity prices, freight rates, and contract behaviour. Uses Mesa (Apache2.0), Rust ABM. v6.3: includes financing market simulation for co-financing scenarios.

Network Science Analytics – Analyses the global trade network to identify critical nodes, vulnerabilities, and new opportunities. Uses NetworkX (BSD), GraphRAGRS. v6.3: adds distressed cargo network analysis (identifies most frequent distressed buyers).

Monte Carlo Risk Simulation – Runs probabilistic scenarios for contract profitability, FX exposure, and default risk. Uses numpy, scipy. v6.3: now includes multi-shipment cash flow simulations.

Disaster Recovery Simulator – Simulates full node failure and tests RTO/RPO in a sandbox. Uses K3s + Chaos Mesh (Apache2.0). v6.3: includes scenarios where the multi-shipment coordinator fails.

Carbon Market Forecaster – Predicts carbon credit prices and optimal offset strategies for carbon-neutral shipments. Uses LightGBM, free EUETS data. v6.3: integrated with CBAM reporting.

Tier 4: Security & Cryptography (Addons 19–24)

Zero-Knowledge Proofs – ZKproofs for payment verification, reserve proofs, and confidential trade terms. Uses Plonky3, SP1 (Rust). v6.3: enhanced for financing reserve proofs (financier can prove liquidity without revealing wallet).

Secure Multi-Party Computation (MPC) – Enables blind bidding and encrypted credit scoring without revealing individual data. Uses MPyC (MIT), TFHE. v6.3: used for co-financing bid aggregation without revealing individual financier bids until window closes.

Differential Privacy – Protects any aggregated analytics with mathematical guarantees. Uses OpenDP (MIT). v6.3: applied to Anonymous Market Intelligence Panel ($\varepsilon=0.1$), also to financier benchmarking.

Post-Quantum Cryptography – Full integration of Dilithium3 and Kyber1024 for all long-term secrets and identity material. Uses rustpqcrypto, OQS. v6.3: applied to multi-shipment master contract archives.

Self-Sovereign Identity Backup – Shamir Secret Sharing of master keys among trusted contacts, with biometric recovery. Uses sharks (Rust), Vosk. v6.3: extended to financier master keys and platform multisig members.

Hardware Security Module (HSM) Emulator – Software-based HSM for key storage and signing, with remote attestation. Uses SoftHSMv2 (BSD), tpm2-tss. v6.3: deployed for all sovereign nodes to secure Governor signing keys.

Tier 5: Decentralisation & Web3 (Addons 25–30)

DePIN Integration – Integrates with decentralised physical infrastructure networks (Helium, Streamr) for connectivity and sensor data. v6.3: used for low-cost IoT temperature monitoring on perishable goods.

DAO Governance – Allows tenants to form multisignature governance groups for shared decisions (e.g., logistics consortia). Uses Aragon (OSx), Snapshot (free). v6.3: extended to multi-shipment buyer-seller DAOs for shared freight contracts.

Self-Executing Smart Escrow – Conditional FeeLock on Polygon with SGTX oracle attestation. Uses ink!/Solidity, ethers-rs. v6.3: now supports per-shipment escrow for multi-shipment contracts.

Tokenised Carbon Credits – Issues and trades tokenised carbon offsets tied to verified shipments. Uses ERC1155, Toucan Protocol (free bridge). v6.3: integrated with CBAM reports.

Decentralised Identity Bridge – Allows tenants to link their GTID to a W3C DID for cross-platform trust. v6.3: used for financier identity verification across multiple platforms.

On-Chain Audit Trail – Anchors the Loom hash chain to Polygon for public verifiability. v6.3: now includes financing agreement hashes.

Tier 6: Climate & Sustainability (Addons 31–36)

Climate Intelligence – Aggregates IPCC scenarios, local weather risks, and long-term climate trends into trade risk models. v6.3: used for commodity-specific margin adjustments (e.g., drought risk affecting coffee prices).

Geopolitical Radar – Realtime dashboard of sanctions, conflicts, and regulatory changes, with predictive risk scores. v6.3: integrated with RIA; triggers emergency operations mode automatically.

Commodity Intelligence – Global production and consumption forecasts for major agricultural and industrial commodities. v6.3: feeds into margin estimation and financing LTV recommendations.

Biodiversity Impact Assessment – Estimates biodiversity footprint of a trade lane, aiding ESG compliance. v6.3: displayed in Trade Command Center for eco-conscious buyers.

Water Footprint Calculator – Computes virtual water content of shipped goods. v6.3: used in ESG reports for agricultural commodities.

Circular Economy Tracker – Monitors reusable packaging, reverse logistics, and waste reduction across trades. v6.3: integrated with distressed cargo (e.g., packaging recovery after distressed sale).

Tier 7: User Experience & Accessibility (Addons 37–42)

Conversational AI – Full natural-language interface to all portal functions, available in all languages. v6.3: now understands dual-mode context (“Switch me to Seller mode”).

AR Field Operations – Augmented reality for container loading, pallet picking, and damage inspection. v6.3: enhanced with multi-shipment container selection.

Accessibility-First Design – WCAG 2.2 Level AAA compliance, screen-reader optimised, high-contrast modes. v6.3: tested with dual-mode toggle, Smart Inbox, and Decision Panel.

Voice-Driven Workflows – Complete voice control across the platform, offline or online. v6.3: includes voice-activated dual-mode switching.

Haptic Feedback for Mobile – Vibration patterns for different alerts and confirmations during field operations. v6.3: used for scanning pallets belonging to different multi-shipment shipments.

Personalised Dashboards – AI-generated custom dashboards based on each user’s role, preferences, and most used features. v6.3: adapts to dual-mode context; shows Buyer or Seller view accordingly.

Tier 8: Infrastructure & DevOps (Addons 43–48)

Self-Healing Infrastructure – AI-driven automatic scaling, failover, and repair of K3s clusters and services. v6.3: includes multi-shipment coordinator self-healing.

Chaos Engineering – Continuously tests platform resilience by randomly terminating pods, injecting latency, and simulating disk failures. v6.3: tests multi-shipment per-shipment lock reliability.

Continuous Compliance – RIA agent extended to continuously validate platform infrastructure against regulatory frameworks (e.g., GDPR, PDPL). v6.3: now validates data residency for multi-shipment contract data.

Predictive Capacity Planning – LSTM-based forecast of hardware needs for the next 6 months, preventing overprovisioning. v6.3: accounts for growth in multi-shipment contracts and financing volumes.

Immutable Infrastructure with Verifiable Boot – All nodes boot from a signed, verifiable OS image; any tampering is detected. v6.3: includes sovereign node identity attestation.

Automated Penetration Testing – Open-source tools run weekly pentests and generate reports. v6.3: includes tests for financing API endpoints.

Tier 9 : Advanced Analytics & Business Intelligence (Addons 49–62)

Trade Finance Analytics – Aggregated, anonymised statistics on financing rates, success rates, and market trends for financiers. v6.3: includes co-financing metrics.

Carbon Accounting Suite – Full Scope 1/2/3 tracking for every shipment, integrated with EU CBAM reporting. v6.3: includes multi-shipment aggregated carbon reports.

Proprietary Indices – SGTX-specific commodity price index and trade lane risk index, published daily. v6.3: used as benchmark for fee calculations.

Fraud Pattern Library – A growing knowledge base of detected fraud patterns, shared anonymously across sovereign nodes via federated learning. v6.3: includes financing fraud patterns.

Tenant Health Scoring – Composite score combining trust, ESG, and payment behaviour, updated in real time. v6.3: used by financiers to assess borrower risk.

Logistics Performance Benchmarks – Publicly available (opt-in) benchmarks for carrier on-time rates, damage rates, etc. v6.3: includes multi-shipment performance.

Marketplace Partner Analytics – Provides marketplace partners with anonymised conversion rates, average SGTX FEES, and trend reports. v6.3: includes attribution dispute analytics.

AI-Generated Board Summaries – Groq automatically produces a monthly executive summary of platform metrics and notable events. v6.3: includes financing and distressed summaries.

Custom Report Builder – Allows tenants to design their own reports using a drag-and-drop interface, with AI assistance. v6.3: includes multi-shipment and financing report widgets.

Scenario-Based Forecasting – Tenants can run “what-if” scenarios (e.g., new tariff, port strike) on their trade pipeline. v6.3: includes financing scenarios (e.g., interest rate hike).

Anomaly-Driven Investigation Workbench – When an anomaly is detected, all related data is automatically gathered into a timeline for investigators. v6.3: includes financing and distressed anomalies.

Regulatory Readiness Dashboard – Shows upcoming regulatory changes that may affect the tenant’s trade lanes, with AI-generated impact assessments. v6.3: includes financing regulations (e.g., MiCA).

Continuous Auditing for Financial Controls – Automated testing of internal controls (e.g., segregation of duties) with evidence generation. v6.3: includes dual-mode toggle access auditing.

Global Trade Observatory – A public-facing, privacy-preserving visualisation of global trade flows (opt-in), powered by SGTX data. v6.3: includes anonymised financing flow visualisation.

15.2 Implementation Priority Matrix (v6.3 Updated)

The matrix ranks modules by Impact, Effort, and Priority (P1 : immediate-term, months 1-6; P2: short-term, months 7-12; P3: medium-term, months 13-18; P4: future). All modules can be activated independently, no dependencies except those noted.

Example of high-priority (P1) modules for v6.3:

Federated Learning Network (P2 – depends on multi-node deployment)

GNN Risk Engine (P2)

Digital Twin Simulator (P1 – already part of Phase 5)

ZKproofs (P1 – used for financing reserve proofs)

DeFi Protocol Risk Oracle (P1 – already part of Phase 4)

Conversational AI (P1 – Smart Inbox AI summary)

Self-Healing Infrastructure (P1 – core platform reliability)

Tenant Health Scoring (P1 – used in Company Admin)

Global Trade Observatory (P3)

15.3 Total Cost of Ownership (TCO) – $0

Cost Category	Why It's $0
AI/ML Training & Inference	All models open-source; run on local hardware (vLLM, ONNX, PyTorch). Groq free tier for fast summarisation; Ollama fallback.
Data Sources	Free public APIs (AISHub, OpenWeatherMap, ECB, FAO, ESA Sentinel, Copernicus, The Graph). Sanctions lists from OFAC/UN/EU.
Infrastructure	Self-hosted on bare-metal K3s; no cloud VMs. Server hardware is capital expense, not operational.
Blockchain / Web3	Polygon PoS gas fees negligible (~$0.01-0.10 per contract). Self-deployed contracts; no SaaS platform fees.
Security & Cryptography	All crypto libraries open-source (Plonky3, sharks, OpenDP, SoftHSM). No HSM appliances needed.
Monitoring & Observability	Prometheus, Grafana, Loki, Jaeger, ClickHouse – all OSS.
Development & Maintenance	Internal team using open-source IDEs, CI/CD (Gitea + Drone, self-hosted).

No billing details ever required.

15.4 Example Activation Process (v6.3 Enhanced)

All modules are activated via a single CLI or Admin Portal toggle. An Ansible playbook (or shell script) provisions the required services. Activation is recorded in Configuration

History (Part 6.2.7.3.9). Groq generates a notification for the Admin Portal's Smart Inbox summarising the activation.

Example : Activating the Digital Twin Simulator (Addon 13) – already part of Phase 5, but for completeness:

CLI command : ./addons/activate.sh --addon digital-twin --node cairo-prod --scale 3

Builds the digitaltwin engine from source.

Deploys as a K3s deployment.

Registers with AI orchestrator.

Groq notification : "Digital Twin Simulator activated on Cairo node (3 replicas). You can now simulate port congestion, weather delays, and 'what-if' scenarios from the Trade Command Center."

All addons follow the same self-service, zero-cost activation pattern. Tenants can request activation of tenant-visible addons (e.g., personalised dashboards) from their Company Admin tab; the Platform Governance Authority approves via multisig.

END OF PART 15 – v6.3 WITH ALL MODIFICATIONS

Part 16: KYB/KYC & Registration – Aidriven, Jurisdiction-Dynamic (v6.3 – Fully Upgraded, Zerocost, with All Modifications: Three-Tier AI, Multi-Shipment, Dual-Mode Toggle, Onboarding Wizard Integration, Manual Review SLA, Private Financier Support, and Removal of Operating Modes)

16.0 Purpose & Scope

Every tenant on the SGTX platform – corporate trader, logistics provider, financier, QC, government, marketplace partner – must be verified before they can initiate or participate in any trade. The KYB/KYC layer acts as the platform's immune system, ensuring that only legitimate, sanction-cleared, and risk-profiled entities gain access.

The process is jurisdiction-dynamic : the required documentation, verification tier, and acceptable identity providers vary by country and are continuously updated by the Regulatory Intelligence Agent (RIA). It is AI-driven: Hugging Face Donut and PaddleOCR extract data from uploaded documents, local NLP models screen against sanctions and PEP lists, and XGBoost models compute initial trust scores. It is privacy-preserving: all document processing runs locally, biometric liveness is handled via ZITADEL's standard WebAuthn (no third-party service), and sensitive data never leaves the tenant's sovereign node.

In v6.3, the KYB/KYC process is tightly integrated with the Tenant Onboarding Wizard (Part 2.7) to provide a smooth, plain-language, step-by-step experience. A fallback manual review queue is provided for jurisdictions where government registries are unavailable or document extraction confidence is low. SLA transparency and queue position are displayed in the Smart Inbox. Operating modes (SIMPLE/ADVANCED/ENTERPRISE) have been removed – all tenants have uniform access, with permissions controlled solely by RBAC and data scopes. Dual-mode toggle(Buyer/Seller) is configured after onboarding, not during KYB.

16.1 KYB/KYC Tiers & Jurisdiction Requirements

KYB/KYC is not one-size-fits-all. Four tiers are defined, and each country's required tier is stored in jurisdictions.kyc_tier_required, continuously updated by the RIA.

Tier	Typical Requirements	Applicable To
Tier 1 – Basic	Valid business registration, proof of address, tax ID.	Low-risk jurisdictions, SMEs with transaction limits.

	Liveness check via ZITADEL passkey.	
Tier 2 – Standard	Tier 1 + government-issued photo ID of UBOs and directors, PEP/sanctions screening, ownership structure disclosure.	Most corporate tenants, logistics providers, QC agencies.
Tier 3 – Enhanced	Tier 2 + certified financial statements, bank reference letter, enhanced due diligence (EDD) questionnaire for high-risk jurisdictions or politically exposed persons.	Traders in high-risk corridors, large financial institutions, government contractors.
Tier 4 – Maximum	Tier 3 + on-site audit (where possible), additional regulatory licences, background checks on all key employees.	Central banks, sovereign wealth funds, entities in special economic zones.

The jurisdiction matrix dynamically assigns the required tier. The RIA may upgrade a country's requirement based on new regulations or downgrade it when a country's risk profile improves. When a change occurs, all existing tenants in that jurisdiction are re-evaluated within 24h; if they no longer meet the new tier, they receive a Smart Inbox COMPLIANCE item and a grace period to upload additional documentation (default 30 days). During the grace period, they can continue trading but new trade requests are blocked. Reminders are sent at 15 and 5 days.

16.2 Document Extraction & Verification (Zero-Cost AI Pipeline)

The automated verification pipeline runs entirely on the platform's sovereign infrastructure.

16.2.1 Document Types & Extraction

Tenants upload documents through the Onboarding Wizard (Step 3) or the Company Admin tab. Supported formats: PDF, JPEG, PNG, TIFF. Typical documents:

Business Registration Certificate

Proof of Address (utility bill or bank statement)

Tax ID Certificate

UBO Declaration (list of ultimate beneficial owners with percentages)

Government-Issued Photo ID (passport or national ID) for directors/UBOs

Extraction Pipeline:

Preprocessing : PaddleOCR (local) detects text regions and performs OCR if the document is an image. HF Donut (local) handles structured and semi-structured documents (e.g., registration certificates with tables).

Field Extraction : Extracted text parsed using rule-based templates + a finetuned DistilBERT model that maps free-text fields to structured keys (legal_name, registration_number, etc.). Confidence scores (0-100) assigned.

Cross-Validation : The system attempts to validate extracted data against free government registries (e.g., UK Companies House API, Egyptian MCDR via scraping). If registry reachable and data matches, confidence boosted. If registry unreachable, system falls back to extracted data and queues for manual review (see 16.5).

Sanctions & PEP Screening : Legal name and all UBO names are embedded using multilingual sentence-transformer (HF all-MiniLM-L6-v2) and compared against the platform's continuously updated sanctions and PEP caches using pgvector cosine similarity. Match >0.92 triggers human review flag.

Trust Score Calculation : The trust-score-service (XGBoost) computes an initial trust score (0-100) based on verification completeness, sanctions status, jurisdiction risk, and company age. Visible only to the tenant and to platform governance; counterparties see a summarised trust portrait (Part 18).

16.3 Biometric Liveness & KYC

For Tier 2 and above, individual KYC is required for directors and UBOs. SGTX uses the existing ZITADEL WebAuthn infrastructure – no third-party biometric service.

Process : Individual receives email invitation, logs into a dedicated KYC portal, registers a WebAuthn passkey (fingerprint/face), and uploads a photo of their government-issued ID. The passkey registration provides FIDO2-level biometric liveness.

Document Verification : ID processed by Donut/PaddleOCR; photo checked for basic tampering using a local ViT model. Extracted name and ID number cross-referenced with UBO declaration.

Cost: No per-verification fee.

16.4 PEP & Sanctions Screening (Real-Time, Continuous)

All entities and individuals screened at registration and continuously thereafter.

Data Sources (free):

OFAC SDN, UN Consolidated List, EU Sanctions List, UK Sanctions List, and hundreds of national sanctions lists, scraped daily.

PEP lists compiled from open government portals and Wikipedia-derived datasets (AI-verified to reduce false positives). RIA refreshes PEP cache weekly.

Screening Process:

Fuzzy Name Matching : Name embeddings compared via pgvector; similarity threshold 0.92 flags for human review.

Graph Proximity : corporate-graph-service runs GraphRAG traversal from tenant's UBOs to detect indirect links to sanctioned entities (distance ≤3 hops, high-confidence path triggers risk alert).

Continuous Rescreening : Every 6 hours, all active tenants rescreened against latest lists. If a new sanction names an existing tenant, Governor automatically suspends trading ability and notifies compliance team.

16.5 Manual Review Fallback & SLA Transparency

Automated verification is efficient, but no AI is perfect. The platform provides a manual review queue accessible to the Platform Governance Authority and designated compliance officers.

Triggers for manual review:

Any extracted field with confidence <85%.

Sanctions/PEP screening flag (fuzzy match).

Government registry unreachable (e.g., MCDR down).

Document tampering detection confidence >70% (from ViT model).

Tenant voluntarily requests manual review (“Request Human Review” button in Onboarding Wizard).

SLA Transparency (v6.3) : When a tenant’s KYB/KYC status enters PENDING_MANUAL, the tenant’s Smart Inbox immediately receives a high-priority item (score 85) that includes:

Estimated response time (from jurisdictions.kyb_review_sla, default 48 hours).

Queue position (number of pending tenants ahead in the same jurisdiction).

Responsible team: “Compliance Team – [Jurisdiction Name]”.

Escalation : “If not resolved within 48 hours, escalate to Platform Governance Authority.”

The tenant can click “View status” to see a live dashboard. If SLA is exceeded, a follow-up Smart Inbox item (priority 95) is sent, and the Platform Governance Authority is notified. All SLA breaches are logged in compliance_events with severity MODERATE.

16.6 Integration with Onboarding Wizard (v6.3)

KYB/KYC is Step 3 of the Tenant Onboarding Wizard (Part 2.7). The wizard:

Guides the tenant to upload required documents based on jurisdiction and entity type.

Displays real-time progress : “KYB Tier 2: 75% complete – upload proof of address.”

Provides plain-language explanations : “We need your business registration to verify your company’s legal identity. This document is never shared with other traders.”

Gracefully handles manual review fallback : if registry unreachable, the wizard proceeds but shows a banner: “We couldn’t verify your registration automatically. Your documents will be manually reviewed within 48 hours. You may continue setup while we verify.”

16.7 Governance Gates (KYB/KYC)

Gate ID	Check	Enforcer	Failure Action
KYB1	All required documents for jurisdiction's tier are uploaded and extracted with confidence ≥85% or flagged for manual review	KYB Validator (A2)	Set status to PENDING_MANUAL; block trade initiation for Tier 3+ until resolved
KYB2	Entity passes sanctions and PEP screening (fuzzy and graph)	Sanctions Agent (A2/A3)	DENY or CONDITIONAL; block trade initiation
KYB3	Biometric KYC completed for all required individuals (Tier 2+)	ZITADEL + KYC Agent (A2)	Set KYC status to PENDING; restrict employee login until verified
KYB4	Tenant's KYB/KYC tier meets or exceeds jurisdiction's current required tier	KYB Threshold (WasmEdge)	Block all new trade requests until upgrade completed; provide grace period and Smart Inbox notification
KYB5	All changes in tenant details (UBO, address, legal name) trigger automatic reverification	Compliance Triggers (A2)	Rescreen sanctions and PEP; if risk score changes by >10 points, flag for review
KYB6	Manual review completed within SLA (48h default)	Governor (A3)	Auto-escalate to senior compliance officer; notify tenant of delay

16.8 Data Retention & Privacy

All uploaded documents stored on tenant's sovereign node, encrypted at rest (AES256GCM).

Tenants can request deletion of their documents upon account closure; processed within 30 days, subject to statutory retention (logged immutably).

Trust score and verification status shared with counterparties as part of GTID resolution; raw documents never shared.

16.9 Zero-Cost Tech Stack

Component	Technology	Cost
Document OCR & Extraction	HF Donut, PaddleOCR	$0 (local)
Field Parsing &	DistilBERT (finetuned)	$0 (local)

Classification		
Sanctions/PEP Screening	Sentence-transformers, pgvector, GraphRAGRS	$0 (local)
Biometric Liveness	ZITADEL WebAuthn (built-in)	$0
Trust Score Calculation	XGBoost	$0 (local)
Government Registry Scraping	curl + HF Donut	$0 (free APIs)
Manual Review Queue	Admin Portal (Governor decisions)	$0
Data Encryption	AES256GCM (built-in)	$0

No external identity verification services, no per-check fees, no cloud dependencies.

16.10 Implementation Checklist (macOS/Linux – ZeroCost)

Summary:

Build kyb-kyc-service.

Seed sanctions cache and set up RIA monitor.

Apply KYB-specific database migrations (if any).

Test document extraction pipeline.

Verify ZITADEL passkey integration (manual test with a real device).

END OF PART 16 – v6.3 WITH ALL MODIFICATIONS

Part 17: Marketplace API & Revenue Share (v6.3 – Fully Upgraded, Global, Zerocost, with All Modifications: Auto-Attribution, Rate Limit Gauges, Sandbox, Webhook Management, Dispute Resolution, and Partner Self-Service)

17.0 Philosophy

Marketplaces are not competitors – they are powered by SGTX. The Marketplace API allows any registered partner to embed SGTX's execution engine into their own platform, providing their users with cryptographic trade certainty, AI-optimised logistics, and zero-counterparty-risk settlement while earning a negotiated share of the trade fee. The API is fully sovereign: all endpoints pass through the Governor, all revenue share attribution is cryptographically signed, and every action is immutably logged. The partner never touches the underlying trade funds; they simply connect buyers and sellers and are rewarded for successful matches.

In v6.3, the Marketplace Partner Portal (Part 6.2.9) provides a self-service interface for these partners. The API now includes transparent rate limits with per-endpoint monitoring, a gauge chartin the partner dashboard, and a mechanism to request limit increases. Attribution is automatic (first-touch) with a dispute window. All modifications from earlier (removal of operating modes, three-tier AI, SGTX fee paid by seller, etc.) are reflected in the partner revenue share calculation – the partner receives a percentage of the SGTX trade fee (not the financing fee).

17.1 Architecture & Zero-Cost Stack

All partner interactions use standard HTTPS REST endpoints served from the SGTX API gateway (Nginx + WasmEdge filters). No special SDK is required – partners send signed JWTs obtained from ZITADEL (self-hosted) and receive chain-verified responses.

Component	Technology	Purpose
API Gateway	Nginx + Lua + WasmEdge	Validates partner JWT and routes requests to marketplace-api-service.
Authentication	ZITADEL (OIDC, Ed25519 public key pinned)	Partners obtain an API key and sign all requests using their Ed25519 private key.
API Server	Rust, Axum	Handles all partner endpoints (/v1/partner/*).

AI Routing	Rig (local)	Orchestrates calls to Groq / HF / Ollama when needed.
Revenue Attribution	PostgreSQL, NATS JetStream	Stores partner leads, tracks conversions, calculates revenue splits.
Analytics	ClickHouse + OpenDP	Generates privacy-safe, real-time dashboards for partners.
Document Generation	Tera + headless Chrome	Produces signed revenue-share agreements.

Zero-cost and no external billing required. All services run on the same bare-metal K3s cluster as the rest of SGTX.

17.2 Marketplace API – Endpoint Specifications (v6.3, with Rate Limits, Idempotency, and Standard Errors)

All endpoints are served under /v1/partner. Requests must include an Authorization: Bearer <partner_jwt> header. The JWT contains the partner_id and is signed with the partner’s Ed25519 private key. The Governor validates every request.

Standard Error Responses (all endpoints):

All errors return a JSON object with an error field containing code, message, and optionally retry_after_seconds and governor_decision_id. Common HTTP status codes:

400 INVALID_REQUEST – malformed JSON, missing required field.

401 UNAUTHORIZED – missing or invalid JWT (expired, wrong signature).

403 FORBIDDEN – partner suspended, jurisdiction blocked, IP not whitelisted.

404 NOT_FOUND – resource does not exist.

409 CONFLICT – duplicate attribution, state mismatch, or idempotency key replayed with different payload.

429 RATE_LIMIT_EXCEEDED – too many requests in current window; includes Retry-Afterheader.

500 INTERNAL_ERROR – platform error (rare); retry with exponential backoff.

503 SERVICE_UNAVAILABLE – platform temporarily unavailable (maintenance, overload).

Rate Limit Headers : Every response includes X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset. For 429 responses, Retry-After (seconds) is also provided.

Idempotency Keys : All POST endpoints support an optional Idempotency-Key header (UUID v4). The platform stores the first response for 24 hours; replaying the same key returns the original response. Replaying with a different payload returns 409 CONFLICT.

17.2.1 POST /v1/partner/intent/analyze

Purpose : Submit a raw trade intent from the marketplace's end user (buyer). The system extracts structured data, scores the lead's viability, and returns a prefilled trade request ready for the buyer to confirm.

Request Headers : Authorization, Idempotency-Key (optional), Content-Type: application/json.

Request Body:

json

{

"raw_text": "I want 20,000 kg of Valencia oranges from Vietnam, CFR Alexandria",

"source_language": "en",

"marketplace_user_id": "mp-12345",

"additional_context": {

"buyer_gtid": "SGTX-EG-TRD-000123", // optional, if known

"preferred_currency": "USD"

}

}

Processing:

Intent Parser (A2, HF local + Groq) extracts structured commodities, quantities, incoterm.

Lead Scorer (A2, LightGBM) evaluates viability (0-100) based on commodity, trade lane, current market conditions, and partner's historical success.

Container Advisor (A1, Groq/Ollama) suggests container type and count.

Compliance prescreen (G1U5) returns ALLOW or CONDITIONAL.

Response (200 OK):

json

{

"intent_id": "int-9876",

"status": "ACCEPTED",

"viability_score": 87,

"parsed_specs": { ... },

"recommendations": {

"container_type": "40ft High-Cube Reefer",

"estimated_containers": 2,

"suggested_next_action": "Redirect buyer to SGTX to confirm and enter seller GTID."

},

"ai_explanation": "This trade lane is active. Viability score 87 indicates high probability of successful match."

}

Status values:

ACCEPTED (score ≥80, compliant) – partner should redirect buyer to SGTX.

CONDITIONAL (score 50-79 or compliance warning) – partner should collect missing information (e.g., HS code, quantity).

REJECTED (score <50, sanctions, unsupported corridor) – partner should not proceed; reason in ai_explanation.

17.2.2 POST /v1/partner/trade/initiate

Purpose : Called after the buyer confirms the trade request on the SGTX portal. Ties the confirmed trade to the marketplace partner for revenue attribution.

Request Body : { "intent_id": "int-9876", "trade_request_id": "tr-12345" }

Response (200 OK):

json

{

"attribution_id": "att-5678",

"status": "ATTRIBUTED",

"revenue_share_pct": 0.50,

"ustn": "SGTX-EG-20260415120000-ABC12345-V1"

}

Error scenarios : 404 NOT_FOUND (intent not found), 409 CONFLICT (trade already attributed to another partner, or idempotency key mismatch).

17.2.3 GET /v1/partner/ratelimits (v6.3)

Purpose : Return current rate limit configuration and usage for the partner.

Response:

json

{

"partner_id": "partner-001",

"limits": [

{ "endpoint": "/v1/partner/intent/analyze", "limit": 100, "window_seconds": 3600, "current_usage": 85, "remaining": 15, "resets_at": "2026-06-14T14:00:00Z" },

{ "endpoint": "/v1/partner/trade/initiate", "limit": 50, "window_seconds": 3600, "current_usage": 12, "remaining": 38, "resets_at": "2026-06-14T14:00:00Z" }

]

}

17.2.4 POST /v1/partner/ratelimits/increase (v6.3)

Purpose : Submit a request to increase the rate limit for a specific endpoint. Requires partner admin role.

Request Body:{ "endpoint": "/v1/partner/intent/analyze", "requested_limit": 200, "justification": "Our user base has grown 40% this quarter." }

Response (202 Accepted):{ "request_id": "rlr-001", "status": "PENDING_MULTISIG", "message": "Your request has been submitted. The Platform Governance Authority will review it within 48 hours." }

Status values : PENDING_MULTISIG, APPROVED, REJECTED. Partner can track status via a subsequent GET endpoint (not detailed).

17.2.5 POST /v1/partner/webhook/register

Purpose : Register a webhook URL to receive real-time events for attributed trades.

Request Body:{ "webhook_url": "https://partner.example.com/sgtx-events", "events": ["trade.locked", "SGTX FEES.settled", "dispute.filed"], "secret": "optional_hmac_secret" }

Response (200 OK):{ "webhook_id": "wh-abc123", "status": "ACTIVE", "verification": { "ping_sent_at": "...", "ping_response_status": 200 } }

Events : trade.locked (contract signed, USTN generated), SGTX FEES.settled (trade fee collected and settled to SGTX), dispute.filed (dispute involving an attributed trade), lead.rejected (intent rejected). Webhook payloads are signed with HMAC-SHA256 if a secret is provided.

17.3 Revenue Attribution Logic & Auto-Attribution

First-touch attribution is the default : the marketplace partner that introduced the buyer and seller receives a share of the SGTX trade fee for that trade and for any subsequent direct trade between the same parties within 12 months (if the parties did not already have prior trading history before the introduction).

Auto-attribution (v6.2/v6.3) : When a new trade request is created, the system checks partner_lead_attributions for the most recent marketplace-sourced trade between the same buyer and seller. If found, it automatically sets marketplace_partner_id and revenue_share_pct on the trade request. Both parties see a banner explaining the attribution and can dispute within 72 hours.

Dispute resolution : If disputed, AI (A2) analyses the case; if confidence <70 or dispute is complex, it escalates to Platform Governance Authority (multisig). Attribution can be upheld, removed, or adjusted.

Revenue share calculation : Partner’s share = (negotiated percentage) × (total SGTX trade fee collected). For multi-shipment contracts, the attribution applies to the master contract, and the partner receives a share of each shipment’s trade fee proportionally. Financing fees are not shared.

17.4 Governance Gates (Marketplace-Specific)

Gate ID	Check	Enforcer	Failure Action
GM1	Partner JWT valid, rate limit not exceeded	API Gateway (WasmEdge)	429 / 401
GM2	Lead score below critical threshold → reject	Lead Scorer (A2)	Return REJECTED with reason
GM3	Sanctions / jurisdiction compliance passed	Governor (A3)	Block trade initiation
GM4	Revenue share split within negotiated bounds	Governor (WasmEdge)	Revert to default; alert admin
GM5	All attributed trades have a Loom hash	Governor (A3)	Hold attribution
GM6	Webhook delivery retried up to 3 times; if still fails, logged	Webhook Manager (A2)	Alert partner and Platform Governance

17.5 AIPowered Partner Features (v6.3)

AI Revenue Optimizer : XGBoost model analyses partner performance (conversion rate, deal size, dispute rate) and recommends revised revenue-share splits within bounds. Groq drafts proposal.

AI-Generated Legal Addenda : Clause Forge (HF Mixtral) drafts custom revenue-share agreements per partner, translated to local language via Groq.

Synthetic Sandbox Data : Sandbox environment uses Groq-generated realistic, anonymised trade scenarios for testing, reset daily.

Automated Revenue Share Dispute Resolution : Disputes trigger AI analysis; if confident misattribution, recommended reversal; otherwise escalates to multisig.

17.6 Implementation Checklist (macOS/Linux – ZeroCost)

Summary:

Build marketplace-api-service.

Seed test partner (ZITADEL client).

Generate test JWT and call intent/analyze.

Test rate limit endpoint and increase request.

Register webhook and simulate test ping.

END OF PART 17 – v6.3 WITH ALL MODIFICATIONS

Part 18: Saved Contacts / Network Feature (v6.3 – Fully Upgraded, Global, Zerocost, with All Modifications: Dual-Mode Aware, Multi-Shipment, Financing, Distressed, Voice-Driven, Predictive Health, Organization Memory Layer, and Indirect Connections)

18.0 Philosophy

The Saved Contacts feature transforms the SGTX platform from a mere execution engine into a relationship intelligence hub. Every time a tenant interacts with another – a trade, a quote, a message, a distressed cargo notification, a financing agreement – the relationship is automatically recorded and enriched with AI-driven insights. This is not a marketplace: the platform never suggests or ranks contacts for cold outreach. It only provides deep, dynamic intelligence on entities a tenant already knows or has directly interacted with. All data is stored privately per tenant. Contacts are never shared between tenants.

All features are built on the zero-cost, sovereign stack. In v6.3, the Network tab is fully integrated into the Buyer, Seller, Logistics, Financier, and Government portals, and includes:

Delivery performance dashboards (for buyers)

Predictive relationship health scores (trend prediction)

Indirect connection discovery via GraphRAG (shared logistics providers, common UBOs, sanctions proximity)

Voice-driven contact management (hands-free)

Privacy-preserving verifiable credentials (ZKproofs for certifications)

Organization Memory Layer (learns recurring trade patterns to pre-fill workflows – see 18.7)

The feature respects the dual-mode toggle : an employee in Buyer mode sees sellers, logistics providers, financiers; switching to Seller mode shows buyers, logistics partners, QC providers. For multi-shipment contracts, contacts are linked to the master contract and each shipment’s counterparty.

18.1 Core Data Model (Extended)

Each relationship is stored in the tenant_contacts table (see Part 5). Key columns with v6.3 additions:

Column	Type	Description
tenant_id	UUID	The tenant that owns this contact relationship.
contact_gtid	TEXT	GTID of the other tenant.
relationship_type	TEXT	BUYER, SUPPLIER, LOGISTICS_PROVIDER, FINANCIER, QC_PROVIDER, INSURER, GENERAL.
trade_count	INTEGER	Number of completed trades between the two (including multi-shipment shipments).
total_value	DECIMAL	Total trade value (USD) across all trades.
first_interaction	TIMESTAMPTZ	Date of first trade, financing, or message.
last_interaction	TIMESTAMPTZ	Date of last activity.
is_favorite	BOOLEAN	Manually starred by user.
is_blocked	BOOLEAN	Prevent further contact (blocks new trade requests, financing, distressed outreach).
trust_snapshot	JSONB	AI-generated 360° portrait (refreshed daily).
relationship_health_score	DECIMAL(5,2)	Predicted future strength (0-100).
health_summary	TEXT	Groq-generated plain-language explanation.
smart_labels	TEXT[]	e.g., RELIABLE_SUPPLIER, HIGH_RISK_JURISDICTION, INACTIVE_12M, FREQUENT_FINANCING.
indirect_connections	JSONB	GraphRAG-discovered 2-hop links.
last_ai_update	TIMESTAMPTZ	Last refresh timestamp.

18.2 Automatic Contact Saving (No User Action Required)

Contacts are saved automatically on the following triggers (dual-mode aware : only contacts relevant to the current mode are shown, but all are saved):

Trade request created (buyer ↔ seller).

Logistics provider quote accepted (seller ↔ logistics provider).

Financing agreement completed (financier ↔ borrower).

Direct message exchanged via Trade Room (both parties).

Distressed cargo notification sent (sender ↔ recipient).

QC inspection booking confirmed (seller ↔ QC provider).

No notification is sent; the contact appears silently in the Network tab. The relationship type is inferred from the interaction context.

18.3 AI-Enriched 360° Trust Portrait (Daily Refresh)

For each saved contact, the network-service spawns a daily AI update job (or on-demand when viewed):

Internal data aggregation : trade count, total value, average payment delay, dispute rate, on-time delivery %, document accuracy, financing repayment performance (if applicable), distressed purchase history.

External intelligence (privacy-safe) : A local HF NLP model scrapes free public news headlines (RSS) about the contact's company, analyses sentiment, flags major events (expansion, lawsuits, regulatory actions). Only public, non-PII data is used.

Synthesis (Groq, fallback Ollama) : All data is fed to Groq, which produces a concise, plain-language portrait stored as trust_snapshot JSONB. Example:

json

{

"summary": "Mekong Fresh has been a supplier for 14 months. Completed 12 trades totaling $480,000. Average payment time net+2 days, zero disputes. Recent expansion into European markets. New GOTS certification. Overall trust score: 91/100. Relationship stable and growing.",

"trust_score": 91,

"payment_reliability": "Excellent (100% on time)",

"dispute_rate": "0%",

"recent_news": "Positive expansion into EU; new organic certification",

"recommendation": "Continue standard terms; consider volume discount."

}

The portrait is displayed in the Saved Contacts detail panel. Users see the last update timestamp.

18.4 Predictive Relationship Health Score

An LSTM model (trained on historical interaction patterns across the platform, including financing and distressed interactions) predicts the relationship's trajectory over the next 90 days.

Input features:

Trade frequency slope (increasing, flat, declining).

Payment behaviour trend (faster/slower).

Communication responsiveness (from chat timestamps).

Dispute frequency trend.

Market conditions for shared commodity.

Financing repayment trend (if applicable).

Distressed purchase frequency.

Output : Health score (0-100) and a one-sentence Groq summary, visible only to the tenant (never shared). Example: “Health score 68 – relationship has cooled slightly. Nile Foods has reduced order volume by 20% and replies to messages 40% slower than six months ago. Consider a check-in call to reengage.”

18.5 GraphRAG-Powered Indirect Connections

Using GraphRAGRS (Rust), the system traverses the corporate graph (entity_nodes, relationship_edges) and trade interactions to discover indirect links between the tenant and the saved contact. Displayed in an expandable panel within the contact detail view.

Examples:

“Mekong Fresh shares a logistics provider (Fast Trucking Co.) with your contact Cairo Logistics.”

“A director of Mekong Fresh is also a UBO of a company that was recently added to a sanctions list – 3-hop distance. Risk: LOW (reviewed by compliance).”

“Your competitor Nile Foods also trades with Mekong Fresh (discovered via shared port usage).”

“Mekong Fresh has financed with Trade Finance Bank, which you also have a financing agreement with – potential syndication opportunity.”

Connections help assess business synergy and hidden risk. Data refreshed weekly.

18.6 Smart Labels & Dynamic Grouping

An unsupervised clustering model (k-means / DBSCAN) runs weekly over all contacts of a tenant, grouping by:

Trade frequency and value

Payment reliability

Jurisdiction risk

Commodity type

Financing frequency

Distressed purchase behaviour

Each contact receives automatic smart labels : RELIABLE_SUPPLIER, HIGH_RISK_JURISDICTION, INACTIVE_12M, PERISHABLE_SPECIALIST, FREQUENT_FINANCING, DISTRESSED_BUYER. Tenants can also define custom labels manually. Labels are used for filtering in the Network tab and for the predictive relationship model. Example filter: “Show all contacts labelled RELIABLE_SUPPLIER for citrus.”

18.7 Organization Memory Layer (v6.3 – New)

The platform learns from recurring trade patterns across the entire organisation and proactively pre-fills workflows, suggests defaults, and reduces repetitive data entry. This is opt-in (tenant admin enables in Company Admin). Learned patterns are stored per tenant and never shared.

What is remembered for each combination of (commodity HS chapter, origin country, destination country):

Preferred incoterm (most frequently used).

Typical packaging (carton dimensions, pallet type, stacking height).

Common logistics providers (GTIDs of providers most often selected, with success rates).

Average EXW price range (mean ± std dev).

Preferred financing terms (typical tenor, LTV, collateral type).

Required documents (almost always requested).

Where memory is used:

New Trade Request : When buyer enters seller GTID and commodity, system prefills incoterm, packaging, container type based on memory. Banner: “Based on 12 similar trades, we prefilled CFR incoterm and 3×40′ reefers. Adjust if needed.”

EXW Price Lock : AI-recommended price range uses remembered average as an additional signal.

Logistics Builder : Highlights providers used successfully before with a “Frequent partner” badge.

Financing : Prefills terms for financiers (“Your typical APR for citrus is 5.2%”).

Clear memory : Tenant admin can click “Clear All Memory” (requires Governor approval, logged).

18.8 Voice-Driven Contact Management (Hands-Free)

Integrated with the VoiceCommandButton (Part 6.5), users can manage their network entirely by voice, respecting dual-mode context (e.g., in Buyer mode, voice commands only apply to seller contacts). Examples:

“Show details for Mekong Fresh.” – opens contact card.

“Add Nile Foods to my favourites.” – toggles is_favorite.

“Block any contact from Russia.” – filters and updates is_blocked for matching contacts.

“List all suppliers who haven’t traded in 6 months.” – runs query, displays results.

“Show me financiers with whom I have active loans.”

Processing : Vosk (offline) transcribes; HF Mixtral extracts intent and parameters; network-serviceexecutes. Available in web, mobile, and desktop.

18.9 Privacy-Preserving Verifiable Credentials

A saved contact can prove certain attributes (e.g., “ISO 9001 certified”, “GOTS certified organic”) without uploading full documents. Using Plonky3 (ZK proofs), the contact’s tenant generates a proof that a specific credential is valid and issued by a recognised authority. The proof is stored alongside the contact and can be verified with a single click, displaying a green checkmark and the issuing authority.

Example : Mekong Fresh holds a GOTS organic certification. They generate a ZKproof that the certificate is valid and issued by an accredited body, without revealing the certificate document itself. Nile Foods, viewing Mekong Fresh’s profile, sees a badge “□ GOTS Organic – verified”. Clicking shows the issuing authority and proof’s creation date. Nile Foods never accesses the original document.

18.10 Network-Wide Anomaly Alerts

An Isolation Forest model (scikit-learn, local) checks all saved contacts daily. If a contact exhibits a sharp deviation – sudden payment delays, negative news sentiment spike, new PEP link, trade volume collapse, financing default – the tenant receives a real-time Smart Inbox alert (A1). Groq summarises: “Alert: Nile Foods has had 3 late payments in the last month (previously 0). A recent news article reports financial restructuring. Consider reducing credit exposure until clarified.”

Alert is advisory only – does not auto-block trades. Tenant can dismiss; if situation improves, alert clears automatically.

18.11 Portal Integration – Network Tab (Dual-Mode Aware)

The Network tab appears in every portal as follows:

Portal	What is shown
Buyer	Saved sellers, logistics providers, financiers, QC providers. Each seller includes Delivery Performance Dashboard (ontime %, avg delay, document accuracy, dispute rate). “Start Trade” button prefills New Trade Request.
Seller	Buyers, logistics partners, QC providers, financiers. Buyers’ performance includes order frequency, negotiation speed, payment behaviour, financing history.
Logistics	Clients and other logistics entities limited to specific services provided.
Financier	Borrowers and lenders. Includes repayment history, current exposure, and co-financing partners.
Government	Tab may be hidden or renamed “Counterparties” with limited features (only other government agencies).
Admin	Not used.

All Network tabs share:

Searchable, sortable table with name, trust score, relationship health, last trade.

Quick actions : “View Profile”, “Start Trade” (if applicable), “Notify Distressed”.

Detail panel with full trust portrait, health trend graph, indirect connections.

Filters by smart labels, jurisdiction, commodity, relationship type.

18.12 Governance & Privacy Gates

Gate ID	Check	Enforcer	Failure Action
GN1	Only direct interactions autosave a contact; no scraping of other tenants’ data	Governor (OPA)	Block save
GN2	Contact’s trust portrait uses only public info + internal data tenant is authorised to see	Data Scope (WasmEdge)	Remove unauthorised data
GN3	Voice commands cannot delete or permanently alter critical data without confirmation	Governor (A2)	Require visual confirmation
GN4	All AI-generated summaries logged with confidence scores and can be reviewed	Loom	–

18.13 Zero-Cost Technology Stack

Component	Technology
Contact CRUD & sync	Rust (network-service)
Graph relationship discovery	GraphRAGRS (Rust)
Trust portrait & summaries	Groq (free tier) + HF NLP (local)
Relationship health prediction	LSTM (PyTorch, local)
Smart labels	scikit-learn (k-means/DBSCAN)
Anomaly detection	Isolation Forest (scikit-learn)
Verifiable credentials	Plonky3 (ZK proofs)
Voice commands	Vosk (offline) + HF Mixtral
Real-time alerts	NATS JetStream
Organization Memory	LightGBM (local) + frequency counters

No external contact management or CRM service required.

18.14 Implementation Checklist (macOS/Linux – ZeroCost)

Summary:

Build network-service.

Ensure GraphRAGRS is built.

Run daily contact enrichment job (cron).

Test voice command via API.

Verify automatic contact saving on trade creation.

END OF PART 18 – v6.3 WITH ALL MODIFICATIONS

Part 19: Enhanced Commodity & Packing Specifications (Multicommodity) – v6.3 Fully Extended, with Structured Container Data, Multi-Shipment Support, AI-Driven Packaging Advisor, Collaborative Editing, Capacity Heatmap, Ecological Packaging, Carbon Footprint, and All Modifications

19.0 Philosophy

The Commodity & Packing module is the physical brain of the platform. It takes the buyer's structured per-container data (from Phase 1) and transforms it into a millimetre-accurate, regulation-compliant, transport-optimised packing plan. Every detail, from the HS code to the biodegradable twine on a pallet, is verified, documented, and traceable. The seller can edit, collaborate, and lock the plan, and the system generates SSCC barcodes, loading guides, and ecological suggestions.

In v6.3, this module is fully integrated with:

Buyer's structured container form (Country of Origin, Destination, Port of Discharge, Palletized status, Pallet Size, Commodities, Packaging, Number of Pallets per commodity, etc.)

Multi-shipment contracts – each shipment has its own packing plan, but they share commodity data.

Dual-mode toggle – the seller sees the packing UI; the buyer sees a read-only view.

Three-tier AI – Groq for advisory (packaging suggestions, loading guides), HF local for compliance checks (ISPM 15, food contact), Ollama as fallback.

Collaborative editing (Yjs + WebRTC) for multiple seller employees.

Capacity heatmap (opt-in, differential privacy) showing historical loading density.

Ecological Packaging Advisor and Carbon Footprint Calculation Methodology (ISO 14067-compliant, CBAM ready).

Removal of operating modes – all features available to all verified sellers (subject to permissions).

19.1 Multi-Commodity Data Representation (Consuming Buyer’s Structured Form)

The buyer’s parsed_specs JSONB (from Part 3, Phase 1) contains per-container arrays. The seller’s packing module reads this structure directly.

Example parsed_specs for a single container with two commodities:

json

```json
{
"containers": [
{
"container_index": 1,
"country_of_origin": "VN",
"destination_country": "EG",
"port_of_discharge": "Alexandria",
"palletized": true,
"pallet_size": "EUR 800×1200 mm",
"destination_override": null,
"commodities": [
{
"commodity_type": "Fresh Fruits",
"product": "Valencia Oranges",
"hs_code": "0805.10",
"specification": "Grade A, Size 72-80 mm",
"packaging": "Boxes (10 kg cartons)",
"number_of_pallets": 22,
"notes": "Stack max 5 layers"
},
{
"commodity_type": "Fresh Fruits",
"product": "Eureka Lemons",
"hs_code": "0805.50",
"specification": "Grade B, Size 50-60 mm",
"packaging": "Mesh bags (8 kg)",
"number_of_pallets": 14,
"notes": null
}
],
```

"notes": "Expedite customs clearance"

}

],

"global_notes": "Seller to provide phyto certificate."

}

For multi-shipment contracts, the parsed_specs includes a shipments array, each with its own containers array.

19.2 AI-Driven HS Code Classification & Dual-Use Screening (A2)

The system already classified HS codes during Phase 1; during packing, it does not re-classify but may validate against updated WCO lists. The HS Code Monitor Agent (RIA submodule) scrapes the World Customs Organization Harmonized System database and national tariff notices daily. If a change affects an already-locked packing plan, the seller is notified via Smart Inbox and may be required to adjust the plan (if the change is significant, e.g., new dual-use restriction). The Dual-Use Screening (Wassenaar Arrangement) runs periodically and can block the packing plan lock if a new restriction is detected. The PlainLanguage Governor Decision Panel explains the issue and provides a link to the relevant regulation.

19.3 Commodity Compatibility & Global Mixing Rules (A2)

The Commodity Compatibility Graph (GraphRAGRS) is queried for each pair of commodities within the same container. If an incompatibility is found (e.g., apples (high ethylene) and kiwis (sensitive)), the Governor returns a CONDITIONAL warning. The seller must either separate the commodities into different containers or acknowledge the warning with a written justification. The warning includes a Groq-generated explanation: “Apples produce ethylene which accelerates ripening in kiwis. Separate containers or ethylene absorbers are required. Please confirm your decision.” The justification is stored.

19.4 Palletisation & Container Loading (ORTools + AI)

19.4.1 Core Solver (ORTools CP-SAT)

The packing-containerisation-service (Rust + Python ORTools) takes as input:

Carton dimensions (from buyer’s packaging field, e.g., “Boxes (10 kg cartons)” – system maps to standard dimensions; seller can override).

Pallet type and dimensions (from buyer’s pallet size, e.g., EUR 800×1200 mm).

Max stacking height (from commodity notes or default from regulations).

Container internal dimensions (from container_types table, pre-loaded).

Maximum payload weight.

Outputs:

Cartons per layer, number of layers, total cartons per pallet.

Total pallets per commodity.

3D loading plan (pallet positions within container).

SSCC18 assignment per pallet.

The solver runs automatically when the seller clicks “Calculate Palletisation” or can be triggered after any change.

19.4.2 Dynamic Loading Patterns from IoT Shock Data (v6.3)

The solver incorporates a feedback loop from Phase 5 IoT sensors (shock, vibration) and satellite wave-height data (Copernicus). For routes where historical shock data shows that top-layer pallets frequently shift, the solver adds a soft constraint to place heavier pallets at the bottom and increase strapping. Groq explains the adjustment: “Based on 30 trips, rear pallets are at higher risk. Placing dense orange pallets in the center will reduce damage probability by 15%.” This is advisory; the seller can override.

19.4.3 Voice & Image Input (Optional, A1)

For non-standard packaging, the seller can use voice or camera:

Voice : “Use EUR pallets, 5 layers max, carton weight 10 kilos.” Vosk transcribes; HF Mixtral extracts parameters; solver runs.

Image : Take a photo of a sample carton on a pallet. The mobile app uses AR dimensioning (MediaPipe + ARKit/ARCore) to estimate carton dimensions and pallet utilisation. The extracted dimensions are prefilled for confirmation.

19.4.4 Loading Guide Generation (A1)

After the packing plan is locked, Groq (fallback Ollama) produces a step-by-step loading guide in the seller’s preferred language. The guide includes container precooling, pallet order, air gap instructions, and seal verification steps. Stored in packing_plans.loading_guide. Can be printed or sent to the loading crew via the SGTX mobile app.

19.5 Collaborative Packing Plan Editing (v6.3 – Extended)

Multiple employees of the same seller tenant can edit the same packing plan simultaneously. Uses Yjs (CRDT) for conflict-free merging and WebRTC (Pion, self-hosted) for peer-to-peer communication. Each user’s cursor is visible with their name. A chat sidebar allows discussion. Permissions are enforced by OPA: a warehouse operator may only edit pallet positions and seal numbers, not commercial fields (EXW price, etc.). For multi-shipment contracts, each shipment’s packing plan is separate; collaborators can work on different shipments simultaneously.

Edge Cases:

Race condition on lock : First request to lock wins; subsequent request receives a CONDITIONALverdict with a Decision Panel explanation: “Packing plan was locked by another user while you were editing. Please refresh.”

Loss of connectivity : Local changes queued; on reconnect, Yjs automatically merges. Conflicts resolved deterministically (last write wins based on timestamp). Conflict log stored.

Large team editing : Maximum 10 concurrent editors per packing plan (configurable); beyond that, new users get readonly view.

19.6 Container Capacity Heatmap (v6.3 – Opt-In, Differential Privacy)

On the 3D container viewer (Three.js), a toggle enables a capacity heatmap overlay. The heatmap displays historical loading density for the same commodity and route, derived from opt-in anonymised data using OpenDP (ε=0.1). High-density zones appear green, underutilised spaces red. The seller can immediately see if their pallet layout leaves significant gaps where previous shipments have successfully added extra cartons. The underlying data never reveals any competitor’s configuration; only aggregated, privacy-safe patterns are shown. The heatmap is advisory and can be disabled per user.

19.7 Automated Packaging Compliance Checker (Global, Dynamic, A2)

Every packing plan is validated against the destination country’s packaging regulations – a living knowledge base updated continuously by the RIA (scraping official gazettes, EU directives, etc.). Regulations covered include:

ISPM 15 (wood packaging) : Checks that wooden pallets/crates are heat-treated and stamped. If not, blocks export; Decision Panel explains: “Wood packaging material for EU must be ISPM 15 compliant. Please upload a heat-treatment certificate or switch to plastic pallets.”

EU 10/2011 & FDA (food contact materials) : Verifies that plastic cartons, bags, and coatings have required migration-limit certifications.

EU Single-Use Plastics Directive : If packaging includes prohibited single-use plastics, the system suggests alternatives (see 19.8).

Country-specific rules (e.g., France’s ban on non-recyclable plastic packaging, Germany’s Packaging Act). The RIA updates rules within hours of publication.

Compliance result : PASSED, WARNING (with suggested actions), or FAILED (blocking contract lock until resolved). The seller can request a manual override (A3) with a justification, which is logged and requires Platform Governance Authority approval.

19.8 Ecological Packaging Advisor & Carbon Footprint Calculation Methodology (v6.3 – Expanded)

19.8.1 Ecological Packaging Advisor (A1)

After the packing solver generates a loading plan, the system automatically analyses the packaging materials (cartons, pallets, strapping, cushioning, bags) and suggests sustainable alternatives that reduce carbon footprint, comply with regulations, and often save costs.

Data sources:

Material databases : scraped from open-source lifecycle assessment (LCA) databases (Ecoinvent free summary, Federal LCA Commons).

Regulatory rules : RIA-maintained table of destination-country packaging regulations.

Cost data : local material prices scraped from free industry portals or estimated from historical SGTX purchase data (anonymised).

Suggestion generation (Groq, fallback Ollama) : Compares current materials against alternatives, considering carbon footprint difference (kg CO_2e), cost difference, regulatory compliance, and suitability for the commodity and route (validated by Packaging Compliance Checker). Example:

text

```
┌──────────────────────────────────────────────────────────────┐
│  Ecological Packaging Suggestion │
│ Switching from virgin plastic strapping to recycled PET │
│ strapping reduces your carbon footprint by 12 kg CO₂e per │
│ container and saves approximately $45 in packaging tax in │
│ Germany. The alternative meets all strength requirements for │
│ this route (validated). │
│ │
│ [Apply to all containers] [Apply to container 1 only] [Dismiss]│
└──────────────────────────────────────────────────────────────┘
```

If seller applies, the packing plan's material specifications are updated and logged.

19.8.2 Carbon Footprint Calculation Methodology (ISO 14067-compliant, CBAM ready)

Scopes : Calculated per shipment (or per multi-shipment shipment) in accordance with GHG Protocol and EU CBAM.

Scope	Description	Data Sources
Scope 1	Direct emissions (vessel/truck fuel, reefer generator fuel)	AIS fuel consumption estimates (IMO EEXI), OSRM distance × EPA SmartWay factors

Scope 2	Indirect emissions from purchased electricity (reefer containers, warehouse)	IEA grid emission factors (free, per country), IoT power sensors (if available)
Scope 3 (upstream)	Packaging material production, pallet manufacturing, dunnage	Ecoinvent free summary, Federal LCA Commons, industry-average factors
Scope 3 (downstream)	Port handling, warehousing, last-mile delivery, disposal of packaging	Port authority data (if available), EPA WARM model defaults

CBAM reporting : For EU-bound shipments, the system automatically generates a CBAM report (XML format) containing embedded emissions (Scope 1 and Scope 2 of production process). The report is stored in generated_documents with a digital signature. Tenants can download it from the Trade Command Center’s Carbon Card.

API Endpoint : GET /v1/shipment/{ustn}/carbonfootprint returns total emissions, breakdown, CBAM embedded emissions, confidence interval, and calculation method.

19.9 Generative AI for Custom Packaging (A1)

For non-standard goods (machinery, art, irregular shapes), the seller can describe the product in natural language, and Groq (fallback Ollama) proposes a complete packaging solution. Example: “We have 500 ceramic tiles, each 30×30×1 cm, weight 1.2 kg. Need to fit in a 20′ container.” Groq returns: “Suggested box size: 60×30×20 cm, 10 tiles per box, 50 boxes. Use 2 EUR pallets, 25 boxes each, with foam interleaving. Total pallet weight 620 kg. Use corner protectors and vertical strapping. Estimate pallet height 152 cm. The remaining space can accommodate 100 more tiles if needed.” The proposal is then verified by the packing solver.

19.10 Governance Gates (Commodity & Packing)

Gate ID	Check	Enforcer	Failure Action
GCP1	HS code classification confidence ≥0.90, dual-use check passed	HS Classifier (A2) + WasmEdge	Block or require human waiver
GCP2	Commodity mixing rules satisfied (no blocked combination)	Commodity Compatibility Agent (A2)	Require human waiver or separate containers
GCP3	Packaging compliance with destination regulations (ISPM 15, food contact, etc.)	Compliance Checker (A2)	Block lock; suggest correction
GCP4	Palletisation solution feasible (ORTools found)	Packing Solver (A2)	Require manual adjustment
GCP5	Loading plan respects weight limits and route-specific shock constraints	Solver + IoT constraint engine (A2)	Reconfigure load; Groq explanation

GCP6	Ecological packaging advisory displayed (advisory)	Eco Advisor (A1)	–
GCP7	Perishable degradation model consulted; if predicted shelf life < minimum, alert	Digital Twin (A2)	Unsuitable packing – require change
GCP8	Packing plan locked with a valid USTN (master contract or per-shipment) before any settlement instruction can reference it	Governor (WasmEdge)	Hold settlement instruction until USTN populated

19.11 Zero-Cost Technology Stack

Component	Technology	Cost
HS code monitoring	HF DistilBERT + RIA (scraped WCO)	$0
Commodity compatibility graph	GraphRAGRS (Rust)	$0
Palletisation & container loading	ORTools (Python) + Rust service	$0
Collaborative editing	Yjs (MIT), Pion (Apache2.0), WebRTC	$0
Capacity heatmap	Three.js shader + OpenDP (differential privacy)	$0
Loading guide	Groq (free tier) / Ollama	$0
Packaging compliance	HF Donut (regulation scraping) + WasmEdge	$0
Ecological advisor	Groq + LightGBM (local)	$0
Carbon footprint	IMO EEXI, EPA SmartWay, IEA factors, Ecoinvent (free summaries)	$0
CBAM XML generation	Tera templates + Groq	$0

No external packaging or carbon accounting service is required.

19.12 Worked Example – Seller Uses Structured Buyer Data, Collaborative Editing, and Ecological Suggestion

Context : Mekong Fresh (seller) receives a structured request from Nile Foods for a container of oranges and lemons (22 pallets oranges, 14 pallets lemons). The buyer specified EUR pallets, boxes for oranges, mesh bags for lemons.

Loading plan : The packing solver automatically calculates 5 layers for oranges (8 cartons per layer, 40 cartons per pallet) and 6 layers for lemons (8 cartons per layer, 48 cartons per pallet). Total pallets match buyer’s request.

Collaborative editing : The warehouse operator and trade manager both open the packing plan. The trade manager adjusts the stacking height for lemons to 5 layers (to reduce risk). The operator sees the change in real time.

Capacity heatmap : The operator toggles the heatmap; it shows that the front 1.5m of the container is typically packed at 95% density; they add two extra cartons of oranges to fill a gap.

Ecological advisor : Groq suggests switching from virgin plastic strapping to recycled PET, saving 12 kg CO_2e per container and €45 in German packaging tax. The seller applies the suggestion.

Loading guide : After locking, Groq generates a loading guide in Vietnamese: “Precool container to 5°C. Load pallets in order: first 22 orange pallets, then 14 lemon pallets. Ensure 2 cm air gap between pallets.”

Carbon footprint : The system calculates total CO_2e = 7,700 kg (transport 2,800, reefer electricity 1,100, packaging 3,800). A CBAM report for the EU (if destination Germany) is generated and stored.

END OF PART 19 – v6.3 WITH ALL MODIFICATIONS

Part 20: Location-Based Trucking & Route Intelligence (v6.3 – Fully Extended, Global, Zerocost, with All Modifications: Multi-Shipment Support, Three-Tier AI, OSRM Offline Routing, Predictive Dispatching, Automated E-Way Bills, Proof-of-Location Consent, Voice Navigation, and Integration with Distressed Cargo & Financing)

20.0 Philosophy

Every container's journey on land is as critical as its ocean voyage. The Location-Based Trucking & Route Intelligence module transforms raw GPS pings into a self-learning, predictive, and regulation-aware logistics brain. It plans optimal routes, detects anomalies in real time, generates all necessary transit documents, and even feeds proofs of location into smart contracts – all while running on free, self-hosted infrastructure and open-source routing engines. In v6.3, the module is fully integrated with multi-shipment contracts (each shipment has its own route plan), distressed cargo (route deviations may trigger distressed declaration), financing (proof-of-location can be used as collateral verification), and the dual-mode toggle (drivers in field do not see the toggle; dispatchers see only relevant mode). All router-related alerts appear in Smart Inbox; if a required transit document is missing, the PlainLanguage Governor Decision Panel explains the block.

20.1 Core Technology Stack (Zero-Cost, Offline-First)

Component	Technology	Purpose
Geocoding	Nominatim (self-hosted, OSM data)	Convert addresses ↔ coordinates
Routing Engine	OSRM (self-hosted) or Valhalla	Base routing, travel time computation
Map Rendering	Leaflet + OpenStreetMap tiles (pre-cached)	Web and mobile map display
Offline Routing	OSRM offline tiles in mobile app	Driver navigation in dead zones
GPS Trace Storage	PostgreSQL (PostGIS) + TimescaleDB	Historical trip data
Correction Factors	route_correction_factors table	AI-tuned multipliers per

		segment/time
ML/AI	XGBoost, LSTM, Isolation Forest, PyTorch	Travel time prediction, anomaly detection, maintenance forecasting
Fast Advisory	Groq (free tier), fallback Ollama	Natural-language explanations and alerts
Speech	Vosk (offline)	Voice-guided navigation and driver commands
Document Generation	HF Donut (local), Tera + headless Chrome	Automated e-way bills, transit permits
Weather Data	OpenWeatherMap (free tier), Copernicus	Realtime and forecast weather for routing
Proof-of-Location	Plonky3 (self-hosted)	ZKproofs for smart contract triggers
Consent Management	PostgreSQL + Governor logging	Driver consent for location tracking
Smart Inbox	inbox-service (Rust)	All route alerts and overdue milestones as prioritised items
Decision Panel	Governor + Groq/Ollama	Blocks explained in plain language
Multi-shipment	contract_shipments table	Each shipment has its own route plan

No billing details required; all routing and navigation work offline once tiles are cached.

20.2 AI-Driven Dynamic Route Optimization

The routing engine is never static. It continuously learns from every completed shipment.

20.2.1 Continuous Learning Loop

When a truck trip finishes, the trucking_gps_traces table is populated with the full anonymised trace and actual distance/duration.

A LSTM model (trained on all traces for that region) is retrained weekly to predict travel time for each OpenStreetMap segment, factoring time-of-day, day-of-week, historical congestion patterns, weather, and truck weight/type.

The model updates the route_correction_factors table (distance_multiplier and time_multiplier per geohash/time bucket).

Groq summarises notable changes for logistics providers, and the respective Smart Inbox item appears: *“Based on the last 200 trips, the Cairo-Alexandria desert road has a new average speed of 72 km/h (down from 80 km/h) due to construction until Oct 2026. ETA adjusted accordingly.”*

20.2.2 Real-Time Rerouting (v6.3)

OSRM/Valhalla computes the initial route with learned factors.

During transit, if a disruption is predicted (accident, sandstorm, flooding) or an update from Phase 5 disruption service arrives, the route is recalculated and pushed to the driver's mobile app.

The dispatcher's Smart Inbox receives an alert : *"⚠ Sandstorm warning on desert highway from 16:00. Recommending coastal route (+45 km, +35 min). [Apply Reroute]"* – clicking applies the new route. If the dispatcher does not act within 15 minutes, the system automatically applies the safer route (if auto-reroute is enabled in the tenant's preferences).

20.3 Predictive Dispatching & Smart Load Consolidation (VRP)

Every 30 minutes, the platform runs a Vehicle Routing Problem (VRP) solver (ORTools) for all pending pickups and dropoffs in a region, respecting multi-shipment schedules. Inputs: open container moves, available trucks, current positions, loading windows, port cutoffs. Output: an optimised dispatch plan for each trucking company, minimising empty miles. Groq generates a dispatch note that appears in the trucking company's Smart Inbox and pushes to the driver's app. Example: "Truck FL772 can pick up both TCNU123 (oranges, 14:00) and TCNU456 (lemons, 15:30) at Saigon port. Combined route saves 120 km and $85 fuel. Arrival at both cutoffs on time. [Accept Plan] [Adjust]"

20.4 Real-Time Anomaly Detection & Geofencing

An Isolation Forest model (scikit-learn, local) trained on normal route behaviour creates an envelope for each scheduled truck trip. Geofencing defines allowed corridors; if the truck exits the corridor without authorisation, an alert fires. Anomaly detection flags unscheduled stops >5 min, excessive speed, or major route deviation. Example alert appears in Smart Inbox of logistics provider, cargo owner, and SGTX compliance: "⚠ Truck carrying Container TCNU8901 deviated 12 km off route near Mansoura at 14:23. No scheduled stop. Contact driver immediately." If deviation coincides with a known secure rest area (OSM tags), the alert severity is lowered to informational.

20.5 Voice-Guided Turn-by-Turn Navigation (Offline)

The SGTX mobile app becomes a truck-specific GPS navigator:

Driver enters destination (or receives automatically from assigned job). OSRM routes pre-downloaded for offline use.

App provides visual turn-by-turn directions and voice prompts via device TTS.

Voice commands : "Show next turn", "Find nearest truck stop", "Report accident ahead".

Respects bridge heights, weight limits, hazardous material restrictions using OSM tags.

For multi-shipment, the driver can switch between shipments' destinations via voice: "Navigate to next pickup for Shipment 2".

20.6 Carbon-Minimising Route Options (A1)

When a trucking route is computed, the system evaluates the CO_2 footprint of multiple alternatives using the ESG model (LightGBM). The “greenest” route is presented alongside the fastest route. Example: “Standard route: 250 km, 280 kg CO_2. Eco-route: 265 km, 260 kg CO_2 (saving 7%). Arrives 35 min later. Select?” The driver or logistics manager can choose; the choice is recorded for sustainability reporting (CBAM, etc.).

20.7 Automated E-Way Bill & Transit Document Generation (v6.3)

Many jurisdictions require electronic waybills, transit permits, or customs transit documents for road freight. The RIA scrapes and maintains the latest requirements per country (e.g., India’s e-Way Bill, East Africa’s RCA transit, EU’s T2L). When a trucking job is confirmed, the system auto-fills the official PDF/API form using route, vehicle registration, cargo details, container number, and USTN. HF Donut validates the filled PDF against the official template. Groq drafts any required declarations. The document is stored in generated_documents and can be printed or handed to border authorities via the driver’s app. Governor Gate G-EW1: If the required document is not generated and verified before scheduled departure, the truck’s departure milestone is blocked. The PlainLanguage Governor Decision Panel appears for the dispatcher: “Truck FL772 cannot depart because the e-Way Bill for USTN ... has not been generated. [Generate Now]”

20.8 AIPredicted Maintenance Alerts for Fleets (A1)

For trucking companies that opt in, an XGBoost model (local) is trained on anonymised fleet data (odometer, trip distances, average loads, engine hours). It predicts when a vehicle is likely to need maintenance (oil change, brake pads, tyres). Groq generates Smart Inbox alerts: “Truck FL772 is due for an oil change after the current trip (predicted 1200 km over schedule). Recommend scheduling maintenance in Alexandria.” This is purely advisory.

20.9 Dynamic Loading Dock Scheduling with AI (A2)

The system collects historical gate wait time data from AIS and provider-reported statuses; trains an LSTM model to predict congestion per terminal, day, and hour. When a loading window is requested, the system suggests the optimal 1-hour slot to minimise waiting. Groq provides a summary in the dispatch note: “Booked 08:30-09:30 slot at Alexandria Terminal. Predicted wait: 20 min. Average wait at 10:00 is 90 min.” The booking is made via the terminal’s free portal/API (if available) or suggested to the trucking company.

20.10 Decentralised Proof-of-Location with Consent & Privacy Controls (v6.3 – New)

20.10.1 Proof Generation
For DeFi-financed shipments or high-value cargo, the driver's mobile app generates a ZKproof (Plonky3) that the device was inside a geofence at a specific time, without revealing exact GPS. The proof is verified by the SGTX oracle and recorded on-chain (Polygon). The smart escrow automatically releases a payment tranche.
Example: "Container TCNU123 gated into Saigon port at 06:23. ZKproof verified. Releasing 10% of escrow to carrier."

20.10.2 Consent & Privacy Controls
Location tracking is off by default. Before any location-based ZKproof is generated, the driver must grant explicit consent via the mobile app. Consent modal explains: what data is used (device location), that it is processed on-device, that only ZKproofs are transmitted, and that consent can be withdrawn at any time. The driver can withdraw consent via Settings → Privacy → Location Tracking. Upon withdrawal: app stops collecting location, no new ZKproofs are generated (existing proofs remain valid). The driver's employer (carrier) receives a Smart Inbox notification that consent has been withdrawn but cannot override. Consent status stored in employees.location_consent_granted and location_consent_timestamp. Every consent grant or withdrawal is logged as a Governor decision (decision_type = 'location_consent_change').

20.11 Governance & Reliability Gates

Gate ID	Check	Enforcer	Failure Action
GT1	GPS trace received within expected interval; if missing >15 min, trigger alert	Telemetry Monitor (A2)	Alert logistics provider; start investigation
GT2	Route deviation exceeds geofence without authorisation	Anomaly Detector (A2)	Immediate alert; possible theft escalation
GT3	Required e-way bill or transit permit generated and verified before departure	Doc Compliance (A2)	Hold truck at origin; display PlainLanguage Decision Panel with direct generation link
GT4	Proof-of-Location ZKproof verified before smart contract release	Governor (A3)	Hold release
GT5	GPS trace data anonymised and stored in compliance with local privacy laws	Privacy Guardian (WasmEdge)	Block storage; alert DPO
GT6	Location consent must be granted before any	Governor (A3)	Block proof generation; show

	ZKproof is generated		consent modal
GT7	Consent withdrawal logged and immediately enforced	Governor (A3)	If withdrawal not honoured, escalate to Admin Portal
GT8	For multi-shipment, each shipment's route plan is independent; delays in one shipment do not block others	Governor (A3)	Enforced by USTN-level scoping

20.12 Worked Example – Dispatcher Uses Predictive Rerouting, e-Way Bill, and Proof-of-Location

Context : Fast Trucking Co. is transporting a container of oranges to Alexandria for a multi-shipment contract (Shipment 1). ETA 14:00. A sandstorm warning is issued for the desert highway.

Dispatcher receives Smart Inbox alert : "Sandstorm warning – reroute via coastal road (+45 km, +35 min)." Clicks "Apply Reroute". New route automatically pushed to driver's mobile app.

The system detects that an e-Way Bill for this route is missing. Decision Panel appears: "Truck cannot depart because e-Way Bill not generated. [Generate Now]". Dispatcher clicks, system auto-fills form, submits to government API, stores document. Milestone cleared.

As the truck enters the port geofence, the driver's app (with consent granted) generates a ZKproof of location. The smart escrow contract releases 10% of the freight payment to Fast Trucking Co. The financier (if any) sees the proof verified.

The driver arrives, delivers container. Sensor data shows no temperature excursion. Shipment completes.

END OF PART 20 – v6.3 WITH ALL MODIFICATIONS

Part 21: Auto-Generated Barcodes for Shipment Tracking (v6.3 – Fully Extended, Global, Zerocost, with All Modifications: Multi-Shipment Support, Predictive Scanning Reliability, Reprint Policy, Blockchain Anchoring, Self-Sovereign Pallet Identity, Voice & AR Integration, and Enhanced Print Workflow)

21.0 Philosophy

Every pallet, carton, or item that moves through a trade must leave an unforgeable, instantly verifiable digital fingerprint. Barcodes are not just stickers – they are the physical anchor of the platform's digital twin. In v6.3, the barcode is intelligent, self-sovereign, and resilient. It carries exactly the data each party needs, can be verified even without network access, and can be read by voice or camera when physical labels fail. All generation, scanning, and verification runs on open-source libraries and zero-cost infrastructure. The system supports multiple symbologies (GS1-128, Data Matrix, QR) and automatically assigns globally unique SSCC18 codes to every logistic unit (pallet). All scans trigger Governor-gated milestone confirmations and are immutably logged via Loom.

The barcode system fully supports multi-shipment contracts : each shipment has its own set of pallets, each with a unique SSCC, and the barcode includes the shipment USTN (or microUSTN for distressed splits). The predictive scanning reliability agent (A2, XGBoost) proactively warns warehouse managers if a batch of labels is likely to fail in the intended environment. A detailed label print workflow ensures flawless physical application from PDF to pallet, with a reprint policyenforced by the Governor after pallets have been loaded. Blockchain anchoring (Polygon) provides an immutable, publicly verifiable chain of custody for high-value cargo. Self-sovereign pallet identity (W3C Verifiable Credentials) allows offline verification using a pre-cached SGTX public key.

21.1 Core Technology Stack (Zero-Cost)

Component	Technology	Purpose
GS1-128 / Code 128	Rust barcoders (or zxing-cpp-bindings)	Linear barcode for pallets (SSCC18)
QR / Data Matrix	Rust qrcode crate	High-density 2D codes with

	or barcoders	dynamic content
SSCC18 allocation	packing-containerisation-service(Rust)	Unique serial numbers per logistic unit
Computer vision (barcode reading)	ZXingC++ (WASM for web, native for mobile)	Robust scanning even if code is damaged
AI-powered text recognition	HF Donut (local) or ONNX OCR	Reads printed text on labels when barcode unreadable
Voice command	Vosk (offline) + HF Mixtral (local)	Handsfree pallet identification
AR overlay	AR.js + WebGPU (client-side)	Visual scan status in camera view
Verifiable Credentials / ZK	ssi crate (Rust) + Plonky3	Self-sovereign pallet identity and offline proofs
Predictive analytics	XGBoost (local)	Forecasting scan failure risk and printer anomalies
Document rendering	Tera + headless Chrome	Printable label sheets with intelligent layouts
Blockchain anchoring	Polygon PoS + ethers-rs	Immutable hash anchoring for high-value pallets
Data storage	shipment_barcodes and pallet_details tables (PostgreSQL)	Persistent mapping of SSCC to shipment data

No cloud APIs, no subscription fees.

21.2 Barcode Generation & Dynamic Content (A1)

21.2.1 SSCC18 Allocation

During the packing plan lock (Phase 2), the packing-containerisation-service generates a globally unique Serial Shipping Container Code (SSCC18) for every pallet. The SSCC conforms to GS1 standard: 1 (extension digit) + 6-digit company prefix (reserved for SGTX) + 9-digit serial (unique per pallet) + 1 check digit. Example: 1 0614141 123456789 0. The SSCC is stored in pallet_details.sscc and is used as the primary key for all tracking events.

For multi-shipment contracts, the SSCC includes an internal reference to the shipment number (encoded in the serial range). For distressed micro-contracts, a new microUSTN is associated, and new SSCCs may be generated for the split pallets (or existing ones retained with a flag).

21.2.2 QR Code Payload (Dynamic per Stakeholder)

In addition to the linear barcode, a QR code is generated that encodes a JSON payload. The AI (Groq, A1) decides which attributes to include based on the intended audience of that specific label copy (the seller can select from predefined templates or customise). Options:

For warehouse staff : SSCC, container number, product name, lot number, carton count, gross weight.

For customs brokers : SSCC, HS code, origin country, net weight, shipper GTID, USTN.

For consignee : SSCC, product, lot, quantity, link to the full digital twin dashboard, plus a verifiable credential proof.

The same physical label can contain a GS1-128 for universal scanning and a tailored QR for human users.

21.2.3 Self-Sovereign Pallet Identity (Verifiable Credential)

Each SSCC is also issued a W3C Verifiable Credential signed by the SGTX platform. This credential contains the pallet's immutable identity data (product, lot, weight, origin, USTN, GTIDs) and can be embedded in the QR code as a compact URL or raw VC. A recipient can scan the QR and verify the pallet's provenance without internet access, using only the SGTX public key (pre-cached in the mobile app). This is achieved via a lightweight ZKproof (Plonky3) that the data matches the platform's signature. Example: at a rural warehouse in Egypt with intermittent connectivity, a receiver scans a lemon pallet's QR, the app verifies the cryptographic proof, and displays a green checkmark: "Verified – Lot VN2406L, 48 cartons, 384 kg. No internet required."

21.2.4 AIPlanned Label Layout

When generating printable label sheets (PDF), the system uses an AI planner (LightGBM) to optimise label placement on A4 sheets to minimise waste. It also adjusts font sizes and data density based on the label size and the amount of information requested by the user.

21.3 Multi-Modal Pallet Identification (Offline & Online)

Three independent identification methods work seamlessly together:

Camera-Based Barcode Scanning (ZXingC++) : Decodes GS1-128 and QR codes in real time. Continuous scanning mode; successful decode triggers haptic feedback and advances to the next pallet automatically.

AI-Powered Visual Recognition (Offline) : If a barcode is torn, smudged, or partially covered, the worker switches to visual recognition mode. The device camera captures the pallet label, and a tiny ONNX model (finetuned on warehouse label images) recognises and extracts the printed SSCC number and key text. Even if only the human-readable digits are visible, the pallet is identified.

Voice-Activated Hands-Free Identification (Offline) : Worker speaks: "Pallet OR005 loaded into container TCNU1234567." Vosk transcribes; HF Mixtral extracts pallet ID and action; triggers the same milestone as a scan. Works in noisy environments with a headset.

All three methods produce the same shipment_milestone entry with the same governance gates.

21.4 Augmented Reality (AR) Scan Assistant

In the mobile app, AR mode overlays a live camera view with colour-coded bounding boxes around each pallet's label:

Red: Not yet scanned (urgent)

Blue: Pending (in this session's queue)

Green: Successfully scanned and verified

Yellow: Damaged barcode detected – try voice or visual mode

The worker can instantly see which pallets are still to be processed, reducing errors and speeding up loading operations. Rendered entirely with AR.js, no network required.

21.5 Predictive Scanning Reliability Agent (v6.3 – Enhanced)

A background agent (A2, XGBoost) monitors all scan events across the platform, tracking:

Scan success rate per barcode batch (by printer, label material, print date).

Ambient lighting conditions (from mobile metadata, if available, with user consent).

Pallet handling environment (e.g., outdoor vs. warehouse).

It predicts the probability that a specific batch of barcodes will fail to scan in the intended environment. If the risk exceeds a threshold (default 70%), the system proactively sends a Smart Inbox alert to the warehouse manager and the seller:

"Pallet labels printed on 'Printer B' have a 72% probability of scan failure in outdoor conditions at destination (Alexandria port). Recommend reprinting with higher contrast or adding a protective sleeve. Affected pallets: OR012 to OR022. [Reprint Labels]"

The manager can click "Reprint Labels" to automatically queue a new print job for the affected pallets, using the same SSCCs but with enhanced contrast settings. The original labels remain valid; the reprint is logged and does not generate new SSCCs.

21.6 Label Print Workflow (v6.3 – Expanded)

21.6.1 Printer Compatibility

Two print paths:

ZPL (Zebra Programming Language) – for thermal label printers: The barcode-servicegenerates ZPL code directly, optimised for standard 100×150 mm label rolls. The desktop app (Tauri) or mobile app (via network print server) sends the ZPL stream to the printer's IP address over raw TCP (port 9100). No third-party drivers needed.

PDF – for laser/inkjet printers: A PDF sheet is rendered using Tera templates + headless Chrome, formatted for standard A4 label sheets (e.g., 2 columns × 11 rows for 100×150 mm labels). The user prints directly from the browser or desktop app.

21.6.2 Reprint Policy & Governance (Governor-Enforced)

Before pallet is scanned as LOADED : Labels can be reprinted freely. The "Reprint" button is always active. Each reprint is logged in the audit log with user ID and timestamp, but no Governor decision required.

After pallet is scanned as LOADED : The “Reprint” button is disabled. A “Request Reprint” button appears. Clicking it opens a modal requiring a mandatory reason (e.g., “Label was damaged during transit to port”). This creates a Governor decision (decision_type = 'barcode.reprint.post_loading') with the reason logged. If the Governor returns ALLOW, the reprint proceeds. The PlainLanguage Governor Decision Panel explains if the reprint is denied (e.g., “This pallet has already been delivered; reprinting is not allowed.”)

21.6.3 Label Content Customisation

The seller can customise label content before printing:

Select language for human-readable text (Groq translates product name and handling instructions).

Choose which data fields to display (e.g., hide lot number, show buyer name).

Predefined templates : “Standard”, “Customs-Ready” (includes HS code and origin in large font), “Consignee” (includes delivery address).

21.7 Blockchain-Anchored Barcode Hashes (Immutable Timeline)

For high-value or regulated shipments (optional, selected by seller), the platform anchors the hash of each pallet’s identity data on Polygon for an additional layer of public verifiability. At packing plan lock, the pallet_details record is hashed (SHA256) along with the SSCC, shipment USTN, and a timestamp. The hash is included in a batch transaction to a simple smart contract (self-deployed, minimal gas). The transaction hash is stored in shipment_barcodes or a dedicated blockchain_verifications table. This creates an immutable, publicly auditable chain of custody that proves the pallet existed with exactly those specifications at that point in time. Cost: negligible (~$0.01-0.10 per batch, paid by the platform operator as part of infrastructure cost). No per-transaction fee charged to tenants.

21.8 Scan-Triggered Milestone Confirmation & Fee Release

Every successful scan (barcode, visual, or voice) triggers the standard Phase 5 milestone flow:

Mobile app sends a signed pallet.loaded event to shipment-service via NATS.

Governor (OPA + WasmEdge) verifies device identity (ZITADEL certificate), employee permissions, and current FeeLock status.

If allowed, milestone confirmed, and pallet_details.status updated to LOADED.

When the last pallet of a container is loaded, the container milestone (“Loaded”) is auto-confirmed (A4 if multisource consensus reached).

FeeLock release conditions are evaluated (but note : SGTX fee already paid upfront – no further release; if financing fee, it was deducted at disbursement). The milestone confirmation is still logged for audit.

All events are logged in shipment_milestones with scanned_barcode_id linking back to the pallet_details row.

21.9 Governance & Verification Gates

Gate ID	Check	Enforcer	Failure Action
GB1	SSCC generated is globally unique and conforms to GS1 standard	Barcode Service (A4)	Reject label; regenerate
GB2	Every scan event originates from an authenticated device (ZITADEL certificate)	Governor (A3)	Reject scan; log anomaly
GB3	Visual/Voice identification must match an existing pallet SSCC with >95% confidence	AI Verifier (A2)	Require manual override
GB4	Printed label output passes quality control (resolution, contrast)	Predictive Agent (A2)	Flag for reprint
GB5	Blockchain anchor transaction (if used) is confirmed on-chain	Blockchain Verifier (A2)	Retry; alert if still fails
GB6	Post-loading reprint requires Governor decision with mandatory reason	Governor (A3)	Deny reprint; display explanation in Decision Panel
GB7	For multi-shipment, barcode includes shipment reference; scanning a pallet from wrong shipment is rejected	Governor (A3)	Reject scan; warn user

21.10 Worked Example – Barcode Printing, Scanning, and Post-Loading Reprint

Context : Mekong Fresh locks a packing plan for a multi-shipment contract (Shipment 1, oranges). 22 pallets.

Label generation : System generates SSCCs, QR codes, and ZPL. Seller selects “Customs-Ready” template, language Arabic. Prints via Zebra printer.

Loading : Warehouse worker scans each pallet with mobile app (AR overlay shows green bounding boxes). All scans succeed.

Post-loading damage : One pallet’s label is torn during transport. At destination, the receiver cannot scan. They use visual recognition: app extracts SSCC from printed text, verifies, and confirms delivery.

Reprint request : Seller realises another pallet’s label faded. Clicks “Request Reprint” (after loading). Provides reason: “Label faded due to humidity.” Governor decision created, auto-approved (risk low). New labels printed with same SSCC.

Blockchain anchor (optional) : Seller had enabled blockchain anchoring. A batch transaction with all 22 pallet hashes is submitted to Polygon. Transaction hash stored.

END OF PART 21 – v6.3 WITH ALL MODIFICATIONS

Part 22: Implementation Roadmap & Scaling Strategy (v6.3 – Fully Upgraded, Global, Zerocost, with All Modifications: Tenantcentric Enhancements, Multi-Shipment, Financing, Distressed, Dual-Mode Toggle, and All Prior Features Included)

22.0 Philosophy

The SGTX platform is not built once and then operated – it evolves continuously, driven by real-time feedback from the market, from developers, and from the global regulatory landscape. The implementation roadmap is itself an AI-managed, self-correcting process. Priorities shift automatically when a new law passes in a target jurisdiction, when a microservice shows unexpected latency, or when user growth in a region demands immediate infrastructure expansion. All planning, deployment, and scaling use zero-cost infrastructure (bare-metal K3s, open-source CI/CD, self-hosted project management). No cloud vendor lock-in, no proprietary project management tools, no billing details required.

The roadmap explicitly includes the TenantCentric Enhancement Pack (Smart Inbox, Trade Command Center, PlainLanguage Decision Panel, Onboarding Wizard, Dual-Mode Toggle, Internal Organization Graph) as part of the MVP (Phase 1). All v6.3 modifications (multi-shipment, structured container form, seller-paid SGTX fee, flat 0.25% financing fee, co-financing, distressed triage dashboard, etc.) are integrated into the phases. The three-tier AI stack (Groq + HF local + Ollama) is used throughout.

22.1 Core Phases (AIDriven Feature Slicing, v6.3 Updated)

Phase 1 – MVP (Months 1-6)

Goal : Launch a fully functional, non-custodial trade execution pipeline for at least 3 pilot corridors, proving the zero-cost model, and delivering the complete v6.3 tenant experience.

Core Deliverables:

Identity & Governance : ZITADEL integration, GTID generation, Governor with OPA + WasmEdge, basic RBAC, tenant-friendly decision messages (Groq/Ollama), PlainLanguage Governor Decision Panel.

Trade Initiation & Quoting : Phase 1 (GTID entry, structured container form, multi-shipment request option) and Phase 2 (EXW lock with live market chart, post-lock price watch, manual logistics cost entry, alternative delivery ports, RFQ mode, SGTX fee paid by seller and added to quote).

Packing & Containerisation : Full packing solver (ORTools + AI advisor), container loading plan, SSCC generation, loading guide, collaborative editing (single-user fallback), capacity heatmap (opt-in).

Contract & Fee Collection : Phase 3 (contract genesis, AI clause verification, digital signatures, FeeLock). Amendments, mutual confirmation, SGTX Witness Clause. Multi-shipment master contract with per-shipment fee collection.

Financing (Basic) : Conventional financing (no DeFi) with full disclosure, co-financing enabled, flat 0.25% financing fee deducted at disbursement, financier preferences (countries, amount, settlement type), automatic RFQ broadcast. Financier historical data storage.

Execution & Tracking : Phase 5 with pallet-level barcode scanning (camera, AI vision, voice), IoT ingestion (optional LoRaWAN), milestone autoconfirmation. QC override accountability (mandatory reason). Predictive cold-chain alerts (LSTM). Container release confirmation after fee payment.

Settlement & Fee Collection Infrastructure : Phase 6 (trade principal settlement) and Phase 9 (PSP routing, gross-up calculation, webhook verification, OpenBanking reconciliation). No separate “SGTX FEES” phase.

Distressed Cargo (Basic) : Phase 7 (declaration, AI condition assessment, dynamic pricing, microcontract) and Phase 8 (direct outreach to saved contacts, privacy notice). “Check Buyers” advisory included. Triage dashboard with three paths.

Dispute Resolution : Phase 10 (filing, AI triage, evidence autocompiler (including QC overrides), mediation log, settlement proposal).

Compliance : KYB/KYC tier 1-2, sanctions screening (OFAC/UN/EU), basic jurisdiction matrix, integrated into Onboarding Wizard.

Tenant Experience (v6.3 mandatory) : Smart Inbox, Trade Command Center, PlainLanguage Governor Decision Panel, Onboarding Wizard & Sandbox, Dual-Mode Toggle for trading companies, Internal Tenant Organization Graph (basic – business units, approval chains), Workflow Recovery & Draft States.

Marketplace API (Basic) : Intent analysis, trade initiation attribution, basic webhooks. Rate limit gauges.

Pilot Corridors : Egypt-Vietnam (citrus), India-UAE (textiles), Germany-Turkey (automotive parts). One PSP per region (Stripe, Payoneer, Fawry).

Infrastructure : 3 sovereign nodes (Cairo, Dubai, Frankfurt), all on bare-metal K3s. All v6.3 services (inbox, tenant-experience, workflow-recovery, data-export, financier-preference, multi-shipment-coordinator, fee-split-orchestrator) deployed alongside core mesh.

KPIs:

100 verified tenants across all corridors.

50 completed trades ($5M GMV).

Fee collection rate ≥ 98%.

Governor uptime 99.9%.

Average trade cycle time < 14 days.

Tenant Experience KPIs : Average Smart Inbox time-to-action < 4h, onboarding completion rate > 90% within first session, <5% support tickets related to “can’t find feature”.

Phase 2 – Scale (Months 7-12)

Goal : Expand to 15+ trade lanes, deploy DeFi financing options, harden infrastructure for enterprise use, roll out advanced Tenant Organization Graph features.

New Capabilities:

DeFi & Advanced Financing : Phase 4 DeFi options (protocol risk oracle, stablecoin monitor, automated smart contract auditor, liquidation early warning, DeFi PlainLanguage Risk Summary modal). Cross-chain interoperability with bridge risk scoring.

Advanced Logistics : Logistics Bundle Optimiser with ReOptimise diff view, Dynamic Carrier Risk Score, Logistics Provider Warm-Up Program (including unsubscribe from anonymous RFQs).

Digital Twin & Cold-Chain : Full digital twin simulation, cold-chain predictive quality (LSTM), shelf-life forecasting, one-tap smart escrow claim.

Voice Everywhere : Voice-activated workflows in all portals and mobile (offline Vosk + local LLM + Groq).

Distressed Cargo (Full) : Accelerated Outreach Mode with real-time ranking, express negotiation bot, AI-insurer integration.

Marketplace API (Full) : Webhook management console, sandbox with synthetic data, dispute management, revenue share agreement management.

Compliance Expansion : KYB/KYC tier 3, continuous PEP monitoring, regulatory reporting automation (CBE, ECB, CBAM), expanded jurisdiction matrix to 100+ countries.

Additional PSPs : 35+ PSPs integrated via AI Rail Discovery Agent.

Infrastructure Scaling : 2 additional sovereign nodes (Singapore, Mumbai), predictive autoscaling on K3s. inbox-service scaled horizontally based on user demand data from Phase 1.

Advanced Tenant Organization Graph : Delegated administration, cost centre tracking, cross-BU reporting.

KPIs:

1,000+ verified tenants.

2,000+ completed trades ($200M GMV).

Financing volume $50M.

AI accuracy (credit scoring, ETA) > 85%.

Platform uptime 99.95%.

Phase 3 – Enterprise (Months 13-18)

Goal : Full global coverage, advanced AI addons, enterprise-grade compliance, complete self-healing operations, and full decentralisation features.

New Capabilities:

All 62 AddOns (from Part 15) : Federated learning, GNN risk engine, ZKproof settlements, chaos engineering, continuous compliance, global trade observatory, carbon accounting suite, etc.

Enterprise RBAC & Multisig : Tenant-level multisig for high-value contract approvals, advanced data scopes, granular cost hiding.

CBDC & Instant Payments : Full integration with dynamic CBDC rails discovered by the Rail Discovery Agent.

Full eBL Interoperability : Automatic bill of lading issuance, transfer, AI-based compliance for all major carriers.

Post-Quantum Cryptography : All long-term identity and contract records encrypted/signed with Dilithium3 / Kyber1024.

Federated Learning : Cross-node model training without data sharing (fraud detection, credit scoring, margin estimation).

Self-Healing Infrastructure : All K3s clusters managed by AIOps; nodes added, removed, healed without manual intervention.

Global Regulatory Auto-Adaptation : RIA maintains complete, real-time compliance for 195+ countries; all rules fed directly into WasmEdge gates.

DAO Governance : Optional tenant consortia can form DAOs for shared logistics purchasing.

Global Trade Observatory : Public-facing, privacy-preserving visualisation of trade flows (opt-in), powered by SGTX data.

KPIs:

10,000+ verified tenants.

50,000+ completed trades ($5B+ GMV).

$5M annual recurring revenue from fees.

AI-driven operations > 90% (incidents automatically resolved).

100% global sanctions coverage, updated in real time.

RTO < 5 min, RPO = 0 for all critical services.

22.2 AIPowered Project Management & Dynamic Resource Allocation

Project Oracle agent (Rig + XGBoost) monitors the entire development pipeline : source control (Gitea), CI/CD (Drone), developer activity, production metrics

(Prometheus), and Smart Inbox engagement data (time-to-action, snooze rates). Outputs a revised priority list every Monday. Example: “The barcode visual recognition model is taking 40% longer to train than estimated due to insufficient labelled data. Allocate Developer Y to data labelling for 3 days, delaying AR feature by 2 days but keeping Phase 2 schedule on track.” Groq drafts weekly progress report for Platform Governance Authority.

22.3 Self-Healing CI/CD with Intelligent Rollback

Canary deployments : new version to 10% of traffic. Prometheus metrics (latency, error rate, throughput, Smart Inbox latency, tenant message generation success rate) compared to baseline. Isolation Forest detects deviation >10% → automatic rollback to previous stable version. Root-cause analysis creates Gitea issue with Groq summary.

22.4 Predictive Scaling Engine (A2)

LSTM model trained on historical trade volumes, user registration rates, system resource metrics, and partner intent leads. Outputs 12-month capacity plan per sovereign node. Example: “Based on current growth and upcoming citrus season, Cairo node will reach 85% CPU utilisation by November 2026. Provision a second node in Cairo by 2026-10-01.” Provisioning automated via Ansible playbooks triggered by AI.

22.5 Dynamic Geographic Expansion Optimizer (A2)

Multi-armed bandit (contextual) balances exploration (new regions) and exploitation (scaling existing regions) to maximise platform revenue. Features: pending tenant registrations, RIA-detected regulatory friendliness, existing trade volume, latency from nearest node, availability of free peering. Outputs monthly expansion recommendations. Example: “Priority 1: São Paulo, Brazil. 230 registered users, regulatory environment stable, no sanctions. Estimated first-year GMV $12M. Deploy sovereign node.”

22.6 Zero-Cost Infrastructure Lifecycle Management

LSTM model trained on public hardware failure datasets (Backblaze) predicts component failure (SSD, power supply). When failure predicted, system automatically creates replacement order, schedules maintenance window during lowest activity, migrates workloads using K3s cordon/drain. Groq notification: “Cairo node SSD 3 predicted to fail in 45 days. Maintenance window scheduled for 2026-08-15 02:00-04:00 EET. Workloads shifted to Cairo node 2.”

22.7 AIGenerated Partnership Proposals (A1)

RIA compiles list of potential marketplace partners from public trade registries, fintech databases (scraped, privacy-safe). Groq drafts personalised, jurisdiction-specific partnership proposal with expected SGTX FEES revenue, integration effort, compliance alignment. Output PDF stored in generated_documents for business development team.

22.8 Continuous Compliance Roadmap Alignment (A2)

When RIA detects significant regulatory change (e.g., EU MiCA, India DPDP), it creates a compliance milestone in project management (Gitea), sets deadline based on effective date, assigns priority (P1 if within 3 months). Groq explains impact: “New EU General Product Safety Regulation effective 13 Dec 2026 requires additional documentation for certain consumer goods. Document Requirement Service must be updated. P1 issue created; due 1 Dec 2026.”

22.9 Scaling Metrics (Living Dashboard, v6.3 Enhanced)

Admin Portal’s “Platform Health” dashboard (Grafana) shows:

Volume: GMV per corridor per day/week/month.

Users : New tenant registrations, active users, churn, dual-mode toggle usage.

Revenue : Total trade fees, financing fees, partner share, trends.

Collection Rate: % of fees collected vs. due.

Dispute Rate: disputes per 100 trades.

AI Accuracy : per model (credit scoring, ETA, shelf-life, margin estimation) with drift indicators.

Infrastructure : node CPU/RAM/disk, predicted saturation dates.

Compliance : number of jurisdictions covered, pending regulation changes, overdue reports.

Tenant Engagement : average Smart Inbox time-to-action, snooze rate, onboarding completion %, multi-shipment contract usage, distressed cargo usage.

All data fed to Groq for natural-language monthly board summary : “This month, GMV grew 12% driven by the new Brazil corridor. Platform uptime 99.97%. Two compliance items resolved; three new EU regulations flagged for Q1 2027. Tenant experience: 92% of items actioned within 4h, onboarding completion rate 88%.”

22.10 Implementation Checklist for the Roadmap Itself (ZeroCost)

Summary steps:

Set up self-hosted project management (Gitea + Drone).

Deploy Project Oracle agent.

Enable Predictive Scaling Engine (Ansible integration).

Train hardware failure model (Backblaze data).

Deploy compliance roadmap agent (RIA extension).

Generate first AI board summary.

Verify tenant experience KPIs are being collected.

END OF PART 22 – v6.3 WITH ALL MODIFICATIONS

Part 23: Global Country Coverage & Settlement Matrix (v6.3 – Fully Upgraded, Dynamic, Zerocost, with All Modifications: Three-Tier AI, Multi-Shipment, Distressed Country Factor, Financing Regulations, Dual-Mode Aware, and Real-Time Updates)

23.0 Philosophy

The SGTX platform operates in a world where regulations, payment rails, currencies, and country risk change by the hour. The Global Country Coverage & Settlement Matrix is not a static Excel sheet – it is a living, AI-driven knowledge graph that continuously maps every jurisdiction on Earth onto a dynamic tier, selects the optimal settlement path for every trade corridor, and ensures compliance with local laws before a single cent moves. Every country (195+ recognised jurisdictions) is assigned a real-time coverage tier. Every PSP, instant-payment system, CBDC pilot, and traditional banking rail is monitored, ranked, and integrated automatically. All of this runs on zero-cost data feeds, open-source ML models, and the three-tier AI (Groq + HF local + Ollama).

In v6.3, the settlement matrix is deeply integrated with the Smart Inbox – when a country's tier changes, or a new payment rail becomes available, or a sanction is imposed, the affected tenants receive prioritised inbox items with direct links to the relevant trade or compliance action. The PlainLanguage Governor Decision Panel explains any blocked payment attempts due to jurisdiction rules, and the Admin Portal's PSP Manager tab displays the live matrix for platform governance. The matrix also supplies distressed country factors (used in Phase 7) and DeFi eligibility flags(used in Phase 4). Multi-shipment contracts are validated against the matrix per shipment; financing regulations are applied per agreement.

23.1 Core Technology Stack (Zero-Cost)

Component	Technology	Purpose
Country risk & tiering model	XGBoost (local)	Daily dynamic tier assignment (Full/Standard/Limited/Restricted/Blocked)
PSP & rail ranking	LightGBM (local)	Real-time settlement path ranking per corridor
Regulatory intelligence	HF NLP + RIA (Rust/Rig)	Monitor sanctions, embargoes, banking regulations, CBDC announcements
FX arbitrage & multi-hop	LSTM + 0x API (free) + ECB	Cheapest multi-hop currency

path		conversion
Sanctions lists	OFAC/UN/EU free XML/JSON, updated every 15 min	Instant propagation to matrix and Governor
Country data fabric	World Bank, IMF, UN Comtrade, central bank sites (scraped via HF Donut)	Economic indicators, population, trade volume
Voice query	Vosk (offline) + HF Mixtral + Groq	Natural-language questions about country coverage
Document generation	Tera + headless Chrome	Auto-generated regulatory reports per country
Infrastructure	PostgreSQL, pgvector, ClickHouse, WasmEdge, OPA	Storage, enforcement, audit
Smart Inbox (v6.3)	inbox-service (Rust)	All jurisdiction changes, new rail alerts, sanction notifications as actionable items
Decision Panel (v6.3)	Governor + Groq/Ollama	If payment blocked due to country rules, plain-language explanation and alternative rail suggestion

No billing details required.

23.2 Dynamic Coverage Tiers (AIAssigned, Real-Time)

Every country is categorised into one of five tiers, recalculated daily by the Country Tiering Agent (A2). The model ingests sanctions status (RIA), political stability (World Bank Governance Indicators), currency volatility (ECB), PSP availability & health, regulatory friendliness (RIA), and platform historical data (default/dispute rates). If a country's score changes beyond a threshold, the tier is automatically updated. Groq generates an explanation logged in Admin Portal and broadcast via NATS to Smart Inboxes of affected tenants.

Tier	Meaning	Example Countries (dynamic)	Allowed Actions
Full	All platform features available; multiple local PSPs; DeFi allowed if permitted locally; full document verification	UAE, Singapore, Germany, UK, Netherlands, Canada	Unrestricted trading, financing, DeFi, advanced KYB
Standard	Core trade execution and fee collection supported; may lack advanced financing options	Egypt, India, Brazil, Kenya, Vietnam, Indonesia	Full trading, standard PSP settlement, basic financing
Limited	Trade execution possible only via pre-approved corridors or PSPs; enhanced due diligence required	Nigeria, Bangladesh, Pakistan, Lebanon (post-crisis)	Trading allowed with escrow or LC; restricted PSP list

Restricted	Only specific commodities or corridors allowed; requires high-level KYB and manual compliance review	Russia, Belarus, Myanmar, Venezuela	Conditional trade only; no DeFi; enhanced monitoring
Blocked	No transactions permitted under any circumstances (sanctions, platform policy)	North Korea, Iran, Syria, Cuba, Crimea region	No access; all requests DENY automatically

Tier changes are irreversible without a Governor decision; a downgrade automatically freezes pending trades in that jurisdiction until reviewed. The affected party sees the block explained in the PlainLanguage Governor Decision Panel.

23.3 Global Settlement Rail & PSP Matrix (Dynamic)

The matrix maps every trade corridor (origin-destination currency pair) to the optimal settlement path, computed in real time by the PSP Router (A2) and displayed in the Admin Portal's PSP Manager tab.

23.3.1 Rail Types Supported (Auto-Discovered)

Rail Type	Examples	Integration
Traditional SWIFT	MT103, MT202	Existing PSPs (Stripe, Payoneer, etc.)
Instant Payment Systems	SEPA Instant, Faster Payments (UK), UPI (India), PIX (Brazil), Fawry (Egypt), MPesa (Kenya)	Scraped central bank APIs, tested via sandboxes
Regional Clearing	Buna (Arab), PAPSS (Africa), SADC-RTGS	Discovered by RIA; added if free API available
CBDC Pilots	eCNY (China), Sand Dollar (Bahamas), ePound (Egypt)	Monitored; integrated when publicly accessible
Stablecoin Rails	USDC, USDT (on Polygon/Ethereum)	Via Ethereum RPC (self-hosted) and 0x API
OpenBanking (PSD2)	EU, UK	Sandbox APIs, self-hosted

23.3.2 Real-Time Matrix Example (Partial, Dynamic)

Buyer Country	Seller Country	Preferred Rail	Fallback Rail	PSP (Primary)	PSP (Fallback)	Fee Rail	DeFi Allowed?
Egypt	Vietnam	SWIFT + Fawry	Stablecoin (USDC)	Fawry	Payoneer	Stripe	No (CBE ban)
UAE	India	Buna + UPI	SWIFT	Buna	Razorpay X	Payoneer	Yes (ADGM)
Germany	Turkey	SEPA Instant	SWIFT	Adyen	Stripe	Stripe	Restricted (EU MiCA)
Kenya	China	MPesa + PAPSS	SWIFT	Flutterwave	–	Payoneer	No

Brazil	Nigeria	PIX + PAPSS	Stablecoin (USDC)	PIX	Flutterwave	Stripe	Limited (CBN sandbox)
Singapore	Switzerland	Stablecoin (USDC)	SWIFT	Aave (DeFi)	Payoneer	Payoneer	Yes

This matrix is a live view from the database; it changes automatically when rails are added, PSP health degrades, or sanctions hit. Admin Portal's Configuration History records every change.

23.3.3 Multi-Hop FX Settlement (Optimal Path)

When direct settlement is not possible or expensive, the system computes the cheapest multi-hop path in real time using the FX Arbitrage Hunter (A2). Example: KES → USDC (Flutterwave on-ramp) → USDT (Uniswap V3, 0.001% fee) → IDR (Indodax off-ramp). Total cost 0.35% vs SWIFT 3.5%. Groq summary: "Optimal path saves 3.15% (630ona630ona20k trade). Recommended." User approves; platform generates settlement instruction with all three steps. Fallback rail used automatically if any step fails.

23.4 Global Sanctions & Embargoes – Instant Propagation

RIA ingests OFAC, UN, EU, UK, and 300+ national sanctions lists every 15 minutes. When a new sanction is detected:

Affected country/entity flagged in sanctions_cache.

Country Tiering Agent immediately re-evaluates tier. If downgrade to Blocked, Governor freezes all pending trades involving that country.

All relevant jurisdictions rows updated (e.g., defi_allowed = false, distressed_sale_allowed = false).

Groq drafts compliance alert sent to affected tenants via Smart Inbox.

Admin Portal's PSP Manager tab and Jurisdiction Matrix Editor reflect the change immediately.

When sanction is lifted (detected by same process), tier restored automatically unless other risk factors persist. Affected tenants receive follow-up Smart Inbox alert.

23.5 Automated Country-Specific Regulatory Reporting

The jurisdictions.reporting_requirements JSONB field is maintained by RIA. Example for Egypt:

json

{

"reports": [

{ "name": "CBE Monthly Remittance Report", "frequency": "monthly", "deadline": "10th day after month end", "format": "CBE_Form_10.xml", "auto_generate": true },

{ "name": "VAT Settlement Report", "frequency": "quarterly", "deadline": "45 days after quarter", "format": "VAT_Return_2025.pdf", "auto_generate": true }

]

}

Reporting Agent compiles data from settled trades, fills official templates (HF Donut), generates ready-to-submit PDF. Smart Inbox item: “Monthly CBE remittance report for June 2026 is ready. [Download]”

23.6 Distressed Country Factor & Financing Regulations

For each country, jurisdictions stores:

distressed_sale_allowed (YES/CONDITIONAL/NO)

distressed_country_factor (multiplier for SGTX fee on distressed sales, e.g., 0.85 for UAE due to mandatory liquidator fee)

defi_allowed and defi_restrictions

stablecoins_allowed array

The RIA updates these from local laws. When a distressed listing is created (Phase 7), the Governor applies the country factor. For financing (Phase 4), the DeFi eligibility gate uses defi_allowed and stablecoins_allowed.

23.7 Voice-Queried Country Intelligence (A1)

Through any portal or mobile app, users can ask natural-language questions about country coverage, tiers, settlement options. Vosk transcribes offline (or Web Speech API), HF Mixtral extracts intent, Groq generates response. Examples:

“What tier is Kenya?” → “Kenya is Standard tier. You can trade freely, settlement via MPesa or PAPSS. DeFi is not yet allowed per Central Bank of Kenya regulations as of June 2026.”

“Can I settle a trade with Nigeria via USDC?” → “USDC settlement to Nigeria is currently Limited. The Central Bank of Nigeria allows USDC only for transactions below $10,000. For larger amounts, SWIFT via Flutterwave is recommended.”

*“What’s the best rail for a 50kpaymentfromEgypttoGermany?”∗→Querieslivematrix,computesoptimalpathviaSEPAInstantwithAdyen(cost50kpaymentfromEgypttoGermany?”∗→Querieslivematrix,computesoptimalpathviaSEPAInstantwithAdyen(cost12), presents result.

23.8 Zero-Cost Country Data Fabric

For every jurisdiction, platform continuously enriches a profile using free public data:

Population, GDP, trade volume – World Bank API, IMF, UN Comtrade

Currency, central bank rate – ECB, national central banks (scraped)

Regulatory body contacts – RIA-scraped official gazettes

Trade agreements (GAFTA, COMESA, etc.) – WTO RTA database

Political stability, corruption index – World Bank Governance Indicators, Transparency International

Telecom infrastructure – ITU (for mobile money readiness)

This data feeds the Country Risk Score, tiering model, and settlement matrix. Refreshed daily, stored in jurisdictions metadata JSONB.

23.9 Expansion Scoring for Uncovered Regions

Expansion Optimizer (Phase 22) calculates an Expansion Score (0-100) for countries not yet at Full or Standard tier. Features: pending tenant registrations, trade volume with existing SGTX countries, regulatory readiness, infrastructure availability.
Example: *"Bangladesh scores 82/100: 120 registered users waitlisted, strong textile trade with existing EU tenants, no sanctions, new instant-payment rail (bKash). Recommended to deploy a sovereign node and integrate bKash as PSP. Estimated first-year GMV $8M."* Recommendation appears in Admin Portal's Smart Inbox; multisig can approve expansion with one click, triggering Ansible provisioning.

23.10 Governance Gates (Coverage & Settlement)

Gate ID	Check	Enforcer	Failure Action
GC1	Every country tier recalculated at least daily	Country Tiering Agent (A2)	Use last known tier; alert Admin via Smart Inbox
GC2	Settlement rail selection passes jurisdiction & sanctions checks	Governor (WasmEdge)	Block transaction; display PlainLanguage Decision Panel with fallback rail suggestion
GC3	FX path slippage prediction ≤1% vs agreed rate	FX Oracle (A2)	Hold payment; alert treasury via Smart Inbox
GC4	All required regulatory reports generated for each completed trade in that jurisdiction	Reporting Agent (A2)	Block new trades in that jurisdiction until resolved; tenant receives COMPLIANCE Smart Inbox item
GC5	New sanctions automatically propagate to relevant matrices within 15 min	RIA (A2)	Manual override by compliance officer if delayed; affected tenants notified via Smart Inbox

23.11 Implementation Checklist (macOS/Linux – ZeroCost)

Summary:

Build jurisdiction & matrix services.

Seed initial jurisdiction data (195+ countries).

Start RIA sanctions monitor.

Deploy country tiering model.

Query matrix via voice (test).

Generate monthly regulatory reports for all active jurisdictions.

Verify Smart Inbox items generated on tier change.

END OF PART 23 – v6.3 WITH ALL MODIFICATIONS

Part 23: Global Country Coverage & Settlement Matrix (v6.3 – Fully Upgraded, Dynamic, Zerocost, with All Modifications: Three-Tier AI, Multi-Shipment, Distressed Country Factor, Financing Regulations, Dual-Mode Aware, and Real-Time Updates)

23.0 Philosophy

The SGTX platform operates in a world where regulations, payment rails, currencies, and country risk change by the hour. The Global Country Coverage & Settlement Matrix is not a static Excel sheet – it is a living, AI-driven knowledge graph that continuously maps every jurisdiction on Earth onto a dynamic tier, selects the optimal settlement path for every trade corridor, and ensures compliance with local laws before a single cent moves. Every country (195+ recognised jurisdictions) is assigned a real-time coverage tier. Every PSP, instant-payment system, CBDC pilot, and traditional banking rail is monitored, ranked, and integrated automatically. All of this runs on zero-cost data feeds, open-source ML models, and the three-tier AI (Groq + HF local + Ollama).

In v6.3, the settlement matrix is deeply integrated with the Smart Inbox – when a country's tier changes, or a new payment rail becomes available, or a sanction is imposed, the affected tenants receive prioritised inbox items with direct links to the relevant trade or compliance action. The PlainLanguage Governor Decision Panel explains any blocked payment attempts due to jurisdiction rules, and the Admin Portal's PSP Manager tab displays the live matrix for platform governance. The matrix also supplies distressed country factors (used in Phase 7) and DeFi eligibility flags(used in Phase 4). Multi-shipment contracts are validated against the matrix per shipment; financing regulations are applied per agreement.

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Component	Technology	Purpose
Country risk & tiering model	XGBoost (local)	Daily dynamic tier assignment (Full/Standard/Limited/Restricted/Blocked)
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Regulatory intelligence	HF NLP + RIA (Rust/Rig)	Monitor sanctions, embargoes, banking regulations, CBDC announcements
FX arbitrage & multi-hop	LSTM + 0x API (free) + ECB	Cheapest multi-hop currency

path		conversion
Sanctions lists	OFAC/UN/EU free XML/JSON, updated every 15 min	Instant propagation to matrix and Governor
Country data fabric	World Bank, IMF, UN Comtrade, central bank sites (scraped via HF Donut)	Economic indicators, population, trade volume
Voice query	Vosk (offline) + HF Mixtral + Groq	Natural-language questions about country coverage
Document generation	Tera + headless Chrome	Auto-generated regulatory reports per country
Infrastructure	PostgreSQL, pgvector, ClickHouse, WasmEdge, OPA	Storage, enforcement, audit
Smart Inbox (v6.3)	inbox-service (Rust)	All jurisdiction changes, new rail alerts, sanction notifications as actionable items
Decision Panel (v6.3)	Governor + Groq/Ollama	If payment blocked due to country rules, plain-language explanation and alternative rail suggestion

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Every country is categorised into one of five tiers, recalculated daily by the Country Tiering Agent (A2). The model ingests sanctions status (RIA), political stability (World Bank Governance Indicators), currency volatility (ECB), PSP availability & health, regulatory friendliness (RIA), and platform historical data (default/dispute rates). If a country's score changes beyond a threshold, the tier is automatically updated. Groq generates an explanation logged in Admin Portal and broadcast via NATS to Smart Inboxes of affected tenants.

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Standard	Core trade execution and fee collection supported; may lack advanced financing options	Egypt, India, Brazil, Kenya, Vietnam, Indonesia	Full trading, standard PSP settlement, basic financing
Limited	Trade execution possible only via pre-approved corridors or PSPs; enhanced due diligence required	Nigeria, Bangladesh, Pakistan, Lebanon (post-crisis)	Trading allowed with escrow or LC; restricted PSP list

Restricted	Only specific commodities or corridors allowed; requires high-level KYB and manual compliance review	Russia, Belarus, Myanmar, Venezuela	Conditional trade only; no DeFi; enhanced monitoring
Blocked	No transactions permitted under any circumstances (sanctions, platform policy)	North Korea, Iran, Syria, Cuba, Crimea region	No access; all requests DENY automatically

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UAE	India	Buna + UPI	SWIFT	Buna	Razorpay X	Payoneer	Yes (ADGM)
Germany	Turkey	SEPA Instant	SWIFT	Adyen	Stripe	Stripe	Restricted (EU MiCA)
Kenya	China	MPesa + PAPSS	SWIFT	Flutterwave	–	Payoneer	No

Brazil	Nigeria	PIX + PAPSS	Stablecoin (USDC)	PIX	Flutterwave	Stripe	Limited (CBN sandbox)
Singapore	Switzerland	Stablecoin (USDC)	SWIFT	Aave (DeFi)	Payoneer	Payoneer	Yes

This matrix is a live view from the database; it changes automatically when rails are added, PSP health degrades, or sanctions hit. Admin Portal's Configuration History records every change.

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When direct settlement is not possible or expensive, the system computes the cheapest multi-hop path in real time using the FX Arbitrage Hunter (A2). Example: KES → USDC (Flutterwave on-ramp) → USDT (Uniswap V3, 0.001% fee) → IDR (Indodax off-ramp). Total cost 0.35% vs SWIFT 3.5%. Groq summary: "Optimal path saves 3.15% (630ona630ona20k trade). Recommended." User approves; platform generates settlement instruction with all three steps. Fallback rail used automatically if any step fails.

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All relevant jurisdictions rows updated (e.g., defi_allowed = false, distressed_sale_allowed = false).

Groq drafts compliance alert sent to affected tenants via Smart Inbox.

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For each country, jurisdictions stores:

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distressed_country_factor (multiplier for SGTX fee on distressed sales, e.g., 0.85 for UAE due to mandatory liquidator fee)

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stablecoins_allowed array

The RIA updates these from local laws. When a distressed listing is created (Phase 7), the Governor applies the country factor. For financing (Phase 4), the DeFi eligibility gate uses defi_allowed and stablecoins_allowed.

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For every jurisdiction, platform continuously enriches a profile using free public data:

Population, GDP, trade volume – World Bank API, IMF, UN Comtrade

Currency, central bank rate – ECB, national central banks (scraped)

Regulatory body contacts – RIA-scraped official gazettes

Trade agreements (GAFTA, COMESA, etc.) – WTO RTA database

Political stability, corruption index – World Bank Governance Indicators, Transparency International

Telecom infrastructure – ITU (for mobile money readiness)

This data feeds the Country Risk Score, tiering model, and settlement matrix. Refreshed daily, stored in jurisdictions metadata JSONB.

23.9 Expansion Scoring for Uncovered Regions

Expansion Optimizer (Phase 22) calculates an Expansion Score (0-100) for countries not yet at Full or Standard tier. Features: pending tenant registrations, trade volume with existing SGTX countries, regulatory readiness, infrastructure availability.
Example: *“Bangladesh scores 82/100: 120 registered users waitlisted, strong textile trade with existing EU tenants, no sanctions, new instant-payment rail (bKash). Recommended to deploy a sovereign node and integrate bKash as PSP. Estimated first-year GMV $8M.”* Recommendation appears in Admin Portal’s Smart Inbox; multisig can approve expansion with one click, triggering Ansible provisioning.

23.10 Governance Gates (Coverage & Settlement)

Gate ID	Check	Enforcer	Failure Action
GC1	Every country tier recalculated at least daily	Country Tiering Agent (A2)	Use last known tier; alert Admin via Smart Inbox
GC2	Settlement rail selection passes jurisdiction & sanctions checks	Governor (WasmEdge)	Block transaction; display PlainLanguage Decision Panel with fallback rail suggestion
GC3	FX path slippage prediction ≤1% vs agreed rate	FX Oracle (A2)	Hold payment; alert treasury via Smart Inbox
GC4	All required regulatory reports generated for each completed trade in that jurisdiction	Reporting Agent (A2)	Block new trades in that jurisdiction until resolved; tenant receives COMPLIANCE Smart Inbox item
GC5	New sanctions automatically propagate to relevant matrices within 15 min	RIA (A2)	Manual override by compliance officer if delayed; affected tenants notified via Smart Inbox

23.11 Implementation Checklist (macOS/Linux – ZeroCost)

Summary:

Build jurisdiction & matrix services.

Seed initial jurisdiction data (195+ countries).

Start RIA sanctions monitor.

Deploy country tiering model.

Query matrix via voice (test).

Generate monthly regulatory reports for all active jurisdictions.

Verify Smart Inbox items generated on tier change.

END OF PART 23 – v6.3 WITH ALL MODIFICATIONS

This response is AI-generated and for reference purposes only.

proceed

We have reached the end of the blueprint modifications. The user has confirmed all changes and we have now regenerated Parts 0 through 23 with full expansions and all requested modifications. The remaining parts (24-29) are less likely to need extensive modification, but the user asked to "proceed" implying continue through the rest. I will produce the remaining parts concisely, incorporating any v6.3 modifications (threat model, SLA, glossary, OpenAPI, appendices, manifesto) as needed. I will ensure consistency with earlier changes (removal of operating modes, three-tier AI, SGTX fee paid by seller, etc.). I will produce them without code blocks, in plain text as requested.

Part 24: Threat Model & Attack Surface Analysis (STRIDE + MITRE) – v6.3 Fully Updated, with All Modifications

24.1 Threat Model Methodology

SGTX applies STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) per component and maps high-level adversary tactics to MITRE ATT&CK for Enterprise & Cloud. Every microservice, database, WASM module, and Governor decision point is treated as a ministate with explicit trust boundaries. The model is continuously verified by a self-hosted Rig agent (threat-scanner) that runs weekly STRIDE rescans, quarterly red-team simulations, and daily vulnerability checks using open-source scanners (Trivy, ZAP, nuclei). All findings are logged in a threat_findings table and must be resolved with a Governor-approved remediation plan.

In v6.3, the threat model is extended to cover new tenant-centric surface areas : the Smart Inbox (realtime feed of prioritised actions), the tenant_message JSONB in Governor decisions (plain-language explanations shown to users), tenant impersonation by platform admins, dual-mode toggle, multi-shipment per-shipment locking, financing co-bidding, distressed triage dashboard, and the expanded API endpoints for export, onboarding, and organisation management.

24.2 STRIDE Analysis Summary (Key Threats & Mitigations)

STRIDE Category	Asset	Primary Threat	Mitigation	Residual Risk
Spoofing	GTID, Governor signatures	Impersonation of Tenant Root or Governor	Ed25519 signatures + ZITADEL WebAuthn (biometric liveness). Every request carries a signed JWT verified by Governor. GTID generation cryptographically unique.	Low
Spoofing	Pallet barcode (SSCC)	Counterfeit physical label	Each SSCC linked to pallet_details record with SHA256 hash; scans verify database. For	Very Low

			high-value goods, hash anchored on-chain (Polygon). AI visual recognition detects tampering.	
Spoofing	Voice commands	Impersonation via recorded speech	Vosk offline STT + HF Mixtral intent extraction; biometric voiceprint for high-risk actions; stress flagging. Voice data never leaves device.	Low
Spoofing	Smart Inbox items	Injection of fake items into tenant's inbox	inbox-service only creates items in response to NATS events from trusted services, each with cryptographic signature. Frontend subscribes to authenticated user's topic. RLS prevents cross-tenant injection.	Low
Spoofing	Tenant message (tenant_message JSONB)	Altered explanation displayed to user	Message stored in governor_decisions, append-only and Loom-chained. Frontend fetches via Governor-gated endpoint.	Very Low
Tampering	governor_decisions, Loom log	Alteration of decision history	Cryptographic hash chaining (Loom). Every decision signed with Ed25519, stored in append-only ClickHouse. Hourly chain integrity verification.	None
Tampering	WASM modules (constitution, jurisdiction)	Malicious constitutional bypass	WASM binaries sandboxed in WasmEdge, signed by multisig, verified	None

			by Governor before loading. Any change requires Governor decision.	
Tampering	Packing plan (container loading)	Unauthorised modification	Packing plan locked (irreversible) with Governor decision and Loom hash. Any subsequent change creates new version with full audit trail.	None
Tampering	Smart Inbox item content	Altering title/description to mislead user	Items generated server-side from trusted NATS events; frontend does not modify. Updates require signed API call. Groq-generated text logged.	Low
Tampering	Administrator impersonation session	Admin modifies user data during session	Impersonation is readonly. All write buttons disabled clientside and any write API calls blocked by serverside check. All page views logged.	Very Low
Repudiation	Milestone confirmations	Denial of action	Every milestone confirmation signed by confirming device (ZITADEL certificate) and linked to Governor decision ID. Full audit trail replayable via Loom.	None
Repudiation	Contract signatures	“I never signed” claim	ZITADEL passkeys provide FIDO2-level non-repudiation. Cryptographic signature stored in contracts table and chained.	None

Repudiation	Smart Inbox action taken	"I didn't dismiss that" claim	Each snooze/dismiss action logged in inbox_history with employee ID and timestamp (append-only, auditable). Not Loom-chained (UI action), but sufficient for audit.	Low
Information Disclosure	FeeLock KV (NATS)	Leak of payment split instructions	NATS JetStream KV encrypted at rest (AES256GCM), short TTLs. Access restricted via NATS credentials and network policies (Cilium).	Low
Information Disclosure	Tenant trade data (cross-tenant)	One tenant reading another's data	PostgreSQL Row-Level Security (RLS) enforced by app.current_tenant_id. OPA policies evaluated client-side before rendering. Data scope columns further limit visibility.	None
Information Disclosure	Voice/biometric raw data	Eavesdropping or storage leak	All voice/AR processing on-device. Only extracted intents (text) transmitted. Raw audio/photos never leave device unless explicitly uploaded.	None
Information Disclosure	tenant_message content	Leaking sensitive compliance reasons to unauthorised parties	Message accessible only via authenticated user's session and Governor-gated /v1/decisions/{decision_id}/explanation.	None
Denial of Service	Governor Service	Overload leading to default-DENY	Active-active K3s deployment with	Low

			autoscaling (HPA). Circuit-breaker pattern (fail-closed) with WasmEdge. Rate-limiting at Nginx/WasmEdge layer.	
Denial of Service	Payment Orchestrator	PSP flood or timeout	Multi-PSP fallback with AI-driven retry and circuit breaker. Fallback PSPs ranked by real-time health.	Low
Denial of Service	Smart Inbox (inbox-service)	Overwhelming event stream	Service stateless, horizontally scalable. NATS JetStream provides backpressure; events queued if service overwhelmed.	Low
Elevation of Privilege	WasmEdge modules	Malicious code escaping sandbox	WasmEdge runs in strict sandbox with no filesystem or network access (cap-based). Modules pre-compiled with known hashes. Runtime integrity checks.	None
Elevation of Privilege	Employee RBAC	Privilege escalation via role manipulation	OPA policies evaluate permissions in real time; RBAC changes require tenant admin approval and are logged. Administrative functions require multisig (3/5).	None
Elevation of Privilege	Admin impersonation	Abuse of impersonation for unauthorised access	Requires multisig approval (3/5); session time-limited (30 min), readonly, fully logged. Tenant notified after session ends.	Low
Elevation of Privilege	Dual-mode toggle override	Gaining access to hidden features	Dual-mode context enforced	Low

		by manipulating API calls	both client-side (UI hiding) and server-side (API endpoints reject or filter fields based on JWT claim).	

24.3 MITRE ATT&CK Coverage (High-Priority Tactics)

Tactic	Technique	Mitigation	Example
TA0001 – Initial Access	Phishing, Valid Accounts	ZITADEL WebAuthn (FIDO2) with biometric liveness. Voice actions require continuous biometric verification. KYB/KYC blocks unverified entities.	Phishing attempt to steal password fails without physical security key; failed login logged, tenant admin alerted.
TA0002 – Execution	Command & Scripting	No user-facing shell or code execution. AI agents run sandboxed; all irreversible actions go through Governor, which only executes pre-compiled WASM modules.	Malformed API call from compromised logistics provider account rejected by Governor input validation; IP rate-limited.
TA0004 – Privilege Escalation	Exploitation for Privilege Escalation	Constitutional layer (WASM) forbids self-escalation. Admin functions gated by multisig. OPA policies versioned, immutable without multisig approval.	Employee attempts employee.in vitewithout permission; OPA returns DENY, logged, tenant admin notified.
TA0005 – Defense Evasion	Impair Defenses, Indicator Removal	Loom tamper-evident logging cannot be altered or deleted. Continuous compliance agent verifies log integrity hourly.	Attacker deletes governor_deci sions row; Loom hash chain breaks, detected within 60 minutes, CRITICAL incident created.
TA0006 – Credential Access	Unsecured Credentials	All secrets at rest encrypted (AES256GCM). Tenant keys stored in SoftHSMv2. API keys hashed. No plaintext credentials in config files.	Developer accidentally commits API key; pre-commit hook detects pattern and refuses commit.
TA0008 – Lateral Movement	Remote Services	Microsegmentation via Cilium network policies; each microservice can only	Compromised barcode -service pod attempts to connect to PostgreSQL; Cilium

		talk to explicitly allowed peers. SSH access restricted.	drops packet, logs attempt, raises alert.
TA0009 – Collection	Data from Information Repositories	Data minimisation: no trade principal stored; voice raw data stays on device. Access to PostgreSQL strictly via Governor proxy enforcing RLS.	Insider runs SQL query to dump cross-tenant trade values; RLS returns only rows for their tenant; query logged, flagged.
TA0010 – Exfiltration	Exfiltration Over Web Service	All outbound traffic from K3s pods blocked by default (NetworkPolicy deny-all). Only approved egress endpoints (PSP webhooks, free data feeds) whitelisted. Data-in-transit TLS 1.3.	Malware on notification-serviceattempts to exfiltrate data to external IP; Cilium egress policy denies connection, logs, pod terminated.
TA0040 – Impact	Data Destruction, Service Stop	Self-healing infrastructure with chaos-engineered resilience. Backups continuous (WAL, daily pg_dump), encrypted. Multi-region active-active failover.	Cairo node storage fails; WAL-shipping to Dubai ensures zero data loss (RPO=0). Anycast DNS redirects traffic to Dubai within seconds (RTO<5 min).

24.4 Attack Surface Inventory (v6.3)

#	Surface	Protection
1	Public API (/v1/*)	Nginx rate-limiting, WasmEdge JWT validation, OPA + Governor gating, input validation (Rust typesafety), WAF (OWASP Coraza). New endpoints (inbox, export, tenant experience) gated by same pipeline.
2	Governor Proxy (single ingress)	Active-active, circuit-breaker, deep request inspection, TLS 1.3, mutual TLS for inter-service calls.
3	NATS JetStream (internal, not exposed)	CiliumNetworkPolicy; only Governor and specific services can publish/subscribe. Encrypted at rest.
4	Temporal Server (internal)	Only accessible from K3s service mesh. Workflows run with least-privilege service accounts.

5	WasmEdge Runtime (internal sandbox)	No network or filesystem access; pre-compiled modules with known hashes; runtime integrity checks.
6	PostgreSQL (internal)	Only reachable from configured pods; RLS enforced; audit logging via pgaudit; TLS connections.
7	ClickHouse (internal analytics)	Separate credentials, read-only access for analytics services; network policies restrict.
8	ZITADEL (identity endpoint)	Standalone, hardened deployment; WebAuthn/RSA; rate-limiting; brute-force protection.
9	Mobile & Web Frontends	CSP headers, no sensitive logic client-side, all data validated server-side, OPA-compiled WASM policies run in browser for UI hiding only.
10	Partner API (marketplace)	Dedicated API keys with Ed25519 signatures, rate-limiting per partner, scope-limited JWT. Rate limit gauges and increase requests governed.
11	PSP SDK Integrations	Only outbound to known IPs; credentials encrypted at rest; webhook signatures verified cryptographically.
12	DNS / Infrastructure	DNSSEC (where supported), Anycast for resilience, infrastructure as code (Ansible) with immutable configs.
13 (v6.3)	Smart Inbox WebSocket	Authenticated NATS WebSocket (JWT); messages scoped to user’s tenant; no cross-tenant leakage.
14 (v6.3)	Data Export Endpoint	Governor-gated; exports only data belonging to authenticated tenant; download URLs signed and expire.
15 (v6.3)	Tenant Impersonation Session	Requires multisig approval; readonly mode enforced server-side; session timeout 30 min; full audit logging; tenant notified after session.

24.5 Continuous Controls & Verification

Weekly STRIDE rescan (Rig agent) generates fresh STRIDE matrix, compares with baseline, alerts Admin Portal.

Daily vulnerability scan : Trivy scans container images; OWASP ZAP runs against staging APIs; nuclei scans for misconfigurations. CRITICAL findings block CI/CD.

Quarterly Red-Team Exercise : Rig agent simulates advanced adversary using predefined MITRE attack graphs in sandboxed staging. Post-mortem auto-generated.

Model & Data Poisoning Detection : Federated learning validator checks for anomalous updates; drift monitor detects data poisoning.

Incident Response : Prometheus Alertmanager sends alerts to SGTX mobile app/push notification. Groq-generated incident summary posted to Admin Portal Smart Inbox.

Smart Inbox Integrity Check (v6.3) : Periodic job verifies every inbox_items entry corresponds to a valid NATS event and that priority_score correctly computed. Orphaned or miscalculated items flagged.

All findings stored in threat_findings table (PostgreSQL), resolved within SLAs (Critical: 4h, High: 24h). Resolutions require Governor decision with explanation.

24.6 Residual Risk Acceptance

The Platform Governance Authority (multisig 3/5) reviews residual risk register quarterly. Any residual risk rated Medium or above explicitly accepted with signed justification and remediation timeline. Published (anonymised) in Admin Portal security dashboard. v6.3 register includes Smart Inbox data integrity (Low), tenant impersonation misuse (Low given controls), and Groq API unavailability for tenant message generation (Low due to Ollama fallback).

24.7 Zero-Cost Toolchain

All threat modelling and testing tools are OSS : OWASP Threat Dragon (or manual STRIDE sheets), MITRE ATT&CK Navigator, Trivy, OWASP ZAP, nuclei, Rig agent (built in-house), Prometheus/Alertmanager, Cilium (eBPF), Plonky3 (for ZKproofs). No external penetration-testing services required.

END OF PART 24 – v6.3 WITH ALL MODIFICATIONS

Part 25: SLA & Uptime Guarantees (v6.3 – Fully Upgraded, Zerocost, with All Modifications)

25.0 Purpose & Scope

This document defines the service-level agreements (SLAs) that govern the SGTX platform's production environment. These commitments cover availability, latency, recovery time (RTO), and recovery point (RPO) for all critical services that directly affect a tenant's ability to initiate, track, settle, or manage trades, financing agreements, or distressed cargo. The SLAs are contractually binding under the Master Service Agreement and are monitored entirely by self-hosted, zero-cost open-source tools (Prometheus, Grafana, Loki, and a custom sla-monitor Rust service). No external monitoring service or paid APM is used. In v6.3, the SLA framework is extended to include the new tenant-facing services – Smart Inbox, Governor tenant-message generation, Tenant Data Export, Financing Service, and Multi-shipment Coordinator – reflecting their criticality. All SLA metrics are published on a public, self-hosted status page (status.sgtx.io), and tenants can view their own SLA performance history in the Admin Portal (read-only for tenant admins).

25.1 Core Platform SLAs (Production Bare-Metal Cluster)

The following SLAs apply to the production environment, which is deployed on a minimum of three sovereign nodes (multi-region). All percentages are calculated on a monthly basis.

Component	Availability Target	Latency (p95)	RTO	RPO	Credits if Breached (Monthly)
Governor Service	99.95%	≤800 ms	5 min	0 sec	10% of monthly platform fee
End-to-End Workflow (API gateway → decision)	99.90%	≤3 sec	15 min	0 sec	15% of monthly platform fee
FeeLock KV (NATS JetStream)	99.99%	≤50 ms	2 min	0 sec	5% of monthly platform fee
GTID Resolution	99.99%	≤200 ms	2 min	0 sec	None (best-effort)
Audit Log (Loom)	100% (immutable by design)	N/A	N/A	0 sec	Full month refund if integrity

					breach detected
Smart Inbox Service (v6.3)	99.90%	≤500 ms (item list retrieval)	5 min	0 sec (items can be regenerated from NATS events)	5% of monthly platform fee
Governor Tenant-Message Generation (v6.3)	99.50% (success rate within fallback timeout)	≤3 s (Groq) / ≤50 ms (template fallback)	N/A (degrades to template)	N/A	None; template fallback ensures users always see explanation. Persistent Groq unavailability >24h triggers 2% platform fee credit.
Tenant Data Export Service (v6.3)	99.50%	≤15 min (generation time for exports up to 10 MB)	30 min	0 sec (requests requeued)	2% of monthly platform fee if export queue stalled for >1h
Financing Service (bidding, disburseme nt)	99.90%	≤2 sec (bid submission)	10 min	0 sec	5% of monthly platform fee
Multi-shipment Coordinator	99.90%	≤1 sec (per-shipment lock)	5 min	0 sec	5% of monthly platform fee

Definitions:

Availability : Measured as (total_minutes_in_month – downtime_minutes) / total_minutes_in_month. Downtime is any period where the service responds with 5xx errors for >90% of requests, as measured by sla-monitor from at least two geographic regions.

RTO (Recovery Time Objective) : Maximum time from declaration of a disaster (automatically by AIOps or manually by Platform Governance Authority) to full restoration of the service in at least one region.

RPO (Recovery Point Objective) : Maximum acceptable data loss measured in time. For most services, zero due to synchronous replication and JetStream mirroring. For Tenant Data Export Service, RPO means any export request submitted during downtime will be automatically requeued and processed when service recovers.

25.2 Maintenance Windows

Announcement : All scheduled maintenance announced via Tenant Portal (banner in Smart Inbox), public status page (status.sgtx.io), and email to tenant admins at least 14 calendar daysin advance.

Duration : Maximum 4 hours per quarter, per sovereign node. Maintenance executed during low-activity periods (e.g., Sunday 02:00-06:00 UTC for global, local off-peak for regional nodes). sla-monitor automatically excludes announced windows from availability calculations.

Emergency Maintenance : For critical security patches, notice period reduced to 24 hours. Affected tenants receive push notifications via SGTX mobile app and high-priority Smart Inbox item. Emergency maintenance exceeding 1 hour counts against SLA unless directly attributable to a third-party zero-day with no workaround.

25.3 Status Page & Incident Response

25.3.1 Public Status Page (status.sgtx.io)
Built with cState or Upptime (both open-source). Displays:

Realtime component status (Operational, Degraded Performance, Partial Outage, Major Outage).

Incident history for last 90 days, including post-mortem summaries (Groq-generated, reviewed by Platform Governance Authority).

Upcoming scheduled maintenance.

Per-component uptime percentage for current and previous month.
Tenants can subscribe via email or RSS feed. Infrastructure runs on a separate lightweight VPS, independent of main SGTX nodes.

25.3.2 Incident Response Process

Detection : Prometheus Alertmanager detects SLA breach (e.g., Governor 5xx >60 sec) → triggers alert to on-call engineer via push notification and self-hosted webhook. AIOps agent (A2) attempts automated remediation (cordoning failing node, scaling replicas).

Triage : On-call engineer acknowledges within 5 min. Incident logged in incidents table with unique ID. If automated remediation fails or incident persists >10 min, engineer escalates to Platform Governance Authority via Smart Inbox critical alert.

Resolution : Root cause fixed. incident.end_time set. Post-Mortem Generator (Groq) drafts summary within 1h.

Review : Platform Governance Authority approves post-mortem within 5 business days; published to status page and incident history. If breach resulted in SLA credits, sla-monitor automatically calculates credit amount and posts notification to tenant's Smart Inbox and Admin Portal.

25.4 Credit & Compensation Model

SLA credits are calculated as a percentage of the platform's fixed monthly usage fee (the "platform fee" agreed in the MSA). If no fee charged (promotional period), credits applied as service credits toward future transactions.

Calculation : Credit amount = (breach percentage from table) × monthly platform fee. Multiple SLAs breached in same month → credits applied independently (not capped).

Application : Automatically applied to tenant's next monthly statement. No manual claim required. Tenants can view SLA credit balance and history on "SLA Dashboard" (Company Admin, read-only, powered by ClickHouse).

Exclusions : No credits for force majeure, third-party PSP outages, scheduled maintenance, or tenant's own infrastructure failures. For Tenant Data Export Service, delays caused by exceptionally large exports (>100 MB) that exceed standard SLA are exempt (communicated to tenant at time of request).

Maximum Credit : Total SLA credits in any month cannot exceed 100% of monthly platform fee.

25.5 Zero-Cost Monitoring & Enforcement

sla-monitor Rust service runs on each sovereign node and from an external independent region. Sends synthetic requests to each service's health endpoint, measures latency, records 5xx errors.

Every hour, aggregates probe results into uptime percentages and latency percentiles, publishing to Prometheus (sgtx_sla_uptime_percent{component=...}) and ClickHouse.

Alerting : sla-monitor pushes to Alertmanager → routes to on-call engineer. Alert includes component, current uptime, link to Grafana dashboard.

Transparency : Public API endpoint /v1/sla/status returns current month's uptime for all covered components (authenticated, scoped to tenant's own data).

Audit : All SLA calculations, credit issuance events, and incident records logged immutably in ClickHouse; audit-chain-verifier ensures no tampering.

25.6 Example SLA Incident Flow (v6.3)

Scenario : Network partition between Cairo and Dubai nodes causes Smart Inbox service partially unavailable for EU users routed to Frankfurt. External probe detects 5xx errors for 12 consecutive minutes.

Detection (03:12 UTC): External probe fails 4 of 5 requests to inbox-service on Frankfurt node. Alertmanager fires. On-call engineer receives push notification: "CRITICAL: Smart Inbox service – 80% failure rate on Frankfurt node for 5 min."

AIOps Response : AIOps agent detects NATS connection flapping, cordons Frankfurt pods, redirects traffic to Dubai node. Service restored within 2 min.

Manual Verification (03:14 UTC): Engineer acknowledges alert, verifies failover, investigates root cause (faulty switch). Total downtime: 12 min (0.028% of monthly minutes). 99.90% availability SLA breached.

Credit Calculation : sla-monitor records downtime, automatically computes credit. End of month, tenant's statement shows credit of 5% of monthly fee (~50for50for1,000 fee).

Post-Mortem : Groq drafts: "Smart Inbox service experienced 12-minute partial outage on Frankfurt node due to network hardware failure. Automatic failover restored service in 2 min. Root cause: faulty switch. Mitigation: additional network redundancy ordered. Credits issued to all affected tenants." Reviewed and published.

Tenant Notification : Affected tenants receive Smart Inbox item: "On 14 Jun, Smart Inbox was unavailable for 12 min. Service restored. A credit of $50 has been applied. [View Incident Report]"

END OF PART 25 – v6.3 WITH ALL MODIFICATIONS

Part 26: Glossary & Acronyms (v6.3 – Quick Reference – with All Modifications)

26.1 Core Terms

Term	Definition	Example / Context				
A0-A4	AI Authority Levels: Observational (A0), Advisory (A1), Constraining (A2), Escalation (A3), Governance (A4). A5 forbidden.	Negotiation Bot at A2; escalates to A3 for deadlock.				
Accelerated Outreach Mode	Time-boxed, simultaneous notification to eligible saved contacts for distressed cargo, with floor price and real-time ranking.	Seller sets 6h window, floor $810; notifies 5 contacts; system ranks offers.				
Activity Feed	Chronological, plain-language log of every significant trade/financing/distressed event, displayed in Trade Command Center.	“2026-06-10 08:30 – Seller accepted trade request.”				
AI Agent	Rig-based software component performing	Carrier Risk Scorer updates logistics				

	a specific AI function (Intent Parser, Credit Oracle, etc.).	provider risk scores from news.				
Anonymous Market Intelligence Panel	Opt-in, differentially private (ε=0.1) aggregated market statistics (freight rates, APRs, etc.).	Seller sees average freight rate for corridor, ±$300 confidence.				
Approval Chains (Internal)	Tenant-defined multi-signature requirements for high-value actions (contract.sign, quote.submit, etc.).	Contract >$100k requires 2 trade managers + 1 finance approver.				
AR Inspection Mode	Augmented reality overlay for QC inspectors, showing pallet positions, AI defect detection, and override accountability.	Inspector overrides AI mould detection with mandatory reason.				
Barcode (SSCC)	GS1-128 serial shipping container code, uniquely identifying each pallet. Includes QR with verifiable credential.	Pallet OR-001 has SSCC 106141411234567890.				

Biometric Stress Flag	Off-by-default, on-device voice stress analysis; consent required. Advisory only, never blocks actions.	Financier's voice stress flag triggers advisory note to tenant admin.				
Bot Decision Log	Post-negotiation summary of all offers/counteroffers made by AI negotiation bot.	"Round 1 – offered 47, 500;Round 2–offered47,500;Round2–offered47,800 and accepted."				
Business Unit (BU)	Divisions within a tenant's organization graph, enabling delegated administration and approval chains.	"Fresh Produce BU" and "Textiles BU" under same GTID.				
Capacity Heatmap	Overlay on 3D container viewer showing historical loading density (opt-in, differential privacy).	Heatmap shows rear 2m underutilised; seller adjusts pallet placement.				
CBDC	Central Bank Digital Currency (e.g., ePound, Digital Rupee). Monitored and integrated dynamically.	Rail Discovery Agent adds "CBDC-ePound" as settlement rail.				

Check Buyers (Distressed)	Advisory feature allowing seller to see eligible saved contacts before starting outreach.	Seller clicks “Check Buyers”, sees Juice Factory (trust 91) and selects.				
Co-Financing	Multiple financiers each financing a portion of the same financing request.	Bank X 60k,DeFiPoolY60k,DeFiPoolY40k → total $100k.				
Collaborative Packing Plan	Realtime multi-user editing of packing plan using Yjs (CRDT) and WebRTC.	Warehouse operator and trade manager adjust pallet layout simultaneously.				
SGTX FEES Calculator Preview	Removed; replaced by SGTX fee display (seller pays, added to quote). Buyer sees total price and “includes SGTX fee of $X”.	Buyer sees 30,360totalincluding30,360totalincluding 360 fee.				
Configuration History	Version-controlled manifest of all platform configuration (OPA policies, WASM modules, PSP lists), with diff and rollback.	Admin rolls back jurisdiction matrix to Version 12; Governor hot-reloads.				
Container	After fee	Shipping				

Release Confirmation	payment, SGTX sends email/API to logistics provider with one-time token; container cannot be released without acknowledgment.	line acknowledges token; “Container Released” milestone unlocked.				
Customs Readiness Tab	Buyer tab with dynamic document checklist, traffic light status, one-click document requests.	Missing phytosanitary certificate flagged red 72h before ETA.				
DeFi PlainLanguage Risk Summary	Mandatory modal before any DeFi financing bid; 5 bullet points explaining stablecoins, health factor, collateral risk, no guarantee.	Financier reads and clicks “I understand”.				
DeFi Protocol Risk Oracle	Rust agent monitoring protocol audits, TVL, exploits → risk score (0-100).	Aave V3 score 94; Compound score 88.				
Digital Twin	Realtime simulation of shipment physical state (position, temperature, ETA, shelf-life).	Predicts 12h ETA delay due to storm; alerts buyer via Smart Inbox.				
Distressed Country	Multiplier applied to	UAE factor 0.85				

Factor	SGTX fee for distressed sales in countries with mandatory costs (e.g., liquidator fees).	reduces fee from 1.03% to 0.8755%.				
Distressed Triage Dashboard	Three-path UI: “Sell Quickly”, “Comply with Local Law”, “File Insurance Claim”.	Seller selects “Comply” and generates CBE notification automatically.				
Draft State	Incomplete trade requests, packing plans, quotes, contracts, financing requests saved with status = 'DRAFT'. Autosaves every 30 sec; expires after 14 days.	Seller resumes packing plan from Smart Inbox “Draft” item.				
Dual-Mode Toggle	Header switch for trading companies (trader_mode = DUAL) to toggle between Buyer and Seller dashboards within the Trader Portal.	User clicks “Seller”; portal reloads showing seller tabs.				
Evidence Package Auto-Compiler	Automatically gathers all trade, milestone, sensor, document,	Dispute evidence includes inspector’s override reason and				

	AI decision data into a cryptographically sealed package for disputes. QC overrides flagged.	original AI confidence.				
FeeLock (formerly SGTX FEESLock)	Non-custodial instruction record in NATS KV that instructs PSP to split fee payment; SGTX never holds funds.	Marked ACTIVE after trade fee payment; for multi-shipment, per-shipment.				
Financier Preferences	Pre-set filters (countries, amount, settlement type) that determine which RFQs a financier receives.	Barclays accepts only UK, EU, USA; trust score ≥75.				
Financing Fee (SGTX)	Flat 0.25% of financed amount, deducted from disbursement via PSP split. Paid by borrower.	100,000financing→fee100,000financing→fee250, borrower receives $99,750.				
Governor	Authoritative decision engine (OPA + WasmEdge + Loom). Generates tenant_message for any DENY/CONDITIONAL verdict.	Blocked contract lock due to missing KYC → Decision Panel explains.				

GTID	Global Trade Identity: SGTX-{COUNTRY}-{TYPE}-{SEQ}-{CHECKSUM}. Unique, verifiable identifier for every tenant.	SGTX-EG-TRD-002139-7F3A resolves to Nile Foods.				
HS Code	Harmonized System code for commodity classification. AI-classified, dynamic updates from WCO.	“Valencia Oranges” → HS 0805.10.				
Incote (Incoterms)	International Commercial Terms (2020). Incoterms engine validates logistics cost requirements.	CFR requires seller to pay sea freight; FOB does not.				
Logistics Addendum	Mandatory signed agreement by each logistics provider, binding them not to release container without SGTX confirmation, and to include USTN/GTIDs on documents.	Shipping line signs addendum before contract lock.				
Loom	Deterministic immutable	Each decision’s loom_hash		decision_json		signature).

	hash chain for all Governor decisions.	= SHA256(previous				
Margin Estimator (Commodity-Specific)	XGBoost + LLM refinement model that estimates profit margin per commodity and country pair. Used to calculate SGTX trade fee rate.	Oranges Vietnam→Egypt margin 18.6% → fee rate 1.03%.				
Marketplace Auto-Attribution	Automatic revenue share attribution for repeated direct trades between parties introduced by a marketplace partner (first-touch, 12-month window).	Banner: “This trade attributed to TradeBridge – dispute within 72h.”				
Memory Layer (Organization)	Opt-in pattern learning: remembers preferred incoterm, packaging, providers, price ranges for commodity-route. Prefills forms.	“Based on 12 similar trades, we prefilled CFR incoterm.”				
MicroUSTN	New USTN generated for distressed partial sale (split from original). Linked via	Original USTN retained; microUSTN for 3 damaged pallets.				

	parent-child relationship.					
Multi-Shipment Contract	Master contract with multiple future shipments; each shipment locks independently after per-shipment fee payment.	6 monthly shipments; each shipment pays its own fee before locking.				
Network Tab	Saved contacts with trust portraits, relationship health scores, performance dashboards, indirect connections. Dual-mode aware.	Buyer sees sellers' delivery performance; seller sees buyers' payment behaviour.				
Operating Modes (SIMPLE/ADVANCED/ENTERPRISE)	REMOVED – all tenants have uniform access; visibility controlled by RBAC, data scopes, and dual-mode toggle.	–				
PlainLanguage Governor Decision Panel	Universal UI component that translates every DENY/CONDITIONAL verdict into plain language with	"Your trade is on hold because seller jurisdiction requires enhanced KYC. [Upload Certificate]".				

	remediation links.					
Proof-of-Location (ZK)	Zero-knowledge proof that a truck entered a geofence at a specific time, without revealing exact GPS. For smart escrow release.	Driver's app generates ZKproof; escrow releases payment.				
QC Override Accountability	Inspector overriding AI defect detection must provide mandatory reason (min 10 chars); override flagged in dispute evidence.	Inspector overrides mould to bruising; reason stored.				
Rail Discovery Agent	RIA submodule that scrapes central bank websites and regulatory feeds to discover new payment rails (CBDC, instant systems).	Discovers ePound in Egypt; proposes integration.				
ReOptimise Diff View	Side-by-side comparison of previous vs new logistics bundle recommendation after carrier risk scores	FF Y replaces FF X, saving $150, risk score improved.				

	change.					
SGTX Trade Fee	Dynamic fee (0.1%-2.5%) paid solely by seller, added to quoted price. Collected upfront (single-shipment) or per-shipment (multi-shipment).	Trade fee 360add edto360add edto30,000 quote; seller pays before lock.				
Smart Inbox	Unified, prioritised action feed; landing page for all portals. Items include title/description generated by Groq (fallback Ollama).	“Sign contract before 14:00 UTC – your trade will be cancelled.”				
SSCC	Serial Shipping Container Code (GS1-128) uniquely identifies pallets.	Each pallet gets unique SSCC.				
Stuck Trade Recovery	Automated SLA-driven reminders, escalations, and autocancellations for stalled trades or shipments.	Trade pending seller response 7 days → autocancelled.				
Tenant Activity Log	Self-service audit trail of all employee actions, with	Admin exports filtered log as CSV, verifies signature.				

	cryptographically signed exports.					
Tenant Branding (Whitelabel)	Free for all tenants: custom logo, colours, portal name, email sender name.	Nile Foods portal shows their logo and “Nile Foods Trade Desk”.				
Tenant Lifecycle	State machine (REGISTERED → ONBOARDING → KYB_PENDING → VERIFIED → etc.) controlling access and features.	KYB_PENDING blocks real trade creation; sandbox still allowed.				
Trade Command Center (TCC)	Single page view for each USTN, financing agreement, or distressed microUSTN: timeline, pending action, summary cards, activity feed.	Clicking USTN opens TCC with all trade information.				
USTN	Unique SGTX Transaction Number: SGTX-{BUYER_SUFFIX}-{SELLER_SUFFIX}-{TIMESTAMP}-{RANDOM8}. No version	SGTX-1397F3A-456ABC-20260415120000-A1B2C3D4.				

	suffix.					
Verifiable Credential	W3C standard ZKproof-based credential for pallet attributes (e.g., organic certification). Verified offline.	QR scan shows "□ GOTS Organic – verified".				
Workflow Recovery Service	Rust microservice monitoring SLA timers for stuck trades; generates reminders, escalations, autocancellations.	Trade unanswered for 5 days → reminder; 7 days → cancellation.				

26.2 Acronyms (Selected)

AIS – Automatic Identification System (vessel tracking)

AML – Anti-Money Laundering

API – Application Programming Interface

CBAM – Carbon Border Adjustment Mechanism (EU)

CBE – Central Bank of Egypt

CBDC – Central Bank Digital Currency

CFR, CIF, FOB, EXW – Incoterms

DDL – Data Definition Language

eBL – electronic Bill of Lading

ECB – European Central Bank

ESG – Environmental, Social, Governance

FAO – Food and Agriculture Organization

FATF – Financial Action Task Force

GDPR – General Data Protection Regulation (EU)

GMV – Gross Merchandise Value

GTID – Global Trade Identity

HF – Hugging Face

HS – Harmonized System

ICC – International Chamber of Commerce

IoT – Internet of Things

ISO – International Organization for Standardization

KYB/KYC – Know Your Business / Know Your Customer

LCA – Life Cycle Assessment

LC – Letter of Credit

LTV – Loan-to-Value

MQTT – Message Queuing Telemetry Transport

MSA – Master Service Agreement

NATS – Neural Autonomic Transport System (messaging)

OPA – Open Policy Agent

OSRM – Open Source Routing Machine

PEP – Politically Exposed Person

PSP – Payment Service Provider

RAG – Retrieval-Augmented Generation

RIA – Regulatory Intelligence Agent

RLS – Row-Level Security

RPO/RTO – Recovery Point / Time Objective

SAR – Suspicious Activity Report

SGTX – Sovereign Governed Trade Execution

SLA – Service Level Agreement

SSCC – Serial Shipping Container Code

TLS – Transport Layer Security

UBO – Ultimate Beneficial Owner

USTN – Unique SGTX Transaction Number

VC – Verifiable Credential

VRP – Vehicle Routing Problem

WASM – WebAssembly

WCO – World Customs Organization

ZPL – Zebra Programming Language

END OF PART 26 – GLOSSARY & ACRONYMS (v6.3)

Part 27: Openapi 3.1 Specification Index (v6.3 – API Overview – with All Modifications)

27.0 Overview

All SGTX public and partner endpoints are documented in OpenAPI 3.1 format. The specification files are served from /api/v1/openapi.json (public) and /api/v1/partner/openapi.json(partner). The interactive Swagger UI is available to verified tenants only. This index lists every endpoint, grouped by service, with a summary description and, where applicable, a brief example or note. Endpoints new or modified in v6.3 are marked with ★. All endpoints are prefixed with /v1unless explicitly noted as public (/health, /metrics) or internal (/admin/*).

27.1 Governor & Platform Endpoints

Method	Path	Description	Notes
GET	/health	Health check for Governor and backends	Returns 200 if all engines (OPA, WasmEdge, Loom) responsive
GET	/v1/openapi.json	Public OpenAPI specification	–
POST	/v1/governor/decision	Submit an action for Governor approval (internal)	–
GET	/v1/gtid/resolve	Resolve a GTID to full tenant details	?gtid=... → legal name, jurisdiction, trust score, KYB tier
GET	/v1/ustn/resolve	Resolve a USTN to full shipment details	?ustn=... → current status, milestones, containers
POST	/v1/SGTX FEES/verify	Verify a fee payment receipt (PSP webhook)	–
★ GET	/v1/decisions/{trade_id}/pending	Returns all pending Governor decisions for a given trade/financing/distressed entity	Used by “What’s blocking me?” and Decision Panel
★ GET	/v1/decisions/{decision_id}/explanation	Returns full tenant_message for a specific Governor decision	Rendered by PlainLanguage Decision Panel

★ **POST**	/v1/decisions/{decision_id}/escalate	Triggers A3 escalation for the decision, notifying Admin Portal	Body: {"reason": "..."}

27.2 Trade & Quote Endpoints (Including Structured Container and Multi-Shipment)

Method	Path	Description	Notes
POST	/v1/trade/initiate	Create a new trade request (Phase 1) using structured per-container form or free-text (Express Mode). Body includes buyer GTID, structured containers (country of origin, destination, port of discharge, commodities, packaging, pallet count, etc.), optional multi-shipment schedule.	Returns trade_request_id.
GET	/v1/trade/{id}	Retrieve a trade request by ID	–
PATCH	/v1/trade/{id}	Update trade status (phase transition)	–
★ **POST**	/v1/trade/amend	Submit amendment request (buyer)	Body includes changes to weight, packaging, port, etc.
★ **POST**	/v1/trade/amend/respond	Seller responds to amendment (accept/partial/counter)	–
POST	/v1/quote/exwlock	Phase 2.1 – Lock EXW price for a trade (seller)	Body includes EXW price per commodity. Governor gate G2UA1.
★ **POST**	/v1/quote/loading-origin	Set seller’s loading country and port	Required before EXW lock.
★ **POST**	/v1/quote/logistics/manual	Phase 2.3 – Manual logistics cost entry (per incoterm)	Body includes trucking, customs, THC, freight, etc.
POST	/v1/quote/logistics/rfq	Phase 2.3 – Generate and send RFQ to logistics providers	Body specifies sourcing mode (directed/anonymous) per service.
POST	/v1/quote/logistics/bundle	Phase 2.3 – AI bundle optimiser: get recommended bundle	Returns optimal provider combination with cost and on-time

			probability.
★ POST	/v1/quote/logistics/reoptimise	Phase 2.3 – Trigger bundle reoptimisation with updated risk scores and new quotes	Returns diff view. Logged as Governor decision.
★ POST	/v1/quote/alternative-ports	Add alternative delivery ports to a quote	Body includes port, transit days, extra cost.
POST	/v1/quote/logistics/select	Phase 2.3 – Select a provider for a service	–
★ POST	/v1/quote/submit	Phase 2.7 – Submit final assembled quote to buyer	Body includes total delivered price (including SGTX fee added by seller). For multi-shipment, includes per-shipment prices.
★ GET	/v1/quote/delivery-options/{quote_id}	Get all delivery options (primary + alternatives) with total prices	–

27.3 Packing & Containerisation Endpoints

Method	Path	Description	Notes
POST	/v1/packing/optimise	Phase 2.2 – Run palletisation and container loading solver	Body includes buyer's structured container data (carton dimensions, pallet type, etc.). Returns pallet plan and 3D layout.
POST	/v1/packing/collaborative/session	Start a collaborative packing plan editing session (Yjs + WebRTC)	Returns session token.
POST	/v1/packing/lock	Phase 2.2 – Lock the final packing plan	Governor validates container type legality, weight limits.
GET	/v1/packing/{id}	Retrieve a packing plan by ID	–
POST	/v1/packing/visuallayout	Get 3D visual layout data for a container	Returns JSON for Three.js rendering.
POST	/v1/packing/generatelabels	Generate printable PDF of SSCC barcodes	Body includes language, template, printer type (ZPL or PDF).
★ POST	/v1/packing/reprintlabels	Request reprint after pallet has been	Body: {"pallet_ids": [...], "reason": "..."}.

		loaded	Creates Governor decision.
★ **POST**	/v1/packing/carbonfootprint	Calculate carbon footprint for the packing plan (advisory)	Returns CO_2e breakdown, CBAM-ready report.

27.4 Contract & Fee Endpoints

Method	Path	Description	Notes
POST	/v1/contract/genesis	Phase 3 – Initiate contract genesis workflow	–
POST	/v1/contract/upload-own	Upload own contract (PDF)	Must also sign SGTX FEES Addendum. AI extracts fields.
POST	/v1/contract/sign	Phase 3 – Digitally sign a contract (ZITADEL passkey)	–
GET	/v1/contract/{id}	Retrieve contract details and status	–
★ **POST**	/v1/contract/multi-shipment/activate	Activate a specific shipment in a multi-shipment contract (seller)	Triggers per-shipment fee payment request.
★ **POST**	/v1/contract/multi-shipment/confirm	Buyer confirms readiness for a specific shipment	After mutual confirmation, fee payment required.
POST	/v1/fee/calculate	Compute SGTX trade fee (dynamic, commodity- & country-pair-specific)	Returns rate and amount.
POST	/v1/fee/pay	Initiate SGTX fee payment (seller for trade fee, borrower for financing fee)	Redirect to PSP.
★ **GET**	/v1/fee/breakdown/{trade_id}	Returns full adjustment breakdown for the SGTX trade fee	Response: base rate, country boost, seasonality, etc.

27.5 Financing Endpoints (Including Co-Financing and Financier Preferences)

Method	Path	Description	Notes
POST	/v1/finance/request	Phase 4.1 – Create a financing request	Body includes USTN, amount, tenor,

			financing type, preferred settlement method.
★ POST	/v1/finance/preferences	Set financier preferences (countries, amount, settlement type)	For financier portal.
★ GET	/v1/finance/opportunities	List open financing requests matching financier's preferences (auto-RFQ)	–
POST	/v1/finance/bid	Phase 4.2 – Submit an encrypted bid (including portion for co-financing)	Encrypted with borrower's public key. Rate-limited.
★ GET	/v1/finance/bids/{request_id}	Borrower views all bids (decrypted) after window closes	–
POST	/v1/finance/award	Phase 4.4 – Award a winning bid or combination of bids (co-financing)	Creates financing agreement with annexes.
★ POST	/v1/finance/co-financing/accept	Borrower accepts a combination of bids	Sum of accepted bids ≤ requested amount.
GET	/v1/finance/agreement/{id}	Retrieve financing agreement details	–
POST	/v1/finance/disburse	Phase 4.5 – Initiate disbursement (PSP split instruction)	Deducts 0.25% financing fee automatically.
★ GET	/v1/finance/historical/{borrower_gtid}	Financier view of historical financing data (repayments, credit scores)	–
★ POST	/v1/finance/defi/acknowledge	Record DeFi Risk Summary acknowledgement	Stores in employees.defi_risk_acknowledged_at, protocols, stablecoins.
★ GET	/v1/finance/defi/risksummary/{protocol_id}	Returns the five-bullet-point DeFi risk summary	Used in modal.

27.6 Shipment & Milestone Endpoints

Method	Path	Description	Notes
GET	/v1/shipment/{ustn}	Retrieve shipment details by USTN	–
GET	/v1/shipment/{ustn}/milestones	List all milestones for a shipment	–

POST	/v1/shipment/milestone/confirm	Phase 5 – Confirm a milestone (e.g., pallet loaded)	Body includes confirmation method (barcode/voice/manual).
POST	/v1/shipment/barcode/scan	Phase 5 – Report a barcode scan (mobile)	Body: {"sscc": "...", "device_id": "..."}
POST	/v1/shipment/barcode/recognize	Phase 5 – AI visual recognition of damaged label	Body: {"image_hash": "sha256:..."}
POST	/v1/shipment/voice-milestone	Phase 5 – Voice-activated milestone confirmation	Body: {"transcript": "Pallet OR005 loaded into TCNU1234567"}
POST	/v1/shipment/document/upload	Phase 5.3 – Upload a required document	Triggers AI validation (Donut) and comparison with contract.
★ **POST**	/v1/shipment/release/acknowledge	Logistics provider acknowledges container release confirmation	Body: {"release_token": "...", "ustn": "..."}
★ **GET**	/v1/shipments-vault	Aggregated endpoint for Shared Shipments Vault (paginated, filterable, dual-mode aware)	Query params: ?status_filter=active&search=...&page=1&per_page=25
★ **GET**	/v1/shipments-vault/multi-shipment/{contract_id}	List all shipments for a multi-shipment contract	–

27.7 Settlement & Payment Orchestrator Endpoints

Method	Path	Description	Notes
POST	/v1/settlement/instruction	Phase 6 – Create a settlement instruction (trade principal)	Body includes USTN, amount, rail.
POST	/v1/settlement/verify	Phase 6 – Verify payment receipt (PSP webhook, manual upload)	–
POST	/v1/settlement/fxpath	Phase 6 – Get optimal FX conversion path	Returns multi-hop path with estimated costs.
★ **POST**	/v1/settlement/split-instruction	Create PSP split instruction for fee collection (trade fee or financing fee)	Used by fee split orchestrator.
POST	/v1/payment/psp/health	Phase 9 – Report PSP health (internal)	–
POST	/v1/payment/fee/reconcile	Phase 9 – Trigger OpenBanking	–

		reconciliation for fee payments	
★ GET	/v1/payment/psp/fallback/{country}	Get current fallback chain for a country	–

27.8 Distressed Cargo Endpoints (Including "Check Buyers")

Method	Path	Description	Notes
POST	/v1/distressed/declare	Phase 7 – Declare distressed cargo (create listing)	Body includes USTN, pallet IDs, photos, condition description.
★ GET	/v1/distressed/check-buyers/{listing_id}	Advisory: return eligible saved contacts for this distressed lot	Ranked by score. No notifications sent.
POST	/v1/distressed/offer	Phase 7 – Submit an offer for distressed cargo	–
★ POST	/v1/distressed/outreach/accelerated	Phase 8 – Start Accelerated Outreach (requires privacy notice opt-in)	Body includes selected buyer GTIDs, window, floor price.
POST	/v1/distressed/outreach/standard	Phase 8 – Standard outreach (one-by-one notifications)	–
POST	/v1/distressed/microcontract/lock	Phase 7 – Lock microcontract for distressed sale (after offer accepted)	Creates new microUSTN, separate SGTX fee.
GET	/v1/distressed/listing/{id}	Get distressed listing details	–

27.9 Dispute Endpoints (Including QC Fast-Track and Fee Dispute)

Method	Path	Description	Notes
POST	/v1/dispute/file	Phase 10 – File a new dispute (trade, financing, fee)	Body includes entity type (USTN/financing ID), dispute type, description.
POST	/v1/dispute/mediate	Phase 10 – Post a message to the mediation log	–
★ GET	/v1/dispute/evidence/package/	Compile automated evidence package	Returns Loom-hashed JSON.

	{dispute_id}	(including QC overrides)	
★ GET	/v1/dispute/evidence/verify/{package_id}	Public verification page (clientside)	Returns only hashes and signature, no trade data.
POST	/v1/dispute/arbitration/prepare	Phase 10 – Prepare arbitration case filing (PDF)	–
★ POST	/v1/dispute/fee	File a dispute specifically about SGTX fee calculation	Body includes USTN, reason. Escalates to multisig.

27.10 Network & Saved Contacts Endpoints

Method	Path	Description	Notes
GET	/v1/contacts	List saved contacts for current tenant (dual-mode aware)	Supports filtering by smart labels, jurisdiction.
GET	/v1/contacts/{gtid}	Get enriched profile (trust portrait, health score, performance metrics)	–
★ POST	/v1/contacts/voice-command	Execute a voice command on contacts (e.g., “show all suppliers with trust score below 80”)	Body: {"raw_text": "..."}
★ POST	/v1/contacts/verifiable-credential/request	Request a verifiable credential (ZKproof) from a contact	–
★ POST	/v1/contacts/organization-memory/enable	Enable Organization Memory Layer (opt-in)	–

27.11 Marketplace Partner Endpoints (v6.3 – Enhanced)

Method	Path	Description	Notes
POST	/v1/partner/intent/analyze	Submit a raw trade intent for AI analysis	Returns viability score, parsed specs, container recommendation.
POST	/v1/partner/trade/initiate	Attribute a confirmed trade to the partner	Body: intent_id, trade_request_id.
POST	/v1/partner/suppliers/match	Get matching suppliers for an intent (optional)	–
GET	/v1/partner/analytics	Get partner	All benchmarks use

		conversion analytics (privacy-safe)	OpenDP differential privacy.
POST	/v1/partner/webhook/register	Register a webhook for trade events (trade.locked, SGTX FEES.settled, dispute.filed)	–
★ **GET**	/v1/partner/ratelimits	Return current rate limit configuration and usage for the partner	Used by the API Usage gauge in Partner Portal.
★ **POST**	/v1/partner/ratelimits/increase	Request an increase to a rate limit; requires Platform Governance Authority multisig approval	Body: {"endpoint": "...", "requested_limit": 200, "justification": "..."}
★ **POST**	/v1/partner/agreement/propose	Propose a revenue share agreement amendment	Body includes new split and justification.

27.12 KYB/KYC & Onboarding Endpoints

Method	Path	Description	Notes
POST	/v1/kyc/onboard	Submit initial KYB/KYC documents for a tenant	Handles document upload and extraction via Donut/PaddleOCR.
GET	/v1/kyc/status/{tenant_id}	Check verification status	–
POST	/v1/kyc/reverify	Trigger reverification (e.g., tier upgrade, PEP hit)	–
★ **GET**	/v1/onboarding/state	Returns current step, saved drafts, sandbox status	–
★ **POST**	/v1/onboarding/state	Saves progress for an onboarding step	Body: {"step": 2, "data": {...}}
★ **POST**	/v1/onboarding/skip	Skips an optional step (e.g., Step 5)	–
★ **POST**	/v1/onboarding/sandbox/reset	Clears sandbox data for the tenant	–
★ **POST**	/v1/onboarding/exit-sandbox	Transitions tenant from sandbox to production mode	Requires KYB approval or pending manual review.

27.13 Country & Jurisdiction Endpoints

Method	Path	Description	Notes
GET	/v1/country/{code}	Get the current	–

		dynamic tier and settlement rails for a country	
POST	/v1/country/voice-query	Natural-language query about country coverage	Body: {"raw_text": "What are the settlement options for Egypt to Germany?"}
GET	/v1/country/reports	List pending regulatory reports for a jurisdiction	–
POST	/v1/country/reports/generate	Generate a regulatory report (e.g., CBE monthly)	–

27.14 Admin & Observability Endpoints (Internal)

Method	Path	Description	Notes
GET	/metrics	Prometheus metrics endpoint	–
★ **GET**	/admin/decisions	Query Governor decision log (natural-language via Groq)	Supports ?query=...
GET	/admin/threats	List current threat findings	–
POST	/admin/policy/update	Propose an OPA policy update (multisig)	–
★ **POST**	/admin/policy/simulate	Simulate impact of a proposed Rego change by replaying historical decisions	Returns affected decisions count, revenue impact, dispute rate change.
★ **POST**	/admin/config/rollback	Roll back platform configuration to a previous version (multisig)	Creates new version; full audit.
★ **POST**	/admin/tenant/impersonate	Request impersonation of a tenant (readonly, time-limited, multisig approval)	Body: {"tenant_id": "...", "reason": "..."}

27.15 Inbox & Preferences Endpoints (v6.3)

Method	Path	Description	Notes
GET	/v1/inbox	Returns current list of active Smart Inbox items for the authenticated user	Query params: ?min_score=50&categories=NEEDS_SIGNATURE

GET	/v1/inbox/history	Returns last 200 inbox items (including dismissed/snoozed)	–
POST	/v1/inbox/{item_id}/snooze	Snoozes an item	Body: {"duration_minutes": 120}
POST	/v1/inbox/{item_id}/dismiss	Permanently dismisses an item (still logged in history)	–
POST	/v1/inbox/preferences	Saves user preferences (min score, muted categories)	Body: {"min_score": 60, "muted_categories": ["GENERAL"]}

27.16 Tenant Data Export & Statement Endpoints (v6.3)

Method	Path	Description	Notes
POST	/v1/tenant/export	Creates a data export job (trade, financing, distressed data)	Body: {"format": "CSV", "date_range": {...}, "scope": "ALL"}
GET	/v1/tenant/export/{job_id}	Returns job status and, if completed, a signed download URL	–
GET	/v1/tenant/statement	Returns monthly reconciliation statement (JSON or PDF)	Query: ?month=2026-06&format=pdf
POST	/v1/tenant/webhook/register	Registers an accounting webhook for settlement events	–

27.17 Trade Command Center Endpoint (v6.3)

Method	Path	Description	Notes
★ GET	/v1/trade/{ustn}/command-center	Returns aggregated data for the Trade Command Center: timeline, pending action, summary cards, activity feed, trade room, permissions map (filtered by dual-mode context).	Also works for financing agreements (/v1/financing/{id}/command-center) and distressed microUSTNs.

27.18 Dual-Mode & Employee Context Endpoint (v6.3)

Method	Path	Description	Notes

★ **POST**	/v1/employee/switch-context	Switches the active trader mode context (BUY = Buyer, SELL = Seller) for dual-mode tenants. Returns new JWT.	Returns new JWT.

END OF PART 27 – v6.3 WITH ALL MODIFICATIONS

Part 28: Appendices (v6.3 – Fully Upgraded, Zerocost, with All Modifications)

A1: USTN Format Specification (Company-Anchored, No Version)

The USTN (Unique SGTX Transaction Number) is generated at contract lock (single-shipment) or per-shipment lock (multi-shipment). It incorporates the identities of the involved companies to prevent replay and ensure non-repeatability.

Format : SGTX-{BUYER_SUFFIX}-{SELLER_SUFFIX}-{YYYYMMDDHHMMSS}-{RANDOM8}

SGTX: fixed prefix

BUYER_SUFFIX : last 6 alphanumeric characters of the buyer’s GTID (after removing hyphens)

SELLER_SUFFIX : last 6 alphanumeric characters of the seller’s GTID

YYYYMMDDHHMMSS: UTC timestamp of contract/shipment lock

RANDOM8: 8 random alphanumeric characters

Example: SGTX-1397F3A-456ABC-20260415120000-A1B2C3D4

No version suffix is needed because the combination of parties, timestamp, and random ensures uniqueness.

Usage : USTN is stored in shipments.ustn and appears on all documents (invoices, packing lists, bills of lading, customs declarations). For distressed sales, a microUSTN is generated with a new timestamp but same party suffixes, linked via parent_ustn.

A2 : Incoterms 2020 Reference Table (Simplified for Cost Rules)

Incoterm	Seller provides logistics	Seller includes ocean/air freight	Seller includes destination charges	Default SGTX trade fee payer (seller always – no split)
EXW	No (buyer arranges)	No	No	Seller (but no logistics to add)
FOB	Up to loading on vessel	No	No	Seller
CFR	All up to destination port	Yes	No	Seller
CIF	All up to destination port + insurance	Yes	No	Seller
DAP	All up to named place	Yes (if ocean/air)	Yes	Seller
DDP	All including duties	Yes	Yes (including duties)	Seller

Governance : The incoterms engine (WasmEdge) validates that manual logistics cost entries match the incoterm (e.g., ocean freight disabled for FOB). The SGTX trade fee is always paid by the seller and added to the quoted price.

A3 : AI Agent Registry & Authority Mapping (v6.3 – Three-Tier, Removed Operating Modes)

All agents use the three-tier strategy : Groq (A1) with Ollama fallback, Hugging Face local (A2/A3) with Ollama fallback. The table below lists key agents. Full registry is in the repository.

Agent Name	Authority	Phases	Model(s)	Zero-Cost Stack
Intent Parser	A2	1	HF Mixtral (local) + Groq	vLLM, Ollama
Commodity Compatibility	A2	1	WasmEdge rules + HF NLP	Custom rules + local
Container Advisor	A1	1,2	Groq/Ollama	Groq free tier
Margin Estimator (commodity-specific)	A2	2,7	XGBoost + LLM refinement	Local ML + Ollama
Pricing Dynamics	A2	2	XGBoost	Local ML
Carrier Risk Scorer	A2	2,5	LightGBM + HF sentiment	Local ML + news feeds
Logistics Bundle Optimiser	A2	2	Neural CF (PyTorch)	Local training
QC Inspection Recommender	A2	2,5	HF + rules	Local inference
Clause Forge / Checker	A2	3	HF Mixtral (local)	vLLM, Ollama
Negotiation Bot	A2/A3	3,7	Groq (generation) + HF (strategy)	Rig + dual API + Ollama
Credit Intelligence	A2	4	XGBoost	Local ML
DeFi Risk Oracle	A2	4,9	Bayesian + HF Donut + The Graph	Local compute
Milestone Verifier	A2/A4	5	Bayesian + WasmEdge	Python/Rust
ColdChain Quality Predictor	A2	5	LSTM (PyTorch)	Local training
Dispute Triage	A2	10	XGBoost + HF Mixtral	Local ML
Regulatory Intelligence (RIA)	A2/A3	All	HF NLP + Donut + Groq	Rig + free feeds
Smart Inbox Text Generator	A1	All	Groq/Ollama	Groq free tier

Tenant Message Generator	A1	All	Groq/Ollama	Groq free tier

A4: Technical Specifications (Key Services)

Governor Service : Rust 1.88, Axum 0.8, WasmEdge 0.14, OPA 0.63, Loom (custom), Ed25519, port 8080. Generates tenant_message JSONB for DENY/CONDITIONAL.

Identity Service : ZITADEL (self-hosted), GTID generation, onboarding state, dual-mode toggle, lifecycle state.

Database : PostgreSQL 18 + pgvector + TimescaleDB; ClickHouse 24.12 for analytics; Valkey 8.0 optional.

Messaging : NATS 2.11 JetStream (KV for FeeLock, event streams); Temporal 1.26.

Frontend : Next.js 15 (React 19), Tailwind CSS 4, Radix UI, Zustand, Chart.js, Three.js, Leaflet.

Mobile : React Native 0.76 (Expo), WatermelonDB, Vosk, ONNX Runtime, ZXingC++.

Desktop: Tauri v2 (Rust), OS WebView, auto-updater.

AI : vLLM (local), Hugging Face Transformers (local), Ollama, Groq API (free tier).

A5: Distressed Cargo Matrix (Extended, Dynamic)

Country	Distressed Sale Allowed	Restrictions	Reporting	AI-Adjusted Fee Factor (applied to standard trade fee)
USA	Yes	File customs notice if reexported	None	1.0
Germany	Yes	Buyer must be VAT-registered	VAT report	1.0
UAE	Conditional	Only through licensed liquidators	Central Bank notification	0.85
China	No	Must be returned or destroyed	N/A	–
Egypt	Yes	CBE notification required	Monthly report	1.0
India	Yes	Customs bond required	RBI notification	0.95
Saudi Arabia	Conditional	Only after SFDA inspection	SFDA filing	0.90
Singapore	Yes	None	None	1.0
France	Yes	Environmental disposal rules	DGCCRF report	0.92
Nigeria	Conditional	NAFDAC inspection for food	NAFDAC report	0.88

The matrix is updated in real time by RIA. The fee factor is applied to the distressed micro-contract's SGTX fee.

A6 : AI Provider Fallback Chain (Zero-Cost, No Billing Details)

A1 tasks (advisory) : Groq → if rate limited or error → Ollama/LocalAI.

A2/A3 tasks (privacy-critical) : Hugging Face local → if model not loaded or overloaded → Ollama/LocalAI.

A3 formatting : HF for deep analysis (fallback Ollama) then Groq for formatting (fallback Ollama).

Fallback template : If all AI providers fail, the Governor uses a static English template for tenant_message with a notice: "This message is not available in your language. Please contact support."

All calls logged with provider and fallback status. No credit card or billing details required.

A7 : Tenant Quick Start Guide (No Operating Modes – Uniform Experience)

For first-time users:

Onboarding Wizard (Step 1-6) : Complete KYB/KYC, set trader mode (BUY/SELL/DUAL), add first resource (optional), enter sandbox.

Smart Inbox : Landing page. Prioritised actions. Click any USTN to open Trade Command Center.

Buyer (trader mode BUY or DUAL with Buyer active) : New Trade Request (structured container form), Quote Review, Contract Signing, Customs Readiness, Shipments (inbound).

Seller (trader mode SELL or DUAL with Seller active) : Pending Requests, EXW Price Lock, Containerisation & Packing, Logistics Builder (manual or RFQ), Quote Submission (SGTX fee added automatically), Barcode Print, Shipments (outbound).

Financing (optional) : Request financing (seller or buyer). Financiers see full trade details, co-financing available, 0.25% financing fee deducted from disbursement.

Distressed Cargo : Declare distress, use triage dashboard, "Check Buyers" advisory, accelerated outreach with privacy notice.

Disputes : File dispute, evidence autocompiled, mediation log, AI settlement proposal.

Need help? AI Assistant (chat), Decision Panel explains blocks, Smart Inbox always shows next action.

A8 : Terminal Automation Reference & cd Path Index (Linux x64 / macOS)

Root directory: /opt/sgtx (or /Users/.../Desktop/sgtx)

Core service directories:

core/governor

core/identity

core/trade-service

core/contract-service

core/shipment-service

core/payment-orchestrator

core/finance-service

core/multi-shipment-coordinator

core/inbox-service

core/tenant-experience-service

core/workflow-recovery-service

core/tenant-data-export-service

core/financier-preference-service

core/distressed-service

core/network-service

Portals : portals/buyer-portal, portals/seller-portal, portals/logistics-portal, portals/qc-portal, portals/financier-portal, portals/government-portal, portals/admin-portal, portals/marketplace-partner-portal

Common automation scripts (no coding – reference only):

./scripts/dev_start.sh – start all services (infrastructure + Governor + microservices)

./scripts/health_check.sh – curl all health endpoints

./scripts/build_all.sh – build all Rust services with --locked

./scripts/db_migrate.sh – apply PostgreSQL migrations

./scripts/verify_deps.sh – check that all required binaries are present (no reinstall)

Environment variables :
SGTX_DEV_PATH, DATABASE_URL, NATS_URL, GROQ_API_KEY (optional), HF_TOKEN (optional), RUST_BACKTRACE=1, SQLX_OFFLINE=true

END OF PART 28 – v6.3 WITH ALL MODIFICATIONS

Part 29: the Sovereign Trade Operating System – a Manifesto & Living Document (v6.3 – with All Modifications)

29.1 The Problem: Friction as a Service

Global trade is the bloodstream of human civilisation, yet its infrastructure remains trapped in the 19th century. Letters of credit require manual document checking at five banks over seven days. Bills of lading are still printed, stamped, couriered, and lost. A shipment of oranges from Vietnam to Egypt involves at least 17 paper documents, 9 intermediaries, and an average of $1,200 in avoidable demurrage, insurance, and FX costs – before the seller sees payment.

The platforms that have attempted to digitise this process fall into one of two traps:

Marketplaces hold funds, take title, and insert themselves as counterparties, concentrating risk and extracting monopoly rents.

ERP systems digitise internal processes but never bridge the trust gap between counterparties. They lack cryptographic certainty, sovereign governance, and jurisdictional awareness.

The result is a world where small and medium traders – the backbone of emerging economies – are systematically excluded from the most efficient trade finance instruments, forced to pay cash upfront or accept usurious factoring rates, simply because there is no neutral, non-custodial execution layer they can trust.

SGTX was designed to be that layer.

29.2 The SGTX Thesis: Invisible Rails, Visible Trust

SGTX is not a marketplace. It is not a bank. It is not a logistics company. It is the operating systemthat powers trade execution beneath all of them.

Like an operating system, SGTX:

Manages resources (documents, milestones, fees, payments) without consuming them.

Enforces rules (constitutional, jurisdictional, contractual) without exception.

Provides an interface (portals, APIs, voice) that abstracts complexity into simple actions.

Runs anywhere (bare-metal, air-gapped, sovereign nodes) without depending on any single cloud or jurisdiction.

The “invisible rails” metaphor is deliberate : a marketplace or an ERP provider should be able to integrate with SGTX and offer their users cryptographic trade execution without their users ever needing to know SGTX exists. The rails are the API; the trust is the Governor; the experience is the Smart Inbox.

In v6.3, we proved that this operating system could be as intuitive as a consumer app without sacrificing an iota of constitutional rigour. A farmer in the Mekong Delta, a customs officer in Alexandria, and a trade finance banker in Singapore all see the same platform, but each sees only what they need and understands every word.

29.3 Three Unshakable Pillars

Every feature, every service, every policy in this blueprint is anchored to three pillars. Any future change – by the Platform Governance Authority, by a contributor, or by a fork – that violates any of these pillars is, by definition, not SGTX.

Pillar I: Non-Custodial by Structure, Not by Policy
SGTX never holds trade principal or loan funds. Not because a policy says so – because the code makes it structurally impossible. There is no funds table, no escrow bank account in the platform's name, no wallet controlled by the Governor. The platform issues instructions; banks, PSPs, and blockchains execute them. The FeeLock is an instruction to split, not a temporary holding of money. This is not a marketing claim. It is enforced by the WasmEdge constitutional engine, and it is verifiable by any tenant at any time by auditing the Loom chain.

Pillar II: AI May Block, Never Force
The AI Authority Ladder (A0-A4, A5 forbidden) is the constitution's most important safeguard. AI can observe, advise, flag, and escalate – but it can never autonomously execute an irreversible action. A negotiation bot can propose a price; a credit intelligence agent can warn of default risk; a document fraud detector can reject an upload – but no AI can sign a contract, move a dollar, or lock a fee. In v6.3, we extended this pillar to the tenant experience: every AI-driven block is accompanied by a plain-language explanation that tells the human exactly why the block occurred and what to do next.

Pillar III: Sovereign Jurisdiction Supremacy
Laws are code, and code is law – but neither should override the sovereignty of the jurisdictions in which trade occurs. SGTX always applies the strictest applicable rule among buyer, seller, logistics, financier, and governing law. The jurisdiction matrix is not a static configuration file; it is a living knowledge graph updated every 15 minutes by the Regulatory Intelligence Agent from over 300 official sources. A trade that is legal under Egyptian law but would violate an EU sanction is blocked. A distressed sale that is permitted in Germany but requires a liquidator in the UAE is automatically adjusted with the correct fee factor. The platform never asks a tenant to choose which law to follow; it applies the strictest, and explains why.

29.4 Who SGTX Serves (and Who It Does Not)

SGTX serves every legitimate participant in global trade, regardless of size:

The small seller in Vietnam who has never used a letter of credit and is afraid of being cheated. They complete a wizard, use the structured container form (or express mode), and ship their first container within an hour.

The midsized buyer in Egypt who needs to know exactly what documents are missing before the vessel arrives. Their Customs Readiness tab shows a red traffic light and a one-click “Request Document” button.

The logistics provider who receives anonymous RFQs and competes solely on performance, not on relationships. Their match score is transparent, and they see exactly why they were eligible. They can opt out of anonymous RFQs with one click.

The financier who wants to deploy capital into short-term trade finance but lacks the credit intelligence to price risk. The platform gives them 200+ signals, a default probability, a recommended LTV, and full trade disclosure – all computed locally, none shared with competitors.

The government official who needs to approve a customs declaration in 30 seconds, not 3 days, because the AI has already verified every document and the risk score is green.

SGTX does not serve:

Speculators who want to trade commodity futures without physical delivery.

Platforms that want to disguise a marketplace as an execution engine.

Any entity that seeks to circumvent sanctions, evade taxes, or obscure beneficial ownership.

Bad-faith actors who litigate every milestone and refuse mediation. The platform provides irrefutable evidence; it does not replace courts.

29.5 The Zero-Cost Imperative

The platform’s zero-cost commitment is not a cost-saving measure – it is a strategic weaponagainst vendor lock-in, geopolitical risk, and economic exclusion. Every line of code, every model, every data feed is open-source and free. A sovereign node can be deployed on a refurbished server in a shipping container in Djibouti, and it will function identically to a node in Frankfurt. No cloud provider can deplatform SGTX. No proprietary AI API can hold the platform’s intelligence hostage (all models have local fallbacks – Ollama/LocalAI). No PSP can capture monopoly pricing (the Rail Discovery Agent continuously finds new rails). This zero-cost architecture ensures that SGTX can operate in sanctions-adjacent jurisdictions, in countries with capital controls, and in regions where even $10/month SaaS fees are prohibitive. It is, in the most literal sense, trade infrastructure for the entire planet.

29.6 The TenantCentric Revolution (v6.3)

The original v6.2 blueprint was architecturally complete but operationally intimidating. Three independent professional audits confirmed what our own testing showed: tenants were getting lost. They didn’t know what to do next, they didn’t understand why actions were blocked, and they couldn’t find the information they needed without clicking through five tabs.

v6.3 was the systematic resolution of every one of those friction points:

Smart Inbox : A single, prioritised feed that tells every user exactly what to do next, in their language, with a single click to the exact screen.

Trade Command Center : One page per trade. No more tab-switching. Timeline, pending action, documents, finances, risk – all in one place.

PlainLanguage Governor Decision Panel : No more “DENY due to G2UA1.” Tenants see “Your trade is on hold because the seller’s jurisdiction requires enhanced KYC. Upload a Certificate of Origin to proceed.”

Onboarding Wizard & Sandbox : A new user can go from zero to a completed practice trade in under 10 minutes, without reading a manual.

Structured Container Form : Buyers define containers, origins, destinations, ports, commodities, packaging, pallets – all via dropdowns, with cloning and multi-commodity support.

Multi-Shipment Contracts : Flexible per-shipment fee collection, amendments, and independent execution.

Dual-Mode Toggle : Trading companies switch seamlessly between Buyer and Seller dashboards.

No Operating Modes : All tenants get full capabilities from the start; only RBAC and data scopes control visibility.

29.7 The Governance Compact

SGTX is governed by a multisig (3-of-5) of individuals who are bound by the same constitutional rules they enforce. No single person can change the constitution, alter the fee bounds, or override a DENY verdict. This governance compact is not a legal document filed in a drawer; it is live code. The Admin Portal’s Constitutional Policies tab allows the multisig to edit Rego policies and simulate their impact on 10,000 historical decisions before approval. The Configuration History tab records every change, and any change can be rolled back with a single click, producing a new version with full audit. The compact extends to tenants: every tenant admin can define internal approval chains for their own company, requiring dual signatures for high-value contracts or restricting which employees can see commercial invoices.

29.8 The Future (As Defined by the RIA and Project Oracle)

SGTX is not finished. The Implementation Roadmap (Part 22) is continuously reprioritised by the Project Oracle agent, which ingests real-time development data, production metrics, and regulatory changes to suggest the optimal next sprint. The future includes:

Full CBDC integration as more central banks launch digital currencies.

Post-quantum cryptography for all long-term identity and contract records.

Federated learning across sovereign nodes to train fraud-detection models without sharing raw data.

Decentralised governance via DAOs for tenant consortia.

Global Trade Observatory – a public, privacy-safe visualisation of trade flows, powered by SGTX data.

All 62 AddOns from Part 15, activated progressively as tenants demand them and resources allow.

The exact timeline is unknowable because the platform's growth is, by design, reactive to the real world. When a new sanction hits, the RIA updates the jurisdiction matrix within 15 minutes, and the roadmap automatically reprioritises compliance over features. When a new instant-payment rail opens between two corridors, the Rail Discovery Agent proposes integration, and the Admin Portal can activate it with a multisig click.

What will never change : the three pillars. Non-custodial structure. AI that blocks but never forces. Sovereign jurisdiction supremacy. Everything else is subject to evolution.

29.9 A Note to Developers, Auditors, and Future Maintainers

You hold in your hands (or on your screen) the complete DNA of a sovereign, AI-governed, non-custodial global trade execution platform. It is long. It is detailed. It is, in places, uncompromisingly technical. That is by design. This blueprint is meant to be the single authoritative source. If a behaviour is not described here, it is not intended. If a policy contradicts the constitution, the constitution wins. If a tenant asks "why can't I do X?" the answer must be traceable to a principle in this document.

SGTX is open-source. You can fork it, modify it, deploy it in your own jurisdiction. But if you modify it, you must also modify this blueprint. The Loom chain of documentation should be as immutable as the Loom chain of decisions.

One final request : as you build, maintain, or audit this platform, remember the farmer in the Mekong Delta. She does not know what WasmEdge is. She does not care about the difference between A2 and A3. She cares that her oranges arrive in Alexandria on time, that she gets paid, and that the platform does not cheat her. Every dialog, every notification, every default setting should honour her. If it does not, it is a bug – perhaps the most serious kind.

29.10 Closing Words

SGTX began as an idea : that international trade could be executed with the same cryptographic certainty as a blockchain transaction, but without requiring any participant to understand cryptography. That AI could protect traders from fraud and insolvency, but never replace human judgment. That the laws of nations could be respected by code, not bypassed by it. That all of this could be done at zero marginal cost, on infrastructure the operators control, with no external dependencies. v6.3 proves that idea is not only possible – it is ready for production. The rest is not in the blueprint. It is in the trades that will flow through it, the disputes it will prevent, the small businesses it will empower, and the trust it will build between strangers who will never meet but who will, through this platform, honour their contracts.

END OF BLUEPRINT – SGTX PLATFORM v6.3

Total parts : 0 through 29 inclusive, with all subsections, appendices A1–A8 (within Part 28), and comprehensive documentation. This document constitutes the complete, authoritative developer specification for the SGTX sovereign, AI-governed, non-custodial global trade execution platform, built entirely on zero-cost, open-source technology with no billing details ever required. All requested modifications (structured container form, multi-shipment per-shipment fee collection, dual-mode toggle, removal of operating modes, seller-paid SGTX fee added to quote, three-tier AI with fallback, co-financing, distressed “Check Buyers”, logistics addenda, container release control, own-contract upload with addendum, etc.) have been fully integrated. The blueprint is ready for implementation on your own server.

SGTX PLATFORM – COMPLETE IMPLEMENTATION PLAN (FOR ENGINEERS)

This plan sequences all work from the blueprint across 7 phases (0 through 6). Each phase lists the blueprint parts that must be implemented, the core deliverables, and dependencies. Effort estimates are for a team of 3–5 engineers (backend, frontend, DevOps). Adjust based on your team size.

Key principle : Build the execution backbone first, then the tenant experience, then financing & advanced features. Never bypass the Governor.

Phase 0 – Infrastructure & Shared Libraries (2 Weeks)

Goal : Set up development environment, version control, CI/CD, and core data stores.

Blueprint Parts	Sections to implement
Part 6.6	Development Setup & Zero-Cost Stack (all)
Part 5	Initial schema: governor_decisions, audit_log, tenants, employees, roles, gtid_sequences (plus later extensions)
Part 12	Sovereign Tech Stack – deployment topology (development only)
Part 4	Service definitions – but only build shared crates, not full services

Deliverables:

PostgreSQL 18 + pgvector + TimescaleDB running

NATS 2.11 with JetStream enabled

Valkey/Redis for caching

Workspace : /sgtx with core/, portals/, database/, config/, scripts/

Shared Rust crates : sgtx-common (types, error handling, Loom client, Governor client)

CI pipeline (Gitea/Drone or GitHub Actions) with lint, test, security scan

Version-locked dependencies (Cargo.lock, pnpm-lock.yaml)

Key tasks:

Install all system dependencies (Rust, PostgreSQL, NATS, WasmEdge, OPA, Node, pnpm).

Create database schemas (start with minimal tables for identity and governance).

Implement loom logging crate (deterministic hash chain).

Implement governor-client – a Rust trait that all services will use to call Governor.

Write idempotent setup scripts (./scripts/dev_start.sh, ./scripts/health_check.sh).

Dependencies: None – this is pure infrastructure.

Phase 1 – Core Services (MVP Backbone) (4 Weeks)

Goal : Build the Governor, Identity, Trade, Quote, Contract, Payment, and Settlement services – enough to execute a simple trade (free-text, no web portals yet).

Blueprint Parts	Sections to implement
Part 13	Governor Service – full (including OPA, WasmEdge, Loom, tenant_message)
Part 2.1, 2.2, 2.3	Identity – GTID generation, resolution, tenant types (minimal: CORPORATE only)
Part 3, Phase 1	Trade Initiation (simplified: free-text only, not structured form)
Part 3, Phase 2	Seller Quote & Packing (static EXW price, manual logistics cost entry, no alternative ports)
Part 3, Phase 3	Contracting (static clauses, SGTX Witness Clause, digital signatures, FeeLock)
Part 3, Phase 6	Settlement (basic – one PSP, manual reconciliation)
Part 10	Payment Orchestrator (Stripe integration only, gross-up, webhook)
Part 3, Phase 5	Execution (milestones via API, barcode generation – no mobile)
Part 5	All tables needed for above: trade_requests, seller_quotes, contracts, shipments, fee_payment_requests, payment_attempts, pallet_details, etc.

Deliverables:

Governor service (ports 8080) – can evaluate policies, call mock AI, return verdicts.

Identity service (port 8081) – GTID generation, resolution.

Trade service (port 8082) – create trade request (free-text only), status updates.

Quote service (port 8083) – EXW lock, manual logistics cost entry.

Contract service (port 8084) – contract assembly, signature, USTN generation.

Payment orchestrator (port 8085) – Stripe fee collection, webhook verification.

Shipment service (port 8086) – milestones, barcode SSCC generation.

Settlement service (port 8087) – principal settlement (buyer → seller).

All services successfully pass through Governor (intercept middleware).

Integration test : create trade → accept → lock EXW → add logistics → sign contract → pay fee → confirm loading → settle → all logged in Loom.

Key tasks:

Implement Governor's OPA evaluator with basic policies (permissions.rego, contract.rego).

Implement Governor's WasmEdge sandbox (load a simple jurisdiction rule).

Implement tenant_message generation (initially template-based; add Groq/Ollama later).

Build identity service with ZITADEL (self-hosted) and GTID generation.

Build trade service : accept free-text, store in parsed_specs (JSON).

Build contract service : static clause template + mandatory SGTX Witness Clause.

Implement FeeLock in NATS KV.

Integrate Stripe test mode for fee collection.

Build basic shipment milestones (via REST).

Testing Strategy:

Unit tests for each service.

Integration test with Postman collection that exercises the whole flow.

Governor policy tests (OPA opa test).

Dependencies: Phase 0 complete.

Phase 2 – Tenant Experience Layer (Web Portals) (6 Weeks)

Goal : Build the Smart Inbox, Trade Command Center, Buyer Portal, Seller Portal, and dual-mode toggle – making the platform usable by real tenants.

Blueprint Parts	Sections to implement
Part 6.1.1	Smart Inbox (backend service + frontend component)
Part 6.1.2	Trade Command Center (backend aggregation endpoint + frontend)
Part 6.1.3	PlainLanguage Governor Decision Panel (frontend component)
Part 6.2.1	Buyer Portal (full: structured container form, multi-shipment request, customs readiness, etc.)
Part 6.2.2	Seller Portal (full: loading origin, EXW chart, manual logistics, alternative ports, quote submission, barcode print)
Part 6.2.8	MultiTenant Administration – roles, data scopes (minimal for Phase 2)
Part 2.8	Dual-Mode Toggle (global header, context

	switching)
Part 2.7	Onboarding Wizard & Sandbox (basic version)
Part 3.6	Stuck Trade Recovery & Draft States
Part 11	Smart Inbox text generator (A1) and tenant message generator (integrated with Governor)

Deliverables:

inbox-service (port 8090) – creates items from NATS events, urgency scoring, Groq/Ollama titles.

tenant-experience-service (port 8091) – aggregates data for TCC.

Next.js 15 portals (Buyer, Seller) with all specified tabs.

Structured container form (dropdowns, cloning, multi-commodity).

Multi-shipment request UI (Phase 1) and seller response UI (Phase 2).

Alternative delivery ports UI (seller).

Manual logistics cost entry UI (incoterm-aware).

SGTX fee display (added to total quote, shown to buyer as “includes fee”).

Dual-mode toggle in header, with /switch-context endpoint.

Onboarding wizard (6 steps – GTID, legal details, KYB/KYC, profile, optional resources, sandbox).

Sandbox environment (separate PostgreSQL schema, synthetic data).

Key tasks:

Implement inbox-service : subscribe to NATS events, compute score, call Groq/Ollama for title/description.

Build TCC endpoint – join trade, contract, shipment, payment data.

Build Next.js portals with shared components (Smart Inbox, Shipments Vault, Decision Panel).

Implement structured container form (React react-hook-form, dynamic fields).

Implement multi-shipment schedule builder (tabular input).

Implement seller’s alternative ports and manual logistics forms.

Extend Governor policies to enforce dual-mode context.

Implement sandbox data isolation (row-level security based on sandbox_active flag).

Build draft auto-save (every 30s) and draft history tables.

Testing Strategy:

E2E tests with Cypress covering typical user journeys (buyer-seller pair).

Accessibility audits (axe-core, Lighthouse).

Load test for inbox-service (item generation).

Dependencies: Phase 1 services must be stable.

Phase 3 – Financing & Co-Financing (4 Weeks)

Goal : Add conventional and DeFi financing, co-financing, financier preferences, and financier portal.

Blueprint Parts	Sections to implement
Part 3, Phase 4	Universal Trade Finance (all steps)
Part 6.2.5	Financier Portal (full)
Part 9	Hybrid DeFi & Crypto Financing Framework (including DeFi PlainLanguage Risk Summary)
Part 2.3	Tenant types: FINANCIAL, private financier onboarding
Part 5	Tables: financing_requests, financing_offers, financing_agreements, financier_preferences, financier_historical_data, defi_tranche_positions
Part 11	Credit Intelligence Agent (A2), DeFi Risk Oracle
Part 10	Payment orchestrator – financing fee split (deduct 0.25%)

Deliverables:

finance-service (port 8088) – handles requests, RFQ broadcast, bids, disbursement.

defi-service (port 8089) – protocol monitoring, smart contract deployment (testnet).

Financier Portal (React) – opportunities, bidding, co-financing, collateral dashboard.

DeFi PlainLanguage Risk Summary modal (mandatory before any DeFi bid).

Automatic RFQ broadcast based on financier preferences (country, amount, settlement type).

Co-financing acceptance (multiple financiers, sum of bids ≤ requested amount).

Financing fee collection (0.25% deducted at disbursement via PSP split).

Financier historical data (saved repayment performance, credit score trend).

Key tasks:

Implement finance-service endpoints : request, broadcast, bid (encrypted), award.

Implement financier preferences storage and matching logic.

Integrate PSP split instruction for financing fee.

Build DeFi Risk Oracle (fetch protocol info from The Graph, HF Donut for audits).

Build collateral monitoring dashboard (LTV gauge, history chart, margin call workflow).

Extend the payment orchestrator to handle split instructions for financing.

Implement co-financing accounting (separate SGTX FEESLocks per financier).

Testing Strategy:

Unit tests for bid encryption and decryption.

Integration test : financier preferences → RFQ broadcast → co-financing → disbursement split → repayment.

DeFi protocol mock with known risk scores.

Dependencies : Phase 2 portals (to test financing as part of trade flow).

Phase 4 – Distressed Cargo (3 Weeks)

Goal : Implement distressed cargo declaration, AI assessment, triage dashboard, outreach, and microcontract.

Blueprint Parts	Sections to implement
Part 3, Phase 7	Distressed Cargo Resolution (all steps)
Part 3, Phase 8	Distressed Cargo Outreach (including “Check Buyers”, accelerated outreach, privacy notice)
Part 6.2.2.3.11	Distressed Cargo & Notifications tab (seller portal)
Part 6.1.8	Tenant Exit & Data Portability – includes distressed records
Part 5	Tables: distressed_cargo_listings, distressed_cargo_offers, distressed_contact_checks, micro_contracts, distressed_contact_notifications
Part 11	AI Condition Assessment (HF ViT), Dynamic Pricing (XGBoost)

Deliverables:

distressed-service (port 8092) – declaration, condition assessment, pricing, triage, outreach.

Triage dashboard UI (three paths : Sell Quickly, Comply with Local Law, File Insurance Claim).

“Check Buyers” advisory modal (shows eligible saved contacts with ranking).

Accelerated Outreach (time-boxed, simultaneous notifications, privacy notice opt-in).

Microcontract creation with separate SGTX fee (adjusted by distressed country factor).

Co-branded email notifications for distressed outreach.

Unsubscribe mechanism for logistics anonymous RFQs.

Key tasks:

Implement distressed declaration with photo upload and AI condition analysis (HF ViT).

Implement pricing engine (XGBoost) using original value, condition, shelf life.

Build triage dashboard UI (three cards).

Add “Check Buyers” – query tenant_contacts, rank by past distressed purchase success and trust score.

Implement privacy notice modal and Governor log for opt-in.

Implement microcontract creation (new USTN, separate FeeLock).

Integrate with existing notification service for co-branded emails.

Testing Strategy:

Unit tests for condition scoring and pricing.

E2E test : seller declares distressed → “Check Buyers” → selects contacts → accelerated outreach → buyer makes offer → microcontract signed.

Dependencies : Phase 2 (Saved Contacts feature) must be available.

Phase 5 – Advanced Features & Multi-Shipment (4 Weeks)

Goal : Add multi-shipment contracts, commodity-specific margin estimator, logistics addendum & container release, and QC override accountability.

Blueprint Parts	Sections to implement
Part 3, Phase 1 (multi-shipment request)	Buyer toggles multi-shipment schedule
Part 3, Phase 2 (seller response)	Seller modifies schedule, per-shipment pricing
Part 3, Phase 3 (multi-shipment contract)	Master contract, per-shipment mutual confirmation, per-shipment fee payment
Part 5	contract_shipments table, contracts.SGTX FEES_collection_model
Part 7	Commodity-specific & country-pair margin estimator (XGBoost + LLM)
Part 3, Phase 2 (logistics addendum signing)	Logistics providers sign addendum before contract
Part 3, Phase 3 (container release confirmation)	Email/API token after fee payment
Part 6.2.4 (QC override accountability)	Mandatory reason (≥10 chars) when overriding AI
Part 4, Part 6.6	Multi-shipment coordinator service, workflow-recovery-service extended

Deliverables:

Multi-shipment contract UI (buyer schedule builder, seller schedule editor).

Per-shipment activation : “Ready for Shipment” and “Confirm Shipment” buttons.

Per-shipment fee payment and USTN generation.

Margin estimator integrated into SGTX fee calculation (replaces static rate).

Logistics addendum signature workflow (in logistics portal).

Container release confirmation email/API token.

QC override enforcement (≥10 chars, stored in DB).

Key tasks:

Extend trade_requests.parsed_specs to include multi_shipment_schedule.

Extend seller_quotes to store counter_schedule (seller’s modifications).

Implement contract_shipments table and per-shipment activation endpoints.

Train XGBoost model on historical data (or simulate) and integrate into margin estimator.

Add LLM refinement (HF local or Ollama) for real-time adjustments.

Build logistics addendum PDF generation and signing (ZITADEL).

Implement release-confirmation service (email/API with one-time token).

Enforce QC override reason in mobile app and web.

Testing Strategy:

Integration test : multi-shipment contract from request to final shipment.

A/B test for margin estimator (compare with static rate).

Security audit for container release token generation.

Dependencies : Phase 2 (portals) and Phase 4 (distressed) should be stable.

Phase 6 – Mobile & Desktop Apps (4 Weeks)

Goal : Build offline-first mobile app (drivers, QC) and desktop app (logistics, trade managers).

Blueprint Parts	Sections to implement
Part 6.3	Mobile Application (React Native + Expo) – full specification
Part 6.4	Desktop Application (Tauri v2) – full specification
Part 6.3.2.3	AR Inspection Mode (QC)
Part 6.3.2.1	Barcode scanner (offline)
Part 6.3.2.2	Voice commands (Vosk offline)
Part 6.3.2.4	Offline queue & sync engine
Part 6.4.2	System tray, native notifications, direct ZPL printing
Part 20	Location-based trucking (offline OSRM, voice navigation) – optional

Deliverables:

iOS and Android mobile apps (Expo) – installable via TestFlight / internal distribution.

Desktop app (macOS, Windows, Linux) – Tauri shell.

Offline barcode scanning and milestone confirmation.

Voice commands (offline) for loading and QC.

AR inspection overlay with defect detection and override accountability.

Offline queue (WatermelonDB) with automatic sync.

Desktop system tray with notification badges.

Direct ZPL printing to Zebra printers.

Key tasks:

Set up React Native (Expo) project, configure WatermelonDB.

Integrate ZXingC++ for barcode scanning.

Integrate Vosk (offline) + on-device NLU (HF DistilBERT quantised).

Build offline sync engine (queue, replay).

Build AR inspection module (AR.js, camera, HF ViT on-device).

Build Tauri desktop wrapper (Next.js build as frontend, Rust backend).

Implement system tray, notifications, and deep links.

Implement ZPL label printing via raw TCP socket.

Testing Strategy:

Test offline scenarios (airplane mode) on real devices.

Stress test sync conflict resolution.

Automated UI tests (Detox for mobile, Tauri integration for desktop).

Dependencies : Phase 2 portals (APIs must be stable and rate-limited).

Phase 7 – Hardening, Compliance, Production Deployment (4 Weeks)

Goal : Make the platform production-ready: security, SLAs, scalability, compliance, and disaster recovery.

Blueprint Parts	Sections to implement
Part 8	Legal, Compliance & Risk Framework (full) – RIA, SAR drafting, regulatory reports
Part 14	Comprehensive Audit, Observability & Compliance (Prometheus, Grafana, Loki, Jaeger, Loom verifier)
Part 24	Threat Model & Attack Surface Analysis (implement mitigations)
Part 25	SLA & Uptime Guarantees (status page, sla-monitor)
Part 22	Implementation Roadmap & Scaling Strategy (auto-scaling, chaos testing)
Part 12	Production deployment (K3s, Cilium, Firecracker, backup)
Part 2.10	Tenant Lifecycle Engine (complete state machine)
Part 2.9	Internal Tenant Organization Graph (full)
Part 6.2.7	Admin Portal (multisig, policy simulation, impersonation, config rollback)
Part 6.1.4	Tenant Data Export & Reconciliation Statements (cryptographically signed)
Part 6.1.8	Tenant Exit & Data Portability Center
Part 6.1.10	SLA Transparency & Human Escalation Directory
Part 6.1.11	Global UX Rule – The Four Questions (audit all screens)
Part 15	ZeroCost AddOns Suite (activate desired

	addons)

Deliverables:

Production K3s cluster with ≥3 nodes (sovereign nodes as needed).

Full observability stack (Grafana dashboards, alerting).

Loom chain verifier running every hour.

RIA scraper active for 195+ jurisdictions.

Automated regulatory reports (CBE, ECB, CBAM) as PDF/XML.

Tenant Activity Log with cryptographic signing.

Admin Portal with all governance tabs.

Multi-region failover and backup restoration tested.

Key tasks:

Deploy K3s on bare-metal (or VPS) with Cilium and Firecracker.

Configure auto-scaling (HPA) and predictive scaling (LSTM).

Deploy RIA as a cron job; integrate with Governor for real-time updates.

Set up ClickHouse for audit logs; configure 7-year retention.

Implement tenant exit workflow and transfer ownership.

Build Admin Portal (React) – all tabs described in Part 6.2.7.

Implement multisig (3/5) for all constitutional changes.

Run security scans (Trivy, OWASP ZAP, nuclei) in CI and production.

Perform chaos experiments (pod termination, network delay) to validate resilience.

Generate final SLAs and publish status page.

Testing Strategy:

Monthly disaster recovery drill (restore from backup).

Quarterly red-team exercise.

Penetration test by external firm (optional, but recommended).

Dependencies: All previous phases complete.

CROSS-CUTTING CONCERNS (Throughout All Phases)

Concern	Implementation Action
Governor integration	Every service must use the Governor proxy for any write or irreversible read. Use the governor-client crate.
Loom logging	Every Governor decision must call Loom before returning. Include tenant_message in JSON.
Three-tier AI	Implement ai-router with fallback chain: Groq → Ollama, HF → Ollama. Log provider and fallback in ai_inference_records.
Row-level security	Add tenant_id and employee_id to every table; enable RLS and set session variables via

	Governor proxy.
Dual-mode toggle	JWT must include active_trader_mode_context. OPA policies enforce actions against this context.
Testing	Maintain >80% unit test coverage; integration tests for each phase; E2E tests for critical paths.
Version locking	Use --locked for cargo, frozen-lockfile for pnpm, and uv lockfor Python.

SUMMARY OF MILESTONES (Weeks)

Phase	Duration (weeks)	Cumulative	Key Deliverable
0 – Infrastructure	2	2	Databases, NATS, shared libs
1 – Core Services	4	6	Bare trade execution (API-only)
2 – Tenant Experience	6	12	Full web portals (Buyer/Seller)
3 – Financing	4	16	Financier portal, co-financing
4 – Distressed Cargo	3	19	Distressed workflow
5 – Advanced Features	4	23	Multi-shipment, margin estimator
6 – Mobile & Desktop	4	27	Mobile and desktop apps
7 – Hardening & Production	4	31	Production-ready platform

Total estimated team effort : 31 weeks for a team of 3–5 full-time engineers. Adjust based on parallel work.

Landing page: SGTX — The Invisible Rails. Visible Trust.
A Sovereign Trade Execution Operating System — non-custodial infrastructure for the TradeTech and FinTech industries.

[Create your free GTID →] [Explore how it works ↓]

Why SGTX exists

Every year, $32 trillion moves across borders. The infrastructure underneath? 19th-century paper, 17-document chains, and 9 intermediaries who each take a cut — before the seller gets paid. Existing platforms either hoard your funds, lock you into their walled garden, or digitise spreadsheets without solving the actual problem: no one can prove what happened, to whom, when.
SGTX is different. We don't hold your money. We don't compete with marketplaces, ERPs, or logistics platforms. We provide the provably neutral execution layer — the sovereign

operating system that makes every trade self-verifying, self-enforcing, and jurisdiction-aware. Marketplaces integrate us via API; traders, banks, and governments use us directly. The result: trust that doesn't depend on a phone call — or on giving up control of your goods.

What's on SGTX that doesn't exist anywhere else

27. Your Global Trade Identity (GTID)

A single, verifiable, jurisdiction-tagged identity for every company on the planet.
Format: SGTX-EG-TRD-002139-7F3A. Cryptographic. Unforgeable. Resolves instantly to your KYB-verified profile, trust score, and sanctions clearance — no separate credentials per bank, carrier, or counterparty.

28. Your Trade's DNA — the USTN

Every contract locks into a Unique SGTX Transaction Number:
SGTX-1397F3A-456ABC-20260415120000-A1B2C3D4
It embeds both companies' GTIDs + a deterministic timestamp. No two trades can ever collide. The USTN follows your goods across documents, milestones, payments, and disputes — a single thread of truth from offer to settlement.

29. The Governor — AI that protects, never controls

Every irreversible action (sign, lock, pay, release) must pass the Governor. It enforces:

Your permissions

Sanctions and jurisdiction rules (updated every 15 minutes across 195+ countries)

FeeLock (non-custodial instruction that tells your PSP to split the fee – we never touch your funds)

If a rule blocks you, you instantly see a plain-language explanation — never a cryptic error code.

30. Structurally Non-Custodial FeeLock

No escrow account. No wallet. The FeeLock is code that instructs the payment rail to split the fee at the moment of payment. The platform cannot abscond with funds because it never holds them. This is verifiable on-chain or via your own audit.

31. Zero-Knowledge Proofs Embedded in Every Pallet and Payment

SGTX uses Plonky3 zero-knowledge proofs natively across the platform:

A buyer can prove they paid without revealing the amount, the bank, or the counterparty.

A pallet can prove it's organic-certified without sharing the underlying certificate.

A financier can prove liquidity reserves without exposing their wallet balance.
Privacy and compliance coexist — and no other trade platform does it.

32. Smart Inbox & Trade Command Center

Login → see exactly what needs your attention, prioritised by urgency. Click any USTN → everything about that trade (timeline, docs, finance, risk, logistics) on one screen. No dashboard fatigue.

33. Universal Trade Finance with Full Disclosure & Co-Financing

Financiers see the actual trade, not a credit score. They get the full documents, the counterparty's repayment history, and an AI-generated risk summary — then bid blind. Multiple financiers can co-finance a single shipment. The 0.25% flat fee is deducted from disbursement, not added to your bill.

34. Distressed Cargo Resolution That Respects Local Law

Cargo damaged? AI assesses condition, suggests a price, and lets you check your saved contacts for potential buyers — all before sending a single notification. The micro-contract comes with a separate USTN and a fee adjusted by the destination country's legal requirements.

35. Zero-Cost, Open-Source, Sovereign

Run it on your own bare-metal servers. No cloud dependency. No subscription. No credit card required. The entire stack — Rust, PostgreSQL, WasmEdge, OPA, Loom, Plonky3 — is open-source. You can fork the constitution itself.

Who it's for

Marketplace & platform operators who want to offer their users cryptographic trade certainty, AI-optimised logistics, and zero-counterparty-risk settlement — without building it themselves. Integrate SGTX via API and earn a revenue share on every executed contract.

Exporters who want a single source of truth from EXW price lock to final payment — without the execution layer taking custody of their goods or interfering in commercial relationships.

Importers who need to know exactly what documents are missing before the vessel arrives, and who want a verifiable, neutral witness for every milestone.

Logistics providers who want to compete on performance, not relationships — and who can opt out of anonymous RFQs with one click.

Banks and private credit funds seeking pre-vetted, fully documented trade finance opportunities with AI-powered credit intelligence, full trade disclosure, and co-financing capabilities.

Governments that need a sovereign, auditable, real-time view of cross-border trade without exposing commercial data — complete with zero-knowledge proof options for sensitive transactions.

Built-in add-on value (no extra charge)

62 advanced modules — digital twin, carbon accounting (CBAM-ready), federated fraud detection, zero-knowledge proofs — all activated via toggle.

Automatic sanctions propagation from 300+ official sources in under 15 minutes.

Container release control : no container moves without the logistics provider acknowledging the platform's token.

Multi-shipment contracts with independent per-shipment fees — pay only when each shipment locks.

Dual-mode toggle for trading companies : switch between Buyer and Seller dashboards instantly.

Organisation memory layer : learns your trade patterns to prefill forms (opt-in).

What happens when you click “Create my free GTID”

Onboarding wizard — 6 steps, 10 minutes.

You practice a full trade in your private sandbox (synthetic data, no risk).

Go live. Your Smart Inbox will show your first action.

NEXT STEPS FOR ENGINEERS

Read the full blueprint – especially Parts 0, 1, 2, 3, 5, 6, 13, 14.

Set up the development environment (Phase 0) on each developer’s machine.

Start with Phase 1 – Governor and Identity services.

Use the issue tracker to break down each task into subtasks (e.g., “GTID generation”, “OPA policy for contract.sign”).

Create a shared API specification (OpenAPI) as you build.

Do weekly integration tests to ensure services work together.

Good luck – you are building the future of global trade execution.

APPENDIX A — Portals & End-to-End Workflow Summary

This appendix consolidates every portal in the SGTX platform and the end-to-end trade workflow, cross-referenced back to the precise sections of this blueprint where each component is fully specified.

A.1 Portals — One-Line Summary & Reference Map

Portal / Component	Purpose (one-line)	Blueprint Reference
Buyer Portal (inside Trader Portal)	Structured container-level trade request, multi-shipment scheduler, AI Container Advisor, dual-mode toggle awareness, Smart Inbox integration.	Part 6.2.1, Part 6.2.2 (buyer tabs), Part 3 Phase 1
Seller Portal (inside Trader Portal)	Loading origin, EXW price lock, manual logistics entry or RFQ, alternative ports, packing & containerisation, multi-shipment response, SGTX Fee added to quote (paid by seller).	Part 6.2.1, Part 6.2.2, Part 3 Phase 2
Logistics Portal	Role-based views for Freight Forwarder, Shipping Line, Trucking Company and Customs Broker; addendum signing; container release confirmation; unsubscribe from anonymous RFQs; warm-up program.	Part 6.2.3, Part 3 Phase 2 (addendum), Part 3 Phase 3 (release), Part 3 Phase 5
QC Portal	Dynamic quality specifications, AR-assisted inspection, AQL sampling, mandatory override accountability, QC dispute fast-track.	Part 6.2.4, Part 3 Phase 5, Part 3 Phase 10 (QC fast-track)

Financier Portal	Full disclosure of the underlying trade, financier preferences engine, automatic RFQ broadcast, co-financing, collateral dashboard, historical performance data, flat 0.25% financing fee deducted at disbursement.	Part 6.2.5, Part 3 Phase 4, Part 9 (DeFi/Crypto)
Government Portal	Dynamic module configuration per jurisdiction, anonymous-trade management, customs API integration, regulatory reporting.	Part 6.2.6, Part 8 (Compliance), Part 16 (KYB/KYC)
Admin Portal	Multisig governance oversight, OPA policy simulation, natural-language audit, configuration rollback, tenant impersonation, customer-care chatbot, special-rate management.	Part 6.2.7, Part 13 (Governor), Part 14 (Audit)
Multi-Tenant Administration (per portal)	Employee invitation, RBAC, data scopes, approval chains, tenant activity log, configuration versioning & rollback, tenant branding (free white-label).	Part 6.2.8
Marketplace Partner Portal	Rate-limit gauges, usage analytics, sandbox, revenue-share agreement management, webhook management, lead management.	Part 6.2.9, Part 17 (Marketplace API)
Mobile Application (React Native + Expo)	Offline-first, barcode scanning, voice commands (Vosk offline), AR inspection, offline queue, dual-mode aware, multi-shipment aware.	Part 6.3
Desktop Application (Tauri v2)	System tray, native notifications, offline caching, direct ZPL label printing, auto-update, deep links.	Part 6.4
Common Shared Components	Smart Inbox, Trade Command Center, PlainLanguage Governor Decision Panel, Tenant	Part 6.1.1 – 6.1.12

	Data Export & Reconciliation, Shared Shipments Vault, Accessibility Standards, Tenant Exit & Data Portability, Guided Recovery, SLA Transparency, Four-Questions UX rule, Dual-Mode Toggle.	
Cross-Cutting Features	Self-Sovereign Identity Backup, One-Tap Smart Escrow Claim, Anonymous Market Intelligence Panel, Voice Language Matrix & TTS Fallback, Biometric Stress Flag Consent, Non-Marketplace Footer, Role-Specific Value Cards, Timezone & Holiday Adjustment, Emergency Operations Mode, Operating Complexity Score.	Part 6.5

A.2 End-to-End Trade Workflow — Phase-by-Phase Summary

The SGTX platform implements a ten-phase end-to-end trade workflow with a governance overlay applied at every step. The table below summarises each phase, its inputs, outputs, and the blueprint section where it is specified in full.

Phase	What Happens	Outputs	Blueprint Reference
Phase 1 — Trade Initiation (Buyer Portal)	Buyer creates a structured container-level request with multi-shipment schedule, AI Container Advisor, draft auto-save, marketplace relationship detection, and Governor pre-screen.	Direct GTID entry, container & commodity entry, AI advisory, multi-shipment request, draft submission.	Part 3 Phase 1, Part 6.2.1, Part 6.2.2
Phase 2 — Seller Quote, Packing & Logistics Orchestration	Seller sets loading origin, locks EXW price, enters logistics costs	EXW price, logistics cost, alternative ports, packing plan, multi-shipment	Part 3 Phase 2, Part 19 (Packing), Part 6.2.3 (Logistics)

	(manual or via RFQ), proposes alternative delivery ports, plans packing & containerisation, responds to multi-shipment, and adds SGTX Fee to the quote.	response, fee added to quote.	
Phase 3 — Contracting, Negotiation & Fee Collection	Amendment / negotiation cycle, mutual confirmation, contract assembly with SGTX Witness Clause (or upload own contract with SGTX FEES addendum), per-shipment fee collection, logistics provider signature on addendum, digital signatures & contract lock, container release confirmation.	Signed contract, signed addenda, fee paid, release token issued.	Part 3 Phase 3, Part 8 (Legal)
Phase 4 — Universal Trade Finance	Buyer or seller initiates financing; AI Credit Intelligence; financier preferences + automatic RFQ broadcast; full disclosure to financiers; bidding & co-financing; agreement with SGTX Witness Clause; 0.25% financing fee deducted at disbursement; repayment & release; historical performance data; optional DeFi path.	Financing offers, signed financing agreement, disbursement, repayment.	Part 3 Phase 4, Part 6.2.5 (Financier Portal), Part 9 (DeFi)
Phase 5 — Physical Execution &	Loading window selection, booking confirmation & AI	Booking confirmation, BL, AIS tracking,	Part 3 Phase 5, Part 21 (Barcodes), Part 20 (Trucking)

Multiparty Tracking	extraction, dynamic document requirements, granular provider status tracking, provider invoice upload & AI comparison, barcode scanning & predictive reliability, stuck-trade recovery.	predictive ETA, milestone updates.	
Phase 6 — Settlement & Payment Orchestration	Settlement instruction generation, AI PSP router, buyer approval (voice or Smart Inbox), payment execution & monitoring, AI-powered reconciliation, multi-shipment settlement.	Payment executed, ledger reconciled, USTN closed (or partially).	Part 3 Phase 6, Part 10 (Payment Orchestrator)
Phase 7 — Distressed Cargo Resolution	Proactive demurrage alert, declaration of distressed cargo, AI condition assessment (HF ViT), dynamic AI pricing (XGBoost), triage dashboard, “Check Buyers” advisory, partial distress & USTN splitting, microcontract + separate SGTX fee, accelerated outreach mode, AI-mediated express negotiation, AI-insurer integration.	Distressed listing, microcontract, accelerated buyer outreach.	Part 3 Phase 7
Phase 8 — Distressed Cargo Outreach	Access distressed listing, “Check Buyers”, recipient selection, outreach mode & privacy notice, AI-composed notification, send &	Outreach campaign + microcontract conclusion.	Part 3 Phase 8

	track, recipient response & microcontract.		
Phase 9 — Global Payment Orchestrator & Fee Collection Infrastructure	Fee collection trigger & responsibility, AI-driven PSP router, dynamic fee gross-up, payment initiation & webhook verification, PSP health monitoring & automatic fallback, OpenBanking reconciliation, voice-activated fee payment approval.	Verified fee payment, non-custodial split, reconciled ledger.	Part 3 Phase 9, Part 10
Phase 10 — Dispute Resolution & Arbitration	Filing, AI triage & categorisation, evidence package autocompiler & external verification, predictive dispute outcome, mediation log & multi-language support, AI-generated settlement proposal, smart FeeLock management, document authenticity check, escalation to international arbitration, SGTX FEES dispute workflow, QC dispute fast-track.	Resolved dispute, FeeLock release, optional arbitration award.	Part 3 Phase 10, Part 8.7

A.3 Cross-Cutting Governance Overlay

Every phase above runs through the Governor Service (single point of truth) with the four governing principles G1–G4: Human Supremacy, AI May Block (Never Force), Jurisdiction Supremacy, and Cryptographic Enforcement. Decisions are translated by the PlainLanguage Decision Panel into tenant-friendly explanations generated by Groq (primary) with Ollama/LocalAI fallback. References: Part 0.2 – 0.5, Part 1 (full), Part 6.1.3, Part 13 (Governor).

APPENDIX B — AI Models & API Keys Reference

The SGTX platform follows a three-tier zero-cost AI strategy: a fast cloud advisory tier (Groq, free tier), a privacy-critical self-hosted tier (Hugging Face local models served via vLLM), and a universal fallback tier (Ollama / LocalAI / GPT4All). No paid AI providers, no billing details, and no external secrets are strictly required — every API key listed below is OPTIONAL because each tier has a self-hosted fallback.

B.1 AI Authority Ladder & Provider Tiers

Authority Level	Definition	Recommended Provider & Fallback	Used For (Examples)
A0 — Observation	Read-only inference, no effect on the trade.	Any tier (default Groq if available, otherwise Ollama).	Logging summarisation, dashboard insights.
A1 — Advisory	Suggestions only; cannot block.	Groq (primary, free tier). Fallback: Ollama / LocalAI with quantised Llama 3 8B.	Smart Inbox titles, tenant messages, AI Container Advisor, live market suggestions, AI Loading Guide, AI Port Suggester, Trust Portrait summaries.
A2 — Constraining	Can block, delay, or escalate — never force.	Hugging Face local inference (Mixtral 8x7B, Donut, ViT) via vLLM. Fallback: Ollama / LocalAI.	Compliance pre-screen, sanctions/PEP screening, document extraction, contract clause analysis, Credit Intelligence Agent (A2), Risk Oracle, Liquidation Watch (LSTM), QC override review, dispute triage.
A3 — Escalation	Forces human review; cannot decide autonomously.	Hybrid: HF local for deep analysis + Groq for fast formatting. Fallback: Ollama / LocalAI for both.	Governor orchestrator final routing, high-risk trade approval, dispute escalation, Emergency Multisig path.

A4 — Reserved	Future authority level for fully autonomous (only used after multisig and ratified policy).	Not currently active.	Reserved.

B.2 AI Used per Blueprint Part / Feature

Blueprint Part / Feature	AI Model(s) / Provider(s) Used	Purpose
Part 0.7 / Part 1.2 — Three-Tier AI Strategy	Groq (free, advisory) + HF local (vLLM, privacy-critical) + Ollama/LocalAI/GPT4All (fallback).	Authority ladder A0–A4 implementation.
Part 1.5 / Part 6.1.3 — Governor Decision Explanations	Groq (primary) → Ollama/LocalAI (fallback) → template fallback if both unavailable.	PlainLanguage tenant_message generation for every DENY/CONDITIONAL verdict.
Part 2 — Tenant Onboarding & Lifecycle	Groq (state-change explanations) + Ollama fallback; XGBoost local for trust score (training optional).	Onboarding wizard, lifecycle state explanations, Smart Inbox state-change alerts.
Part 3 Phase 1 — Trade Initiation / AI Container Advisor	Groq (primary) + Ollama (fallback) for container advisory; Vosk offline + Web Speech API for voice intent; HF Donut for document parsing in Express Mode.	Container Advisor banner, voice trade entry, AI Intent Parser (A2).
Part 3 Phase 2 — Seller Quote / Loading & Port Suggestions	Groq (Port Suggester, Loading Guide) + Ollama fallback; XGBoost margin estimator + LLM refinement (HF Mixtral local + Ollama fallback).	AI Port Suggester (A1), AI Loading Guide (A1), Dynamic Margin Estimator (A2).
Part 3 Phase 3 — Contract Assembly	HF Clause Forge agent (local, fallback Ollama); Risk Oracle; Jurisdiction Harmonizer; Groq for negotiation bot messages.	Contract assembly, SGTX Witness Clause, negotiation bot.
Part 3 Phase 4 — Universal Trade Finance	Credit Intelligence Agent (A2, HF local, Ollama fallback); Groq for plain-language risk summary; LSTM Liquidation Watch (A2).	Credit scoring, default probability, recommended LTV, financier RFQ explanations, DeFi PlainLanguage Risk Summary.

Part 3 Phase 5 — Physical Execution	HF Donut (document extraction & eBL parsing); HF ViT (photo condition assessment); Groq (alerts & summaries); PyTorch LSTM (cold-chain shelf-life); XGBoost (demurrage probability); Vosk (offline voice updates); HF Mixtral local (intent extraction).	Booking AI extraction, milestone tracking, predictive cold-chain, demurrage prediction, voice updates.
Part 3 Phase 6 — Settlement & Payment	Groq + Ollama for reconciliation explanations; HF local for AI-powered reconciliation (A2); voice-activated approval (Vosk + HF Mixtral).	AI PSP Router, AI Reconciliation, Voice Approval.
Part 3 Phase 7 — Distressed Cargo Resolution	HF ViT (condition assessment); XGBoost (dynamic pricing); Groq (negotiation summaries); HF Mixtral local (express negotiation, fallback Ollama).	AI Condition Assessment, Dynamic Pricing Engine, AI-Mediated Express Negotiation.
Part 3 Phase 8 — Distressed Cargo Outreach	Groq (AI-composed notifications) + Ollama fallback.	AI-Composed Notification, Recipient Response triage.
Part 3 Phase 9 — Global Payment Orchestrator	AI-driven PSP Router (XGBoost local) + Groq for explanations.	PSP routing, gross-up calculation explanations, OpenBanking reconciliation.
Part 3 Phase 10 — Dispute Resolution	HF local for AI Triage & Evidence Autocompiler (A2); Groq for plain-language settlement proposals; XGBoost for predictive outcome modelling.	AI Triage, Evidence Autocompiler, Predictive Outcome, AI Settlement Proposal.
Part 9 — Hybrid DeFi & Crypto Financing	DeFi Risk Oracle (HF local, fallback Ollama); LSTM Liquidation Watch; Groq for PlainLanguage Risk Summary.	Protocol Risk, Stablecoin Monitor, Liquidation Early Warning, Tokenised Trade Assets.
Part 10 — Global Payment Orchestrator	AI Reconciliation (HF local, Ollama fallback) + Groq for tenant messages.	Non-custodial PSP splits, multi-rail redundancy.
Part 11 — AI Intelligence Layer (deep dive)	All providers: Groq, HF local (vLLM), Ollama/LocalAI, Rig orchestrator (Rust).	Authority ladder routing, fallback chain, agent registry.
Part 14 — Audit & Observability	Loom verifier (local), Prometheus/Grafana/Loki/Ja	Audit log summarisation, natural-language audit

	eger; AI summarisation via Groq + Ollama fallback.	queries.
Part 16 — KYB/KYC & Registration	HF Donut + PaddleOCR (document extraction); local ML models for sanctions/PEP screening; Groq for trust portrait summaries.	Document OCR, sanctions screening, trust score.
Part 18 — Saved Contacts / Network	Vosk (offline voice) + HF Mixtral local (intent extraction); Groq for trust portrait refresh.	Voice-driven contact management, AI-Enriched 360° Trust Portrait.
Part 19 — Commodity & Packing	Groq (advisor) + Ollama (fallback); HF local for packaging analysis.	AI-Driven Packaging Advisor, Capacity Heatmap, Carbon Footprint.
Part 20 — Trucking & Route Intelligence	OSRM (offline routing); XGBoost (predictive dispatching); Vosk (voice navigation) + HF Mixtral (intent); Groq (alerts).	Offline routing, predictive dispatch, voice navigation, e-way bills.
Part 21 — Auto-Generated Barcodes	XGBoost (predictive scanning reliability); Groq (alerts).	Barcode reprint policy, voice & AR integration.
Part 6.3 — Mobile App	Vosk (offline voice), HF Mixtral local (intent), HF ViT (AR inspection), Groq (advisory text).	Offline voice, barcode scanning, AR inspection.
Part 6.4 — Desktop App	Same as web; local model cache for offline use.	System tray notifications, ZPL printing.

B.3 API Keys & Environment Variables

All listed API keys are OPTIONAL. The platform is fully operational with self-hosted models alone (Ollama / LocalAI + Hugging Face local models served via vLLM). External keys merely accelerate certain advisory paths. No paid AI provider, no cloud secrets, and no billing details are required to run the platform.

Variable / Key	Status	How to Obtain	Used For
GROQ_API_KEY	Optional	Free tier (30 req/min, 14,400 req/day). Sign up at groq.com → developer console → create API key. No billing.	Used for A1 advisory inferences (Smart Inbox titles, tenant messages, AI Container Advisor, Port Suggester, Loading Guide, plain-language Governor decision explanations, trust

			portrait summaries).
HF_TOKEN	Optional	Free tier from huggingface.co. Needed only if pulling private/gated models. Public Mixtral / Donut / ViT weights work without a token.	Used by vLLM to download / serve HF models for A2 constraining tasks: document extraction (Donut), contract clauses (Mixtral), image analysis (ViT), credit intelligence.
OLLAMA_URL	Optional (default http://localhost:11434)	Self-hosted Ollama / LocalAI / GPT4All instance. No external API key.	Universal fallback for ALL tiers when Groq rate-limited or HF model unavailable.
DATABASE_URL	Required	PostgreSQL connection string (self-hosted).	Backbone for every service (RLS, tenant scoping, ai_inference_records, governor_decisions, Loom logs).
NATS_URL	Required	Self-hosted NATS messaging.	Inter-service communication, Smart Inbox events, workflow recovery.
SGTX_DEV_PATH	Required	Absolute path to local development tree.	Used by tooling, signing key paths, OPA policy directory.
RUST_BACKTRACE	Recommended	Set to 1 for clearer diagnostics in Rust services.	Debugging.
SQLX_OFFLINE	Recommended	Set to true for reproducible Rust builds with sqlx.	Build pipeline.
Marketplace Partner API Keys (Ed25519)	Auto-generated	Issued by Admin Portal after multisig approval (Part 17). Not external — internally generated, stored hashed.	Marketplace API authentication, webhook signing.

Tenant / Internal API Keys	Auto-generated	Internal only, never external. Stored AES256-GCM encrypted, keys in SoftHSMv2.	Inter-service authentication where JWT is insufficient.
AISHub Feed (optional)	Free / optional	Public AIS receiver feed for vessel positions. No key required for public endpoints; community keys via aishub.net.	Vessel ETA prediction (Part 3 Phase 5).
Nominatim / OSM Tile Server (optional)	Free / self-hosted	Self-hosted OSM tile server + Nominatim geocoder, or public free tier with rate limits.	Trucking, port locations, distress logistics (Part 20).
PSP / OpenBanking Credentials	Per integration	Provided by each PSP and OpenBanking provider per jurisdiction. Stored encrypted; managed under Government Portal & Admin Portal.	Payment execution (Part 10), reconciliation (Part 3 Phase 9).

B.4 Sample .env File

```
# === Required ===
DATABASE_URL=postgres://sgtx:strong-pwd@localhost:5432/sgtx
NATS_URL=nats://localhost:4222
SGTX_DEV_PATH=/srv/sgtx

# === Recommended ===
RUST_BACKTRACE=1
SQLX_OFFLINE=true

# === Optional AI (system runs without these via Ollama fallback) ===
GROQ_API_KEY=gsk_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
HF_TOKEN=hf_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
OLLAMA_URL=http://localhost:11434

# === Optional integrations ===
AISHUB_USERNAME=your_username
NOMINATIM_URL=http://localhost:8080
```

B.5 Closing Note

The SGTX platform is intentionally engineered so that no commercial AI provider, no SaaS API key, and no recurring cloud bill is required to operate the full production stack. The optional Groq free tier and Hugging Face token simply accelerate the advisory path. Every workflow, every portal, and every governance decision falls back gracefully to fully self-hosted models, preserving sovereignty, privacy, and zero-cost guarantees.